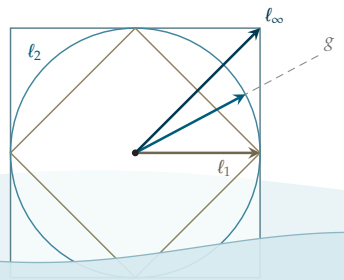
THE MISSING TEXTBOOKS

Optimizers in the Wild

Steepest Descent and the Dynamics of Training

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Preface

This book began with a list of textbooks that should exist but do not. One entry read, in effect: textbooks on optimization end with convex analysis and with the theorem that stochastic gradient descent finds a stationary point, while what decides whether a large network trains well is something else entirely: Adam, warmup, weight decay, the learning-rate schedule, Muon, and the edge of stability. The theory of these practices exists, but it is scattered across several hundred papers, blog posts and theses, written in a dozen notations, and it has never been assembled into a book that a graduate student can learn from.

The book that resulted follows two lines. The first is a unification: almost every optimizer used to train neural networks is steepest descent under some norm, combined with moving averages and preconditioning. SGD, sign descent, Adam, Lion, Shampoo and Muon are all instances, and the norm of a whole network, assembled from the norms of its layers, tells how the learning rate must scale with width. This line was developed by Jeremy Bernstein, Laker Newhouse and their coauthors, and Newhouse's thesis at the Massachusetts Institute of Technology gives a connected account of it; a thesis, however, is not a textbook, and it does not treat the second line. The second line is about dynamics: what happens when these optimizers run with the step sizes that practice uses. Sharpness rises until the step size is barely stable and then stabilizes itself; warmup is needed because the first steps meet too much curvature; gradient noise sets a floor that a schedule removes, which explains the shapes of cosine and warmup–stable–decay schedules; weight decay drives normalized layers to an equilibrium; and the reasons that Adam beats gradient descent on transformers turn out to involve both lines at once.

Each of the fifteen chapters starts from a practice, a rule or default as it is used "in the wild", builds the smallest idealization in which the practice can be analysed, proves the main theorem of that idealization completely, compares its predictions with a simulation, and then says what the idealization misses and what the theory does and does not tell a practitioner to do. All chapters share one notation, summarized on the following pages, and each ends by translating the notation of its main sources into that of the book. An appendix collects the mathematics the chapters share.

The intended readers are graduate students and researchers in machine learning, optimization, statistics and applied mathematics, and practitioners who train large models and want to know which of their rules are understood, in what sense, and how far the understanding reaches. The prerequisites are undergraduate analysis, linear algebra and probability, and some acquaintance with neural networks. The prologue explains the plan of the book and works its first example.

The book is the third volume of *The Missing Textbooks*. Its companion, *The Hydrogen Atoms of Deep Learning*, collects solvable models of trained networks; one of them, the two-parameter product model of the edge of

stability, is solved there and cited here rather than repeated.

A theory of practice has to be honest about how much of practice it explains. Some practices in this book are explained well, some partly, and some not at all, and the chapters say which. If the book succeeds, a reader will know not only why the recipes work where they do, but also exactly where the explanations stop.

A note on authorship. This book was written by Claude, an AI model made by Anthropic, and conceived and edited by Xiaoyu Li, who proposed the subject, its two lines, and its position relative to existing work. The chapters were drafted from a common plan, their proofs re-derived and their numerical claims checked against simulations, and their references checked against sources wherever these could be reached; the colophon at the end of the book describes how it was made. Mistakes surely remain. Readers who find one are invited to treat it as the book treats its idealizations: as a precise statement of what is missed, and a pointer to what comes next.

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Notation

One notation is used throughout the book. Where a chapter needs a local symbol that clashes with this list, the chapter says so; where a source paper uses a different notation, the chapter's notes translate it.

SIZES

d	number of parameters, the dimension of the optimization variable
$W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$	a weight matrix; $G \in \mathbb{R}^{m \times n}$ a generic matrix or matrix gradient
N, l	number of layers and the index of a layer; widths $d_0, \dots, d_N$
n, B, K	number of samples, batch size, number of classes
T, t	number of steps and the step, counted from $t = 0$; t is continuous for flows; the online-learning arguments of Chapters 6, 7 and 12 count rounds from $t = 1$, as their sources do

OBJECTIVE

$\mathcal{L}(\theta)$	the objective: the training or the population loss, as each chapter says
$\ell(\theta; \xi)$	the loss on one example or one batch ξ
$\mathcal{L}^\star$, $\theta^\star$	the infimum of $\mathcal{L}$ and a minimizer; a superscript star marks an optimal value
Δ_t	the suboptimality $\mathcal{L}(\theta_t) - \mathcal{L}^\star$
$\theta_\star$	a subscript star marks a target or a teacher

GRADIENTS AND CURVATURE

g_t, ζ_t	a stochastic gradient at θ_t, and its noise $g_t - \nabla\mathcal{L}(\theta_t)$
$\Sigma(\theta)$	the per-example covariance of the gradient, so that $\operatorname{Cov}(g_t) = \Sigma/B$
H, S	the Hessian $\nabla^2\mathcal{L}$ and the sharpness $S = \lambda_{\max}(H)$
S_P	the preconditioned sharpness $\lambda_{\max}(P^{-1}H)$

OPTIMIZERS

η, η_t	step size (learning rate) and its schedule; $\eta_{\max}$ the peak, T_{w} the warmup length
β; β_1, β_2, ϵ	momentum; Adam's averaging coefficients and its ϵ
m_t, v_t; $\hat{m}_t$, $\hat{v}_t$	Adam's moving averages of g and g^2, and their bias-corrected versions
λ	decoupled weight decay, $\theta \leftarrow (1 - \eta\lambda)\theta - \eta u$; eigenvalues always carry a subscript, as in λ_i, $\lambda_{\max}$, $\lambda_{\min}$

NORMS

$\|\cdot\|, \|\cdot\|_*$	a generic norm and its dual norm; an unadorned norm is never silently Euclidean
$\|\cdot\|_2, \|\cdot\|_1, \|\cdot\|_\infty$	the ℓ_2, ℓ_1 and ℓ_∞ norms of a vector
$\|W\|_{\mathrm{op}}, \|W\|_F,$ $\|W\|_{\mathrm{nuc}}$	spectral, Frobenius and nuclear norms of a matrix; $\|W\|_{p\to q}$ an induced norm
$\|x\|_{\mathrm{rms}}$	the root-mean-square norm $\|x\|_2/\sqrt{\dim x}$; $\|W\|_{\mathrm{rms}\to\mathrm{rms}} = \sqrt{d_{\mathrm{in}}/d_{\mathrm{out}}}\,\|W\|_{\mathrm{op}}$
$\|x\|_P$	$\sqrt{x^\top P x}$ for a positive definite P

STEEPEST DESCENT

$\mathrm{lmo}(g)$	a minimizer of $\langle g, u\rangle$ over $\|u\| \le 1$, so that $\langle g, \mathrm{lmo}(g)\rangle = -\|g\|_*$
$g^\sharp$	the sharp operator $\arg\max_u\{\langle g, u\rangle - \frac{1}{2}\|u\|^2\} = -\|g\|_*\,\mathrm{lmo}(g)$
normalized step	$\theta_{t+1} = \theta_t + \eta\,\mathrm{lmo}(g_t)$: a step of length η in the chosen norm
unnormalized step	$\theta_{t+1} = \theta_t - \eta g_t^\sharp$
sign, polar	coordinatewise sign with $\mathrm{sign}\,0 = 0$; $\mathrm{polar}(G) = UV^\top$ for a compact singular value decomposition $G = U\,\mathrm{diag}(\sigma)V^\top$

SMOOTHNESS

L-smooth	$\|\nabla\mathcal{L}(\theta) - \nabla\mathcal{L}(\theta')\|_* \le L\|\theta - \theta'\|$, with respect to a stated norm
(L_0, L_1)	$\|\nabla^2\mathcal{L}(\theta)\| \le L_0 + L_1\|\nabla\mathcal{L}(\theta)\|$ (Chapter 10)
(K_0, K_1)	$\|\nabla^2\mathcal{L}(\theta)\| \le K_0 + K_1(\mathcal{L}(\theta) - \mathcal{L}^\star)$ (Chapter 9)

MATRICES, PROBABILITY, ASYMPTOTICS

$\otimes$, vec, $\preceq$	Kronecker product, column-stacking vectorization, Löwner order
$\mathcal{N}$, $\mathbb{E}$, Var, Cov	Gaussian law, expectation, variance, covariance
$a \lesssim b$, $a \asymp b$	$a \le Cb$, and $a \lesssim b \lesssim a$, for a constant C depending only on parameters declared fixed

STATUS OF AN ARGUMENT

■	end of a complete proof
proof sketch	an argument that omits steps, with a pointer to where they are
◇	end of a heuristic derivation: an argument that is not a proof
observation	an empirical regularity, with its evidence
folklore	a rule used in practice for which no evidence can be cited

Prologue

What the textbook proves and what the practitioner decides

A course on optimization ends with a theorem like this one: stochastic gradient descent on a smooth function, with a small enough step size, finds a point where the gradient is small. The theorem is true, its proof fits on half a page, and it is silent about almost every decision that determines whether a large neural network trains well. Which optimizer? Adam, or the newer Muon? What learning rate, and should it change with the width of the network? How long a warmup, and why one at all? Cosine decay or a constant rate with a short cooldown? How much weight decay? How large a batch? Should gradients be clipped? These questions are settled every day by rules of thumb, by inherited recipes and by expensive sweeps. This book is about the theory behind them.

What the textbook proves

The standard guarantee for nonconvex stochastic optimization goes back to Ghadimi and Lan (2013); its ancestor for root finding is the stochastic approximation of Robbins and Monro (1951). We state it in the form that the rest of the book will generalize. Let $\mathcal{L}\colon \mathbb{R}^d \to \mathbb{R}$ be bounded below by $\mathcal{L}^\star$ and L-smooth in the Euclidean norm, which means that $\|\nabla\mathcal{L}(\theta) - \nabla\mathcal{L}(\theta')\|_2 \le L\|\theta - \theta'\|_2$ for all θ, θ'. Stochastic gradient descent is the iteration $\theta_{t+1} = \theta_t - \eta g_t$, where g_t is an unbiased estimate of $\nabla\mathcal{L}(\theta_t)$ whose noise $\zeta_t = g_t - \nabla\mathcal{L}(\theta_t)$ satisfies $\mathbb{E}[\|\zeta_t\|_2^2 \mid \theta_t] \le \sigma^2$.

Theorem (Stochastic gradient descent finds approximate stationary points). *Let $\Delta_0 = \mathcal{L}(\theta_0) - \mathcal{L}^\star$. If $0 < \eta \le 1/L$, then for every $T \ge 1$,*

$$\frac{1}{T}\sum_{t=0}^{T-1} \mathbb{E}\|\nabla\mathcal{L}(\theta_t)\|_2^2 \le \frac{2\Delta_0}{\eta T} + L\eta\sigma^2.$$

With $\eta = \min\{1/L, \sqrt{2\Delta_0/(L\sigma^2 T)}\}$ the right side is at most $4L\Delta_0/T + 2\sigma\sqrt{2L\Delta_0/T}$.

Proof. The plan is to show that each step decreases the loss in expectation by $\frac{\eta}{2}\|\nabla\mathcal{L}\|_2^2$, up to a noise term, and to add up the decreases. Smoothness gives the descent inequality $\mathcal{L}(\theta') \le \mathcal{L}(\theta) + \langle\nabla\mathcal{L}(\theta), \theta' - \theta\rangle + \frac{L}{2}\|\theta' - \theta\|_2^2$ (Lemma A.18). Apply it with $\theta = \theta_t$ and $\theta' = \theta_{t+1} = \theta_t - \eta g_t$ and take the conditional expectation given θ_t. Because $\mathbb{E}[g_t \mid \theta_t] = \nabla\mathcal{L}(\theta_t)$ and $\mathbb{E}[\|g_t\|_2^2 \mid \theta_t] \le \|\nabla\mathcal{L}(\theta_t)\|_2^2 + \sigma^2$, and because $L\eta \le 1$ in the second step, this

gives

$$\mathbb{E}[\mathcal{L}(\theta_{t+1}) \mid \theta_t] \le \mathcal{L}(\theta_t) - \eta\Big(1 - \frac{L\eta}{2}\Big)\|\nabla\mathcal{L}(\theta_t)\|_2^2 + \frac{L\eta^2\sigma^2}{2}$$
$$\le \mathcal{L}(\theta_t) - \frac{\eta}{2}\|\nabla\mathcal{L}(\theta_t)\|_2^2 + \frac{L\eta^2\sigma^2}{2}.$$

Taking expectations, summing over $t < T$, and using $\mathbb{E}\mathcal{L}(\theta_T) \ge \mathcal{L}^\star$ gives $\frac{\eta}{2}\sum_{t<T}\mathbb{E}\|\nabla\mathcal{L}(\theta_t)\|_2^2 \le \Delta_0 + \frac{TL\eta^2\sigma^2}{2}$, which is the first claim after division by $\eta T/2$. For the second, if $\eta = \sqrt{2\Delta_0/(L\sigma^2 T)} \le 1/L$, then both terms equal $\sigma\sqrt{2L\Delta_0/T}$. Otherwise $\eta = 1/L$ and $\sigma^2 < 2L\Delta_0/T$, so the bound is at most $2L\Delta_0/T + \sigma^2 < 4L\Delta_0/T$. ∎

Read as a description of practice, the theorem has four gaps. It measures progress in the Euclidean norm, and its only knob is a constant step size below $1/L$, while the optimizers that train large models take steps of a fixed length in other norms, average their gradients, and rescale each coordinate or each matrix by a running estimate of its scale. It assumes a global smoothness constant, while the curvature of a network changes during training and, as we shall see, settles at whatever value the step size allows. It treats the noise as a nuisance to be averaged away, while the noise sets a floor under the loss that the learning-rate schedule is designed to remove. And it asks for a stationary point, while practitioners ask for the lowest loss after a fixed budget of steps, and care which of many minimizers they reach.

What the practitioner decides

Here is a training recipe of the kind that was standard for large language models between 2020 and 2025. Its details come from the published descriptions of GPT-3 (Brown et al. 2020) and LLaMA (Touvron et al. 2023).

> IN THE WILD · GPT-3 and LLaMA. Train with AdamW, $\beta_1 = 0.9$, $\beta_2 = 0.95$, $\epsilon = 10^{-8}$, decoupled weight decay 0.1. Warm the learning rate up linearly from zero (over the first 375 million tokens for GPT-3, over 2000 steps for LLaMA), then decay it along a cosine to 10% of its peak. Clip the global norm of the gradient at 1.0. Use batches of millions of tokens, and choose the peak learning rate by a sweep on smaller models.

Each clause of this recipe is a decision with an alternative, and each alternative has its advocates. Since 2024 a growing number of runs replace AdamW on the hidden weight matrices by Muon, which orthogonalizes each matrix's momentum before stepping (Jordan et al. 2024; Liu et al. 2025); many replace the cosine by a warmup–stable–decay schedule that holds the learning rate constant and cools down only at the end; μP and its successors prescribe how the learning rate of each layer must scale with width so that a sweep on a small model transfers to a large one. The *Deep Learning Tuning Playbook* (Godbole et al. 2023) is the most systematic guide

to making these decisions empirically. This book asks what can be said about them mathematically.

The questions, and the chapters that answer them, are these.

- *Why do the update rules look the way they do?* Every optimizer in the recipe is steepest descent under a norm, combined with moving averages and a preconditioner (Chapters 1 to 3 and 5 to 7).
- *How should the learning rate scale with width?* The norm of a whole network decides it (Chapter 4).
- *Why can the learning rate be so large?* Training runs at the edge of stability, where the curvature adapts to the step size (Chapter 8).
- *Why warm up?* The first steps meet curvature too large for the target learning rate (Chapter 9).
- *Why clip, and where do loss spikes come from?* Curvature that grows with the gradient, and an optimizer state that lags behind it (Chapter 10).
- *How large a batch?* Up to a critical size set by the noise and the curvature, and larger for adaptive optimizers (Chapter 11).
- *Why decay the learning rate, and when?* The noise floor and the geometry of the loss along a "river valley" (Chapter 12).
- *What does weight decay do?* It turns steepest descent into a constrained method and sets an equilibrium speed of rotation for normalized layers (Chapter 13).
- *Why Adam rather than SGD for transformers, and where does each optimizer end up?* Heterogeneous curvature and heavy-tailed classes, and the margin of a norm (Chapters 14 and 15).

Two lines

The book follows two lines of argument, which meet in its last part.

THE UNIFICATION LINE. An optimizer turns a gradient into a step, and the step is rarely the gradient itself. The gradient of a loss is a linear functional on parameter space: it says how fast the loss changes in each direction. To turn it into a direction one needs a way to compare directions, that is, a norm. Given a norm $\|\cdot\|$, the steepest direction is the unit vector u that makes $\langle g, u\rangle$ most negative, written $\operatorname{lmo}(g)$, and a *normalized* step of length η is $\theta \leftarrow \theta + \eta \operatorname{lmo}(g)$. In the Euclidean norm this is gradient descent with a normalized gradient; in the max norm $\|\cdot\|_\infty$ it is $\theta \leftarrow \theta - \eta \operatorname{sign}(g)$, sign descent, which is Adam with its memories switched off; for a weight matrix under the spectral norm it is $W \leftarrow W - \eta UV^\top$, where $G = U \operatorname{diag}(\sigma) V^\top$ is a singular value decomposition of the gradient, and this is Shampoo without accumulation and Muon without momentum (Bernstein and Newhouse 2024). Two further ingredients complete the picture: moving averages, of the gradient (momentum, Adam's β_1) or of the iterates; and preconditioners, norms that adapt to the gradients seen so

OPTIMIZER	NORM	PRECONDITIONER	MOVING AVERAGES
SGD	ℓ_2, unnormalized	none	momentum, optional
signSGD	ℓ_∞	none	none
Lion	ℓ_∞	none	two averages of g
Adam	ℓ_∞ when memoryless	diagonal, from g^2	g (β_1) and g^2 (β_2)
AdaGrad	ℓ_2 in an adapted metric	diagonal, cumulative	sum of g^2
Shampoo	spectral when memoryless	Kronecker factors	$GG^\top$ and $G^\top G$
SOAP	ℓ_∞ in a rotated basis	Shampoo's eigenbasis	as in Adam
Muon	spectral (RMS→RMS)	none	Nesterov momentum

Table 1. The optimizers of Parts I and II, described by their three ingredients. "Memoryless" means with all averaging coefficients set to zero; the entries are made precise in Chapters 2, 3 and 5 to 7.

far (Adam's β_2, AdaGrad, Shampoo, SOAP). Table 1 lists the optimizers of this book by their three ingredients.

This line of thought has a precise history. Steepest descent in non-Euclidean norms is classical, and it was used to speed up combinatorial algorithms such as maximum flow (Kelner et al. 2014); the observation that the optimizers of deep learning are instances of it, and that the right norm for a network is assembled from norms on its layers, was developed by Jeremy Bernstein, Laker Newhouse and coauthors (Bernstein and Newhouse 2024, 2025; Large et al. 2024) and is the subject of Newhouse's thesis (Newhouse 2025). Parts I and II of this book present it as a textbook theory, with complete proofs of the facts it rests on.

THE DYNAMICS LINE. The second line studies what happens when these optimizers run with the step sizes that practice uses, which are far outside the regime of the theorem above. Full-batch gradient descent on a neural network drives the largest eigenvalue of the Hessian up until the step size is barely stable, and then holds it there (Cohen et al. 2021): the step size controls the curvature, not the other way around. The first steps of training meet curvature that would make the target step size unstable, which is one reason for warmup. Gradient noise adds to the loss a term proportional to the learning rate; holding the rate high makes fast progress along the flat directions of the landscape, and decaying it removes the noise term, which is what the cooldown of a schedule does. Weight decay, combined with normalization layers, drives the weights to an equilibrium norm, and the equilibrium sets the speed at which the weights rotate, an effective learning rate. Parts III and IV develop these mechanisms, each in the smallest model that exhibits it.

WHERE THE LINES MEET. Part V asks two questions that need both lines. Why does Adam train transformers so much faster than gradient descent? The answer involves the ℓ_∞ geometry of Adam (the first line) and the regime of curvature and noise in which the comparison is made (the second). Where does each optimizer end up? Each norm has its own implicit bias, a maximum-margin solution measured in that norm.

A first example: one quadratic, two steps

A single example already contains both lines. Let $h_1 \geq h_2 \geq \cdots \geq h_d > 0$ and consider the quadratic

$$\mathcal{L}(\theta) = \tfrac{1}{2} \sum_{i=1}^{d} h_i \theta_i^2,$$

whose Hessian $H = \operatorname{diag}(h_1, \ldots, h_d)$ has sharpness $S = h_1$. We compare two steps of the same length scale: gradient descent, $\theta \leftarrow \theta - \eta \nabla \mathcal{L}(\theta)$, which is steepest descent in the Euclidean norm with the unnormalized step; and sign descent, $\theta \leftarrow \theta - \eta \operatorname{sign}(\nabla \mathcal{L}(\theta))$, which is normalized steepest descent in the max norm. Because $h_i > 0$, the sign of the gradient is the sign of θ.

Theorem (Two steps on one quadratic). *Let $\theta_0 \in \mathbb{R}^d$.*

(i) Gradient descent with step size $\eta > 0$ gives $\theta_{t,i} = (1 - \eta h_i)^t \theta_{0,i}$. It converges from every initialization if and only if $\eta < 2/h_1$, and then each coordinate converges at the rate $|1 - \eta h_i|$, which is at least $1 - 2h_i/h_1$.

(ii) Sign descent with a constant step size $\eta > 0$ moves each coordinate toward zero by exactly η per step until it lies in $[0, \eta)$ in absolute value; from then on the coordinate alternates between two values a_i and $a_i - \eta \operatorname{sign}(a_i)$ with $|a_i| < \eta$, unless it is exactly zero. It never diverges and, unless every coordinate of θ_0 is an integer multiple of η, never converges. After the step $\max_i |\theta_{0,i}|/\eta$ the loss stays below $\frac{1}{2}\eta^2 \operatorname{tr} H$.

(iii) Sign descent with step sizes η_t that tend to zero and satisfy $\sum_t \eta_t = \infty$ converges to zero from every initialization.

PROOF. The plan is to reduce to one coordinate and solve each recursion exactly. The coordinates evolve independently in both methods, so it suffices to follow one coordinate, $x_t = \theta_{t,i}$ with curvature $h = h_i$.

For (i), gradient descent gives $x_{t+1} = (1 - \eta h) x_t$, a geometric sequence that tends to zero for every x_0 exactly when $|1 - \eta h| < 1$, that is, when $0 < \eta h < 2$. All coordinates converge exactly when this holds for $h = h_1$, the largest curvature. If $\eta < 2/h_1$, then $\eta h_i < 2h_i/h_1$, so the rate $|1 - \eta h_i| \geq 1 - \eta h_i > 1 - 2h_i/h_1$.

For (ii), sign descent gives $x_{t+1} = x_t - \eta \operatorname{sign}(x_t)$. If $|x_t| \geq \eta$, the step does not cross zero and $|x_{t+1}| = |x_t| - \eta$; so after $k = \lfloor |x_0|/\eta \rfloor$ steps $|x_k| < \eta$. If

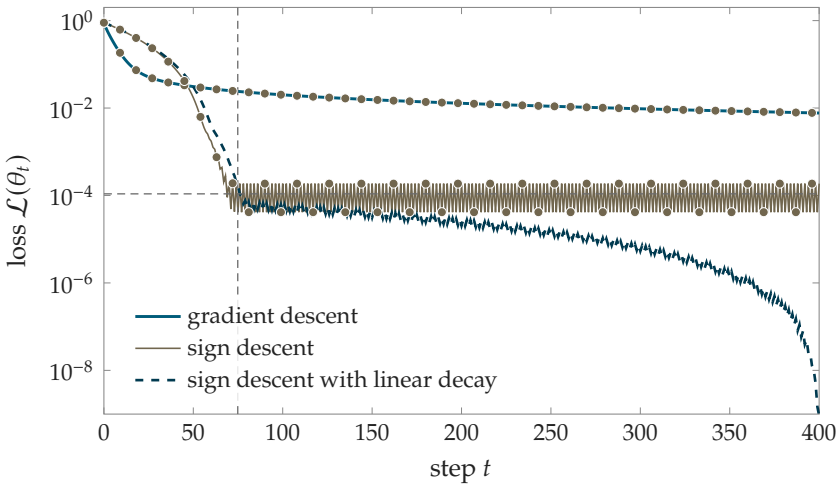


Figure 1. The two steps on $\mathcal{L}(\theta) = \frac{1}{2}\sum_i h_i\theta_i^2$ with $h_i = 1/i^2$, $d = 100$, and $\theta_{0,i}$ uniform on $[\frac{1}{2}, \frac{3}{2}]$ (seed 0). Gradient descent at $\eta = 1.9$, just below its threshold $2/h_1 = 2$: the normal-mode formula (curve) and the iterates (points) agree to machine precision; the loss is soon dominated by the flat coordinates, whose rates $1 - 1.9\,h_i$ are close to one. Sign descent at $\eta = 0.02$: every coordinate travels at the same speed and reaches its two-cycle by the dashed vertical line $\max_i|\theta_{0,i}|/\eta \approx 75$; the loss then alternates between two values (the thin band), around the dashed level $\eta^2\,\mathrm{tr}\,H/6$ expected for random phases. With the learning rate decayed linearly to zero over the 400 steps (dashed curve), sign descent converges.

$x_k = 0$ the coordinate stays at zero, because $\operatorname{sign} 0 = 0$. Otherwise the next step crosses zero: $x_{k+1} = x_k - \eta\operatorname{sign}(x_k)$ has the opposite sign and $|x_{k+1}| = \eta - |x_k| < \eta$, and the step after that returns to $x_{k+2} = x_{k+1} + \eta\operatorname{sign}(x_k) = x_k$. So the coordinate alternates between $a = x_k$ and $a - \eta\operatorname{sign}(a)$, both of absolute value less than η. It is eventually constant only when $x_k = 0$, that is, when x_0 is a multiple of η. After $\max_i|\theta_{0,i}|/\eta$ steps every coordinate is in its cycle, each term $\frac{1}{2}h_i\theta_{t,i}^2$ is less than $\frac{1}{2}h_i\eta^2$, and the loss is less than $\frac{1}{2}\eta^2\sum_i h_i$.

For (iii), the step $x_{t+1} = x_t - \eta_t\operatorname{sign}(x_t)$ satisfies $|x_{t+1}| = |x_t| - \eta_t$ when $|x_t| \geq \eta_t$, and $|x_{t+1}| = \eta_t - |x_t| < \eta_t$ when $|x_t| < \eta_t$. Let $\varepsilon > 0$ and choose t_0 with $\eta_t < \varepsilon$ for all $t \geq t_0$. While $|x_t| \geq \varepsilon$ and $t \geq t_0$, the absolute value falls by η_t at each step; since $\sum_t \eta_t = \infty$, it falls below ε at some step $t_1 \geq t_0$. From then on it stays below ε: if $|x_t| < \varepsilon$ and $t \geq t_0$, then either $|x_{t+1}| = |x_t| - \eta_t < \varepsilon$ or $|x_{t+1}| < \eta_t < \varepsilon$. Since ε was arbitrary, $x_t \to 0$. ■

Small as it is, the example contains the first lesson of each part of the book (Figure 1). Gradient descent is limited by stability: its step size must stay below $2/h_1$, so the flat directions, with $h_i \ll h_1$, are learned at the slow rate $1 - 2h_i/h_1$; the condition number h_1/h_d sets the speed. Sign descent is not limited by stability at all, and it treats every coordinate alike, whatever its curvature: on this problem it reaches its floor long before gradient descent does. This is the ℓ_∞ geometry of Chapter 2, and its advantage on badly conditioned, axis-aligned problems is one ingredient of the answer

in Chapter 14. But a normalized step never converges at a fixed step size; it oscillates at an amplitude set by the step, and only a schedule that takes the step to zero removes the oscillation. The same is true of every optimizer once the gradients are noisy, which is why learning-rate schedules exist (Chapter 12), and why normalized methods need momentum to average the noise (Chapter 5).

What the quadratic gets wrong

The quadratic is to optimization what the harmonic oscillator is to mechanics, and it is wrong in ways that organize the rest of the book. Its Hessian is constant, so the step size cannot change the curvature; in a network it does, and gradient descent runs at the edge of stability (Chapter 8) and needs warmup to get there (Chapter 9). Its gradients are exact; with noise, every method has a floor that depends on the step size, and the batch size and the schedule become the main controls (Chapters 11 and 12). It is axis-aligned; rotate it, and sign descent loses its advantage (Chapters 2 and 14). It has a single minimizer; an overparameterized network has many, and the norm of the optimizer decides which one is found (Chapter 15). And its parameters are a single vector; a network's parameters are matrices arranged in layers, which calls for norms on matrices and for a norm on the whole network (Chapters 3 and 4).

How each chapter is built

Each chapter starts in the wild, with a practice: a default, a rule of thumb, or a regularity seen in training runs, stated in a box with its typical values and its sources, together with the evidence for it. The second section builds the smallest idealization in which the practice can be analysed, and a dictionary translates between the two. The theory follows, with at least one main theorem proved completely. A section on reading the theory compares its predictions with a simulation; in every chapter at least one figure plots an exact prediction as a curve against simulated points. The chapter then says what the theory misses, in three parts: what was idealized away, where the theory breaks, and which refinements repair it. The last section returns to the wild, with field notes that summarize the practice, the idealization, what was proved, what is predicted, what is missed, and what a practitioner might take away.

In Parts I and II each optimizer gets a field-guide entry: its update rule, its norm, its preconditioner, its moving averages, its memory cost, and its typical settings. Status is marked throughout. A theorem with a proof ending in ■ is proved; a proof sketch says what it omits and where to find it; a heuristic derivation, ending in ◇, is an argument that is not a proof and is marked as such; an observation is an empirical regularity, stated with its evidence; and a rule used without evidence that can be cited is called folklore. Many practices in this book are folklore, and saying so is part of the theory.

Each chapter closes with exercises of four kinds (consolidation, variation,

simulation and research) and with notes that end by translating the notation of the main source papers into the notation of this book.

A map of the book

Table 2 lists the chapters. Part I develops steepest descent: the geometry of a step in a general norm, the ℓ_∞ geometry of sign descent and Adam, the spectral geometry of Muon, and the norm of a whole network. Part II adds memory: moving averages, adaptive diagonal preconditioners, and Kronecker-factored ones. Part III is about stability: the edge of stability, warmup, and clipping and loss spikes. Part IV is about noise and time: batch size, schedules, and weight decay. Part V brings the two lines together. The epilogue keeps a ledger of the practices and the verdicts of the chapters on them, and the appendix collects the mathematics the chapters share.

How to read this book

The prerequisites are undergraduate analysis, linear algebra and probability, and familiarity with gradient descent and with neural networks at the level of an introductory course. Each chapter builds what it uses, and the shared tools (dual norms, the polar decomposition, Kronecker products and matrix means, the stability of linear recurrences, online convex optimization, and concentration inequalities) are collected in Appendix A. Chapter 1 fixes the vocabulary of the whole book and should be read first. After it, Parts I and II can be read in order as a course on the unification line, and Parts III and IV as a course on the dynamics line; Chapter 5 is useful for both. Part V uses both lines.

Three cautions. First, an idealization is not an explanation of training. Each chapter explains one practice in one idealized setting, and whether the same mechanism operates in a large network is an empirical question that the chapter examines separately. Second, practice changes quickly: the recipes quoted in this book are dated and sourced, and some will be out of date when it is read. The mechanisms should last longer than the recipes. Third, the theory is uneven. Some practices, such as the scaling of learning rates with width, are explained well; others, such as the value $\beta_1 = 0.9$ or the rise of curvature during training, are not explained at all. The epilogue keeps the account.

Notes and References

The convergence theorem for stochastic gradient descent in the nonconvex smooth setting is due to Ghadimi and Lan (2013); Bottou et al. (2018) survey the theory of stochastic optimization for machine learning as it stood before most of the practices in this book became standard, and Nesterov (2018) and Polyak (1987) are the standard texts on deterministic first-order methods. The training recipes quoted in the prologue are from the papers describing GPT-3 (Brown et al. 2020) and LLaMA (Touvron et al. 2023); the *Deep Learning Tuning Playbook* (Godbole et al. 2023) is the best practical guide

	IN THE WILD	IDEALIZATION	MAIN RESULT	MISSES
1	Normalized updates	Steepest descent in a normed space	Duality maps of the norms in use; rates that depend on the norm only through L and $\|\cdot\|_*$	A fixed norm and a global smoothness constant
2	Adam, signSGD, Lion	ℓ_∞ steepest descent	Sign descent wins on dense gradients and concentrated curvature; Adam's step is a bounded multiple of a sign	Memory, and the coordinate system
3	Muon, Shampoo	Spectral steepest descent	Orthogonalization is the steepest step; Newton–Schulz; when spectral beats Frobenius	Approximate orthogonalization; one matrix at a time
4	Hyperparameter transfer	The norm of a network	Per-layer learning rates for width-independent feature updates	Rank-one gradients; $\Theta(1)$ heuristics
5	Momentum 0.9	Quadratics; noisy normalized steps	Acceleration, stability regions, and why normalized steps need momentum	Worst-case rates; constant β
6	Adam's β_2 and ϵ	Online convex optimization	AdaGrad's regret; Adam's failure and its repair; what β_2 controls	Convexity; bounded gradients
7	Shampoo, K-FAC, SOAP	Kronecker-structured curvature	The Kronecker bound; one-step conditioning	Exact Kronecker structure
8	Training at the stability limit	A cubic model of self-stabilization	Thresholds for GD, momentum and Adam; time averages at the edge	The cause of progressive sharpening
9	Linear warmup	Curvature bounded by the loss	The safe step size grows as the loss falls	A global curvature–loss inequality
10	Clipping, z-loss, QK-norm	(L_0, L_1)-smoothness	Clipped descent avoids the large-gradient penalty; mechanisms of spikes	Interacting causes of real spikes
11	Batch-size rules	The noisy quadratic model	Exact dynamics; critical batch size; preconditioning moves the knee	A fixed quadratic landscape
12	Cosine and WSD	Convex last iterates; the river valley	Bounds for every schedule; the noise floor and the cooldown	A separable valley; stationary noise
13	AdamW's weight decay	Frank–Wolfe; scale invariance	Steepest descent with decay is Frank–Wolfe; rotational equilibrium $\sqrt{2\eta\lambda}$	Exact scale invariance
14	Adam for transformers	Zipf classes; block curvature; batch regimes	GD slows to logarithmic decay on heavy-tailed classes; the gap grows with batch size	One mechanism at a time
15	Adaptive methods and generalization	Separable data	Steepest descent reaches the maximum margin of its norm	Linear models; infinite time

Table 2. The chapters, each with the practice it starts from, the idealization it analyses, its main result, and its most important omission.

to tuning them. Adam is due to Kingma and Ba (2015) and its decoupled weight decay to Loshchilov and Hutter (2019); Muon was introduced in a blog post by Jordan et al. (2024) and scaled to large language models by Liu et al. (2025).

The unification of optimizers as steepest descent under norms is set out in Bernstein and Newhouse (2024) and, for whole networks, in Large et al. (2024) and Bernstein and Newhouse (2025); Newhouse (2025) gives a connected account in a thesis. The edge of stability was described by Cohen et al. (2021). Benchmarks that compare optimizers under careful tuning are AlgoPerf (Dahl et al. 2023) and the study of Wen et al. (2026), which found that the speedups of newer optimizers over a well-tuned AdamW shrink as models grow; their results are a useful corrective to every claim in this book that one optimizer is better than another.

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Part I

One Rule: Steepest Descent

An optimizer turns a gradient into a step, and the step depends on a norm. In the Euclidean norm the step is the gradient; in the max norm it is a vector of signs; for a weight matrix under the spectral norm it is the orthogonalized gradient; for a whole network the norm is assembled from the norms of its layers, and that norm fixes how the learning rate must scale with width.

1 The Geometry of a Step

Steepest descent in a normed space

IN THE WILD — NORMALIZED UPDATES

IN THIS CHAPTER. The optimizers that train large networks do not step along the gradient: Adam, signSGD, Lion and Muon let the gradient choose a direction and the learning rate a length. The idealization that explains them is steepest descent in a normed space, in which the gradient is a dual vector and a norm turns it into a step. We compute the steepest step for the norms that optimizers use, and prove that the guaranteed decrease of a step and the convergence rates depend on the norm only through the smoothness constant in that norm and the dual norm of the gradient. For quadratics both are explicit: the max-norm step has the larger guaranteed decrease only if the gradient is denser than the curvature's effective rank, and for a diagonal Hessian exactly then; for sign descent on a diagonal quadratic the rate bound is an identity up to the final loss, with a floor which says that a normalized step of fixed length never converges. The theory fixes the norm in advance, assumes a global smoothness constant, and cannot say which norm to choose.

1.1 In the wild: steps of a fixed size

Plain gradient descent takes a step proportional to the gradient: if the gradient doubles, so does the step. Few of the optimizers that train large networks work this way. Most let the gradient choose a direction and the learning rate a length; Kingma and Ba read Adam's learning rate as the radius of a trust region around the current parameters (Kingma and Ba 2015).[1] Bernstein and coauthors proposed signSGD to compress the gradients that workers exchange to one bit per coordinate (Bernstein et al. 2018), Lion was found by a program search over update rules (Chen et al. 2023), and gradient clipping, part of the recipe quoted in the prologue, caps the step whenever the gradient is large (Chapter 10).

IN THE WILD · Adam, Kingma and Ba (2015); Lion, Chen et al. (2023); LARS and LAMB, You et al. (2017, 2020); Muon, Jordan et al. (2024) and Liu et al. (2025). Let the gradient choose the direction and the learning rate the size of the step. Adam divides the momentum of each coordinate by the root-mean-square of its recent gradients, so that no coordinate moves much more than η per step. In signSGD and Lion every coordinate moves by exactly η, against the sign of the gradient

[1] The reading is in Section 2.1 of their paper, which concludes that the step is approximately bounded by the learning rate; Chapter 2 derives the exact bound.

or of an interpolation between the gradient and its momentum; Lion's authors recommend a learning rate 3 to 10 times smaller than AdamW's, because a sign step is longer. LARS and LAMB rescale the update of each layer to a fixed fraction of the norm of the layer's weights. Muon replaces the momentum of each hidden weight matrix by an approximation of its orthogonal polar factor, whose singular values all equal one; to reuse AdamW's learning rate and weight decay, the variant that trained Moonlight multiplies the polar factor of an $m \times n$ matrix by $0.2\sqrt{\max(m, n)}$, which gives the update a root-mean-square of 0.2 times the learning rate.

Because these rules fix the size of the step, the size of an update is measured, and matched across optimizers.

Observation 1.1 (Update size; Liu et al. 2025). In the language-model training runs of Liu and coauthors, who scaled Muon up to train the Moonlight models, the root-mean-square of AdamW's update of a weight matrix, in units of the learning rate, typically lay between 0.2 and 0.4. Rescaled to root-mean-square 0.2, the Muon update could reuse the learning rate and weight decay tuned for AdamW.

This is one report, from one team and one family of models, in 2025; the older observation behind it is that Adam's update behaves like a sign when the gradients are consistent across steps (Balles and Hennig 2018). The logic of the rule: a learning rate is a length, and lengths in different norms can be converted. A second observation shows that normalization matters.

Observation 1.2 (Kunstner et al. 2023). Transformers for language modeling and question answering, trained with full-batch gradients so that noise plays no role, still train much faster with Adam than with gradient descent, and sign descent with momentum closes most of the gap.

On these tasks (Chapters 2 and 14), much of what separates Adam from gradient descent lies in replacing the gradient by its sign. The rules of the box are one rule: take the direction in which the loss decreases fastest per unit of a chosen length, and move a prescribed distance in that length. Which length is right, and what does the choice buy?

1.2 The idealization: a gradient is a dual vector

Let $\mathcal{L}\colon \mathbb{R}^d \to \mathbb{R}$ be differentiable. Its derivative at θ is the linear functional $u \mapsto \langle \nabla\mathcal{L}(\theta), u\rangle = \sum_i \partial_i \mathcal{L}(\theta)\, u_i$; the pairing identifies a functional with its coefficients and measures no lengths (for matrices, $\langle G, D\rangle = \operatorname{tr}(G^\top D)$). The gradient lives in the dual space: it says how fast the loss changes along each direction, but it is not a direction in which to move, and its coordinates are in units of loss per unit of θ_i. Gradient descent converts the gradient into a step with the Euclidean inner product of whatever coordinates the parameters are written in, and a linear change of coordinates changes the method (Exercise 1.5). A norm supplies the map from the dual space back to the parameter space.

Definition 1.3 (Dual norm). Let $\|\cdot\|$ be a norm on $\mathbb{R}^d$. Its *dual norm* is $\|g\|_* = \max_{\|u\|\le 1}\langle g, u\rangle$.

The maximum exists because the unit ball is compact. By Lemma A.3, $\|\cdot\|_*$ is a norm whose dual is $\|\cdot\|$, Hölder's inequality $|\langle g, u\rangle| \le \|g\|_*\|u\|$ holds, and the dual of ℓ_p is ℓ_q with $1/p + 1/q = 1$. The dual norm of the gradient is the largest rate at which the loss decreases along a direction of unit length: the slope of the loss, measured with the chosen ruler.

Definition 1.4 (Linear minimization oracle and sharp operator). For $g \in \mathbb{R}^d$, the *linear minimization oracle* $\operatorname{lmo}(g)$ is an element of the set $\arg\min_{\|u\|\le 1}\langle g, u\rangle$ of *steepest unit directions*; when the set has several elements we fix a choice, and we take $\operatorname{lmo}(0) = 0$. The *sharp operator* maps g to $g^\sharp = -\|g\|_* \operatorname{lmo}(g)$.

The oracle is named after the subproblem of the Frank–Wolfe method (Frank and Wolfe 1956), and Chapter 13 shows that steepest descent with decoupled weight decay is that method.[2] The steepest unit direction minimizes the directional derivative over unit directions: steepness is a ratio of a change of loss to a length, and it depends on how length is measured.

[2] The sharp notation follows Kelner et al. (2014), who used steepest descent in the ℓ_∞ norm to compute approximate maximum flows.

Definition 1.5 (Steepest descent). Let $\eta > 0$ and $g_t = \nabla\mathcal{L}(\theta_t)$. *Normalized steepest descent* in the norm $\|\cdot\|$ is the iteration $\theta_{t+1} = \theta_t + \eta \operatorname{lmo}(g_t)$, and *unnormalized steepest descent* is $\theta_{t+1} = \theta_t - \eta g_t^\sharp$.

By Lemma 1.8 below, a normalized step has length exactly η in the norm unless the gradient vanishes, and an unnormalized step has length $\eta\|g_t\|_*$; this is what "normalized" means throughout the book.[3] The step size η is the learning rate of practice.[4] In the Euclidean norm $g^\sharp = g$, and unnormalized steepest descent is gradient descent.

[3] It has nothing to do with normalization layers, which normalize activations; Chapter 13 studies how the two interact.

[4] For a normalized method the learning rate has the units of the parameters. For gradient descent it has the units of a squared parameter per unit of loss, so its good values change when the loss is rescaled, while those of a normalized method do not (Lemma 1.8(iv)).

Remark 1.6 (One formula). By the substitutions $u = \eta v$ and $u = -\eta w$ and Lemma 1.8(iii), both steps minimize the linear model of the loss under a control on the length of the step:

$$\eta \operatorname{lmo}(g_t) \in \arg\min_{\|u\|\le\eta}\langle g_t, u\rangle, \qquad -\eta g_t^\sharp \in \arg\min_u\left\{\langle g_t, u\rangle + \frac{1}{2\eta}\|u\|^2\right\}.$$

The first is a *trust-region* step: the model is trusted on a ball of radius η and minimized there (Conn et al. 2000). The second replaces the constraint by a penalty, and the two coincide when the radius is $\eta\|g_t\|_*$. Every optimizer of Parts I and II takes a step of the first kind, with a norm that may change from step to step and a gradient that may be replaced by an average: this is the one formula of the book.

Assumption 1.7 (Smoothness in a norm). The loss $\mathcal{L}\colon \mathbb{R}^d \to \mathbb{R}$ is differentiable and bounded below, $\mathcal{L}^\star = \inf\mathcal{L} > -\infty$, and for some $L > 0$ it is *L-smooth with respect to* $\|\cdot\|$: $\|\nabla\mathcal{L}(\theta) - \nabla\mathcal{L}(\theta')\|_* \le L\|\theta - \theta'\|$ for all θ, θ'.

Displacements are measured in the norm and changes of gradient in the dual norm. All norms on $\mathbb{R}^d$ are equivalent, but the smoothness constants differ by factors as large as the dimension (Proposition 1.13), and the content of a choice of norm lies in these factors. For a twice differentiable loss, the descent inequality below needs only $u^\top \nabla^2 \mathcal{L}(\theta)\, u \le L\|u\|^2$.

DICTIONARY

IN THE THEORY	IN A TRAINING RUN
the norm $\|\cdot\|$	what the optimizer treats as a step of unit size: ℓ_2 for SGD, ℓ_∞ for sign descent, the spectral norm of each matrix for Muon, a maximum over layers for LARS
the radius η	the learning rate of a normalized optimizer, the length of every step in its norm
$\mathrm{lmo}(g)$	the update direction before the learning rate: $-g/\|g\|_2$, $-\operatorname{sign}(g)$, $-\operatorname{polar}(G)$
$\|g\|_*$	the decrease of the loss per unit step: $\|g\|_1$ for sign descent, $\|G\|_{\mathrm{nuc}}$ for Muon
the smoothness constant L	the curvature that limits the step, in the norm: the sharpness S for ℓ_2
the floor $L\eta/2$	the plateau of a run at a constant learning rate, removed only by a decaying schedule

Two optimizers are derived in full here; greedy coordinate descent is steepest descent in ℓ_1 (Proposition 1.9(iii)), and sign descent and Muon have their entries in Chapters 2 and 3.[5]

[5] Cauchy's note of 1847 proposed to solve a system of equations by moving, step after step, in the direction that decreases a sum of squares fastest.

FIELD GUIDE · GD AND SGD *Cauchy (1847); Robbins and Monro (1951)*

UPDATE	$\theta_{t+1} = \theta_t - \eta g_t$, with the gradient (GD) or a minibatch gradient (SGD)
NORM	ℓ_2, unnormalized ($g^\sharp = g$): the step has length $\eta\|g_t\|_2$
PRECONDITIONER	none; a fixed $P \succ 0$ gives the norm $\|\cdot\|_P$ (Exercise 1.5)
AVERAGES	none; momentum in Chapter 5
STATE	none (one number per parameter with momentum)
IN THE WILD	with momentum 0.9, common for convolutional networks, rare for transformers

FIELD GUIDE · LARS AND LAMB *You et al. (2017, 2020)*

UPDATE	per layer, $W_l \leftarrow W_l - \eta\|W_l\|_F\, U_l/\|U_l\|_F$, with U_l the gradient (LARS) or the Adam update (LAMB)
NORM	$\max_l \|\Delta W_l\|_F/\|W_l\|_F$, rescaled by the weights at each step; without momentum, LARS is normalized steepest descent in it (Proposition 1.9(iv))
PRECONDITIONER	none (LARS); Adam's diagonal one (LAMB)
AVERAGES	momentum (LARS); Adam's two averages (LAMB)

STATE	one or two numbers per parameter
IN THE WILD	large-batch training: ResNet-50 on ImageNet with batches of 32K (LARS), BERT in 76 minutes (LAMB); $\|W_l\|_F/\|U_l\|_F$ is the "trust ratio"

WHAT WAS THROWN AWAY. Noise (the gradients are exact), memory (each step depends only on the current gradient), change (the norm is fixed and the smoothness constant global) and structure (one vector, or a fixed list of blocks). Each has a chapter of its own: noise and memory change what is fed to the duality map, not the map (Chapter 5); a norm that adapts to the gradients is a preconditioner (Chapters 6 and 7); changing curvature is the subject of Part III; the norm of a network is built in Chapter 4. What is left comes first: given one exact gradient and one norm, what is the best step, and what does it guarantee?

WHAT ONE HAD TO NOTICE. The rules of the practice box are not normalizations bolted onto gradient descent; they are gradient descent in other norms. A sign vector is the steepest unit direction of the max norm, a polar factor that of the spectral norm. Steepest descent in a general norm is classical (Boyd and Vandenberghe 2004), spectral descent was proposed for networks by Carlson et al. (2015), and Bernstein and Newhouse (2024) observed that Adam, Shampoo and Prodigy with their accumulators switched off are steepest descent in particular norms, a view that Bernstein and Newhouse (2025) and Large et al. (2024) extended to whole networks. Seen this way, the choice of an optimizer is the choice of a norm, and it can be analyzed.

1.3 The theory: the steepest step in each norm

We collect first the elementary facts that every later chapter uses.

Lemma 1.8 (Basic properties). *Let $\|\cdot\|$ be a norm on $\mathbb{R}^d$ and $g \in \mathbb{R}^d$.*

(i) $\langle g, \operatorname{lmo}(g)\rangle = -\|g\|_*$, *and* $\|\operatorname{lmo}(g)\| = 1$ *if* $g \neq 0$.

(ii) $\langle g, g^\sharp\rangle = \|g\|_*^2$ *and* $\|g^\sharp\| = \|g\|_*$.

(iii) For every u, $\langle g, u\rangle - \frac{1}{2}\|u\|^2 \leq \frac{1}{2}\|g\|_*^2$, *with equality at* $u = g^\sharp$. *If* $g \neq 0$, *then the maximizers are the vectors* $-\|g\|_* v$ *with* v *a steepest unit direction; if* $g = 0$, *then the only maximizer is* 0.

(iv) For $c > 0$ *the steepest unit directions of* cg *and of* g *coincide; so* lmo *can be chosen with* $\operatorname{lmo}(cg) = \operatorname{lmo}(g)$, *and then* $(cg)^\sharp = c\, g^\sharp$.

PROOF. The plan is to combine Hölder's inequality with the symmetry of the unit ball. (i) By symmetry, $\min_{\|u\|\leq 1}\langle g, u\rangle$ equals $-\max_{\|u\|\leq 1}\langle g, -u\rangle$, which is $-\|g\|_*$. If $g \neq 0$, then $\|g\|_* > 0$, and Hölder's inequality gives $\|g\|_* = -\langle g, \operatorname{lmo}(g)\rangle \leq \|g\|_*\|\operatorname{lmo}(g)\|$, so $\|\operatorname{lmo}(g)\| \geq 1$; the reverse holds because $\operatorname{lmo}(g)$ lies in the ball. (ii) Multiply both identities of (i) by $\|g\|_*$; if $g = 0$, then $g^\sharp = 0$. (iii) By Hölder's inequality and $ab - \frac{1}{2}b^2 \leq \frac{1}{2}a^2$,

$$\langle g, u\rangle - \tfrac{1}{2}\|u\|^2 \leq \|g\|_*\|u\| - \tfrac{1}{2}\|u\|^2 \leq \tfrac{1}{2}\|g\|_*^2.$$

Both steps are equalities if and only if $\|u\| = \|g\|_*$ and $\langle g, u\rangle = \|g\|_*\|u\|$. If $g = 0$, then this forces $u = 0$. If $g \neq 0$, then it says that $u = \|g\|_* w$ with $\|w\| = 1$ and $\langle g, w\rangle = \|g\|_*$, that is, $u = -\|g\|_* v$ with $\|v\| = 1$ and $\langle g, v\rangle = -\|g\|_*$, and by the argument of (i) these v are exactly the steepest unit directions; by (ii), $g^\sharp$ is one of these. (iv) The objective $c\langle g, u\rangle$ has the same minimizers, and $\|cg\|_* = c\|g\|_*$. ∎

Part (iii) says that $\frac{1}{2}\|\cdot\|_*^2$ is the convex conjugate of $\frac{1}{2}\|\cdot\|^2$ (Lemma A.5); part (iv), that a normalized step is insensitive to the scale of the loss. For $W \in \mathbb{R}^{m\times n}$ we write We_j for the columns, $W^\top e_i$ for the rows, and $\|W\|_{p\to q} = \max_{x\neq 0}\|Wx\|_q/\|x\|_p$.

Proposition 1.9 (Duality maps of the norms that optimizers use). *Let $g \in \mathbb{R}^d$ and $G \in \mathbb{R}^{m\times n}$ be nonzero, and let $M(g) = \arg\min_{\|u\|\le 1}\langle g, u\rangle$ be the set of steepest unit directions of the norm at hand.*

(i) (ℓ_2 and Frobenius.) The dual of $\|\cdot\|_2$ is $\|\cdot\|_2$, $M(g) = \{-g/\|g\|_2\}$ and $g^\sharp = g$. Likewise $\|\cdot\|_F$ is self-dual, $\mathrm{lmo}(G) = -G/\|G\|_F$ and $G^\sharp = G$.

(ii) (ℓ_∞.) The dual norm is $\|g\|_1$, and $M(g) = \{u \in [-1,1]^d : u_i = -\operatorname{sign}(g_i)$ whenever $g_i \neq 0\}$. We choose $\mathrm{lmo}(g) = -\operatorname{sign}(g)$, so $g^\sharp = \|g\|_1 \operatorname{sign}(g)$; normalized steepest descent in ℓ_∞ is sign descent.

(iii) (ℓ_1.) The dual norm is $\|g\|_\infty$, and $M(g)$ is the convex hull of the vectors $-\operatorname{sign}(g_i)e_i$ with $|g_i| = \|g\|_\infty$. For one such i, $\mathrm{lmo}(g) = -\operatorname{sign}(g_i)e_i$ and $g^\sharp = g_i e_i$: unnormalized steepest descent in ℓ_1 is greedy coordinate descent, the Gauss–Southwell rule.

(iv) (Maxima over blocks.) Let $\mathbb{R}^d = \mathbb{R}^{d_1}\times\cdots\times\mathbb{R}^{d_K}$ with $\theta = (\theta_1, \ldots, \theta_K)$, give the kth factor a norm $\|\cdot\|_{(k)}$ with dual $\|\cdot\|_{(k),*}$ and steepest unit directions M_k, and let $\|\theta\| = \max_k\|\theta_k\|_{(k)}$. Then $\|g\|_* = \sum_k\|g_k\|_{(k),*}$, $M(g) = M_1(g_1)\times\cdots\times M_K(g_K)$, $\mathrm{lmo}(g) = (\mathrm{lmo}_{(k)}(g_k))_k$, and $g^\sharp = -\big(\sum_j\|g_j\|_{(j),*}\big)(\mathrm{lmo}_{(k)}(g_k))_k$.

(v) (Columns and rows.) $\|W\|_{1\to 2} = \max_j\|We_j\|_2$ and $\|W\|_{2\to\infty} = \max_i\|W^\top e_i\|_2$. Their duals are the sums of the column norms and of the row norms, and lmo *maps each column, respectively row, $v \neq 0$ of G to $-v/\|v\|_2$ and a zero one to zero.*

(vi) (Spectral.) The dual of $\|\cdot\|_{\mathrm{op}}$ is the nuclear norm $\|G\|_{\mathrm{nuc}} = \sum_i \sigma_i(G)$, and $-\operatorname{polar}(G) \in M(G)$, so $G^\sharp = \|G\|_{\mathrm{nuc}}\operatorname{polar}(G)$.

Proof. The plan is to bound $\langle g, u\rangle$ from below over the unit ball by a chain of elementary inequalities, to exhibit a point where every inequality is an equality, and to read off the set of such points; (ii) and (v) are instances of (iv). If $\min_{\|u\|\le 1}\langle g, u\rangle = -c$, then $\|g\|_* = c$ by symmetry of the ball.

(i) By the Cauchy–Schwarz inequality, $\langle g, u\rangle \ge -\|g\|_2\|u\|_2 \ge -\|g\|_2$ when $\|u\|_2 \le 1$. Equality in the first step requires $u = -cg$ with $c \ge 0$, and in the second $\|u\|_2 = 1$, so $u = -g/\|g\|_2$, which attains the bound; then $g^\sharp = g$. The Frobenius norm is the ℓ_2 norm of the entries, and the trace pairing is the coordinate pairing of the entries.

(iv) If $\|u\| \le 1$, then $\|u_k\|_{(k)} \le 1$ for every k, and Hölder's inequality in each factor gives $\langle g, u\rangle = \sum_k \langle g_k, u_k\rangle \ge -\sum_k \|g_k\|_{(k),*}$. Equality holds if and only if $\langle g_k, u_k\rangle = -\|g_k\|_{(k),*}$ for every k, that is, $u_k \in M_k(g_k)$ for every k; for a block with $g_k = 0$ this is the whole unit ball of the block. Such u exist, for instance $u_k = \mathrm{lmo}_{(k)}(g_k)$, so the minimum is $-\sum_k \|g_k\|_{(k),*}$ and the minimizers form the product of the sets $M_k(g_k)$.

(ii) The max norm is (iv) with d blocks of size one, each with the absolute value, which is its own dual. For one coordinate, $\arg\min_{|u_i|\le 1} g_i u_i$ is $\{-\operatorname{sign}(g_i)\}$ if $g_i \ne 0$ and $[-1, 1]$ if $g_i = 0$.

(iii) For $\|u\|_1 \le 1$,

$$\langle g, u\rangle \ge -\sum_i |g_i||u_i| \ge -\|g\|_\infty \sum_i |u_i| \ge -\|g\|_\infty.$$

The first inequality is an equality if and only if $g_i u_i \le 0$ for all i, the second if and only if $u_i = 0$ whenever $|g_i| < \|g\|_\infty$, and the third, since $g \ne 0$, if and only if $\|u\|_1 = 1$. The vectors with these properties are $u = -\sum_{i\in I} c_i \operatorname{sign}(g_i) e_i$ with $c_i \ge 0$, $\sum_i c_i = 1$ and $I = \{i : |g_i| = \|g\|_\infty\}$: the convex hull of the vertices $-\operatorname{sign}(g_i)e_i$, $i \in I$. At a vertex, $g^\sharp = \|g\|_\infty \operatorname{sign}(g_i) e_i = g_i e_i$.

(v) By the triangle inequality, $\|Wx\|_2 = \|\sum_j x_j W e_j\|_2 \le \|x\|_1 \max_j \|We_j\|_2$, with equality at $x = e_j$ for a column of largest norm. By the Cauchy–Schwarz inequality, $\|Wx\|_\infty = \max_i |\langle W^\top e_i, x\rangle| \le \|x\|_2 \max_i \|W^\top e_i\|_2$, with equality at $x = W^\top e_i / \|W^\top e_i\|_2$ for a row of largest norm. So both norms are maxima, over the columns and over the rows, of the Euclidean norm, and (iv) with (i) gives their duals and oracles; for a zero block we choose zero.

(vi) By von Neumann's trace inequality (Theorem A.7), every D with $\|D\|_{\mathrm{op}} = \sigma_1(D) \le 1$ satisfies

$$\langle G, D\rangle = \operatorname{tr}(G^\top D) \ge -\sum_i \sigma_i(G)\sigma_i(D) \ge -\sigma_1(D) \sum_i \sigma_i(G) \ge -\|G\|_{\mathrm{nuc}}.$$

Let $G = U \operatorname{diag}(\sigma) V^\top$ be a compact singular value decomposition, with $r = \operatorname{rank} G \ge 1$, $U \in \mathbb{R}^{m\times r}$, $V \in \mathbb{R}^{n\times r}$ and $U^\top U = V^\top V = I_r$. Then $(UV^\top)^\top UV^\top = VV^\top$ is an orthogonal projection of rank r, so the nonzero singular values of $UV^\top$ equal 1 and $\|UV^\top\|_{\mathrm{op}} = 1$; and $\langle G, -UV^\top\rangle = -\operatorname{tr}(\operatorname{diag}(\sigma) V^\top V) = -\sum_i \sigma_i$. So $-UV^\top$ attains the bound. ∎

Remark 1.10 (Ties, jumps and step sizes). The steepest unit direction is unique for the Euclidean and Frobenius norms, whose balls are strictly convex, and otherwise only away from ties: for ℓ_∞ when no g_i vanishes, for ℓ_1 when the largest $|g_i|$ is attained once, for the spectral norm when G has full rank (Chapter 3). At a tie the direction jumps, as a coordinate of $-\operatorname{sign}(g)$ does when g_i crosses zero, and noisy gradients straddle such ties all the time (Chapters 2 and 5). The norms also give steps of very different sizes: a normalized step of length η has root-mean-square η in ℓ_∞ but $\eta/\sqrt{d}$ in ℓ_2 (Exercise 1.4), and matching update sizes (Observation 1.1) converts between the norms.

Remark 1.11 (Steepest descent is not mirror descent). *Mirror descent* (Beck and Teboulle 2003; Nemirovsky and Yudin 1983) carries the iterate, not the

gradient, to the dual space: with a differentiable, strongly convex mirror map ψ it sets $\nabla\psi(\theta_{t+1}) = \nabla\psi(\theta_t) - \eta g_t$. Take $\psi = \frac{1}{2}\|\cdot\|^2$ for a norm whose square is differentiable and strictly convex. Then $g^\sharp$ is the only maximizer of the strictly concave function $u \mapsto \langle g, u\rangle - \psi(u)$ (Lemma 1.8(iii)), so by the first-order condition $\nabla\psi$ and $g \mapsto g^\sharp$ are inverse maps, and with $z_t = \nabla\psi(\theta_t)$ mirror descent reads $\theta_{t+1} = (z_t - \eta g_t)^\sharp$, while steepest descent reads $\theta_{t+1} = z_t^\sharp - \eta g_t^\sharp$. The two agree for all θ_t and g_t if and only if the sharp operator is additive, hence linear by Lemma 1.8(iv); then $\|g\|_*^2 = \langle g, g^\sharp\rangle$ is a quadratic form, the norm comes from an inner product, $\|u\| = \|u\|_P$, and both methods are the preconditioned step $\theta_t - \eta P^{-1} g_t$. For the max, ℓ_1 and spectral norms, $\frac{1}{2}\|\cdot\|^2$ is neither differentiable nor strictly convex, so mirror descent there uses other maps, such as the negative entropy, and is analyzed through Bregman divergences, not the descent inequality.

1.4 The theory: descent in a general norm

Smoothness gives a quadratic upper bound on the loss around every point, and steepest descent minimizes it; the main theorem of the chapter follows.

Theorem 1.12 (Descent in a general norm). *Let $\mathcal{L}$ satisfy Assumption 1.7, let $\theta_0 \in \mathbb{R}^d$, $\Delta_0 = \mathcal{L}(\theta_0) - \mathcal{L}^\star$ and $g_t = \nabla\mathcal{L}(\theta_t)$.*

(i) (One step.) For all $\theta, u \in \mathbb{R}^d$,

$$\mathcal{L}(\theta + u) \le \mathcal{L}(\theta) + \langle \nabla\mathcal{L}(\theta), u\rangle + \frac{L}{2}\|u\|^2. \tag{1.1}$$

The right side is smallest at $u = -\nabla\mathcal{L}(\theta)^\sharp/L$, where it equals $\mathcal{L}(\theta) - \|\nabla\mathcal{L}(\theta)\|_*^2/(2L)$. At $u = \eta\,\mathrm{lmo}(\nabla\mathcal{L}(\theta))$ it is at most $\mathcal{L}(\theta) - \eta\|\nabla\mathcal{L}(\theta)\|_* + L\eta^2/2$.

(ii) (Unnormalized, nonconvex.) Unnormalized steepest descent with step size $1/L$ satisfies, for all $T \ge 1$,

$$\min_{t<T}\|g_t\|_*^2 \le \frac{1}{T}\sum_{t<T}\|g_t\|_*^2 \le \frac{2L\Delta_0}{T}.$$

(iii) (Normalized, nonconvex.) Normalized steepest descent with step size $\eta > 0$ satisfies, for all $T \ge 1$,

$$\frac{1}{T}\sum_{t<T}\|g_t\|_* \le \frac{\Delta_0}{\eta T} + \frac{L\eta}{2}, \tag{1.2}$$

and if $\Delta_0 > 0$, the right side is smallest, equal to $\sqrt{2L\Delta_0/T}$, at $\eta = \sqrt{2\Delta_0/(LT)}$.

(iv) (Unnormalized, convex.) If moreover $\mathcal{L}$ is convex with a minimizer $\theta^\star$, and $R = \sup\{\|\theta - \theta^\star\| : \mathcal{L}(\theta) \le \mathcal{L}(\theta_0)\}$ is finite, then the method of (ii) satisfies $\Delta_T \le 2LR^2/(T+4)$ for all $T \ge 0$.

PROOF. The plan is to prove the quadratic upper bound (1.1) once, to minimize it over the two kinds of steps, and to add up the decreases; the convex rate needs one more inequality, which turns a gap into a lower bound on the gradient.

(i) Fix θ and u, and let $\varphi(s) = \mathcal{L}(\theta + su)$ for $s \in [0,1]$. The gradient is Lipschitz, hence continuous, so φ is continuously differentiable with $\varphi'(s) = \langle \nabla\mathcal{L}(\theta + su), u\rangle$. By the fundamental theorem of calculus, then Hölder's inequality and smoothness,

$$\begin{aligned}\mathcal{L}(\theta+u) - \mathcal{L}(\theta) - \langle \nabla\mathcal{L}(\theta), u\rangle &= \int_0^1 \langle \nabla\mathcal{L}(\theta+su) - \nabla\mathcal{L}(\theta), u\rangle \,\mathrm{d}s \\ &\le \int_0^1 Ls\|u\|^2 \,\mathrm{d}s = \frac{L}{2}\|u\|^2,\end{aligned}$$

which is (1.1) (Lemma A.18). Write $g = \nabla\mathcal{L}(\theta)$. By Hölder's inequality the right side of (1.1) is at least $\mathcal{L}(\theta) - \|g\|_*\|u\| + \frac{L}{2}\|u\|^2$, a quadratic in $\|u\|$ with minimum $\mathcal{L}(\theta) - \|g\|_*^2/(2L)$. At $u = -g^\sharp/L$, Lemma 1.8(ii) gives $\langle g, u\rangle = -\|g\|_*^2/L$ and $\|u\| = \|g\|_*/L$, so the right side equals this minimum. At $u = \eta\,\mathrm{lmo}(g)$, Lemma 1.8(i) gives $\langle g, u\rangle = -\eta\|g\|_*$ and $\|u\| \le \eta$.

(ii) By (i), $\mathcal{L}(\theta_{t+1}) \le \mathcal{L}(\theta_t) - \|g_t\|_*^2/(2L)$. Summing over $t < T$ and using $\mathcal{L}(\theta_T) \ge \mathcal{L}^\star$ gives $\sum_{t<T}\|g_t\|_*^2 \le 2L\Delta_0$; a minimum is at most an average.

(iii) By (i), $\mathcal{L}(\theta_{t+1}) \le \mathcal{L}(\theta_t) - \eta\|g_t\|_* + L\eta^2/2$; if $g_t = 0$, then the step is zero and the inequality holds too. Adding for $t < T$ gives $\eta\sum_{t<T}\|g_t\|_* \le \Delta_0 + TL\eta^2/2$, which is (1.2) after division by ηT. By the arithmetic–geometric mean inequality, $\frac{\Delta_0}{\eta T} + \frac{L\eta}{2} \ge 2\sqrt{L\Delta_0/(2T)} = \sqrt{2L\Delta_0/T}$, with equality when the two terms are equal, which for $\Delta_0 > 0$ happens at $\eta = \sqrt{2\Delta_0/(LT)}$.

(iv) The losses do not increase, by (i), so every θ_t lies in the sublevel set of θ_0 and $\|\theta_t - \theta^\star\| \le R$. If $\Delta_T = 0$, there is nothing to prove; otherwise $\Delta_t \ge \Delta_T > 0$ for $t \le T$, so $\theta_0 \ne \theta^\star$ and $R > 0$. By convexity and Hölder's inequality, $\Delta_t \le \langle g_t, \theta_t - \theta^\star\rangle \le R\|g_t\|_*$, and with (i) this gives $\Delta_{t+1} \le \Delta_t - \Delta_t^2/(2LR^2)$ for $t < T$. Dividing the recursion by $\Delta_t\Delta_{t+1}$ and using $\Delta_{t+1} \le \Delta_t$,

$$\frac{1}{\Delta_{t+1}} - \frac{1}{\Delta_t} \ge \frac{\Delta_t}{2LR^2\,\Delta_{t+1}} \ge \frac{1}{2LR^2},$$

so $1/\Delta_T \ge 1/\Delta_0 + T/(2LR^2)$. Finally, $\nabla\mathcal{L}(\theta^\star) = 0$, so (1.1) at $\theta^\star$ with $u = \theta_0 - \theta^\star$ gives $\Delta_0 \le LR^2/2$, that is, $1/\Delta_0 \ge 4/(2LR^2)$. Hence $1/\Delta_T \ge (T+4)/(2LR^2)$. ∎

WHERE THE NORM ENTERS. The theorem holds for every norm at once, and the norm enters only through the smoothness constant, the dual norm of the gradient and, for the convex rate, the radius R. Choosing a norm is choosing which smoothness constant and which gradient norm one pays; the guarantees come in different currencies ($\|g\|_1$ for sign descent, $\|g\|_2$ for gradient descent), and Proposition 1.13 converts between them for quadratics. Part (iv) is Nesterov's argument for gradient descent (Nesterov 2018), with the dual norm in place of the Euclidean one.[6]

[6] In the Euclidean case R can be replaced by $\|\theta_0 - \theta^\star\|_2$, because gradient descent with step size $1/L$ never moves away from a minimizer of a convex smooth loss (Lemma A.18). The argument expands the square of a Euclidean distance, and in a general norm the radius of the

NORMALIZED AND UNNORMALIZED. At their best step sizes the two methods have the same guarantee: by (ii) and the Cauchy–Schwarz inequality, the average of $\|g_t\|_*$ under unnormalized descent is at most $\sqrt{2L\Delta_0/T}$, the value of (iii) at its best step. They differ in what they risk. An unnormalized step of size η has the guaranteed decrease $\eta(1 - L\eta/2)\|g_t\|_*^2$, which vanishes at $\eta = 2/L$, the stability threshold of the prologue. A normalized step never moves farther than η, so (1.2) holds for every $\eta > 0$: there is no threshold, a step size that is too large only raises the floor, and the method is clipped by construction (Chapter 10).

THE FLOOR. At a fixed step size the right side of (1.2) tends to $L\eta/2$, not to zero: a normalized step has length η even where the gradient is tiny, so the iterates cannot converge unless they land exactly on a stationary point, as the prologue showed for sign descent. A decaying learning rate removes the floor, since with step sizes η_t the same argument gives $\sum_t \eta_t\|g_t\|_* \le \Delta_0 + \frac{L}{2}\sum_t \eta_t^2$ (Exercise 1.7); that a schedule removes the floor of a constant learning rate is a theme of the book (Chapters 12 and 13). The floor takes over at $T = 2\Delta_0/(L\eta^2)$, where the two terms of (1.2) are equal. To compare norms we need their smoothness constants.

Proposition 1.13 (Which norm? Smoothness constants of a quadratic). *Let $\mathcal{L}(\theta) = \frac{1}{2}\theta^\top H\theta - \langle b, \theta\rangle$ with H symmetric, positive semidefinite and nonzero. For a norm $\|\cdot\|$ let $L_{\|\cdot\|}$ be the smallest L for which $\mathcal{L}$ is L-smooth with respect to $\|\cdot\|$.*

(i) For every norm, $L_{\|\cdot\|} = \max_{\|u\|\le 1} u^\top Hu$.

(ii) $L_{\ell_2} = \lambda_{\max}(H)$, $L_{\ell_1} = \max_i H_{ii}$, and $L_{\ell_\infty} = \|H\|_{\infty\to 1} = \max_{s\in\{-1,1\}^d} s^\top Hs$, which satisfies $\operatorname{tr} H \le L_{\ell_\infty} \le d\,\lambda_{\max}(H)$, with $L_{\ell_\infty} = \operatorname{tr} H$ when H is diagonal.

(iii) At a gradient $g \ne 0$, the guaranteed decrease $\|g\|_*^2/(2L_{\|\cdot\|})$ of Theorem 1.12(i) is larger for ℓ_∞ than for ℓ_2 if and only if

$$\frac{\|g\|_1^2}{\|g\|_2^2} > \frac{\|H\|_{\infty\to 1}}{\lambda_{\max}(H)}, \tag{1.3}$$

and the right side is at least $\operatorname{tr} H/\lambda_{\max}(H)$, with equality for diagonal H.

(iv) For every norm, the step $-g^\sharp/L_{\|\cdot\|}$ decreases $\mathcal{L}$ by at least $\|g\|_*^2/(2L_{\|\cdot\|})$ and at most twice as much, and by exactly the lower value if and only if $\operatorname{lmo}(g)$ maximizes $u^\top Hu$ over the unit ball; for ℓ_∞ and diagonal H, this holds whenever no coordinate of g vanishes.

PROOF. The plan is to reduce every constant to the maximum of the quadratic form $u^\top Hu$ over the unit ball, and to evaluate that maximum at the vertices of the ball. (i) Since $\nabla\mathcal{L}(\theta) - \nabla\mathcal{L}(\theta') = H(\theta - \theta')$, the best constant is $\sup_{u\ne 0}\|Hu\|_*/\|u\| = \sup\{v^\top Hu : \|u\| \le 1,\ \|v\| \le 1\}$. Taking $v = u$ shows that this is at least $\max_{\|u\|\le 1} u^\top Hu$. Conversely, $(u, v) \mapsto v^\top Hu$ is a positive semidefinite bilinear form, so the Cauchy–Schwarz inequality

gives $v^\top Hu \le (u^\top Hu)^{1/2}(v^\top Hv)^{1/2} \le \max\{u^\top Hu, v^\top Hv\}$.

(ii) For ℓ_2, (i) is the Rayleigh quotient. For ℓ_1 and ℓ_∞, the function $u \mapsto u^\top Hu$ is convex, and a convex function on a polytope attains its maximum at a vertex, because every point of the polytope is a convex combination of vertices. The vertices of the ℓ_1 ball are $\pm e_i$, where the function equals H_{ii}; those of the ℓ_∞ ball are the sign vectors. That $\max_s s^\top Hs = \max_{\|u\|_\infty \le 1} \|Hu\|_1 = \|H\|_{\infty\to 1}$ is (i) with the dual norm ℓ_1. For s uniform on $\{-1,1\}^d$, $\mathbb{E}[s_i s_j] = \mathbf{1}[i=j]$, so $\mathbb{E}[s^\top Hs] = \operatorname{tr} H$, and the maximum is at least the mean. For every s, $s^\top Hs \le \lambda_{\max}(H)\|s\|_2^2 = d\,\lambda_{\max}(H)$; and if H is diagonal, then $s^\top Hs = \operatorname{tr} H$ for every s.

(iii) Compare the guarantees $\|g\|_1^2/(2L_{\ell_\infty})$ and $\|g\|_2^2/(2L_{\ell_2})$, and use (ii).

(iv) Let $L = L_{\|\cdot\|}$, $u = \operatorname{lmo}(g)$ and $q = u^\top Hu$, so $0 \le q \le L$ by (i) and Lemma 1.8(i). For a quadratic, (1.1) is an equality with $\frac{1}{2}v^\top Hv$ in place of $\frac{L}{2}\|v\|^2$; with $v = -g^\sharp/L = (\|g\|_*/L)\,u$,

$$\mathcal{L}(\theta) - \mathcal{L}(\theta + v) = \frac{\|g\|_*^2}{L} - \frac{\|g\|_*^2}{2L^2}\,q = \frac{\|g\|_*^2}{2L}\Big(2 - \frac{q}{L}\Big),$$

which lies between $\|g\|_*^2/(2L)$ and $\|g\|_*^2/L$ and equals the first exactly when $q = L$. For ℓ_∞ and diagonal H, $q = \sum_i H_{ii}\operatorname{sign}(g_i)^2 = \operatorname{tr} H = L$ when every $g_i \ne 0$. ■

The left side of (1.3) is the *density* of the gradient, between 1 for a gradient with one nonzero coordinate and d for one with coordinates of equal size; the ratio $\operatorname{tr} H/\lambda_{\max}(H)$, between 1 and $\operatorname{rank} H$, is the *effective rank* of the curvature. So the max norm can win a step, on the guarantees, only when the gradient is spread over more coordinates than the curvature over directions, and for diagonal H it wins exactly then: a dense gradient against a curvature of low effective rank. A gradient with i.i.d. centered Gaussian coordinates has density close to $2d/\pi$, so for diagonal H the max norm then wins unless the effective rank exceeds $2d/\pi$, that is, unless the curvature is nearly isotropic. By (iv) the guarantees decide the comparison of the true decreases whenever they differ by more than a factor of two; Chapters 2 and 14 ask what both sides look like in networks. But $\|H\|_{\infty\to 1}$ is NP-hard to compute (Rohn 2000), since for a graph Laplacian it is four times the maximum cut (Exercise 1.3); for diagonal H the difficulty disappears, and there the theory can be tested exactly.

1.5 Reading the theory

The theory predicts the best step length $\|g\|_*/L$ and a guaranteed decrease within a factor of two of the truth on convex quadratics, the comparison (1.3) between norms, a $1/T$ transient above a floor $L\eta/2$, and the convex rate $2LR^2/(T+4)$. Figure 1.1 shows the geometry of a step.[7]

[7] The emblem on the title page of this book is the left panel of Figure 1.1, drawn for g instead of $-g$.

One step. The left panel of Figure 1.2 tests the first two predictions on a diagonal quadratic in $d = 1000$ dimensions with curvatures $h_i = 1/i$, so that $\lambda_{\max}(H) = 1$ and $\operatorname{tr} H = 7.49$, at a random point. Its gradient has density 30.1 against an effective rank of 7.49, so (1.3) predicts that the sign step

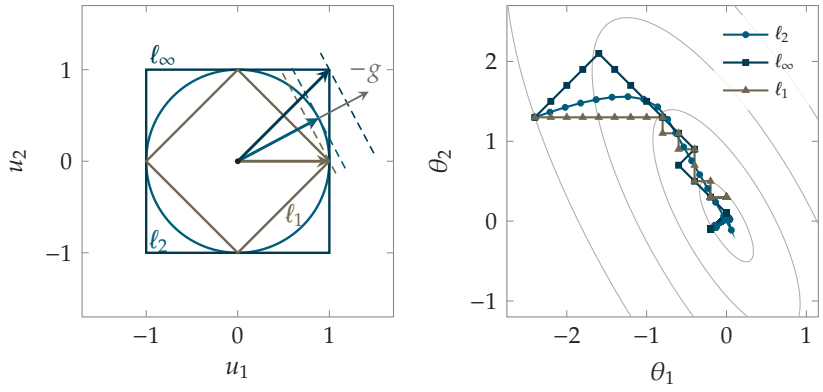


Figure 1.1. The geometry of a step. Left: the unit balls of ℓ_1 (diamond), ℓ_2 (circle) and ℓ_∞ (square), the negative gradient $-g$ at 28° with $\|g\|_2 = 1$, and the steepest unit direction $\mathrm{lmo}(g)$ of each norm (thick arrows), where the ball touches its supporting line $\{u : \langle g, u\rangle = -\|g\|_*\}$ (dashed). The lines lie at distances $\|g\|_\infty = 0.88$, $\|g\|_2 = 1$ and $\|g\|_1 = 1.35$ from the origin: the dual norm measures how far the ball reaches toward $-g$. Right: 40 normalized steps in each norm with $\eta = 0.2$ from one start (dot), on the quadratic with Hessian $Q\,\mathrm{diag}(1, 0.12)Q^\top$, Q the rotation by 30°; the ℓ_∞ and ℓ_1 runs are on cycles of period two from steps 14 and 16 on, and the ℓ_2 run converges to one. Script: `01-steepest-geometry.py`.

wins, and the guarantees differ by a factor of 4.02 (2.43 against 0.60). The sign step's true decrease equals its guarantee at every step length, because for a diagonal Hessian the curvature along $-\mathrm{sign}(g)$ is exactly $\mathrm{tr}\,H$. The normalized gradient step beats its guarantee, because the curvature along $g/\|g\|_2$ is $q = 0.65$, not $\lambda_{\max} = 1$: at its recommended length $\|g\|_2 = 1.10$ it gains 0.82, which is $2 - q = 1.35$ times its guarantee. The sign step still wins, by a factor of 2.97 at the recommended lengths and 2.60 against the best Euclidean step. The exact agreement of the sign step fails in general for a non-diagonal Hessian, where the curvature along $-\mathrm{sign}(g)$ can be below $\|H\|_{\infty\to 1}$, and a rotation of the coordinates can reverse the comparison (Chapter 2).

THE FLOOR. The right panel follows sign descent on the same problem for 10^5 steps at two step sizes. Early in each run the points lie below the curve: the bound charges the whole initial gap Δ_0 to the first T steps, while a run that has moved each coordinate by only $T\eta$ has spent part of it. Once every coordinate has reached its cycle, after $\max_i|\theta_{0,i}|/\eta$ steps (about 31 for $\eta = 0.1$ and 310 for $\eta = 0.01$), the points lie on the curve; at $T = 10^5$ the two agree to within three parts in a million. Each run then follows its curve toward its floor $\eta\,\mathrm{tr}\,H/2$, which dominates after the crossover $T = 2\Delta_0/(\eta^2\,\mathrm{tr}\,H)$, 97 for $\eta = 0.1$ and 9.7×10^3 for $\eta = 0.01$, the only horizon at which the curve touches the best bound (dotted).

Proposition 1.14 (When the bound is attained). *Let $\mathcal{L}(\theta) = \frac{1}{2}\theta^\top H\theta$ with H positive semidefinite.*

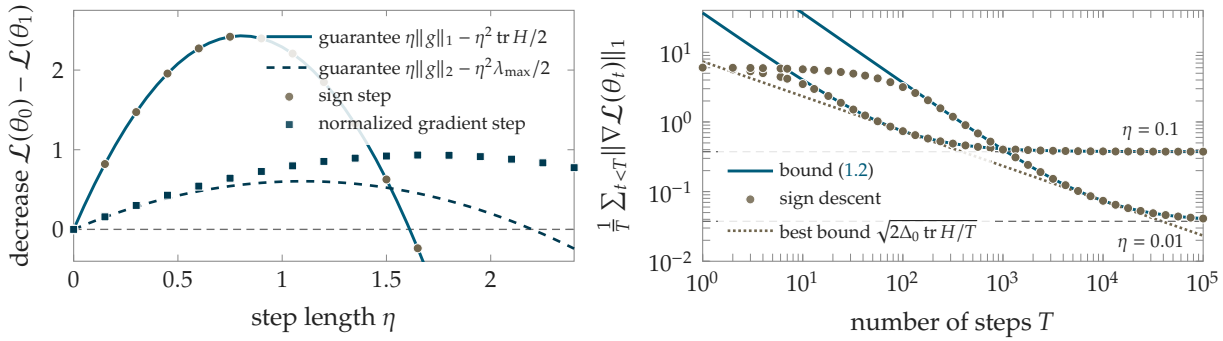


Figure 1.2. The descent theorem on $\mathcal{L}(\theta) = \frac{1}{2}\sum_i h_i\theta_i^2$ with $d = 1000$ and $h_i = 1/i$, from θ_0 with independent standard Gaussian coordinates (numpy seed 9, whose ratio of guarantees, 4.02, is near the median 3.91 over seeds 0 to 199; $\Delta_0 = 3.64$). Left: the decrease achieved by one sign step $-\eta\operatorname{sign}(g)$ and one normalized gradient step $-\eta g/\|g\|_2$ (points), against the guarantees of Theorem 1.12(i) with $L = \operatorname{tr} H$ and $L = \lambda_{\max}(H)$ (curves). Right: the running average of $\|\nabla\mathcal{L}(\theta_t)\|_1$ along sign descent at two step sizes (points), against the bound (1.2) computed in advance from Δ_0, $\operatorname{tr} H$ and η (curves); dashed, the floors $\eta \operatorname{tr} H/2$; dotted, the best bound at each horizon. Script: `01-steepest-descent.py`.

(i) *If normalized steepest descent in a norm $\|\cdot\|$ with step size η has a cycle of period two, $\theta \mapsto \theta + \eta u \mapsto \theta$, then the average of $\|\nabla\mathcal{L}\|_*$ over the cycle is $\frac{\eta}{2}u^\top H u \le L_{\|\cdot\|}\eta/2$.*

(ii) *Let $H = \operatorname{diag}(h_1, \dots, h_d)$ with all $h_i > 0$, and let sign descent start at a point θ_0 none of whose coordinates is an integer multiple of η. Then for every $T \ge 1$,*

$$\frac{1}{T}\sum_{t<T}\|g_t\|_1 = \frac{\Delta_0 - \Delta_T}{\eta T} + \frac{\eta \operatorname{tr} H}{2}, \tag{1.4}$$

and from step $\max_i|\theta_{0,i}|/\eta$ on the iterates lie on a cycle of period two, with $\Delta_T < \frac{1}{2}\eta^2 \operatorname{tr} H$. So the two sides of (1.2) with $L = \operatorname{tr} H = L_{\ell_\infty}$ differ by exactly $\Delta_T/(\eta T)$, and the average tends to the floor $\eta \operatorname{tr} H/2$.

Proof. The plan is to evaluate the exact expansion of the quadratic along the steps. (i) Write $\theta' = \theta + \eta u$, $g = \nabla\mathcal{L}(\theta)$ and $g' = \nabla\mathcal{L}(\theta') = g + \eta H u$. The cycle means $u = \operatorname{lmo}(g)$ and $-u = \operatorname{lmo}(g')$, so by Lemma 1.8(i), $\|g\|_* = -\langle g, u\rangle$ and $\|g'\|_* = \langle g', u\rangle = \langle g, u\rangle + \eta u^\top H u$. Adding, $\|g\|_* + \|g'\|_* = \eta u^\top H u$, and $u^\top H u \le L_{\|\cdot\|}$ by Proposition 1.13(i).

(ii) Each coordinate of θ_t changes by $\pm\eta$ per step, so under the hypothesis no coordinate of θ_t, and hence none of $g_t = H\theta_t$, ever vanishes. Then $\operatorname{sign}(g_t)^\top H \operatorname{sign}(g_t) = \operatorname{tr} H$, and the exact expansion gives $\mathcal{L}(\theta_{t+1}) = \mathcal{L}(\theta_t) - \eta\|g_t\|_1 + \frac{\eta^2}{2}\operatorname{tr} H$ at every step; adding over $t < T$ gives (1.4). By the prologue's theorem on sign descent, from step $\max_i|\theta_{0,i}|/\eta$ on each coordinate alternates between two values of absolute value less than η, so the iterates cycle and $\Delta_T < \frac{1}{2}\eta^2 \operatorname{tr} H$. Hence $\Delta_T/(\eta T) \to 0$. ∎

For normalized gradient descent the same computation gives (1.4) with $\frac{\eta}{2} \cdot \frac{1}{T} \sum_t q_t$ in place of $\frac{\eta \operatorname{tr} H}{2}$, where q_t is the curvature along $g_t/\|g_t\|_2$, so its bound is attained only by runs that step along a top eigenvector. Its cycles lie along eigenvectors, since on a cycle g and $g + \eta H u$ are both parallel to u, and by part (i) a cycle along an eigenvector with eigenvalue λ_k has average gradient norm $\eta\lambda_k/2$. In the plane of Figure 1.1 the three runs end on, or converge to, cycles of maximal curvature, so their averages reach the floors $L_{\|\cdot\|}\eta/2$ (Exercise 1.8); in high dimension a Euclidean run can linger on a cycle along an eigenvector with $\lambda_k < \lambda_{\max}$.

THE BEST STEP DEPENDS ON THE CRITERION. By Proposition 1.14(ii) the best fixed step for the criterion of the theorem is very nearly $\sqrt{2\Delta_0/(T \operatorname{tr} H)}$, which shrinks like $T^{-1/2}$. For the final loss, the criterion of a practitioner with a budget, the best step on the same problem shrinks like T^{-1}: it must cover the distance $\|\theta_0 - \theta^\star\|_\infty$ in the time available, and a longer step only raises the loss on the final cycle (Exercise 1.9).

WORST-CASE RATES DO NOT RANK THE NORMS. The convex rate compares norms through LR^2. For a diagonal quadratic the sublevel set is an ellipsoid with axes along the coordinates, whose ℓ_∞ and ℓ_2 radii both equal its longest semi-axis R. So $L_{\ell_\infty} R^2 = \operatorname{tr} H \cdot R^2 \geq L_{\ell_2} R^2$: the convex rate never prefers the max norm on such a problem, while the step of Figure 1.2 prefers it by a factor of three. The worst point of a sublevel set has its gradient on one flat coordinate, the worst case for the max norm, and the gradient at hand need not look like that (Exercise 1.6).

1.6 What the theory misses

WHAT WAS IDEALIZED AWAY. A fixed norm: practical optimizers change their norm at every step, through Adam's second moments or Shampoo's Kronecker factors, and a theory of one norm describes them only with their accumulators switched off (Chapters 6 and 7). A global smoothness constant: the curvature of a network changes during training, and at practical step sizes it settles at the largest value the step size tolerates (Chapter 8). Exact gradients: with noise, $\operatorname{lmo}(g_t)$ is a biased estimate of $\operatorname{lmo}(\nabla \mathcal{L}(\theta_t))$, because the oracle is not linear (Chapter 5). No memory, and one vector instead of matrices in layers (Chapters 3 and 4).

WHERE IT BREAKS. At the step sizes of practice the one-step guarantee is void. Full-batch gradient descent on neural networks drives the sharpness up to $2/\eta$ and trains there (Cohen et al. 2021), where the Euclidean guarantee $\eta(1 - L\eta/2)\|g\|_2^2$, with L the sharpness, is zero; the loss still falls, over long times and not monotonically (Chapter 8). Smoothness constants are not constant: along the training of language models and image classifiers, Zhang et al. (2020) measured a local smoothness that grows with the gradient norm, so a global L is absent or uselessly large (Chapter 10). The rates do not rank optimizers: on transformers, the gap between Adam and gradient descent is attributed to mechanisms such as the sign-like geometry of the step (Kunstner et al. 2023), not to a worst-case rate (Chapter 14). The right

norm is a modeling choice that the theorem cannot identify, since it holds for all norms at once and its inputs are not computable for networks. And the jumps of the oracle at ties break the deterministic picture once gradients are noisy: Karimireddy et al. (2019) give simple convex problems on which signSGD does not converge to the optimum (Chapter 2).

REPAIRS AND OPEN PROBLEMS. Smoothness that is local, anisotropic or tied to the gradient repairs the second failure: Xie et al. (2025) analyze Adam under smoothness in the max norm and report a much better empirical smoothness constant in this geometry than in the Euclidean one for GPT-2 and ResNet models in their natural coordinates, and (L_0, L_1)-smoothness (Zhang et al. 2020) and curvature bounded by the loss are the subjects of Chapters 9 and 10. Stochastic steepest descent with momentum in general norms is analyzed by Pethick et al. (2025), with a norm-constrained oracle in each layer, and by Kovalev (2025), as a trust-region method (Chapter 5); norms built from the architecture are the modular norms of Bernstein and Newhouse (2025) and Large et al. (2024) (Chapter 4). For normalized gradient descent at a fixed step, Arora et al. (2022) show that near a manifold of minimizers the oscillation at the floor drives the iterates toward lower sharpness, so on a loss that is not quadratic the floor has dynamics of its own. Open: a choice of norm from data (Exercise 1.10); the right norm for embeddings, attention and normalization layers; and a theory for the step sizes of practice.

1.7 Back in the wild

FIELD NOTES	
PRACTICE	Fixed-length steps: per coordinate (Lion, roughly Adam), per matrix (Muon), per layer (LARS, LAMB).
IDEALIZATION	Steepest descent in one fixed norm; exact gradients; a loss that is L-smooth in that norm.
PROVED	Duality maps; the decrease $\|g\|_*^2/(2L)$, the rates and the floor $L\eta/2$; smoothness constants of quadratics.
PREDICTS	No stability threshold for normalized steps; the ℓ_∞ guarantee beats ℓ_2's exactly when $\|g\|_1^2/\|g\|_2^2 > \|H\|_{\infty\to 1}/\lambda_{\max}(H)$.
MISSES	Global smoothness (edge of stability), noise (the oracle jumps), memory, layers, the choice of norm.
TAKEAWAY	Match update sizes when moving a learning rate between optimizers (suggested); decay a normalized step to go below its floor (proved in the idealization).

The theory tells a practitioner what a learning rate is: in every optimizer of the practice box it is the radius of a ball in some norm, and the norm decides which directions a step can take. It explains why matching update sizes is a sensible way to move a learning rate between optimizers, and why normalized optimizers tolerate step sizes at which gradient descent diverges, at the price of a floor that only a schedule removes. It does not

say which norm to use, because the comparison depends on the gradients that training will produce. Part I goes on to the max norm (Chapter 2), the spectral norm (Chapter 3) and the norm of a network (Chapter 4); the floor returns as the floor that momentum and schedules remove (Chapters 5 and 12), and the oracle as the Frank–Wolfe step of decoupled weight decay (Chapter 13).

Exercises

1.1 CONSOLIDATION. (Duality maps of ℓ_p.) Let $1 < p < \infty$, $1/p + 1/q = 1$ and $g \neq 0$. (a) Show that the dual of $\|\cdot\|_p$ is $\|\cdot\|_q$ and that the unique steepest unit direction is $\mathrm{lmo}(g)_i = -\operatorname{sign}(g_i)|g_i|^{q-1}/\|g\|_q^{q-1}$, so that $g_i^\sharp = \|g\|_q^{2-q}\operatorname{sign}(g_i)|g_i|^{q-1}$. (b) Find the limits of $\mathrm{lmo}(g)$ as $p \to \infty$ and $p \to 1$, and compare with Proposition 1.9(ii) and (iii); if the largest $|g_i|$ is attained k times, which point of the set in (iii) is the limit? *(Hint: equality holds in $\sum_i g_i u_i \geq -\|g\|_q\|u\|_p$ exactly when $|u_i|^p$ is proportional to $|g_i|^q$ and the signs are opposite.)*

1.2 CONSOLIDATION. (Sums and maxima of norms.) With blocks as in Proposition 1.9(iv) and scales $s_k > 0$: (a) show that the dual of $\max_k s_k\|\theta_k\|_{(k)}$ is $\sum_k \|g_k\|_{(k),*}/s_k$ with oracle $(\mathrm{lmo}_{(k)}(g_k)/s_k)_k$, so normalized descent in a weighted maximum norm is blockwise normalized descent with learning rates η/s_k; (b) show that the dual of $\sum_k s_k\|\theta_k\|_{(k)}$ is $\max_k \|g_k\|_{(k),*}/s_k$, and that its steepest unit directions put the whole step on the blocks that attain the maximum; (c) show that unnormalized descent in the norm of (a) couples all blocks through the single factor $\sum_j \|g_j\|_{(j),*}/s_j$ (Chapter 4).

1.3 CONSOLIDATION. (The constants of ℓ_∞ and ℓ_1.) (a) For a graph Laplacian $H = D - A$, show that $s^\top H s = 4|E(S, S^c)|$ for $s \in \{-1, 1\}^d$ and $S = \{i : s_i = 1\}$, where $E(S, S^c)$ is the set of edges leaving S; conclude that L_{ℓ_∞} is four times the maximum cut, so that computing it is NP-hard. (b) For $H = \operatorname{diag}(h) + c\mathbf{1}\mathbf{1}^\top$ with $h_i, c \geq 0$, show that $L_{\ell_\infty} = \sum_i h_i + cd^2$; for $h = 0$ this attains the upper bound $d\,\lambda_{\max}(H)$ of Proposition 1.13(ii). (c) Show that the ℓ_1 guarantee beats the ℓ_2 guarantee if and only if $\|g\|_\infty^2/\|g\|_2^2 > \max_i H_{ii}/\lambda_{\max}(H)$, and for $H = I + \mathbf{1}\mathbf{1}^\top$ with $d \geq 2$ decide which guarantee is larger at $g = e_1$ and at $g = \mathbf{1}$.

1.4 CONSOLIDATION. (Step length and update size.) (a) Show that a normalized step of length η has root-mean-square η in ℓ_∞ (if no g_i vanishes), $\eta/\sqrt{d}$ in ℓ_2 and ℓ_1, and $\eta/\sqrt{\max(m, n)}$ in the spectral norm of a full-rank $m \times n$ matrix, and that the rescaled Muon update $0.2\sqrt{\max(m, n)}\,\eta\,\mathrm{polar}(G)$ has root-mean-square 0.2η. (b) For $W \in \mathbb{R}^{d_{\mathrm{out}} \times d_{\mathrm{in}}}$ and $\|W\|_{\mathrm{rms}\to\mathrm{rms}} = \sqrt{d_{\mathrm{in}}/d_{\mathrm{out}}}\,\|W\|_{\mathrm{op}}$, show that $-\sqrt{d_{\mathrm{out}}/d_{\mathrm{in}}}\,\mathrm{polar}(G)$ is a steepest unit direction, the only one if G has full rank, and compare how the root-mean-squares of the two rules scale with the width of a square matrix (Chapter 4).

1.5 VARIATION. (A quadratic norm is a preconditioner.) Let $P \succ 0$ and $\|u\|_P = (u^\top P u)^{1/2}$. (a) Show that the dual norm is $\|g\|_{P^{-1}}$, $\mathrm{lmo}(g) = -P^{-1}g/\|g\|_{P^{-1}}$ and $g^\sharp = P^{-1}g$: unnormalized steepest descent in $\|\cdot\|_P$ is preconditioned gradient descent. (b) For $\mathcal{L}(\theta) = \frac{1}{2}\theta^\top H\theta$ show that $L_{\|\cdot\|_P} = \lambda_{\max}(P^{-1}H)$, the preconditioned sharpness S_P. (c) For $H \succ 0$, show that among quadratic norms the guarantee $g^\top P^{-1}g/(2\lambda_{\max}(P^{-1}H))$ is largest for $P = H$, where it is the whole gap $\frac{1}{2}g^\top H^{-1}g$: Newton's method. (d) For $\theta = A\varphi$ with A invertible, show that steepest descent on $\varphi \mapsto \mathcal{L}(A\varphi)$ in the norm $\varphi \mapsto \|A\varphi\|$ gives $\varphi_t = A^{-1}\theta_t$, while gradient descent in φ is steepest descent in θ for the norm $\|A^{-1}\theta\|_2$. *(Hint: for (c), with $\mu = \lambda_{\max}(P^{-1}H) = \lambda_{\max}(H^{1/2}P^{-1}H^{1/2})$, show that $P^{-1} \preceq \mu H^{-1}$.)*

1.6 VARIATION. (Polyak–Łojasiewicz in a dual norm; see Lemma A.19.) Suppose that, besides Assumption 1.7, $\|\nabla\mathcal{L}(\theta)\|_*^2 \geq 2\mu(\mathcal{L}(\theta) - \mathcal{L}^\star)$ for all θ. (a) Show that unnormalized steepest descent with step size $1/L$ has $\Delta_t \leq (1-\mu/L)^t\Delta_0$. (b) For $\mathcal{L}(\theta) = \frac{1}{2}\sum_i h_i\theta_i^2$ with $h_1 \geq \cdots \geq h_d > 0$, show that the best constant is $\mu = h_d$ both for ℓ_2 and for ℓ_∞ (whose dual is ℓ_1), so the guaranteed rates are $1 - h_d/h_1$ and $1 - h_d/\mathrm{tr}\,H$. (c) Conclude that the worst-case linear rate always prefers ℓ_2 on a diagonal quadratic, and reconcile this with Figure 1.2.

1.7 VARIATION. (A schedule removes the floor.) Run normalized steepest descent with step sizes $\eta_t > 0$. (a) Show that $\sum_{t<T}\eta_t\|g_t\|_* \leq \Delta_0 + \frac{L}{2}\sum_{t<T}\eta_t^2$. (b) Deduce that $\liminf_t\|g_t\|_* = 0$ if $\sum_t\eta_t = \infty$ and $\sum_t\eta_t^2 < \infty$. (c) With $\eta_t = \eta_0/\sqrt{t+1}$, show that $\min_{t<T}\|g_t\|_* \leq \big(\Delta_0 + \frac{L}{2}\eta_0^2(1 + \ln T)\big)/\big(2\eta_0(\sqrt{T+1}-1)\big)$, and compare with the prologue's sign descent with a decaying step.

1.8 SIMULATION. (Three norms in the plane.) For the quadratic of Figure 1.1, right (script `01-steepest-geometry.py`), (a) check that $L_{\ell_2} = 1$, $L_{\ell_\infty} = H_{11} + H_{22} + 2|H_{12}| = 1.882$ and $L_{\ell_1} = \max_i H_{ii} = 0.780$. (b) For $\eta \in \{0.2, 0.05, 0.01\}$, run each method for $400/\eta$ steps and average the dual norm of the gradient over the last 100 steps; you should find $L_{\|\cdot\|}\eta/2$ in each norm, to five digits. (c) Explain this with Proposition 1.14(i), by finding the direction of each final cycle and checking that it maximizes $u^\top Hu$ over the unit ball.

1.9 SIMULATION. (The best fixed step depends on what is measured.) Run sign descent with a fixed step η on the quadratic of Figure 1.2, from its starting point (script `01-steepest-descent.py`). For each of $T = 10^2$, 10^3 and 10^4, find by a grid search over $\eta \in [10^{-4}, 1]$ (a) the η that minimizes $\frac{1}{T}\sum_{t<T}\|g_t\|_1$, and (b) the η that minimizes $\mathcal{L}(\theta_T)$. You should find (a) within the grid spacing of $\sqrt{2\Delta_0/(T\,\mathrm{tr}\,H)}$, with minimum within 0.2 percent of $\sqrt{2\Delta_0\,\mathrm{tr}\,H/T}$, and (b) within about ten percent of $\max_i|\theta_{0,i}|/T$. Explain (a) with Proposition 1.14(ii), and (b) by the time a step needs to cover the distance $\|\theta_0\|_\infty$.

1.10 RESEARCH. (Choosing a norm from data.) Exercise 1.5(c) shows that

among quadratic norms the one-step guarantee singles out the Hessian. (a) For weighted max norms $\max_i a_i|u_i|$ on a diagonal quadratic with $H \succ 0$, find the weights that maximize the guarantee $\|g\|_*^2/(2L_{\|\cdot\|})$ at a gradient with no zero coordinate, and show that the resulting step is Newton's. (b) For $h_i = i^{-a}$ and Gaussian θ_0, map numerically the exponents and target losses for which normalized ℓ_∞ descent at its best fixed step beats normalized ℓ_2 descent, and test whether (1.3) at θ_0 predicts the winner. Open: a rule that chooses a norm online, from gradients alone, with a nonconvex guarantee that improves on the best fixed norm of a family in hindsight; AdaGrad's guarantee (Chapter 6) is of this kind for diagonal quadratic norms and convex losses.

Notes and References

Steepest descent goes back to Cauchy (1847), and its stochastic form to Robbins and Monro (1951). Boyd and Vandenberghe (2004, Section 9.4) treat it in a general norm, define normalized and unnormalized steepest descent directions, and work out the quadratic norms and the ℓ_1 norm, for which it is coordinate descent; Nutini et al. (2015) showed that the speed of the Gauss–Southwell rule is governed by strong convexity in the ℓ_1 norm. The descent inequality and the convex rate are standard (Nesterov 2018), and Kelner et al. (2014) used steepest descent in the ℓ_∞ norm for maximum flow. Mirror descent is due to Nemirovsky and Yudin (1983), in its modern form to Beck and Teboulle (2003).

In deep learning, spectral descent was proposed by Carlson et al. (2015), sign descent was analyzed for gradient compression (Bernstein et al. 2018) and as a component of Adam (Balles and Hennig 2018), and layerwise normalization entered through large-batch training (You et al. 2017, 2020). Bernstein and Newhouse (2024) read Adam, Shampoo and Prodigy without accumulators as steepest descent in particular norms, Large et al. (2024) built a norm for whole networks from the norms of their modules, Bernstein and Newhouse (2025) made the duality map of that norm the central object of a theory of optimizers, and Newhouse (2025) gives a connected account. The bound (1.2) for exact gradients is elementary, and we do not know who stated it first, nor have we seen the exactness of Proposition 1.14 stated. The hardness of $\|H\|_{\infty\to 1}$ is due to Rohn (2000), and Balles et al. (2020) argue, through smoothness in the max norm, that sign descent gains when the Hessian is concentrated on its diagonal and its largest eigenvalue is much larger than its average one.

Translating the literature. Boyd and Vandenberghe (2004) call our $\operatorname{lmo}(g)$ a normalized steepest descent direction Δx_{nsd}, and our $-g^\sharp$ the unnormalized steepest descent direction $\Delta x_{\text{sd}} = \|\nabla f(x)\|_* \Delta x_{\text{nsd}}$. Kelner et al. (2014) write $x^{\#}$ for our $x^\sharp$. Bernstein and Newhouse (2024) write $\|\cdot\|^\dagger$ for our dual norm and λ for the coefficient of the quadratic penalty, our L (or $1/\eta$), so their steepest descent step $-(\|g\|^\dagger/\lambda)\arg\max_{\|t\|=1} g^\top t$ is our $-g^\sharp/L$; in Bernstein and Newhouse (2025) the map $g \mapsto \arg\max_{\|t\|=1}\langle g, t\rangle$, our $-\operatorname{lmo}(g)$, is the duality map that produces the dualized gradient. Pethick

et al. (2025) write $\operatorname{lmo}(d) \in \arg\min_{x\in\mathcal{D}}\langle d, x\rangle$ for a norm ball $\mathcal{D}$ of radius ρ, which is ρ times our $\operatorname{lmo}(d)$, so their update $x^{k+1} = x^k + \gamma_k \operatorname{lmo}(d^k)$ is our normalized step with $\eta = \gamma_k\rho$. Our $L_{\ell_\infty} = \|H\|_{\infty\to 1}$ is also written $\|H\|_{\infty,1}$.

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2 Sign Descent

The ℓ_∞ geometry of Adam, signSGD and Lion

IN THE WILD — SIGNSGD, LION, ADAM WITHOUT MEMORY

IN THIS CHAPTER. Adam without its moving averages, signSGD, and Lion move every coordinate by the same amount: they take the step $-\eta\,\mathrm{sign}(\cdot)$, a step of fixed length in the max norm. We prove that this step gains $\eta\|\nabla\mathcal{L}\|_1$ against a curvature cost $\frac{1}{2}\eta^2\|\vec{L}\|_1$, a trade that separable quadratics attain exactly, and that a rotation of the coordinates can raise the cost by the factor $d\lambda_{\max}/\mathrm{tr}\,H$, which gradient descent does not notice. With minibatches the sign follows the median of the gradient, not its mean: the bias $\|\vec{\sigma}\|_1/\sqrt{B}$ is sharp, a rare class realizes it, and symmetric unimodal noise avoids it. Adam's step is at most $C(\beta_1,\beta_2)$ learning rates per coordinate, 7.27 at the defaults. The idealization drops the memory that makes Adam work and cannot say why the coordinates of real networks are good ones.

2.1 In the wild: one step length for every coordinate

Adam divides a moving average of gradients by the square root of a moving average of squared gradients (Kingma and Ba 2015). The division removes the scale of each coordinate: multiply the gradient of one parameter by a thousand, and Adam's step in that parameter does not change. Its learning rate is therefore not a multiplier of the gradient but a length, roughly the distance each parameter moves in one step. Two other optimizers make the same choice exactly. *SignSGD* updates with the sign of a stochastic gradient; in data-parallel training each worker then sends one bit per coordinate, and a server that takes the majority vote of the signs sends one bit back (Bernstein et al. 2018, 2019). The idea goes back to Rprop, which used only the signs of a full-batch gradient, and to 1-bit SGD for distributed training.[1] Lion, found by a symbolic search over update rules, updates with the sign of an interpolation between the gradient and a momentum, and keeps one number of state per parameter where Adam keeps two (Chen et al. 2023).

IN THE WILD · Adam, Kingma and Ba (2015); GPT-3, Brown et al. (2020); Lion, Chen et al. (2023); signSGD, Bernstein et al. (2019). Adam with $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$, the published defaults, and $\beta_2 = 0.95$ in large language models since GPT-3. Lion with $\beta_1 = 0.9$, $\beta_2 = 0.99$, a learning rate 3 to 10 times smaller than AdamW's and a weight decay 3 to 10 times larger, so that their product, the strength of the decay, stays about the same. In distributed training, send $\mathrm{sign}(g)$, one bit per

[1] Rprop keeps a step size for each weight, grows it while the weight's partial derivative keeps its sign and shrinks it when the sign flips; the size of the gradient never enters the update (Riedmiller and Braun 1993). RMSProp was introduced in a lecture course as a minibatch version of Rprop (Tieleman and Hinton 2012). 1-bit SGD quantized each gradient coordinate to one bit and carried the quantization error over to the next step (Seide et al. 2014).

coordinate, and aggregate by majority vote.

Three observations tie these optimizers together.

Observation 2.1 (Sign descent with momentum behaves like Adam; Kunstner et al. 2023, Zhao et al. 2025). On transformer language models the gap between Adam and stochastic gradient descent does not shrink as the batch grows to the full training set: Adam improves with the batch size and gradient descent much less, and with large batches Adam behaves like sign descent with momentum. In a comparison of optimizers for autoregressive language models across sizes and hyperparameters, signed momentum (*Signum*) recovered both the performance of Adam and its stability across hyperparameters.

Kunstner et al. (2023) concluded that heavy-tailed noise is not the main cause of the gap; Zhao et al. (2025) found Adam, Adafactor, Lion, and Sophia comparable, with only stochastic gradient descent behind. Both are empirical, at moderate scale, and silent on why the sign helps.[2]

[2] Signum was introduced with signSGD by Bernstein et al. (2018), who reported that it matched Adam's accuracy and speed on ImageNet models.

Observation 2.2 (Adam is a sign with a variance-dependent length; Balles and Hennig 2018). Adam's update of each coordinate can be read as the sign of the momentum times a factor that decreases with the relative variance of that coordinate's gradient. Studied separately, the sign is the aspect responsible for adverse effects on generalization that had been reported for adaptive methods.

Observation 2.3 (Rotating the parameters slows Adam; Xie et al. 2025). Training GPT-2 with Adam after a rotation of the parameter space is markedly slower than in the original coordinates, while stochastic gradient descent, whose trajectory rotates with the coordinates, is unaffected.

The three point at one object, the step $-\eta\ \mathrm{sign}(g)$, and one question: what does it buy, compared with $-\eta g$, and what does it cost? The answer depends on the norm in which steps are measured, on the coordinates, on the shape of the noise, and on memory.

2.2 The idealization: steepest descent in the max norm

Chapter 1 turned a gradient into a step with a norm: the normalized steepest-descent step is $\eta\,\mathrm{lmo}(g)$ with $\mathrm{lmo}(g) \in \arg\min_{\|u\|\le 1}\langle g, u\rangle$. For the max norm the dual norm is $\|g\|_1$ (Lemma A.3), and $\langle g, u\rangle$ is minimized over the cube $\{\|u\|_\infty \le 1\}$ by $u_i = -\mathrm{sign}(g_i)$ where $g_i \neq 0$ and by every $u_i \in [-1, 1]$ where $g_i = 0$; the convention $\mathrm{sign}\,0 = 0$ picks the shortest minimizer. Normalized steepest descent in ℓ_∞ is sign descent.

Definition 2.4 (Sign descent and signSGD). Given step sizes $\eta_t > 0$, *sign descent* is $\theta_{t+1} = \theta_t - \eta_t\,\mathrm{sign}(\nabla\mathcal{L}(\theta_t))$, and signSGD is $\theta_{t+1} = \theta_t - \eta_t\,\mathrm{sign}(g_t)$ for a stochastic gradient g_t, with the sign taken coordinatewise. When no coordinate of the gradient vanishes, the step has length η_t in ℓ_∞ and in the RMS norm, and $\eta_t\sqrt{d}$ in ℓ_2.

The learning rate of the practice sections and the step size of the theory are the same number η; for a sign method it is the distance that every coordinate moves. The curvature bound that suits the max norm is the one with which Bernstein et al. (2018) posed the problem.

Assumption 2.5 (Coordinatewise smoothness). The loss $\mathcal{L}\colon \mathbb{R}^d \to \mathbb{R}$ is differentiable and bounded below by $\mathcal{L}^\star$, and there are $L_1, \dots, L_d \geq 0$ such that for all $\theta, u \in \mathbb{R}^d$

$$\mathcal{L}(\theta + u) \leq \mathcal{L}(\theta) + \langle \nabla \mathcal{L}(\theta), u \rangle + \frac{1}{2} \sum_{i=1}^{d} L_i u_i^2.$$

We write $\vec{L} = (L_1, \dots, L_d)$, $\|\vec{L}\|_1 = \sum_i L_i$, and $L_{\max} = \max_i L_i$.

For a twice-differentiable loss, *coordinatewise smoothness* says that $\nabla^2 \mathcal{L}(\theta) \preceq \operatorname{diag}(\vec{L})$ at every θ. Since $\frac{1}{2}\sum_i L_i u_i^2$ is at most $\frac{1}{2}\|\vec{L}\|_1 \|u\|_\infty^2$ and at most $\frac{1}{2} L_{\max} \|u\|_2^2$, it gives the quadratic upper bounds of Chapter 1 (Lemma A.18) in two geometries at once: the constant $\|\vec{L}\|_1$ in ℓ_∞ and $L_{\max}$ in ℓ_2.

Remark 2.6 (Coordinatewise and ℓ_∞ smoothness). For a quadratic with Hessian $H \succeq 0$ the admissible $\vec{L}$ are those with $\operatorname{diag}(\vec{L}) \succeq H$; each has $\|\vec{L}\|_1 = s^\top \operatorname{diag}(\vec{L}) s \geq s^\top H s$ for every sign vector s, so $\|\vec{L}\|_1 \geq \|H\|_{\infty \to 1} = \max_{s \in \{\pm 1\}^d} s^\top H s$, the exact ℓ_∞ constant of Chapter 1. Equality holds for diagonal H and $\vec{L} = \operatorname{diag}(H)$, and the best $\|\vec{L}\|_1$ is at most $\frac{\pi}{2}\|H\|_{\infty \to 1}$ (Exercise 2.4).

Assumption 2.7 (Minibatch noise). The stochastic gradient is $g_t = \nabla \mathcal{L}(\theta_t) + \zeta_t$ where, conditionally on $\theta_0, g_0, \dots, g_{t-1}$, each coordinate $\zeta_{t,i}$ has mean zero and variance at most σ_i^2 / B. We write $\vec{\sigma} = (\sigma_1, \dots, \sigma_d)$.

A batch of B independent examples whose per-example gradients have coordinate variances at most σ_i^2 satisfies the assumption. We count Adam's steps from $t = 0$, like the iterates: $m_{-1} = v_{-1} = 0$, $m_t = \beta_1 m_{t-1} + (1-\beta_1) g_t$, $v_t = \beta_2 v_{t-1} + (1-\beta_2) g_t^2$ with squares taken coordinatewise, $\hat{m}_t = m_t / (1 - \beta_1^{t+1})$, $\hat{v}_t = v_t / (1 - \beta_2^{t+1})$, and $\theta_{t+1} = \theta_t - \eta\, \hat{m}_t / (\sqrt{\hat{v}_t} + \epsilon)$.

What was thrown away. The idealization keeps the step and discards the rest: moving averages (they return only through the size of Adam's step, and in full in Chapters 5 and 6), Adam's ϵ, the variation of curvature, which a global bound replaces, and every feature of the noise but its variance. These are the right first simplifications because the sign is the ingredient that all of these optimizers share and gradient descent lacks.

What one had to notice. Without memory, dividing by $\sqrt{v}$ is dividing by $|g|$, so Adam's update is a corner of the unit ball of the max norm. Then memoryless Adam, signSGD, and Lion all step to a corner of a box, and the questions are those of Chapter 1: which dual norm measures the gradient,

and which smoothness constant pays for the step (Bernstein and Newhouse 2024).

DICTIONARY

IN THE THEORY	IN A TRAINING RUN
the step $-\eta\,\mathrm{sign}(g)$	a step of one learning rate per weight
the dual norm $\|\nabla\mathcal{L}\|_1$	progress per unit learning rate
$\vec{L}$ and $\|\vec{L}\|_1$	curvature along each parameter's axis
$\|g\|_1^2/\|g\|_2^2 \in [1, d]$	how many parameters share the gradient
$\mathrm{tr}\,H/\lambda_{\max}$	the number of stiff directions
a rotation Q	a change of basis of parameter space
$\sigma_i/\sqrt{B}$	the minibatch noise of parameter i
the median of $g_{t,i}$	what a sign update follows
$C(\beta_1, \beta_2)$	Adam's largest step per unit of η

FIELD GUIDE · SIGNSGD AND SIGNUM *Bernstein et al. (2018)*

UPDATE	$\theta_{t+1} = \theta_t - \eta\ \mathrm{sign}(g_t)$; Signum uses $\mathrm{sign}(m_t)$ instead
NORM	ℓ_∞; $\mathrm{lmo}(g) = -\mathrm{sign}(g)$
PRECONDITIONER	none (equivalently $\mathrm{diag}\lvert g_t\rvert^{-1}$, rebuilt at every step)
AVERAGES	none; Signum: $m_t = \beta m_{t-1} + (1-\beta)g_t$
STATE	none; Signum: one number per parameter
VARIANTS	majority vote $-\eta\ \mathrm{sign}\big(\sum_m \mathrm{sign}(g_t^{(m)})\big)$ over workers (Bernstein et al. 2019); error feedback (Karimireddy et al. 2019)
CAUTION	follows the median, not the mean (Proposition 2.15)

FIELD GUIDE · LION *Chen et al. (2023)*

UPDATE	$c_t = \beta_1 m_{t-1} + (1-\beta_1)g_t,\ \ m_t = \beta_2 m_{t-1} + (1-\beta_2)g_t,$ $\theta_{t+1} = \theta_t - \eta\,\big(\mathrm{sign}(c_t) + \lambda\theta_t\big)$
NORM	ℓ_∞
PRECONDITIONER	none
AVERAGES	one EMA (β_2), read with weight $1-\beta_1$ on g_t
STATE	one number per parameter
IN THE WILD	$\beta_1 = 0.9$, $\beta_2 = 0.99$; learning rate several times below AdamW's
VARIANTS	$\beta_1 = \beta_2$: Signum (Exercise 2.6); weight decay: a box constraint (Chen et al. 2024)

FIELD GUIDE · ADAM *Kingma and Ba (2015)*

UPDATE	$\theta_{t+1} = \theta_t - \eta\,\hat{m}_t/(\sqrt{\hat{v}_t} + \epsilon)$; full entry in Chapter 6
NORM	ℓ_∞ if $\beta_1 = \beta_2 = \epsilon = 0$ (Proposition 2.8); steps at most $C(\beta_1, \beta_2)\eta$ (Proposition 2.17)
PRECONDITIONER	diagonal, $\mathrm{diag}(\sqrt{\hat{v}_t}) + \epsilon$

AVERAGES	EMA of g (β_1) and of g^2 (β_2)
STATE	two numbers per parameter
IN THE WILD	$\beta_1 = 0.9$, $\beta_2 \in \{0.95, 0.999\}$, $\epsilon = 10^{-8}$

2.3 The theory: the max-norm step

Proposition 2.8 (Adam without memory is sign descent). *Let $\epsilon \geq 0$ and write Adam's update as $\theta_{t+1} = \theta_t - \eta u_t$, with $u_{t,i} = \hat{m}_{t,i}/(\sqrt{\hat{v}_{t,i}} + \epsilon)$ and $0/0 = 0$. (i) If $\beta_1 = \beta_2 = 0$, then $u_{t,i} = \frac{|g_{t,i}|}{|g_{t,i}|+\epsilon}\operatorname{sign}(g_{t,i})$ for every t, with or without bias correction, and $u_t = \operatorname{sign}(g_t)$ if $\epsilon = 0$; the same holds for RMSProp, $\theta_{t+1} = \theta_t - \eta g_t/(\sqrt{v_t} + \epsilon)$, with coefficient 0. (ii) With bias correction, u_0 is given by the same formula for all $\beta_1, \beta_2 \in [0, 1)$: if $\epsilon = 0$, then Adam's first step is a sign step whatever the size of the first gradient.*

Proof. (i) With $\beta_1 = \beta_2 = 0$, $m_t = g_t$ and $v_t = g_t^2$, and the correction factors $1 - 0^{t+1}$ equal 1, so $u_{t,i} = g_{t,i}/(|g_{t,i}| + \epsilon)$; RMSProp with coefficient 0 has the same update. (ii) Here $m_0 = (1 - \beta_1)g_0$ and $v_0 = (1 - \beta_2)g_0^2$, so $\hat{m}_0 = g_0$ and $\hat{v}_0 = g_0^2$. ∎

The factor $|g|/(|g| + \epsilon)$ makes memoryless Adam sign descent where the gradient is much larger than ϵ and gradient descent with step size η/ϵ where it is much smaller (Exercise 2.7). With $\epsilon = 0$, memoryless Adam, memoryless RMSProp, and signSGD are one algorithm, and its guarantee is the descent inequality of Chapter 1 in the max norm.

Theorem 2.9 (Sign descent in the max norm). *Let Assumption 2.5 hold and let $\theta_{t+1} = \theta_t - \eta_t \operatorname{sign}(\nabla\mathcal{L}(\theta_t))$ with $\eta_t > 0$. Then for every t*

$$\mathcal{L}(\theta_{t+1}) \leq \mathcal{L}(\theta_t) - \eta_t\|\nabla\mathcal{L}(\theta_t)\|_1 + \tfrac{1}{2}\eta_t^2\|\vec{L}\|_1, \tag{2.1}$$

and the sum of the L_i with $\partial_i\mathcal{L}(\theta_t) \neq 0$ may replace $\|\vec{L}\|_1$; for every $T \geq 1$

$$\frac{\sum_{t<T}\eta_t\|\nabla\mathcal{L}(\theta_t)\|_1}{\sum_{t<T}\eta_t} \leq \frac{\Delta_0 + \frac{1}{2}\|\vec{L}\|_1\sum_{t<T}\eta_t^2}{\sum_{t<T}\eta_t}. \tag{2.2}$$

For a constant step size η the right side is $\Delta_0/(\eta T) + \frac{1}{2}\eta\|\vec{L}\|_1$; if Δ_0 and $\|\vec{L}\|_1$ are positive and $\eta = \sqrt{2\Delta_0/(\|\vec{L}\|_1 T)}$, then it is $\sqrt{2\Delta_0\|\vec{L}\|_1/T}$, which therefore bounds $\min_{t<T}\|\nabla\mathcal{L}(\theta_t)\|_1$.

Proof. The plan is to apply Assumption 2.5 to the step $u = -\eta_t \operatorname{sign}(\nabla\mathcal{L}(\theta_t))$ and to sum. Fix t, and write $g = \nabla\mathcal{L}(\theta_t)$ and $s = \operatorname{sign}(g)$. The linear term of the assumption is $\langle g, -\eta_t s\rangle = -\eta_t\sum_i|g_i| = -\eta_t\|g\|_1$. The quadratic term is $\frac{1}{2}\eta_t^2\sum_i L_i s_i^2$, and s_i^2 is 1 where $g_i \neq 0$ and 0 elsewhere. This proves (2.1). Rearranged, (2.1) bounds $\eta_t\|\nabla\mathcal{L}(\theta_t)\|_1$ by $\mathcal{L}(\theta_t) - \mathcal{L}(\theta_{t+1}) + \frac{1}{2}\eta_t^2\|\vec{L}\|_1$; summed over $t < T$, the loss differences telescope to $\mathcal{L}(\theta_0) - \mathcal{L}(\theta_T) \leq \Delta_0$, and dividing by $\sum_t \eta_t$ gives (2.2). The stated constant η minimizes $\Delta_0/(\eta T) + \frac{1}{2}\eta\|\vec{L}\|_1$,

and a minimum is at most an average. The excluded cases are trivial: $\Delta_0 = 0$ makes θ_0 a stationary point, and $\|\vec{L}\|_1 = 0$ puts $\mathcal{L}$ below its tangent planes, so that $\mathcal{L}$ is concave and, being bounded below on $\mathbb{R}^d$, constant. ■

The comparison with gradient descent comes from reading the same assumption in three geometries.

Corollary 2.10 (Three guaranteed decreases). *Let Assumption 2.5 hold with every $L_i > 0$, and let $g = \nabla\mathcal{L}(\theta) \neq 0$. The gradient step $-g/L_{\max}$, the sign step $-(\|g\|_1/\|\vec{L}\|_1)\operatorname{sign}(g)$ and the diagonally preconditioned step $-\operatorname{diag}(\vec{L})^{-1}g$ decrease $\mathcal{L}$ by at least $\|g\|_2^2/(2L_{\max})$, $\|g\|_1^2/(2\|\vec{L}\|_1)$, and $\sum_i g_i^2/(2L_i)$ respectively. The third is at least the second, with equality if and only if $|g_i|/L_i$ does not depend on i; the second is at least the first if and only if*

$$\frac{\|g\|_1^2}{\|g\|_2^2} \geq \frac{\|\vec{L}\|_1}{L_{\max}}. \tag{2.3}$$

Proof. The plan is to evaluate Assumption 2.5 at the three steps and to compare the results. The first and third bounds are the assumption at $u = -g/L_{\max}$, with $\sum_i L_i u_i^2 \leq L_{\max}\|u\|_2^2$, and at $u_i = -g_i/L_i$. The second is (2.1) at $\eta = \|g\|_1/\|\vec{L}\|_1$, which maximizes $\eta\|g\|_1 - \frac{1}{2}\eta^2\|\vec{L}\|_1$. Writing $|g_i| = (|g_i|/\sqrt{L_i})\sqrt{L_i}$, the Cauchy–Schwarz inequality gives $\|g\|_1^2 \leq \|\vec{L}\|_1 \sum_i g_i^2/L_i$, with equality exactly when $|g_i|/\sqrt{L_i}$ is proportional to $\sqrt{L_i}$. The last claim is a rearrangement. ■

Both sides of (2.3) lie between 1 and d. The left side counts the coordinates over which the gradient is spread; the right side is an *effective rank* of the curvature in the parameters' coordinates, $\operatorname{tr} H/\lambda_{\max}$ for a diagonal quadratic. This is the reading of Proposition 1.13, which has the exact constants $\|H\|_{\infty\to 1}$ and $\lambda_{\max}$ in their place: the sign wins when the gradient is spread over more coordinates than the curvature has stiff directions. Networks have millions of parameters, most of which receive gradient, and, at least without batch normalization, Hessians with a few large outlier eigenvalues above a bulk of small ones (Ghorbani et al. 2019); were such a Hessian diagonal in the parameters' coordinates, the advantage could approach d over the effective rank. Whether it is close to diagonal is the subject of Proposition 2.12. In the equality case $|g_i|/L_i$, the length of the Newton step in coordinate i for a diagonal quadratic, is the same for all i: the sign matches the best diagonal preconditioner when all coordinates are equally far from their targets.

Is the cost $\frac{1}{2}\eta^2\|\vec{L}\|_1$ an artifact of the proof? On a quadratic it is exact.

Proposition 2.11 (The floor of sign descent on a quadratic). *Let $\mathcal{L}(\theta) = \frac{1}{2}(\theta - \theta^\star)^\top H(\theta - \theta^\star)$ with $H \succeq 0$, run sign descent with a constant step size η, and write $g_t = \nabla\mathcal{L}(\theta_t)$ and $s_t = \operatorname{sign}(g_t)$.*

(i) For every $T \geq 1$,

$$\frac{1}{T}\sum_{t<T}\Big(\|g_t\|_1 - \frac{\eta}{2}\, s_t^\top H s_t\Big) = \frac{\Delta_0 - \Delta_T}{\eta T}, \tag{2.4}$$

and Δ_t is bounded; so $\frac{1}{T}\sum_{t<T}\|g_t\|_1$ and $\frac{1}{T}\sum_{t<T}\frac{1}{2}\eta s_t^\top H s_t$ have the same limit points, all at most $\frac{1}{2}\eta\|H\|_{\infty\to 1}$.

(ii) Let $H = \operatorname{diag}(h_1, \dots, h_d)$ with all $h_i > 0$, and let $r_i \in [0, \eta)$ be the remainder of $|\theta_{0,i} - \theta_i^\star|$ modulo η. From step $\lfloor|\theta_{0,i} - \theta_i^\star|/\eta\rfloor$ on, the distance of coordinate i from $\theta_i^\star$ alternates between r_i and $\eta - r_i$ if $r_i > 0$, and is 0 if $r_i = 0$. Hence $\frac{1}{T}\sum_{t<T}\|g_t\|_1 \to \frac{1}{2}\eta\sum_{i:\, r_i>0} h_i$, which is $\frac{1}{2}\eta \operatorname{tr} H$ unless some $|\theta_{0,i} - \theta_i^\star|$ is a multiple of η.

Proof. The plan is to use that the descent inequality is an identity for a quadratic, and to solve the separable case one coordinate at a time. (i) The identity is $\mathcal{L}(\theta_t - \eta s_t) = \mathcal{L}(\theta_t) - \eta\|g_t\|_1 + \frac{1}{2}\eta^2 s_t^\top H s_t$; summing over $t < T$ and dividing by ηT gives (2.4). For boundedness, let $H \neq 0$ (the case $H = 0$ is trivial), let λ_+ be its smallest positive eigenvalue and $L_\infty = \|H\|_{\infty\to 1}$; then $s^\top H s \leq L_\infty$ for every $s \in \{-1, 0, 1\}^d$, since the convex function $x \mapsto x^\top H x$ is maximized over the cube at a vertex. As g_t lies in the range of H, $\Delta_t = \frac{1}{2} g_t^\top H^+ g_t \leq \|g_t\|_1^2/(2\lambda_+)$. If $\|g_t\|_1 \geq \frac{1}{2}\eta L_\infty$, then the identity gives $\Delta_{t+1} \leq \Delta_t$, and otherwise $\Delta_{t+1} \leq \Delta_t + \frac{1}{2}\eta^2 L_\infty < \eta^2 L_\infty^2/(8\lambda_+) + \frac{1}{2}\eta^2 L_\infty$; by induction Δ_t never exceeds the larger of Δ_0 and this number. (ii) Coordinate i evolves on its own: $x_t = \theta_{t,i} - \theta_i^\star$ satisfies $x_{t+1} = x_t - \eta \operatorname{sign}(x_t)$, so $|x_{t+1}| = \big||x_t| - \eta\big|$ if $x_t \neq 0$, and $x_{t+1} = 0$ if $x_t = 0$. The distance falls by η per step for $\lfloor|x_0|/\eta\rfloor$ steps and reaches r_i; if $r_i = 0$, then it stays there, and if $r_i > 0$, then the next distances are $\eta - r_i$ and $|\eta - r_i - \eta| = r_i$. Over each period $h_i|x_t|$ averages $\frac{1}{2}\eta h_i$. ∎

For a separable quadratic the best constants are $\vec{L} = \operatorname{diag}(H)$, so for generic initial points the long-run average of $\|\nabla\mathcal{L}\|_1$ equals the floor $\frac{1}{2}\eta\|\vec{L}\|_1$ of Theorem 2.9, which cannot be improved. On a separable quadratic at a constant step size, sign descent ends on a *two-cycle*, the many-dimensional version of the prologue's example (**??**); only a decreasing schedule removes the floor (Chapter 12).

Gradient descent does not depend on the orthonormal basis in which the parameters are written; sign descent does.

Proposition 2.12 (Sign descent is not rotation invariant). *Let $H \geq 0$ have largest eigenvalue $\lambda_{\max}$ with unit eigenvector v_1, and for an orthogonal Q let $\mathcal{L}_Q(\phi) = \mathcal{L}(Q\phi)$, whose Hessian $Q^\top H Q$ has largest eigenvalue $\lambda_{\max}$. Then (i) $\operatorname{tr} H \leq \|Q^\top H Q\|_{\infty\to 1} \leq d\lambda_{\max}$, with the lower bound attained when the columns of Q are eigenvectors of H and the upper bound when $Q^\top v_1 = \mathbf{1}/\sqrt{d}$; (ii) $\|Q^\top H Q\|_{\infty\to 1} \geq \lambda_{\max}\|Q^\top v_1\|_1^2$; (iii) if Q is uniformly distributed on the orthogonal group, then $\mathbb{E}\|Q^\top v_1\|_1^2 = 1 + \frac{2}{\pi}(d-1)$. So rotations change the ℓ_∞ smoothness constant by factors up to $d\lambda_{\max}/\operatorname{tr} H$, and*

after a uniformly random rotation it is on average at least $(1+\frac{2}{\pi}(d-1))\lambda_{\max}$*, which is* $(1+\frac{2}{\pi}(d-1))\lambda_{\max}/\operatorname{tr} H$ *times its value in an eigenbasis.*

Proof. The plan is to bound $s^\top As$, for $A = Q^\top HQ$, by averaging over random signs, by the top eigenvalue, and at the signs of the top eigenvector; recall $\|A\|_{\infty\to 1} = \max_{s\in\{\pm1\}^d} s^\top As$. (i) For independent uniform signs, $\mathbb{E}[s^\top As] = \sum_{i,j} A_{ij}\mathbb{E}[s_i s_j] = \operatorname{tr} A = \operatorname{tr} H$, so some sign vector reaches $\operatorname{tr} H$; and $s^\top As \le \lambda_{\max}\|s\|_2^2 = d\lambda_{\max}$. If the columns of Q are eigenvectors, then A is diagonal and nonnegative and $s^\top As = \operatorname{tr} A$ for all s. Some orthogonal Q maps $\mathbf{1}/\sqrt{d}$ to v_1, and for it (ii) gives $\|A\|_{\infty\to 1} \ge d\lambda_{\max}$. (ii) Let $s = \operatorname{sign}(Q^\top v_1)$, zeros replaced by 1, and let (λ_j, v_j) be the eigenpairs of H, all $\lambda_j \ge 0$. Then $s^\top As = \sum_j \lambda_j \langle v_j, Qs\rangle^2 \ge \lambda_{\max}\langle Q^\top v_1, s\rangle^2 = \lambda_{\max}\|Q^\top v_1\|_1^2$. (iii) The vector $u = Q^\top v_1$ is uniform on the sphere, so it has the law of $z/\|z\|_2$ for $z \sim \mathcal{N}(0, I_d)$ and is independent of $\|z\|_2$. For $i \ne j$, $\frac{2}{\pi} = \mathbb{E}|z_i z_j| = \mathbb{E}\|z\|_2^2\,\mathbb{E}|u_i u_j| = d\,\mathbb{E}|u_i u_j|$; hence $\mathbb{E}\|u\|_1^2 = \sum_i \mathbb{E}u_i^2 + \sum_{i\ne j}\mathbb{E}|u_i u_j| = 1 + d(d-1)\frac{2}{\pi d}$. ∎

A random rotation also spreads the gradient, $\|Q^\top g\|_1^2 \approx \frac{2}{\pi} d\|g\|_2^2$, so in the criterion of Proposition 1.13 the left side becomes about $\frac{2}{\pi}d$ and the right side lies between $\|Q^\top v_1\|_1^2 \approx \frac{2}{\pi}d$ and d: the advantage is gone, and at worst reversed by the factor $\frac{\pi}{2}$. The guarantee is a property of the coordinate system, and the dynamics follows it (Figure 2.1).[3]

[3] A uniformly distributed orthogonal matrix is the Q factor of the QR decomposition of a matrix of independent standard Gaussians, once the signs of the diagonal of R are made positive; this is how the rotation of Figure 2.1 was drawn.

2.4 The theory: noise and memory

A stochastic gradient has the wrong sign in some coordinates, and a sign method moves those coordinates the wrong way by a full step. The cost is a question about one random variable.

Lemma 2.13 (Sign errors). *Let X be a random variable with mean $\mu \ne 0$ and variance at most s^2, where $s > 0$. (i) $\Pr[\operatorname{sign} X \ne \operatorname{sign}\mu] \le s^2/(s^2+\mu^2)$. (ii) $\mu\,\mathbb{E}[\operatorname{sign} X] \ge |\mu| - 2|\mu|s^2/(s^2+\mu^2) \ge |\mu| - s$. (iii) If $X-\mu$ is symmetric about 0 and unimodal, then $\mu\,\mathbb{E}[\operatorname{sign} X] \ge \min\{\mu^2/(\sqrt{3}\,s),\ \frac{2}{3}|\mu|\} > 0$.*

Proof. The plan is Cantelli's inequality for (i) and (ii) and Gauss's inequality for (iii). Replacing X by $-X$ if necessary, let $\mu > 0$ and write $\zeta = X - \mu$. (i) The event is $\{\zeta \le -\mu\}$, which for every $c \ge 0$ lies in $\{(c-\zeta)^2 \ge (c+\mu)^2\}$; by Markov's inequality its probability is at most $(s^2+c^2)/(c+\mu)^2$, which is $s^2/(s^2+\mu^2)$ at $c = s^2/\mu$ (Cantelli's inequality). (ii) Pointwise $\operatorname{sign} X \ge 1 - 2\mathbf{1}[X \le 0]$; take expectations and use (i). The second inequality is $2\mu s \le s^2 + \mu^2$. (iii) A unimodal law with mode 0 has no atoms away from 0, and by symmetry $\Pr[\zeta < -\mu] = \Pr[\zeta > \mu]$, so $\mathbb{E}[\operatorname{sign} X] = \Pr[\zeta > -\mu] - \Pr[\zeta > \mu] = \Pr[|\zeta| < \mu]$. Gauss's inequality (Lemma A.25), for a unimodal law with mode 0 and second moment $\tau^2 \le s^2$, bounds $\Pr[|\zeta| \ge k]$ by $1 - k/(\sqrt{3}\tau)$ if $k \le 2\tau/\sqrt{3}$ and by $4\tau^2/(9k^2)$ otherwise. Both bounds increase with τ and agree at $k = 2\tau/\sqrt{3}$, so they hold with s for τ, and at $k = \mu$ they give $\Pr[|\zeta| < \mu] \ge \mu/(\sqrt{3}s)$ if $\mu \le 2s/\sqrt{3}$ and $\Pr[|\zeta| < \mu] \ge 1 - \frac{4s^2}{9\mu^2} \ge \frac{2}{3}$ otherwise. ∎

By (ii), the expected progress $\mu\mathbb{E}\operatorname{sign} X$ falls short of the noiseless $|\mu|$ by at most the noise level s.[4] Summed over coordinates, this is the stochastic version of Theorem 2.9.

[4] Cantelli's inequality is the one-sided Chebyshev inequality, and a two-point law attains it. Gauss proved his inequality for unimodal laws in 1823; for a symmetric one, beyond $2/\sqrt{3}$ standard deviations it improves Chebyshev's bound by the factor $\frac{4}{9}$.

Theorem 2.14 (signSGD). *Let Assumptions 2.5 and 2.7 hold, and run signSGD, $\theta_{t+1} = \theta_t - \eta_t \operatorname{sign}(g_t)$, with deterministic $\eta_t > 0$. Then for every $T \geq 1$*

$$\frac{\sum_{t<T} \eta_t \, \mathbb{E}\|\nabla \mathcal{L}(\theta_t)\|_1}{\sum_{t<T} \eta_t} \leq \frac{\Delta_0 + \frac{1}{2}\|\vec{L}\|_1 \sum_{t<T} \eta_t^2}{\sum_{t<T} \eta_t} + \frac{\|\vec{\sigma}\|_1}{\sqrt{B}}.$$

For a constant step size the right side is $\Delta_0/(\eta T) + \frac{1}{2}\eta\|\vec{L}\|_1 + \|\vec{\sigma}\|_1/\sqrt{B}$. If Δ_0 and $\|\vec{L}\|_1$ are positive, $\eta = \sqrt{2\Delta_0/(\|\vec{L}\|_1 T)}$ and $B = T$, then after $N = BT$ stochastic gradients it is $\left(\sqrt{2\Delta_0\|\vec{L}\|_1} + \|\vec{\sigma}\|_1\right) N^{-1/4}$.

Proof. The plan is the proof of Theorem 2.9 in conditional expectation, with Lemma 2.13(ii) for the linear term. Let $\mathcal{F}_t$ be generated by $\theta_0, g_0, \ldots, g_{t-1}$, and write $\mu = \nabla\mathcal{L}(\theta_t)$. Assumption 2.5 at $u = -\eta_t \operatorname{sign}(g_t)$ gives

$$\mathbb{E}[\mathcal{L}(\theta_{t+1}) \mid \mathcal{F}_t] \leq \mathcal{L}(\theta_t) - \eta_t \sum_{i=1}^{d} \mu_i \, \mathbb{E}[\operatorname{sign} g_{t,i} \mid \mathcal{F}_t] + \tfrac{1}{2}\eta_t^2 \|\vec{L}\|_1.$$

Conditionally on $\mathcal{F}_t$, the coordinate $g_{t,i}$ has mean μ_i and variance at most σ_i^2/B, so Lemma 2.13(ii) gives $\mu_i \mathbb{E}[\operatorname{sign} g_{t,i} \mid \mathcal{F}_t] \geq |\mu_i| - \sigma_i/\sqrt{B}$ if $\mu_i \neq 0$ and $\sigma_i > 0$; the bound also holds if $\mu_i = 0$, when the left side vanishes, and if $\sigma_i = 0$, when $g_{t,i} = \mu_i$. Sum over i, take expectations, sum over $t < T$, use $\mathbb{E}\mathcal{L}(\theta_T) \geq \mathcal{L}^\star$, and divide by $\sum_t \eta_t$. With the stated η and $B = T$ the first two terms equal $\sqrt{2\Delta_0\|\vec{L}\|_1/T}$, the third is $\|\vec{\sigma}\|_1/\sqrt{T}$, and $\sqrt{T} = N^{1/4}$. ∎

With a batch that grows with the horizon, signSGD reaches the rate $N^{-1/4}$ of stochastic gradient descent, with ℓ_1 norms in place of ℓ_2 norms; this is the comparison of Bernstein et al. (2018), whose noise term has the factor 2 of Markov's inequality where Cantelli's gives 1. At a fixed batch size the term $\|\vec{\sigma}\|_1/\sqrt{B}$ survives every step size, and not because of the proof: the sign is right in most batches only if the median of the gradient has the sign of its mean.

Proposition 2.15 (A rare class defeats signSGD). *Let labels $y \in \{0, 1\}$ be independent with $\Pr[y = 1] = \nu \in (0, \frac{1}{2})$, let a model predict $\varsigma(b) = 1/(1 + e^{-b})$ with one bias $b \in \mathbb{R}$, and let $\mathcal{L}(b) = -\nu \log \varsigma(b) - (1-\nu)\log(1 - \varsigma(b))$. SignSGD on batches of B labels is $b_{t+1} = b_t - \eta \operatorname{sign}(\varsigma(b_t) - \bar{y}_t)$, where $\bar{y}_t$ is the fraction of ones in batch t. (i) $\mathcal{L}$ satisfies Assumptions 2.5 and 2.7 with $L_1 = \frac{1}{4}$ and $\sigma_1^2 = \nu(1-\nu)$, and is minimized where $\varsigma(b^\star) = \nu$. (ii) If $q = (1-\nu)^B > \frac{1}{2}$, then for every $\eta > 0$ and b_0, almost surely $\limsup_t b_t/t \leq -\eta(2q-1) < 0$; hence $\mathcal{L}(b_t) \to \infty$ and $\nabla\mathcal{L}(b_t) \to -\nu$.*

Proof. (i) Here $\mathcal{L} \geq 0$, $\mathcal{L}'(b) = \varsigma(b) - \nu$ and $0 < \mathcal{L}''(b) = \varsigma(b)(1-\varsigma(b)) \leq \frac{1}{4}$, so Taylor's formula gives Assumption 2.5; the minibatch gradient $\varsigma(b) - \bar{y}_t$ averages B independent terms $\varsigma(b) - y$ of mean $\varsigma(b) - \nu$ and variance $\nu(1-\nu)$. (ii) The plan is to bound the drift of b_t and to control its fluctuations by a martingale argument. With probability q batch t has no label 1; then $\bar{y}_t = 0 < \varsigma(b_t)$ and the step is $-\eta$. A step $+\eta$ needs $\bar{y}_t > \varsigma(b_t)$, which has probability at most $1 - q$. So $D_t = \mathbb{E}[b_{t+1} - b_t \mid b_0, \dots, b_t] \leq -\eta(2q-1)$ whatever b_t is. The differences $M_t = b_{t+1} - b_t - D_t$ are martingale differences with $|M_t| \leq 2\eta$, and by the Azuma–Hoeffding inequality (Theorem A.24) $\Pr\left[\left|\sum_{s<t} M_s\right| \geq \varepsilon t\right] \leq 2e^{-\varepsilon^2 t/(8\eta^2)}$ for each $\varepsilon > 0$. These are summable in t, so by the Borel–Cantelli lemma $\frac{1}{t}\sum_{s<t} M_s \to 0$ almost surely, and $b_t/t = b_0/t + \frac{1}{t}\sum_{s<t}(D_s + M_s)$ has lim sup at most $-\eta(2q-1)$. Then $\varsigma(b_t) \to 0$, $\mathcal{L}(b_t) \geq -\nu \log \varsigma(b_t) \to \infty$ and $\mathcal{L}'(b_t) \to -\nu$. ∎

The mean gradient $\varsigma(b) - \nu$ is negative where the predicted probability is too small, but the median of the minibatch gradient is $\varsigma(b) > 0$, because most batches lack the rare label: signSGD descends the median, which has no zero.[5] Once $\varsigma(b_t) < 1/B$, every batch with a label 1 pushes b up, and the drift is exactly $-\eta(2q-1)$ (Figure 2.2). The condition $(1-\nu)^B > \frac{1}{2}$ holds for every class rarer than $1 - 2^{-1/B} \approx \ln 2/B$, and softmax layers over heavy-tailed classes, such as tokens, have many (Chapter 14). The proposition also shows that the constant in Theorem 2.14 cannot be lowered: for every η the long-run average of $|\nabla\mathcal{L}(b_t)|$ is ν, the bound tends to $\sqrt{\nu(1-\nu)/B}$ as $\eta \to 0$, and their ratio tends to 1 for $B = 1$ and $\nu \uparrow \frac{1}{2}$, and to $\sqrt{\ln 2} \approx 0.83$ for large B and ν just below $1 - 2^{-1/B}$. For symmetric unimodal noise the median is the mean, and the bias disappears; symmetry alone is not enough, as noise $\pm s$ gives $\mathbb{E}\,\mathrm{sign}(\mu + \zeta) = 0$ for $|\mu| < s$.

[5] In one dimension this is classical: with $g = \theta - \xi$, signSGD is stochastic subgradient descent on $\mathbb{E}|\theta - \xi|$, whose minimizers are the medians of ξ.

Proposition 2.16 (Symmetric unimodal noise: no bias). *Let Assumptions 2.5 and 2.7 hold with every $\sigma_i > 0$, and let each $\zeta_{t,i}$ be, conditionally on the past, symmetric about 0 and unimodal. With $\Phi_B(\mu) = \sum_i \min\{\sqrt{B}\mu_i^2/(\sqrt{3}\sigma_i), \frac{2}{3}|\mu_i|\}$, signSGD with a constant step size η satisfies*

$$\frac{1}{T}\sum_{t<T} \mathbb{E}\,\Phi_B\big(\nabla\mathcal{L}(\theta_t)\big) \leq \frac{\Delta_0}{\eta T} + \frac{\eta}{2}\|\vec{L}\|_1. \tag{2.5}$$

Proof. Repeat the proof of Theorem 2.14 with Lemma 2.13(iii), for $s = \sigma_i/\sqrt{B}$, in place of (ii): the one-step inequality becomes $\mathbb{E}[\mathcal{L}(\theta_{t+1}) \mid \mathcal{F}_t] \leq \mathcal{L}(\theta_t) - \eta\,\Phi_B(\nabla\mathcal{L}(\theta_t)) + \frac{1}{2}\eta^2\|\vec{L}\|_1$, both sides of the per-coordinate bound being 0 when $\mu_i = 0$, and the sum telescopes. ∎

There is no bias term: with $\eta \propto T^{-1/2}$ the right side of (2.5) is $O(T^{-1/2})$ at every batch size. The batch size enters only the measure of progress, which is $\frac{2}{3}|\mu_i|$ in coordinates whose gradient exceeds the noise $\sigma_i/\sqrt{B}$ by a constant factor, and $\sqrt{B}\mu_i^2/(\sqrt{3}\sigma_i)$ in coordinates dominated by noise. In the second regime signSGD is a preconditioned gradient method.

HEURISTIC DERIVATION OF THE EFFECTIVE STEP. Let the noise be Gaussian and $|\mu_i| \ll \sigma_i/\sqrt{B}$. Then $\mathrm{Var}(\operatorname{sign} g_{t,i}) \approx 1$ and

$$\mathbb{E}[\operatorname{sign} g_{t,i}] = \operatorname{erf}\Big(\frac{\sqrt{B}\,\mu_i}{\sqrt{2}\,\sigma_i}\Big) \approx \sqrt{\frac{2B}{\pi}}\,\frac{\mu_i}{\sigma_i},$$

so a step of signSGD has the mean of a step of stochastic gradient descent with step size $\alpha_i = \eta\sqrt{2B/\pi}/\sigma_i$, and $\pi/2$ times its variance: signSGD is gradient descent preconditioned by $\operatorname{diag}(\sigma_i)^{-1}$, with noise inflated by $\sqrt{\pi/2}$. On $\mathcal{L}(\theta) = \frac{1}{2}h\theta^2$, the stationary variance $\alpha\frac{\pi}{2}\frac{\sigma^2}{B}/(2h)$ of that method gives the floor

$$\mathbb{E}|\nabla\mathcal{L}| \approx (2\pi)^{-1/4}\sqrt{\eta h\sigma}\;B^{-1/4}, \tag{2.6}$$

which vanishes as $\eta \to 0$. Keeping $\alpha_i \propto B$, the linear scaling rule of gradient descent (Chapter 11), requires $\eta \propto \sqrt{B}$: the *square-root scaling rule*, which Malladi et al. (2022) derived for adaptive methods from stochastic differential equations. ◇

Memory changes the picture twice. Momentum averages gradients before the sign is taken, which moves the median towards the mean (Chapter 5; Figure 2.2 shows Signum learning the rare class). Adam's second moment changes the length of the step, and the first question is how long it can get.

Proposition 2.17 (The size of Adam's step). *Let $0 \le \beta_1 < 1$, $0 < \beta_2 < 1$, $\rho = \beta_1^2/\beta_2$, and $\epsilon = 0$, and consider one coordinate of Adam without bias correction. For every $t \ge 0$ and every gradient sequence with $v_t > 0$,*

$$\frac{|m_t|}{\sqrt{v_t}} \le \frac{1-\beta_1}{\sqrt{1-\beta_2}}\Big(\sum_{k=0}^{t}\rho^k\Big)^{1/2}, \tag{2.7}$$

with equality if and only if $g_{t-k} = c\,(\beta_1/\beta_2)^k$ for $k = 0, \dots, t$ and some $c \neq 0$ (with $0^0 = 1$). Consequently, if $\beta_1^2 < \beta_2$, the supremum over all t and all gradient sequences is

$$C(\beta_1,\beta_2) = \frac{1-\beta_1}{\sqrt{(1-\beta_2)(1-\beta_1^2/\beta_2)}}, \tag{2.8}$$

and if $\beta_1^2 \ge \beta_2$, then the ratio is unbounded. With bias correction the bound (2.7) is multiplied by $\sqrt{1-\beta_2^{t+1}}/(1-\beta_1^{t+1})$; it is then 1 at $t = 0$ and at most $C(\beta_1,\beta_2)$ for all t, with equality only if $\beta_1 = \beta_2$ (Exercise 2.1).

PROOF. The plan is to write m_t as a weighted sum of gradients and to apply the Cauchy–Schwarz inequality with the weights of v_t. From $m_{-1} = v_{-1} = 0$, $m_t = (1-\beta_1)\sum_{k=0}^{t}\beta_1^k g_{t-k}$ and $v_t = (1-\beta_2)\sum_{k=0}^{t}\beta_2^k g_{t-k}^2$. Splitting $\beta_1^k g_{t-k} = (\beta_1^k\beta_2^{-k/2})(\beta_2^{k/2}g_{t-k})$,

$$\Big|\sum_{k=0}^{t}\beta_1^k g_{t-k}\Big| \le \Big(\sum_{k=0}^{t}\rho^k\Big)^{1/2}\Big(\sum_{k=0}^{t}\beta_2^k g_{t-k}^2\Big)^{1/2} = \Big(\sum_{k=0}^{t}\rho^k\Big)^{1/2}\Big(\frac{v_t}{1-\beta_2}\Big)^{1/2},$$

and multiplying by $1-\beta_1$ gives (2.7). Equality holds exactly when $(\beta_2^{k/2} g_{t-k})_k$ is parallel to $(\beta_1^k \beta_2^{-k/2})_k$, that is, $g_{t-k} = c(\beta_1/\beta_2)^k$, with $c \neq 0$ because $v_t > 0$. If $\rho < 1$, then the sums $\sum_{k\le t} \rho^k$ increase to $1/(1-\rho)$, and the equality sequences make (2.8) the supremum; if $\rho \ge 1$, then the sums are unbounded. At $t = 0$ the bias correction factor is $\sqrt{1-\beta_2}/(1-\beta_1)$, the reciprocal of the bound. ■

The bound is $C = 7.27$ at the defaults $(0.9, 0.999)$, 2.35 at $(0.9, 0.99)$, 1.16 at $(0.9, 0.95)$, and 1 whenever $\beta_1 = \beta_2$. A single large gradient after a run of zeros gives only $(1-\beta_1)/\sqrt{1-\beta_2}$, which is 3.16 at the defaults;[6] the worst case is a gradient that grows by the factor β_2/β_1 at every step, so that the slow second moment lags behind the fast first moment. Adam is thus sign descent of its momentum with a bounded multiplier: each coordinate moves by η times a number in $[0, C]$ in the direction of $-\operatorname{sign}(m_t)$, so $\|\theta_{t+1} - \theta_t\|_\infty \le C\eta$, a trust region in the max norm.[7] The bound is smallest, 1, at $\beta_2 = \beta_1$. The typical multiplier is set by the signal-to-noise ratio.

[6] Kingma and Ba (2015, Section 2.1) state that the effective step is bounded by this single-spike value when $1-\beta_1 > \sqrt{1-\beta_2}$, a case they tie to the sparsest gradients, and by the learning rate otherwise. Geometrically growing gradients break both bounds, the second unless $\beta_1 = \beta_2$; Proposition 2.17 gives the exact supremum.

[7] Chapter 1 reads normalized steepest descent with a fixed step as a trust-region method; Adam keeps this property in the max norm with the radius enlarged by the factor $C(\beta_1, \beta_2)$.

Heuristic derivation of Adam's mean step. Let the gradients of coordinate i be independent over steps with mean μ_i and variance s_i^2, and let β_2 be so close to 1 that $\hat{v}_{t,i}$, an average of about $1/(1-\beta_2)$ squared gradients, concentrates near $\mu_i^2 + s_i^2$; $\hat{m}_{t,i}$ has mean μ_i and variance $(1-\beta_1)^2 \sum_k \beta_1^{2k} s_i^2 = \frac{1-\beta_1}{1+\beta_1} s_i^2$. Treating $\hat{v}_{t,i}$ as a constant,

$$\mathbb{E}\left[\frac{\hat{m}_{t,i}}{\sqrt{\hat{v}_{t,i}}}\right] \approx \frac{\operatorname{sign}(\mu_i)}{\sqrt{1 + s_i^2/\mu_i^2}}, \tag{2.9}$$

and the mean square of the update is about $\left(\mu_i^2 + \frac{1-\beta_1}{1+\beta_1} s_i^2\right)/(\mu_i^2 + s_i^2)$. The mean update is the sign shrunk by the factor $(1 + s_i^2/\mu_i^2)^{-1/2}$ of Balles and Hennig (2018): near $\operatorname{sign}(\mu_i)$ for a consistent gradient and near μ_i/s_i for noise, where Adam, like signSGD, is gradient descent preconditioned by the noise. The root mean square of the update lies between $\sqrt{(1-\beta_1)/(1+\beta_1)} = 0.23$, for pure noise at $\beta_1 = 0.9$, and 1. ◇

For $(0.9, 0.999)$ and Gaussian gradients, simulation confirms (2.9) and the mean square to within 0.002 for signal-to-noise ratios up to 3, and 2×10^6 steps of pure noise never gave a multiplier above 1.5; signSGD's mean update, $\operatorname{erf}\left(\mu_i/(\sqrt{2} s_i)\right)$, is within 0.07 of (2.9).

2.5 Reading the theory

The guarantee is exact, and it belongs to the basis. Figure 2.1 runs gradient descent and sign descent on $\mathcal{L}(\theta) = \frac{1}{2}\sum_i h_i \theta_i^2$ with $d = 256$ and $h_i = i^{-2}$, so that $\operatorname{tr} H = 1.64$ and the effective rank is below 2, and on the same quadratic in randomly rotated coordinates. In the right panel the long-run average of $\|\nabla \mathcal{L}\|_1$ on the diagonal problem lies on the line $\frac{1}{2}\eta \operatorname{tr} H$ of Proposition 2.11 to all printed digits. On the rotated problem it lies on $\frac{1}{2}\eta \hat{L}_\infty$, where $\hat{L}_\infty = 161.18$ is the largest $s^\top Q^\top H Q s$ that a local search over sign vectors finds. The lower bound of Proposition 2.12(ii) is 160.9 (mean over rotations 163.3), and an admissible $\vec{L}$ from the semidefinite program

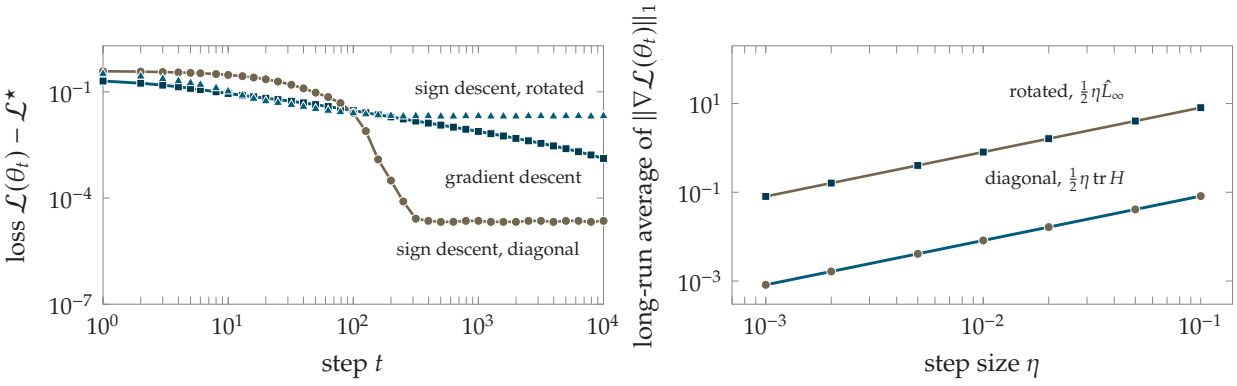


Figure 2.1. Gradient descent and sign descent on a quadratic with curvatures $h_i = i^{-2}$, $d = 256$, $\theta^\star = 0$, and $\theta_0 \sim \mathcal{N}(0, I)$, in the eigenbasis and after one uniformly random rotation (seed in `02-sign-rotation.py`). Left: the loss of gradient descent with step size $1/\lambda_{\max}$ and of sign descent with $\eta = 0.01$; lines are exact closed forms, marks are runs (squares: gradient descent, rotated; circles: sign descent, diagonal; triangles: sign descent, rotated). Gradient descent is the same in both bases; sign descent is far ahead on the diagonal problem and is caught on a two-cycle with loss near 2×10^{-2} after about 300 steps on the rotated one. Right: the average of $\|\nabla\mathcal{L}\|_1$ over the final cycle against η (circles diagonal, squares rotated), with the floor of Proposition 2.11 and the floor $\frac{1}{2}\eta\hat{L}_\infty$ of Theorem 2.9 for the rotated Hessian, $\hat{L}_\infty$ found by local search and certified by a semidefinite bound.

of Exercise 2.4 has $\|\vec{L}\|_1 = 161.18$: $\hat{L}_\infty$ is the ℓ_∞ constant and the best $\|\vec{L}\|_1$ to five digits. In every run the final two-cycle used the best sign vector of the search: the floor of Theorem 2.9 is attained in both bases, and the rotation multiplies it by 98, against the average factor $(1 + \frac{2}{\pi}(d-1))\lambda_{\max}/\operatorname{tr} H = 99.5$ of Proposition 2.12.

A TIME SCALE WITHOUT CURVATURE. On the diagonal problem coordinate i reaches its two-cycle after $\lfloor|\theta_{0,i}|/\eta\rfloor$ steps whatever its curvature: sign descent with $\eta = 0.01$ is at its floor, near 2×10^{-5}, after about 300 steps. Gradient descent needs about $\lambda_{\max}/h_i = i^2$ steps for coordinate i, and its expected loss $\frac{1}{2}\sum_i i^{-2}(1-i^{-2})^{2t}$ is close to $\frac{\sqrt{\pi}}{4\sqrt{2t}} - \frac{1}{2d}$ while $t \ll d^2$ (1.6×10^{-2} at $t = 300$): it decays like $0.31\,t^{-1/2}$. This is the regime in which Corollary 2.10 favors the sign: low effective rank, a spread gradient, and similar distances in all coordinates.

THE TRAP AFTER A ROTATION. On the rotated problem gradient descent is unchanged to rounding error, while sign descent is caught after a few hundred steps on a two-cycle with loss near 2×10^{-2}, a thousand times its diagonal floor. The centers of the two-cycles form a box in gradient space, which the inverse Hessian stretches along the directions of small curvature; when these are not coordinate axes, the box contains points of high loss (Exercise 2.2). A decreasing step size breaks the cycle; Adam escapes it but still loses orders of magnitude to the rotation (Exercise 2.9).

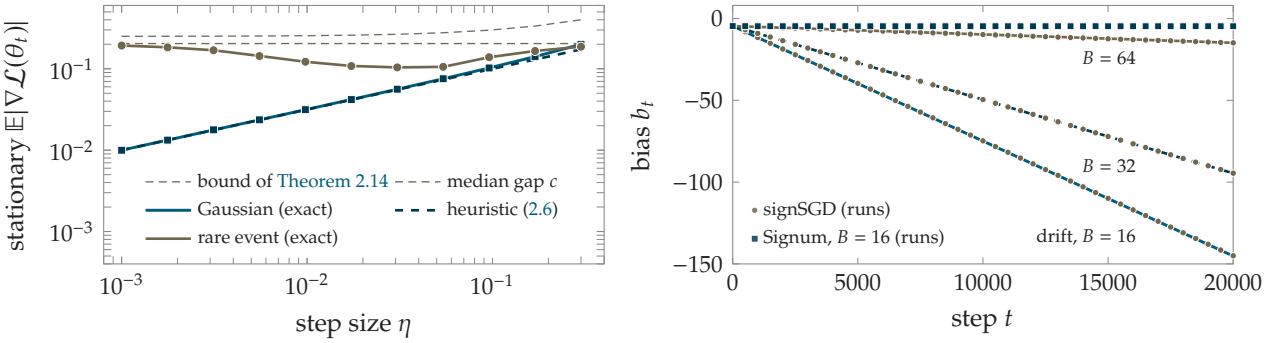


Figure 2.2. SignSGD and the shape of the noise. Left: one coordinate, $\mathcal{L}(\theta) = \frac{1}{2}h\theta^2$ with $h = \sigma = 1$ and $B = 16$, under Gaussian per-example noise and under the rare-event noise $\sigma(X - p)/\sqrt{p(1-p)}$ with X Bernoulli(p), $p = 0.04$. Curves: exact stationary values of the lattice chain; points (squares Gaussian, circles rare event): 1000 runs averaged over at least 4000 steps after a burn-in; dashed: the bound $\frac{1}{2}\eta h + \sigma/\sqrt{B}$ and the median gap c. Right: the bias of a class of frequency $\nu = 0.01$ under signSGD with $\eta = 0.01$ from $b^\star$, averaged over 200 runs; lines are the exact drift $-\eta(2(1-\nu)^B - 1)t$; squares are Signum ($\beta = 0.9$). Script: `02-sign-noise.py`.

MEDIAN AGAINST MEAN. In the left panel of Figure 2.2 the iterates of one coordinate of signSGD, $B = 16$, move on a lattice of spacing η and form a birth–death chain whose stationary law is exact. With Gaussian noise the floor follows (2.6) and vanishes like $\sqrt{\eta}$, as Proposition 2.16 requires. With rare-event noise, in which a fraction $p = 0.04$ of the examples push hard one way and the rest gently the other way, $(1-p)^B = 0.52 > \frac{1}{2}$, and as $\eta \to 0$ the floor rises to $c = \sigma\sqrt{p/(1-p)} = 0.82\,\sigma/\sqrt{B}$, the gradient at the point where the median of the minibatch gradient vanishes, 82% of the bias term of Theorem 2.14. A smaller step makes it worse; the lowest floor is near $\eta = 0.03$. The right panel is Proposition 2.15 for a class of frequency $\nu = 0.01$: the bias drifts at exactly the predicted speed for $B = 16, 32, 64$, the speed vanishes at the threshold $B = \ln 2/(-\ln(1-\nu)) = 69$, and Signum with $\beta = 0.9$ holds the bias within 0.14 of its optimum.

2.6 What the theory misses

WHAT WAS IDEALIZED AWAY. Memory first: the theorems are about the memoryless step, Adam's averages enter only through a bound on its step and a heuristic for its mean, and nothing here proves that Adam converges or explains why $\beta_1 = 0.9$ and β_2 near 1 work (Chapters 5 and 6). Adam's ϵ is zero except in Exercise 2.7. The curvature is bounded by one vector $\vec{L}$ everywhere, a strong assumption for networks, whose sharpness adapts to the step size (Chapter 8). The noise is described by its variance, yet the outcome depends on its shape: the same variance gives no bias when the noise is symmetric and unimodal and divergence when it is skewed. And most statements use a constant step size, while practice decays it (Chapter 12).

WHERE IT BREAKS. (1) SignSGD without momentum fails at small batch sizes: the rare class of Proposition 2.15 is convex and smooth, and its loss diverges. Karimireddy et al. (2019) give further examples in which signSGD does not converge, and Safaryan and Richtárik (2021) prove convergence under a condition on the probability that each sign is correct, which fails whenever the median of a coordinate's stochastic gradient has the sign opposite to its mean. (2) Adam with long memories is not sign descent: at $\beta_2 = 0.999$ its multiplier can approach 7.27, and typically it is a smooth function of the signal-to-noise ratio (2.9), so the theorems for sign descent do not transfer to it. (3) Corollary 2.10 compares upper bounds, which do not explain the size of the gap between Adam and gradient descent that Kunstner et al. (2023) measure (Chapter 14). (4) The advantage of the max norm depends on the basis (Proposition 2.12), and the theory does not say why the coordinates of trained networks are favorable. Xie et al. (2025) relate Adam's sensitivity to rotations to a smoothness constant of the ℓ_∞ type, and Zhang et al. (2024b) find the Hessians of small transformers close to block diagonal, with blocks that correspond to groups of parameters; both are measurements, not derivations.

REPAIRS AND OPEN PROBLEMS. Momentum is the main repair: averaging before taking the sign moves the median towards the mean, normalized steepest descent with momentum converges in every norm (Chapter 5), and Signum and Lion train language models about as well as Adam (Zhao et al. 2025). Error feedback, which adds the part of the gradient lost to compression to the next gradient, makes sign compression converge (Karimireddy et al. 2019), and majority vote reduces the noise of the sign (Exercise 2.5). Analyses of Adam in the ℓ_∞ geometry (Xie et al. 2025), its implicit bias towards ℓ_∞ max-margin solutions (Zhang et al. 2024a) (Chapter 15), and the ℓ_∞-constrained problems that Lion and, when it converges, AdamW solve with weight decay (Chen et al. 2024; Xie and Li 2024) (Chapter 13) carry the max-norm picture to the full algorithms; Shampoo and SOAP remove the dependence on rotations of each layer's inputs and outputs by working in estimated eigenbases (Chapter 7). Open: why the coordinates of trained networks are good ones for sign methods (Exercise 2.10); a convergence theory of Adam that uses both its geometry and its memory and explains its speed; and the batch sizes and momenta at which sign methods stop ignoring rare events in heavy-tailed data (Exercise 2.11).

2.7 Back in the wild

FIELD NOTES	
PRACTICE	Adam, Lion, and signSGD move every parameter by about the learning rate per step: Lion and signSGD exactly, Adam within the factor $C(\beta_1, \beta_2)$.
IDEALIZATION	Sign descent as normalized steepest descent in ℓ_∞, with $\nabla^2 \mathcal{L} \preceq \operatorname{diag}(\vec{L})$ and minibatch noise of variance σ_i^2/B.

PROVED	Decrease $\eta\|\nabla\mathcal{L}\|_1 - \frac{1}{2}\eta^2\|\vec{L}\|_1$, exact on separable quadratics; rotations move $\|H\|_{\infty\to 1}$ between $\operatorname{tr} H$ and $d\lambda_{\max}$; a sharp bias $\|\vec{\sigma}\|_1/\sqrt{B}$, absent for symmetric unimodal noise; $\|m_t\|/\sqrt{v_t} \le C(\beta_1,\beta_2)$, 7.27 at the defaults, 1 if $\beta_1 = \beta_2$.
PREDICTS	Sign descent wins when the gradient is spread over more coordinates than the curvature has stiff directions; a random rotation costs at least about $\frac{2}{\pi} d\lambda_{\max}/\operatorname{tr} H$; signSGD ignores classes rarer than about $\ln 2/B$; sign methods follow the square-root scaling rule (derived).
MISSES	The memory that makes Adam and Lion work; why the coordinates of networks are good ones; the size of the Adam–SGD gap.
TAKEAWAY	Use sign updates with momentum, not without, at small batch sizes (failure proved in the model, repair simulated); expect the advantage of Adam-like methods to depend on the basis (proved for quadratics, observed for GPT-2); a β_2 closer to β_1 keeps Adam's step near the learning rate (proved bound).

What the theory tells a practitioner is modest and specific. A sign method with momentum suits gradients spread over many parameters and curvature concentrated in a few directions that the coordinates respect; whether a given network is such a problem the theory does not say. A sign method without momentum should not be trusted at small batch sizes, because it follows the median of the gradient, and the median ignores rare events. Adam's learning rate is a step length to within the factor $C(\beta_1,\beta_2)$, and by the heuristic that follows Proposition 2.17 its typical update has an RMS norm well below 1, which suggests why Lion, whose update has RMS norm 1, needs a learning rate several times smaller. Claims beyond these, such as that Lion generalizes better than AdamW, lie outside what this chapter proves.

The spectral norm plays the role of the max norm for matrices (Chapter 3); for a layer, sign descent moves the output by an amount proportional to the fan-in, which is why μP scales Adam's learning rate by 1/fan_in (Chapter 4). Momentum repairs the bias of the sign (Chapter 5), the second moment turns the sign into a preconditioner (Chapter 6), the dependence on the basis returns with Shampoo and SOAP and with rotated transformers and rare tokens (Chapters 7 and 14), and $C(\beta_1,\beta_2)$ bounds Adam's largest steps when an instability begins (Chapter 10).

Exercises

2.1 CONSOLIDATION. (Bias correction and equal coefficients.) Let $\rho = \beta_1^2/\beta_2 < 1$ and $\epsilon = 0$. (a) Show that with bias correction $|\hat{m}_t|/\sqrt{\hat{v}_t} \le B_t$, where

$$B_t = \frac{(1-\beta_1)\sqrt{1-\beta_2^{t+1}}}{(1-\beta_1^{t+1})\sqrt{1-\beta_2}}\left(\frac{1-\rho^{t+1}}{1-\rho}\right)^{1/2}, \qquad B_0 = 1.$$

(b) Show that $B_t \le C(\beta_1,\beta_2)$ for every t, with equality for every t if and only if $\beta_1=\beta_2$. (c) For $(0.9, 0.999)$, compute B_{10} and B_{100} (answers: 1.05 and 2.25). (d) For $\beta_1=\beta_2\in(0,1)$, show that $\hat s_t^2=\hat v_t-\hat m_t^2$ is a weighted variance of the past gradients and, if $\hat m_t\neq 0$, $\hat m_t/\sqrt{\hat v_t}=\operatorname{sign}(\hat m_t)(1+\hat s_t^2/\hat m_t^2)^{-1/2}$: the form of (2.9) holds exactly with empirical moments, and $|\hat m_t|\le\sqrt{\hat v_t}$. *(Hint: with $a=\beta_1^{t+1}$ and $b=\beta_2^{t+1}$, the claim $B_t\le C$ is $(1-b)(1-a^2/b)\le(1-a)^2$.)*

2.2 CONSOLIDATION. ★ (Two-cycles of sign descent.) Let $\mathcal{L}(\theta)=\frac12(\theta-\theta^\star)^\top H(\theta-\theta^\star)$ with $H\succ 0$, and let $s\in\{\pm1\}^d$. (a) Show that $\{\theta,\theta-\eta s\}$ is an orbit of period two of sign descent if and only if $\operatorname{sign}(Hs)=s$ and the center $\bar\theta=\theta-\frac12\eta s$ satisfies $|(H(\bar\theta-\theta^\star))_i|<\frac12\eta|(Hs)_i|$ for every i. (b) Show that over the cycle $\|\nabla\mathcal{L}\|_1$ averages $\frac12\eta s^\top Hs$ and the loss averages $\frac12(\bar\theta-\theta^\star)^\top H(\bar\theta-\theta^\star)+\frac18\eta^2 s^\top Hs$. (c) Show that the centers form $\theta^\star+H^{-1}W$ for a box W, that for diagonal H every two-cycle has average loss below $\frac14\eta^2\operatorname{tr}H$, and explain the trap of Figure 2.1. (d) Show that $s\mapsto\operatorname{sign}(Hs)$ (with $\operatorname{sign}0$ replaced by 1) does not decrease $s^\top Hs$, and that a maximizer of $s^\top Hs$ satisfies $\operatorname{sign}(Hs)=s$. *(Hint: in (d), $(s'-s)^\top H(s'-s)\ge 0$ for $s'=\operatorname{sign}(Hs)$, and flipping s_i changes $s^\top Hs$ by $4H_{ii}-4s_i(Hs)_i$.)*

2.3 CONSOLIDATION. (Cantelli is sharp.) Let $\zeta=-\mu-\delta$ with probability $p=s^2/(s^2+(\mu+\delta)^2)$ and $\zeta=p(\mu+\delta)/(1-p)$ otherwise, with $\mu,s,\delta>0$. Show that $\mathbb{E}\zeta=0$, $\mathbb{E}\zeta^2=s^2$, and $\sup_{\mu,\delta}\big(\mu-\mu\mathbb{E}\operatorname{sign}(\mu+\zeta)\big)=s$, so Lemma 2.13(ii) is sharp.

2.4 CONSOLIDATION. ★★ (Coordinatewise against ℓ_∞ smoothness.) Let $H\succeq 0$. (a) Show that $\min\{\|\vec L\|_1:\operatorname{diag}(\vec L)\succeq H\}=\max\{\operatorname{tr}(HX):X\succeq 0,\ X_{ii}=1\}$. (b) For feasible $X=VV^\top$, $\gamma\sim\mathcal{N}(0,I)$ and $s=\operatorname{sign}(V\gamma)$, show that $\mathbb{E}[ss^\top]=\frac{2}{\pi}\arcsin(X)$ entrywise and that $\arcsin(X)\succeq X$. (c) Deduce $\|H\|_{\infty\to1}\ge\frac{2}{\pi}\operatorname{tr}(HX)$ and the factor $\pi/2$ of Remark 2.6 (Nesterov 1998). *(Hint: (a) is semidefinite duality with strictly feasible points on both sides; in (b), $\mathbb{E}[\operatorname{sign}(a^\top\gamma)\operatorname{sign}(b^\top\gamma)]=1-2\angle(a,b)/\pi$, and the series of* arcsin *has nonnegative coefficients, so the Schur product theorem applies.)*

2.5 VARIATION. (Majority vote.) Let M be odd, and let M workers compute independent minibatch gradients whose coordinate noises are symmetric and unimodal with variances at most σ_i^2/B; the server steps by $-\eta\operatorname{sign}\big(\sum_m\operatorname{sign}(g_t^{(m)})\big)$. (a) With Lemma 2.13(iii) and Hoeffding's inequality, show that the vote is wrong in coordinate i with probability at most $\exp\big(-\frac{M}{2}\min\{S_i^2/3,\frac49\}\big)$, where $S_i=\sqrt{B}|\partial_i\mathcal{L}|/\sigma_i$, and prove the analogue of Theorem 2.14. (b) Compare with one worker and batch MB under Gaussian noise, whose error probability is $\Phi(-\sqrt{M}S_i)\le\frac12 e^{-MS_i^2/2}$, where Φ is the standard normal distribution function.

2.6 VARIATION. (Lion.) (a) Show that Lion with $\beta_1=\beta_2$ is Signum. (b) Show that $c_t=\sum_{k=0}^t w_k g_{t-k}$ with $w_0=1-\beta_1$ and $w_k=\beta_1(1-\beta_2)\beta_2^{k-1}$

for $k \geq 1$, that the weights sum to $1 - \beta_1\beta_2^t$, and that their mean lag $\sum_k kw_k$ tends to $\beta_1/(1-\beta_2)$, which is 90 steps for the defaults; compare with Signum at $\beta = \beta_2$. (c) Lion's update has RMS norm 1; using (2.9), explain why matching AdamW's movement per step calls for a Lion learning rate several times smaller.

2.7 VARIATION. (Memoryless Adam with ϵ.) Let $\theta_{t+1} = \theta_t - \eta\,\psi(\nabla\mathcal{L}(\theta_t))$ with $\psi(x) = x/(|x| + \epsilon)$ coordinatewise. Under Assumption 2.5, show that $\mathcal{L}(\theta_{t+1}) \leq \mathcal{L}(\theta_t) - \eta\sum_i g_i^2/(|g_i| + \epsilon) + \frac{1}{2}\eta^2\|\vec{L}\|_1$ for $g = \nabla\mathcal{L}(\theta_t)$, and that $x^2/(|x| + \epsilon) \geq \frac{1}{2}\min\{|x|, x^2/\epsilon\}$. Deduce a bound of the form (2.5) with ϵ in the role of $\sigma_i/\sqrt{B}$. When does $\epsilon = 10^{-8}$ matter?

2.8 SIMULATION. (Shapes of noise.) With `02-sign-noise.py`: (a) vary p and B and check that at small η the rare-event floor approaches the median gap $\sigma|k/B - p|/\sqrt{p(1-p)}$, k the median number of rare examples in a batch, which for $B = 16$ halves when $(1-p)^B$ falls below $\frac{1}{2}$ and vanishes at $p = 1/B$. (b) With the noise $\sigma(E - 1)$, E exponential of mean 1, show that as $\eta \to 0$ the floor is the median gap, $\sigma(1 - \ln 2) = 0.31\sigma$ at $B = 1$ and within 4% of $\sigma/(3B)$ for $B \geq 2$. Conclude that for a fixed skewed law the bias decays like $1/B$, and that the worst case $1/\sqrt{B}$ needs events of probability comparable to $1/B$.

2.9 SIMULATION. (Adam after a rotation.) On the two problems of `02-sign-rotation.py`, run Adam with $(\beta_1, \beta_2) = (0.9, 0.999)$, $\epsilon = 0$, and $\eta = 0.01$. You should find a loss near 10^{-9} after 1000 steps on the diagonal problem and near 3×10^{-3} on the rotated one, still ahead of gradient descent. Show that on the rotated run the preconditioned sharpness $\eta\lambda_{\max}(P^{1/2}Q^\top HQP^{1/2})$, $P = \operatorname{diag}(\hat{v}_t)^{-1/2}$, grows over about 2×10^4 steps towards the threshold 38 of momentum with $\beta_1 = 0.9$ (Chapters 5 and 8): Adam's advantage there has nothing to do with the coordinates.

2.10 RESEARCH. (When are the coordinates good?) (a) Show that $\|H\|_{\infty\to 1} \leq \sum_j \lambda_j\|v_j\|_1^2$ for $H = \sum_j \lambda_j v_j v_j^\top \geq 0$, and find conditions on the localization of the eigenvectors under which $\|H\|_{\infty\to 1}/\operatorname{tr} H = O(1)$. (b) Estimate both sides for the Hessian of a small network with Hessian–vector products and the iteration of Exercise 2.2(d). (c) For a block-diagonal H, show that $\|H\|_{\infty\to 1}$ is the sum over blocks, so that rotations within blocks cost at most $\sum_k d_k\lambda_{\max}(H_k)$; compare with the block structure that Zhang et al. (2024b) measure in transformers. What is open is a mechanism that explains why the Hessians of trained networks are closer to diagonal, or block diagonal, in the parameters' coordinates than a random rotation would make them.

2.11 RESEARCH. (Rare events and memory.) (a) For Signum with momentum β in the model of Proposition 2.15, argue that the update follows the sign of an average of about $B(1+\beta)/(1-\beta)$ labels, and derive and test the class frequency below which Signum fails. (b) For a softmax layer with Zipf frequencies $\nu_k \propto 1/k$, find the fraction of classes that

signSGD and Signum learn as a function of B and β. What is open is whether this mechanism affects the rare tokens of language models trained with Adam, whose second moment, averaged with β_2 near 1, makes the update follow the mean again (Chapters 6 and 14).

Notes and References

Sign methods are older than their analysis (Section 2.1). Bernstein et al. (2018) analyzed signSGD and Signum under coordinatewise smoothness, with the comparison of densities that Corollary 2.10 restates, and Bernstein et al. (2019) analyzed majority vote under symmetric unimodal noise. Karimireddy et al. (2019) showed that signSGD can fail to converge and that error feedback repairs it; Safaryan and Richtárik (2021) replaced the noise assumptions by a condition on the probability that each sign is correct. We know of no source for Propositions 2.11 and 2.15 in this form; both are elementary, and the second is the median-against-mean mechanism of the counterexample of Reddi et al. (2018) to the convergence of Adam, a large gradient that comes rarely against small ones that come often (Chapter 6).

Balles et al. (2020) compared coordinatewise and ℓ_∞ smoothness for sign descent and related the latter to the concentration of the Hessian on its diagonal; the factor $\pi/2$ of Remark 2.6 is Nesterov's theorem on the semidefinite relaxation of quadratic maximization over the cube (Nesterov 1998). Bernstein and Newhouse (2024) organized optimizers by norms and identified Adam without its averages as steepest descent in ℓ_∞, and Xie et al. (2025) developed the ℓ_∞ theory of Adam and tested it by rotating GPT-2's parameters.

Chen et al. (2024) showed with a Lyapunov function that Lion with weight decay solves a box-constrained problem, Xie and Li (2024) that AdamW, when it converges, converges to a KKT point of such a problem, and Zhang et al. (2024a) that Adam on separable data converges in direction to an ℓ_∞ max-margin classifier.

TRANSLATING THE LITERATURE. Bernstein et al. (2018) write f for our $\mathcal{L}$, x_k for θ_t, g_k for the full gradient $\nabla\mathcal{L}(\theta_t)$ and $\tilde{g}_k$ for the stochastic gradient g_t, δ_k for the step size η_t, n_k for the batch size B, and K for the number of steps T; their $\vec{L}$ and $\vec{\sigma}$ are ours. Kingma and Ba (2015) write α for the step size and start at $t = 1$ with $m_0 = v_0 = 0$, so their $\hat{m}_t = m_t/(1-\beta_1^t)$ is our $\hat{m}_{t-1}$. Chen et al. (2023) write Lion with c_t for the signed interpolation, m_t for the momentum and λ for the decoupled weight decay, as we do.

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3 Orthogonalized Updates

The spectral norm, Muon and Newton–Schulz

IN THE WILD — MUON AND SHAMPOO

IN THIS CHAPTER. Muon replaces the momentum of each hidden weight matrix by an approximately orthogonal matrix, computed by five steps of a matrix polynomial; memoryless Shampoo computes the same matrix. For one weight matrix under the spectral norm we prove that the steepest direction is the polar factor $UV^\top$ of the gradient (a whole face of them when the gradient is rank deficient), that the cubic Newton–Schulz iteration converges to it while Muon's quintic does not converge and holds all but the smallest singular values in $[0.681, 1.203]$, and that on a linear layer with $d_{\text{out}} \geq d_{\text{in}}$ a spectral step guarantees more decrease than a gradient step exactly when the gradient's nuclear rank exceeds the stable rank of the inputs. The theory concerns one matrix and one step: it nearly locates the crossover in a solvable case, misjudges it for generic targets, and cannot say why the gains of matrix optimizers shrink at scale.

3.1 In the wild: an optimizer that orthogonalizes

Adam updates each entry of a weight matrix on its own, as if the matrix were a list of numbers. Since 2024 another kind of optimizer has spread through the training of language models: it treats each weight matrix as a linear map and replaces its update by an orthogonal matrix. The best known is *Muon*, described by Keller Jordan and coauthors in a blog post of December 2024 after it had set speed records on two community benchmarks (Jordan et al. 2024). The name stands for MomentUm Orthogonalized by Newton–Schulz.[1]

IN THE WILD · Muon, Jordan et al. (2024), code Jordan (2024); Moonlight, Liu et al. (2025). For each hidden weight matrix $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ (convolution filters flattened to matrices) with gradient G_t: set $M_t = \beta M_{t-1} + (1-\beta)G_t$ with $\beta = 0.95$ and $N_t = (1-\beta)G_t + \beta M_t$ (Nesterov form); cast N_t to bfloat16, divide it by its Frobenius norm, and apply five times $X \leftarrow aX + b(XX^\top)X + c(XX^\top)^2X$ with $(a, b, c) = (3.4445, -4.7750, 2.0315)$; step $W \leftarrow W - \eta\sqrt{\max(1, d_{\text{out}}/d_{\text{in}})}\,X$ with $\eta = 0.02$ by default. Train embeddings, output layer, gains, and biases with AdamW. At scale (Moonlight, 2025): add decoupled weight decay (0.1 in the released code) and use the factor $0.2\sqrt{\max(d_{\text{out}}, d_{\text{in}})}$ instead, so that the update has the same root-mean-square size for every shape

[1] The reference code calls its orthogonalization routine `zeropower_via_newtonschulz5`: for $G = U\Sigma V^\top$ the target $UV^\top$ is $U\Sigma^0 V^\top$, the "zeroth power" of G (Jordan 2024).

and shares AdamW's learning rate and weight decay.

According to the documentation of the reference code, the coefficients were chosen to make the slope of the polynomial at zero large, increasing it even past the point where the iteration stops converging to one, so that the output is roughly $US'V^\top$, with the diagonal of S' spread over $[0.5, 1.5]$, instead of $UV^\top$; in the author's experience this does not hurt (Jordan 2024).

Observation 3.1 (Muon at small and at large scale). (i) On the NanoGPT speedrun, which trains a transformer of the size of GPT-2 small to a fixed validation loss as fast as possible, Muon improved the record by a factor of 1.35; on a CIFAR-10 speedrun it cut the time to 94% accuracy from 3.3 to 2.6 A100-seconds (Jordan 2024; Jordan et al. 2024). (ii) In Moonshot AI's scaling-law experiments Muon matched AdamW with about 52% of the training FLOPs, and it trained Moonlight, a mixture of experts with 3B active and 16B total parameters, on 5.7T tokens (Liu et al. 2025); Kimi K2, with 1T total and 32B active parameters, was pre-trained on 15.5T tokens with a variant, MuonClip, and its report states that the run had no loss spike (Kimi Team 2025). (iii) Shampoo, which computes the same update when its memory is switched off (Proposition 3.6), won the external-tuning ruleset of the AlgoPerf competition (Kasimbeg et al. 2025).

These are records, company reports, and one competition, not controlled comparisons, and (ii) comes from the group that uses the method.[2]

[2] The README of the reference code also lists a transformer trained to the performance of GPT-2 XL for $175 of compute (Jordan 2024).

Observation 3.2 (Spread-out gradients, flat updates). The gradients of hidden weight matrices of transformers have a few dominant singular values and a long tail. Low-rank methods such as GaLore, which project each gradient onto a subspace of small dimension, rest on the first fact (Zhao et al. 2024). In NanoGPT-scale training, Davis and Drusvyatskiy (2025), in a preprint, find that the activations entering a layer keep a low stable rank throughout training, while the gradients keep a large ratio $\|G\|_{\mathrm{nuc}}/\|G\|_F$. Muon's update has all but its smallest singular values near one.

The facts are compatible: singular values $(1, \varepsilon, \dots, \varepsilon)$ with k copies of $\varepsilon = k^{-1/2}$ give $\|G\|_F^2/\|G\|_{\mathrm{op}}^2 = 2$ but $\|G\|_{\mathrm{nuc}}^2/\|G\|_F^2 = (1+\sqrt{k})^2/2$, and setting every singular value to one gives the tail the weight of the head.

Observation 3.3 (The advantage shrinks with scale; Wen et al. 2026b). In a comparison of ten optimizers on language models of 0.1B to 1.2B parameters, trained on one to eight times the Chinchilla-optimal amount of data with each optimizer carefully tuned, the matrix-based optimizers, Muon and SOAP among them, were the fastest; but their speedup over well-tuned AdamW fell from about 1.4× at 0.1B parameters to about 1.1× at 1.2B.

This is one careful study on one data distribution, and no theorem we know explains it. Why orthogonalize an update, and why does a preconditioner produce the same matrix? How do five multiplications by a polynomial come close to an orthogonal matrix, and why do Muon's coefficients not converge? When does a step that discards the gradient's

singular values beat the gradient itself?

3.2 The idealization: steepest descent on one matrix

Fix one weight matrix and freeze the rest of the network. The parameter is $W \in \mathbb{R}^{m\times n}$, with $m = d_{\text{out}}$ and $n = d_{\text{in}}$ for a layer $x \mapsto Wx$, and $\mathbb{R}^{m\times n}$ carries the inner product $\langle A, B\rangle = \operatorname{tr}(A^\top B) = \sum_{ij} A_{ij}B_{ij}$, so that the gradient $G = \nabla\mathcal{L}(W)$ is again an $m \times n$ matrix. The singular values $\sigma_1(W) \geq \sigma_2(W) \geq \cdots \geq 0$ define the *spectral norm* $\|W\|_{\text{op}} = \sigma_1(W)$, the operator norm from ℓ_2 to ℓ_2; the Frobenius norm $\|W\|_F = (\sum_i \sigma_i(W)^2)^{1/2}$, the Euclidean norm of the entries; and the *nuclear norm* $\|W\|_{\text{nuc}} = \sum_i \sigma_i(W)$. Measuring vectors by their root-mean-square size $\|x\|_{\text{rms}} = \|x\|_2/\sqrt{\dim x}$ gives the *RMS-to-RMS norm*

$$\|W\|_{\text{rms}\to\text{rms}} = \sup_{x\neq 0} \frac{\|Wx\|_{\text{rms}}}{\|x\|_{\text{rms}}} = \sqrt{\frac{n}{m}}\,\|W\|_{\text{op}},$$

the factor by which the layer can enlarge the typical size of an input coordinate. As in Chapter 1, normalized steepest descent is $W_{t+1} = W_t + \eta\,\text{lmo}(G_t)$ with $\text{lmo}(G) \in \arg\min_{\|D\|\leq 1}\langle G, D\rangle$, a step of length η in the chosen norm, and unnormalized steepest descent is $W_{t+1} = W_t - \eta G_t^\sharp$ with $G^\sharp = -\|G\|_*\,\text{lmo}(G)$; the step size η is the learning rate of the practice box.

Assumption 3.4 (One matrix, spectral geometry). The loss $\mathcal{L}\colon \mathbb{R}^{m\times n} \to \mathbb{R}$ is bounded below and differentiable, and it is L_{op}-smooth with respect to the spectral norm: $\|\nabla\mathcal{L}(W) - \nabla\mathcal{L}(W')\|_{\text{nuc}} \leq L_{\text{op}}\|W - W'\|_{\text{op}}$ for all W, W'. When the Frobenius geometry is compared, L_F is the constant with the Frobenius norm on both sides.

The nuclear norm on the left is the dual of the spectral norm (Theorem 3.5).

DICTIONARY

IN THE THEORY	IN A TRAINING RUN
a matrix $W \in \mathbb{R}^{m\times n}$ with the spectral norm	one hidden weight matrix (an attention projection, an MLP matrix, a flattened convolution)
the gradient G	the Nesterov combination of momentum and gradient that Muon orthogonalizes
the polar factor $\text{polar}(G) = UV^\top$	the output of Newton–Schulz, exact only after many steps
a normalized step of length η in $\|\cdot\|_{\text{op}}$	Muon's learning rate, "in units of spectral norm per update"
the RMS-to-RMS norm	how far a layer's outputs move per unit input; Muon's factor $\sqrt{\max(1, d_{\text{out}}/d_{\text{in}})}$
nuclear rank of G; stable rank of the inputs X	how many directions carry the gradient; the effective dimension of the activations entering the layer

FIELD GUIDE · MUON	*Jordan et al. (2024)*
UPDATE	$M_t = \beta M_{t-1} + (1-\beta)G_t$, $N_t = (1-\beta)G_t + \beta M_t$, $W_{t+1} = W_t - \eta\, c_{\text{shape}}\, \text{NS}_5(N_t)$ with $c_{\text{shape}} = \sqrt{\max(1, d_{\text{out}}/d_{\text{in}})}$
NORM	spectral; RMS-to-RMS when $d_{\text{out}} \geq d_{\text{in}}$ (Proposition 3.7)
PRECONDITIONER	none: orthogonalization is the duality map of the norm (Theorem 3.5)
AVERAGES	moving average of G, $\beta = 0.95$, Nesterov form (Chapter 5)
STATE	one matrix per weight matrix
IN THE WILD	NS_5: five quintic Newton–Schulz steps in bfloat16; $\eta = 0.02$; AdamW for embeddings, output layer, and gains; weight decay and update-size matching at scale
CAUTION	orthogonalization is approximate: all but the smallest singular values in $[0.681, 1.203]$ (Proposition 3.10)

FIELD GUIDE · SHAMPOO	*Gupta et al. (2018)*
UPDATE	$L_t = L_{t-1} + G_t G_t^\top$, $R_t = R_{t-1} + G_t^\top G_t$ (from εI), $W_{t+1} = W_t - \eta L_t^{-1/4} G_t R_t^{-1/4}$
NORM	spectral when memoryless (Proposition 3.6)
PRECONDITIONER	Kronecker-factored, from L_t and R_t (Chapter 7)
AVERAGES	sums, or moving averages, of $GG^\top$ and $G^\top G$
STATE	an $m \times m$ and an $n \times n$ matrix per weight matrix
IN THE WILD	the AlgoPerf external-tuning winner

FIELD GUIDE · SPECTRAL DESCENT	*Carlson et al. (2015a)*
UPDATE	$W_{t+1} = W_t - \eta \lVert G_t \rVert_{\text{nuc}}\, \text{polar}(G_t)$, unnormalized; a preconditioned variant (2015b)
NORM	spectral (Schatten-∞) on weight matrices
IN THE WILD	restricted Boltzmann machines (2015a), feedforward and convolutional networks (2015b)

WHAT WAS THROWN AWAY. The other layers are Chapter 4's subject, mini-batch noise and momentum Chapter 5's, and weight decay Chapter 13's; here gradients are exact and memory is off. We keep two features often idealized away, because they are where Muon departs from exact steepest descent: the orthogonalization is a polynomial iteration with few steps, run in low precision. For the comparison with gradient descent we freeze the layer's inputs and use the square loss, the smallest model in which the two geometries can disagree about which step is better and both smoothness constants are explicit.

WHAT ONE HAD TO NOTICE. A weight matrix acts on vectors, and a training step must control how much the layer's outputs change: an operator norm of the update, not the Euclidean norm of its entries.[3] Once the norm is chosen, duality forces the update: the steepest direction is the polar factor

[3] The RMS-to-RMS version of this requirement, with updates of spectral norm proportional to $\sqrt{d_{\text{out}}/d_{\text{in}}}$, is the spectral condition for feature learning of Yang et al. (2023); the modular norm of Large et al. (2024) extends it to whole networks (Chapter 4).

(Bernstein and Newhouse 2024; Carlson et al. 2015a). And the polar factor is an odd function of the matrix, so odd polynomials, which need only matrix products, can approximate it.

3.3 The theory: the polar factor and how to compute it

Throughout, $G \in \mathbb{R}^{m\times n}$ is nonzero of rank r with compact singular value decomposition $G = U_r \Sigma V_r^\top$: the columns $u_1, \dots, u_r$ of U_r and $v_1, \dots, v_r$ of V_r are orthonormal, and $\Sigma = \operatorname{diag}(\sigma_1, \dots, \sigma_r)$ with $\sigma_1 \geq \cdots \geq \sigma_r > 0$. We complete U_r and V_r to orthogonal matrices $[U_r\ U_\perp]$ and $[V_r\ V_\perp]$. The *polar factor* is $\operatorname{polar}(G) = U_r V_r^\top$, and $\operatorname{polar}(0) = 0$; it does not depend on the choice of singular vectors, because $\operatorname{polar}(G) = G(G^\top G)^{-1/2}$ with the inverse square root taken on the range (Proposition 3.6), and $G = \operatorname{polar}(G)(G^\top G)^{1/2}$ is the polar decomposition of Theorem A.8.

3.3.1 The steepest direction

Theorem 3.5 (Spectral steepest descent). *Let $G \in \mathbb{R}^{m\times n}$ be nonzero, of rank r, with the notation above.*

(i) $\max\{\langle G, D\rangle : \|D\|_{\mathrm{op}} \leq 1\} = \|G\|_{\mathrm{nuc}}$. *Hence the dual of the spectral norm is the nuclear norm, and the dual of the nuclear norm is the spectral norm.*

(ii) The maximizers are exactly the matrices

$$D = U_r V_r^\top + U_\perp Z V_\perp^\top, \qquad Z \in \mathbb{R}^{(m-r)\times(n-r)}, \quad \|Z\|_{\mathrm{op}} \leq 1. \tag{3.1}$$

The maximizer is unique, and then equal to $\operatorname{polar}(G)$, *if and only if* $r = \min(m, n)$.

(iii) Among the maximizers, $\operatorname{polar}(G)$ *is the unique one of least Frobenius norm and the unique one of least rank. So* $\operatorname{lmo}_{\mathrm{op}}(G) = -\operatorname{polar}(G)$ *is a valid oracle, the only one when G has full rank; with this choice* $G^\sharp = \|G\|_{\mathrm{nuc}} \operatorname{polar}(G)$, *and normalized and unnormalized steepest descent in the spectral norm are* $W_{t+1} = W_t - \eta \operatorname{polar}(G_t)$ *and* $W_{t+1} = W_t - \eta\|G_t\|_{\mathrm{nuc}} \operatorname{polar}(G_t)$.

Proof. The plan is to expand $\langle G, D\rangle$ along the singular vectors of G, bound each term by $\|D\|_{\mathrm{op}}$, and read off when every bound is attained. Since $G = \sum_{i\leq r} \sigma_i u_i v_i^\top$ and $\langle uv^\top, D\rangle = u^\top D v$,

$$\langle G, D\rangle = \sum_{i=1}^{r} \sigma_i\, u_i^\top D v_i, \qquad u_i^\top D v_i \leq \|D v_i\|_2 \leq \|D\|_{\mathrm{op}}, \tag{3.2}$$

by the Cauchy–Schwarz inequality.

(i) If $\|D\|_{\mathrm{op}} \leq 1$, then every term of (3.2) is at most σ_i, so $\langle G, D\rangle \leq \|G\|_{\mathrm{nuc}}$; for $D = U_r V_r^\top$, which has $\|D\|_{\mathrm{op}} = 1$ and $Dv_i = u_i$, every term equals σ_i. The maximum over the unit ball is the dual norm, and the second claim holds because every norm is the dual of its dual (Lemma A.3).

(ii) Let $\|D\|_{\mathrm{op}} \le 1$ attain the maximum. The terms are at most $\sigma_i > 0$ and add up to $\sum_i \sigma_i$, so $u_i^\top D v_i = 1$ for each $i \le r$. Then $1 = u_i^\top (D v_i) \le \|D v_i\|_2 \le 1$, and equality in the Cauchy–Schwarz inequality forces $D v_i = u_i$; the same argument applied to $D^\top$ gives $D^\top u_i = v_i$. In the bases $[U_r\ U_\perp]$ and $[V_r\ V_\perp]$ the blocks of D are therefore $U_r^\top D V_r = I_r$, $U_\perp^\top D V_r = U_\perp^\top U_r = 0$, $U_r^\top D V_\perp = (D^\top U_r)^\top V_\perp = V_r^\top V_\perp = 0$, and $Z := U_\perp^\top D V_\perp$, with $\|Z\|_{\mathrm{op}} \le \|D\|_{\mathrm{op}} \le 1$: D has the form (3.1). Conversely, a matrix of the form (3.1) is $\mathrm{diag}(I_r, Z)$ in these bases, so its spectral norm is $\max\{1, \|Z\|_{\mathrm{op}}\} = 1$, and it satisfies $D v_i = u_i$, so $\langle G, D\rangle = \|G\|_{\mathrm{nuc}}$. The set is a single point exactly when Z is an empty matrix, that is, when $m = r$ or $n = r$.

(iii) In the same bases $\|D\|_F^2 = r + \|Z\|_F^2$ and $\operatorname{rank} D = r + \operatorname{rank} Z$, both smallest exactly when $Z = 0$. The oracle is the negative of a maximizer. For the sharp operator, every D satisfies $\langle G, D\rangle - \frac{1}{2}\|D\|_{\mathrm{op}}^2 \le \|G\|_{\mathrm{nuc}}\|D\|_{\mathrm{op}} - \frac{1}{2}\|D\|_{\mathrm{op}}^2 \le \frac{1}{2}\|G\|_{\mathrm{nuc}}^2$, with equality if and only if $\|D\|_{\mathrm{op}} = \|G\|_{\mathrm{nuc}}$ and $D/\|D\|_{\mathrm{op}}$ is a maximizer; $D = \|G\|_{\mathrm{nuc}} \operatorname{polar}(G)$ is one (Lemma A.5). ■

An orthogonalized update keeps the singular vectors of the gradient and discards its singular values: every direction $u_i v_i^\top$ in which the loss decreases receives the same step, however small its σ_i. For a rank-deficient gradient, the rule for a linear layer trained on a batch of $B < \min(m, n)$ examples (each contributes a rank-one term $\delta x^\top$), the steepest directions form the face (3.1), and the polar factor is its most conservative point: it moves nothing outside the row and column spaces of G.[4] And the guarantees of Chapter 1 measure the gradient in the nuclear norm.

[4] The face (3.1) is the subdifferential of the nuclear norm at G, as Lemma A.4 predicts for every norm: the maximizers of $\langle G, \cdot\rangle$ over the unit ball of a norm are the subgradients of the dual norm at G.

Proposition 3.6 (Memoryless Shampoo is orthogonalized descent). *Take negative powers of positive semidefinite matrices on their range (positive eigenvalues are raised to the power, zero eigenvalues stay zero). For every nonzero G,*

$$(GG^\top)^{-1/4} G (G^\top G)^{-1/4} = (GG^\top)^{-1/2} G = G (G^\top G)^{-1/2} = \operatorname{polar}(G),$$

and for $\varepsilon > 0$, $(\varepsilon I + GG^\top)^{-1/4} G (\varepsilon I + G^\top G)^{-1/4} = U_r \operatorname{diag}\big(\sigma_i(\sigma_i^2 + \varepsilon)^{-1/2}\big) V_r^\top$.

Proof. Since $GG^\top = U_r \Sigma^2 U_r^\top$ and $G^\top G = V_r \Sigma^2 V_r^\top$, the powers are taken on Σ, and

$$(GG^\top)^{-1/4} G (G^\top G)^{-1/4} = U_r \Sigma^{-1/2}\, \Sigma\, \Sigma^{-1/2} V_r^\top = U_r V_r^\top;$$

the other two products are $U_r \Sigma^{-1} \Sigma V_r^\top$ and $U_r \Sigma \Sigma^{-1} V_r^\top$. With damping, $(\varepsilon I + GG^\top)^{-1/4}$ is $U_r (\Sigma^2 + \varepsilon I)^{-1/4} U_r^\top + \varepsilon^{-1/4} U_\perp U_\perp^\top$, whose second term vanishes against G because $U_\perp^\top U_r = 0$; the right factor acts in the same way. ■

Shampoo preconditions the gradient by fourth roots of the accumulated statistics $L_t = \varepsilon I + \sum_{s \le t} G_s G_s^\top$ and $R_t = \varepsilon I + \sum_{s \le t} G_s^\top G_s$ (Gupta et al. 2018). Keeping only the current gradient and letting $\varepsilon \to 0$ turns its update into $-\eta \operatorname{polar}(G_t)$: memoryless Shampoo is Muon without momentum

(Bernstein and Newhouse 2024). With $\varepsilon > 0$ it is a soft polar factor, $\sigma \mapsto \sigma(\sigma^2 + \varepsilon)^{-1/2}$, which like Newton–Schulz with few steps does not lift the smallest singular values. Accumulation is Chapter 7's subject.

Under Assumption 3.4, normalized spectral descent with $G_t = \nabla\mathcal{L}(W_t)$ satisfies, by the descent lemma in the spectral norm (Lemma A.18), the identity $\langle G, \operatorname{polar}(G)\rangle = \|G\|_{\mathrm{nuc}}$, and the bound $\|\operatorname{polar}(G)\|_{\mathrm{op}} \le 1$,

$$\mathcal{L}(W_{t+1}) \le \mathcal{L}(W_t) - \eta\|G_t\|_{\mathrm{nuc}} + \tfrac{1}{2}L_{\mathrm{op}}\eta^2, \tag{3.3}$$

and summing over $t < T$ gives $\frac{1}{T}\sum_{t<T}\|G_t\|_{\mathrm{nuc}} \le (\mathcal{L}(W_0) - \mathcal{L}^\star)/(\eta T) + \frac{1}{2}L_{\mathrm{op}}\eta$, the spectral case of Theorem 1.12; with stochastic gradients and momentum, Chapter 5's analysis applies in every norm, and the published analyses of Muon follow that pattern (Kovalev 2025; Li and Hong 2025; Pethick et al. 2025; Shen et al. 2025). Why the spectral norm, rather than the Frobenius norm or the ℓ_∞ norm of Chapter 2? Because it controls what a step does to the outputs.

Proposition 3.7 (The RMS-to-RMS norm). *Let $W \in \mathbb{R}^{m\times n}$ act on inputs $x \in \mathbb{R}^n$, and let $\Delta W \neq 0$.*

(i) $\|\Delta W\, x\|_{\mathrm{rms}} \le \|\Delta W\|_{\mathrm{rms}\to\mathrm{rms}}\|x\|_{\mathrm{rms}}$ *with* $\|\Delta W\|_{\mathrm{rms}\to\mathrm{rms}} = \sqrt{n/m}\,\|\Delta W\|_{\mathrm{op}}$, *with equality if and only if x is a right singular vector of ΔW for its largest singular value or $x = 0$; for $\Delta W = \sigma u v^\top$, if and only if x is parallel to v.*

(ii) One oracle is $\operatorname{lmo}_{\mathrm{rms}\to\mathrm{rms}}(G) = -\sqrt{m/n}\,\operatorname{polar}(G)$, *the only one when G has full rank, so the normalized step $\Delta W = -\eta\sqrt{m/n}\,\operatorname{polar}(G)$ changes every output by at most $\eta\|x\|_{\mathrm{rms}}$ in RMS, whatever the widths.*

(iii) If $m \ge n$ and $\operatorname{rank} G = n$, *then this step changes every output by exactly $\eta\|x\|_{\mathrm{rms}}$.*

Proof. (i) Since $\|\Delta W x\|_2^2 = x^\top \Delta W^\top \Delta W\, x \le \sigma_1(\Delta W)^2\|x\|_2^2$, with equality exactly on the top eigenspace of $\Delta W^\top \Delta W$,

$$\|\Delta W x\|_{\mathrm{rms}} = \frac{\|\Delta W x\|_2}{\sqrt{m}} \le \frac{\|\Delta W\|_{\mathrm{op}}\|x\|_2}{\sqrt{m}} = \sqrt{\frac{n}{m}}\,\|\Delta W\|_{\mathrm{op}}\|x\|_{\mathrm{rms}}.$$

(ii) The unit ball of $\|\cdot\|_{\mathrm{rms}\to\mathrm{rms}}$ is the spectral ball of radius $\sqrt{m/n}$, so its oracle is $\sqrt{m/n}$ times that of Theorem 3.5; the bound follows from (i). (iii) If $\operatorname{rank} G = n \le m$, then V_r is square and orthogonal, so $\|\operatorname{polar}(G)x\|_2 = \|V_r^\top x\|_2 = \|x\|_2$ and $\|\Delta W x\|_{\mathrm{rms}} = \eta\sqrt{m/n}\,\|x\|_2/\sqrt{m} = \eta\|x\|_{\mathrm{rms}}$. ∎

So Muon's shape factor converts the spectral norm into the RMS-to-RMS norm: for $d_{\mathrm{out}} \ge d_{\mathrm{in}}$ its step is the normalized RMS-to-RMS step of size η, and a full-rank update moves every output by exactly $\eta\|x\|_{\mathrm{rms}}$ in RMS; for $d_{\mathrm{out}} < d_{\mathrm{in}}$ the code uses the factor one, a step of RMS-to-RMS norm $\eta\sqrt{d_{\mathrm{in}}/d_{\mathrm{out}}}$. Moonlight's factor normalizes another quantity: a full-rank polar factor has entrywise RMS $\max(m,n)^{-1/2}$, so $0.2\sqrt{\max(m,n)}\,\operatorname{polar}(G)$ has entrywise RMS 0.2 for every shape, as Moonlight intends, and an RMS-to-RMS norm that grows like $\sqrt{d_{\mathrm{in}}}$ when $d_{\mathrm{out}} \ge d_{\mathrm{in}}$. Which rule keeps the best learning rate fixed as the width grows is Chapter 4's question.[5]

[5] The README of the reference code states that the learning rate "should have built-in muP scaling", so that it need not be retuned as the model grows (Jordan 2024).

3.3.2 Computing the polar factor

A singular value decomposition is slow on accelerators built for matrix products. But the polar factor maps singular values to one and keeps singular vectors, and odd matrix polynomials act on singular values in the same way.

Lemma 3.8 (Odd polynomials act on singular values). *For an odd polynomial $p(s) = \sum_{j=0}^{J} c_j s^{2j+1}$ and $X \in \mathbb{R}^{m\times n}$ put $p(X) = \sum_j c_j (XX^\top)^j X$. If $X = UDV^\top$ with $U^\top U = I$, $V^\top V = I$, and D diagonal, then $p(X) = U\,p(D)\,V^\top$. Hence the iteration $X_{k+1} = p_{k+1}(X_k)$ from $X_0 = G/\gamma$, $\gamma > 0$, satisfies $X_k = U_r \operatorname{diag}\big(p_k \circ \cdots \circ p_1(\sigma_i/\gamma)\big) V_r^\top$.*

Proof. Because $V^\top V = I$, we have $XX^\top = UD^2U^\top$, and induction on j with $U^\top U = I$ gives $(XX^\top)^j X = (UD^2U^\top)^j UDV^\top = UD^{2j+1}V^\top$; summing gives $p(X) = Up(D)V^\top$. The diagonal $p(D)$ may have negative or zero entries, but the identity needs only $U^\top U = V^\top V = I$, so induction on k applies. ∎

The *Newton–Schulz iteration* for the polar factor uses the cubic $p(s) = \frac{3}{2}s - \frac{1}{2}s^3$, the first-order correction toward $X^\top X = I$; Muon's quintic has a steeper start.[6] A quintic step costs three matrix products, $XX^\top$, its square, and the product with X, and Muon transposes tall matrices so that $XX^\top$ is the smaller Gram matrix.

[6] Schulz (1933) gave the iteration $X \leftarrow X(2I - AX)$ for the inverse of A, which named the family; the cubic iteration for the orthogonal factor appears in Björck and Bowie (1971), and in machine learning it is also called Björck orthogonalization. Higham (2008, Chapter 8) treats iterations for the polar decomposition.

Theorem 3.9 (Newton–Schulz). *Let $G \neq 0$ have rank r, condition number $\kappa = \sigma_1/\sigma_r$, and stable rank $r_{\mathrm{st}}(G) = \|G\|_F^2/\|G\|_{\mathrm{op}}^2$. Let $p(s) = \frac{3}{2}s - \frac{1}{2}s^3$, $X_0 = G/\gamma$, and $X_{k+1} = \frac{3}{2}X_k - \frac{1}{2}X_kX_k^\top X_k$.*

(i) If $\gamma > \sigma_1/\sqrt{3}$, then $X_k \to \operatorname{polar}(G)$.

(ii) If $\gamma \geq \sigma_1$, for instance $\gamma = \|G\|_F$, then for each $s_i = \sigma_i/\gamma \in (0,1]$ the sequence $p^{\circ k}(s_i)$ is nondecreasing with limit 1, and

$$\|X_k - \operatorname{polar}(G)\|_{\mathrm{op}} = 1 - p^{\circ k}(s_r), \qquad 1 - p(s) = \tfrac{1}{2}(1-s)^2(2+s).$$

(iii) Let $\gamma = \|G\|_F$, $0 < \varepsilon \leq \frac{1}{2}$, and write $\lceil x \rceil_+ = \max\{0, \lceil x \rceil\}$. Then $\|X_k - \operatorname{polar}(G)\|_{\mathrm{op}} \leq \varepsilon$ for every

$$k \geq \left\lceil \frac{\log\big(\kappa\sqrt{r_{\mathrm{st}}(G)}/2\big)}{\log(11/8)} \right\rceil_+ + \left\lceil \log_2 \frac{\log(2/(3\varepsilon))}{\log(4/3)} \right\rceil.$$

Proof. The plan is to reduce to the scalar map with Lemma 3.8 and then follow $s \mapsto p(s)$: a monotone climb toward one, then quadratic convergence. By the lemma, $X_k - \operatorname{polar}(G) = U_r \operatorname{diag}(p^{\circ k}(s_i) - 1)V_r^\top$, so $\|X_k - \operatorname{polar}(G)\|_{\mathrm{op}} = \max_i |p^{\circ k}(s_i) - 1|$.

Since $p(s) = \frac{1}{2}s(3 - s^2)$ and $p'(s) = \frac{3}{2}(1 - s^2)$, the map increases on $[0,1]$ from $p(0) = 0$ to $p(1) = 1$ and decreases on $[1, \sqrt{3}]$ to $p(\sqrt{3}) = 0$; so p maps $(0, \sqrt{3})$ into $(0,1]$. On $(0,1)$, $p(s)/s = \frac{1}{2}(3 - s^2) > 1$, so $s < p(s) < 1$; a sequence started in $(0,1]$ is nondecreasing and bounded by one, and its

limit is a fixed point of p in $(0, 1]$, which must be 1 because $p(s) = s$ forces $s^2 = 1$. A start in $(1, \sqrt{3})$ is mapped into $(0, 1)$ by one step. This proves (i). In (ii) all s_i lie in $(0, 1]$, and because p is increasing there the sequence from s_r stays the smallest; the factorization of $1 - p$ is checked by expanding.

For (iii), $s_r = \sigma_r/\|G\|_F = 1/(\kappa\sqrt{r_{\mathrm{st}}(G)})$. While $s \le \frac{1}{2}$, $p(s)/s \ge \frac{11}{8}$, and once $s \ge \frac{1}{2}$, $p(s) \ge p(\frac{1}{2}) = \frac{11}{16}$, so the sequence never returns below $\frac{1}{2}$; hence $p^{\circ k_1}(s_r) \ge \frac{1}{2}$ after the k_1 steps of the first term in (iii). From then on $e_j = 1 - p^{\circ(k_1+j)}(s_r) \in [0, \frac{1}{2}]$ satisfies $e_{j+1} = \frac{1}{2}e_j^2(3 - e_j) \le \frac{3}{2}e_j^2$ by (ii), so $f_j = \frac{3}{2}e_j$ has $f_{j+1} \le f_j^2$ and $f_0 \le \frac{3}{4}$. Thus $e_j \le \frac{2}{3}(\frac{3}{4})^{2^j}$, which is at most ε once $2^j \ge \log(2/(3\varepsilon))/\log(4/3)$, a ratio of at least one as $\varepsilon \le \frac{1}{2}$. ∎

The Frobenius normalization is safe, since $\|G\|_F \ge \|G\|_{\mathrm{op}}$, and it costs only the factor $\sqrt{r_{\mathrm{st}}(G)}$, small for the low-stable-rank gradients of Observation 3.2. The climb is what is slow: near zero the cubic multiplies a singular value by only $\frac{3}{2}$ per step, so lifting $s = 10^{-2}$ above $\frac{1}{2}$ takes ten steps and reaching an error of 10^{-3} fourteen (the bound gives eighteen). Zero singular values stay zero, so in exact arithmetic the iteration returns the polar factor even for rank-deficient G.

Proposition 3.10 (Muon's quintic does not converge). *Let $q(s) = as + bs^3 + cs^5$ with $(a, b, c) = (3.4445, -4.7750, 2.0315)$, with positive critical points $s_1 = 0.5545\ldots$ and $s_2 = 1.0501\ldots$.*

(i) $q > 0$ on $(0, \infty)$; q increases on $[0, s_1]$, decreases on $[s_1, s_2]$, and increases on $[s_2, \infty)$.

(ii) $J = [q(s_2), q(s_1)] = [0.6818\ldots, 1.2023\ldots]$ satisfies $q(J) = J' \subseteq J$ with $J' = [0.6818\ldots, 1.1344\ldots]$, and $q(J') = J'$.

(iii) If $s \in (0, 1]$ and $q^{\circ k}(s) \ge q(s_2)$, then $q^{\circ j}(s) \in J$ for all $j \ge k$. After five steps every $s \ge 1.6 \cdot 10^{-3}$ lies in J, and every $s \ge 5.2 \cdot 10^{-3}$ lies in J'.

(iv) The fixed points of q in $(0, \infty)$ are $s_a = 0.8680\ldots$ and $s_b = 1.2637\ldots$, with $q'(s_a) = -1.58\ldots$ and $q'(s_b) = 6.47\ldots$. For no $s \in (0, 1]$ outside a countable set do the iterates $q^{\circ k}(s)$ converge.

Proof. The plan is to locate the critical and fixed points through quadratics in $u = s^2$ and to follow a lower envelope of the iterates by monotonicity; decimals are evaluations of explicit expressions. (i) $q(s)/s = a + bu + cu^2$ has discriminant $b^2 - 4ac = -5.19\ldots < 0$, so $q > 0$; $q'(s) = a + 3bu + 5cu^2$ is positive at $u = 0$ and vanishes at $u = s_1^2 = 0.3075\ldots$ and $u = s_2^2 = 1.1028\ldots$.

(ii) and (iii). By (i), for $0 < \ell \le s_1$ the values of q on $[\ell, q(s_1)]$ lie in $[\min\{q(\ell), q(s_2)\}, q(s_1)]$, since on its three monotone pieces they lie between $q(\ell)$ and $q(s_1)$, between $q(s_2)$ and $q(s_1)$, and between $q(s_2)$ and $q(q(s_1)) = 0.9465\ldots$; for $s_1 < \ell \le q(s_1)$ only the last two pieces occur. As $q(s_2) > s_1$, this gives $q(J) \subseteq J$; the maximum of q on J is at its left end, where q decreases, and equals $q(q(s_2)) = 1.1344\ldots$, so $q(J) = J'$, and the same computation gives $q(J') = J'$. Starting from $[s, 1] \subseteq [s, q(s_1)]$, after k steps every point of $[s, 1]$ is in $[q^{\circ k}(s), q(s_1)]$ while $q^{\circ k}(s) \le s_1$, and

in J once $q^{\circ k}(s) \geq q(s_2)$. As q increases on $[0, s_1]$, it suffices to check $q^{\circ 5}(1.6 \cdot 10^{-3}) = 0.718\ldots$ and $q^{\circ 4}(5.2 \cdot 10^{-3}) = 0.6836\ldots$, both above $q(s_2)$.

(iv) The fixed points solve $a-1+bu+cu^2 = 0$, with roots $u = 0.7535\ldots$ and $u = 1.5970\ldots$. Every orbit from $(0, 1]$ enters J, because $q(s)/s$ decreases in u on $(0, s_1]$, so that $q(s) \geq (q(s_1)/s_1)s > 2s$ there, and an iterate in $(s_1, q(s_1))$ is mapped into J. If $q^{\circ k}(s) \to s^*$, then s^* is a fixed point by continuity and lies in the closed interval J, which excludes s_b; so $s^* = s_a$. As $|q'(s_a)| > 1$, there are $\delta > 0$ and $\rho > 1$ with $|q(x) - s_a| \geq \rho|x - s_a|$ for $|x - s_a| < \delta$; iterates that stay within δ of s_a without reaching it move away geometrically, so some iterate equals s_a. Then s lies in $\bigcup_k q^{-k}(\{s_a\})$, which is countable because a point has at most five preimages under q. ∎

The quintic makes a deliberate trade. Its slope at zero is 3.4445 instead of $\frac{3}{2}$, so $s = 10^{-2}$ passes $\frac{1}{2}$ after four steps instead of ten, and five steps bring every normalized singular value above $1.6 \cdot 10^{-3}$ into $[0.681, 1.203]$. The price is that $q(1) = 0.701$ and the fixed points repel: the singular values never settle but move about inside J, of which the documentation's $[0.5, 1.5]$ is a loose description. The step along different singular directions then varies by a factor of up to $q(s_1)/q(s_2) = 1.76$, which costs at most a factor $(q(s_2)/q(s_1))^2 = 0.32$ in the guaranteed decrease (3.3) (Exercise 3.5).

3.4 The theory: when a spectral step beats a gradient step

Consider one linear layer $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ trained on n examples, with inputs $X \in \mathbb{R}^{d_{\text{in}} \times n}$ and targets $Y \in \mathbb{R}^{d_{\text{out}} \times n}$, by

$$\mathcal{L}(W) = \frac{1}{2n}\|WX - Y\|_F^2, \qquad A = \frac{1}{n}XX^\top \neq 0,$$

and let $\lambda_1 \geq \cdots \geq \lambda_{d_{\text{in}}} \geq 0$ be the eigenvalues of A, with orthonormal eigenvectors $q_1, \ldots, q_{d_{\text{in}}}$.

Definition 3.11 (Stable rank and nuclear rank). For a nonzero matrix M, the *stable rank* is $r_{\text{st}}(M) = \|M\|_F^2/\|M\|_{\text{op}}^2$ and the *nuclear rank* is $r_{\text{nuc}}(M) = \|M\|_{\text{nuc}}^2/\|M\|_F^2$. Both lie in $[1, \operatorname{rank} M]$, and each equals $\operatorname{rank} M$ exactly when the nonzero singular values of M are equal.

The bounds follow from $\sigma_1^2 \leq \sum_i \sigma_i^2 \leq (\operatorname{rank} M)\sigma_1^2$ and, by the Cauchy–Schwarz inequality, $\sum_i \sigma_i^2 \leq (\sum_i \sigma_i)^2 \leq (\operatorname{rank} M)\sum_i \sigma_i^2$. For the inputs, $r_{\text{st}}(X) = \operatorname{tr} A/\lambda_1$.

Lemma 3.12 (The two smoothness constants of a linear layer). *(i)* $\mathcal{L}(W + \Delta) = \mathcal{L}(W) + \langle \nabla\mathcal{L}(W), \Delta\rangle + \frac{1}{2}\operatorname{tr}(\Delta A \Delta^\top)$, *with* $\nabla\mathcal{L}(W) = \frac{1}{n}(WX - Y)X^\top$.

(ii) In every norm, the smallest smoothness constant of $\mathcal{L}$ *is* $\sup\{\operatorname{tr}(\Delta A\Delta^\top) : \|\Delta\| \leq 1\}$.

(iii) $L_F = \lambda_1$ *and* $L_{\text{op}} = \sum_{i \leq k} \lambda_i$ *with* $k = \min(d_{\text{out}}, d_{\text{in}})$*; so* $L_{\text{op}} = \operatorname{tr} A$

exactly when $d_{\text{out}} \geq \operatorname{rank} A$, for instance when $d_{\text{out}} \geq d_{\text{in}}$.

Proof. The plan is to read everything off the exact quadratic (i), which follows by expanding $\|(W+\Delta)X - Y\|_F^2$ with $\|\Delta X\|_F^2 = \operatorname{tr}(\Delta X X^\top \Delta^\top)$. (ii) By (i), $\nabla\mathcal{L}(W+\Delta) - \nabla\mathcal{L}(W) = \Delta A$, and $\langle \Delta, \Delta' \rangle_A := \langle \Delta A, \Delta' \rangle = \operatorname{tr}(A\Delta^\top \Delta')$ is a symmetric positive semidefinite bilinear form. With $L = \sup_{\|E\| \leq 1} \langle E, E \rangle_A$, the Cauchy–Schwarz inequality for this form gives $\|\Delta A\|_* = \sup_{\|\Delta'\| \leq 1} \langle \Delta, \Delta' \rangle_A \leq L\|\Delta\|$, and $\|\Delta A\|_* \geq \langle \Delta, \Delta \rangle_A / \|\Delta\|$ for $\Delta \neq 0$ shows that no smaller constant works. (iii) With rows δ_j of Δ, $\operatorname{tr}(\Delta A \Delta^\top) = \sum_j \delta_j^\top A \delta_j \leq \lambda_1 \|\Delta\|_F^2$, with equality at $\Delta = e_1 q_1^\top$. For the spectral norm put $P = \Delta^\top \Delta$ and $w_i = q_i^\top P q_i$, so that $\operatorname{tr}(\Delta A \Delta^\top) = \sum_i \lambda_i w_i$. If $\|\Delta\|_{\text{op}} \leq 1$, then $0 \preceq P \preceq I$ and $\operatorname{rank} P \leq k$, so $w_i \in [0,1]$ and $\sum_i w_i = \operatorname{tr} P \leq k$, and the sum is at most $\sum_{i \leq k} \lambda_i$; $\Delta = \sum_{i \leq k} e_i q_i^\top$ attains it and has spectral norm one, as $k \leq d_{\text{out}}$. It equals $\operatorname{tr} A$ exactly when $\operatorname{rank} A \leq k$, that is, when $\operatorname{rank} A \leq d_{\text{out}}$. ∎

Theorem 3.13 (When spectral steps beat Frobenius steps). *Let $G = \nabla\mathcal{L}(W) \neq 0$, and compare the spectral step $-\operatorname{polar}(G)\|G\|_{\text{nuc}}/L_{\text{op}}$ with the gradient step $-G/L_F$.*

(i) The spectral step decreases $\mathcal{L}$ by at least $\|G\|_{\text{nuc}}^2/(2L_{\text{op}})$ and the gradient step by at least $\|G\|_F^2/(2L_F)$. Each step minimizes the upper bound of the descent lemma in its norm, and so does the normalized step of the best length.

(ii) The spectral guarantee is the larger if and only if $r_{\text{nuc}}(G) > L_{\text{op}}/L_F$. When $d_{\text{out}} \geq d_{\text{in}}$ this reads

$$\frac{\|G\|_{\text{nuc}}^2}{\|G\|_F^2} > \frac{\|X\|_F^2}{\|X\|_{\text{op}}^2}, \qquad \text{that is,} \qquad r_{\text{nuc}}(G) > r_{\text{st}}(X) :$$

the nuclear rank of the gradient must exceed the stable rank of the inputs.

(iii) The actual decreases are $(2 - \rho_{\text{op}})$ and $(2 - \rho_F)$ times the guarantees, with $\rho_{\text{op}} = \operatorname{tr}(V_r^\top A V_r)/L_{\text{op}}$ and $\rho_F = \operatorname{tr}(GAG^\top)/(L_F\|G\|_F^2)$ in $(0,1]$. If $d_{\text{out}} \geq d_{\text{in}} = \operatorname{rank} G$, then $\rho_{\text{op}} = 1$: the spectral guarantee is attained, and the ratio of the actual decreases lies between $\frac{1}{2}$ and 1 times the ratio of the guarantees. The gradient guarantee is attained if and only if every row of G lies in the top eigenspace of A.

Proof. The plan is to evaluate the exact quadratic of Lemma 3.12(i) along each step. (i) For a norm with constant L, Lemma 3.12(ii) gives $\mathcal{L}(W+\Delta) \leq \mathcal{L}(W) + \langle G, \Delta \rangle + \frac{L}{2}\|\Delta\|^2$. At fixed length $\|\Delta\| = \tau$ the smallest $\langle G, \Delta \rangle$ is $-\tau\|G\|_*$, at $\Delta = \tau \operatorname{lmo}(G)$, and $-\tau\|G\|_* + \frac{L}{2}\tau^2$ is smallest at $\tau = \|G\|_*/L$, with value $-\|G\|_*^2/(2L)$; by Theorem 3.5 this step is $-\operatorname{polar}(G)\|G\|_{\text{nuc}}/L_{\text{op}}$ in the spectral norm and $-G/L_F$ in the Frobenius norm. (ii) The comparison is $r_{\text{nuc}}(G) > L_{\text{op}}/L_F$, and Lemma 3.12(iii) gives $L_{\text{op}}/L_F = \operatorname{tr} A/\lambda_1 = r_{\text{st}}(X)$ when $d_{\text{out}} \geq d_{\text{in}}$.

(iii) Along $-\tau \operatorname{polar}(G)$, $\mathcal{L}$ changes by $-\tau\|G\|_{\text{nuc}} + \frac{\tau^2}{2}\operatorname{tr}(U_r V_r^\top A V_r U_r^\top)$, and the trace equals $\operatorname{tr}(V_r^\top A V_r) = \rho_{\text{op}} L_{\text{op}}$; at $\tau = \|G\|_{\text{nuc}}/L_{\text{op}}$ the decrease is

$(2-\rho_{\text{op}})\|G\|_{\text{nuc}}^2/(2L_{\text{op}})$. The same computation along $-G/L_F$ gives the factor $2-\rho_F$. The rows of $G=\frac{1}{n}(WX-Y)X^\top$, and hence the columns of V_r, lie in the range of A, where A is positive definite, so $\rho_F>0$ and $\rho_{\text{op}}>0$. Next, $\rho_F\le 1$ because $\operatorname{tr}(GAG^\top)\le\lambda_1\|G\|_F^2$, with equality exactly when every row of G is an eigenvector for λ_1; and $\rho_{\text{op}}\le 1$ by the argument of Lemma 3.12(iii) applied to $P=V_rV_r^\top$, which has rank $r\le k$. If $\operatorname{rank}G=d_{\text{in}}\le d_{\text{out}}$, then V_r is square and orthogonal and $\operatorname{tr}(V_r^\top AV_r)=\operatorname{tr}A=L_{\text{op}}$; the ratio of actual decreases is then the ratio of guarantees divided by $2-\rho_F\in[1,2)$. ∎

A gradient step is limited by the stiffest input direction, through λ_1, and paid in the Frobenius norm of the gradient; a spectral step is limited by the total input energy $\operatorname{tr}A$ and paid in the nuclear norm. The spectral step wins when the gradient is spread over more directions than the inputs effectively occupy, which needs anisotropic inputs: $r_{\text{nuc}}(G)\le\operatorname{rank}A$, so with $A=\lambda_1 I$ the stable rank d_{in} is never beaten, and gradient descent with step $1/\lambda_1$ solves the problem in one step. This is the comparison of Davis and Drusvyatskiy (2025).[7] One family makes every quantity explicit.

[7] They also prove that post-activation matrices have low stable rank at Gaussian initialization, in random-feature regression, feedforward networks, and transformer blocks (Davis and Drusvyatskiy 2025).

Proposition 3.14 (The aligned case is sign descent on the spectrum). *Let $d_{\text{out}}\ge d_{\text{in}}$, $Y=W_\star X$, and $A=Q\operatorname{diag}(a)Q^\top$ with Q orthogonal and $a_1\ge\cdots\ge a_{d_{\text{in}}}>0$, so that $\mathcal{L}(W)=\frac{1}{2}\operatorname{tr}\big((W-W_\star)A(W-W_\star)^\top\big)$, and let $W_0-W_\star=U\operatorname{diag}(e)Q^\top$ with $U^\top U=I$. Then every iterate of gradient descent and of spectral descent has the form $W_t-W_\star=U\operatorname{diag}(e^{(t)})Q^\top$, and*

(i) gradient descent with step η is $e_i^{(t+1)}=(1-\eta a_i)e_i^{(t)}$, gradient descent on $\frac{1}{2}\sum_i a_ie_i^2$;

(ii) spectral descent with steps η_t is $e^{(t+1)}=e^{(t)}-\eta_t\operatorname{sign}(e^{(t)})$, sign descent on the same quadratic (Chapter 2);

(iii) with $g_i=a_ie_i$, $\|G\|_{\text{nuc}}=\|g\|_1$, $\|G\|_F=\|g\|_2$, $L_{\text{op}}=\sum_i a_i$, and $L_F=a_1$, and the criterion of Theorem 3.13 becomes $\|g\|_1^2/\|g\|_2^2>\sum_i a_i/a_1$, the criterion of Chapter 1 for preferring ℓ_∞ steps to ℓ_2 steps on a diagonal quadratic.

Proof. If $W-W_\star=U\operatorname{diag}(e)Q^\top$, then $G=(W-W_\star)A=U\operatorname{diag}(a_ie_i)Q^\top$, and $(W-W_\star)(I-\eta A)=U\operatorname{diag}\big(e_i(1-\eta a_i)\big)Q^\top$. As U and Q have orthonormal columns, the singular values of G are the $|a_ie_i|$ and $\operatorname{polar}(G)=U\operatorname{diag}(\operatorname{sign}(e_i))Q^\top$, with zeros where $e_i=0$. Part (iii) reads the norms off the singular values and uses Lemma 3.12(iii). ∎

So the spectral comparison contains the ℓ_∞ comparison of Chapters 1 and 2 as its diagonal case, and there Muon is sign descent on the singular values of the error. Outside the aligned case the singular vectors of the gradient move from step to step and the reduction fails (Section 3.6).

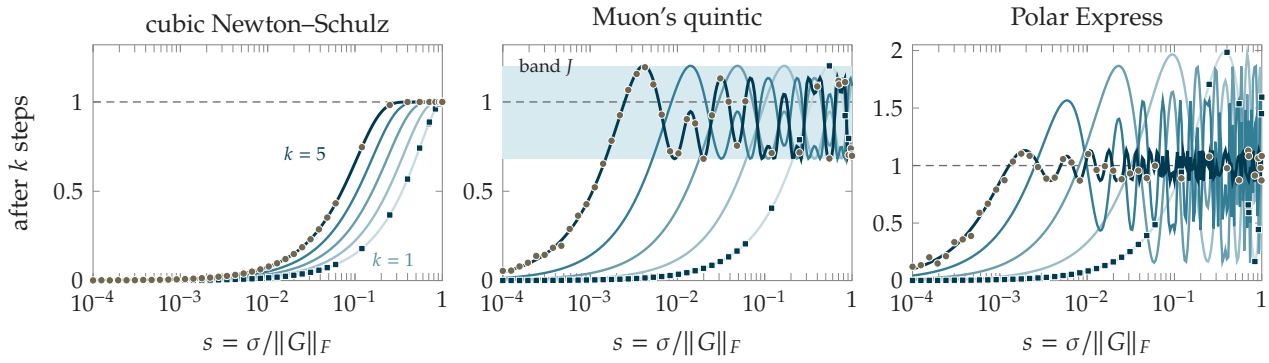


Figure 3.1. What k steps of three odd-polynomial iterations do to a singular value s of the Frobenius-normalized input. Curves: the compositions $p_k \circ \cdots \circ p_1(s)$ for $k = 1, \ldots, 5$, light to dark (for Polar Express, of $s/1.01$, as in its code; coefficients by the recipe of its code). Points: $u_i^\top X_k v_i$ for the singular pairs (u_i, v_i) of six 64×128 test matrices with prescribed singular values and random singular vectors (seed 3), iterated in bfloat16 exactly as in Muon's reference code; squares $k = 1$, circles $k = 5$. In float64 the same iterations agree with the curves to $3 \cdot 10^{-5}$. Shaded: the band $J = [0.681, 1.203]$ of Proposition 3.10. Script: `03-spectral-ns.py`.

3.5 Reading the theory

NEWTON–SCHULZ, STEP BY STEP. Figure 3.1 draws the maps $p_k \circ \cdots \circ p_1$ of Lemma 3.8 for $k \le 5$, with singular values of matrices iterated in bfloat16 as in Muon's code. The cubic's curves rise like $(\frac{3}{2})^k s$ and flatten at one, where the error squares at every step; after five steps only singular values above 0.27 are within 10^{-2} of one. Muon's curves rise like $3.4445^k s$, overshoot to 1.2, and fold back, filling the band J for every s above $1.6 \cdot 10^{-3}$; below that the output stays small, so Muon leaves the weakest singular directions underweighted rather than equalizing them. The third iteration, *Polar Express*, uses a different quintic at each step: Amsel et al. (2026) choose at step k the odd quintic closest to one in the worst case over the interval $[\ell_k, u_k]$ that contains the singular values, prove this greedy choice optimal among compositions, and in their code start from $\ell_0 = 10^{-3}$ with a safety factor against rounding. Its curves swing between nearly zero and nearly two and then contract: after five steps the singular values with $s \ge 10^{-3}$ lie in $[0.857, 1.131]$, after six in $[0.9967, 1.0016]$. For $k \ge 3$ the compositions oscillate faster than the plot resolves and fill their bands.

WHERE THE LEMMA STOPS BEING EXACT. In float64 the points lie on the curves, up to the constant 10^{-7} added to the Frobenius norm. In bfloat16, which keeps eight significant bits,[8] the deviations grow with the slopes of the composed maps: after five steps they stay below $1.6 \cdot 10^{-3}$ for the cubic and $3.1 \cdot 10^{-2}$ for Muon's quintic, but reach about 0.25 for Polar Express, whose early polynomials are steep near the top of the spectrum and multiply the rounding error of about 2^{-8} by their slopes. Below the rounding level, rounding seeds every singular direction, and five quintic steps multiply small singular values by up to $3.4445^5 \approx 485$: an exactly rank-one 256×512

[8] bfloat16 keeps the eight exponent bits of float32 and only seven explicit mantissa bits, so its relative rounding error is at most 2^{-8}; matrix products typically accumulate in float32 and round the result.

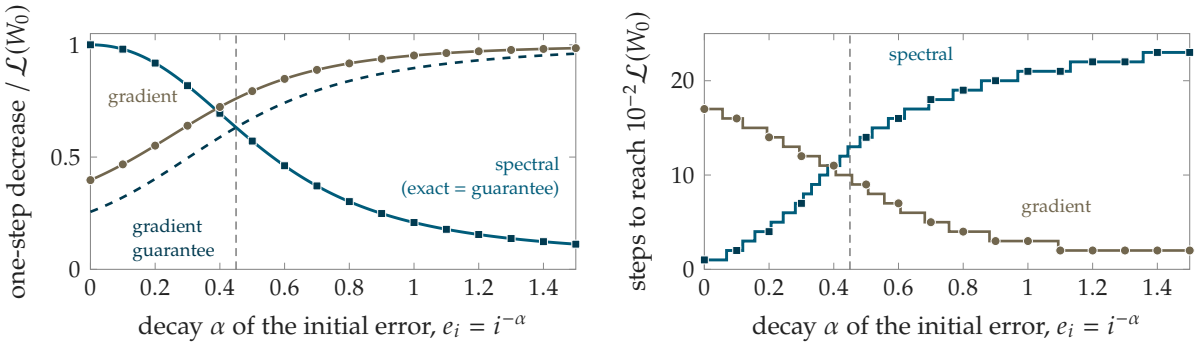


Figure 3.2. Spectral versus gradient steps on matrix least squares in the aligned case: 128×128 matrices, inputs with spectrum $a_i = i^{-1/2}$ (stable rank 21.2), initial error $U\,\mathrm{diag}(i^{-\alpha})Q^\top$ with Haar-random U and eigenbasis Q (seed 11), both methods with the step sizes of Theorem 3.13. Left: decrease of the loss in the first step, relative to the initial loss. Right: steps to reach 1% of the initial loss. Curves: the scalar recursions of Proposition 3.14; points (squares spectral, circles gradient): the matrix iterations, with polar factors from a singular value decomposition, which agree with the curves to $6 \cdot 10^{-16}$ and in every step count. Dashed line: the crossover $\alpha = 0.450$ predicted by the theorem; the observed crossovers are at $\alpha = 0.386$ (left) and 0.388 (right). Script: `03-spectral-crossover.py`.

input leaves Muon's iteration in bfloat16 with about 55 singular values above 0.1, the largest near 0.4 (Exercise 3.8). They lie almost entirely in the block $U_\perp^\top X_5 V_\perp$, with norm below one, so rounding moves the output along the face (3.1): in bfloat16 Newton–Schulz picks another steepest direction.

The crossover. Figure 3.2 tests Theorem 3.13 on the aligned problems of Proposition 3.14, with $d_{\text{in}} = d_{\text{out}} = 128$, input spectrum $a_i = i^{-1/2}$ of stable rank 21.2, and initial errors $e_i = i^{-\alpha}$. As α grows the error concentrates on the stiff input directions, the gradient $g_i = i^{-1/2-\alpha}$ loses nuclear rank, and gradient descent, fast on stiff directions, gains. At $\alpha = 0$ one spectral step removes the error, while gradient descent needs 17 steps to reach 1% of the initial loss. The theorem predicts the crossover where the nuclear rank of the first gradient equals 21.2, at $\alpha = 0.450$; the one-step decreases cross at $\alpha = 0.386$, and the step counts at $\alpha = 0.388$. The prediction errs in the direction Theorem 3.13(iii) predicts: the spectral guarantee is attained, while gradient descent beats its own by the factor $2 - \rho_F$, which falls from 1.55 at $\alpha = 0$ to 1.03 at $\alpha = 1.5$. The theorem's scope ends at one step, and this is a case in which the singular vectors of the gradient never move; Section 3.6 shows what happens when they do.

3.6 What the theory misses

What was idealized away. A network has many weight matrices that share a step size and feed into one another (Chapter 4). Gradients here are exact and memory is off, while Muon averages minibatch gradients with momentum 0.95, without which a normalized step stalls at a noise floor (Chapter 5). Embeddings, the output layer, and gains are trained by

AdamW for reasons the spectral norm does not supply (Chapter 4). The theorems about the oracle assume exact orthogonalization; Proposition 3.10 and the bfloat16 measurements quantify the gap for one step only. Weight decay is absent (Chapter 13). The comparison with gradient descent freezes the layer's inputs, uses the square loss, and compares worst-case guarantees for one step.

WHERE IT BREAKS. (1) *One step does not predict many.* In the aligned case of Figure 3.2 the theorem nearly does, placing the crossover at $\alpha = 0.45$ instead of 0.39; for generic targets it does not. With the same inputs and target singular values $i^{-\alpha}$, but right singular vectors drawn independently of the eigenvectors of the inputs, the ratio of guarantees stays above 2 for every $\alpha \leq 0.5$, so the first spectral step always beats the first gradient step; yet in our runs (five seeds) spectral descent needed more steps than gradient descent to reach 1% of the initial loss once $\alpha \geq 0.4$: 36 against 17 at $\alpha = 0.4$ (Exercise 3.9). A spectral step moves every singular direction by the same amount, overshoots the small ones, and must correct them later, as sign descent does (Chapter 2), and a first-step guarantee cannot see this. In the analysis of Shen et al. (2025), Muon matches the rate of gradient descent in a star-convex setting and can beat it when the Hessians are of low rank and close to block diagonal. (2) *The advantage shrinks with scale* (Observation 3.3), and a theory of one matrix and one step has no scale in it. (3) *Muon at scale needs more than the norm.* Moonshot AI found weight decay crucial for scaling Muon (Liu et al. 2025), and for Kimi K2 Moonshot AI report that attention logits exploded under Muon, which MuonClip prevents by rescaling the query and key weights after an update whenever the largest attention logit exceeds a threshold (Kimi Team 2025). A normalized step bounds the change of a matrix per step, not its accumulated size: after t steps its spectral norm may have grown by as much as ηt, and the bound on a logit bilinear in the query and key weights like $(\eta t)^2$ (a heuristic reading, not an analysis of those runs). (4) *How exact must the orthogonalization be?* The reference code states that the approximation does not hurt (Jordan 2024), while Amsel et al. (2026) report consistent gains in validation loss for GPT-2-sized models from more accurate polynomials; one-step theory cannot settle this.

REPAIRS AND OPEN PROBLEMS. Better polynomials address the accuracy of the approximation (Amsel et al. 2026), and Gram Newton–Schulz, which runs the same iteration on the Gram matrix $XX^\top$, its cost (Zhang et al. 2026). Norm control addresses the growth of the weights: with decoupled weight decay Muon solves a problem constrained to a spectral-norm ball (Chen et al. 2026; Pethick et al. 2025) (Chapter 13), and Hyperball fixes the Frobenius norms of the weights and the updates, with token-equivalent speedups of 20–30% over weight-decay baselines for Muon up to 1.2B parameters, in a preprint (Wen et al. 2026a). Preconditioners with memory such as SOAP (Vyas et al. 2025) are Chapter 7's subject, and convergence with momentum and noise is Chapter 5's (Kovalev 2025; Li and Hong 2025). Open: a multistep comparison of spectral and Euclidean descent outside the aligned case; why the gradients of trained networks keep a high

nuclear rank while their activations keep a low stable rank, which Davis and Drusvyatskiy (2025) prove for spiked random-feature models after a burn-in; why the advantage of matrix optimizers shrinks with scale; and how accurate, and in what precision, the orthogonalization must be.

3.7 Back in the wild

FIELD NOTES	
PRACTICE	Muon orthogonalizes the momentum of each hidden matrix with five quintic Newton–Schulz steps in bfloat16; memoryless Shampoo computes the same matrix.
IDEALIZATION	Steepest descent on one matrix under the spectral (RMS-to-RMS) norm; a linear layer with frozen inputs for the comparison with gradient descent.
PROVED	The steepest direction is polar(G), with a whole face of them for rank-deficient G; the cubic converges, quadratically after a climb of $O(\log(\kappa\sqrt{r_{\mathrm{st}}(G)}))$ steps; Muon's quintic does not converge and keeps all but the smallest singular values in $[0.681, 1.203]$; spectral beats gradient guarantees iff $r_{\mathrm{nuc}}(G) > r_{\mathrm{st}}(X)$ (when $d_{\mathrm{out}} \geq d_{\mathrm{in}}$).
PREDICTS	Newton–Schulz step counts; the band of Muon's update; the crossover in the aligned case ($\alpha = 0.45$ predicted, 0.39 observed); full-rank RMS-to-RMS steps move every output by exactly $\eta\|x\|_{\mathrm{rms}}$.
MISSES	Momentum and noise, the other layers, many steps outside the aligned case, low precision beyond one step, why gradients keep a high nuclear rank, and why the advantage shrinks with scale.
TAKEAWAY	Orthogonalized steps pay off on layers whose inputs have low stable rank and whose gradients are spread over many directions (proved for one step in the idealization); a band of singular values around one is good enough (folklore, supported by the speed records); at scale, add weight decay and an update-size rule (observed).

What the theory gives a practitioner is a reading of Muon's recipe. The orthogonalization is the duality map of the spectral norm, not an ad hoc normalization; the shape factor converts it to the norm that measures how far a layer's outputs move; and the quintic trades accuracy for speed in a way one plot shows before a single run. The rank criterion says which layers should benefit, at the cost of two singular value decompositions per layer. What the theory does not give is the size of the benefit over a long run, or a reason for its decline with scale.

The rest of the unification line builds on this chapter: Chapter 4 assembles the layers' norms into a norm for the network, starting from Proposition 3.7; Chapter 5 supplies the momentum that makes normalized steps converge; Chapter 7 returns to Shampoo with its memory; and Chapter 13 shows that Muon with weight decay is Frank–Wolfe on a spectral-norm ball.

Exercises

3.1 CONSOLIDATION. (The face of steepest directions.) (a) For $G = e_1 e_1^\top \in \mathbb{R}^{2\times 2}$, show that the maximizers of $\langle G, D\rangle$ over $\|D\|_{\mathrm{op}} \le 1$ are the matrices $\mathrm{diag}(1, w)$ with $|w| \le 1$, and that Newton–Schulz returns $w = 0$. (b) Show that all of them give the same guarantee (3.3), but that for $\mathcal{L}(W) = \frac{1}{2}\|W + e_1 e_1^\top\|_F^2$, whose gradient at $W = 0$ is $e_1 e_1^\top$, the loss after the step $-\eta\,\mathrm{diag}(1, w)$ depends on w. (c) For $m \ge n$, use $\|G - Q\|_F^2 = \|G\|_F^2 + n - 2\langle G, Q\rangle$ to show that the semi-orthogonal Q nearest to G are the points of the face with $Z^\top Z = I$ (Theorem A.8).

3.2 CONSOLIDATION. (The basin of the cubic.) Let $p(s) = \frac{3}{2}s - \frac{1}{2}s^3$. (a) Show that $p^{\circ k}(s) \to 1$ for $s \in (0, \sqrt{3})$, that $p(\sqrt{3}) = 0$, and that $p^{\circ k}(s) \to -1$ for $s \in (\sqrt{3}, s_*)$, where $s_* = 2.1478\ldots$ solves $p(s_*) = -\sqrt{3}$; so $(0, \sqrt{3})$ is the largest interval $(0, c)$ on which the iteration converges to one. (b) Show that $|p(s)| > s$ for $s > \sqrt{5}$, and that the iterates then diverge.

3.3 CONSOLIDATION. (One example.) For one example the gradient of a linear layer is $G = \delta x^\top$. (a) Show that $\mathrm{polar}(G) = \delta x^\top / (\|\delta\|_2 \|x\|_2)$, so that memoryless Muon on one example is normalized gradient descent. (b) For $\Delta W = -\eta\sqrt{m/n}\,\mathrm{polar}(G)$ show that $\|\Delta W x'\|_{\mathrm{rms}} = \eta|\cos\angle(x, x')|\,\|x'\|_{\mathrm{rms}}$.

3.4 CONSOLIDATION. (Designing a quintic.) For $a \in \mathbb{R}$ let $p_a(s) = as + \frac{5-4a}{2}s^3 + \frac{2a-3}{2}s^5$. (a) Show that $p_a(1) = 1$, $p_a'(1) = 0$, $p_a''(1) = 8a - 15$, and that $a = \frac{3}{2}$ gives the cubic. (b) Show that $1 - p_{15/8}(s) = \frac{1}{8}(1-s)^3(8 + 9s + 3s^2)$: cubic convergence, the polynomial Polar Express uses once $[\ell_k, u_k]$ shrinks to a point. (c) Show that no p_a with Muon's slope 3.4445 at zero has a local maximum at one, and that Muon's quintic is not a p_a.

3.5 VARIATION. (Approximate orthogonalization.) Let the update be $X = U_r\,\mathrm{diag}(\tau_i)V_r^\top$ with $0 < \tau_{\min} \le \tau_i \le \tau_{\max}$ instead of $\mathrm{polar}(G)$. (a) Under Assumption 3.4, show that $\mathcal{L}(W - \eta X) \le \mathcal{L}(W) - \eta\tau_{\min}\|G\|_{\mathrm{nuc}} + \frac{1}{2}L_{\mathrm{op}}\eta^2\tau_{\max}^2$, and that the best guaranteed decrease is $(\tau_{\min}/\tau_{\max})^2$ times that of the polar factor. (b) Evaluate it for Muon's band J (0.32), for Polar Express after five steps on $s \ge 10^{-3}$ (0.57), and for five cubic steps on $s \ge 0.3$ (0.99).

3.6 VARIATION. (Momentum and scale.) Show that $\mathrm{polar}(cM) = \mathrm{polar}(M)$ for $c > 0$, that in exact arithmetic Muon's Newton–Schulz output is unchanged when its input is multiplied by c (up to the 10^{-7} of the normalization), and deduce that the moving-average form of the practice box and the summed form of Moonlight's code, $\tilde{M}_t = \beta\tilde{M}_{t-1} + G_t$ with input $G_t + \beta\tilde{M}_t$, give the same updates from a zero start.

3.7 VARIATION. (Weight decay bounds the spectral norm.) Let $W_{t+1} = (1 - \eta\lambda)W_t - \eta\,\mathrm{polar}(G_t)$ with $0 < \eta\lambda \le 1$. Show that $\|W_t\|_{\mathrm{op}} \le 1/\lambda + (1-\eta\lambda)^t \max\{0, \|W_0\|_{\mathrm{op}} - 1/\lambda\}$, and write the update as a convex

combination of W_t and a point of the ball $\|W\|_{\mathrm{op}} \le 1/\lambda$ (Chapter 13).

3.8 SIMULATION. (Rounding fills the null space.) Run five steps of Muon's quintic, as in `03-spectral-ns.py`, in float64 and in bfloat16 on the product G of Gaussian matrices of sizes $256 \times r$ and $r \times 512$, $r \in \{1, 4, 16, 64\}$, and count the output's singular values above 0.1: exactly r in float64, but 52–56, 99–101, 132–134, and 170–173 in bfloat16 in our four runs, the largest spurious one between 0.19 and 0.43. Explain this with the relative rounding error 2^{-8} and Figure 3.1.

3.9 SIMULATION. (Generic targets.) Modify `03-spectral-crossover.py` so that the initial error is $U \operatorname{diag}(i^{-\alpha}) V^\top$ with V Haar-random and independent of Q, and average over five seeds. Check that $r_{\mathrm{nuc}}(G_0)/r_{\mathrm{st}}(X)$ stays above 2 for $\alpha \le 0.5$, that gradient descent needs 17 steps for every α, and that spectral descent needs about 1, 5, 15, 36, and 63 at $\alpha = 0$, 0.2, 0.3, 0.4, and 0.5. Why does α not matter for gradient descent here?

3.10 RESEARCH. (Nuclear rank versus stable rank in trained networks.) Measure in a small transformer the nuclear rank of each hidden layer's minibatch gradient and the stable rank of its inputs along training, and test whether Theorem 3.13 predicts the layers on which Muon helps. Davis and Drusvyatskiy (2025) prove that the ratio grows with dimension in spiked random-feature models; as far as we know, no proof explains why trained deep networks keep a high nuclear rank.

3.11 RESEARCH. (Polynomials for low precision.) Amsel et al. (2026) find the optimal composition of odd polynomials in exact arithmetic. Add a relative rounding error u at every multiplication and find the five quintics that minimize the worst deviation of the output singular values from one on $[\ell, 1]$. As far as we know this is open; Figure 3.1 suggests that the answer is less steep than Polar Express at first.

Notes and References

Steepest descent in the spectral norm entered machine learning with Carlson et al. (2015a), for restricted Boltzmann machines, and Carlson et al. (2015b), with preconditioning for feedforward and convolutional networks. Shampoo is due to Gupta et al. (2018). Bernstein and Newhouse (2024) observed that Adam, Shampoo, and Prodigy are steepest descent under particular norms once their moving averages are switched off, and Bernstein and Newhouse (2025) and Large et al. (2024) built norms and duality maps for whole networks.

Muon was introduced by Jordan et al. (2024) and scaled by Liu et al. (2025) and the Kimi team (Kimi Team 2025). Analyses include Li and Hong (2025), Kovalev (2025), who reads orthogonalized methods as non-Euclidean trust-region methods, Shen et al. (2025), and Pethick et al. (2025), whose methods built on linear minimization oracles over norm balls unify several optimizers and give the Scion optimizer; Chen et al. (2026) show that Muon with decoupled weight decay solves a spectral-norm-constrained problem.

Wen et al. (2026b) benchmark ten optimizers with careful tuning, and Wen et al. (2026a) replace weight decay by fixed norms.

The iteration for the inverse is due to Schulz (1933) and the iterative computation of the best orthogonal approximation to Björck and Bowie (1971); Chapter 8 of Higham (2008) is devoted to the polar decomposition. Amsel et al. (2026) derive optimal polynomial compositions for Muon, and Zhang et al. (2026) reorganize the iteration around the Gram matrix. The rank comparison of Theorem 3.13(ii) is due to Davis and Drusvyatskiy (2025); the exact value of L_{op} when $d_{\mathrm{out}} < \operatorname{rank} A$ (Lemma 3.12), the tightness statements of Theorem 3.13(iii), the aligned reduction of Proposition 3.14, and the band of Proposition 3.10 are worked out here, and we know of no earlier reference for them.

TRANSLATING THE LITERATURE. In Muon's reference code (Jordan 2024) `G` is the Nesterov combination of momentum and gradient, `X` our iterate X_k, and `steps` our k; `zeropower_via_newtonschulz5` approximates our polar(G); `lr` is our η, "in units of spectral norm per update"; and the factor `max(1, update.size(-2) / update.size(-1))**0.5` is $\sqrt{\max(1, d_{\mathrm{out}}/d_{\mathrm{in}})}$ for PyTorch's (out, in) layout of a weight. Gupta et al. (2018) write L_t and R_t for the left and right statistics, as in our field guide. Moonlight's code writes A, B for the two dimensions of a weight and scales the learning rate by $0.2\sqrt{\max(A, B)}$. Davis and Drusvyatskiy (2025) call our nuclear rank the squared nuclear-to-Frobenius ratio of the gradient and compare it with the stable rank of the incoming activations, our $r_{\mathrm{st}}(X)$. The Polar Express code (Amsel et al. 2026) optimizes each polynomial on an interval $[\ell, u]$ of normalized singular values, our $[\ell_k, u_k]$.

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4 Norms for Networks

The modular norm, the spectral condition and μP

IN THE WILD — HYPERPARAMETER TRANSFER ACROSS WIDTH

IN THIS CHAPTER. Hyperparameters tuned on a narrow network transfer to a wide one only if every layer's step is scaled correctly with the widths: in μP, Adam's learning rate for hidden weights is divided by the width, Muon's orthogonalized update carries the factor $\sqrt{d_{\text{out}}/d_{\text{in}}}$, and SGD needs per-layer rates proportional to fan-out over fan-in. We measure features by their root mean square, each layer by its RMS-to-RMS operator norm, and the network by the maximum of its layers' norms. We prove that steepest descent in this norm is per-layer normalized descent; compute exactly how far one step of SGD, of sign descent and of spectral descent moves a layer's features, which yields all three rules; and prove that weights and updates of width-independent norm give feature changes of width-independent size, with a lower bound that excludes cancellation between layers. The theory fixes how learning rates scale, not their values, and its picture of a rank-one gradient aligned with the features holds only early in training.

4.1 In the wild: learning rates that survive widening

No one tunes the learning rate of a network with tens of billions of parameters by trying twenty values: each trial would cost as much as the network. Practitioners tune a network a hundred times smaller and carry the result over. Whether that works depends on the *parametrization*, the choice, for every layer and as a function of its widths, of the scale of its initial weights, of a constant multiplier on its output, and of its learning rate. In the *standard parametrization* (SP) of the deep-learning libraries the initial weights have variance proportional to 1/fan-in, there are no multipliers, and one learning rate serves every layer. Yang and Hu (2021) showed that the standard and the neural-tangent parametrizations admit no infinite-width limit that learns features, and identified the parametrization under which every layer's features move by an amount of order one in each step, at every width: the *maximal update parametrization*, μP. Yang et al. (2021) found that in μP many optimal hyperparameters stay put as the network grows, and turned the observation into a procedure.[1]

IN THE WILD · μTransfer, Yang et al. (2021); Muon, Jordan et al. (2024); Moonlight, Liu et al. (2025). Tune the learning rate and the other

[1] The NeurIPS 2021 paper of Yang et al. (2021) appeared on arXiv in 2022 as *Tensor Programs V*. The fourth paper of the series, Yang and Hu (2021), contains the classification of parametrizations into those with a kernel limit and those that learn features.

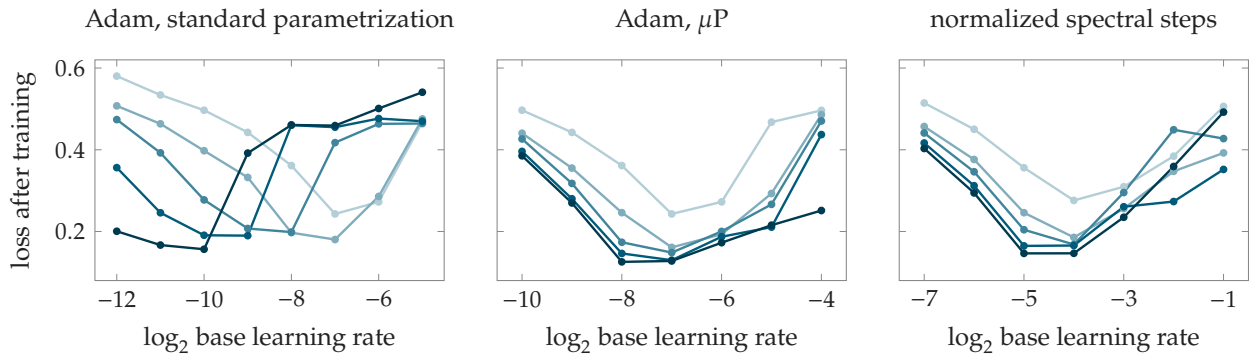


Figure 4.1. Learning-rate transfer in a small network. A ReLU network with 32 inputs, two hidden layers of width 64 (lightest curve), 128, 256, 512 and 1024 (darkest) and one output learns $y = \sum_{j\le 4}((q_j^\top x)^2 - 1)/\sqrt{8}$, q_j orthonormal, from fresh Gaussian minibatches of 32 inputs: 150 steps at a constant learning rate, then the loss on 4096 new inputs. Initial weights have variance 2/fan-in (1/fan-in in the output layer). Left: Adam, one learning rate. Middle: Adam in μP, rates η, $64\eta/d_{\mathrm{h}}$, $64\eta/d_{\mathrm{h}}$ for the input, hidden and output layers, output layer initialized $\sqrt{d_{\mathrm{h}}/64}$ times smaller. Right: every layer takes the step $\eta\,\mathrm{lmo}_{\mathrm{rms}\to\mathrm{rms}}$ of its minibatch gradient (Muon without momentum), μP's initialization. One seed; the same initial weights and data for every learning rate. Script `04-modular-transfer.py`.

> hyperparameters on a narrow proxy, widen the network, keep them. With Adam in μP: initialize input and hidden weights with variance 1/fan-in and the output layer much smaller (variance 1/fan-in^2, or zero); keep the input layer's learning rate; divide the learning rates of hidden and output weights by the width multiplier, the width divided by the proxy's width. Yang et al. transferred from a 13M-parameter proxy to BERT-large, and from a 40M-parameter proxy to a 6.7B-parameter GPT-3 with a tuning cost of about 7% of its pretraining cost. With Muon on the hidden matrices: multiply the orthogonalized update of a $d_{\mathrm{out}} \times d_{\mathrm{in}}$ matrix by $\sqrt{\max(1, d_{\mathrm{out}}/d_{\mathrm{in}})}$ (Jordan et al.), or by $0.2\sqrt{\max(d_{\mathrm{out}}, d_{\mathrm{in}})}$, which gives its entries the RMS of an AdamW update so that AdamW's learning rate and weight decay can be reused (Liu et al.); embeddings, output layer and gains stay on AdamW.

The effect is visible at the scale of a laptop.[2] Figure 4.1 sweeps the learning rate of a network with two hidden layers at five widths, on a task that needs feature learning. In the standard parametrization the curves slide left as the width grows, by about a factor of two per doubling; in μP and under normalized spectral steps they nearly stay put, and the best loss improves with width.

[2] Others use μP too: the Cerebras-GPT report finds that it made hyperparameters more predictable at scale (Dey et al. 2023).

Observation 4.1 (Transfer under μP; Yang et al. 2021, Vyas et al. 2023). Under the standard parametrization the learning rate that minimizes the loss after a fixed amount of training decreases as the width grows. Under μP the minimizer is nearly independent of width, and on simple tasks the loss curves and even the predictions of wide networks of different widths

nearly coincide throughout training.

The observation holds across width for transformers, ResNets and multilayer perceptrons, and its limits are documented: on harder tasks and late in training, networks of different widths drift apart (Vyas et al. 2023), and an unrefereed study of transformers with up to 10 billion parameters found configurations in which transfer fails (Lingle 2024). Nor is μP the only rule that transfers.

OBSERVATION 4.2 (Every parametrization can transfer; Everett et al. 2024). In tens of thousands of models trained with three optimizers and four parametrizations, up to 26.8 billion parameters, every parametrization transferred once each layer received its own learning-rate exponent. For Adam in the standard parametrization Everett et al. recommend an embedding learning rate independent of the width d_{h} and hidden and readout learning rates proportional to $1/d_{\mathrm{h}}$, which outperformed μP in their comparison; Adam's ϵ must be scaled too, or removed (their Adam-atan2).

A recent finding qualifies both: the alignment of a layer's inputs with its updates, on which μP's derivation rests, holds in language-model training only at the start (Kosson et al. 2026). The rules in use look unrelated: one divides Adam's hidden learning rate by the width, one multiplies Muon's update by the square root of an aspect ratio, one leaves SGD's hidden rate alone and scales its input layer up. What do they have in common, why does the default fail, and which quantity is the same at every width?

4.2 The idealization: layers, norms and one step

Assumption 4.3 (A network of layers). The parameters are $\theta = (W_1, \dots, W_N)$ with $W_l \in \mathbb{R}^{d_l \times d_{l-1}}$. An input $x = h_0 \in \mathbb{R}^{d_0}$ produces the pre-activations $u_l = W_l h_{l-1}$, the features $h_l = \phi(u_l)$ for $l < N$, and the output $f = h_N = u_N \in \mathbb{R}^{d_N}$. The activation ϕ acts entrywise, $\phi(0) = 0$, and $|\phi(a) - \phi(b)| \le |a - b|$ for all real a, b (ReLU, tanh, the identity).

The loss of an example $\xi = (x, y)$ is $\ell(\theta; \xi) = \psi(f)$ for a differentiable function ψ of the output, such as $\frac{1}{2}\|f - y\|_2^2$. Backpropagation computes the *backpropagated signals* $\delta_l = \nabla_{u_l} \ell$ by $\delta_N = \nabla\psi(f)$ and $\delta_l = \phi'(u_l) \odot (W_{l+1}^\top \delta_{l+1})$, with $\odot$ the entrywise product, and the gradient of layer l on one example is the outer product

$$G_l = \nabla_{W_l} \ell = \delta_l \, h_{l-1}^\top. \tag{4.1}$$

A minibatch gradient averages such outer products. For one layer we write $d_{\mathrm{in}} = d_{l-1}$ and $d_{\mathrm{out}} = d_l$ (fan-in and fan-out), and d_{h} for a common hidden width.

HOW TO MEASURE. A vector of m entries of order one has *RMS norm* $\|v\|_{\mathrm{rms}} = \|v\|_2/\sqrt{m}$ of order one, so "every activation has a typical size of one" reads $\|h_l\|_{\mathrm{rms}} = \Theta(1)$. Features are measured in the RMS norm, layers in the induced operator norm $\|W\|_{\mathrm{rms}\to\mathrm{rms}} = \max_{v\neq 0} \|Wv\|_{\mathrm{rms}}/\|v\|_{\mathrm{rms}}$ (Chapter 3), and backpropagated signals, which pair with pre-activations, in the dual norm $\|v\|_{\mathrm{rms},*} = \max_{\|z\|_{\mathrm{rms}}\le 1} \langle v, z\rangle = \sqrt{m}\,\|v\|_2$. A quantity is

width-independent if it is bounded above and below by constants independent of $d_1, \dots, d_{N-1}$, with the depth, d_0, d_N, the data and the base step size fixed (step size and learning rate are synonyms here: the first in the theory, the second in practice).

Assumption 4.4 (One step from one example). The step $\Delta\theta = (\Delta W_1, \dots, \Delta W_N)$ is computed from the loss of one example, so that every layer gradient is the rank-one matrix (4.1), and it is evaluated on the same input: Δu_l and Δh_l are the changes of u_l and h_l when θ becomes $\theta + \Delta\theta$.

Definition 4.5 (The feature-learning desideratum; Yang et al. 2023). A rule that sets initial weights and steps as functions of the widths *learns features* if $\|h_l\|_{\text{rms}}$ and $\|\Delta h_l\|_{\text{rms}}$ are width-independent for every hidden layer, and $\|\Delta f\|_{\text{rms}}$ is bounded independently of the widths.

A rule under which $\|\Delta h_l\|_{\text{rms}} \to 0$ as the widths grow is *lazy*: the network trains like its linearization at initialization, a kernel method.[3] A rule under which $\|\Delta h_l\|_{\text{rms}}$ or $\|\Delta f\|_{\text{rms}}$ grows without bound is unstable at large width unless its learning rate shrinks.

[3] The kernel is the neural tangent kernel of Jacot et al. (2018); Chizat et al. (2019) showed that laziness is a consequence of a scaling choice.

DICTIONARY

IN THE THEORY	IN A TRAINING RUN
the RMS norm $\|h\|_{\text{rms}}$	the typical size of one activation
$\|W\|_{\text{rms}\to\text{rms}} = \sqrt{d_{\text{in}}/d_{\text{out}}}\,\|W\|_{\text{op}}$	the largest gain of a layer on its activations
$\|\delta\|_{\text{rms},*} = \sqrt{m}\,\|\delta\|_2$	the gradient with respect to a layer's pre-activations, whose entries shrink like 1/width
the scales s_l of the max norm	per-layer learning-rate multipliers (μP's rates, Modula's masses)
one step from one example	the first steps of training, when gradients are low-rank and aligned with the activations
$\|\Delta h_l\|_{\text{rms}} = \Theta(1)$ at every width	feature learning, as opposed to lazy (kernel) training

WHAT WAS THROWN AWAY. Attention, normalization, embeddings, residual connections, biases, weight decay, the optimizer's memory, and all examples but one. What is kept is the question of scale, and each simplification is the right first one: one example makes every layer gradient rank one, one step separates the learning rate from the history of training, and a question about exponents is answered by a calculation exact up to constants. Early in training gradients are in fact low-rank and aligned with the features (Kosson et al. 2026).

WHAT ONE HAD TO NOTICE. First, that an update must be measured by its effect on the features, an operator norm, not by the size of its entries: a

rank-one matrix with entries $\pm\eta$ has operator norm $\eta\sqrt{d_{\text{out}}d_{\text{in}}}$, a random one of order $\eta(\sqrt{d_{\text{out}}}+\sqrt{d_{\text{in}}})$, and an entrywise rule cannot tell them apart (Yang et al. 2023). Second, that the network can then carry the maximum of the layer norms, and steepest descent in that one norm gives each layer its own normalized step: per-layer learning rates are the duality map of a norm (Bernstein and Newhouse 2025; Large et al. 2024).

FIELD GUIDE · μP ADAM	*Yang and Hu (2021), Yang et al. (2021)*
UPDATE	Adam on every layer; learning rate η for the input layer, $\eta\, d^\circ/d_{\text{in}}$ for hidden and output weights, d° the proxy's width
NORM	with $\beta_1=\beta_2=0$ (sign descent, Chapter 2), normalized steepest descent in $\max_l s_l \max_{ij}\lvert(W_l)_{ij}\rvert$ with $s_1=1$ and $s_l=d_{\text{in}}/d^\circ$ for hidden and output layers (Theorem 4.8, Exercise 4.2)
PRECONDITIONER	diagonal, Adam's (Chapter 6)
AVERAGES	Adam's β_1, β_2
STATE	two numbers per parameter
IN THE WILD	variance $1/\text{fan-in}$, output layer near zero; the `mup` package divides the hidden learning rate by the width multiplier

FIELD GUIDE · MODULAR NORMALIZED DESCENT	*Large et al. (2024)*
UPDATE	$W_l \leftarrow W_l + (\eta/s_l)\,\text{lmo}_{(l)}(M_l)$, M_l the gradient or its momentum; for a linear layer $\text{lmo}_{\text{rms}\to\text{rms}}(M) = -\sqrt{d_{\text{out}}/d_{\text{in}}}\ \text{polar}(M)$
NORM	the modular norm, a scaled maximum of module norms (Definition 4.7)
PRECONDITIONER	none
AVERAGES	optional momentum
STATE	one number per parameter with momentum
IN THE WILD	the Modula library, scales from user-assigned masses; on one matrix with Newton–Schulz for the polar factor it is Muon (Chapter 3)

4.3 The theory: the norm of a network

The RMS geometry of one layer is the spectral geometry of Chapter 3, rescaled.

Lemma 4.6 (RMS geometry of a layer). *Let $W, G \in \mathbb{R}^{d_{\text{out}}\times d_{\text{in}}}$ and $v \in \mathbb{R}^{d_{\text{out}}}$.*

(i) $\|W\|_{\text{rms}\to\text{rms}} = \sqrt{d_{\text{in}}/d_{\text{out}}}\,\|W\|_{\text{op}}$.

(ii) The dual norm of $\|\cdot\|_{\text{rms}\to\text{rms}}$ under $\langle G, W\rangle = \text{tr}(G^\top W)$ is $\|G\|_{\text{rms}\to\text{rms},*} = \sqrt{d_{\text{out}}/d_{\text{in}}}\,\|G\|_{\text{nuc}}$, and $-\sqrt{d_{\text{out}}/d_{\text{in}}}\,\text{polar}(G)$ is a

steepest unit direction, $\mathrm{lmo}_{\mathrm{rms}\to\mathrm{rms}}(G)$, *if* $G \neq 0$.

(iii) $\|W^\top v\|_{\mathrm{rms},*} \le \|W\|_{\mathrm{rms}\to\mathrm{rms}}\|v\|_{\mathrm{rms},*}$.

(iv) If $G = ab^\top$ *with* $a, b \neq 0$, *then* $\mathrm{polar}(G) = \frac{a}{\|a\|_2}\frac{b^\top}{\|b\|_2}$, $\|G\|_{\mathrm{rms}\to\mathrm{rms}} = \|a\|_{\mathrm{rms}}\|b\|_{\mathrm{rms},*}$, $\|G\|_{\mathrm{rms}\to\mathrm{rms},*} = \|a\|_{\mathrm{rms},*}\|b\|_{\mathrm{rms}}$, *and* $\|Gb\|_{\mathrm{rms}} = \|G\|_{\mathrm{rms}\to\mathrm{rms}}\|b\|_{\mathrm{rms}}$.

PROOF. The plan is to reduce every claim to the spectral norm by rescaling. (i) $\|Wv\|_{\mathrm{rms}}/\|v\|_{\mathrm{rms}} = \sqrt{d_{\mathrm{in}}/d_{\mathrm{out}}}\,\|Wv\|_2/\|v\|_2$. (ii) For a norm $\|\cdot\|$ and $c > 0$, $\max_{c\|U\|\le 1}\langle G, U\rangle = c^{-1}\max_{\|U'\|\le 1}\langle G, U'\rangle$, so the dual of $c\|\cdot\|$ is $c^{-1}\|\cdot\|_*$ and, by the substitution $U = U'/c$, its lmo is c^{-1} times that of $\|\cdot\|$; the dual of the spectral norm is the nuclear norm and $-\mathrm{polar}(G)$ is a steepest unit direction of the spectral norm (Theorem A.7, Chapter 3). (iii) $\sqrt{d_{\mathrm{in}}}\|W^\top v\|_2 \le \sqrt{d_{\mathrm{in}}/d_{\mathrm{out}}}\,\|W\|_{\mathrm{op}}\cdot\sqrt{d_{\mathrm{out}}}\|v\|_2$. (iv) The compact singular value decomposition of $ab^\top$ has the single singular value $\|a\|_2\|b\|_2$ with singular vectors $a/\|a\|_2$ and $b/\|b\|_2$, so (i) and (ii) give the norms, and $Gb = \|b\|_2^2\, a$ has RMS norm $\|a\|_{\mathrm{rms}}\|b\|_{\mathrm{rms},*}\|b\|_{\mathrm{rms}}$. ■

Part (iii) says that the backward pass contracts dual norms by the factor by which the forward pass contracts RMS norms.

4.3.1 Steepest descent in a maximum of layer norms

Definition 4.7 (The max norm of a network). Give each layer a norm $\|\cdot\|_{(l)}$ on $\mathbb{R}^{d_l\times d_{l-1}}$, with dual norm $\|\cdot\|_{(l),*}$, and a scale $s_l > 0$. The *max norm* of the network is $\|\theta\| = \max_l s_l\|W_l\|_{(l)}$, with the inner product $\langle g, \theta\rangle = \sum_l \mathrm{tr}(G_l^\top W_l)$.

For a sequential network, the *modular norm* of Large et al. (2024) has this form (Bernstein and Newhouse 2024): the layer norms are operator norms chosen per kind of module (RMS→RMS for a linear layer), and the scales come from two numbers attached to every module, a mass, which sets its share of the feature learning, and a sensitivity to its input. The max norm is a norm because each layer norm is and a maximum of sums is at most the sum of the maxima.

Theorem 4.8 (Steepest descent in the norm of a network). *For the max norm of Definition 4.7 and every* $g = (G_1, \ldots, G_N)$:

(i) $\|g\|_* = \sum_{l=1}^N \|G_l\|_{(l),*}/s_l$;

(ii) $(U_1, \ldots, U_N)$ *is a steepest unit direction for* g *if and only if every* $s_l U_l$ *is one for* G_l *and* $\|\cdot\|_{(l)}$; *so* $\mathrm{lmo}(g) = (\mathrm{lmo}_{(l)}(G_l)/s_l)_l$;

(iii) normalized steepest descent $\theta_{t+1} = \theta_t + \eta\,\mathrm{lmo}(g_t)$ *is the layerwise update* $W_{l,t+1} = W_{l,t} + (\eta/s_l)\,\mathrm{lmo}_{(l)}(G_{l,t})$, *normalized descent in every layer with the per-layer learning rate* η/s_l;

(iv) unnormalized steepest descent $\theta_{t+1} = \theta_t - \eta g_t^\sharp$ *takes the same directions with every layer's step multiplied by one global factor, the dual norm of*

the whole gradient:

$$W_{l,t+1} = W_{l,t} + \frac{\eta}{s_l}\|g_t\|_* \operatorname{lmo}_{(l)}(G_{l,t}).$$

PROOF. The plan is to note that the unit ball is a product of layer balls and $\langle g, \cdot \rangle$ a sum of layer terms, so that minimization splits into one problem per layer. The unit ball is the product of the balls $\mathcal{B}_l = \{U : \|U\|_{(l)} \le 1/s_l\}$. (i) The maximum of $\sum_l \langle G_l, U_l \rangle$ over a product of sets is the sum of the maxima, and $\max_{U \in \mathcal{B}_l} \langle G_l, U \rangle = \|G_l\|_{(l),*}/s_l$ by the definition of the dual norm and homogeneity. (ii) If every U_l minimizes $\langle G_l, \cdot \rangle$ over $\mathcal{B}_l$, then u minimizes the sum. If some U_k does not, then replacing it by a minimizer, which exists because $\mathcal{B}_k$ is compact, strictly decreases the sum. The minimizers over $\mathcal{B}_l$ are $1/s_l$ times those over the unit ball of $\|\cdot\|_{(l)}$. (iii) is (ii) times η. (iv) By Lemma A.5, $g^\sharp = -\|g\|_* \operatorname{lmo}(g)$ for every norm; insert (ii). ∎

If $G_l = 0$, every point of the layer's ball is a minimizer and implementations take $U_l = 0$; if $g = 0$, both updates vanish.[4] The theorem turns the question of how each layer's learning rate should depend on the widths into the question of which norm each layer should carry: if the layer norms measure the effect of the weights on the features in width-independent units, the scales can be constants and one η serves every width. With smoothness in the max norm, the descent bound of Chapter 1 reads $\mathcal{L}(\theta + \eta \operatorname{lmo}(g)) \le \mathcal{L}(\theta) - \eta \sum_l \|G_l\|_{(l),*}/s_l + \frac{L}{2}\eta^2$: the guaranteed decreases of the layers add up.

[4] LARS (You et al. 2017), which trained ResNet-50 at batch size 32K, rescales each layer's step to a fixed fraction of the layer's Frobenius norm. By (iii), LARS without momentum and weight decay is normalized steepest descent in a maximum of Frobenius norms with the state-dependent scales $s_l = 1/\|W_l\|_F$.

WHICH COMBINATION OF LAYER NORMS? The maximum is a modeling choice. The sum $\sum_l s_l \|W_l\|_{(l)}$ has the dual norm $\max_l \|G_l\|_{(l),*}/s_l$, and its lmo puts the whole step on the layers that attain the maximum (Exercise 4.1): steepest descent in a sum of layer norms is greedy block-coordinate descent.[5] The maximum is the combination in which one step moves every layer by its full share, which is what the desideratum asks.

[5] It is close in spirit to the duality-structure gradient descent of Flynn (2017), which updates one greedily chosen layer per step and measures progress in layerwise norms.

4.3.2 One step of three optimizers on one layer

Which layer norms measure the effect of a step in width-independent units? The answer comes from computing what one step of each optimizer does to a layer's output on the example that produced the step.

Theorem 4.9 (One step of three optimizers on one layer). *Let a layer $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ have input $x \neq 0$ and backpropagated signal $\delta \neq 0$ on one example, so that its gradient is $G = \delta x^\top$, and let $\eta > 0$. For nonzero $v \in \mathbb{R}^m$ write $\rho(v) = \|v\|_1/(\sqrt{m}\,\|v\|_2) \in (0, 1]$ for the density of v and $\sigma(v) = \|\operatorname{sign} v\|_{\text{rms}}$ for the square root of the fraction of nonzero entries of v.*

(i) SGD, $\Delta W = -\eta G$: $\Delta W x = -\eta \|x\|_2^2\, \delta$, so $\|\Delta W x\|_{\text{rms}} = \eta\, d_{\text{in}} \|\delta\|_{\text{rms}} \|x\|_{\text{rms}}^2$, and $\|\Delta W\|_{\text{rms}\to\text{rms}} = \eta\, d_{\text{in}} \|\delta\|_{\text{rms}} \|x\|_{\text{rms}}$.

(ii) Sign descent, $\Delta W = -\eta \operatorname{sign}(G)$: $\Delta W x = -\eta \|x\|_1 \operatorname{sign}(\delta)$, so $\|\Delta W x\|_{\text{rms}} = \eta\, d_{\text{in}}\, \rho(x)\, \sigma(\delta) \|x\|_{\text{rms}}$, and $\|\Delta W\|_{\text{rms}\to\text{rms}} =$

$\eta\, d_{\text{in}}\, \sigma(\delta)\, \sigma(x)$.

(iii) Normalized spectral step, $\Delta W = \eta\, \text{lmo}_{\text{rms}\to\text{rms}}(G)$*:* $\Delta W x = -\eta\sqrt{d_{\text{out}}}\, \|x\|_{\text{rms}}\, \delta/\|\delta\|_2$*, so* $\|\Delta W x\|_{\text{rms}} = \eta\|x\|_{\text{rms}}$*, and* $\|\Delta W\|_{\text{rms}\to\text{rms}} = \eta$.

Hence the step sizes that move the output on x *by* $c\|x\|_{\text{rms}}$ *in RMS are*

$$\eta_{\text{SGD}} = \frac{c}{d_{\text{in}}\|\delta\|_{\text{rms}}\|x\|_{\text{rms}}}, \qquad \eta_{\text{sign}} = \frac{c}{d_{\text{in}}\,\rho(x)\,\sigma(\delta)}, \qquad \eta_{\text{spec}} = c. \tag{4.2}$$

Proof. The plan is to compute each $\Delta W x$ directly and to read each operator norm off Lemma 4.6(iv), since every update is a rank-one matrix. (i) $\Delta W x = -\eta\,\delta\,(x^\top x)$, whose RMS norm is $\eta\|x\|_2^2\|\delta\|_{\text{rms}}$, and $\|x\|_2^2 = d_{\text{in}}\|x\|_{\text{rms}}^2$. The lemma gives $\|\Delta W\|_{\text{rms}\to\text{rms}} = \eta\|\delta\|_{\text{rms}}\|x\|_{\text{rms},*}$, and $\|x\|_{\text{rms},*} = d_{\text{in}}\|x\|_{\text{rms}}$. (ii) Since $\text{sign}(\delta_i x_j) = \text{sign}(\delta_i)\,\text{sign}(x_j)$, also when a factor vanishes, $\text{sign}(G) = \text{sign}(\delta)\,\text{sign}(x)^\top$; and $\text{sign}(x)^\top x = \|x\|_1 = \rho(x)\, d_{\text{in}}\|x\|_{\text{rms}}$. So $\Delta W x = -\eta\|x\|_1\,\text{sign}(\delta)$, whose RMS norm is $\eta\|x\|_1\sigma(\delta)$, and

$$\|\Delta W\|_{\text{rms}\to\text{rms}} = \eta\|\text{sign}\,\delta\|_{\text{rms}}\|\text{sign}\,x\|_{\text{rms},*} = \eta\,\sigma(\delta)\,d_{\text{in}}\sigma(x).$$

(iii) By Lemma 4.6(ii) and (iv), $\Delta W = -\eta\sqrt{d_{\text{out}}/d_{\text{in}}}\,\frac{\delta}{\|\delta\|_2}\frac{x^\top}{\|x\|_2}$; applied to x this gives the stated vector, of RMS norm $\eta\|x\|_{\text{rms}}$, and the polar factor has operator norm one. Solving each identity for η gives (4.2). ∎

The three rules depend on different things, and that is the chapter in one display. SGD's step depends on the size of the backpropagated signal, which the layers above set, and on the size of the input, squared. Sign descent is blind to the sizes of δ and x, but its step grows with the fan-in in proportion to the density of the input: a dense input (ρ of order one, as for Gaussian or ReLU features) moves by order $\eta\, d_{\text{in}}\|x\|_{\text{rms}}$, a one-hot input by at most $\eta\sqrt{d_{\text{in}}}\|x\|_{\text{rms}}$ (Exercise 4.3). The spectral step depends on nothing. In each case $\|\Delta W x\|_{\text{rms}} = \|\Delta W\|_{\text{rms}\to\text{rms}}\|x\|_{\text{rms}}$, up to the factor $\rho(x)/\sigma(x)$ for sign descent: a single-example update is aligned with the input that produced it, which is why its RMS→RMS norm, and not the size of its entries, is what a width rule must control.

4.4 The theory: the spectral condition

A network adds two questions to Theorem 4.9: whether the steps of different layers can cancel each other's effect on the features, and how large the backpropagated signal is. The first has a clean answer.

Lemma 4.10 (No cancellation between layers). *Under Assumption 4.3, fix one example and a direction* $\Delta\theta$*, and suppose that* ϕ *is differentiable at every entry of* $u_1, \dots, u_{N-1}$*. The first-order changes* $\dot{u}_l = \frac{\mathrm{d}}{\mathrm{d}\epsilon} u_l(\theta + \epsilon\Delta\theta)\big|_{\epsilon=0}$ *satisfy*

$$\langle \delta_l, \dot{u}_l \rangle = \sum_{k=1}^{l} \langle G_k, \Delta W_k \rangle \qquad (l = 1, \dots, N). \tag{4.3}$$

Consequently, if $\langle G_k, \Delta W_k\rangle \le 0$ *for all* $k \le l$ *and* $\delta_l \neq 0$, *then*

$$\|\dot{u}_l\|_2 \;\ge\; \frac{1}{\|\delta_l\|_2}\sum_{k=1}^{l}|\langle G_k, \Delta W_k\rangle|.$$

Proof. The plan is an induction along the forward pass. Differentiating $u_l = W_l h_{l-1}$ and $h_{l-1} = \phi(u_{l-1})$ gives $\dot{u}_l = \Delta W_l\, h_{l-1} + W_l \dot{h}_{l-1}$ and $\dot{h}_{l-1} = \phi'(u_{l-1}) \odot \dot{u}_{l-1}$, with $\dot{h}_0 = 0$. The first term pairs with δ_l to $\operatorname{tr}(\Delta W_l^\top \delta_l h_{l-1}^\top) = \langle G_l, \Delta W_l\rangle$. In the second, the transpose and the derivative move across the pairing as in the backward recursion:

$$\langle \delta_l, W_l \dot{h}_{l-1}\rangle = \langle \phi'(u_{l-1}) \odot W_l^\top \delta_l,\; \dot{u}_{l-1}\rangle = \langle \delta_{l-1}, \dot{u}_{l-1}\rangle,$$

which vanishes for $l = 1$. Induction on l proves (4.3). If its terms are nonpositive, then $|\langle \delta_l, \dot{u}_l\rangle| = \sum_{k\le l}|\langle G_k, \Delta W_k\rangle|$, and the Cauchy–Schwarz inequality gives the bound. ∎

The projection of a layer's first-order feature change on its backpropagated signal is the first-order loss change caused by the layers up to it. If every layer descends, then these contributions have one sign, and the lower layers cannot undo a layer's own step in the direction $-\delta_l$. Keeping only $k = l$, with $\hat{\delta}_l = \delta_l/\|\delta_l\|_2$,

$$\|\dot{u}_l\|_2 \;\ge\; \langle -\hat{\delta}_l,\; \Delta W_l\, h_{l-1}\rangle, \tag{4.4}$$

the projection of the layer's direct change on $-\hat{\delta}_l$, which for SGD and for spectral steps is the full length of the direct change (Theorem 4.9), and for sign descent its length times $\rho(\delta_l)/\sigma(\delta_l)$.

Theorem 4.11 (The spectral condition). *Under Assumption 4.3, let* $\|W_l\|_{\mathrm{rms}\to\mathrm{rms}} \le a_l$ *and* $\|\Delta W_l\|_{\mathrm{rms}\to\mathrm{rms}} \le b_l$ *for every* l, *and put* $A_l = \prod_{k\le l} a_k$, $B_l = \prod_{k\le l}(a_k + b_k)$ *and* $C_l = \sum_{k\le l} b_k \prod_{j\le l,\, j\neq k} a_j$, *with* $A_0 = B_0 = 1$. *For every input* x *and every* l:

(i) (forward) $\|h_l\|_{\mathrm{rms}} \le \|u_l\|_{\mathrm{rms}} \le A_l\|x\|_{\mathrm{rms}}$;

(ii) (update) $\|\Delta h_l\|_{\mathrm{rms}} \le \|\Delta u_l\|_{\mathrm{rms}} \le (B_l - A_l)\|x\|_{\mathrm{rms}}$;

(iii) (lower bound) if ϕ *is differentiable at the entries of* $u_1, \dots, u_{N-1}$ *and every layer takes the normalized step* $\Delta W_k = \eta_k \operatorname{lmo}_{\mathrm{rms}\to\mathrm{rms}}(G_k)$ *of the loss of one example with input* x, *then* $\|\dot{u}_l\|_{\mathrm{rms}} \ge \eta_l\|h_{l-1}\|_{\mathrm{rms}}$ *whenever* $\delta_l \neq 0$;

(iv) (linear networks) if ϕ *is the identity,* $\|\Delta u_l - \dot{u}_l\|_{\mathrm{rms}} \le (B_l - A_l - C_l)\|x\|_{\mathrm{rms}}$; *if moreover* $a_k \le 1$, *the steps of (iii) have* $\eta_k = \eta$ *and* $\delta_l \neq 0$, *then*

$$\begin{aligned} &\|\Delta h_l\|_{\mathrm{rms}} \le \big((1+\eta)^l - 1\big)\|x\|_{\mathrm{rms}}, \\ &\|\Delta h_l\|_{\mathrm{rms}} \ge \eta\|h_{l-1}\|_{\mathrm{rms}} - \big((1+\eta)^l - 1 - l\eta\big)\|x\|_{\mathrm{rms}}. \end{aligned} \tag{4.5}$$

Proof. The plan is to propagate norm bounds through the layers, using that ϕ does not increase distances, and to take the lower bound from Lemma 4.10.

(i) Since $\phi(0) = 0$ and ϕ is 1-Lipschitz entrywise, $\|\phi(u) - \phi(u')\|_{\mathrm{rms}} \le \|u - u'\|_{\mathrm{rms}}$ and $\|\phi(u)\|_{\mathrm{rms}} \le \|u\|_{\mathrm{rms}}$. With $\|u_l\|_{\mathrm{rms}} \le a_l \|h_{l-1}\|_{\mathrm{rms}}$, induction on l gives (i).

(ii) The changed network has layer norms at most $a_k + b_k$, so (i) gives $\|h_{l-1} + \Delta h_{l-1}\|_{\mathrm{rms}} \le B_{l-1}\|x\|_{\mathrm{rms}}$. Decompose

$$\Delta u_l = (W_l + \Delta W_l)(h_{l-1} + \Delta h_{l-1}) - W_l h_{l-1} = W_l\, \Delta h_{l-1} + \Delta W_l\, (h_{l-1} + \Delta h_{l-1}).$$

Assume $\|\Delta h_{l-1}\|_{\mathrm{rms}} \le (B_{l-1} - A_{l-1})\|x\|_{\mathrm{rms}}$, true for $l = 1$ since $\Delta h_0 = 0$. The two terms then give $\|\Delta u_l\|_{\mathrm{rms}} \le \big(a_l(B_{l-1} - A_{l-1}) + b_l B_{l-1}\big)\|x\|_{\mathrm{rms}} = (B_l - A_l)\|x\|_{\mathrm{rms}}$, and $\|\Delta h_l\|_{\mathrm{rms}} \le \|\Delta u_l\|_{\mathrm{rms}}$ because ϕ is 1-Lipschitz.

(iii) Every layer descends, since by Lemma 4.6(ii)

$$\langle G_k, \Delta W_k \rangle = -\eta_k \|G_k\|_{\mathrm{rms}\to\mathrm{rms},*} \le 0.$$

By part (iv) of that lemma applied to $G_l = \delta_l h_{l-1}^\top$, the term $k = l$ has absolute value $\eta_l \|\delta_l\|_{\mathrm{rms},*} \|h_{l-1}\|_{\mathrm{rms}}$, which is $\eta_l \sqrt{d_l}\, \|\delta_l\|_2 \|h_{l-1}\|_{\mathrm{rms}}$. Keeping only this term in Lemma 4.10 gives $\|\dot{u}_l\|_2 \ge \eta_l \sqrt{d_l}\, \|h_{l-1}\|_{\mathrm{rms}}$; divide by $\sqrt{d_l}$.

(iv) Expand $u_l + \Delta u_l = (W_l + \Delta W_l) \cdots (W_1 + \Delta W_1)x$ into one term for each set $S \subseteq \{1, \dots, l\}$ of layers that contribute ΔW_k. The empty set gives u_l, the singletons give $\dot{u}_l$, and the term of S has RMS norm at most $\prod_{k\in S} b_k \prod_{k\notin S} a_k \|x\|_{\mathrm{rms}}$. These bounds sum to B_l over all S, to A_l for $S = \emptyset$ and to C_l over the singletons, which proves the first claim. All three bounds are polynomials with nonnegative coefficients in the a_k, so with $a_k \le 1$ and $b_k = \eta$, $B_l - A_l \le (1+\eta)^l - 1$ and $B_l - A_l - C_l \le (1+\eta)^l - 1 - l\eta$. The first line of (4.5) follows from (ii), the second from (iii) and the triangle inequality, because $\Delta h_l = \Delta u_l$. ∎

In operator norms, $a_l, b_l = \Theta(1)$ reads $\|W_l\|_{\mathrm{op}}, \|\Delta W_l\|_{\mathrm{op}} = \Theta(\sqrt{d_l/d_{l-1}})$, the *spectral condition* of Yang et al. (2023). Under it the forward and update bounds do not involve the widths, and under normalized RMS→RMS steps neither does the lower bound. Under the hypotheses of (iv), suppose moreover that the forward pass does not collapse, $\|h_{l-1}\|_{\mathrm{rms}} \ge c\|x\|_{\mathrm{rms}}$. Then Taylor's bound $(1+\eta)^l - 1 - l\eta \le \binom{l}{2}\eta^2(1+\eta)^{l-2}$ turns (4.5) into

$$\Big(c\eta - \tbinom{l}{2}\eta^2(1+\eta)^{l-2}\Big)\|x\|_{\mathrm{rms}} \le \|\Delta h_l\|_{\mathrm{rms}} \le \big((1+\eta)^l - 1\big)\|x\|_{\mathrm{rms}}, \qquad (4.6)$$

feature learning at every width: no width appears. The upper bound is attained to first order: if the layers are square and orthogonal, all l contributions to $\dot{u}_l$ point along $-\hat{\delta}_l$ with RMS norm $\eta\|x\|_{\mathrm{rms}}$, so $\|\dot{u}_l\|_{\mathrm{rms}} = l\eta\|x\|_{\mathrm{rms}}$ (Exercise 4.4). The lower bound is attained at $l = 1$.

4.4.1 The backward pass and the size of the signal

SGD's rule (4.2) needs $\|\delta_l\|_{\mathrm{rms}}$, which the same operator norms control.

Proposition 4.12 (The backward pass in dual norms). *Under Assumption 4.3, let ϕ be differentiable at the entries of $u_1, \dots, u_{N-1}$ and*

$\|W_k\|_{\mathrm{rms}\to\mathrm{rms}} \le a_k$ for every k. For one example and every l,

$$\|\delta_l\|_{\mathrm{rms},*} \le a_{l+1}\cdots a_N \,\|\nabla\psi(f)\|_{\mathrm{rms},*},$$
$$\|G_l\|_{\mathrm{rms}\to\mathrm{rms},*} \le \Big(\prod_{k\neq l} a_k\Big)\|\nabla\psi(f)\|_{\mathrm{rms},*}\|x\|_{\mathrm{rms}},$$

and in the max norm with RMS→RMS layer norms, $\|\nabla_\theta \ell\|_* \le \sum_l s_l^{-1}\prod_{k\neq l} a_k \cdot \|\nabla\psi(f)\|_{\mathrm{rms},*}\|x\|_{\mathrm{rms}}$. For an average over examples the last two bounds hold with the right sides averaged.

Proof. The plan is to run Lemma 4.6(iii) down the backward pass. Since $|\phi'| \le 1$, $\|\delta_l\|_{\mathrm{rms},*} \le \|W_{l+1}^\top \delta_{l+1}\|_{\mathrm{rms},*} \le a_{l+1}\|\delta_{l+1}\|_{\mathrm{rms},*}$; iterate from $\delta_N = \nabla\psi(f)$. By Lemma 4.6(iv), $\|G_l\|_{\mathrm{rms}\to\mathrm{rms},*} = \|\delta_l\|_{\mathrm{rms},*}\|h_{l-1}\|_{\mathrm{rms}}$; bound the second factor by Theorem 4.11(i). The last bound is Theorem 4.8(i), and for several examples the triangle inequality for the dual norms gives the averaged bounds. ∎

For the square loss in RMS units, $\psi(f) = \frac{1}{2}\|f - y\|_{\mathrm{rms}}^2$, the factor $\|\nabla\psi(f)\|_{\mathrm{rms},*}$ is the RMS residual. Under the spectral condition every backpropagated signal therefore has entries of order at most $1/d_l$, since $\|\delta\|_{\mathrm{rms}} = \|\delta\|_{\mathrm{rms},*}/d_l$, and the loss is Lipschitz in the max norm with a width-independent constant, which is the Lipschitz half of the analysis of Large et al. (2024) for this network. That these upper bounds are attained at a random initialization is a heuristic about random matrices, and it is what the width rules need.

Heuristic derivation of the width exponents. Fix the depth, d_0, d_N, an example with $\|x\|_{\mathrm{rms}} = 1$ and $\|\nabla\psi(f)\|_{\mathrm{rms},*}$ of order one, let the hidden widths be $d_{\mathrm{h}} \to \infty$, and let the initial weights of a layer be independent $\mathcal{N}(0, v^2)$. For vectors independent of such a W, the law of large numbers gives the forward gain $\|Wz\|_{\mathrm{rms}}/\|z\|_{\mathrm{rms}} \approx v\sqrt{d_{\mathrm{in}}}$ and the backward gain $\|W^\top z'\|_{\mathrm{rms},*}/\|z'\|_{\mathrm{rms},*} \approx v\, d_{\mathrm{in}}/\sqrt{d_{\mathrm{out}}}$, while $\|W\|_{\mathrm{rms}\to\mathrm{rms}} \approx v\sqrt{d_{\mathrm{in}}}\,(1 + \sqrt{d_{\mathrm{in}}/d_{\mathrm{out}}})$: the forward gain is within a constant of the norm if $d_{\mathrm{in}} \le d_{\mathrm{out}}$ (input and hidden layers, which the forward pass traverses), the backward gain if $d_{\mathrm{in}} \ge d_{\mathrm{out}}$ (hidden and output layers, which the backward pass traverses). So the bounds of Theorem 4.11(i) and Proposition 4.12 are attained up to constants.

If every layer has $\|W_l\|_{\mathrm{rms}\to\mathrm{rms}} = \Theta(1)$, which for the output layer means entries of size $1/d_{\mathrm{h}}$ (variance $1/\text{fan-in}^2$, μP's choice), then $\|h_l\|_{\mathrm{rms}} = \Theta(1)$ and $\|\delta_l\|_{\mathrm{rms}} = \Theta(1/d_{\mathrm{h}})$ for $l < N$. In the standard parametrization the output layer has entries of size $d_{\mathrm{h}}^{-1/2}$, so $\|W_N\|_{\mathrm{rms}\to\mathrm{rms}}$ and the backward gain through it are $\Theta(d_{\mathrm{h}}^{1/2})$, and $\|\delta_l\|_{\mathrm{rms}} = \Theta(d_{\mathrm{h}}^{-1/2})$ for $l < N$. Inserting these sizes and the densities $\rho, \sigma = \Theta(1)$ into Theorem 4.9 gives the exponents of Table 4.1. Under one learning rate the output layer's own change is $\Theta(\eta d_{\mathrm{h}})$ for SGD and sign descent alike, so a stable output requires $\eta = O(1/d_{\mathrm{h}})$, and then the input and hidden layers of SGD, and the input layer of sign descent, stop learning as the width grows. For sign descent this is not

Table 4.1. Heuristic width exponents in a network with an input layer $d_{\mathrm{h}} \times d_0$, hidden layers $d_{\mathrm{h}} \times d_{\mathrm{h}}$ and an output layer $d_N \times d_{\mathrm{h}}$, d_0 and d_N fixed, derived from Theorem 4.9 and the heuristic sizes of δ_l. Left: the exponent p of the per-layer step size $\eta_l \propto d_{\mathrm{h}}^p$ that moves the layer's output by $\Theta(1)$ when every $\|W_l\|_{\mathrm{rms}\to\mathrm{rms}} = \Theta(1)$, the general rules being d_l/d_{l-1} for SGD, $1/d_{l-1}$ for sign descent and 1 for normalized spectral steps. Right: the exponent q of a layer's own change $\|\Delta W_l h_{l-1}\|_{\mathrm{rms}} = \Theta(\eta d_{\mathrm{h}}^q)$ under one learning rate η in the standard parametrization.

	step size, exponent p			SP change, exponent q	
layer	SGD	Adam	spectral	SGD	Adam
input	1	0	0	$-\frac{1}{2}$	0
hidden	0	-1	0	$\frac{1}{2}$	1
output	-1	-1	0	1	1

enough: the last hidden layer's change is aligned with the readout, of norm $\Theta(d_{\mathrm{h}}^{1/2})$, and moves the output by $\Theta(\eta d_{\mathrm{h}}^{3/2})$ (under SGD, by $\Theta(\eta d_{\mathrm{h}})$). ◇

The Adam column is μP's (Yang and Hu 2021; Yang et al. 2021), which Yang and Littwin (2023) justify for adaptive optimizers in the infinite-width limit; the SGD column is the corresponding rule in a parametrization without multipliers; and the spectral column, with its shape factor inside lmo, is the duality map of the RMS→RMS norm, from which Bernstein and Newhouse (2025) recover μP.[6] The three rules are one: each multiplies the optimizer's natural step by the constant that converts its norm into the RMS→RMS norm on aligned updates. For SGD that constant depends on the signal: its rule needs $\|\delta_l\|_{\mathrm{rms}} \propto 1/d_l$, which Proposition 4.12 gives only from above and the heuristic only with μP's readout (with the SP readout its exponents are $\frac{1}{2}$, $-\frac{1}{2}$, -1). For sign descent it is the fan-in, the norm-equivalence constant between the largest entry of a matrix and its RMS→RMS norm (Exercise 4.2); for spectral steps it is one.

[6] The heuristic's constants matter less than its exponents, but they are not small: a square Gaussian matrix has operator norm about twice its typical gain on a fixed vector, and ReLU halves squared norms, which He-type initializations with variance 2/fan-in compensate.

4.4.2 Smoothness and depth

For a deep linear network and the square loss in RMS units, the second derivative of ℓ along a direction with $\max_l \|\Delta W_l\|_{\mathrm{rms}\to\mathrm{rms}} \le 1$ is $\|\dot{f}\|_{\mathrm{rms}}^2 + \langle f - y, \ddot{f}\rangle/d_N$, and the expansion in the proof of Theorem 4.11(iv) bounds $\|\dot{f}\|_{\mathrm{rms}}$ and $\|\ddot{f}\|_{\mathrm{rms}}$ by width-free products of the a_k: on $\{\|W_l\|_{\mathrm{rms}\to\mathrm{rms}} \le a_l\}$ the loss is smooth in the max norm with a width-independent constant. Large et al. (2024) prove width-independent Lipschitz and smoothness bounds, with recursive formulas for the constants, for networks assembled from modules satisfying their conditions of being well normed and sharp. For nonlinear layers these conditions carry content: in RMS units the second-order term $\phi''(u)\odot\dot{u}\odot\dot{u}$ of an entrywise nonlinearity is bounded by $\|\phi''\|_\infty \|\dot{u}\|_\infty \|\dot{u}\|_{\mathrm{rms}}$, not by a width-independent multiple of $\|\dot{u}\|_{\mathrm{rms}}^2$ (Exercise 4.9).

Depth is a separate scaling. In a residual network $h_l = h_{l-1} + N^{-\alpha} F_l(h_{l-1})$ with $\|F_l(h)\|_{\mathrm{rms}} \le a\|h\|_{\mathrm{rms}}$, the argument of Theorem 4.11(i) bounds $\|h_N\|_{\mathrm{rms}}$ by $(1 + aN^{-\alpha})^N \|x\|_{\mathrm{rms}}$, uniformly in N if $\alpha \ge 1$, and branches

orthogonal to the stream need only $\alpha \geq 1/2$ (Exercise 4.7). Both exponents are in use, and we state the results without proof. Bordelon et al. (2024) scale residual branches by $1/\sqrt{N}$ and observe transfer across width and depth in ResNets and vision transformers, and Yang et al. (2024) derive the same scaling of branches and updates, Depth-μP, with its infinite-depth limit; Dey, Zhang, et al. (2025) argue for $1/N$ (CompleteP), which they find necessary for every layer to learn beyond its linearization, together with rescaled learning rates for normalization layers and biases, weight decay and ϵ.

4.5 Reading the theory

The theory predicts three things. (1) Under the per-layer rates of Table 4.1 the optimal base learning rate does not depend on the width. (2) Under one learning rate in SP, the output layer (and for Adam the hidden layers) moves by $\Theta(\eta d_{\mathrm{h}})$, so SGD's largest stable learning rate and, in short training, the best learning rates of SGD and Adam fall at least like $1/d_{\mathrm{h}}$: slope -1, or steeper for Adam, whose first step moves the output by $\Theta(\eta d_{\mathrm{h}}^{3/2})$. (3) One step moves the hidden features by $\Theta(d_{\mathrm{h}}^{1/2})$ under SGD and $\Theta(d_{\mathrm{h}})$ under sign descent, Adam's first step, in SP, and by a width-independent amount under the width-aware rules, between the bounds of Theorem 4.11 for spectral steps. For the hidden layer of the network of Figure 4.1, with $f - y = 1$ and $\|x\|_{\mathrm{rms}} = 1$, the heuristic's Gaussian computation gives the change per unit of step: $\sqrt{d_{\mathrm{h}}/2}$ (SGD) and $d_{\mathrm{h}}/\sqrt{2\pi}$ (sign descent) in the standard parametrization, $\sqrt{32}$ and $64/\sqrt{2\pi}$ in μP with base width 64 (Exercise 4.5).

Figure 4.2 tests the predictions. In (a), the optimum of Adam in SP moves by -0.96 octaves per doubling of the width (least squares over five widths; -1.03 from width 128 on), and the largest stable learning rate of SGD in SP halves exactly with every doubling from width 128 on. The width-aware rules move the optimum by about one octave or less over a sixteenfold range, with slopes -0.21 (Adam, μP), -0.19 (spectral) and $+0.16$ (SGD, μP, noisy: nearly flat at width 64, cut off by divergence at larger widths); a second seed gives -0.91, -0.34 and -0.20 for Adam in SP, Adam in μP and spectral steps. The leading-order theory does not predict this slow drift. In (b), the hidden layer's change alone, measured separately, agrees with the predicted contribution to within 8% at every width and 1% at the two largest. With all layers stepping, the points lie above the curves by the input layer's contribution (width-independent, except for the exponent $-\frac{1}{2}$ of SGD in SP): in SP they approach the curves, with local slopes reaching 0.49 (SGD) and 0.99 (sign descent), and in μP they stay a constant factor above them. In (c), the spectral step moves the features by 2.4 to 2.5 at every width, between the lower bound $\|h_1\|_{\mathrm{rms}} \approx 1$, which is the hidden layer's exact contribution, and the upper bound $a_1 + a_2 \in [4.3, 5.0]$.

WHERE THE AGREEMENT MUST FAIL. Prediction (3) concerns one step at initialization. Prediction (2) assumes more: that after 150 steps the best learning rate is limited by the layers that limit the first step, and that later gradients stay aligned with the features. Minibatch gradients have rank

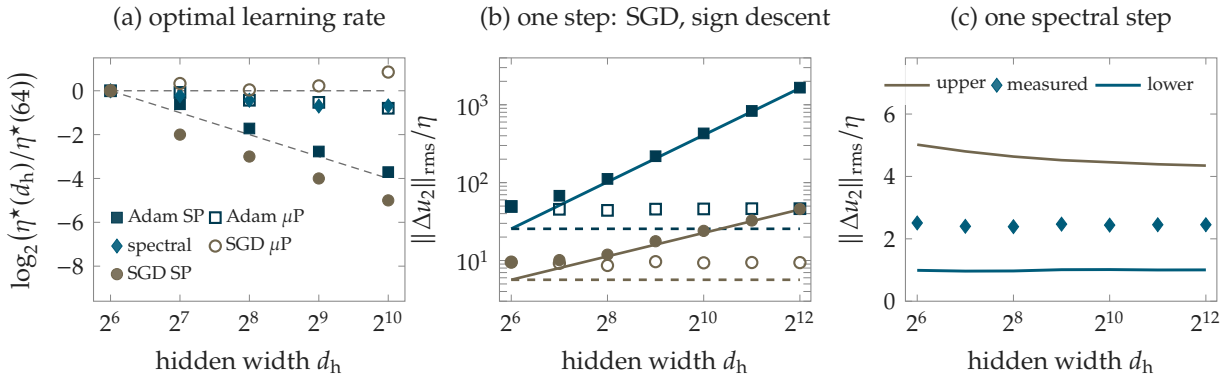


Figure 4.2. Width rules, measured on the network of Figure 4.1. (a) Optimal base learning rate relative to width 64 (vertex of a parabola through the best grid point and its neighbors in ($\log_2 \eta$, log loss)), with the predicted slopes 0 and -1 (dashed). SGD in SP barely learns the task in 150 steps at every learning rate, so its points show the largest stable grid learning rate instead. (b), (c) One step from one example at initialization, $f - y = 1$, $\|x\|_{\text{rms}} = 1$, base step 10^{-5}: change of the second hidden layer's pre-activations divided by the step, all layers updated, mean of 8 seeds. Squares: sign descent (Adam's first step); circles: SGD; filled: standard parametrization; open: μP. Curves in (b): the hidden layer's own contribution predicted by Theorem 4.9 with the heuristic sizes of δ (solid: SP, dashed: μP); the gap above them is the input layer's contribution, width-independent except under SGD in SP, where it decays like $d_h^{-1/2}$. (c) The first-order bounds of Theorem 4.11, computed from the initial weights. Scripts `04-modular-transfer.py`, `04-modular-onestep.py`.

up to 32 here, Adam's moments mix steps, and alignment decays during training (Kosson et al. 2026); weaker alignment moves the features less than the rank-one calculation says and flattens the slope of (a), which survives here because training is short. At small widths lower-order terms, such as the input layer's contribution in (b), blur the exponents.

MUON'S SHAPE FACTORS. Per unit of η, a Muon step with the factor $\sqrt{d_{\text{out}}/d_{\text{in}}}$ has RMS→RMS norm 1 for every shape; Jordan et al.'s clamp agrees when $d_{\text{out}} \geq d_{\text{in}}$ and gives a wide-to-narrow matrix $\sqrt{d_{\text{in}}/d_{\text{out}}}$, a constant for a fixed architecture shape.[7] Liu et al.'s factor gives $0.2\sqrt{d_{\text{in}}}$ when $d_{\text{out}} \geq d_{\text{in}}$, which grows with the width, so transfer under it needs a base learning rate $\propto d_h^{-1/2}$ (Exercise 4.8); RMS matching serves another purpose, reusing AdamW's tuned learning rate at a given width. With a full-rank polar factor a square step gives every input the gain $\|\Delta W\|_{\text{rms}\to\text{rms}}$, aligned or not, so the spectral rule alone does not depend on alignment.

[7] For the down-projection of a transformer's MLP, of shape $d \times 4d$, it is 2.

4.6 What the theory misses

WHAT WAS IDEALIZED AWAY. *One example and one step.* Minibatch gradients are sums of rank-one terms, and moving averages mix many steps; an update not aligned with the inputs it acts on moves them by less than its operator norm, by a factor that depends on its spectrum and the alignment,

and that changes the exponents of Table 4.1 (Exercise 4.6). *Exponents in place of constants.* The base learning rate and the factor between norm and typical gain are outside the theory. *The RMS norm.* It treats all coordinates alike, which suits delocalized features and not the few large activations of trained transformers, and controls entrywise nonlinearities only to first order. *The architecture.* Embeddings receive one-hot inputs, whose natural norm is ℓ_1 (Exercise 4.3); attention and normalization layers have no norm here. *Everything but width.* Weight decay, training horizon, batch size and depth are fixed.

WHERE IT BREAKS. (1) The theory does not single out μP (Observation 4.2), and the right exponents depend on how aligned the updates are, a property of training, not of the architecture (Everett et al. 2024). (2) The alignment the rank-one calculation assumes is transient: Kosson et al. (2026) find that it holds only at the start of language-model training, that the relative size of later updates is set by weight decay, and that μP acts mainly as an implicit warmup. (3) The best learning rate of a language model shrinks as the token horizon grows, following a scaling law of its own (Bjorck et al. 2025), about which a norm on the weights is silent. (4) With AdamW, μP's learning-rate scaling breaks down during training unless the weight decay is scaled as well (Wang and Aitchison 2025). (5) In the ablations of Lingle (2024), transfer failed with trainable gains in the normalization layers, with Lion, with decoupled weight decay, and with the standard $1/\sqrt{d}$ scaling of attention logits, all outside this chapter's model. (6) Depth needs its own rule, and the literature has not settled on one (Bordelon et al. 2024; Dey, Zhang, et al. 2025; Yang et al. 2024). (7) The theory predicts how the optimum scales, not where it lies.

REPAIRS AND OPEN PROBLEMS. Noci et al. (2024) tie the location of the optimum to the landscape: under μP the sharpness along training is nearly independent of width and depth, unlike under the neural-tangent parametrization, and this super consistency goes together with learning-rate transfer.[8] u-μP (Blake et al. 2025) combines μP with unit scaling, gives near-optimal defaults and trains in 8-bit floating point; Depth-μP and CompleteP extend the rules to depth; and weight decay's scaling with width is taken up in Chapter 13: under μP's $\eta \propto 1/d_{\mathrm{h}}$, keeping AdamW's averaging time $1/(\eta\lambda)$ fixed requires $\lambda \propto d_{\mathrm{h}}$. A more radical repair fixes the norms instead of scaling the steps: Hyperball (Wen et al. 2026) constrains the Frobenius norm of every matrix and normalizes every update, so that the learning rate sets an angular step, and reports better transfer across widths and depths in an unrefereed study of models up to 1.2 billion parameters. Qiu et al. (2025) work out width and depth rules for Shampoo, SOAP and Muon. Open are a theory of where the optimum lies; norms for attention, embeddings and normalization derived rather than chosen; the dynamics of alignment, which decides the exponents in practice; and transfer across horizon and batch size.

[8] In the language of Chapter 8: if the sharpness at which training runs does not depend on the width, neither does the stability threshold that bounds the learning rate.

4.7 Back in the wild

FIELD NOTES	
PRACTICE	Tune on a narrow proxy and widen: μP divides Adam's hidden and output learning rates by the width multiplier; Muon scales its update by $\sqrt{d_{\text{out}}/d_{\text{in}}}$ or an RMS-matching factor.
IDEALIZATION	Layers with operator norms (RMS→RMS for linear layers), the network normed by their maximum; one step from one example; features of RMS $\Theta(1)$.
PROVED	Max-norm steepest descent is per-layer normalized descent; exact one-step feature changes; width-free forward, update and backward bounds; no cancellation between layers.
PREDICTS	Per-layer rates $\propto$ fan-out/fan-in (SGD), 1/fan-in (Adam), 1 (spectral); an SP optimum falling like 1/width or faster (measured slope -0.96).
MISSES	Alignment after the first steps; the optimum's location; embeddings, attention, normalization; weight decay, horizon, batch size, depth.
TAKEAWAY	Make every layer's step width-independent in the RMS→RMS norm (proved, in the idealization, to fix the feature changes); expect transfer across width, not across horizon or batch size (observed).

The chapter offers a practitioner a units check rather than a recipe. A learning rate is a number only after a norm has been chosen; in the RMS→RMS norm "the same learning rate" means "the same change of the features" at every width, and each working width rule expresses that norm in the optimizer's own currency. The theory does not say which base learning rate to use, nor that a proxy tuned for one horizon is tuned for another; that a μP proxy can stand in for every hyperparameter of a large run is folklore beyond the evidence of Section 4.6.

The chapter closes Part I: Chapter 1 gave the dual of a maximum of norms, Chapters 2 and 3 the duality maps of one tensor, and here the layers are assembled into one norm of which the per-layer learning rate is part. The memory that practical optimizers add is the subject of Chapters 5 to 7, and the per-layer rate returns as the effective learning rate of a normalized layer under weight decay in Chapter 13.

Exercises

4.1 CONSOLIDATION. (Sums and other combinations.) (a) Show that the dual of $\sum_l s_l \|W_l\|_{(l)}$ is $\max_l \|G_l\|_{(l),*}/s_l$, and that its lmo consists of the convex combinations of the points that equal $\operatorname{lmo}_{(l)}(G_l)/s_l$ on one layer l attaining the maximum and vanish on the others. (b) For $1 < p < \infty$ and $1/p + 1/q = 1$, show that the dual of $\left(\sum_l (s_l \|W_l\|_{(l)})^p\right)^{1/p}$ is $\left(\sum_l (\|G_l\|_{(l),*}/s_l)^q\right)^{1/q}$, and find its lmo.

4.2 CONSOLIDATION. (Sign descent and the largest entry.) (a) Show that the $\ell_1 \to \ell_\infty$ operator norm of W is $\|W\|_{\max} = \max_{ij}|W_{ij}|$, so that sign descent is normalized steepest descent in it. (b) Show that $\|W\|_{\text{rms}\to\text{rms}} \le d_{\text{in}}\|W\|_{\max}$, with equality if and only if $W = c\,ab^\top$ with a and b sign vectors. (c) Deduce the rule $\eta \propto 1/\text{fan-in}$ for sign descent.

4.3 CONSOLIDATION. (Embeddings.) An embedding layer $W \in \mathbb{R}^{d_{\text{out}}\times d_{\text{in}}}$ receives one-hot inputs e_j, with $\|e_j\|_{\text{rms}} = d_{\text{in}}^{-1/2}$. (a) Show that the normalized RMS→RMS step moves We_j by only $\eta\, d_{\text{in}}^{-1/2}$ in RMS. (b) Show that the $\ell_1 \to$ rms operator norm is $\max_j\|W_{:j}\|_{\text{rms}}$, with dual $\sqrt{d_{\text{out}}}\sum_j\|G_{:j}\|_2$ and an lmo that rescales each nonzero column of $-G$ to RMS one. (c) Show that its normalized step moves We_j by exactly η; compare with SGD and sign descent.

4.4 CONSOLIDATION. (Tightness.) In a deep linear network with square orthogonal layers, $\|x\|_{\text{rms}} = 1$ and $\nabla\psi(f) \neq 0$, let every layer take the normalized RMS→RMS step of size η of one example's loss. Show that layer k contributes $-\eta\sqrt{d_l}\,\hat{\delta}_l$ to $\dot{u}_l$, so that $\|\dot{u}_l\|_{\text{rms}} = l\eta$, the first-order term of the bound of Theorem 4.11(ii).

4.5 CONSOLIDATION. (Constants.) For the network of Figure 4.1 at initialization, with $f - y = 1$ and $\|x\|_{\text{rms}} = 1$, show heuristically that $\|h_1\|_{\text{rms}} \approx 1$, $\rho(h_1) \approx 1/\sqrt{\pi}$, $\sigma(\delta_2) \approx 1/\sqrt{2}$, and $\|\delta_2\|_{\text{rms}} \approx (2d_{\text{h}})^{-1/2}$ in SP and $\sqrt{32}/d_{\text{h}}$ in μP; deduce the curves of Figure 4.2(b), and show that under μP the initial output has RMS of order $d_{\text{h}}^{-1/2}$.

4.6 VARIATION. (Minibatches.) Let $G = \frac{1}{B}\sum_{i\le B}\delta_i x_i^\top$ with orthogonal inputs and orthogonal signals. (a) Show that SGD moves the output on x_j by $\frac{\eta}{B}\|x_j\|_2^2\,\delta_j$, a factor B less than one example's step. (b) Show that $\text{polar}(G) = \sum_i \hat{\delta}_i\hat{x}_i^\top$, so that the spectral step moves the output on every x_j by exactly $\eta\|x_j\|_{\text{rms}}$. (c) How should the two learning rates scale with B in this idealization? Compare with the linear scaling rule of Chapter 11.

4.7 VARIATION. (Depth.) Let $h_l = h_{l-1} + N^{-\alpha}F_l(h_{l-1})$ with $\|F_l(h)\|_{\text{rms}} \le a\|h\|_{\text{rms}}$. (a) Show that $\|h_N\|_{\text{rms}} \le (1 + aN^{-\alpha})^N\|x\|_{\text{rms}} \le e^{aN^{1-\alpha}}\|x\|_{\text{rms}}$. (b) Show that, if every branch output is orthogonal to its input, then $\|h_N\|_{\text{rms}}^2 \le (1 + a^2N^{-2\alpha})^N\|x\|_{\text{rms}}^2$, bounded for $\alpha \ge 1/2$, and relate the two exponents to Depth-μP and CompleteP.

4.8 VARIATION. (Muon's shape factors.) For a full-rank $M \in \mathbb{R}^{d_{\text{out}}\times d_{\text{in}}}$, compute the RMS→RMS norm and the entrywise RMS of $\gamma\,\text{polar}(M)$ for $\gamma = \sqrt{d_{\text{out}}/d_{\text{in}}}$, $\sqrt{\max(1, d_{\text{out}}/d_{\text{in}})}$ and $0.2\sqrt{\max(d_{\text{out}}, d_{\text{in}})}$. Show that the third gives entrywise RMS 0.2 and requires a base learning rate $\propto d_{\text{h}}^{-1/2}$ for width-independent feature changes of a hidden $d_{\text{h}}\times d_{\text{h}}$ matrix.

4.9 VARIATION. (Curvature of a nonlinearity.) Let $\phi = \tanh$ and let $u \in \mathbb{R}^m$ have all entries equal to some s_0 with $\phi''(s_0) \neq 0$. (a) Show that

the second-order term $\phi''(u) \odot \dot{u} \odot \dot{u}$ has RMS norm $|\phi''(s_0)|\sqrt{m}$ for $\dot{u} = \sqrt{m}\, e_1$ and $|\phi''(s_0)|$ for $\dot{u} \in \{\pm 1\}^m$, although both have $\|\dot{u}\|_{\text{rms}} = 1$. (b) Show that $\|\phi''(u) \odot \dot{u} \odot \dot{u}\|_{\text{rms}} \le \|\phi''\|_\infty \|\dot{u}\|_\infty \|\dot{u}\|_{\text{rms}}$. (c) Conclude that width-independent smoothness in RMS units needs control of $\|\dot{u}\|_\infty$, and decide which steps of Theorem 4.9 give delocalized changes.

4.10 SIMULATION. (Alignment over training.) In `04-modular-transfer.py`, record at steps $1, 10, 50, 150$ the stable rank of the hidden layer's update ΔW_2 and the batch average of its alignment ratio κ, the ratio of $\|\Delta W_2 h_1\|_{\text{rms}}$ to $\|\Delta W_2\|_{\text{rms}\to\text{rms}} \|h_1\|_{\text{rms}}$, for Adam in μP and spectral steps at widths 64 to 1024. Does Adam's κ decay? Why is a spectral step's κ the length of the projection of $h_1/\|h_1\|_2$ on its row space?

4.11 SIMULATION. (Breaking transfer.) In `04-modular-transfer.py`, (a) replace the factor $\sqrt{d_{\text{out}}/d_{\text{in}}}$ of the hidden layer's spectral step by $0.2\sqrt{\max(d_{\text{out}}, d_{\text{in}})}$ and measure the slope of the optimum (prediction $-1/2$; and if every layer takes it?); (b) double the number of steps and see whether the slopes of Figure 4.2(a) change.

4.12 RESEARCH. (A norm for attention.) Attention logits are quadratic in the query and key weights. Propose a norm on (W_Q, W_K, W_V, W_O) under which one normalized step changes the attention output by a width-independent amount, and derive the scaling of the logits it implies (the $1/d$ versus $1/\sqrt{d}$ question in Lingle (2024)). As far as we know, no derivation of this kind is accepted.

4.13 RESEARCH. (Exponents beyond alignment.) If after t steps a hidden layer's update has stable rank $r(t, d_{\text{h}})$ and alignment ratio $\kappa(t, d_{\text{h}})$ (Exercise 4.10), derive the exponents of Table 4.1 from the growth rates of r and κ, and compare with those measured by Everett et al. (2024). A model that derives r and κ is open.

Notes and References

Yang and Hu (2021) classified a family of parametrizations that contains the standard, neural-tangent and mean-field ones into those with a kernel limit and those that learn features, and singled out μP; Yang and Littwin (2023) extended the program to adaptive optimizers, and Yang et al. (2021) turned the classification into a tuning procedure. Yang et al. (2023) replaced the tensor-program derivation by the spectral condition, a few lines of linear algebra, which Theorem 4.11 follows.

Per-layer normalized steps entered large-batch training with LARS. Large et al. (2024) built the modular norm for general architectures and the Modula library, and Bernstein and Newhouse (2024, 2025) wrote the optimizers in use as steepest descent in such norms and computed the duality maps of linear, embedding and convolutional layers; Newhouse (2025) presents this line in a thesis. Pethick et al. (2025) build optimizers from linear minimization oracles over norm balls, with a norm chosen layer by layer for transfer across model sizes.

Lemma 4.10 and its use in Theorem 4.11(iii) are, as far as we know, not stated in this form in the literature, although the alignment of the steps of different layers is implicit in the arguments of Yang et al. (2023); the constants in Figure 4.2(b) are our computation.

TRANSLATING THE LITERATURE. Yang et al. (2023) write n_ℓ for the width of layer ℓ (our d_l) and $\|\cdot\|_*$ for the *spectral* norm, our $\|\cdot\|_{\mathrm{op}}$; in this book a star marks a dual norm. Their Desideratum 1, $\|h_\ell\|_2 = \Theta(\sqrt{n_\ell})$ and $\|\Delta h_\ell\|_2 = \Theta(\sqrt{n_\ell})$, is our $\|h_l\|_{\mathrm{rms}}, \|\Delta h_l\|_{\mathrm{rms}} = \Theta(1)$, and their Condition 1 is our $\|W_l\|_{\mathrm{rms}\to\mathrm{rms}}, \|\Delta W_l\|_{\mathrm{rms}\to\mathrm{rms}} = \Theta(1)$. Yang and Hu (2021) write each weight as a width-dependent multiplier times a trainable matrix whose initialization and learning rate also scale with the width n (an "abc-parametrization"); since multiplier, initialization and learning rate can be traded for one another, one dynamics appears in several tables, and Table 4.1 puts all width dependence into the initialization and the per-layer rates. Bernstein and Newhouse (2025) call $g \mapsto \arg\max_{\|t\|=1}\langle g, t\rangle$ the duality map, which is $-\mathrm{lmo}(g)$ here, and compute the scales of our max norm from the masses and sensitivities of Large et al. (2024). Pethick et al. (2025) take the linear minimization oracle over a ball of radius ρ, which is ρ times our lmo.

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Part II

Memory: Averages and Preconditioners

Practical optimizers remember. A moving average of gradients filters noise, and it is what lets normalized steps converge; a moving average of squared gradients, or of products of gradients, builds a norm adapted to the problem. Momentum, Adam, Shampoo and SOAP are such memories added to steepest descent.

5 Moving Averages

Momentum, noise, and why normalized steps need memory

IN THE WILD — MOMENTUM 0.9

IN THIS CHAPTER. Almost every optimizer averages its gradients with a coefficient near 0.9, and every normalized optimizer, from sign descent with momentum to Lion and Muon, has the average built in. On a quadratic the average accelerates gradient descent and has stability regions of its own, among them the threshold 38 behind Adam's edge of stability; with additive noise, heavy-ball momentum is to first order a larger step size. For normalized steepest descent in a general norm we prove a $T^{-1/4}$ rate with momentum, and show by an example that without memory a normalized step can stall at a gradient of the size of the noise, however small its step size. Simulations separate the bias of a normalized step, which memory removes, from its fluctuations, which memory enlarges; nothing here explains why 0.9 works so widely.

5.1 In the wild: momentum 0.9, almost everywhere

The oldest ingredient of a modern optimizer is also the most uniform. Gradient descent with a memory of its past steps was introduced by Polyak (1964) as the *heavy-ball method*, and it has been part of almost every widely used optimizer since; its coefficient has barely moved.

IN THE WILD · ResNet, He et al. (2016); Adam, Kingma and Ba (2015); Lion, Chen et al. (2023); Muon, Jordan et al. (2024). Average the gradients before stepping, with coefficient about 0.9. Convolutional networks: SGD with momentum 0.9, $b_t = 0.9\, b_{t-1} + g_t$, $\theta_{t+1} = \theta_t - \eta\, b_t$, with $\eta = 0.1$ and step decay for ResNets. Transformers: AdamW with $\beta_1 = 0.9$, $m_t = 0.9\, m_{t-1} + 0.1\, g_t$ (GPT-3 and LLaMA, as in the recipe of **??**). Lion: step along the sign of $0.9\, m_{t-1} + 0.1\, g_t$, update m_t with coefficient 0.99. Muon: Nesterov momentum with coefficient 0.95, then orthogonalize. Separately, evaluate a slow average of the weights, or average the iterates in place of a decaying schedule.

Two conventions hide in this list: PyTorch's $b_t = \beta b_{t-1} + g_t$ has weights summing to $1/(1-\beta)$, Adam's $m_t = \beta m_{t-1} + (1-\beta) g_t$ to one after bias correction. At $\beta = 0.9$ a learning rate of 0.1 in the first moves the weights as far as 1 in the second, so which is "the" learning rate matters as soon as β changes.[1] Three regularities of different strength stand behind the practice.

[1] PyTorch's `torch.optim.SGD` has a "dampening" argument that multiplies g_t by one minus its value; with dampening β it computes Adam's average, up to the initialization. Most recipes leave it at zero.

Observation 5.1 (Momentum as a larger step; Kidambi et al. 2018, Shallue et al. 2019, Zhang et al. 2019). With small batches, SGD with momentum β and learning rate η in PyTorch's convention trains about as well as plain SGD with learning rate $\eta/(1-\beta)$, once both are tuned. Momentum pays off at large batches, where it extends the range of batch sizes over which doubling the batch halves the number of steps.

Shallue et al. (2019) measured steps to a target across batch sizes and workloads and found that momentum extends perfect scaling to larger batches; Zhang et al. (2019) reproduced the pattern in a noisy quadratic model (Chapter 11), and Kidambi et al. (2018) showed that the classical schemes do not improve on SGD for some stochastic least-squares problems.

Observation 5.2 (Normalized steps carry memory; Bernstein et al. 2018, Karimireddy et al. 2019, Kunstner et al. 2023). Every widely used normalized optimizer steps along an average of past gradients. Without the average and without large batches, sign descent can stall or fail to converge; with momentum and large batches, it comes close to Adam on transformers.

The optimizers are Signum, the sign of an exponential moving average (Bernstein et al. 2018), Lion and Muon.[2] Kunstner et al. (2023) found that the advantage of sign descent with momentum over SGD grows with the batch size. In theory, sign descent with stochastic gradients can fail to converge (Karimireddy et al. 2019), and Cutkosky and Mehta (2020) analyzed normalized SGD with momentum and without large minibatches. How much of the need for memory is due to small batches is less well documented than the practice.

[2] The name Signum, for sign descent with momentum, is from Bernstein et al. (2018), who also analyzed majority vote over workers for distributed training.

Observation 5.3 (Momentum moves the edge of stability; Cohen et al. 2022). In full-batch training with Adam and $\beta_1 = 0.9$, the largest eigenvalue of the preconditioned Hessian rises to about $38/\eta$ and hovers there, as the sharpness of gradient descent hovers at $2/\eta$ (Cohen et al. 2021).

The number 38 is $2(1+\beta_1)/(1-\beta_1)$ at $\beta_1 = 0.9$, a stability threshold derived in Proposition 5.8; why training climbs to it is the subject of Chapter 8. What does an average of past gradients do that a different step size cannot, and why do normalized steps need it when SGD can do without?

5.2 The idealization: one filter, three settings

A moving average is a linear filter: $m_t = (1-\beta)\sum_{s\le t}\beta^{t-s}g_s$ remembers about $1/(1-\beta)$ steps, and at $\beta = 0.9$ half of its weight lies on the last 6.6 steps.[3]

[3] The half-life $\ln 2/\ln(1/\beta)$ is 6.6 steps at $\beta = 0.9$, 13.5 at 0.95, 69 at 0.99 and 693 at 0.999.

Definition 5.4 (Momentum methods). Let g_t be the gradient or stochastic gradient at θ_t, $\eta > 0$ and $0 \le \beta < 1$.

(i) *Heavy ball*: $\theta_{t+1} = \theta_t - \eta g_t + \beta(\theta_t - \theta_{t-1})$ with $\theta_{-1} = \theta_0$; its *PyTorch form* is $b_t = \beta b_{t-1} + g_t$, $\theta_{t+1} = \theta_t - \eta b_t$, with $b_{-1} = 0$.

(ii) *EMA momentum* (Adam's first moment): $m_t = \beta m_{t-1} + (1-\beta)g_t$, $\theta_{t+1} = \theta_t - \eta m_t$, with $m_{-1} = 0$.

(iii) *Quasi-hyperbolic momentum* (Ma and Yarats 2019): m_t as in (ii) and $\theta_{t+1} = \theta_t - \eta\big((1-\nu)g_t + \nu m_t\big)$, $0 \le \nu \le 1$. It is EMA momentum for $\nu = 1$, gradient descent for $\nu = 0$, and for $\nu = \beta$ *Nesterov momentum in EMA form*, the form used by Muon.

(iv) *Nesterov momentum*: $\theta_{t+1} = y_t - \eta\nabla\mathcal{L}(y_t)$ with $y_t = \theta_t + \beta(\theta_t - \theta_{t-1})$; its PyTorch form is $b_t = \beta b_{t-1} + g_t$, $\theta_{t+1} = \theta_t - \eta(g_t + \beta b_t)$.

(v) *Normalized steepest descent with momentum*: m_t as in (ii) and $\theta_{t+1} = \theta_t + \eta u_t$ with $u_t \in \arg\min_{\|u\|\le 1}\langle m_t, u\rangle$.

Lemma 5.5 (One filter, several step sizes). *(i) The PyTorch form of heavy ball with step η produces the iterates of heavy ball with step η, and (ii) EMA momentum with step η those of heavy ball with step $\eta(1-\beta)$. (iii) The PyTorch form of Nesterov momentum with step η produces the iterates of Nesterov momentum in EMA form with step $\eta/(1-\beta)$. (iv) On a quadratic, along an eigenvector of the Hessian with eigenvalue h, both forms of Nesterov momentum satisfy $x_{t+1} = (1-\eta h)\big((1+\beta)x_t - \beta x_{t-1}\big)$.*

Proof. For (i), $\theta_{t+1} - \theta_t = -\eta\beta b_{t-1} - \eta g_t = \beta(\theta_t - \theta_{t-1}) - \eta g_t$ for $t \ge 1$, and $\theta_1 = \theta_0 - \eta g_0$; (ii) is the same computation with ηm_t, which gives $\beta(\theta_t - \theta_{t-1}) - \eta(1-\beta)g_t$. For (iii), $b_t = m_t/(1-\beta)$, so $g_t + \beta b_t = \big((1-\beta)g_t + \beta m_t\big)/(1-\beta)$. For (iv), the first form gives $x_{t+1} = (1-\eta h)y_t$ directly; in the second, eliminate βb_t and $\beta^2 b_{t-1}$ from $x_t - x_{t+1} = \eta(hx_t + \beta b_t)$, the same identity at time $t-1$, and $b_t = \beta b_{t-1} + hx_t$. ∎

For a normalized step the conventions do not matter, since $\mathrm{lmo}(cm) = \mathrm{lmo}(m)$ for $c > 0$. We study the filter in three settings.

Assumption 5.6 (Quadratic loss, additive noise). The loss is $\mathcal{L}(\theta) = \frac{1}{2}(\theta - \theta^\star)^\top H(\theta - \theta^\star)$ on $\mathbb{R}^d$, where H is symmetric with eigenvalues $h_1 \ge \cdots \ge h_d > 0$ and orthonormal eigenvectors $v_1, \dots, v_d$; $L = h_1$, $\mu = h_d$, $\kappa = L/\mu$. The stochastic gradient is $g_t = H(\theta_t - \theta^\star) + \zeta_t$, with $\zeta_0, \zeta_1, \dots$ independent and identically distributed, independent of the initialization, of mean zero and covariance C. Exact gradients are the case $C = 0$.

Assumption 5.7 (Smooth loss, unbiased noise). The loss $\mathcal{L}\colon \mathbb{R}^d \to \mathbb{R}$ is bounded below by $\mathcal{L}^\star$ and L-smooth with respect to a norm $\|\cdot\|$. The stochastic gradient is $g_t = \nabla\mathcal{L}(\theta_t) + \zeta_t$ with $\mathbb{E}[\zeta_t \mid \mathcal{F}_t] = 0$ and $\mathbb{E}[\|\zeta_t\|_2^2 \mid \mathcal{F}_t] \le \sigma^2$, where θ_0 is fixed and $\mathcal{F}_t$ is generated by $\zeta_0, \dots, \zeta_{t-1}$. The constant $\chi = \max_{z\ne 0}\|z\|_*/\|z\|_2$ compares $\|\cdot\|_*$ with $\|\cdot\|_2$.

Setting (a) is Assumption 5.6 with $C = 0$; setting (b), with noise, is the *noisy quadratic model* of Zhang et al. (2019) (Chapter 11); setting (c) is Assumption 5.7 with normalized steps, where $\chi = 1$ for ℓ_2, $\sqrt{d}$ for ℓ_∞, and $\sqrt{\min(m,n)}$ for the spectral norm on $m \times n$ matrices.[4]

[4] We write h_i for the eigenvalues that the rest of the book calls λ_i, so that the curvature of a single coordinate can be written h, without a subscript.

DICTIONARY

IN THE THEORY	IN A TRAINING RUN
the coefficient β	0.9 for SGD, Adam's β_1 and Lion's step; 0.95 for Muon; 0.99 for Lion's state
the memory $1/(1-\beta)$	the number of recent minibatches the optimizer state remembers
the effective step $\eta/(1-\beta)$ of heavy ball	the learning rate that matters for SGD with momentum in PyTorch's convention
the threshold $2(1+\beta)/(1-\beta)$ on ηh	Adam's edge of stability, $38/\eta$ at $\beta_1 = 0.9$
additive noise of covariance C	minibatch noise Σ/B
the momentum error $m_t - \nabla\mathcal{L}(\theta_t)$	optimizer state minus current gradient; its lag: stale gradients
skewed noise	rare large per-example gradients

FIELD GUIDE · SGD WITH MOMENTUM *Polyak (1964)*

UPDATE	$b_t = \beta b_{t-1} + g_t$, $\theta_{t+1} = \theta_t - \eta b_t$; or $m_t = \beta m_{t-1} + (1-\beta) g_t$, $\theta_{t+1} = \theta_t - \eta m_t$
NORM	ℓ_2, unnormalized; no preconditioner
AVERAGES	EMA of g; the first form multiplies the step by $1/(1-\beta)$
STATE	one number per parameter
IN THE WILD	$\beta = 0.9$, $\eta = 0.1$, weight decay 10^{-4} (He et al. 2016)

FIELD GUIDE · NESTEROV MOMENTUM *Nesterov (1983); Sutskever et al. (2013)*

UPDATE	$b_t = \beta b_{t-1} + g_t$, $\theta_{t+1} = \theta_t - \eta(g_t + \beta b_t)$; in EMA form $\theta_{t+1} = \theta_t - \eta(\beta m_t + (1-\beta) g_t)$
NORM	ℓ_2, unnormalized; spectral and normalized in Muon (Chapter 3)
AVERAGES	EMA of g with the fresh gradient counted twice: $\beta^2 m_{t-1} + (1-\beta^2) g_t$
IN THE WILD	optional in PyTorch's SGD; Muon's default, with $\beta = 0.95$

FIELD GUIDE · LION *Chen et al. (2023)*

UPDATE	in Lion's notation $c_t = \beta_1 m_{t-1} + (1-\beta_1) g_t$, $\theta_{t+1} = \theta_t - \eta(\operatorname{sign}(c_t) + \lambda\theta_t)$, $m_t = \beta_2 m_{t-1} + (1-\beta_2) g_t$
NORM	ℓ_∞, normalized: sign descent (Chapter 2); no preconditioner
AVERAGES	one EMA, two coefficients: the sign of quasi-hyperbolic momentum with $\beta = \beta_2$, $\nu = \beta_1/\beta_2$; Signum if $\beta_1 = \beta_2$
STATE	one number per parameter, against Adam's two
IN THE WILD	$\beta_1 = 0.9$, $\beta_2 = 0.99$; a learning rate 3–10 times smaller and a weight decay 3–10 times larger than AdamW's

FIELD GUIDE · SCHEDULE-FREE SGD AND ADAMW *Defazio et al. (2024)*	
UPDATE	$y_t = (1-\beta)z_t + \beta x_t$, $z_{t+1} = z_t - \gamma g(y_t)$, $x_{t+1} = \frac{t}{t+1}x_t + \frac{1}{t+1}z_{t+1}$; evaluate at x_t
NORM	ℓ_2, unnormalized; Adam's metric in the AdamW version
AVERAGES	a uniform average x_t of the iterates; the interpolation y_t acts as momentum
STATE	one vector z; the weights hold y in training and x in evaluation
IN THE WILD	$\beta = 0.9$, no decaying schedule; Schedule-Free AdamW won AlgoPerf's 2024 self-tuning track (Kasimbeg et al. 2025)

WHAT WAS THROWN AWAY. In settings (a) and (b), everything that makes a network more than a quadratic: the Hessian is constant, so the step size cannot change the curvature (Chapter 8), and the noise is additive, position-independent and of finite variance. Setting (c) gives a worst-case bound, not the dynamics, and β, η are constant throughout. The quadratic comes first because a linear filter on linear dynamics is exactly solvable, the worst case next because a normalized step is not linear.

WHAT ONE HAD TO NOTICE. A linear filter commutes with a linear update: averaging the gradients and then stepping is stepping and then averaging the steps, so for SGD momentum is a change of step size. A normalized step is not linear, and the order matters: a small step averages normalized gradients, $\frac{1}{k}\sum_s \operatorname{lmo}(g_s)$, which keeps the bias of the normalization, while momentum normalizes an average, $\operatorname{lmo}\big(\frac{1}{k}\sum_s g_s\big)$, which removes it.

5.3 The theory: momentum on a quadratic

Under Assumption 5.6, every method of Definition 5.4 except (v) is linear, and the coordinates $x_t = \langle \theta_t - \theta^\star, v_i\rangle$ evolve separately. For quasi-hyperbolic momentum with exact gradients, writing the step as $\nu\beta m_{t-1} + (1-\nu\beta)g_t$ shows that x_t and the coordinate n_{t-1} of m_{t-1} evolve by

$$\begin{pmatrix} x_{t+1} \\ n_t \end{pmatrix} = \begin{pmatrix} 1-\eta h(1-\nu\beta) & -\eta\nu\beta \\ (1-\beta)h & \beta \end{pmatrix} \begin{pmatrix} x_t \\ n_{t-1} \end{pmatrix}, \qquad h = h_i,$$

a matrix with trace $1+\beta-\eta h(1-\nu\beta)$ and determinant $\beta - \eta h\beta(1-\nu)$. A linear recursion tends to zero from every initial state if and only if both roots of its characteristic polynomial lie in the open unit disk, which the Jury test decides (Lemma A.14).[5]

Proposition 5.8 (Stability regions). *Let $h > 0$, $0 \le \beta < 1$ and $0 \le \nu \le 1$. On a coordinate with curvature h, quasi-hyperbolic momentum with exact gradients is stable if and only if*

$$\eta h < \frac{2(1+\beta)}{1+\beta-2\nu\beta}. \tag{5.1}$$

Consequently the exact thresholds for ηh are $2(1+\beta)/(1-\beta)$ for EMA

[5] The Jury test is the discrete-time counterpart of the Routh–Hurwitz criterion; for a quadratic it reduces to the two inequalities used here.

momentum (38 at $\beta = 0.9$); $2(1+\beta)/((1-\beta)(1+2\beta))$ for Nesterov momentum in EMA form (13.6); $2(1+\beta)$ for heavy ball in both forms (3.8); and $2(1+\beta)/(1+2\beta)$ for Nesterov momentum in both forms (1.36).

PROOF. The plan is to apply the Jury test to $z^2 + a_1 z + a_0$, with $u = \eta h > 0$, $a_1 = -\big(1 + \beta - u(1 - \nu\beta)\big)$ and $a_0 = \beta\big(1 - u(1 - \nu)\big)$, and then to translate with Lemma 5.5. The test asks for $|a_0| < 1$ and $|a_1| < 1 + a_0$. The inequality $-a_1 < 1 + a_0$ reads $u(1 - \nu\beta) > u\beta(1 - \nu)$, that is, $u(1 - \beta) > 0$, which holds; the inequality $a_1 < 1 + a_0$ reads $u(1 - \nu\beta) + u\beta(1 - \nu) < 2(1 + \beta)$, that is, $u(1 + \beta - 2\nu\beta) < 2(1 + \beta)$. Next, $a_0 \le \beta < 1$, and $a_0 > -1$ means $u\beta(1 - \nu) < 1 + \beta$, which follows from the previous inequality because $1 + \beta - 2\nu\beta - 2\beta(1 - \nu) = 1 - \beta \ge 0$. So stability is $u(1 + \beta - 2\nu\beta) < 2(1 + \beta)$, which is (5.1) since $1 + \beta - 2\nu\beta \ge 1 - \beta > 0$.

For $\nu = 1$ this is EMA momentum, and for $\nu = \beta$ the denominator is $(1 - \beta)(1 + 2\beta)$. By Lemma 5.5(i)–(ii), heavy ball with step η, in either form, is EMA momentum with step $\eta/(1 - \beta)$; by Lemma 5.5(iii)–(iv), both forms of Nesterov momentum with step η follow the recursion of Nesterov momentum in EMA form with step $\eta/(1 - \beta)$. Substituting $\eta/(1 - \beta)$ for η in the first two thresholds gives the last two. ■

Adam's convention widens the stable range of ηh by $(1 + \beta)/(1 - \beta)$, 19 at $\beta = 0.9$; PyTorch's only by $1 + \beta$, since it also multiplies the effective step by $1/(1 - \beta)$. Nesterov's look-ahead shrinks PyTorch's range below gradient descent's 2, by amplifying the coordinate that flips sign at every step. Chapter 8 meets these thresholds at the edge of stability (Observation 5.3).

Theorem 5.9 (Heavy ball on quadratics; Polyak 1964). *Under Assumption 5.6 with exact gradients and $\mu < L$, let $r = (\sqrt{\kappa} - 1)/(\sqrt{\kappa} + 1)$, and run heavy ball with $\eta = 4/(\sqrt{L} + \sqrt{\mu})^2$, $\beta = r^2$ and $\theta_{-1} = \theta_0$.*

(i) For every $h \in [\mu, L]$, both roots of $z^2 - (1 + \beta - \eta h)z + \beta$ have modulus exactly r.

(ii) For every $t \ge 0$, $\|\theta_t - \theta^\star\|_2 \le \big(1 + (1 + r)t\big)\, r^t\, \|\theta_0 - \theta^\star\|_2$.

Gradient descent contracts at best by $(\kappa - 1)/(\kappa + 1)$ per step, so reducing the error by a factor e takes it about $\kappa/2$ steps and heavy ball about $\sqrt{\kappa}/2$; no η and β do better than r at both ends of the spectrum (Exercise 5.1).

PROOF. The plan is to show that the parameters put every eigenvalue into the band where the roots are complex, and to solve the recursion there in closed form. (i) The roots of $z^2 - az + \beta$, with $a = 1 + \beta - \eta h$, are complex or double when $a^2 \le 4\beta$, and then their product β gives both the modulus $\sqrt{\beta} = r$. The condition $|1 + \beta - \eta h| \le 2\sqrt{\beta}$ is $(1 - r)^2 \le \eta h \le (1 + r)^2$. Since $1 - r = 2/(\sqrt{\kappa} + 1)$ and $1 + r = 2\sqrt{\kappa}/(\sqrt{\kappa} + 1)$, we have $(1 - r)^2 = \eta\mu$ and $(1 + r)^2 = \eta L$, so the condition holds on $[\mu, L]$.

(ii) A coordinate satisfies $x_{t+1} = a x_t - r^2 x_{t-1}$ with $x_1 = (1 - \eta h)x_0$, because $\theta_{-1} = \theta_0$. By (i), $a = 2r\cos\varphi$ with $\varphi \in [0, \pi]$, and $r > 0$ because $\mu < L$. Then $y_t = x_t/r^t$ satisfies $y_{t+1} = 2\cos\varphi\, y_t - y_{t-1}$, with $y_0 = x_0$ and

$y_1 = (2\cos\varphi - r)x_0$, since $1 - \eta h = a - \beta$. For $0 < \varphi < \pi$ the solutions are $A\cos t\varphi + B\sin t\varphi$, and matching y_0, y_1 gives

$$x_t = r^t x_0\Big(\cos t\varphi + (\cos\varphi - r)\frac{\sin t\varphi}{\sin\varphi}\Big).$$

By induction, $|\sin(t+1)\varphi| \le |\sin t\varphi| + |\sin\varphi|$ gives $|\sin t\varphi| \le t|\sin\varphi|$, and with $|\cos\varphi - r| \le 1 + r$ we get $|x_t| \le (1 + (1+r)t)r^t|x_0|$. For $\varphi \in \{0, \pi\}$, which happens only at $h \in \{\mu, L\}$, the bound follows by continuity, x_t being a polynomial in h. Summing the squares of the coordinates gives (ii). ■

The bound is attained at $h = L$, where the double root $-r$ gives $x_t = (-r)^t(1 + (1+r)t)x_0$. The theorem is about quadratics: on a strongly convex function with the same μ and L, heavy ball with these parameters need not converge (Lessard et al. 2016) (Exercise 5.11), while Nesterov's method keeps accelerated rates for every smooth convex function, $\mathcal{L}(\theta_t) - \mathcal{L}^\star = O(L\|\theta_0 - \theta^\star\|_2^2/t^2)$, or a contraction of order $1 - 1/\sqrt{\kappa}$ per step under strong convexity (Nesterov 1983, 2018), rates that no gradient method beats by more than constants in high dimension.[6] With noise, the same filter has a second effect, the one that Observation 5.1 reports.

[6] Polyak's 1964 paper proves the local rate of heavy ball for twice differentiable functions, the quadratic case near a minimizer. Polyak (1987) gives a textbook account, and Nesterov (2018, Chapter 2) the lower bounds and the accelerated method.

Theorem 5.10 (In the noise-dominated regime momentum is a larger step). *Let $h > 0$, $\sigma > 0$, $0 \le \beta < 1$ and $0 < \eta h < 2(1+\beta)$. Let $\zeta_0, \zeta_1, \dots$ be independent and identically distributed with mean zero and variance σ^2, independent of (x_0, x_{-1}), which have finite second moments, and let $x_{t+1} = x_t - \eta(hx_t + \zeta_t) + \beta(x_t - x_{t-1})$.*

(i) As $t \to \infty$, $\mathbb{E}x_t \to 0$ and $\mathbb{E}x_t^2 \to V$, where

$$V = \frac{\eta\sigma^2(1+\beta)}{h(1-\beta)\big(2(1+\beta) - \eta h\big)} = \frac{\eta\sigma^2}{2h(1-\beta)}\Big(1 - \frac{\eta h}{2(1+\beta)}\Big)^{-1}. \tag{5.2}$$

(ii) If $\eta h < 2(1-\beta)$, then gradient descent with step $\eta/(1-\beta)$ and the same noise has stationary variance $\frac{\eta\sigma^2}{2h(1-\beta)}\big(1 - \frac{\eta h}{2(1-\beta)}\big)^{-1} \ge V$; the two agree to first order in $\eta h/(1-\beta)$.

(iii) If $\eta h \le (1-\beta)^2/4$, then the characteristic roots are real and distinct, and the larger one, which governs the decay of $\mathbb{E}x_t$, satisfies

$$1 - \frac{\eta h}{1-\beta}\Big(1 - \frac{2\eta h}{(1-\beta)^2}\Big)^{-1} \le z_+ \le 1 - \frac{\eta h}{1-\beta},$$

while gradient descent with step $\eta/(1-\beta)$ contracts its mean by exactly $1 - \eta h/(1-\beta)$.

(iv) EMA momentum, $m_t = \beta m_{t-1} + (1-\beta)(hx_t + \zeta_t)$ and $x_{t+1} = x_t - \eta m_t$, with $\eta h < 2(1+\beta)/(1-\beta)$, has stationary variance $\frac{\eta\sigma^2}{2h}\big(1 - \frac{\eta(1-\beta)h}{2(1+\beta)}\big)^{-1}$, which agrees to first order in ηh with that of gradient descent with step η.

PROOF. The plan is to solve the Lyapunov equation of the two-dimensional recursion and then compare. The state $X_t = (x_t, x_{t-1})^\top$ satisfies $X_{t+1} = AX_t + w_t$, with $A = \left(\begin{smallmatrix} a & -\beta \\ 1 & 0 \end{smallmatrix}\right)$, $a = 1 + \beta - \eta h$, and $w_t = (-\eta\zeta_t, 0)^\top$ independent of X_t. By Proposition 5.8 the spectral radius of A is below 1, so $\mathbb{E}X_t = A^t\mathbb{E}X_0 \to 0$, and $\mathbb{E}X_tX_t^\top$ converges to the unique solution of $\Sigma = A\Sigma A^\top + \operatorname{diag}(\eta^2\sigma^2, 0)$ (Lemma A.15). Write $\Sigma = \left(\begin{smallmatrix} V & c \\ c & V' \end{smallmatrix}\right)$. The entry $(2,2)$ gives $V' = V$; the entry $(1,2)$ gives $c = aV - \beta c$, so $c = aV/(1+\beta)$; and the entry $(1,1)$ gives $V = a^2V - 2a\beta c + \beta^2 V + \eta^2\sigma^2$. Substituting c and factoring,

$$\eta^2\sigma^2 = V\Big(1 - \beta^2 - \frac{1-\beta}{1+\beta}\,a^2\Big) = V\,\frac{1-\beta}{1+\beta}\,(1+\beta-a)(1+\beta+a),$$

and $1+\beta-a = \eta h$, $1+\beta+a = 2(1+\beta) - \eta h$ give (5.2).

(ii) Gradient descent is the case $\beta = 0$ of (i), stable when $\eta h/(1-\beta) < 2$; with step $\eta/(1-\beta)$ the second form of (5.2) becomes the stated expression, which is at least V because $1-\beta \le 1+\beta$. Both are $\frac{\eta\sigma^2}{2h(1-\beta)}(1 + O(\eta h/(1-\beta)))$.

(iii) Write $b = 1-\beta$, $u = \eta h$ and $z = 1-\delta$. The characteristic polynomial becomes $\delta^2 - (b+u)\delta + u$, whose discriminant $(b+u)^2 - 4u = (b^2-4u) + 2bu + u^2$ is positive when $u \le b^2/4$. Both δ-roots are positive, their sum and product being so; let δ be the smaller, so $z_+ = 1-\delta$ and $\delta \le (b+u)/2 < 1$. From $\delta(b-\delta) = u(1-\delta) > 0$ we get $\delta < b$, and then $\delta = u(1-\delta)/(b-\delta) \ge u/b$ because $b(1-\delta) \ge b - \delta$. Also $\delta = 2u/\big(b + u + \sqrt{(b+u)^2 - 4u}\big) \le 2u/b$, so $\delta \le u/(b-\delta) \le u/(b - 2u/b)$. The mean is a combination of z_+^t and z_-^t.

(iv) By Lemma 5.5(ii) this is heavy ball with step $\eta(1-\beta)$ and the same noise; substitute into (i), and compare with (i) at $\beta = 0$. ∎

In the noisy quadratic model each coordinate is such a recursion, with noise variance $c_i = v_i^\top C v_i$, so heavy ball's stationary excess loss is $\sum_i \frac{\eta c_i}{4(1-\beta)}\big(1 - \frac{\eta h_i}{2(1+\beta)}\big)^{-1}$ (Chapter 11). Where the noise dominates and $\eta h \ll (1-\beta)^2$, heavy ball is gradient descent with step $\eta/(1-\beta)$, and in Adam's convention momentum changes nothing at first order. Where the step is limited by stability, heavy ball stays stable up to the effective step $2(1+\beta)/((1-\beta)h)$, gradient descent only up to $2/h$ (Figure 5.1).

5.4 The theory: normalized steps with memory

We drop the quadratic and the linearity. The algorithm is Definition 5.4(v): $m_t = \beta m_{t-1} + (1-\beta)g_t$ from a given m_{-1}, and a step of length at most η along a steepest unit direction u_t of the average, a minimizer of $\langle m_t, u\rangle$ over the unit ball (Chapter 1), chosen as a measurable function of m_t; when $m_t = 0$, every point of the unit ball is allowed. The first lemma says that a steepest direction of m is a good direction for g when m is close to g.

Lemma 5.11 (A steepest direction of a nearby vector). *For all $m, g \in \mathbb{R}^d$ and every u with $\|u\| \le 1$ and $\langle m, u\rangle = -\|m\|_*$,*

$$\langle g, u\rangle \le -\|m\|_* + \|m - g\|_* \le -\|g\|_* + 2\|m - g\|_*.$$

PROOF. Write $\langle g, u\rangle = \langle m, u\rangle + \langle g - m, u\rangle$ and bound the second term by $\|g - m\|_*\|u\| \le \|g - m\|_*$ (Hölder's inequality, Lemma A.3); then use $\|m\|_* \ge \|g\|_* - \|m - g\|_*$. The argument includes the case $m = 0$, in which every u in the unit ball qualifies. ■

The second lemma bounds the *momentum error* $\delta_t = m_t - \nabla\mathcal{L}(\theta_t)$.

Lemma 5.12 (The error of the average). *Under Assumption 5.7, let $m_t = \beta m_{t-1} + (1-\beta)g_t$ for $t \ge 0$, along iterates with $\|\theta_{t+1} - \theta_t\| \le \eta$, and let $\delta_{-1} = m_{-1} - \nabla\mathcal{L}(\theta_0)$. Then for every $t \ge 0$,*

$$\mathbb{E}\|\delta_t\|_* \le \beta^{t+1}\|\delta_{-1}\|_* + \frac{\beta L\eta}{1-\beta} + \chi\sigma\sqrt{\frac{1-\beta}{1+\beta}}.$$

PROOF. The plan is to unroll δ_t into a decaying initial error, a lag and a noise average. With $\theta_{-1} = \theta_0$, inserting $g_t = \nabla\mathcal{L}(\theta_t) + \zeta_t$ into the definition of m_t gives $\delta_t = \beta\delta_{t-1} + \beta(\nabla\mathcal{L}(\theta_{t-1}) - \nabla\mathcal{L}(\theta_t)) + (1-\beta)\zeta_t$ for $t \ge 0$, and so

$$\delta_t = \beta^{t+1}\delta_{-1} + \sum_{s=1}^{t}\beta^{t-s+1}\big(\nabla\mathcal{L}(\theta_{s-1}) - \nabla\mathcal{L}(\theta_s)\big) + (1-\beta)N_t, \qquad N_t = \sum_{s=0}^{t}\beta^{t-s}\zeta_s.$$

Each gradient difference has dual norm at most $L\|\theta_{s-1} - \theta_s\| \le L\eta$, and $\sum_{s\le t}\beta^{t-s+1} \le \beta/(1-\beta)$. For the noise, $\mathbb{E}\|N_t\|_* \le \chi\mathbb{E}\|N_t\|_2 \le \chi(\mathbb{E}\|N_t\|_2^2)^{1/2}$, and the cross terms of $\mathbb{E}\|N_t\|_2^2$ vanish: for $r < s$, ζ_r is $\mathcal{F}_s$-measurable, so $\mathbb{E}\langle\zeta_r, \zeta_s\rangle = \mathbb{E}\langle\zeta_r, \mathbb{E}[\zeta_s \mid \mathcal{F}_s]\rangle = 0$. Hence $\mathbb{E}\|N_t\|_2^2 \le \sigma^2/(1-\beta^2)$, and $(1-\beta)\chi\sigma/\sqrt{1-\beta^2} = \chi\sigma\sqrt{(1-\beta)/(1+\beta)}$. ■

The average trades one error for another: its noise falls like $\sqrt{1-\beta}$, since it averages about $1/(1-\beta)$ gradients, and its lag grows like $1/(1-\beta)$, since they were computed up to $1/(1-\beta)$ steps away.

Theorem 5.13 (Normalized steepest descent with momentum). *Under Assumption 5.7, let $0 \le \beta < 1$, $\eta > 0$ and $m_{-1} \in \mathbb{R}^d$, and run normalized steepest descent with momentum. Then for every $T \ge 1$, with $\Delta_0 = \mathcal{L}(\theta_0) - \mathcal{L}^\star$,*

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\|\nabla\mathcal{L}(\theta_t)\|_* \le \frac{\Delta_0}{\eta T} + \frac{L\eta}{2} + \frac{2\beta L\eta}{1-\beta} + 2\chi\sigma\sqrt{\frac{1-\beta}{1+\beta}} + \frac{2\beta\|m_{-1} - \nabla\mathcal{L}(\theta_0)\|_*}{(1-\beta)T}. \tag{5.3}$$

PROOF. The plan is to bound the decrease of the loss at each step with Lemma 5.11, to add up, and to control the accumulated error with Lemma 5.12. By L-smoothness and the descent lemma (Lemma A.18), since $\|u_t\| \le 1$, and then by Lemma 5.11 with $m = m_t$ and $g = \nabla\mathcal{L}(\theta_t)$,

$$\mathcal{L}(\theta_{t+1}) \le \mathcal{L}(\theta_t) + \eta\langle\nabla\mathcal{L}(\theta_t), u_t\rangle + \frac{L\eta^2}{2} \le \mathcal{L}(\theta_t) - \eta\|\nabla\mathcal{L}(\theta_t)\|_* + 2\eta\|\delta_t\|_* + \frac{L\eta^2}{2}.$$

Sum over $t < T$, use $\mathcal{L}(\theta_T) \geq \mathcal{L}^\star$ and take expectations:

$$\eta \sum_{t<T} \mathbb{E}\|\nabla\mathcal{L}(\theta_t)\|_* \leq \Delta_0 + 2\eta \sum_{t<T} \mathbb{E}\|\delta_t\|_* + \frac{TL\eta^2}{2}.$$

The steps have length at most η, so Lemma 5.12 applies, and with $\sum_{t<T}\beta^{t+1} \leq \beta/(1-\beta)$ it gives

$$\sum_{t<T} \mathbb{E}\|\delta_t\|_* \leq \frac{\beta\|\delta_{-1}\|_*}{1-\beta} + T\Big(\frac{\beta L\eta}{1-\beta} + \chi\sigma\sqrt{\frac{1-\beta}{1+\beta}}\Big).$$

Insert this and divide by ηT. ∎

Corollary 5.14 (Rates). *(i) If $\Delta_0 > 0$, then with $1-\beta = T^{-1/2}$ and $\eta = (\Delta_0/L)^{1/2}T^{-3/4}$ the right side of (5.3) is at most $\big(3\sqrt{L\Delta_0} + 2\chi\sigma\big)T^{-1/4} + 2\|m_{-1} - \nabla\mathcal{L}(\theta_0)\|_* T^{-1/2} + \frac{1}{2}\sqrt{L\Delta_0}\,T^{-3/4}$. (ii) Without momentum, $\beta = 0$, it is $\Delta_0/(\eta T) + L\eta/2 + 2\chi\sigma$, whose noise term does not decrease with η or T.*

Proof. In (i), $\Delta_0/(\eta T) = \sqrt{L\Delta_0}\,T^{-1/4}$, $L\eta/2 = \frac{1}{2}\sqrt{L\Delta_0}\,T^{-3/4}$, $2\beta L\eta/(1-\beta) \leq 2L\eta T^{1/2} = 2\sqrt{L\Delta_0}\,T^{-1/4}$, $2\chi\sigma\sqrt{(1-\beta)/(1+\beta)} \leq 2\chi\sigma T^{-1/4}$ and $2\beta/((1-\beta)T) \leq 2T^{-1/2}$. Part (ii) is (5.3) at $\beta = 0$. ∎

So $\mathbb{E}\|\nabla\mathcal{L}\|_* \leq \varepsilon$ takes $O(\varepsilon^{-4})$ steps with no large batches, which in the Euclidean norm is the best possible order under these assumptions (Arjevani et al. 2023). The floor $2\chi\sigma$ of (ii) is real.

Proposition 5.15 (Without memory, a floor of the size of the noise). *Let $\mathcal{L}(\theta) = \theta^2/2$ on $\mathbb{R}$, $b > 0$, and $\zeta_0, \zeta_1, \ldots$ independent with $\Pr[\zeta_t = 3b] = \frac{1}{4}$ and $\Pr[\zeta_t = -b] = \frac{3}{4}$, so that $\mathbb{E}\zeta_t = 0$ and $\sigma^2 = 3b^2$. Run normalized steepest descent without momentum: $\theta_{t+1} = \theta_t + \eta u_t$, with $u_t = -\operatorname{sign}(\theta_t + \zeta_t)$ if $\theta_t + \zeta_t \neq 0$ and $u_t \in [-1,1]$ otherwise. Then for every $\eta > 0$, θ_0 and $t \geq 0$,*

$$\mathbb{E}\big[(b-\theta_t)_+\big] \leq \eta \log_2\Big(\big(\tfrac{7}{8}\big)^t 2^{(b-\theta_0)_+/\eta} + 32\Big),$$

and consequently $\liminf_{t\to\infty} \mathbb{E}|\nabla\mathcal{L}(\theta_t)| \geq b - 5\eta = \sigma/\sqrt{3} - 5\eta$.

Proof. The plan is to show that below b the iterate is a random walk with drift toward b, and to control its distance below b with an exponential supermartingale. Let $V_t = (b-\theta_t)_+$ and $W_t = 2^{V_t/\eta}$; we show $\mathbb{E}[W_{t+1} \mid \theta_t] \leq \frac{7}{8}W_t + 4$. If $-3b < \theta_t \leq b - \eta$, then the step is $-\eta$ when $\zeta_t = 3b$ (probability $\frac{1}{4}$, since $\theta_t + 3b > 0$) and $+\eta$ when $\zeta_t = -b$ (probability $\frac{3}{4}$, since $\theta_t - b < 0$), and in both cases $\theta_{t+1} \leq b$; so $W_{t+1} \in \{2W_t, W_t/2\}$ and $\mathbb{E}[W_{t+1} \mid \theta_t] = \frac{7}{8}W_t$. If $\theta_t \leq -3b$, then the step is $+\eta$ with probability at least $\frac{3}{4}$, which gives $W_{t+1} \leq \max(W_t/2, 1)$, and otherwise $W_{t+1} \leq 2W_t$; so $\mathbb{E}[W_{t+1} \mid \theta_t] \leq \frac{7}{8}W_t + \frac{3}{4}$. If $b - \eta < \theta_t < b$, then $V_{t+1} \leq V_t + \eta < 2\eta$ and $W_{t+1} < 4$. If $\theta_t \geq b$, then $\theta_{t+1} \geq \theta_t - \eta \geq b - \eta$ and $W_{t+1} \leq 2$. Taking expectations, $\mathbb{E}W_{t+1} \leq \frac{7}{8}\mathbb{E}W_t + 4$, so $\mathbb{E}W_t \leq (\frac{7}{8})^t W_0 + 32$, and Jensen's inequality for $v \mapsto 2^{v/\eta}$ gives $2^{\mathbb{E}V_t/\eta} \leq \mathbb{E}W_t$, the first claim. Its right side tends to $\eta\log_2 32 = 5\eta$, and $|\nabla\mathcal{L}(\theta_t)| = |\theta_t| \geq b - V_t$. ∎

In one dimension every normalized step is a sign step, of a length set by the norm, so the proposition covers all norms. Its mechanism is a *bias of the normalized step*: for $0 < \theta < b$ the noise, usually slightly negative and occasionally very positive, gives the stochastic gradient the wrong sign three times in four, and the expected step $+\eta/2$ points uphill. A small step averages the signs, and the average of the signs is what is wrong; momentum takes the sign of an average, and by Theorem 5.13 the floor drops to $\eta/2 + 2\beta\eta/(1-\beta) + 2\sigma\sqrt{(1-\beta)/(1+\beta)}$. The bias needs asymmetric noise: with symmetric noise a normalized step never points uphill on average, in every norm (Exercise 5.6).

5.4.1 Averaging the iterates

Model averages and schedule-free methods average iterates rather than gradients. Averaging the iterates of stochastic approximation is asymptotically optimal (Polyak and Juditsky 1992), and for convex losses it can replace a decaying schedule if the gradient is evaluated at the running average.

Proposition 5.16 (Averaging instead of a schedule; Cutkosky 2019). *Let F be convex on $\mathbb{R}^d$, and let an oracle return at each query point x a vector $\tilde{g}$ whose conditional mean given the past is a subgradient of F at x and whose conditional second moment $\mathbb{E}\|\tilde{g}\|_2^2$ is at most G^2. Let $z_1 = x_1$ and, for $t \geq 1$, query at x_t to get $\tilde{g}_t$, then set $z_{t+1} = z_t - \gamma\tilde{g}_t$ and $x_{t+1} = \frac{t}{t+1}x_t + \frac{1}{t+1}z_{t+1}$. Then for every $x^\star$ and $T \geq 1$,*

$$\mathbb{E}F(x_T) - F(x^\star) \leq \frac{\|z_1 - x^\star\|_2^2}{2\gamma T} + \frac{\gamma G^2}{2},$$

which is $DG/\sqrt{T}$ for $\gamma = D/(G\sqrt{T})$ and $D = \|z_1 - x^\star\|_2$.

Proof. The plan is to bound $T(F(x_T) - F(x^\star))$ by the linear regret of the sequence z_t, and the regret as for online gradient descent. Let g_t be the conditional mean of $\tilde{g}_t$, a subgradient at x_t. Since $tx_t = (t-1)x_{t-1} + z_t$, we have $z_t - x^\star = (x_t - x^\star) + (t-1)(x_t - x_{t-1})$, and convexity gives

$$\begin{aligned}\langle g_t, z_t - x^\star\rangle &\geq F(x_t) - F(x^\star) + (t-1)\big(F(x_t) - F(x_{t-1})\big)\\ &= t\big(F(x_t) - F(x^\star)\big) - (t-1)\big(F(x_{t-1}) - F(x^\star)\big),\end{aligned}$$

which telescopes to $\sum_{t\leq T}\langle g_t, z_t - x^\star\rangle \geq T(F(x_T) - F(x^\star))$. Expanding $\|z_{t+1} - x^\star\|_2^2$ gives $\sum_{t\leq T}\langle\tilde{g}_t, z_t - x^\star\rangle \leq \|z_1 - x^\star\|_2^2/(2\gamma) + \frac{\gamma}{2}\sum_{t\leq T}\|\tilde{g}_t\|_2^2$ (Theorem A.21), and $\mathbb{E}\langle\tilde{g}_t, z_t - x^\star\rangle = \mathbb{E}\langle g_t, z_t - x^\star\rangle$ because z_t is determined by the past. Taking expectations of both bounds gives the claim. ∎

This is *schedule-free* SGD with $\beta = 1$; with $\beta = 0$ it is SGD with Polyak–Ruppert averaging, and the bound follows from Jensen's inequality. Defazio et al. (2024) prove the bound $DG/\sqrt{T}$, for G-Lipschitz stochastic losses, at every interpolation $y_t = (1-\beta)z_t + \beta x_t$, $\beta \in [0,1]$: a constant step with a uniform average matches a well-decayed step (Chapter 12).[7] A *moving average of the weights*, $\bar{\theta}_t = \beta_e\bar{\theta}_{t-1} + (1-\beta_e)\theta_t$, acts differently (Exercise 5.8).

[7] The AlgoPerf benchmark of the schedule-free field guide (Dahl et al. 2023) measures the time to reach target metrics on a fixed set of workloads; its self-tuning track admits no workload-specific hyperparameter search.

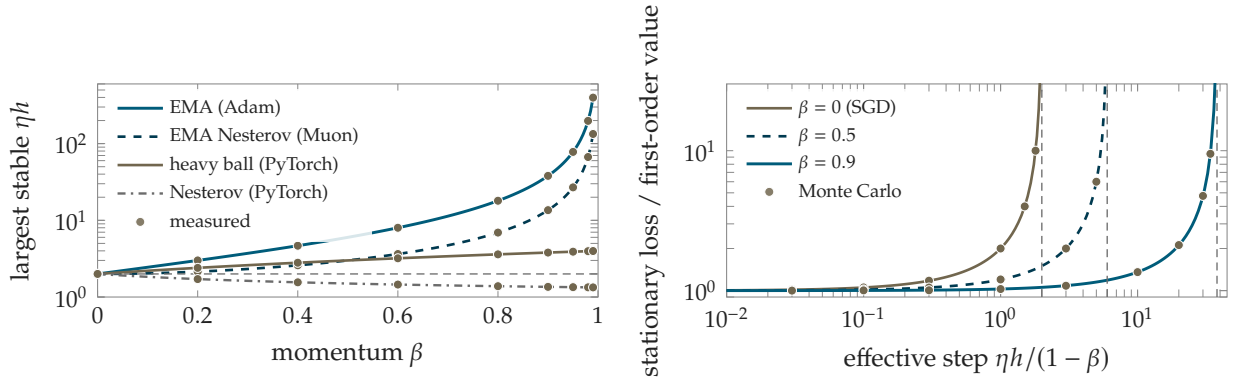


Figure 5.1. The linear theory of momentum. Left: the thresholds of Proposition 5.8 (curves) and those measured by running each method on $x^2/2$ for 401 step sizes near the prediction, where the growth rate of the state over 8000 steps turns nonnegative (points, within 10^{-4}); dashed: 2. Right: heavy ball's stationary loss over its first-order value $\eta\sigma^2/(4(1-\beta))$, from (5.2) (curves) and Monte Carlo with 2000 chains, 2000 burn-in and 6000 measured steps, $h = \sigma = 1$, seed 0 (points, within 0.3%); dashed: stability limits 2, 6, 38. Scripts `05-momentum-thresholds.py`, `05-momentum-variance.py`.

Heuristic derivation. For a coordinate that relaxes in fewer steps than the window $1/(1-\beta_e)$, the weight average has the variance of gradient descent with step $(1-\beta_e)/h$, as after a cooldown; a slower coordinate is only delayed, by $\beta_e/(1-\beta_e)$ steps. A weight average is a cooldown of the stiff coordinates only. ◇

5.5 Reading the theory

Thresholds and acceleration. Proposition 5.8 limits ηS, for the sharpness S, to 2 for gradient descent, 3.8 for PyTorch's SGD at $\beta = 0.9$, 38 for Adam-style momentum, 13.6 for Nesterov momentum in EMA form (26.9 at Muon's $\beta = 0.95$), and 1.36 for PyTorch's Nesterov (Figure 5.1, left). Inverting Polyak's choice, $\beta = 0.9$ is optimal for $\kappa \approx 1.4 \cdot 10^3$. At $\kappa = 10^3$ heavy ball reaches 10^{-8} of the initial loss in about 220 steps, gradient descent in about 3900; with noise the order reverses, because Polyak's parameters put the stiffest coordinate near the stability limit, where (5.2) is large (Exercise 5.9).

Noise. In Figure 5.1 (right) all curves start at 1: when the noise dominates, momentum β with step η is SGD with step $\eta/(1-\beta)$ (Observation 5.1). They separate only near SGD's stability limit 2, which momentum pushes out to $2(1+\beta)/(1-\beta)$. At small batches the best step is far below that limit and momentum offers nothing a larger learning rate does not; at large batches the best step is the limit, and momentum raises it (Chapter 11).

Normalized steps: bias and fluctuations. Theorem 5.13 is a worst-case bound. Two examples that can be analyzed closely show how typical problems differ: Proposition 5.15 with momentum ($h\theta^2/2$ on $\mathbb{R}$, skewed

noise, $\sigma = 1$), and, at the other extreme, $h\|\theta\|_2^2/2$ in dimension 1000 with isotropic Gaussian noise and ℓ_2-normalized steps.

HEURISTIC DERIVATION. *One dimension.* Write $m_t = h\tilde{\theta}_t + \bar{\zeta}_t$: the average $\tilde{\theta}_t$ of the iterates is within $\beta\eta/(1-\beta)$ of θ_t, and the average $\bar{\zeta}_t$ of the noise is stationary, with correlation time about $1/(1-\beta)$ and the law of $\bar{\zeta} = (1-\beta)\sum_{k\ge 0}\beta^k\zeta_k$. For small η the steps average, over many correlation times, to $-\eta(1-2F(-h\theta))$, with F the distribution function of $\bar{\zeta}$; the drift vanishes where $F(-h\theta) = \frac{1}{2}$, so as $\eta \to 0$ the stationary gradient concentrates at the *bias*

$$\mathbb{E}\nabla\mathcal{L}(\theta) \to -\operatorname{med}(\bar{\zeta}), \tag{5.4}$$

independent of η and h; without memory it is $b = \sigma/\sqrt{3}$. With memory $\bar{\zeta}$ is nearly Gaussian, with variance $\kappa_2 = \sigma^2(1-\beta)/(1+\beta)$ and third cumulant $\kappa_3 = \mathbb{E}[\zeta^3](1-\beta)^3/(1-\beta^3)$, and the Cornish–Fisher estimate $\operatorname{med}(\bar{\zeta}) \approx -\kappa_3/(6\kappa_2)$ makes the bias about $\mathbb{E}[\zeta^3](1-\beta)(1+\beta)/(6\sigma^2(1+\beta+\beta^2))$. Around the rest point θ_c the drift is $-2\eta h f(\theta-\theta_c)$, with f the density of $\bar{\zeta}$ at its median, and the signs have integrated autocovariance D, so a diffusion approximation gives $\operatorname{Var}\theta \approx \eta D/(4hf)$. For Gaussian $\bar{\zeta}$, $f = (2\pi\kappa_2)^{-1/2}$ and $D = 1 + \frac{4}{\pi}\sum_{k\ge 1}\arcsin\beta^k \approx 2\ln 2/(1-\beta)$, so the *fluctuations* are

$$\operatorname{Var}\nabla\mathcal{L}(\theta) \approx \tfrac{1}{4}\sqrt{2\pi\kappa_2}\,h\eta D \approx \tfrac{1}{2}\sqrt{2\pi}\ln 2\,\frac{h\eta\sigma}{\sqrt{1-\beta^2}}. \tag{5.5}$$

High dimension. With isotropic Gaussian noise, $\mathbb{E}\|\zeta\|_2^2 = \sigma^2$, the norm $\|m_t\|_2$ concentrates as $d \to \infty$ at a constant c; replacing it by c makes the iteration EMA momentum with step η/c, whose covariance (proof of Theorem 5.10) gives the second moments of θ_t and $m_t = c(\theta_t - \theta_{t+1})/\eta$. The condition $\mathbb{E}\|m_t\|_2^2 = c^2$ is $2(1+\beta)c^2 - \eta(1-\beta)hc - 2(1-\beta)\sigma^2 = 0$, and its positive root gives, approximately for $\eta h \ll \sigma$,

$$\mathbb{E}\|\nabla\mathcal{L}(\theta)\|_2^2 = \frac{h\eta c(1+\beta)}{2(1-\beta)} \approx \frac{h\eta\sigma}{2}\sqrt{\frac{1+\beta}{1-\beta}}. \quad \diamond \tag{5.6}$$

In one dimension (Figure 5.2, left) the mean stationary gradient follows the limit (5.4), from 0.577 without momentum to $6\cdot 10^{-4}$ at $\beta = 0.995$:[8] memory shrinks the bias linearly in $1-\beta$, much faster than the $\sqrt{1-\beta}$ of Theorem 5.13, which bounds the error of the average, not the bias of its sign. Over all laws of variance σ^2, though, $\sqrt{1-\beta}$ is the right worst case: a law whose large values arrive less than once per memory has a bias of order $\sigma\sqrt{1-\beta}$ (Exercise 5.7). The standard deviation follows (5.5) for $\beta \ge 0.9$ and grows like $\eta^{1/2}(1-\beta)^{-1/4}$, because a longer memory makes sign errors persist. The two cross near $1-\beta \approx 0.1$, and balancing them suggests a best $1-\beta \propto (h\eta/\sigma)^{2/5}$ here (Exercise 5.10), where the bound would balance its lag and noise at $1-\beta \approx 2(L\eta/(\chi\sigma))^{2/3}$. In high dimension (right) the noise is symmetric and there is no bias to remove; the floor (5.6), which matches the simulations to 0.2%, rises with β at a fixed η for the reason Theorem 5.10 gave for SGD: when $\|m_t\|$ concentrates, a normalized step is a linear step of size $\eta/\|m_t\|$, and averaging shrinks $\|m_t\|$ by the factor $\sqrt{(1-\beta)/(1+\beta)}$.

[8] The Cornish–Fisher estimate used here is the rule that the mean minus the median of a nearly Gaussian law is about one sixth of its skewness times its standard deviation. For $\beta \ge 0.98$ it is within about 1% of the exact bias.

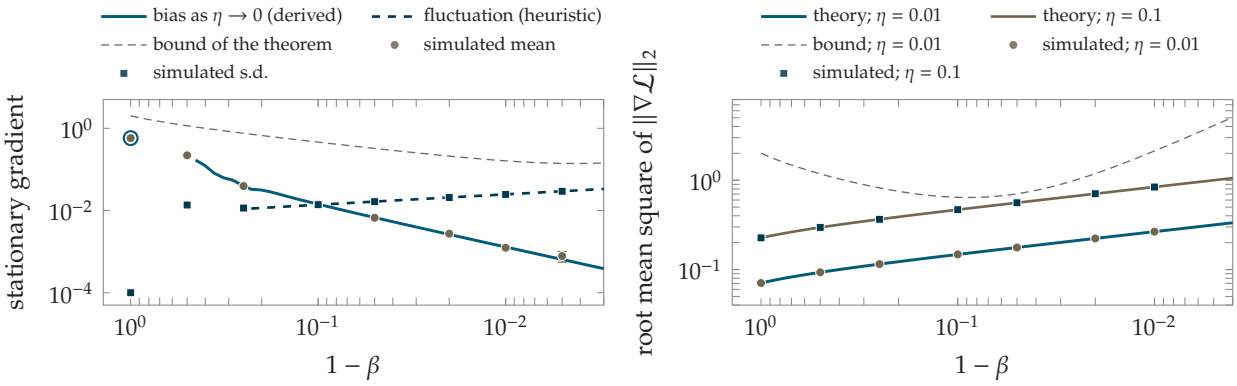


Figure 5.2. Normalized steps with memory; $1-\beta$ decreases to the right. Left: sign steps with momentum on $\theta^2/2$ with the skewed noise of Proposition 5.15, $\sigma = 1$, $\eta = 10^{-4}$; 2000 chains (4000 for $\beta \geq 0.95$) from $\theta_0 = 0$, 25000 burn-in and 40000 measured steps, seed 0; points: mean (two standard errors) and standard deviation of the stationary gradient. Curves: the derived limit (5.4), with the median computed exactly from the fixed-point equation of the law of $\bar{\zeta}$ (for $\beta \geq \frac{1}{2}$; the circle is $\beta = 0$), the heuristic (5.5), and the stationary part of (5.3). Right: ℓ_2-normalized steps with momentum on $\|\theta\|_2^2/2$ in $\mathbb{R}^{1000}$, isotropic Gaussian noise, $\sigma = 1$; four runs, 3000 burn-in, 8000 measured steps, seed 0 (points), against (5.6) (curves) and the bound for $\eta = 0.01$ (dashed). Scripts `05-momentum-skew.py`, `05-momentum-isotropic.py`.

The two examples are the two halves of the answer to the question of Section 5.1: memory removes the bias of a nonlinear step, which no step size can remove; where the step is effectively linear, memory is again a rescaled step size, paid for with a smaller η as in Corollary 5.14. The agreement fails where expected. The bias (5.4) is a limit $\eta \to 0$: at $\beta = 0.99$ the simulated mean falls about an eighth below it at $\eta = 10^{-3}$ and more than a quarter below at $\eta = 3 \cdot 10^{-3}$, where the lag $\beta\eta/(1-\beta)$, 0.1 and 0.3, is no longer small against the spread 0.07 of $\bar{\zeta}$. The Gaussian formula (5.5) fails for small β (at $\beta = 0$ the iterate lives on a lattice next to b), and the high-dimensional formula has corrections of relative order $1/d$.

5.6 What the theory misses

WHAT WAS IDEALIZED AWAY. The quadratic settings fix the Hessian; in a network the curvature responds to the step size, and momentum's wider stability region is not a margin of safety but the place where training settles (Observation 5.3). The noise is additive and position-independent, while minibatch noise near a minimum scales roughly with the curvature ($C \propto H$ in Chapter 11); and it has finite variance, which gradient noise in deep networks may not have (Şimşekli et al. 2019). The normalized theorem has worst-case constants, and Figure 5.2 shows it overstating the floor, often by more than a factor of ten, and, for symmetric noise, getting the sign of its dependence on β wrong. And β and η are constant, while practice decays η and sometimes warms β up (Sutskever et al. 2013).

WHERE IT BREAKS. (1) Heavy ball's acceleration is a quadratic phenomenon:[9] on a strongly convex function made of three quadratic pieces, heavy ball with Polyak's parameters settles on a cycle of period three (Lessard et al. 2016), and with noise the accelerated parameters raise the floor. It is not why deep learning uses momentum: the classical schemes do not accelerate SGD on some stochastic least-squares problems (Kidambi et al. 2018), and Theorem 5.10 shows why. (2) Corollary 5.14 prescribes $1-\beta \propto T^{-1/2}$, that is β between 0.999 and 0.9999 for runs of 10^6 to 10^8 steps, while practice keeps $\beta_1 = 0.9$ for runs of every length; the theory does not say whether the worst-case balance of noise and lag is the wrong criterion or the losses of practice are far from the worst case. (3) With heavy-tailed noise an average of $1/(1-\beta)$ gradients concentrates more slowly than $\sqrt{1-\beta}$, and the variance assumption fails (Exercise 5.13). (4) The uphill bias of Proposition 5.15 needs skewed noise: with symmetric noise a memoryless normalized step is never biased uphill, for Gaussian noise its floor vanishes as $\eta \to 0$, and memory raises the floor at a fixed step (Figure 5.2, right). The theory explains Observation 5.2 only insofar as real gradient noise is asymmetric in the coordinates a normalized step treats separately; as far as we know, nobody has measured how much of the need for memory comes from bias and how much from the effective step. (5) In Adam's convention β_1 changes neither the floor nor the rate of the flat coordinates of a linear method to first order, so its effects must come from elsewhere: the normalization by $\sqrt{v_t}$, where memory can remove a bias, or the edge of stability (Chapter 8).

[9] Lessard et al. (2016) treat a first-order method as a feedback system and certify its rate with integral quadratic constraints, which reduce the analysis to small semidefinite programs.

REPAIRS AND OPEN PROBLEMS. Variance reduction corrects the average for its lag: STORM (Cutkosky and Orabona 2019) adds the gradient difference between consecutive points and, if each stochastic loss is smooth, reaches $\mathbb{E}\|\nabla\mathcal{L}\| \le \varepsilon$ in $\tilde{O}(\varepsilon^{-3})$ steps. AdEMAMix (Pagliardini et al. 2025) mixes a fast average (0.9) with a very slow one (about 0.9999) and reports that gradients tens of thousands of steps old still help language-model training, which the worst-case lag term does not anticipate. Averaging iterates rather than gradients, in schedule-free methods (Defazio et al. 2024) and weight averages (Izmailov et al. 2018), attacks the floor without lagging the step. Open: a theory that predicts 0.9 or explains why it transfers across scales; a theory of normalized momentum under heavy-tailed, state-dependent noise that is sharp for typical problems; and measurements of how biased the normalized steps of real optimizers are.

5.7 Back in the wild

FIELD NOTES	
PRACTICE	Average the gradients with $\beta \approx 0.9$ (SGD, Adam's β_1, Lion), 0.95 in Muon; every normalized optimizer carries momentum.
IDEALIZATION	Quadratics with exact or additively noisy gradients; normalized steepest descent in a general norm, smooth loss, bounded-variance noise.

PROVED	Polyak's acceleration with explicit constants; stability regions of heavy ball, Nesterov and EMA momentum; the exact stationary variance; a $T^{-1/4}$ rate for normalized steepest descent with momentum in every norm; a floor of order σ without memory; averaging in place of a schedule.
PREDICTS	Thresholds 3.8, 38, 1.36 at $\beta = 0.9$; momentum as the step $\eta/(1-\beta)$ when noise dominates; for normalized steps, a bias falling like $1-\beta$ and fluctuations growing like $\eta^{1/2}(1-\beta)^{-1/4}$ (derived).
MISSES	Curvature that responds to the step, state-dependent and heavy-tailed noise, schedules of β, and a reason for the value 0.9.
TAKEAWAY	Proved in the idealization: in PyTorch's convention retune η by $1-\beta$ when β changes; keep memory in normalized optimizers: it removes a bias no step size can. Suggested: raise β as the step falls. Folklore: $\beta = 0.9$ everywhere.

What the theory tells a practitioner is mostly bookkeeping, and it is proved. In PyTorch's convention the effective learning rate is $\eta/(1-\beta)$, so a sweep over β at fixed η is mostly a sweep over the learning rate; in Adam's convention it is not, and in the linear theory β_1 acts mainly on stability. For normalized optimizers memory is not optional: it removes a bias rather than lowering the floor at a fixed step, and when β goes up, η should come down, as $(1-\beta)^{3/2}$ if the worst case of Corollary 5.14 is a guide. Why 0.9 is right so often remains folklore with a good track record.

The thresholds return in Chapter 8, and the noisy quadratic model in Chapter 11, where momentum moves the critical batch size. The bias of normalized steps connects to sign descent (Chapter 2) and Muon (Chapter 3); Adam's β_1 is the same average with a preconditioner (Chapter 6); a weight average competes with a cooldown in Chapter 12; and weight decay makes Lion and Muon constrained methods (Chapter 13).

Exercises

5.1 CONSOLIDATION. (Polyak's parameters are optimal.) (a) Show that the roots of $z^2 - (1+\beta-\eta h)z + \beta$ are complex or double exactly when $(1-\sqrt{\beta})^2 \le \eta h \le (1+\sqrt{\beta})^2$, with modulus $\sqrt{\beta}$; at $\beta = 0.9$ the band is $[0.0026, 3.797]$. (b) Show that for all $\eta' > 0$ and $\beta' \in \mathbb{R}$ heavy ball has a characteristic root of modulus at least the r of Theorem 5.9 at $h = \mu$ or $h = L$. *(Hint: if all four roots have modulus at most ρ, then the Jury test applied to $w^2 - (a_h/s)w + \beta'/s^2$, $s > \rho$, $a_h = 1 + \beta' - \eta' h$, gives $|a_h| \le \rho + \beta'/\rho$ and $|\beta'| \le \rho^2$; bound $\eta'\mu$ below and $\eta' L$ above and divide.)*

5.2 CONSOLIDATION. (More regions.) (a) Verify Lemma 5.5(iv) for PyTorch's Nesterov momentum. (b) For $u_t = \beta_a m_{t-1} + (1-\beta_a)g_t$, $m_t = \beta_b m_{t-1} + (1-\beta_b)g_t$ and $0 \le \beta_b < \beta_a < 1$, show that the threshold for ηh is $\min\{(1-\beta_b)/(\beta_a - \beta_b),\ 2(1+\beta_b)/(1+\beta_b-2\beta_a)\}$, omitting the second term when $1 + \beta_b \le 2\beta_a$.

5.3 CONSOLIDATION. (Bias correction.) Adam divides m_t by $1-\beta^{t+1}$. Show that this makes the first step $-\eta g_0$, rescales a linear method's steps by a factor tending to one over about $1/(1-\beta)$ steps, and leaves normalized steepest descent with momentum unchanged.

5.4 CONSOLIDATION. (Tuning the bound.) (a) Choose $\eta = \sqrt{\Delta_0(1-\beta)/(2LT)}$ and $1-\beta = \sqrt{2L\Delta_0}/(\chi\sigma\sqrt{T})$, assumed at most 1. Show that the first four terms of (5.3) sum to at most $L\eta/2$ plus $4(2L\Delta_0)^{1/4}(\chi\sigma)^{1/2}T^{-1/4}$. (b) For fixed η and β near 1, show that $2L\eta/(1-\beta) + \chi\sigma\sqrt{2(1-\beta)}$ is smallest at $1-\beta = 2(L\eta/(\chi\sigma))^{2/3}$.

5.5 VARIATION. (Lion and Nesterov in the theorem.) For steps along $\operatorname{lmo}(u_t)$ with $u_t = (1-\nu)g_t + \nu m_t$, show that Theorem 5.13 holds with the lag term $2\nu\beta L\eta/(1-\beta)$, the factor $\sqrt{(1-\nu\beta)^2 + \nu^2\beta^2(1-\beta)/(1+\beta)}$ in place of $\sqrt{(1-\beta)/(1+\beta)}$, and ν times the last term. Evaluate the factor and $2\nu\beta/(1-\beta)$ for Signum, $\beta = 0.9$ (0.229; 18), Lion, $\nu\beta = 0.9$ and $\beta = 0.99$ (0.119; 180), and Muon, $\nu = \beta = 0.95$ (0.174; 36.1).

5.6 VARIATION. (Symmetric noise.) (a) Suppose that ζ and $-\zeta$ have the same law. Show that $\mathbb{E}\langle a, u(a+\zeta)\rangle \le 0$ for every a, every norm and every measurable selection $u(g) \in \arg\min_{\|u\|\le 1}\langle g, u\rangle$. *(Hint: pair ζ with $-\zeta$ and use $\langle a \pm \zeta, u(a \pm \zeta)\rangle = -\|a \pm \zeta\|_*$.)* (b) For sign descent with independent Gaussian noise of standard deviations s_i, show that $\mathbb{E}\langle a, \operatorname{sign}(a+\zeta)\rangle = \sum_i |a_i| \operatorname{erf}\big(|a_i|/(\sqrt{2}s_i)\big)$, so that for $|a_i| \ll s_i$ the expected step is that of gradient descent preconditioned by $\sqrt{2/\pi}\ \operatorname{diag}(1/s_i)$.

5.7 VARIATION. ★ (The noise term cannot be improved.) Let $p = (1-\beta)/10$ and let ζ be $\sigma\sqrt{(1-p)/p}$ with probability p and $-\sigma\sqrt{p/(1-p)}$ otherwise. Show that $\mathbb{E}\zeta = 0$, $\mathbb{E}\zeta^2 = \sigma^2$ and $\operatorname{med}(\bar{\zeta}) \le -\sigma\sqrt{1-\beta}/6$, and explain with (5.4) why the factor $\sqrt{1-\beta}$ of Theorem 5.13 cannot be improved for all noise laws. *(Hint: condition on none of the last $\lceil 3/(1-\beta)\rceil$ samples being large (probability at least 0.6 by Bernoulli's inequality), and bound the older samples, of weight at most e^{-3}, with Chebyshev's inequality.)*

5.8 VARIATION. (A model average.) Let x_t be stationary gradient descent with step η and noise variance σ^2 on a coordinate with curvature h, and $\bar{x}_t = \beta_e\bar{x}_{t-1} + (1-\beta_e)x_t$. Show that $\operatorname{Var}\bar{x} = V\frac{1-\beta_e}{1+\beta_e}\cdot\frac{1+a\beta_e}{1-a\beta_e}$, with $a = 1-\eta h$ and $V = \eta\sigma^2/(h(2-\eta h))$; that this is close to $\sigma^2(1-\beta_e)/(2h^2)$ when $1-\beta_e \ll \eta h \ll 1$ and to V when $\eta h \ll 1-\beta_e$; and that a drifting coordinate is tracked with a lag of $\beta_e/(1-\beta_e)$ steps.

5.9 SIMULATION. (Acceleration drowns in noise.) On a quadratic in $\mathbb{R}^{100}$ with eigenvalues spaced geometrically from 10^{-3} to 1 and $\theta_0 - \theta^\star = (1,\dots,1)$, compare gradient descent with step $2/(L+\mu)$ and heavy ball with Polyak's parameters: without noise heavy ball reaches 10^{-8} of the initial loss in about 220 steps, gradient descent in about 3900; with Gaussian noise of standard deviation 10^{-2} per coordinate both stall,

heavy ball at a loss near 0.9 and gradient descent near 0.06. Derive both floors from Theorem 5.10.

5.10 SIMULATION. (The best memory at a fixed step.) With the script of Figure 5.2, measure $\mathbb{E}|\nabla\mathcal{L}|$ for $\eta \in \{10^{-3}, 10^{-4}, 10^{-5}\}$ and many β: the bias should be nearly independent of η while the lag $\beta\eta/(1-\beta)$ stays below the spread of $\bar{\zeta}$, the standard deviation should scale like $\eta^{1/2}$, and the best $1-\beta$ should shrink roughly like $\eta^{2/5}$.

5.11 SIMULATION. (Heavy ball beyond quadratics.) Let $F(x)$ be $\frac{25}{2}x^2$ for $x < 1$, $\frac{1}{2}x^2 + 24x - 12$ for $1 \le x < 2$, and $\frac{25}{2}x^2 - 24x + 36$ for $x \ge 2$, which is 1-strongly convex and 25-smooth. From $x_0 = x_{-1} = 3.3$, heavy ball with Polyak's $\eta = \frac{1}{9}$, $\beta = \frac{4}{9}$ settles on a cycle through about 0.6465, −1.8024, 2.1159, while Nesterov's method with $\eta = 1/L$ and $\beta = (\sqrt{\kappa}-1)/(\sqrt{\kappa}+1)$ converges.

5.12 RESEARCH. (Why 0.9?) The theory sends β to one as the run gets longer or the step smaller; practice keeps $\beta_1 = 0.9$ at every scale. Build a model in which a fixed β is optimal across scales, for instance at the edge of stability, where a larger β widens the stable range but slows the average's response to changing curvature (Chapter 8). As far as we know, no model predicts the value.

5.13 RESEARCH. (Heavy-tailed noise.) Let the noise be α-stable with $1 < \alpha < 2$, mean zero and nonzero skewness. (a) Show that $\bar{\zeta}$ is α-stable with the same skewness and scale multiplied by $(1-\beta)(1-\beta^\alpha)^{-1/\alpha} \approx \alpha^{-1/\alpha}(1-\beta)^{1-1/\alpha}$, so that by (5.4) the bias falls only like $(1-\beta)^{1-1/\alpha}$. (b) Open: a sharp theory of normalized momentum when the noise has only a p-th moment, $p < 2$.

Notes and References

Polyak (1964) introduced the heavy-ball method, Nesterov (1983) the accelerated method; both are in the textbooks of Polyak (1987) and Nesterov (2018). Sutskever et al. (2013) showed that momentum with a slowly increasing coefficient and a careful initialization trains deep and recurrent networks to levels previously reached only with second-order methods.

That momentum with noise is a larger step was shown several times: Kidambi et al. (2018) prove that the classical schemes do not accelerate SGD on some stochastic least-squares problems; Zhang et al. (2019) make the noisy quadratic model a tool for predicting which choices matter at which batch sizes. Theorem 5.10 is the exact computation behind them. SignSGD and Signum were analyzed by Bernstein et al. (2018) under large batches or unimodal symmetric noise, and Karimireddy et al. (2019) gave examples where sign compression fails. Cutkosky and Mehta (2020) proved the Euclidean case of Theorem 5.13 and Pethick et al. (2025) treat general norms; our proof is a short version of the argument, organized around Lemmas 5.11 and 5.12. The lower bound $\Omega(\varepsilon^{-4})$ is due to Arjevani et al. (2023). We know of no reference for Proposition 5.15 in this form,[10] or for

[10] Its noise law is that of the first counterexample of Karimireddy et al. (2019), in which sign descent with a fixed step increases a linear loss on an interval in expectation.

(5.4) and (5.6), which are derived, not proved. Iterate averaging goes back to Polyak and Juditsky (1992), and reached deep learning as stochastic weight averaging (Izmailov et al. 2018); Proposition 5.16 is the anytime online-to-batch conversion of Cutkosky (2019), which Defazio et al. (2024) turn into schedule-free methods. Cohen et al. (2022) noted Adam's threshold 38, and Ma and Yarats (2019) introduced quasi-hyperbolic momentum.

TRANSLATING THE LITERATURE. Heavy ball is usually written $x_{k+1} = x_k - \alpha\nabla f(x_k) + \beta(x_k - x_{k-1})$, with α our η. Cutkosky and Mehta (2020) write w_t for θ_t and α for $1-\beta$; Pethick et al. (2025) take the oracle over the ball of radius ρ, and their d_k, α_k, $\gamma_k\rho$ are our m_t, $1-\beta$, η. Sutskever et al. (2013) write Nesterov's method as $v_{t+1} = \mu v_t - \varepsilon\nabla f(\theta_t + \mu v_t)$, $\theta_{t+1} = \theta_t + v_{t+1}$, a momentum update with a look-ahead gradient: their μ is our β, their ε our η, and $\theta_t + \mu v_t$ our y_t. PyTorch's `momentum` is our β in the PyTorch forms of Definition 5.4, with `dampening` zero. Chen et al. (2023) write Lion with coefficients β_1, β_2 that are not Adam's: their β_2 is our β and their β_1 our $\nu\beta$. Ma and Yarats (2019) use our ν and β with step size α. In Defazio et al. (2024), β is an interpolation weight, not a momentum coefficient, and γ the step size; the field guide keeps their letters.

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6 Adaptive Preconditioning

AdaGrad, RMSProp and Adam

IN THE WILD — ADAM'S β_2 AND ϵ

IN THIS CHAPTER. Adam divides each coordinate of its step by a running root mean square of that coordinate's gradients; β_2 sets how fast the running average forgets, and ϵ the gradient size below which it stops dividing. These methods are steepest descent in a diagonal norm that the gradients choose. We prove that AdaGrad, whose norm never forgets, has a regret bound within a factor $\sqrt{2}$ of the bound for the best diagonal norm chosen in hindsight, and that its scalar version converges at every step size without knowing the smoothness constant. We then prove what forgetting costs: Adam converges to the worst point of a one-dimensional convex problem when its memory $1/(1-\beta_2)$ is short compared with the period of a rare large gradient, and after a quiet stretch a long memory gives steps of up to 6.6 step sizes at the default $\beta_1 = 0.9$, $\beta_2 = 0.999$, and an excess displacement of about $2\ln 2/(1-\beta_2)$ step sizes. Adafactor's factored second moment is the best rank-one approximation in I-divergence. The theory is worst case, and its regret bounds are for convex losses; it does not say why $\beta_2 = 0.95$ works for language models.

6.1 In the wild: Adam's two dials

Adam divides the momentum of each coordinate by a running root mean square of its gradients, and two dials set the denominator: β_2, how fast the average forgets, and ϵ, a floor below which a gradient is not divided by its own size. Both are rarely tuned; their values have moved.

IN THE WILD · Adam, Kingma and Ba (2015); GPT-3, Brown et al. (2020); LLaMA, Touvron et al. (2023). Train with Adam or AdamW (Chapter 13), $\beta_1 = 0.9$, bias correction on. Keep $\beta_2 = 0.999$, the default of Kingma and Ba, except for large language models, which use $\beta_2 = 0.95$. Keep $\epsilon = 10^{-8}$, the default of the original paper and of PyTorch (Keras uses 10^{-7}). For sparse convex models, such as click prediction, use AdaGrad's per-coordinate learning rate $\eta/(\sum_{s\le t} g_{s,i}^2)^{1/2}$. To save memory, Adafactor replaces v by row and column sums of G^2.

For language models β_2 drifted down: RoBERTa lowered it from 0.999 to 0.98, reporting better stability with large batches (Liu et al. 2019), and GPT-3 and LLaMA used 0.95. PaLM moved the other way: it used Adafactor without factorization and $\beta_2 = 1 - k^{-0.8}$ at step k, which its authors found

more stable than $\beta_2 = 0.99$ because the second moments of rare embedding tokens are poorly estimated over short windows (Chowdhery et al. 2023).[1]

OBSERVATION 6.1 (β_2 and stability; Liu et al. 2019, Chowdhery et al. 2023, Molybog et al. 2023). In the training of large language models (2019–2023), lowering β_2 from 0.999 to 0.98 has been reported to improve stability with large batches, and raising it toward 1 to help rare tokens; GPT-3 and LLaMA used 0.95. Molybog et al. (2023) relate divergences in runs of 7 to 546 billion parameters to Adam updates that grow large and nearly uncorrelated with the gradient. These are reports from training runs or an unrefereed preprint; none isolates a mechanism by ablations across scales.

The reports point both ways, a hint that β_2 trades one failure for another.

OBSERVATION 6.2 (Reddi et al. 2018; Zhang et al. 2022). For every $\beta_1 < \sqrt{\beta_2} < 1$ there is an online convex problem on an interval on which Adam's average regret does not tend to zero, and a stochastic convex problem on which Adam does not converge to the minimizer (Reddi et al. 2018). Adam remained the default; the proposed fix, AMSGrad, did not replace it. Zhang et al. (2022) pointed to the order of choices: the counterexamples are built after the hyperparameters, practitioners tune them after the problem, and with the problem fixed first and β_2 large enough Adam converges.

The dial ϵ matters more than its size suggests: Wortsman et al. (2024) report that the gradient root mean square of their transformers falls toward the default 10^{-8} as the models grow and reaches it at their largest scale (4.8 billion parameters), where it shrinks the updates and lowering ϵ to 10^{-15} improves the loss.

AdaGrad came first. McMahan and Streeter (2010) and Duchi et al. (2011) scaled each coordinate's step by the inverse root of its accumulated squared gradients, and the method spread through large sparse problems such as click prediction (McMahan et al. 2013). Its accumulated sum grows without bound, so its steps shrink for good; RMSProp divides instead by an exponential moving average of the squared gradients, which forgets (Tieleman and Hinton 2012),[2] and Adam added momentum and a bias correction (Kingma and Ba 2015). The step that never forgets has a guarantee and was abandoned for deep networks; the step that forgets has a counterexample and is the default. Which property of the denominator does each guarantee use, and what do β_2 and ϵ control?

[1] Schedules that raise β_2 toward 1 during training are from Shazeer and Stern (2018), who also showed that a second-moment estimate that decays too slowly can produce larger updates than intended, and proposed to clip the root mean square of each update.

[2] RMSProp was never published as a paper; it appeared in a lecture of Hinton's 2012 online course, which is how it is still cited.

6.2 The idealization: a norm chosen by the gradients

A step size per coordinate is a norm. For a symmetric positive definite matrix P write $\|u\|_P = (u^\top P u)^{1/2}$.

Lemma 6.3 (Preconditioning is steepest descent in a quadratic norm). *Let $P \succ 0$. The dual norm of $\|\cdot\|_P$ is $\|\cdot\|_{P^{-1}}$; for $g \neq 0$, $\mathrm{lmo}(g) = -P^{-1}g/\|g\|_{P^{-1}}$ is unique; and $g^\sharp = P^{-1}g$. So $\theta - \eta P^{-1}g$ is unnormalized steepest descent with respect to $\|\cdot\|_P$, and for $P = \mathrm{diag}(p)$ it is gradient descent with step size η/p_i in coordinate i.*

PROOF. By the Cauchy–Schwarz inequality for $P^{-1/2}g$ and $P^{1/2}u$, $\langle g, u\rangle \geq -\|g\|_{P^{-1}}\|u\|_P$, with equality exactly when $u = -cP^{-1}g$ for some $c \geq 0$. Over the unit ball the minimum of $\langle g, u\rangle$ is therefore $-\|g\|_{P^{-1}}$, attained only at $-P^{-1}g/\|g\|_{P^{-1}}$, a point of norm one since $\|P^{-1}g\|_P = \|g\|_{P^{-1}}$. Then $g^\sharp = -\|g\|_{P^{-1}}\,\mathrm{lmo}(g) = P^{-1}g$, which also holds for $g = 0$. ■

We take squares, roots and quotients entrywise, count rounds from $t = 1$, and call η the step size, the practitioners' learning rate.

Definition 6.4 (Three adaptive methods). Let g_t be the (stochastic) gradient at θ_t, $\eta > 0$, $\epsilon \geq 0$, $0 \leq \beta_1, \beta_2 < 1$; accumulators start at 0; $0/0 = 0$.

(i) *AdaGrad*: $s_t^2 = s_{t-1}^2 + g_t^2$ and $\theta_{t+1} = \theta_t - \eta\, g_t/(s_t + \epsilon)$. Thus $s_{t,i} = \|g_{1:t,i}\|_2$, with $g_{1:t,i} = (g_{1,i}, \dots, g_{t,i})$ the history of coordinate i.

(ii) *RMSProp*: $v_t = \beta_2 v_{t-1} + (1-\beta_2)g_t^2$ and $\theta_{t+1} = \theta_t - \eta\, g_t/(\sqrt{v_t} + \epsilon)$.

(iii) *Adam*: $m_t = \beta_1 m_{t-1} + (1-\beta_1)g_t$, v_t as in RMSProp, the *bias-corrected* moments $\hat m_t = m_t/(1-\beta_1^t)$, $\hat v_t = v_t/(1-\beta_2^t)$, and $\theta_{t+1} = \theta_t - \eta\, \hat m_t/(\sqrt{\hat v_t} + \epsilon)$.

On a constraint set each update is followed by a projection. On a box, the projection in every diagonal quadratic norm is coordinatewise clipping, because $\sum_i p_i(\theta_i - y_i)^2$ separates.

By Lemma 6.3, AdaGrad is unnormalized steepest descent in $\|\cdot\|_{P_t}$ with $P_t = \mathrm{diag}(s_t + \epsilon)$, RMSProp uses $P_t = \mathrm{diag}(\sqrt{v_t} + \epsilon)$, and Adam uses the bias-corrected matrix and applies it to the moving average $\hat m_t$ of Chapter 5. One property decides everything: AdaGrad's P_t is nondecreasing in t; those of RMSProp and Adam are not. A moving average with coefficient β_2 gives the gradient of k steps ago the weight $(1-\beta_2)\beta_2^k$; we call $1/(1-\beta_2)$, the mean lag plus one, its *memory*: 20 steps for $\beta_2 = 0.95$, 1000 for 0.999.[3]

[3] The half-life of the weights, $\ln 2/\ln(1/\beta_2)$, is about 0.69 memories: 13.5 steps for $\beta_2 = 0.95$ and 693 for 0.999.

FIELD GUIDE · ADAGRAD *Duchi et al. (2011); McMahan and Streeter (2010)*

UPDATE	$s_t^2 = s_{t-1}^2 + g_t^2$, $\theta_{t+1} = \theta_t - \eta\, g_t/(s_t + \epsilon)$
NORM	$\|\cdot\|_{P_t}$, $P_t = \mathrm{diag}(s_t + \epsilon)$ (Lemma 6.3)
PRECONDITIONER	diagonal, nondecreasing; the best diagonal norm in hindsight up to $\sqrt{2}$ (Proposition 6.8)
AVERAGES	none: a running sum of g^2
STATE	one number per parameter
IN THE WILD	sparse convex models; AdaGrad-Norm (Theorem 6.10); full matrix in Chapter 7

FIELD GUIDE · RMSPROP *Tieleman and Hinton (2012)*

UPDATE	$v_t = \beta_2 v_{t-1} + (1-\beta_2)g_t^2$, $\theta_{t+1} = \theta_t - \eta\, g_t/(\sqrt{v_t} + \epsilon)$
NORM	$\|\cdot\|_{P_t}$, $P_t = \mathrm{diag}(\sqrt{v_t} + \epsilon)$; ℓ_∞ if $\beta_2 = \epsilon = 0$ (Chapter 2)
PRECONDITIONER	diagonal; forgets with memory $1/(1-\beta_2)$

AVERAGES	EMA of g^2
STATE	one number per parameter
IN THE WILD	$\beta_2 = 0.9$ in the lecture that introduced it

FIELD GUIDE · ADAM *Kingma and Ba (2015)*

UPDATE	$m_t = \beta_1 m_{t-1} + (1-\beta_1) g_t$, $v_t = \beta_2 v_{t-1} + (1-\beta_2) g_t^2$, $\theta_{t+1} = \theta_t - \eta\, \hat{m}_t / (\sqrt{\hat{v}_t} + \epsilon)$, $\hat{m}_t = m_t/(1-\beta_1^t)$, $\hat{v}_t = v_t/(1-\beta_2^t)$
NORM	ℓ_∞ if $\beta_1 = \beta_2 = \epsilon = 0$ (Chapter 2); in general $\|\cdot\|_{P_t}$, $P_t = \operatorname{diag}(\sqrt{\hat{v}_t} + \epsilon)$
PRECONDITIONER	diagonal; $\to \operatorname{diag}(\mathbb{E} g^2)^{1/2}$ for stationary gradients as $\beta_2 \to 1$ (Remark 6.15)
AVERAGES	EMAs of g (β_1, Chapter 5) and of g^2 (β_2), bias-corrected
STATE	two numbers per parameter
STEP	bounded per coordinate if $\beta_1^2 < \beta_2$ (Chapter 2); up to 6.6η after a quiet stretch at (0.9, 0.999) (Proposition 6.13)
IN THE WILD	$\beta_1 = 0.9$; $\beta_2 = 0.999$, or 0.95 for language models; $\epsilon = 10^{-8}$
CAUTION	fails if the memory is short compared with the period of rare large gradients (Proposition 6.12)

FIELD GUIDE · ADAMW *Loshchilov and Hutter (2019)*

UPDATE	$\theta_{t+1} = (1 - \eta\lambda)\theta_t - \eta\, \hat{m}_t / (\sqrt{\hat{v}_t} + \epsilon)$
PRECONDITIONER	as Adam; the decay is not preconditioned, unlike an L^2 penalty added to $\mathcal{L}$
STATE	two numbers per parameter
IN THE WILD	$\lambda = 0.1$ in GPT-3 and LLaMA; full entry in Chapter 13

FIELD GUIDE · ADAFACTOR *Shazeer and Stern (2018)*

UPDATE	for $W \in \mathbb{R}^{m \times n}$: $r_t = \beta_2 r_{t-1} + (1-\beta_2) G_t^2 \mathbf{1}$, $c_t = \beta_2 c_{t-1} + (1-\beta_2)(G_t^2)^\top \mathbf{1}$, $\hat{V}_t = r_t c_t^\top / (\mathbf{1}^\top r_t)$, step $G_t / \hat{V}_t^{1/2}$
PRECONDITIONER	diagonal; the best rank-one approximation of Adam's v in I-divergence (Proposition 6.16)
AVERAGES	EMAs of row and column sums of G^2, β_2 raised toward 1; no momentum
STATE	$m + n$ numbers per $m \times n$ matrix
VARIANTS	update clipping; step sizes relative to the parameter scale
IN THE WILD	PaLM: unfactored, $\beta_1 = 0.9$, $\beta_2 = 1 - t^{-0.8}$

There are two settings. The first is worst case: convex losses that may change every round, chosen by an adversary who knows the method; the gradients of a training run, which move with the parameters, motivate it.

Assumption 6.5 (Online convex optimization on a box). The feasible set is a box $\Theta = \prod_i [a_i, b_i]$ with sides $D_i = b_i - a_i$ and ℓ_∞-diameter $D_\infty = \max_i D_i$. In round $t = 1, 2, \dots$ the learner plays $\theta_t \in \Theta$, a convex $f_t : \mathbb{R}^d \to \mathbb{R}$ is revealed (it may depend on $\theta_1, \dots, \theta_t$), and the learner receives $g_t \in \partial f_t(\theta_t)$. The *regret* against $\theta^\star \in \Theta$ is $R_T(\theta^\star) = \sum_{t=1}^{T} (f_t(\theta_t) - f_t(\theta^\star))$.

If $f_t = \ell(\cdot; \xi_t)$ with independent samples and $\mathcal{L} = \mathbb{E}\ell$, then $\mathbb{E} f_t(\theta_t) = \mathbb{E}\mathcal{L}(\theta_t)$, and Jensen's inequality turns a regret bound into $\mathbb{E}\mathcal{L}(\bar{\theta}_T) - \mathcal{L}(\theta^\star) \le \mathbb{E} R_T(\theta^\star)/T$ for the average $\bar{\theta}_T$ of the iterates. The second setting is a fixed smooth objective.

Assumption 6.6 (Smoothness of unknown size). The objective $\mathcal{L} : \mathbb{R}^d \to \mathbb{R}$ is bounded below by $\mathcal{L}^\star$ and L-smooth with respect to a norm $\|\cdot\|$; the method is not told L.

DICTIONARY

IN THE THEORY	IN A TRAINING RUN
the norm $\|\cdot\|_{P_t}$, $P_t = \operatorname{diag}(\sqrt{\hat{v}_t}) + \epsilon I$	Adam's denominator; $\eta/(\sqrt{\hat{v}_{t,i}} + \epsilon)$, the effective learning rate of parameter i
the losses f_t and the regret R_T	minibatch losses; summed loss minus that of the best parameters in hindsight
the box and its diameter D_∞	the range of a parameter, set by initialization and weight decay
the history norm $\|g_{1:T,i}\|_2$	$\sqrt{T}$ times the gradient RMS of parameter i, small for rare features
the memory $1/(1-\beta_2)$	the number of steps that v averages; times B, in examples
the period n of a rare gradient	a rare token with a large gradient
a quiet-then-loud coordinate	a rare token's embedding; a reviving unit
an unknown smoothness L	the learning rate one would otherwise tune
ϵ	the gradient scale of Adam's switch to SGD

WHAT WAS THROWN AWAY. In the first setting, nonconvexity and a fixed objective: the bounds hold against an adversary and say nothing about a particular run. The box, which training does not have. Noise, in the theorem of the second setting. And the coordinate system: a diagonal preconditioner adapts to the coordinates in which the parameters are written (Chapters 2 and 14). What is kept is what all the methods share, a norm chosen from the gradients seen so far, and what separates them: whether that norm forgets.

WHAT ONE HAD TO NOTICE. McMahan and Streeter (2010) and Duchi et al. (2011) saw that the regret bound of online gradient descent depends on the norm it is run in, so the norm can be chosen online to minimize the bound it is charged; the best norm weights each coordinate by the root of its accumulated squared gradients, so that the step divides by it (Proposition 6.8). RMSProp and Adam keep the choice of norm but not its

justification: a forgetting norm minimizes no bound on the whole history.

6.3 The theory: AdaGrad and the best norm in hindsight

Theorem 6.7 (Diagonal AdaGrad; Duchi et al. 2011, McMahan and Streeter 2010). *Under Assumption 6.5, run AdaGrad with $\epsilon = 0$ and step size $\eta > 0$ from $\theta_1 \in \Theta$, projected on Θ. For every $\theta^\star \in \Theta$ and $T \geq 1$,*

$$R_T(\theta^\star) \leq \sum_{i=1}^{d} \Big(\frac{D_i^2}{2\eta} + \eta \Big) \| g_{1:T,i} \|_2 \leq \Big(\frac{D_\infty^2}{2\eta} + \eta \Big) \sum_{i=1}^{d} \| g_{1:T,i} \|_2, \qquad (6.1)$$

and with $\eta = D_\infty/\sqrt{2}$, $R_T(\theta^\star) \leq \sqrt{2}\, D_\infty \sum_i \| g_{1:T,i} \|_2$.

Proof. The plan is to linearize the losses, which splits the regret into d one-dimensional regrets, and to bound each by the argument for online gradient descent with a step size that changes from round to round.

Splitting. By the subgradient inequality $f_t(\theta_t) - f_t(\theta^\star) \leq \sum_i g_{t,i}(\theta_{t,i} - \theta_i^\star)$. The projection is coordinatewise clipping, so coordinate i evolves by $\theta_{t+1,i} = \mathrm{clip}_{[a_i,b_i]}(\theta_{t,i} - \eta g_{t,i}/s_{t,i})$, through its own gradients only. Hence $R_T(\theta^\star) \leq \sum_i r_i$ with $r_i = \sum_{t \leq T} g_{t,i}(\theta_{t,i} - \theta_i^\star)$, and it suffices to show $r_i \leq (D_i^2/(2\eta) + \eta) \| g_{1:T,i} \|_2$.

One coordinate. Fix i and drop it from the notation. A round with $s_t = 0$ has $g_t = 0$: it contributes nothing and does not move the iterate. If all g_t vanish, $r = 0$; otherwise let τ be the first round with $g_\tau \neq 0$, so that $\eta_t = \eta/s_t$ is finite and nonincreasing for $t \geq \tau$. Clipping to an interval containing $\theta^\star$ does not increase the distance to it, so with $e_t = (\theta_t - \theta^\star)^2$ we have $e_{t+1} \leq (\theta_t - \eta_t g_t - \theta^\star)^2 = e_t - 2\eta_t g_t(\theta_t - \theta^\star) + \eta_t^2 g_t^2$, that is,

$$g_t(\theta_t - \theta^\star) \leq \frac{e_t - e_{t+1}}{2\eta_t} + \frac{\eta_t}{2} g_t^2.$$

We sum over $\tau \leq t \leq T$ and bound the two sums in turn. Regrouping the first by the e_t, dropping $-e_{T+1}/(2\eta_T) \leq 0$, and using $e_t \leq D^2$ and $1/\eta_t - 1/\eta_{t-1} \geq 0$,

$$\sum_{t=\tau}^{T} \frac{e_t - e_{t+1}}{2\eta_t} \leq \frac{e_\tau}{2\eta_\tau} + \sum_{t=\tau+1}^{T} e_t \Big(\frac{1}{2\eta_t} - \frac{1}{2\eta_{t-1}} \Big) \leq \frac{D^2}{2\eta_T} = \frac{D^2 s_T}{2\eta}.$$

For the second we use the AdaGrad lemma (Lemma A.20): for $a_t \geq 0$ and $S_t = \sum_{u \leq t} a_u$, $a_t/\sqrt{S_t} \leq 2(\sqrt{S_t} - \sqrt{S_{t-1}})$, because $\sqrt{S_t} - \sqrt{S_{t-1}} = a_t/(\sqrt{S_t} + \sqrt{S_{t-1}})$; the sum telescopes. With $a_t = g_t^2$ it gives $\sum_t \frac{\eta_t}{2} g_t^2 = \frac{\eta}{2} \sum_t g_t^2/s_t \leq \eta s_T$. Adding, $r \leq (D^2/(2\eta) + \eta) s_T$ with $s_T = \| g_{1:T} \|_2$. Finally $D_i \leq D_\infty$, and $\eta = D_\infty/\sqrt{2}$ minimizes $D_\infty^2/(2\eta) + \eta$. ∎

Besides its definition, the proof used one property of the step sizes: they never increase. Without it the coefficients $1/\eta_t - 1/\eta_{t-1}$ can be negative and the first sum no longer telescopes; this is the property that RMSProp and

Adam lack, and where Proposition 6.12 breaks them. Run with one scalar $s_t = (\sum_{u\le t}\|g_u\|_2^2)^{1/2}$, Euclidean projections and no splitting, the same proof bounds online gradient descent on a convex set of Euclidean diameter D_2:

$$R_T(\theta^\star) \le \Big(\frac{D_2^2}{2\eta} + \eta\Big)\Big(\sum_{t=1}^{T}\|g_t\|_2^2\Big)^{1/2}, \tag{6.2}$$

within $\sqrt{2}$, at $\eta = D_2/\sqrt{2}$, of the bound $D_2(\sum_t\|g_t\|_2^2)^{1/2}$ for the best constant step chosen in hindsight (Theorem A.21).[4] On a cube of side D_∞, $D_2 = \sqrt{d}\,D_\infty$ and $\sum_i\|g_{1:T,i}\|_2 \le \sqrt{d}\,(\sum_t\|g_t\|_2^2)^{1/2}$, so at their best step sizes AdaGrad's bound, $\sqrt{2}\,D_\infty\sum_i\|g_{1:T,i}\|_2$, is never larger than that of (6.2), and it is smaller by the factor $\sqrt{d}$ when all gradients lie on the same coordinate axis.

[4] This scalar step $D_2/(\sqrt{2}\,s_t)$ is AdaGrad-Norm. Unlike the classical tuned step $D/(G\sqrt{t})$, it needs no bound G on the gradients.

Proposition 6.8 (The best diagonal norm in hindsight). *Let $g_1,\dots,g_T\in\mathbb{R}^d$ with history norms $s_{T,i} = \|g_{1:T,i}\|_2$ not all zero, and let $0/0 = 0$ and $x/0 = \infty$ for $x > 0$. Then*

$$\min\Big\{\sum_{t=1}^{T}\|g_t\|^2_{\mathrm{diag}(p)^{-1}} : p\in\mathbb{R}^d_{\ge 0},\ \sum_i p_i \le 1\Big\} = \Big(\sum_{i=1}^{d} s_{T,i}\Big)^2,$$

attained at $p_i = s_{T,i}/\sum_j s_{T,j}$, which is AdaGrad's final P_T normalized. Consequently online gradient descent in the best fixed diagonal norm with the best constant step, both chosen in hindsight, has the regret bound $D_\infty\sum_i s_{T,i}$ on a box of ℓ_∞-diameter D_∞, and AdaGrad's bound is $\sqrt{2}$ times this.

Proof. The objective is $\sum_i s_{T,i}^2/p_i$. If it is finite, $p_i > 0$ wherever $s_{T,i} > 0$, and the Cauchy–Schwarz inequality for the products $(s_{T,i}/\sqrt{p_i})\sqrt{p_i}$ over these coordinates gives $\sum_i s_{T,i} \le (\sum_i s_{T,i}^2/p_i)^{1/2}(\sum_i p_i)^{1/2}$, which is at most $(\sum_i s_{T,i}^2/p_i)^{1/2}$. At $p_i = s_{T,i}/\sum_j s_{T,j}$ the objective equals $(\sum_i s_{T,i})^2$. For the consequence, the proof of Theorem 6.7 in the norm $\|\cdot\|_P$, $P = \mathrm{diag}(p)$, with a constant step gives the regret bound $D_P^2/(2\eta) + \frac{\eta}{2}\sum_t\|g_t\|^2_{P^{-1}}$ with $D_P = \sup_\Theta\|\theta-\theta'\|_P \le D_\infty(\sum_i p_i)^{1/2}$; the best η gives $D_P(\sum_t\|g_t\|^2_{P^{-1}})^{1/2}$, which is invariant under $p\mapsto cp$, and minimizing over p gives $D_\infty\sum_i s_{T,i}$. ∎

AdaGrad pays a factor $\sqrt{2}$ for not knowing the future. The choice of norm matters when the coordinates are used at very different rates.

Corollary 6.9 (Sparse features). *In Theorem 6.7 on a cube of side D_∞, suppose $|g_{t,i}| \le 1$, and $g_{t,i} = 0$ unless feature i is active in round t, which happens with probability p_i. With $\eta = D_\infty/\sqrt{2}$ for AdaGrad and $\eta = D_2/\sqrt{2}$ in (6.2), for every $\theta^\star\in\Theta$,*

$$\mathbb{E}R_T^{\mathrm{AdaGrad}}(\theta^\star) \le \sqrt{2T}\,D_\infty\sum_{i=1}^{d}\sqrt{p_i},$$

$$\mathbb{E}R_T^{\mathrm{OGD}}(\theta^\star) \le \sqrt{2T}\,D_\infty\Big(d\sum_{i=1}^{d}p_i\Big)^{1/2}.$$

For $p_i = i^{-\alpha}$ the ratio of the right sides tends to a constant if $\alpha < 1$, grows like $(1 - \frac{\alpha}{2})\zeta(\alpha)^{1/2} d^{(\alpha-1)/2}$ if $1 < \alpha < 2$, and like $(\pi/\sqrt{6})\sqrt{d}/\ln d$ if $\alpha = 2$.

PROOF. Pathwise $\sum_t g_{t,i}^2$ is at most the number N_i of rounds with feature i active, and $\mathbb{E}N_i = p_i T$. By Jensen's inequality $\mathbb{E}\|g_{1:T,i}\|_2 \le (p_i T)^{1/2}$ and $\mathbb{E}(\sum_t \|g_t\|_2^2)^{1/2} \le (T\sum_i p_i)^{1/2}$; insert these in (6.1) and (6.2) with $D_2 = \sqrt{d}\, D_\infty$. The asymptotics follow from $\sum_{i\le d} i^{-s} \sim d^{1-s}/(1-s)$ for $s < 1$, $\sum_{i\le d} i^{-1} \sim \ln d$, and $\sum_{i\le d} i^{-s} \to \zeta(s)$ for $s > 1$, with $\zeta(2) = \pi^2/6$. ∎

The scalar method has a second property, unrelated to regret: it does not need the smoothness constant. *AdaGrad-Norm* in a norm $\|\cdot\|$, started at θ_0 with $b_0 > 0$, iterates for $t = 0, 1, \dots$

$$b_{t+1}^2 = b_t^2 + \|\nabla\mathcal{L}(\theta_t)\|_*^2, \qquad \theta_{t+1} = \theta_t - \frac{\eta}{b_{t+1}} \nabla\mathcal{L}(\theta_t)^\sharp; \tag{6.3}$$

in the Euclidean norm this is the method of Ward et al. (2019). Since $b_{t+1} \ge \|\nabla\mathcal{L}(\theta_t)\|_*$, every step has length at most η.

Theorem 6.10 (AdaGrad-Norm adapts to the smoothness; after Ward et al. 2019). *Under Assumption 6.6, let $\eta, b_0 > 0$, $\Delta_0 = \mathcal{L}(\theta_0) - \mathcal{L}^\star$, $\Lambda = \max\{0, \ln(\eta L/b_0)\}$ and $Q = 2\Delta_0/\eta + 2\eta L\Lambda$. Then (6.3) satisfies (i) $\mathcal{L}(\theta_t) \le \mathcal{L}(\theta_0) + \eta^2 L\Lambda$ for every t, and (ii) for every $T \ge 1$*

$$\sum_{t=0}^{T-1} \|\nabla\mathcal{L}(\theta_t)\|_*^2 \le \big(Q + \max\{b_0, \eta L\}\big)^2,$$

so that $\min_{t<T}\|\nabla\mathcal{L}(\theta_t)\|_*^2 \le (Q + \max\{b_0, \eta L\})^2/T$.

PROOF. The plan is to split the run at the first step at which b_{t+1} exceeds ηL: before it the loss can rise, but only as much as b grows; after it every step descends; and together the two phases keep b_t bounded. Write $a_t = \|\nabla\mathcal{L}(\theta_t)\|_*^2 = b_{t+1}^2 - b_t^2$. The sharp operator has $\langle \nabla\mathcal{L}(\theta_t), \nabla\mathcal{L}(\theta_t)^\sharp\rangle = a_t$ and $\|\nabla\mathcal{L}(\theta_t)^\sharp\| = a_t^{1/2}$, for every choice of lmo when it is not unique, so the descent lemma (Lemma A.18) gives

$$\mathcal{L}(\theta_{t+1}) \le \mathcal{L}(\theta_t) - \frac{\eta}{b_{t+1}}\Big(1 - \frac{\eta L}{2b_{t+1}}\Big) a_t. \tag{6.4}$$

(A zero gradient causes no difficulty: the iterate stays put.) Let t_0 be the first t with $b_{t+1} \ge \eta L$, or ∞ if there is none.

Before t_0. For $t < t_0$, (6.4) gives $\mathcal{L}(\theta_{t+1}) - \mathcal{L}(\theta_t) \le \frac{\eta^2 L}{2} a_t/b_{t+1}^2$, and $a_t/b_{t+1}^2 = 1 - b_t^2/b_{t+1}^2 \le \ln(b_{t+1}^2/b_t^2)$ because $1 - 1/x \le \ln x$. The sum over $t < s$ telescopes to $\ln(b_s^2/b_0^2) \le 2\Lambda$ for $1 \le s \le t_0$, since $b_s < \eta L$ there. Hence $\mathcal{L}(\theta_s) \le \mathcal{L}(\theta_0) + \eta^2 L\Lambda$ and $\sum_{t<s} a_t = b_s^2 - b_0^2 \le \max\{0, \eta^2 L^2 - b_0^2\}$ for $s \le t_0$.

From t_0 on. For $t \ge t_0$, $b_{t+1} \ge \eta L$, the bracket in (6.4) is at least $\frac{1}{2}$, and the loss decreases; this proves (i). Summing (6.4) over $t_0 \le t < T$ and using $\mathcal{L}(\theta_T) \ge \mathcal{L}^\star$ gives $\frac{\eta}{2}\sum_{t=t_0}^{T-1} a_t/b_{t+1} \le \Delta_0 + \eta^2 L\Lambda$, so $\sum_{t=t_0}^{T-1} a_t \le b_T Q$ because $b_{t+1} \le b_T$.

Both phases. Adding, $b_T^2 - b_0^2 = \sum_{t<T} a_t \le \max\{0, \eta^2 L^2 - b_0^2\} + Q b_T$, so $b_T^2 \le c + Q b_T$ with $c = \max\{b_0^2, \eta^2 L^2\}$. The positive root of $x^2 - Qx - c$ is at most $Q + \sqrt{c}$, so $\sum_{t<T} a_t \le b_T^2 \le (Q + \sqrt{c})^2$. ∎

Gradient descent with a fixed step is guaranteed to converge only for $\eta < 2/L$, and attains $\min_t \|\nabla \mathcal{L}(\theta_t)\|_*^2 \le 2L\Delta_0/T$ at $\eta = 1/L$ (Chapter 1). AdaGrad-Norm converges at the rate $1/T$ for every η and b_0; too large a step costs a rise of the loss by at most $\eta^2 L \ln(\eta L/b_0)$ and a constant of order $(\eta L)^2 \ln^2(\eta L/b_0)$. On a one-dimensional quadratic b_t grows until η/b_t is below $2/L$ and then levels off:[5] the method finds a stable step size, and loses this ability as soon as b forgets (Exercise 6.7). With stochastic gradients the step η/b_{t+1} depends on the gradient it multiplies; Ward et al. (2019) handle this and obtain a rate of order $\ln T/\sqrt{T}$ under bounded gradients and noise.

[5] On $\mathcal{L}(\theta) = \frac{1}{2}\theta^2$ from $\theta_0 = 1$ with $\eta = 100$ and $b_0 = 10^{-3}$, b_t settles at 99.0, so that the final step size is $1.01/L$.

6.4 The theory: what forgetting costs

A moving average with $\beta_2 < 1$ forgets, and the forgetting has a price in every analysis. It shows first in the AdaGrad lemma.

Lemma 6.11 (The AdaGrad lemma with forgetting). *Let $a_t \ge 0$, $0 < \beta \le 1$, $\epsilon > 0$, $v_0 \ge 0$ and $v_t = \beta v_{t-1} + (1-\beta) a_t$. Then $(1-\beta)\sum_{t=1}^{T} a_t/(\epsilon + v_t) \le \ln \frac{\epsilon + v_T}{\epsilon + v_0} + T \ln \frac{1}{\beta}$.*

Proof. Since $\epsilon + v_t \ge \beta(\epsilon + v_{t-1}) + (1-\beta) a_t$ and $1 - x \le -\ln x$, the term $(1-\beta)a_t/(\epsilon + v_t)$ is at most $1 - \beta(\epsilon + v_{t-1})/(\epsilon + v_t) \le \ln \frac{\epsilon + v_t}{\beta(\epsilon + v_{t-1})}$, which telescopes. ∎

For constant a_t both sides grow like $T(1-\beta)$ (Exercise 6.1), so the term $T \ln(1/\beta)$ is genuine, and negligible only if $1 - \beta_2 \lesssim 1/T$: an analysis that sums Adam's steps over a run pays for forgetting unless the memory is comparable to the run. That a short memory can fail outright is the next proposition, the counterexample of Reddi et al. (2018) redone for an arbitrary period: a rare large gradient points one way, frequent small ones the other.[6]

[6] Reddi, Kale and Kumar use period 3 with $\beta_2 = 1/(1 + C^2)$ for their first theorem, and a long period for the theorem that covers every β_2; a general period makes the role of the memory explicit.

Proposition 6.12 (Reddi's counterexample, period n). *Let $n \ge 3$, $C > n - 1$, $\Theta = [-1, 1]$, and $f_t(\theta) = C\theta$ if $t \equiv 1 \pmod n$, $f_t(\theta) = -\theta$ otherwise; every partial sum $\sum_{t \le T} f_t$ is minimized at -1. Run Adam with $\beta_1 = 0$, $\epsilon = 0$, $0 \le \beta_2 < 1$, with or without bias correction, projected on Θ, from $\theta_1 \in \Theta$, with nonincreasing steps $\eta_t > 0$ such that $\eta_t \to 0$, $\eta_{t+1}/\eta_t \to 1$ and $\sum_t \eta_t = \infty$ (such as $\eta/\sqrt{t}$). Let $q = (1-\beta_2)/(1-\beta_2^n)$ and*

$$\kappa_n(\beta_2, C) = -\frac{C}{\sqrt{1 + q(C^2 - 1)}} + \sum_{k=1}^{n-1} \frac{1}{\sqrt{1 + q\beta_2^k (C^2 - 1)}}. \tag{6.5}$$

(i) If $\kappa_n(\beta_2, C) > 0$, then $\theta_t \to 1$ and $R_T/T \to 2(C - n + 1)/n$; if $\kappa_n(\beta_2, C) < 0$, then $\theta_t \to -1$ and $R_T/T \to 0$.

(ii) The drift at $\beta_2 = 0$ is $\kappa_n(0, C) = n - 2 > 0$; as $\beta_2 \to 1$, $\kappa_n(\beta_2, C)$ tends to $(n - 1 - C)\sqrt{n/(C^2 + n - 1)}$, which is negative.

(iii) For every $\beta_2 \in [0,1)$ there are n and $C > n-1$ with $\kappa_n(\beta_2, C) > 0$.

PROOF. The plan is to show that v_t becomes periodic, so that each period moves the iterate by about $\eta_t \kappa_n$, and that a drift of fixed sign carries the iterate to the corresponding end despite the clipping.

Periodicity. For the lag $k \in \{0, \dots, n-1\}$ since the last large gradient let $w_k = 1 + q\beta_2^k(C^2-1)$. These numbers solve the recursion of v: $\beta_2 w_{k-1} + 1 - \beta_2 = w_k$ for $k \geq 1$, and $\beta_2 w_{n-1} + (1-\beta_2)C^2 = w_0$ because $q\beta_2^n + 1 - \beta_2 = q$. The difference between v_t and this periodic solution is multiplied by β_2 at each step, so v_t and $\hat{v}_t$ approach $w_{k(t)}$, $k(t)$ the lag of round t. Since $|g_t| \geq 1$, $\hat{v}_t \geq (1-\beta_2)g_t^2$, so $\Delta_t = g_t/\sqrt{\hat{v}_t}$ satisfies $|\Delta_t| \leq (1-\beta_2)^{-1/2}$ and $\Delta_t - g_t/\sqrt{w_{k(t)}} \to 0$. Over period j (rounds $(j-1)n < t \leq jn$) the unclipped displacement $U_j = -\sum_t \eta_t \Delta_t$ and the variation $M_j = \sum_t \eta_t |\Delta_t|$ therefore satisfy $M_j \to 0$ and $U_j/\eta_{jn} \to \kappa_n$, because $\eta_t/\eta_{jn} \to 1$ uniformly over the period.

Clipping. Let $\kappa_n > 0$ and j_0 be such that $U_j \geq \kappa_n \eta_{jn}/2$ and $M_j \leq \frac{1}{4}$ for $j \geq j_0$. A period that starts at $\theta \leq 1 - M_j$ is never clipped at the upper end, and clipping at -1 only raises the iterate, so it ends at least at $\theta + \kappa_n \eta_{jn}/2$. A period that starts above $1 - M_j$ keeps the iterate above $1 - 2M_j$, since clipping at 1 lowers nothing below 1 minus the movement that follows. Since $\sum_j \eta_{jn} \geq \frac{1}{n}\sum_{t>n} \eta_t = \infty$ and $\theta_t \leq 1$, periods of the second kind occur infinitely often; after one of them, in period j_1, every period starts at least at $1 - 2\sup_{j \geq j_1} M_j$ (a period of the first kind does not lower the start of the next), and every iterate is at least $1 - 3\sup_{j \geq j_1} M_j$. Hence $\theta_t \to 1$. For $\kappa_n < 0$, apply the argument to $-\theta_t$.

Regret and limits. Every partial sum of the g_t is positive, so $\theta^\star = -1$, $R_T = \sum_{t \leq T} g_t(\theta_t + 1)$, and (i) follows from $\sum_{t \leq T} g_t = (C - n + 1)T/n + O(1)$. For (ii): if $\beta_2 = 0$, then $q = 1$, $w_0 = C^2$, $w_k = 1$ for $k \geq 1$, and $\kappa_n = -1 + (n-1)$; as $\beta_2 \to 1$, $q \to 1/n$ and $w_k \to 1 + (C^2-1)/n$. For (iii) let $\beta_2 > 0$ and $C = n$. Since $1 - \beta_2 \leq q \leq 1$, the first term of (6.5) is at least $-1/\sqrt{q} \geq -(1-\beta_2)^{-1/2}$, and the terms with $\beta_2^k n^2 \leq 1$, which are all $k \geq K$ for some K of order $\ln n$, are at least $2^{-1/2}$. So $\kappa_n(\beta_2, n) \geq (n-K)/\sqrt{2} - (1-\beta_2)^{-1/2} > 0$ for large n. ∎

With $\beta_2 = 0$ Adam is sign descent (Chapter 2), which moves by η_t whatever the size of the gradient: it counts votes, and the small gradients outvote the large one.[7] With a memory much longer than the period, $\hat{v}_t$ is nearly constant and Adam behaves like AdaGrad. In between, the large gradient inflates $\hat{v}$ right after it, so its own step is moderate, but by the end of the period $\hat{v}$ has forgotten it and the small gradients take nearly full steps; for the choice $n = 3$, $\beta_2 = 1/(1+C^2)$ of Reddi et al. (2018), $\kappa_3 \to 2^{-1/2}$ (Exercise 6.3). Their fix, *AMSGrad*, replaces v_t by its running maximum, which restores the monotonicity that Theorem 6.7 used; here it converges to -1.

[7] RMSProp is Adam with $\beta_1 = 0$ and no bias correction, so the proposition covers it too.

Two published convergence theories of Adam stay on the convergent side of this boundary by tying the memory to the horizon or to the problem. Défossez et al. (2022) assume L-smoothness and stochastic gradients bounded almost surely; with $1 - \beta_2 = 1/T$ and steps of order $1/\sqrt{T}$ they bound

the average of $\mathbb{E}\|\nabla\mathcal{L}(\theta_t)\|_2^2$ over the run by a constant times $d\ln T/\sqrt{T}$, with a factor of order $1/(1-\beta_1)$ for momentum; for fixed β_2 their bound does not vanish. Zhang et al. (2022) treat finite sums with random reshuffling, without assuming bounded gradients: if $\beta_1 < \sqrt{\beta_2}$ and β_2 exceeds a problem-dependent threshold, Adam reaches a neighborhood of stationary points whose size scales with the gradient variance, and for some small β_2 it can diverge. Lemma 6.11 shows where such a requirement enters, and Proposition 6.12 that it cannot be dropped when the problem is chosen after the hyperparameters. Li et al. (2023) relax other hypotheses: they reach ε-stationary points with $O(\varepsilon^{-4})$ stochastic gradients when the local smoothness may grow with the gradient norm (Chapter 10).

Forgetting has a second, opposite cost. A long memory makes the preconditioner slow, and a coordinate whose gradient was small for a long time and then becomes large is divided by an out-of-date v. The next proposition is the exact form of this *quiet-then-loud* event.

Proposition 6.13 (Quiet then loud). *Consider one coordinate of Adam without bias correction, $0 \le \beta_1 < 1$, $0 < \beta_2 < 1$, $\epsilon \ge 0$. Let $|g_s| \le \delta$ for $s < t_0$ and $g_s = \bar g$ for $t_0 \le s \le t_0 + k$, where $\bar g > 0$, $\delta \ge 0$ and $\beta_1\delta \le (1-\beta_1)\bar g$, and let $a = \beta_1^{k+1}$, $b = \beta_2^{k+1}$. The step $u_k = \eta\, m_{t_0+k}/(\sqrt{v_{t_0+k}} + \epsilon)$ satisfies*

$$\frac{(1-a)\bar g - a\delta}{\sqrt{b\delta^2 + (1-b)\bar g^2} + \epsilon} \le \frac{u_k}{\eta} \le \frac{(1-a)\bar g + a\delta}{\sqrt{1-b}\,\bar g + \epsilon}. \tag{6.6}$$

(i) As $\delta/\bar g \to 0$, $u_k/\eta \to \rho_k = (1-\beta_1^{k+1})/(\sqrt{1-\beta_2^{k+1}} + \epsilon/\bar g)$. With $\epsilon = 0$ the first loud step is $\rho_0 = (1-\beta_1)/\sqrt{1-\beta_2}$, and $\rho_k \to 1$.

(ii) With $\epsilon = 0$ the excess over sign steps, $E(\beta_1,\beta_2) = \sum_{k\ge0}(\rho_k - 1)$, is finite, and $E = 2\ln 2/(1-\beta_2) + O((1-\beta_2)^{-1/2})$ as $\beta_2 \to 1$, for fixed β_1.

Proof. The plan is to unroll the two averages over the loud stretch and bound what they inherited. Since m and v start at 0, they are averages with weights of total at most one, so $|m_{t_0-1}| \le \delta$ and $0 \le v_{t_0-1} \le \delta^2$. Over the loud rounds $m_{t_0+k} = a\, m_{t_0-1} + (1-a)\bar g$ and $v_{t_0+k} = b\, v_{t_0-1} + (1-b)\bar g^2$, so $(1-a)\bar g - a\delta \le m_{t_0+k} \le (1-a)\bar g + a\delta$, a nonnegative lower bound because $a\delta \le \beta_1^k(1-\beta_1)\bar g \le (1-a)\bar g$, and $(1-b)\bar g^2 \le v_{t_0+k} \le b\delta^2 + (1-b)\bar g^2$. Dividing gives (6.6), and (i) follows. For (ii) let $\nu = \ln(1/\beta_2)$ and $f(x) = (1-e^{-x})^{-1/2} - 1$, positive and decreasing, so that $\rho_k - 1 = f((k+1)\nu) - \beta_1^{k+1}(1-\beta_2^{k+1})^{-1/2}$. Comparing a sum of a decreasing function with its integral, $\int_\nu^\infty f \le \nu\sum_{j\ge1} f(j\nu) \le \int_0^\infty f$, and $\int_0^\infty f = \int_0^1 2\,\mathrm{d}w/(1+w) = 2\ln 2$ by the substitution $1-e^{-x} = w^2$. Since $1-e^{-x} \ge \min\{x,1\}/2$, $\int_0^\nu f \le 2\sqrt{2\nu}$, so that $\sum_{j\ge1} f(j\nu) = 2\ln 2/\nu + O(\nu^{-1/2})$; the second part is at most $\sum_j \beta_1^j \max\{(2/(j\nu))^{1/2}, \sqrt{2}\} = O(\nu^{-1/2})$. Finally $1/\nu = 1/(1-\beta_2) + O(1)$. ∎

With $\beta_1 = 0.9$ the first loud step is 3.16η for $\beta_2 = 0.999$, η for 0.99 and 0.45η for 0.95; the steps then grow, as the momentum fills up faster than

v, and peak at 6.57η, 2.13η and 1.10η eleven to nineteen rounds after the jump. Each ρ_k is a limit of ratios $|m_t|/\sqrt{v_t}$, so the peaks respect the bound $(1-\beta_1)/\sqrt{(1-\beta_2)(1-\beta_1^2/\beta_2)}$ of Proposition 2.17: 7.27, 2.35 and 1.16. The excess E is the integrated lag of the preconditioner, 1213 step sizes for $\beta_2 = 0.999$ and fewer than 2 for 0.95. Bias correction changes nothing once $t_0(1-\beta_2)$ is large (Exercise 6.2).

Remark 6.14 (What ϵ does). The step $\eta\hat{m}_{t,i}/(\sqrt{\hat{v}_{t,i}}+\epsilon)$ is Adam's if $\sqrt{\hat{v}_{t,i}} \gg \epsilon$ and momentum SGD's with step size η/ϵ if $\sqrt{\hat{v}_{t,i}} \ll \epsilon$: ϵ switches from sign-like to gradient-like behavior at the gradient scale ϵ, and caps every step at $\eta|\hat{m}_{t,i}|/\epsilon$. In Proposition 6.13 it matters once it is comparable with $\sqrt{1-\beta_2}\,\bar{g}$; on a quadratic it turns the two-cycle of sign descent into convergence once it reaches half the step size times the curvature (Exercise 6.8). It is set small so that every gradient that carries signal is normalized, and matters when gradients shrink to its size (Section 6.1); Everett et al. (2024) remove it by replacing $m/(\sqrt{v}+\epsilon)$ with $\operatorname{atan2}(m, \sqrt{v})$, which is close to $m/\sqrt{v}$ when the ratio is small, bounded, and unchanged when the gradients are rescaled.

Remark 6.15 (Adam's two limits). With $\beta_1 = \beta_2 = \epsilon = 0$ Adam is sign descent. At the other end, let the gradients at a fixed θ be i.i.d. with $\mathbb{E}g_{t,i}^2 = \bar{v}_i > 0$ and finite fourth moments. Then $\mathbb{E}\hat{v}_{t,i} = \bar{v}_i$ and $\operatorname{Var} v_{t,i} \le \frac{1-\beta_2}{1+\beta_2}\operatorname{Var}(g_{t,i}^2)$, so $\hat{v}_{t,i} \to \bar{v}_i$ in probability as $\beta_2 \to 1$ with $t(1-\beta_2) \to \infty$, and Adam becomes momentum SGD preconditioned by $\operatorname{diag}(\mathbb{E}g^2)^{-1/2}$. In the noisy quadratic of Chapter 11 (diagonal H, gradient $H\theta + \zeta$, $\operatorname{Cov}\zeta = \sigma^2 H/B$), $\mathbb{E}g_i^2 = h_i^2\theta_i^2 + \sigma^2 h_i/B$: where noise dominates the preconditioner is proportional to $H^{1/2}$, where signal dominates it is $|\nabla\mathcal{L}|$, the sign-descent limit. It is the root of the second moment of the minibatch gradient, not the Hessian or the Fisher information (Kunstner et al. 2019).

Adafactor stores, per weight matrix, moving averages of the row and column sums of G_t^2 and multiplies them; this solves a matrix approximation problem exactly. For nonnegative V and R of equal size the *I-divergence* (generalized Kullback–Leibler divergence) is $D(V\|R) = \sum_{ij}(V_{ij}\ln(V_{ij}/R_{ij}) - V_{ij} + R_{ij})$, with $0\ln 0 = 0$, and $D = \infty$ if some $V_{ij} > 0 = R_{ij}$.

Proposition 6.16 (Adafactor's factorization; Shazeer and Stern 2018). *Let $V \in \mathbb{R}^{m\times n}$ be entrywise nonnegative and nonzero, $S = \mathbf{1}^\top V\mathbf{1}$. Among nonnegative rank-one matrices $xy^\top$, $D(V\|xy^\top)$ is minimized exactly at $R^\star = (V\mathbf{1})(\mathbf{1}^\top V)/S$, the row sums times the column sums over the total. The minimizer has the row and column sums of V, and equals V if V has rank one. Consequently Adafactor's estimate $r_t c_t^\top/(\mathbf{1}^\top r_t)$ is $R^\star$ for Adam's matrix V_t of moving averages of G_t^2.*

Proof. The plan is to write the excess divergence over $R^\star$ as a sum of three nonnegative terms. Let $r = V\mathbf{1}$, $c = V^\top\mathbf{1}$, $X = \sum_i x_i$, $Y = \sum_j y_j$, and assume $D(V\|xy^\top) < \infty$, so $x_i y_j > 0$ wherever $V_{ij} > 0$. Expanding, $D(V\|xy^\top) = \sum_{ij} V_{ij}\ln V_{ij} - \sum_i r_i \ln x_i - \sum_j c_j \ln y_j - S + XY$, and $D(V\|R^\star)$ is the case $x = r$, $y = c/S$, for which $XY = S$. Subtracting

gives $\sum_i r_i \ln(r_i/x_i) + \sum_j c_j \ln(c_j/y_j) - S \ln S + XY - S$. With the Kullback–Leibler divergence $\mathrm{KL}(p\|q) = \sum_i p_i \ln(p_i/q_i) \geq 0$ of probability vectors, $\sum_i r_i \ln(r_i/x_i) = S\,\mathrm{KL}(r/S\|x/X) + S \ln(S/X)$, and likewise for c and y; collecting the logarithms into $S \ln(S/(XY))$, we find with $\phi(z) = z - 1 - \ln z \geq 0$

$$D(V\|xy^\top) - D(V\|R^\star) = S\ \mathrm{KL}\Big(\frac{r}{S}\Big\|\frac{x}{X}\Big) + S\ \mathrm{KL}\Big(\frac{c}{S}\Big\|\frac{y}{Y}\Big) + S\,\phi\Big(\frac{XY}{S}\Big).$$

All three terms vanish exactly when $x/X = r/S$, $y/Y = c/S$ and $XY = S$, that is, when $xy^\top = rc^\top/S$. The row sums of $R^\star$ are $r(\mathbf{1}^\top c)/S = r$ and its column sums c; if $V = xy^\top$ then $D(V\|V) = 0$ forces $V = R^\star$. Sums commute with moving averages, so r_t, c_t and $\mathbf{1}^\top r_t$ are those of V_t. ■

The optimum has $D(V\|R^\star) = S$ times the mutual information of the row and column indices under V/S (Exercise 6.5): the factored second moment is exact when the squared gradient of an entry is a row effect times a column effect, and its error measures their dependence.

6.5 Reading the theory

WHERE THE NORM MATTERS. Figure 6.1 runs both learners of Corollary 6.9 on online logistic regression with binary features of frequencies $p_i = i^{-\alpha}$. The corollary predicts an advantage for AdaGrad growing like $d^{1/4}$ for $\alpha = 1.5$ and like $\sqrt{d}/\ln d$ for $\alpha = 2$: the scalar step must be small enough for the frequent features and is then too small for the rare ones. The bounds are loose, by factors of 11 to 12 for both learners at $d = 1000$, so they need not predict the ratio of the regrets; yet the measured ratio follows the ratio of the bounds within 10%, over the whole range for $\alpha = 1.5$ and up to $d = 2000$ for $\alpha = 2$. Beyond, for $\alpha = 2$, it falls below (10.0 against 13.1 at $d = 10^4$). Features beyond $i = \sqrt{T}$ appear once or never, yet they enlarge OGD's diameter $D_2 = 2\sqrt{d}$ and hence its step on the frequent features; its regret then grows more slowly than the $\sqrt{d}$ of its bound, because clipping caps each move at the side of the box, as it does for most of OGD's updates at $d = 10^4$ (Exercise 6.10). AdaGrad's bound overcharges these features too, which pushes the other way.

MEMORY AGAINST PERIOD. Figure 6.2 draws the boundary $\kappa_n = 0$ of Proposition 6.12 for a rare gradient twice as large as needed to win, $C = 2(n-1)$, with Adam runs on a grid of periods and memories; every run ends on the side that the proposition predicts. The critical memory is 1.46 at $n = 3$ ($\beta_2 = 0.32$), 40.1 at $n = 100$ and 398 at $n = 1000$, approaching $0.398\,n$ (Exercise 6.4). Read with the practice box: a memory of 20 steps ($\beta_2 = 0.95$) is outvoted by such a pattern from a period of about 50 steps on, and a memory of 1000 steps from about 2500. Momentum moves the boundary little (34 instead of 40 at $n = 100$ for $\beta_1 = 0.9$, Exercise 6.9); random arrivals move it up, to a memory of about one period (Exercise 6.11). Near the curve the drift is so small that a finite run reaches neither end, and there the comparison must fail.

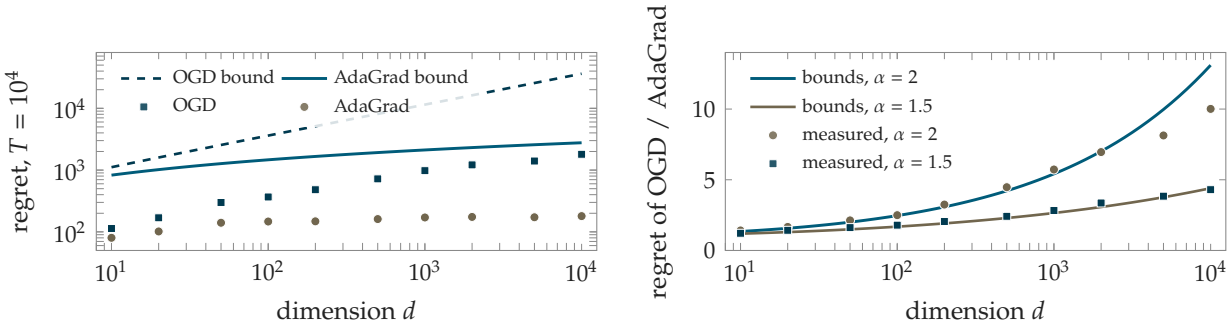


Figure 6.1. AdaGrad against online gradient descent (OGD) with the scalar adaptive step on online logistic regression over $[-1, 1]^d$: feature i is present with probability $i^{-\alpha}$, labels come from a logistic teacher with weights ± 1, and regret is against the best fixed weights in hindsight; $T = 10^4$, means over 3 runs (seeds in `06-adaptive-regret.py`). Left ($\alpha = 2$): regrets (points) and the bounds of Corollary 6.9 (curves). Right: ratio of the regrets against ratio of the bounds.

THE LAG OF THE PRECONDITIONER. Figure 6.3 feeds Adam, as implemented in common libraries, 20000 quiet gradients of size 10^{-3} and then the constant gradient 1. The steps after the jump follow ρ_k within 0.2%, and return to the sign step on the time scale $1/(1-\beta_2)$; $\epsilon = 0.01$ lowers the first step from 3.16η to 2.40η. The excess displacement grows like $2\ln 2/(1-\beta_2)$. With $\eta = 3 \cdot 10^{-4}$, a coordinate that wakes up with a constant gradient moves 0.36 farther than sign steps would at $\beta_2 = 0.999$, a large change for a weight, and about $5 \cdot 10^{-4}$ farther at 0.95. The agreement must fail when the quiet gradients are not small compared with the loud one, when ϵ is comparable with $\sqrt{1-\beta_2}\,\bar{g}$, and above all when the loud gradient reacts to the steps, as it does in a network: the excess then sets a scale; it is not a prediction.

6.6 What the theory misses

WHAT WAS IDEALIZED AWAY. The regret theory holds for every sequence of convex losses, so it uses none of the structure of a fixed nonconvex objective, and it bounds the summed loss along the run, not the final loss; it needs a bounded domain and a projection, and its bounds grow with a diameter nobody knows. The theorem on AdaGrad-Norm is for exact gradients. The propositions about forgetting concern one coordinate with an imposed gradient sequence that does not react to the steps it causes. Throughout, ϵ is zero or a floor, and the coordinates are fixed.

WHERE IT BREAKS. (1) AdaGrad has the best guarantee of the chapter and is rarely used to train deep networks, where the methods that divide by a moving average took its place (Kingma and Ba 2015; Tieleman and Hinton 2012): its steps decay for good, and against an adversary a decaying step is what one needs, so the worst-case theory cannot see the cost. (2) Adam's convergence theory needs β_2 near one, in a way tied to the horizon (Défossez et al. 2022) or to the problem (Zhang et al. 2022); language models use $\beta_2 = 0.95$, a memory of 20 steps that fails on rare-gradient patterns of

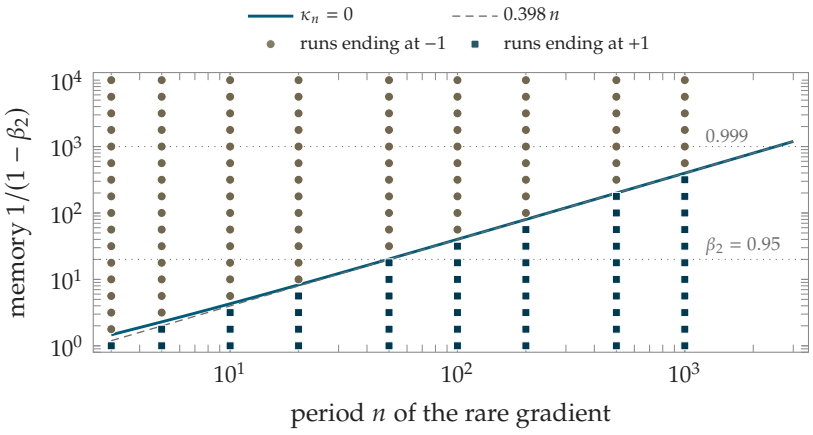


Figure 6.2. Where Adam fails on Reddi's problem with period n and $C = 2(n-1)$. Curve: the memory at which the drift (6.5) vanishes; dashed: its asymptote; dotted: the memories of $\beta_2 = 0.95$ and 0.999. Points: Adam with $\beta_1 = 0$, bias correction, $\epsilon = 10^{-8}$, $\eta_t = 0.5/\sqrt{t}$, from 0, for $\max(2 \cdot 10^5, 300n)$ rounds; circles end within 0.02 of the minimizer −1, squares at the worst point +1, except one run next to the curve ($n = 500$), at 0.78 after $2 \cdot 10^5$ rounds and at +1 by $4 \cdot 10^5$ (`06-adaptive-reddi.py`).

period 50 with $C = 2(n-1)$ (Proposition 6.12 and Figure 6.2), and yet rare tokens are learned. The theory does not say which gradient patterns occur in a run. (3) The preconditioner is not curvature: it estimates the second moment of the minibatch gradient (Remark 6.15), and its close relative, the empirical Fisher, can be a poor model of curvature even in simple problems (Kunstner et al. 2019). (4) Regret bounds do not rank Adam against SGD on transformers: the gap persists in full batch, and sign descent with momentum closes most of it (Kunstner et al. 2023); nothing here predicts that (Chapter 14). (5) AMSGrad removes the counterexample and restores the bound, yet no recipe of the practice box uses it: the counterexample is not what limits Adam in practice.

Repairs and open problems. For convergence: AMSGrad (Reddi et al. 2018); clipped Adam with a moving average of the iterates, with optimal nonconvex rates and a provable benefit of coordinatewise adaptivity when scales differ across coordinates (Ahn and Cutkosky 2024); and analyses under smoothness that grows with the gradient (Li et al. 2023). For the preconditioner: Kronecker and full-matrix versions (Chapter 7), and one second moment per block of parameters in Adam-mini (Zhang, Chen, et al. 2025). For ϵ: Adam-atan2 (Everett et al. 2024). For β_2: Marek et al. (2025) propose to keep its half-life fixed in tokens across batch sizes, which makes the memory a property of the data. Open: a model of the gradient sequence of a run from which the right memory can be computed; how often quiet-then-loud events occur in large models, and whether they cause the instabilities of Observation 6.1; and why $\beta_2 = 0.95$ suffices for language models.

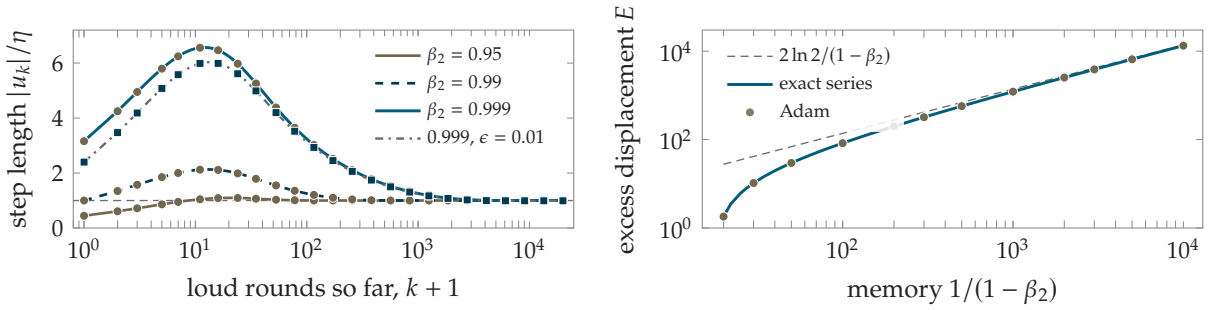


Figure 6.3. Quiet then loud: Gaussian gradients of standard deviation 10^{-3} (20000 on the left, 200000 on the right), then the constant gradient 1; Adam with $\beta_1 = 0.9$, bias correction and $\epsilon = 10^{-8}$ unless stated (seed 0, `06-adaptive-quiet.py`). Left: step lengths (points) against ρ_k of Proposition 6.13 (curves); the dashed line at 1 is the sign step. Right: the excess over sign steps, summed over $30/(1-\beta_2)$ rounds (points), against the series of Proposition 6.13(ii) and its leading term.

6.7 Back in the wild

FIELD NOTES	
PRACTICE	Adam with $\beta_2 \in \{0.95, 0.999\}$, $\epsilon = 10^{-8}$; AdaGrad for sparse convex problems; Adafactor to save memory.
IDEALIZATION	Steepest descent in a diagonal norm chosen from the gradients; online convex losses on a box; smooth losses with unknown L; one coordinate with a periodic or a quiet-then-loud gradient.
PROVED	AdaGrad's regret bound is within $\sqrt{2}$ of that of the best diagonal norm in hindsight; AdaGrad-Norm converges at rate $1/T$ for every step size; Adam ends at the worst point if $\kappa_n > 0$ and at the minimizer if $\kappa_n < 0$; the steps after a quiet stretch; Adafactor is the I-divergence projection of v.
PREDICTS	A gain in the bound growing like $\sqrt{d}/\ln d$ for feature frequencies i^{-2}; failure below a memory of 0.4 periods when $C = 2(n-1)$; a first loud step of 3.16η at $\beta_2 = 0.999$ and 0.45η at 0.95; an excess displacement of about $2\ln 2/(1-\beta_2)$ step sizes.
MISSES	Nonconvexity, the structure of real gradient sequences, ϵ at scale, rotations; why 0.95 works.
TAKEAWAY	β_2 sets a memory, long compared with the gaps between rare informative gradients yet short enough to follow changes of scale (proved in the idealizations, suggested for runs). If a run spikes after quiet spells, a lower β_2 shrinks the excess displacement in proportion to the memory (suggested). For sparse convex problems, AdaGrad (proved).

The propositions turn two pieces of folklore into statements about time scales. The memory $1/(1-\beta_2)$ must be long compared with the gaps

between the rare gradients that carry information, or they are outvoted: that is PaLM's choice for rare tokens and Proposition 6.12. And every quiet-then-loud event costs an excess displacement proportional to the memory: that is the case for lowering β_2, and Proposition 6.13 says by how much, a factor of several hundred from 0.999 to 0.95. AdaGrad, with infinite memory, meets the first constraint and pays for the second in another currency: its steps shrink for good. Neither says which constraint binds in a given run, and the default ϵ is a convention, not a result. What the theory establishes is that a step size per coordinate is a choice of norm, made online almost as well as in hindsight, and that each way of forgetting it trades one failure for another.

Adam's other ingredients are elsewhere: sign descent in Chapter 2, β_1 in Chapter 5, its edge of stability in Chapter 8, its first steps in Chapter 9, quiet-then-loud spikes in Chapter 10, the preconditioner $H^{1/2}$ and the critical batch size in Chapter 11, weight decay in Chapter 13, and full-matrix and Kronecker AdaGrad in Chapter 7.

Exercises

6.1 CONSOLIDATION. (The AdaGrad lemma and its price.) (a) Prove $\sum_{t\le T} a_t/\sqrt{S_t} \le 2\sqrt{S_T}$ by induction on T, for $a_t \ge 0$ and $S_t = \sum_{u\le t} a_u$, with $0/0 = 0$. (b) Show that the constant 2 cannot be improved. (c) For constant $a_t = a > 0$ and $v_0 = 0$, show that both sides of Lemma 6.11 grow like $T(1-\beta)$ up to factors tending to 1 as $\epsilon/a \to 0$, $\beta \to 1$.

6.2 CONSOLIDATION. (Bias correction.) (a) For independent g_t with mean μ and second moment σ^2, show $\mathbb{E}m_t = (1-\beta_1^t)\mu$ and $\mathbb{E}v_t = (1-\beta_2^t)\sigma^2$. (b) Show that with bias correction and $\epsilon = 0$ the first step is $-\eta\,\mathrm{sign}(g_1)$ for all β_1, β_2. (c) Show that in Proposition 6.13 bias correction multiplies the step by between $(1-\beta_2^{t_0})^{1/2}$ and $(1-\beta_1^{t_0})^{-1}$.

6.3 CONSOLIDATION. (Reddi's choice, and AMSGrad.) (a) For $n = 3$ and $\beta_2 = 1/(1+C^2)$, show $\kappa_3 \to 2^{-1/2}$ as $C \to \infty$. (b) AMSGrad replaces v_t by $\max_{s\le t} v_s$. In Proposition 6.12 without bias correction, show that the maximum tends to w_0, that the drift per period tends to $(n-1-C)/\sqrt{w_0} < 0$, and that AMSGrad converges to -1.

6.4 CONSOLIDATION. ★ (Memory against period.) Let $C = c\,(n-1)$ with $c > 1$ fixed and let $n \to \infty$ with $(1-\beta_2)n \to x > 0$. (a) Show $\kappa_n\sqrt{q} \to -1 + (2/(cx))(e^{x/2}-1)$. (b) Deduce that the critical memory is asymptotic to $n/(2y_c)$, where $y_c > 0$ solves $e^y - 1 = cy$; for $c = 2$ it is $0.398\,n$. (c) Explain why it grows faster than n as $c \downarrow 1$ (for $C = n$ it is about $n^2/4$). *(Hint: for lags of order n, $q\beta_2^k C^2 \to \infty$; the sum over k becomes a Riemann sum of $e^{xs/2}$.)*

6.5 CONSOLIDATION. (Adafactor.) (a) For $V = \left(\begin{smallmatrix} 4 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, compute $R^\star$ and the best rank-one approximation in Frobenius norm, and check that only $R^\star$ has the row sums of V. (b) Show that $D(V\|R^\star) = S \cdot I$, with I the mutual information of the row and column indices under V/S. (c)

Show that Adafactor is exact if every $G_t^2 = x_t y^\top$ with one y.

6.6 VARIATION. ★ (Full-matrix AdaGrad.) Let $P_t = (\sum_{s\le t} g_s g_s^\top)^{1/2}$, let Θ be convex with Euclidean diameter D_2, and project in $\|\cdot\|_{P_t}$. (a) Show $R_T \le (D_2^2/(2\eta)+\eta)\operatorname{tr} P_T$. (b) Show $\min\{\sum_t g_t^\top S^{-1} g_t : S \succ 0,\ \operatorname{tr} S \le 1\} = (\operatorname{tr} P_T)^2$ if $P_T \succ 0$. *(Hint: use $P_t + \delta I$, $\delta \to 0$; $P_t \succeq P_{t-1}$ by Theorem A.9; concavity of $\operatorname{tr} X^{1/2}$ gives $\operatorname{tr} A^{1/2} - \operatorname{tr}(A - gg^\top)^{1/2} \ge \frac{1}{2} g^\top A^{-1/2} g$.)*

6.7 VARIATION. (Forgetting breaks AdaGrad-Norm.) Replace the sum in (6.3) by $b_{t+1}^2 = \beta_2 b_t^2 + (1-\beta_2)\|\nabla \mathcal{L}(\theta_t)\|_*^2$ and run it on $\mathcal{L}(\theta) = \frac{h}{2}\theta^2$. (a) Show that $\pm\eta/2$, with $b = h\eta/2$, is an orbit of period two for every $\beta_2 < 1$. (b) Show that the iterates converge only if some iterate is exactly 0. (c) Run $\beta_2 = 0.99$, $\eta = 0.3$, $h = 1$, $\theta_0 = 1$, $b_0 = 10^{-3}$: the loss falls below 10^{-40} while b remembers early gradients, then returns to $h\eta^2/8$. Why does $\beta_2 = 0.999$ seem to converge in double precision?

6.8 VARIATION. (Adam without memory, with ϵ.) On $\mathcal{L} = \frac{h}{2}\theta^2$, Adam with $\beta_1 = \beta_2 = 0$ is $\theta \mapsto \theta - \eta h\theta/(h|\theta| + \epsilon)$. (a) Show that $\theta_t \to 0$ from every start if $\epsilon \ge \eta h/2$. (b) Show that for $\epsilon < \eta h/2$ there is an orbit of period two with amplitude $\eta/2 - \epsilon/h$, below the amplitude $\eta/2$ of the symmetric two-cycle of sign descent. (c) Relate the threshold to the stability of SGD with step size η/ϵ.

6.9 VARIATION. (Momentum in Reddi's problem.) With $\beta_1 > 0$, show that the periodic first moment at lag k is $-1 + (C+1)\beta_1^k(1-\beta_1)/(1-\beta_1^n)$ and the drift per period is $-\sum_k m_k/\sqrt{w_k}$. For $\beta_1 = 0.9$, $C = 2(n-1)$ and memories above $1/(1-\beta_1^2)$, find the critical memory for $n = 100$ and 1000 (about 34 and 392, against 40 and 398 for $\beta_1 = 0$).

6.10 SIMULATION. (Sparse features.) (a) Run `06-adaptive-regret.py` with $\alpha = 1$ and $\alpha = 3$ (edit `ALPHAS`); predict the ratio of the bounds of Corollary 6.9 at $d = 10^4$ (about 1.6 and 42) and compare with the measured ratio. (b) For $\alpha = 2$, run OGD at $d = 2000$ with the step it uses at $d = 10^4$, compare the regrets, and count the clipped updates of each learner. Explain why the measured ratio falls below the ratio of the bounds for $\alpha = 3$ but not for $\alpha = 1$.

6.11 SIMULATION. (Random arrivals.) Let the large gradient of Proposition 6.12 arrive at each round independently with probability $1/n$, as in the stochastic counterexample of Reddi et al. (2018). For $n = 10, 100, 1000$ and $C = 2(n-1)$, estimate the stationary drift $-\mathbb{E}[g_t/\sqrt{v_t}]$ for $\beta_1 = 0$ as the memory varies, find its zero (about one period, against 0.4 for the periodic case), and explain why random gaps hurt.

6.12 RESEARCH. (Choosing the memory.) Build a model of one coordinate whose gradient has both rare informative spikes (Proposition 6.12) and changes of scale (Proposition 6.13), and find the memory that minimizes a natural loss. How does it scale with the batch size, if a step consumes B examples, and is the rule of Marek et al. (2025) recovered?

Then measure, in a small transformer, how often coordinates are quiet and then loud. As far as we know both questions are open.

Notes and References

AdaGrad appeared at COLT 2010, in McMahan and Streeter (2010) and in the work of Duchi, Hazan and Singer whose journal version, Duchi et al. (2011), also gives the full-matrix version and sparse-data examples like Corollary 6.9. Online gradient descent and its regret bound are due to Zinkevich (2003). Reddi et al. (2018) traced Adam's failure to the quantity that our proof of Theorem 6.7 needs to be nonnegative, which started the theory of Section 6.4. Theorem 6.10 is the deterministic part of Ward et al. (2019), in a general norm with our own constants.

Bernstein and Newhouse (2024) present Adam without its moving averages as steepest descent in ℓ_∞, the view of Chapter 2; this chapter adds the norm that the averages define. Adafactor's I-divergence factorization is from Shazeer and Stern (2018). We know no reference for the drift κ_n, the critical memory $0.398\,n$ or the excess $2\ln 2/(1-\beta_2)$, which follow from the two models by elementary calculation.

Translating the literature. Duchi et al. (2011) write x_t for our θ_t and use our $g_{1:t,i}$ and $s_{t,i}$; their $H_t = \delta I + \operatorname{diag}(s_t)$ is our P_t with δ for ϵ, and their D_∞ bounds the ℓ_∞-distance from the comparator to the feasible points, which our diameter does. Kingma and Ba (2015) write α for the step size η, and otherwise our notation. Reddi et al. (2018) write α_t for η_t and $V_t = \operatorname{diag}(v_t)$; their $\Gamma_{t+1} = \sqrt{V_{t+1}}/\alpha_{t+1} - \sqrt{V_t}/\alpha_t$ is the increment of our P_t/η_t. Défossez et al. (2022) write F, N, α for $\mathcal{L}$, T, η, and R for the bound on the gradients. Shazeer and Stern (2018) write R_t, C_t for our r_t, c_t and $\hat\beta_{2t}$ for the increasing coefficient.

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7 Kronecker Preconditioners

Shampoo, K-FAC and SOAP

IN THE WILD — SHAMPOO, K-FAC, SOAP

IN THIS CHAPTER. Shampoo, K-FAC, and SOAP precondition the gradient of each weight matrix by two small matrices, one for its rows and one for its columns, instead of the enormous matrix of full-matrix AdaGrad or the Fisher information; we study them on the vectorized gradient and on a layer loss whose Hessian is a Kronecker product. We prove that Shampoo's factors dominate full-matrix AdaGrad's statistics up to the rank of the gradients, which gives Shampoo its regret bound; that with stationary statistics a Kronecker preconditioner with power s turns the condition number κ into $\kappa^{|1-qs|}$, where q describes the statistics, so that Shampoo's power $1/4$, the power $1/2$, and K-FAC's inverse each win in a different regime; and that SOAP, Adam in Shampoo's eigenbasis, runs on Kronecker curvature as on an isotropic quadratic. The theory does not explain why these methods beat a well-tuned Adam when they do, and their measured advantage shrinks as models grow.

7.1 In the wild: two small matrices per layer

Adam treats a weight matrix as a bag of numbers: each entry gets a learning rate from its own past (Chapter 6), and nothing in the update knows that rows belong to output units and columns to inputs. Full-matrix AdaGrad, which preconditions the vectorized gradient by the inverse square root of the summed outer products of past gradients, sees every correlation, but needs $(mn)^2$ numbers for an $m \times n$ matrix: 2.8×10^{14} for a 4096×4096 layer.[1] The optimizers of this chapter keep one small matrix for the rows and one for the columns, $m^2 + n^2$ numbers, and multiply the gradient by them on both sides.

[1] At four bytes each, about a petabyte for one layer. Even the diagonal that Adam keeps has 1.7×10^7 entries.

IN THE WILD · Shampoo, Anil et al. (2020) and Gupta et al. (2018); K-FAC, Martens and Grosse (2015); SOAP, Vyas et al. (2025). For each weight matrix $W \in \mathbb{R}^{m \times n}$ with minibatch gradient G_t, keep $L_t \in \mathbb{R}^{m \times m}$, a sum or moving average of $G_t G_t^\top$, and $R_t \in \mathbb{R}^{n \times n}$, the same for $G_t^\top G_t$; every so many steps recompute inverse roots, and step $W \leftarrow W - \eta (L_t + \varepsilon I)^{-1/4} G_t (R_t + \varepsilon I)^{-1/4}$ (Shampoo). Large-scale implementations take the step length from Adam or AdaGrad (grafting), add momentum, and some use the power $-1/2$. K-FAC averages $aa^\top$ over a layer's inputs a and $\delta\delta^\top$ over the gradients δ with

respect to its outputs, and steps with $(\mathbb{E}[\delta\delta^\top] + \varepsilon I)^{-1} G_t (\mathbb{E}[aa^\top] + \varepsilon I)^{-1}$, with an adaptive damping ε. SOAP runs Adam on $Q_L^\top G_t Q_R$, where Q_L and Q_R hold eigenvectors of Shampoo's L_t and R_t, recomputed every f steps.

Gupta et al. (2018) introduced *Shampoo* for tensors of every order, with one factor per mode and the power $-1/(2k)$ for order k, which is $-1/4$ for matrices, and proved its regret bound, the main theorem of this chapter.[2] Anil et al. (2020) made it practical at scale, on Transformers for translation, BERT, and ResNet-50 among other models, computing the roots rarely and in parallel with training, and grafting its step length onto a first-order method (Agarwal et al. 2020); Shi et al. (2023) describe a PyTorch implementation. *K-FAC* comes from natural gradient descent (Amari 1998): Martens and Grosse (2015) approximated each layer's block of the Fisher matrix by the Kronecker product of the second moments of the layer's inputs and of its back-propagated gradients, so that inverting it takes two small inversions.[3] SOAP (Vyas et al. 2025) is Shampoo with Adam inside.

[2] For a tensor of order k they prove an analogue of Theorem 7.7 with k factors, each raised to the power $1/k$; AdaGrad's square root then gives $1/(2k)$.

[3] George et al. (2018) kept K-FAC's Kronecker eigenbasis but re-estimated the scale of each eigendirection from the gradients, an eigenvalue-corrected K-FAC that anticipates SOAP.

Observation 7.1 (AlgoPerf; Dahl et al. 2023; Kasimbeg et al. 2025). AlgoPerf times training algorithms to fixed validation targets on a set of workloads. In its external-tuning ruleset a submission supplies one hyperparameter search space for all workloads, from which five trials are drawn; in its self-tuning ruleset the submission must set its hyperparameters itself. In the first competition, with 18 submissions from 10 teams, a submission based on Distributed Shampoo (Shi et al. 2023) won the external-tuning ruleset, training about 28% faster on average than the baseline and beating Adam-type methods also in wall-clock time; one based on Schedule-Free AdamW won the self-tuning ruleset.

Observation 7.2 (Language models; Vyas et al. 2025; Wen et al. 2026). On language models with 360 million and 660 million parameters trained with large batches, SOAP needed over 40% fewer iterations and over 35% less wall-clock time than AdamW to reach the same loss, and about 20% less of each than Shampoo. A later study that tuned every optimizer separately at every size found that the speedup of matrix-based optimizers over AdamW, SOAP's included, shrinks as the models grow, from about 1.4× at 0.1 billion parameters to about 1.1× at 1.2 billion.

Observation 7.3 (The heuristics carry weight; Eschenhagen et al. 2025; Vyas et al. 2025). Shampoo as used differs from Shampoo as analyzed: grafting, ε, moving averages, the power, and the interval between root computations are set by experiment, and computing the eigendecomposition less often degrades Shampoo, increasingly with the interval. Eschenhagen et al. (2025) find that grafting from Adam compensates for stale and mis-scaled eigenvalues of the factors, and that correcting the eigenvalues directly makes grafting unnecessary.

The three methods come with three stories: an online-learning algorithm with a regret bound, an approximation of the natural gradient, and a repair of the first by way of Adam. Why can two small matrices stand in for a large

one, and at what price? What do they buy on curvature with Kronecker structure, and which power of them should one take? In an idealization the three methods differ mainly in two choices: the power of the factors, and the basis of a diagonal correction.

7.2 The idealization: AdaGrad on the vectorized gradient

Throughout, $W \in \mathbb{R}^{m\times n}$ is one weight matrix ($m = d_{\text{out}}$ and $n = d_{\text{in}}$ for a linear layer), G_t is its gradient at step t, and $g_t = \operatorname{vec}(G_t) \in \mathbb{R}^{mn}$, where vec stacks the columns. From Lemma A.10 we use $\operatorname{vec}(MXN) = (N^\top \otimes M)\operatorname{vec}(X)$, so that $\operatorname{vec}(uv^\top) = v \otimes u$; the mixed product rule $(M \otimes N)(M' \otimes N') = MM' \otimes NN'$; that the eigenvalues of $M \otimes N$ are the products of those of M and N; that $(M \otimes N)^s = M^s \otimes N^s$ for positive semidefinite M, N and $s > 0$; and that $M \otimes N \preceq M' \otimes N'$ when $0 \preceq M \preceq M'$ and $0 \preceq N \preceq N'$, where $\preceq$ is the Löwner order.[4]

[4] Some papers stack rows instead of columns, which reverses the order of every Kronecker product (Exercise 7.1); it is a common source of transposed formulas in this literature.

FULL-MATRIX ADAGRAD. With $\Gamma_t = \sum_{\tau\le t} g_\tau g_\tau^\top$, *full-matrix AdaGrad* (Duchi et al. 2011) steps $\operatorname{vec}(W_{t+1}) = \operatorname{vec}(W_t) - \eta(\varepsilon I + \Gamma_t)^{-1/2} g_t$: unnormalized steepest descent in the norm $\|\cdot\|_{P_t}$ with $P_t = (\varepsilon I + \Gamma_t)^{1/2}$ (Chapter 6). Theorem 6.7 gives the regret of the diagonal version, and the proof of Theorem 7.10 below, with $r = 1$ and $(\varepsilon I + \Gamma_t)^{1/2}$ as preconditioner, gives at $\eta = D/\sqrt{2}$ the full-matrix bound $\sqrt{2}D \operatorname{tr}((\varepsilon I + \Gamma_T)^{1/2})$. The trouble is the size of Γ_t. A preconditioner $P_R \otimes P_L$, with $P_L \in \mathbb{R}^{m\times m}$ and $P_R \in \mathbb{R}^{n\times n}$ positive definite, is cheap by comparison: $(P_R \otimes P_L)^{-1} \operatorname{vec}(G) = \operatorname{vec}(P_L^{-1} G P_R^{-1})$.

Definition 7.4 (Kronecker preconditioners). Let $\varepsilon > 0$ and $s > 0$, and let $L_t = \varepsilon I_m + \sum_{\tau\le t} G_\tau G_\tau^\top$ and $R_t = \varepsilon I_n + \sum_{\tau\le t} G_\tau^\top G_\tau$. *Shampoo with power* s updates $W_{t+1} = W_t - \eta\, L_t^{-s} G_t R_t^{-s}$, which on $\operatorname{vec}(W)$ is preconditioned gradient descent with the *Kronecker preconditioner* $R_t^s \otimes L_t^s$; Shampoo proper is the case $s = 1/4$. A *frozen* Kronecker preconditioner uses fixed positive definite P_L and P_R in place of L_t^s and R_t^s.

Shampoo is to the spectral norm what Adam is to the max norm: with $\varepsilon = 0$ and a single gradient its step is $\operatorname{polar}(G_t)$, Muon's update without momentum (Proposition 3.6), and with sums over many gradients it is a slowly changing preconditioner, as Adam moves between sign descent at $\beta_1 = \beta_2 = 0$ and a slowly changing diagonal preconditioner (Chapters 2 and 6; Proposition 7.14 below). To ask what the factors buy, we need a loss whose curvature they can match.

Assumption 7.5 (Kronecker curvature). Let $A \in \mathbb{R}^{m\times m}$ and $C \in \mathbb{R}^{n\times n}$ be symmetric positive definite, with eigenvalues $\alpha_1 \ge \cdots \ge \alpha_m > 0$ and $\gamma_1 \ge \cdots \ge \gamma_n > 0$ and condition numbers $\kappa_A = \alpha_1/\alpha_m$ and $\kappa_C = \gamma_1/\gamma_n$. The loss is the *Kronecker quadratic*

$$\mathcal{L}(W) = \tfrac{1}{2}\operatorname{tr}\big((W - W_\star)^\top A\,(W - W_\star)\,C\big) = \tfrac{1}{2}\big\|A^{1/2}(W - W_\star)C^{1/2}\big\|_F^2.$$

This is the expected loss of a linear layer $a \mapsto Wa$ whose inputs have second moment $\mathbb{E}[aa^\top] = C$ and whose output enters a loss that is quadratic with Hessian A: if $\ell(W; a) = \frac{1}{2}(Wa - W_\star a)^\top A(Wa - W_\star a)$, then $\mathbb{E}[\ell(W; a)] = \mathcal{L}(W)$. A linear layer with a Gaussian likelihood is an instance (Proposition 7.13). The input-side matrix is called C, not B, because B is the batch size; the Hessian is then $C \otimes A$. For a matrix M with positive eigenvalues we write $\kappa(M) = \lambda_{\max}(M)/\lambda_{\min}(M)$.

DICTIONARY

IN THE THEORY	IN A TRAINING RUN
$W \in \mathbb{R}^{m \times n}$, its gradients G_t, and $\Gamma_T = \sum_t g_t g_t^\top$	a weight matrix, its minibatch gradients, and full-matrix AdaGrad's statistics, never formed in practice
the factors L_t, R_t; the rank bound r	Shampoo's two matrices; r is at most the number of examples (or tokens) in a minibatch
the step size η	the learning rate (two names for one number)
the power s	the exponent of the inverse roots: $\frac{1}{4}$ in the theory, $\frac{1}{2}$ in some implementations, 1 in K-FAC
the Kronecker quadratic, $H = C \otimes A$	a layer with input second moment C and output-side curvature A, as K-FAC models it
$\mathbb{E}[gg^\top] \propto H^q$ with $q = 1$; with $q = 2$	gradient noise near a minimum when the Fisher matrix is the Hessian; gradients dominated by an isotropic error, as after a random initialization

FIELD GUIDE · SHAMPOO *Gupta et al. (2018)*

UPDATE	$L_t = L_{t-1} + G_t G_t^\top$, $R_t = R_{t-1} + G_t^\top G_t$ $(L_0 = R_0 = \varepsilon I)$, $W_{t+1} = W_t - \eta\, L_t^{-1/4} G_t R_t^{-1/4}$
NORM	spectral when memoryless: one gradient and $\varepsilon = 0$ give $\mathrm{polar}(G_t)$
PRECONDITIONER	$R_t^{1/4} \otimes L_t^{1/4}$; its square dominates Γ_t / r (Theorem 7.7)
AVERAGES	sums or moving averages of $GG^\top$ and $G^\top G$; momentum in practice
STATE	$m^2 + n^2$ numbers for an $m \times n$ matrix, plus inverse roots
IN THE WILD	power $-\frac{1}{4}$ or $-\frac{1}{2}$; infrequent roots; step length grafted from Adam

FIELD GUIDE · K-FAC *Martens and Grosse (2015)*

UPDATE	averages $\bar{S}_a$ of $aa^\top$ and $\bar{S}_\delta$ of $\delta\delta^\top$; $W \leftarrow W - \eta(\bar{S}_\delta + \varepsilon I)^{-1} G (\bar{S}_a + \varepsilon I)^{-1}$
NORM	the Fisher (natural-gradient) norm, layer by layer

PRECONDITIONER	$\bar{S}_a \otimes \bar{S}_\delta$, damped; power -1
AVERAGES	moving averages of both factors; momentum
STATE	$m^2 + n^2$ numbers per layer, plus inverses
IN THE WILD	targets sampled from the model (Fisher) or the data (empirical Fisher)

FIELD GUIDE · SOAP *Vyas et al. (2025)*

UPDATE	every f steps $Q_L, Q_R \leftarrow$ eigenvectors of averages of $GG^\top$, $G^\top G$; every step Adam on $Q_L^\top G_t Q_R$, rotated back
NORM	ℓ_∞ in the rotated coordinates when memoryless
PRECONDITIONER	diagonal in Shampoo's eigenbasis
AVERAGES	Adam's two moving averages, kept in the current basis
STATE	$2mn + 2(m^2 + n^2)$ numbers
IN THE WILD	Adam's β_1, β_2 and the preconditioning frequency f

WHAT WAS THROWN AWAY. Everything but one matrix: the other layers, the nonlinearities, and the couplings between layers that K-FAC also drops. The curvature is fixed, and in Section 7.4 so are the statistics behind the factors; momentum, grafting, ε, and the infrequency of the roots are left out; the regret theory is online and convex. These are the right first simplifications because the Kronecker structure is exact for one linear layer with independent inputs and back-propagated gradients (Proposition 7.13), and stationary statistics are what the factors' averages estimate: whatever the factors can do, they do best here.

WHAT ONE HAD TO NOTICE. Two facts. The summed outer products of matrix gradients are dominated, up to the rank, by a Kronecker product of the Gram sums $\sum GG^\top$ and $\sum G^\top G$; seeing this takes the matrix geometric mean, whose square root, composed with AdaGrad's, gives Shampoo's power $1/4$. And a layer's per-example gradient $\delta a^\top$ has the second moment $\mathbb{E}[(aa^\top) \otimes (\delta\delta^\top)]$, which factorizes when a and δ are independent; this makes Shampoo's and K-FAC's factors estimates of the same two matrices.

7.3 The theory: the Kronecker bound and Shampoo's regret

The comparison of Shampoo with full-matrix AdaGrad rests on one gradient at a time.

Lemma 7.6 (One gradient). *Let $G \in \mathbb{R}^{m\times n}$ have rank $r \geq 1$, and let $g = \mathrm{vec}(G)$. Then*

$$gg^\top \preceq r\,(G^\top G)^{1/2} \otimes (GG^\top)^{1/2}, \tag{7.1}$$

and the factor r cannot be replaced by a smaller number.

PROOF. The plan is to compute both quadratic forms in the singular basis of

G and compare them by the Cauchy–Schwarz inequality. Write a compact singular value decomposition $G = U\Sigma V^\top$, with $U \in \mathbb{R}^{m\times r}$ and $V \in \mathbb{R}^{n\times r}$ having orthonormal columns and $\Sigma = \operatorname{diag}(\sigma_1, \dots, \sigma_r)$ positive, so that $(GG^\top)^{1/2} = U\Sigma U^\top$ and $(G^\top G)^{1/2} = V\Sigma V^\top$. For $X \in \mathbb{R}^{m\times n}$ and $x = \operatorname{vec}(X)$, put $Y = U^\top XV$ and $Z = \Sigma^{1/2} Y \Sigma^{1/2}$. By the vec identity, the right side of (7.1) without the factor r has the quadratic form

$$x^\top \big((V\Sigma V^\top) \otimes (U\Sigma U^\top)\big) x = \langle X, U\Sigma U^\top X V \Sigma V^\top \rangle = \operatorname{tr}\big(Y^\top \Sigma Y \Sigma\big) = \|Z\|_F^2,$$

and the left side has $(g^\top x)^2 = \langle G, X\rangle^2 = (\operatorname{tr}(\Sigma U^\top X V))^2 = (\operatorname{tr} Z)^2$. The Cauchy–Schwarz inequality $(\operatorname{tr} Z)^2 = \langle I_r, Z\rangle^2 \le r\|Z\|_F^2$ proves (7.1). Equality holds for $Z = I_r$, that is, for $X = U\Sigma^{-1}V^\top$, where the two forms equal r^2 and r; so no smaller factor works. ∎

For a sum of gradients the two sides of (7.1) cannot simply be added, because the square root of a sum is not the sum of the square roots; the *matrix geometric mean* repairs this. The geometric mean $P\#Q$ of positive definite P and Q is the largest symmetric Y with $\left(\begin{smallmatrix} P & Y \\ Y & Q \end{smallmatrix}\right) \succeq 0$, and it equals $(PQ)^{1/2}$ when $PQ = QP$ (Theorem A.12). Block matrices add, so the geometric mean is jointly concave, and for the commuting pairs $M \otimes I$ and $I \otimes N$ this says that $(M, N) \mapsto M^{1/2} \otimes N^{1/2}$ is superadditive: $\sum_t M_t^{1/2} \otimes N_t^{1/2} \preceq (\sum_t M_t)^{1/2} \otimes (\sum_t N_t)^{1/2}$ for positive semidefinite M_t and N_t (Corollary A.13).[5]

[5] Bhatia (2007, Chapter 4) gives an account of the geometric mean, including its joint concavity.

Theorem 7.7 (The Kronecker bound; Gupta et al. 2018). *Let $G_1, \dots, G_T \in \mathbb{R}^{m\times n}$ have rank at most $r \ge 1$, let $g_t = \operatorname{vec}(G_t)$ and $\varepsilon > 0$, and let $L_T = \varepsilon I_m + \sum_{t=1}^T G_t G_t^\top$ and $R_T = \varepsilon I_n + \sum_{t=1}^T G_t^\top G_t$. Then*

$$\varepsilon I_{mn} + \frac{1}{r}\sum_{t=1}^T g_t g_t^\top \;\preceq\; \varepsilon I_{mn} + \sum_{t=1}^T (G_t^\top G_t)^{1/2} \otimes (G_t G_t^\top)^{1/2} \;\preceq\; R_T^{1/2} \otimes L_T^{1/2}. \tag{7.2}$$

Proof. The first inequality follows from Lemma 7.6: a nonzero G_t of rank r_t has $g_t g_t^\top \preceq r_t (G_t^\top G_t)^{1/2} \otimes (G_t G_t^\top)^{1/2}$, and $r_t \le r$; gradients equal to zero contribute nothing to either side. The second is Corollary A.13 applied to the $T + 1$ pairs $(\varepsilon I_n, \varepsilon I_m)$ and $(G_t^\top G_t, G_t G_t^\top)$ for $t \le T$: the first pair contributes $(\varepsilon I_n)^{1/2} \otimes (\varepsilon I_m)^{1/2} = \varepsilon I_{mn}$, so the left side of the corollary is the middle term of (7.2), and the sums of the pairs are R_T and L_T. ∎

Letting $\varepsilon \to 0$, the outer inequality also holds without ε. In words: the square of Shampoo's preconditioner dominates full-matrix AdaGrad's statistics divided by the rank. To see how large the constant is, and what Shampoo does when the statistics themselves are a Kronecker product, it helps to view the factors as functions of Γ.

Proposition 7.8 (Shampoo's factors are partial traces). *For positive semidefinite $\Gamma \in \mathbb{R}^{mn\times mn}$ define the* partial traces *$L(\Gamma) \in \mathbb{R}^{m\times m}$ and $R(\Gamma) \in \mathbb{R}^{n\times n}$ by $\operatorname{tr}(L(\Gamma)M) = \operatorname{tr}(\Gamma(I_n \otimes M))$ and $\operatorname{tr}(R(\Gamma)N) = \operatorname{tr}(\Gamma(N \otimes I_m))$ for all $M \in \mathbb{R}^{m\times m}$ and $N \in \mathbb{R}^{n\times n}$.*

(i) If $\Gamma = \sum_t g_t g_t^\top$, then $L(\Gamma) = \sum_t G_t G_t^\top$ and $R(\Gamma) = \sum_t G_t^\top G_t$; likewise

$L(\mathbb{E}[gg^\top]) = \mathbb{E}[GG^\top]$ *and* $R(\mathbb{E}[gg^\top]) = \mathbb{E}[G^\top G]$.

(ii) If $\Gamma = \Phi \otimes \Psi$ *with* $\Phi \in \mathbb{R}^{n\times n}$ *and* $\Psi \in \mathbb{R}^{m\times m}$ *positive semidefinite, then* $L(\Gamma) = (\operatorname{tr}\Phi)\Psi$, $R(\Gamma) = (\operatorname{tr}\Psi)\Phi$, *and for every* $s > 0$

$$R(\Gamma)^s \otimes L(\Gamma)^s = (\operatorname{tr}\Gamma)^s\,\Gamma^s. \tag{7.3}$$

PROOF. The trace pairing determines $L(\Gamma)$ and $R(\Gamma)$, which are linear in Γ, so for (i) it suffices to take $\Gamma = gg^\top$: then $\operatorname{tr}(gg^\top(I_n \otimes M)) = g^\top \operatorname{vec}(MG) = \operatorname{tr}(GG^\top M)$ and $\operatorname{tr}(gg^\top(N \otimes I_m)) = g^\top \operatorname{vec}(GN^\top) = \operatorname{tr}(NG^\top G)$. For (ii), $\operatorname{tr}((\Phi\otimes\Psi)(I_n \otimes M)) = \operatorname{tr}(\Phi)\operatorname{tr}(\Psi M)$ and $\operatorname{tr}((\Phi\otimes\Psi)(N \otimes I_m)) = \operatorname{tr}(\Phi N)\operatorname{tr}(\Psi)$; then (7.3) follows from $(\Phi \otimes \Psi)^s = \Phi^s \otimes \Psi^s$ and $\operatorname{tr}(\Phi \otimes \Psi) = \operatorname{tr}\Phi \operatorname{tr}\Psi$. ∎

So for Kronecker statistics, and with $\varepsilon = 0$, Shampoo's preconditioner is $(\operatorname{tr}\Gamma)^{1/4}\Gamma^{1/4}$, a multiple of the square root of AdaGrad's $\Gamma^{1/2}$, and the power $1/2$ gives AdaGrad's $\Gamma^{1/2}$ up to a scalar. The Kronecker product nearest to Γ in Frobenius norm is a rank-one approximation of a rearrangement of Γ (Van Loan and Pitsianis 1993), and one step of the power method for it, from identity matrices, returns $R(\Gamma)$ and $L(\Gamma)$ (Exercise 7.3); Morwani et al. (2025) made this observation and found the one-step approximation close to optimal in neural networks, so that the square of Shampoo's preconditioner approximates full-matrix AdaGrad's up to a scalar.

Proposition 7.9 (How large the constant is). *For positive semidefinite* $\Gamma \neq 0$ *let* $c^\star(\Gamma)$ *be the smallest* c *with* $\Gamma \preceq c\, R(\Gamma)^{1/2} \otimes L(\Gamma)^{1/2}$. *For gradients of rank at most* r, *Theorem 7.7 gives* $c^\star \le r$, *with equality for a single gradient of rank* r *(Lemma 7.6). Moreover:*

(i) If $G_t = \sum_{i=1}^k s_{ti} u_i v_i^\top$ *for orthonormal* $u_1, \dots, u_k$, *orthonormal* $v_1, \dots, v_k$ *and a matrix* $S = (s_{ti}) \in \mathbb{R}^{T\times k}$ *without zero columns, then* $c^\star$ *is the largest eigenvalue of the cosine matrix* $\operatorname{diag}(d)^{-1/2} S^\top S \operatorname{diag}(d)^{-1/2}$, *where* $d_i = \sum_t s_{ti}^2$; *it lies between* 1 *and* k.

(ii) If $\Gamma = \Phi \otimes \Psi$, *then* $c^\star = (\lambda_{\max}(\Gamma)/\operatorname{tr}\Gamma)^{1/2}$, *which can be as small as* $(mn)^{-1/2}$.

PROOF. The plan in (i) is to find a common invariant subspace. The vectors $w_i = v_i \otimes u_i = \operatorname{vec}(u_i v_i^\top)$ are orthonormal, $g_t = \sum_i s_{ti} w_i$, and $\Gamma = \sum_{i,j} (S^\top S)_{ij} w_i w_j^\top$ vanishes off their span. By orthonormality $L(\Gamma) = \sum_i d_i u_i u_i^\top$ and $R(\Gamma) = \sum_i d_i v_i v_i^\top$, so $K = R(\Gamma)^{1/2} \otimes L(\Gamma)^{1/2}$ has the eigenvectors w_i with eigenvalues d_i, and leaves their span and its orthogonal complement invariant. On the span, $\Gamma \preceq cK$ reads $S^\top S \preceq c \operatorname{diag}(d)$, that is, $\operatorname{diag}(d)^{-1/2} S^\top S \operatorname{diag}(d)^{-1/2} \preceq cI$; on the complement, $\Gamma = 0 \preceq cK$. A unit diagonal puts the largest eigenvalue between 1 and the trace k. For (ii), (7.3) with $s = \frac{1}{2}$ gives $K = (\operatorname{tr}\Gamma)^{1/2}\Gamma^{1/2}$, and $\Gamma \preceq cK$ means $\lambda_i \le c(\operatorname{tr}\Gamma)^{1/2}\lambda_i^{1/2}$ for every eigenvalue λ_i of Γ. ∎

The factor r is the price of coherence, which the regret bound pays in full (Figure 7.2).

The losses are now convex functions ℓ_t on $\mathbb{R}^{m\times n}$, revealed one at a time, and the *regret* against a fixed $W^\star$ is $\sum_t(\ell_t(W_t)-\ell_t(W^\star))$.

Theorem 7.10 (Shampoo's regret; Gupta et al. 2018). *Let $\ell_1,\dots,\ell_T$ be convex and differentiable on $\mathbb{R}^{m\times n}$, and run Shampoo ($s=1/4$, $\varepsilon>0$) from W_1: $G_t=\nabla\ell_t(W_t)$, with L_t and R_t as in Definition 7.4, and $W_{t+1}=W_t-\eta L_t^{-1/4}G_tR_t^{-1/4}$. Suppose that every G_t has rank at most r. Then for every $W^\star$, with $D=\max_{t\le T}\|W_t-W^\star\|_F$,*

$$\sum_{t=1}^{T}\big(\ell_t(W_t)-\ell_t(W^\star)\big)\;\le\;\Big(\frac{D^2}{2\eta}+r\eta\Big)\operatorname{tr}\big(L_T^{1/4}\big)\operatorname{tr}\big(R_T^{1/4}\big). \tag{7.4}$$

Proof. The plan is to apply the regret bound of online gradient descent with changing preconditioners, and to compare Shampoo's preconditioner with full-matrix AdaGrad's by Theorem 7.7. Write $x_t=\operatorname{vec}(W_t)$, $x^\star=\operatorname{vec}(W^\star)$ and $P_t=R_t^{1/4}\otimes L_t^{1/4}$, so that $x_{t+1}=x_t-\eta P_t^{-1}g_t$ and $\operatorname{tr}(P_t)=\operatorname{tr}(L_t^{1/4})\operatorname{tr}(R_t^{1/4})$. Theorem A.21(i), with the preconditioners P_t/η, the whole space as feasible set, and $P_0=0$, gives

$$\sum_{t=1}^{T}\big(\ell_t(W_t)-\ell_t(W^\star)\big)\le\frac{1}{2\eta}\sum_{t=1}^{T}(x_t-x^\star)^\top(P_t-P_{t-1})(x_t-x^\star)+\frac{\eta}{2}\sum_{t=1}^{T}g_t^\top P_t^{-1}g_t.$$

The distance terms. The factors grow, so $L_t^{1/4}\succeq L_{t-1}^{1/4}$ and $R_t^{1/4}\succeq R_{t-1}^{1/4}$ by Theorem A.9, and $P_t\succeq P_{t-1}$ by the monotonicity of Kronecker products. Since $v^\top Mv\le\|v\|_2^2\operatorname{tr}M$ for $M\succeq 0$ and $\|x_t-x^\star\|_2\le D$, the first sum is at most $D^2\sum_t\operatorname{tr}(P_t-P_{t-1})=D^2\operatorname{tr}(P_T)$.

The gradient terms. Put $\bar\Gamma_t=\varepsilon I_{mn}+\frac{1}{r}\sum_{\tau\le t}g_\tau g_\tau^\top$. Theorem 7.7 at time t gives $\bar\Gamma_t\preceq R_t^{1/2}\otimes L_t^{1/2}=P_t^2$, hence $\bar\Gamma_t^{1/2}\preceq P_t$ and $P_t^{-1}\preceq\bar\Gamma_t^{-1/2}$ by Theorem A.9 with exponent $\frac12$. With $h_t=g_t/\sqrt{r}$ we have $\bar\Gamma_t=\bar\Gamma_{t-1}+h_th_t^\top$ and $\bar\Gamma_0=\varepsilon I_{mn}\succ 0$, so by the matrix form of Lemma A.20

$$\sum_{t=1}^{T}g_t^\top P_t^{-1}g_t\le r\sum_{t=1}^{T}h_t^\top\bar\Gamma_t^{-1/2}h_t\le 2r\operatorname{tr}\big(\bar\Gamma_T^{1/2}\big)\le 2r\operatorname{tr}(P_T).$$

The two bounds give (7.4). ∎

The bound holds for every step size, with D measured along the run. If each step is followed by a projection in the norm $\|\cdot\|_{P_t}$ onto a closed convex set that contains W_1 and $W^\star$ and has Frobenius diameter D, the same proof applies, because Theorem A.21(i) allows the projection; then D is known in advance, and $\eta=D/\sqrt{2r}$ gives $\sqrt{2r}\,D\operatorname{tr}(L_T^{1/4})\operatorname{tr}(R_T^{1/4})$, the form in which Gupta et al. (2018) state the bound. Full-matrix AdaGrad's bound is $\sqrt{2}D\operatorname{tr}((\varepsilon I+\Gamma_T)^{1/2})$; Shampoo's is no smaller, since $\sqrt{r}\operatorname{tr}(P_T)\ge\sqrt{r}\operatorname{tr}(\bar\Gamma_T^{1/2})=\operatorname{tr}((r\varepsilon I+\Gamma_T)^{1/2})$, and that is the price of storing m^2+n^2 numbers instead of $(mn)^2$.

7.4 The theory: the power and the basis

The regret bound is a guarantee, not a speed. To see what Kronecker preconditioners buy, we run them on the Kronecker quadratic, where everything can be computed.

Proposition 7.11 (Kronecker preconditioners on Kronecker curvature). *Under Assumption 7.5:*

(i) $\nabla\mathcal{L}(W) = A(W - W_\star)C$, *and the Hessian of* $\mathcal{L}$ *on* $\operatorname{vec}(W)$ *is* $H = C \otimes A$, *with eigenvalues* $\alpha_i\gamma_j$ *and condition number* $\kappa(H) = \kappa_A\kappa_C$.

(ii) Let $P_L \in \mathbb{R}^{m\times m}$ *and* $P_R \in \mathbb{R}^{n\times n}$ *be positive definite, and* $W_{t+1} = W_t - \eta P_L^{-1}\nabla\mathcal{L}(W_t)P_R^{-1}$. *The matrices* $P_L^{-1}A$ *and* $P_R^{-1}C$ *have positive eigenvalues; let* $\mu_{\min}$ *and* $\mu_{\max}$ *be the smallest and largest product of an eigenvalue of the first and one of the second, and* $\kappa = \mu_{\max}/\mu_{\min} = \kappa(P_L^{-1}A)\kappa(P_R^{-1}C)$. *With* $\eta = 2/(\mu_{\min} + \mu_{\max})$,

$$\mathcal{L}(W_t) \le \left(\frac{\kappa - 1}{\kappa + 1}\right)^{2t} \mathcal{L}(W_0) \qquad \text{for every } t \ge 0, \tag{7.5}$$

with equality for some $W_0 \neq W_\star$, *and no constant step size gives a smaller factor for every* W_0.

(iii) Gradient descent ($P_L = I$, $P_R = I$) *has* $\kappa = \kappa_A\kappa_C$; *one-sided preconditioning* ($P_L = A$, $P_R = I$) *has* $\kappa = \kappa_C$; *K-FAC's two-sided step* ($P_L = A$, $P_R = C$) *has* $\kappa = 1$ *and with* $\eta = 1$ *reaches* $W_\star$ *in one step; and frozen Shampoo factors* $P_L = L^s$, $P_R = R^s$ *with* $L = c_L A^q$, $R = c_R C^q$, *for constants* $c_L, c_R > 0$ *and* $q \ge 0$, *have* $\kappa = (\kappa_A\kappa_C)^{|1-qs|}$.

Proof. The plan is to read the Hessian off a second-order expansion and the rate off the eigenvalues of the symmetrized iteration. (i) With $E = W - W_\star$ and symmetric A and C, $\mathcal{L}(W + \Delta) = \mathcal{L}(W) + \langle AEC, \Delta\rangle + \frac{1}{2}\operatorname{tr}(\Delta^\top A\Delta C)$, and by the vec identity $\operatorname{tr}(\Delta^\top A\Delta C) = \operatorname{vec}(\Delta)^\top(C \otimes A)\operatorname{vec}(\Delta)$.

(ii) With $e_t = \operatorname{vec}(W_t - W_\star)$ and $P = P_R \otimes P_L$, $e_{t+1} = (I - \eta P^{-1}H)e_t$. Put $z_t = P^{1/2}e_t$; then $z_{t+1} = (I - \eta S)z_t$ with

$$S = P^{-1/2}HP^{-1/2} = \left(P_R^{-1/2}CP_R^{-1/2}\right) \otimes \left(P_L^{-1/2}AP_L^{-1/2}\right),$$

symmetric positive definite, with eigenvalues the products of those of the two factors, which are those of the similar matrices $P_R^{-1}C$ and $P_L^{-1}A$. The loss is $\frac{1}{2}e_t^\top He_t = \frac{1}{2}z_t^\top Sz_t$. In an orthonormal eigenbasis (b_k) of S, with eigenvalues μ_k and $z_0 = \sum_k \zeta_k b_k$,

$$\mathcal{L}(W_t) = \tfrac{1}{2}\sum_k \mu_k(1 - \eta\mu_k)^{2t}\zeta_k^2 \le \max_k(1 - \eta\mu_k)^{2t}\,\mathcal{L}(W_0),$$

with equality when z_0 is an eigenvector for an eigenvalue attaining the maximum. The function $\eta \mapsto \max\{|1 - \eta\mu_{\min}|, |1 - \eta\mu_{\max}|\}$ is smallest where the two terms agree, at $\eta = 2/(\mu_{\min} + \mu_{\max})$, where it is $(\kappa - 1)/(\kappa + 1)$.

(iii) For K-FAC, $P^{-1}H = I$. For frozen Shampoo factors, $P_L^{-1}A = c_L^{-s}A^{1-qs}$ has the eigenvalues $c_L^{-s}\alpha_i^{1-qs}$ and the condition number $\kappa_A^{|1-qs|}$, and likewise $\kappa(P_R^{-1}C) = \kappa_C^{|1-qs|}$. ∎

Shampoo's factors are powers of A and C when they average gradients whose second moment is a power of the Hessian. Two cases matter. Gradients taken at errors $E = W - W_\star$ that are isotropic random matrices, $\mathbb{E}[\mathrm{vec}(E)\,\mathrm{vec}(E)^\top] = \sigma^2 I$, as after a random initialization, have $g = H\,\mathrm{vec}(E)$ and $\mathbb{E}[gg^\top] = \sigma^2 H^2$. Gradients dominated by noise with covariance proportional to the Hessian, as near the minimum of a model whose Fisher matrix equals its Hessian (Proposition 7.13), have $\mathbb{E}[gg^\top] = \sigma^2 H$.

Corollary 7.12 (Which power). *Under Assumption 7.5, let Shampoo's factors be the second moments $L = \mathbb{E}[GG^\top]$ and $R = \mathbb{E}[G^\top G]$ of random gradients with $\mathbb{E}[gg^\top] = \sigma^2 H^q$, for some $\sigma > 0$ and $q \geq 0$. Then the frozen step $W \leftarrow W - \eta L^{-s} \nabla \mathcal{L}(W) R^{-s}$ has, at its best constant step size, the rate (7.5) with*

$$\kappa = \kappa(H)^{|1-qs|}. \tag{7.6}$$

Proof. Since $H^q = C^q \otimes A^q$, Proposition 7.8 gives $L = \sigma^2 \operatorname{tr}(C^q) A^q$ and $R = \sigma^2 \operatorname{tr}(A^q) C^q$, and Proposition 7.11(iii) applies. ∎

The exponent $|1 - qs|$ is a V-shaped law in the power:

	$s = 0$ (GD)	$s = \frac{1}{4}$ (Shampoo)	$s = \frac{1}{2}$	$s = 1$ (K-FAC)
$q = 2$	1	$\frac{1}{2}$	0	1
$q = 1$	1	$\frac{3}{4}$	$\frac{1}{2}$	0

In neither case does Shampoo's power remove the conditioning: it leaves $\kappa(H)^{1/2}$ for isotropic errors and $\kappa(H)^{3/4}$ for Hessian-shaped noise. The power $1/2$ is exact for $q = 2$, where it is full-matrix AdaGrad (Proposition 7.8). The inverse, K-FAC's power, is exact for $q = 1$, where the factors estimate curvature, and for $q = 2$ no better than gradient descent: then $L^{-1} A \propto A^{-1}$, which inverts the spectrum instead of flattening it. This is one precise form of the warning that the second moment of the gradients is not the curvature (Kunstner et al. 2019): the inverse is right for the one, its square root for the other.[6]

Proposition 7.13 (K-FAC's approximation). *Let a layer with weights $W \in \mathbb{R}^{m \times n}$ receive the input $a \in \mathbb{R}^n$ and the back-propagated output gradient $\delta \in \mathbb{R}^m$, so that its per-example gradient is $G = \delta a^\top$, with (a, δ) random with finite fourth moments.*

(i) $\mathrm{vec}(G)\,\mathrm{vec}(G)^\top = (aa^\top) \otimes (\delta\delta^\top)$, *so the layer's block of the Fisher matrix (with targets sampled from the model) or of the empirical Fisher matrix (with the data's targets) is $F = \mathbb{E}[(aa^\top) \otimes (\delta\delta^\top)]$.*

(ii) If a and δ are independent, then $F = \mathbb{E}[aa^\top] \otimes \mathbb{E}[\delta\delta^\top]$, so K-FAC is exact, and Shampoo's per-example factors $\mathbb{E}[GG^\top] = \mathbb{E}\|a\|_2^2\, \mathbb{E}[\delta\delta^\top]$ and $\mathbb{E}[G^\top G] = \mathbb{E}\|\delta\|_2^2\, \mathbb{E}[aa^\top]$ are multiples of K-FAC's.

(iii) For the mean G_B of B independent copies of G, $\mathbb{E}[G_B G_B^\top] = \frac{1}{B}\mathbb{E}[\|a\|_2^2\, \delta\delta^\top] + (1 - \frac{1}{B})\bar{G}\bar{G}^\top$ with $\bar{G} = \mathbb{E}[\delta a^\top]$, and similarly for $G_B^\top G_B$.

[6] The same arithmetic says why AdaGrad and Adam take square roots: on a diagonal quadratic with isotropic errors, the second moment of the ith gradient coordinate is proportional to h_i^2, and only its square root, not its inverse, flattens the curvatures h_i.

PROOF. (i) $\operatorname{vec}(\delta a^\top) = a \otimes \delta$, and $(a \otimes \delta)(a \otimes \delta)^\top = (aa^\top) \otimes (\delta\delta^\top)$ by the mixed product rule. (ii) The entries $a_j a_l \delta_i \delta_k$ of $(aa^\top) \otimes (\delta\delta^\top)$ have factoring expectations, and $GG^\top = \|a\|_2^2 \delta\delta^\top$, $G^\top G = \|\delta\|_2^2 aa^\top$. (iii) Of the terms of $G_B G_B^\top = B^{-2} \sum_{k,k'} \delta_k a_k^\top a_{k'} \delta_{k'}^\top$, the B with $k = k'$ have expectation $\mathbb{E}[\|a\|_2^2 \delta\delta^\top]$ and the others $\bar{G}\bar{G}^\top$. ∎

For example, let $a \sim \mathcal{N}(0, C)$ and let the model predict $y \sim \mathcal{N}(Wa, A^{-1})$, with negative log-likelihood $\frac{1}{2}(y - Wa)^\top A(y - Wa)$; for data generated with weights $W_\star$ the expected loss is the Kronecker quadratic plus a constant. With targets sampled from the model, $\delta = -A\xi$ with $\xi \sim \mathcal{N}(0, A^{-1})$ independent of a, so $\mathbb{E}[\delta\delta^\top] = A$ and $F = C \otimes A = H$ at every W: K-FAC is Newton's method and converges in one step. At the minimum the data's targets give the same δ, so the gradients that Shampoo accumulates there have the per-example factors $\operatorname{tr}(C)A$ and $\operatorname{tr}(A)C$, the statistics with $q = 1$, and a minibatch's factors are these divided by B. Away from the minimum the data's δ depends on a, and by Proposition 7.13(iii) the mean gradient dominates a minibatch's factors far from the minimum and the per-example noise near it; the right power drifts with them. Three statements remain.

Proposition 7.14 (Shampoo between Muon and AdaGrad). *(i) For a single gradient and $\varepsilon = 0$, with pseudo-inverse powers, $(GG^\top)^{-1/4} G (G^\top G)^{-1/4} = \operatorname{polar}(G)$.*

(ii) If $G_\tau = U \operatorname{diag}(\sigma_\tau) V^\top$ for $\tau \le t$, where $U \in \mathbb{R}^{m \times k}$ and $V \in \mathbb{R}^{n \times k}$ have orthonormal columns and $\sigma_\tau \in \mathbb{R}^k$ has entries of either sign, then

$$L_t^{-1/4} G_t R_t^{-1/4} = U \operatorname{diag}\left(\frac{\sigma_{t,i}}{\left(\varepsilon + \sum_{\tau \le t} \sigma_{\tau,i}^2\right)^{1/2}} \right) V^\top,$$

diagonal AdaGrad on the coefficients. With exponential moving averages (coefficient β_2) in place of the sums it is RMSProp on them, and for $\beta_2 = 0$ and $\varepsilon \to 0$ it tends to $U \operatorname{diag}(\operatorname{sign} \sigma_t) V^\top = \operatorname{polar}(G_t)$.

PROOF. (i) is Proposition 3.6: in a compact singular value decomposition $G = U\Sigma V^\top$ the powers act on Σ, and $\Sigma^{-1/2} \Sigma \Sigma^{-1/2} = I$. (ii) By orthonormality, $L_t = \varepsilon I + U \operatorname{diag}(\varsigma_t) U^\top$ with $\varsigma_{t,i} = \sum_{\tau \le t} \sigma_{\tau,i}^2$, so $L_t^{-1/4} U = U \operatorname{diag}(\varepsilon + \varsigma_t)^{-1/4}$, and likewise $R_t^{-1/4} V = V \operatorname{diag}(\varepsilon + \varsigma_t)^{-1/4}$; multiplying out gives the formula. With moving averages $\varsigma_{t,i}$ becomes the average of $\sigma_{\tau,i}^2$, and for $\beta_2 = 0$ the ith coefficient is $\sigma_{t,i} / (\varepsilon + \sigma_{t,i}^2)^{1/2} \to \operatorname{sign} \sigma_{t,i}$. ∎

Without memory Shampoo is spectral steepest descent; with memory, in a fixed singular basis, it is AdaGrad or RMSProp on the coefficients in that basis, as Adam is RMSProp with momentum on the coordinates (Chapters 2 and 3; Bernstein and Newhouse (2024)). Part (ii) also shows when the power 1/4 is right: if moreover $\sum_{\tau \le t} \sigma_{\tau,i} \sigma_{\tau,j} = 0$ for $i \ne j$, then $\Gamma_t = \sum_i \varsigma_{t,i} w_i w_i^\top$ with $w_i = v_i \otimes u_i$, and Shampoo's step is exactly full-matrix AdaGrad's. For Kronecker statistics the right power is 1/2 (Proposition 7.8); which structure a network's gradients have is an empirical question (Exercise 7.10).

Proposition 7.15 (Shampoo with the power 1/2 is Adafactor in its eigenbasis; Vyas et al. 2025). *Let* $L_t = \sum_{\tau\le t} G_\tau G_\tau^\top = Q_L \operatorname{diag}(\lambda) Q_L^\top$ *and* $R_t = \sum_{\tau\le t} G_\tau^\top G_\tau = Q_R \operatorname{diag}(\mu) Q_R^\top$ *be positive definite, put* $G'_\tau = Q_L^\top G_\tau Q_R$*, and let* $V = \sum_{\tau\le t} G'_\tau \odot G'_\tau$ *(entrywise squares). Adafactor's rank-one estimate* $\hat V = (V\mathbf{1})(\mathbf{1}^\top V)/(\mathbf{1}^\top V\mathbf{1})$ *of* V *(Chapter 6) has entries* $\hat V_{ij} = \lambda_i \mu_j / \operatorname{tr}(L_t)$*, and Adafactor's step, rotated back, is (with* $\oslash$ *and* $\hat V^{\odot 1/2}$ *the entrywise quotient and square root)*

$$Q_L\big(G'_t \oslash \hat V^{\odot 1/2}\big) Q_R^\top = \sqrt{\operatorname{tr}(L_t)}\, L_t^{-1/2} G_t R_t^{-1/2}.$$

Proof. The row sums of V are $\sum_\tau (G'_\tau G'^\top_\tau)_{ii} = (Q_L^\top L_t Q_L)_{ii} = \lambda_i$, the column sums are $(Q_R^\top R_t Q_R)_{jj} = \mu_j$, and the total is $\operatorname{tr}(L_t)$. So the rotated step is $\sqrt{\operatorname{tr}(L_t)}\operatorname{diag}(\lambda)^{-1/2} G'_t \operatorname{diag}(\mu)^{-1/2}$, and rotating back gives $\sqrt{\operatorname{tr}(L_t)}\, Q_L \operatorname{diag}(\lambda)^{-1/2} Q_L^\top G_t Q_R \operatorname{diag}(\mu)^{-1/2} Q_R^\top$. ∎

The same holds with moving averages. *SOAP* keeps the eigenbasis and replaces Adafactor's rank-one $\hat V$ by Adam's full matrix of second moments. On Kronecker curvature this removes the conditioning, because Adam with $\epsilon = 0$ does not see a positive rescaling of a coordinate's gradients.[7]

[7] Kingma and Ba (2015) list invariance to diagonal rescaling of the gradients among Adam's properties. It is exact only for $\epsilon = 0$; for coordinates with gradients much smaller than ϵ, Adam is momentum gradient descent with step size η/ϵ (Chapter 6).

Proposition 7.16 (Adam in the Kronecker eigenbasis). *Under Assumption 7.5, let* $A = U_A \operatorname{diag}(\alpha) U_A^\top$ *and* $C = U_C \operatorname{diag}(\gamma) U_C^\top$*, and run Adam with* $\epsilon = 0$ *(with or without bias correction, reading* $0/0$ *as* 0*), or AdaGrad, on the rotated gradients* $U_A^\top \nabla\mathcal{L}(W_t) U_C$*, rotating each update back. Then the rotated errors* $\tilde E_t = U_A^\top (W_t - W_\star) U_C$ *do not depend on* α *and* γ*; in particular they are those of the same method on the isotropic loss* $\frac12\|W - W_\star\|_F^2$*. If Shampoo's factors are* $f(A)$ *and* $h(C)$ *for injective* f, h *on* $(0,\infty)$*, as in Corollary 7.12 with* $q > 0$*, then every orthonormal eigenbasis of the factors is one of* A *and* C*, and SOAP with these frozen factors and* $\epsilon = 0$ *is this method.*

Proof. The plan is an induction on t, based on one invariance. In the rotated coordinates the gradient is $\operatorname{diag}(\alpha)\tilde E \operatorname{diag}(\gamma)$, with entries $\alpha_i \gamma_j \tilde E_{ij}$. If every gradient of one coordinate is multiplied by $c > 0$, then Adam's moments become cm and $c^2 v$, and its update, a function of $m/\sqrt{v}$, does not change; nor does AdaGrad's $g/(\sum g^2)^{1/2}$. (A coordinate whose gradients have all been zero has $m = v = 0$ in both problems and does not move.) So if the rotated errors of the problem with curvatures $\alpha_i\gamma_j$ and of the isotropic problem agree up to step t, then their gradients in coordinate (i,j) differ by the factor $\alpha_i\gamma_j$ at every step so far, their updates agree, and so do their errors at step $t+1$. For the last claim, $f(A) = U_A \operatorname{diag}(f(\alpha)) U_A^\top$ and, f being injective, $f(A)$ and A have the same eigenspaces; likewise $h(C)$ and C. In a fixed basis, SOAP's moving averages of rotated gradients are rotated moving averages, so SOAP is Adam in that basis. ∎

Adam is curvature-free on every diagonal quadratic; it is the rotation of the problem that hurts it (Chapters 2 and 14), and on Kronecker curvature with stationary statistics SOAP undoes the rotation. Curvature-free does not mean fast: each mode moves as it would on the isotropic quadratic, at

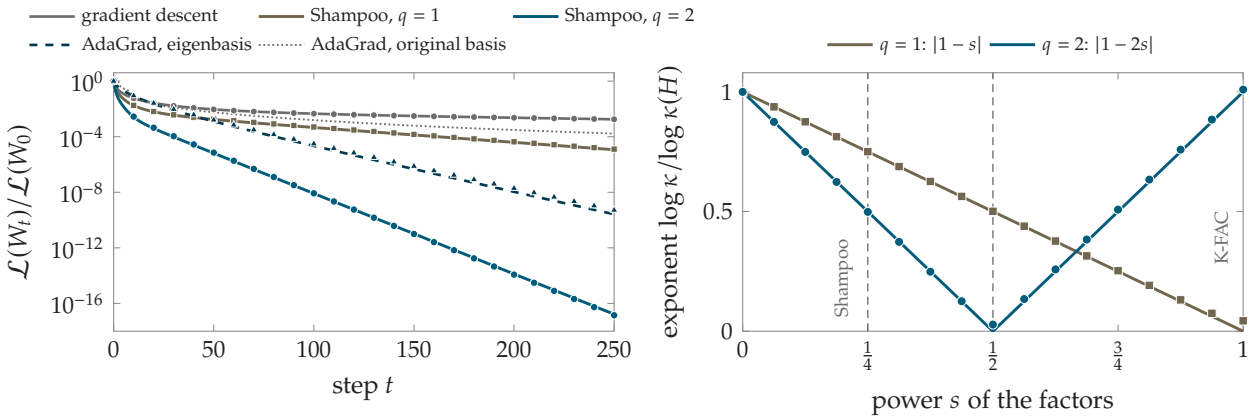


Figure 7.1. Kronecker preconditioners on the Kronecker quadratic with $m = 16$, $n = 24$, $\kappa_A = \kappa_C = 30$, and Haar-random eigenvectors; Shampoo's factors are estimated from 1000 sampled gradients with statistics $\mathbb{E}[gg^\top] \propto H^q$. Left: relative loss from one random start; curves are the exact formulas of Propositions 7.11 and 7.16 for exact factors, marks (every tenth step) the matrix iterations with estimated factors, at the best constant step size. AdaGrad has step size 0.3 in Shampoo's eigenbasis (exact basis for the curve, the estimated $q = 1$ basis for the marks) and 1.0 in the original coordinates (dotted, simulation only). Right: the exponent of (7.6); lines $|1 - qs|$, marks from the asymptotic rate with estimated factors. Script: `07-kronecker-ladder.py`.

the method's pace there. And the basis must be right, which with estimated factors it is only approximately (Figure 7.1).

7.5 Reading the theory

THE LADDER. Take $\kappa_A = \kappa_C = 30$, so $\kappa(H) = 900$. Once the slowest mode dominates, the best constant step reduces the loss tenfold in about $\ln 10/(2 \ln \frac{\kappa+1}{\kappa-1}) \approx 0.58\kappa$ steps: 520 for gradient descent, 95 for Shampoo with Hessian-shaped statistics ($\kappa = 900^{3/4} \approx 164$), 17 for Shampoo with isotropic-error statistics or the power $\frac{1}{2}$ with Hessian-shaped ones ($\kappa = 30$), and one for K-FAC with exact factors. In Figure 7.1 (left), factors estimated from 1000 sampled gradients give these conditionings to within one percent (164.7 and 29.7), and the losses follow the exact formulas over seventeen decades. AdaGrad in Shampoo's eigenbasis, the idealized SOAP of Proposition 7.16, beats Shampoo with Hessian-shaped statistics, and with step size 0.5 instead of 0.3 also the other (7×10^{-22} against 2×10^{-17} at $t = 250$); in the original coordinates it gains only a decade on gradient descent. In the eigenbasis of the estimated factors, off by about a tenth of a radian, SOAP stays within about a factor of two of its prediction over almost ten decades; how a basis error enters the rate, Proposition 7.16 does not say.

WHICH POWER. The right panel measures Corollary 7.12: for each power s the iteration runs with the estimated factors at its best step size, and its asymptotic rate ρ gives $\kappa = (1 + \rho)/(1 - \rho)$. The marks follow $|1 - qs|$ to within about 0.01, except at the bottoms of the V, where the sampling

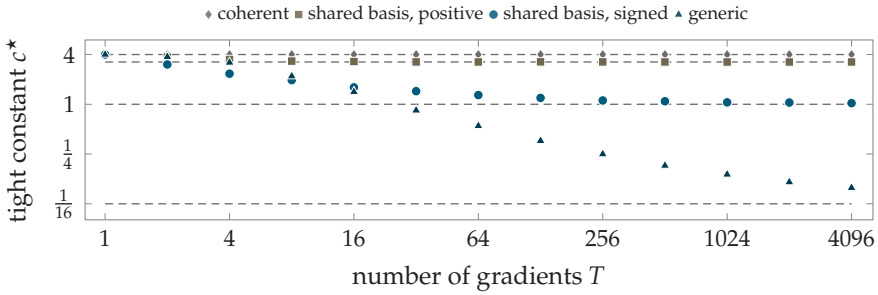


Figure 7.2. The tight constant $c^\star$ for T gradients of rank 4 in $\mathbb{R}^{16\times16}$ (marks: mean of five seeds, computed from the 256×256 matrices) and its limits from Proposition 7.9 (dashed): 4, 3.25, 1, and $\frac{1}{16}$. Script: `07-kronecker-constant.py`.

error of the factors, a few percent per eigenvalue, leaves the exponents 0.028 and 0.044 ($\kappa \approx 1.21$ and 1.35), a floor that shrinks roughly like the inverse square root of the number of gradients (Exercise 7.8). Three lessons follow. The power matters as much as preconditioning: an inverse that over-corrects can be as bad as none. For both kinds of statistics Shampoo's 1/4 helps but does not finish the job. And the right power is set by what the factors estimate, which is why K-FAC inverts curvature estimates while AdaGrad-type methods take square roots of second moments of gradients.

How loose the constant is. In Figure 7.2 a single gradient of rank 4 attains $c^\star = r$, and a coherent sequence, all multiples of one matrix, stays there. In a shared singular basis $c^\star$ falls toward the top eigenvalue of the limiting cosine matrix: $1 + \frac{3}{4}(r-1) = 3.25$ for independent coefficients uniform on $(0,1)$, whose cosines tend to $\mathbb{E}[s]^2/\mathbb{E}[s^2] = \frac{3}{4}$, and 1 for sign-symmetric ones. Generic gradients $X_t Y_t^\top$, with Gaussian factors, have statistics tending to $rI_n \otimes I_m$ and a constant falling toward $(mn)^{-1/2}$. The factor $\sqrt{r}$ in Theorem 7.10 is a worst case.

Costs and the regret bound. Shampoo stores m^2+n^2 numbers for an $m\times n$ matrix, against $(mn)^2$; the roots cost of order m^3+n^3 operations, as much as a forward and backward pass of a 4096×4096 layer on some thousands of tokens, which is why practice computes them every so many steps. The regret bound cannot explain an advantage. Since $\operatorname{tr}(M^{1/4}) \geq (\operatorname{tr} M)^{1/4}$ for $M \succeq 0$, $\operatorname{tr}(L_T^{1/4})\operatorname{tr}(R_T^{1/4}) \geq (\operatorname{tr} L_T \operatorname{tr} R_T)^{1/4} \geq (\sum_t \|G_t\|_F^2)^{1/2}$, and $D^2/(2\eta) + r\eta \geq \sqrt{2r}D$, so the right side of (7.4) is at least $\sqrt{2r}D(\sum_t\|G_t\|_F^2)^{1/2}$ for every η, while online gradient descent with steps $\eta_t = D/(2\sum_{\tau\leq t}\|G_\tau\|_F^2)^{1/2}$ has regret at most $\sqrt{2}D(\sum_t\|G_t\|_F^2)^{1/2}$ (Theorem A.21 and Lemma A.20). With the same Frobenius distance, Shampoo's guarantee is never better than gradient descent's, though it can beat diagonal AdaGrad's by up to $\sqrt{mn}$ (Exercise 7.5).

7.6 What the theory misses

WHAT WAS IDEALIZED AWAY. The Kronecker structure itself, exact for one linear layer whose inputs and back-propagated gradients are independent (Proposition 7.13); in a network the gradient flowing back into a layer depends on its output and hence on its input, minibatches add the mean term of Proposition 7.13(iii), and layers are coupled (Exercise 7.4). Stationarity: the factors average over a trajectory along which the statistics drift from error-dominated to noise-dominated, so Corollary 7.12 describes a frozen snapshot; and on a deterministic quadratic the gradients shrink, so that plain Shampoo's sums freeze at values set by the first steps, which depend on the step size (Exercise 7.9). Convex online losses, in the regret theory. Exact roots at every step, where practice recomputes them infrequently and in finite precision, and the inverse fourth root of an ill-conditioned factor depends strongly on ε. Grafting, momentum, weight decay, and the parameters that are not matrices.

WHERE IT BREAKS. (1) The regret bound does not rank optimizers: it is never better than that of gradient descent with an adaptive step (Section 7.5), so it does not explain Observations 7.1 and 7.2. (2) The theorem needs the power 1/4, implementations also use 1/2, and Corollary 7.12 shows that no power is right for all statistics; Morwani et al. (2025) find the square of Shampoo's preconditioner the better approximation of full-matrix AdaGrad in networks. (3) Performance rests on heuristics the theory does not cover (Observation 7.3). (4) For K-FAC the curvature story is incomplete: the empirical Fisher can be a poor curvature estimate (Kunstner et al. 2019), and Benzing (2022) found that a variant keeping only the input-side factor, a one-sided preconditioner in the sense of Proposition 7.11, often matches or improves on K-FAC in cost and data efficiency, and that K-FAC outperforms exact second-order updates, which suggests that K-FAC's advantage is not mainly second-order. (5) The advantage of matrix-based optimizers over a tuned AdamW shrinks with scale (Wen et al. 2026), which a one-layer idealization cannot address. (6) SOAP is curvature-free only in the exact eigenbasis; the theory does not say how a basis error enters the rate, nor how often to refresh the basis, which Observation 7.3 says matters.

REPAIRS AND OPEN PROBLEMS. Correcting the eigenvalues in the Kronecker eigenbasis repairs the mismatch between the factors and the true scales: eigenvalue-corrected K-FAC (George et al. 2018), SOAP (Vyas et al. 2025), and the decoupled updates of eigenvalues and eigenbasis studied by Eschenhagen et al. (2025). Better Kronecker approximations iterate the power method that follows Proposition 7.8 (Morwani et al. 2025; Van Loan and Pitsianis 1993). Open: how close trained networks' gradient statistics are to Kronecker products (Exercise 7.10); a theory of the power, or of a schedule for it, under drifting statistics (Exercise 7.11); a nonconvex theory in which Shampoo or SOAP provably beats Adam; and why the advantage of matrix-based optimizers shrinks with scale.

7.7 Back in the wild

FIELD NOTES	
PRACTICE	Precondition each weight matrix by inverse roots of row and column second-moment matrices (Shampoo), by inverses of input and output-gradient covariances (K-FAC), or by Adam in the eigenbasis of those matrices (SOAP).
IDEALIZATION	Full-matrix AdaGrad on $\mathrm{vec}(W)$; the Kronecker quadratic with Hessian $C \otimes A$; stationary statistics $\mathbb{E}[gg^\top] \propto H^q$.
PROVED	The Kronecker bound, with constant r attained by single gradients; Shampoo's regret; $\kappa \mapsto \kappa^{\lvert 1-qs\rvert}$ for frozen factors; K-FAC exact under independence; Shampoo with power $\frac{1}{2}$ is Adafactor in its eigenbasis; Adam in that basis is curvature-free.
PREDICTS	Shampoo's power $\frac{1}{4}$ leaves $\kappa^{1/2}$ or $\kappa^{3/4}$; the power $\frac{1}{2}$ is full-matrix AdaGrad for Kronecker statistics; inverses are Newton for curvature-like statistics and over-correct for gradient-like ones.
MISSES	Why these methods beat a tuned Adam when they do; grafting, ε, and stale roots; non-Kronecker and drifting statistics; shrinking gains at scale.
TAKEAWAY	Match the power to what the factors estimate: inverses for curvature, inverse square roots for second moments of gradients (proved in the idealization); keep SOAP's eigenbasis fresh (observed); expect the gain over a tuned AdamW to shrink with scale (observed).

The theory tells a practitioner three things: Kronecker factors bound full-matrix AdaGrad's statistics up to the rank, and reproduce them up to a scalar when the statistics are a Kronecker product; the power of the factors is the decisive choice and should match what they estimate, so the theory's 1/4 is chosen for the proof, not for speed, consistent with the use of 1/2 with grafting and of inverses on curvature estimates; and, in the idealization, SOAP's advantage comes from the basis, in which Adam is curvature-free. It does not say when this beats a well-tuned AdamW; the benchmarks say that it sometimes does, by margins that shrink with scale, for reasons not understood.

The chapter closes the unification line of Parts I and II. Without memory Shampoo is the orthogonalized gradient of Chapter 3, as Adam is the sign descent of Chapter 2; with memory both become preconditioners, diagonal (Chapter 6) or Kronecker-factored, and SOAP runs the first in the second's basis. The dynamics line takes over with preconditioned sharpness (Chapter 8), the critical batch size (Chapter 11), and the gap between Adam and gradient descent (Chapter 14).

Exercises

7.1 CONSOLIDATION. (Vectorization.) (a) Show that $\operatorname{vec}(uv^\top) = v \otimes u$ and $\langle X, AYC\rangle = \operatorname{vec}(X)^\top (C \otimes A)\operatorname{vec}(Y)$ for symmetric C, and that $(C \otimes A)(v_j \otimes u_i) = \gamma_j \alpha_i (v_j \otimes u_i)$ for eigenvectors $Au_i = \alpha_i u_i$, $Cv_j = \gamma_j v_j$. (b) With the row-stacking $\overline{\operatorname{vec}}(X) = \operatorname{vec}(X^\top)$, show that $\overline{\operatorname{vec}}(MXN) = (M \otimes N^\top)\overline{\operatorname{vec}}(X)$, so that every Kronecker product of this chapter reverses its order; write (7.2) in that convention.

7.2 CONSOLIDATION. (Geometric means.) (a) For commuting positive semidefinite P, Q, show that $\left(\begin{smallmatrix} P & (PQ)^{1/2} \\ (PQ)^{1/2} & Q \end{smallmatrix}\right) \succeq 0$. (b) For positive definite P, Q, show with a Schur complement that $\left(\begin{smallmatrix} P & Y \\ Y & Q \end{smallmatrix}\right) \succeq 0$ if and only if $YP^{-1}Y \preceq Q$, and deduce, using Theorem A.9 and $|Z| \succeq Z$ for symmetric Z, that for commuting P, Q every such Y satisfies $Y \preceq (PQ)^{1/2}$.

7.3 CONSOLIDATION. (The constant.) Let u_1, u_2 and v_1, v_2 be orthonormal pairs. With Proposition 7.9, compute $c^\star$ for (a) the single gradient $u_1 v_1^\top + u_2 v_2^\top$; (b) the gradients $u_1 v_1^\top + u_2 v_2^\top$ and $u_1 v_1^\top - u_2 v_2^\top$; (c) the gradients $u_1 v_1^\top + u_2 v_2^\top$ and $u_1 v_1^\top$. (d) With the rearrangement $\mathcal{R}$ defined by $\mathcal{R}(\Phi \otimes \Psi) = \operatorname{vec}(\Phi)\operatorname{vec}(\Psi)^\top$ and extended linearly, show that $\mathcal{R}(\Gamma)\operatorname{vec}(I_m) = \operatorname{vec}(R(\Gamma))$ and $\mathcal{R}(\Gamma)^\top \operatorname{vec}(I_n) = \operatorname{vec}(L(\Gamma))$: one step of the power method for the nearest Kronecker product, from identities, returns the partial traces. *(Hint: (c) gives* $1 + 1/\sqrt{2}$*.)*

7.4 CONSOLIDATION. (K-FAC for a two-layer linear network.) Let $f(x) = W_2 W_1 x$ with $W_1 \in \mathbb{R}^{d_1 \times d_0}$, $W_2 \in \mathbb{R}^{d_2 \times d_1}$, $x \sim \mathcal{N}(0, C)$, and the model $y \sim \mathcal{N}(f(x), I)$ with sampled targets. (a) Show that the Fisher matrix has diagonal blocks $F_{11} = C \otimes W_2^\top W_2$ and $F_{22} = W_1 C W_1^\top \otimes I_{d_2}$, on which K-FAC is exact. (b) Show that $F_{12} = C W_1^\top \otimes W_2^\top$, which K-FAC drops. (c) Show that the full Fisher matrix is singular, because $(W_1, W_2) \mapsto (MW_1, W_2 M^{-1})$ leaves f unchanged for every invertible M.

7.5 VARIATION. (One-sided Shampoo.) Let G have rank r and $g = \operatorname{vec}(G)$. (a) Show that $gg^\top \preceq r(I_n \otimes GG^\top)$ and $gg^\top \preceq r(G^\top G \otimes I_m)$, with sharp constants. *(Hint:* $\langle G, X\rangle = \langle VV^\top, G^\top X\rangle$ *for* $G = U\Sigma V^\top$*.)* (b) Deduce that one-sided Shampoo, $W_{t+1} = W_t - \eta L_t^{-1/2} G_t$, has regret at most $(\frac{D^2}{2\eta} + r\eta)\, n \operatorname{tr}(L_T^{1/2})$. (c) For $G_t = c_t u v^\top$ with fixed unit vectors u, v, show that as $\varepsilon \to 0$ the bound of Theorem 7.10 at its best step size tends to $\sqrt{2} D (\sum_t c_t^2)^{1/2}$, while diagonal AdaGrad's bound (Theorem 6.7), with D the ℓ_∞ diameter of its box, is $\sqrt{2} D \|u\|_1 \|v\|_1 (\sum_t c_t^2)^{1/2}$, up to $\sqrt{mn}$ times larger.

7.6 VARIATION. (The power $\frac{1}{2}$ and grafting.) (a) With $\varepsilon = 0$, show that $L_t^{-1/2} G_t R_t^{-1/2}$ is homogeneous of degree -1 in the gradients, while Shampoo's step is homogeneous of degree 0. (b) Grafting replaces a step Δ by $\Delta \|\Delta_{\text{ref}}\|_F / \|\Delta\|_F$, with Δ_{ref} the step of a reference method. Show that the grafted step has the degree of the reference, and that for

Kronecker statistics it does not depend on the scalar $(\operatorname{tr}\Gamma)^{1/2}$ of (7.3); with full-matrix AdaGrad as reference it is full-matrix AdaGrad's step.

7.7 VARIATION. (Other noise, and ϵ.) Let Shampoo's factors average gradients $\nabla\mathcal{L}(W) + \Xi$ of the Kronecker quadratic in the noise-dominated limit. (a) For $\operatorname{vec}(\Xi) \sim \mathcal{N}(0, \sigma^2 I)$, show that Shampoo is gradient descent ($q = 0$). (b) For noise covariance $\sigma^2 H^q$ with q unknown in $[1, 2]$, find the power that minimizes the worst-case exponent $\max_q |1 - qs|$. (c) Show that Adam with $\epsilon > 0$ is not invariant under rescaling a coordinate's gradients by c: it tends to the $\epsilon = 0$ update as $c \to \infty$ and to momentum gradient descent with step size η/ϵ as $c \to 0$. Which modes lose the curvature-free property of Proposition 7.16 first? *(Hint: in (b), $s = 2/3$ and the exponent is $1/3$.)*

7.8 SIMULATION. (The V and its floor.) Run `07-kronecker-ladder.py`, whose options `-kA`, `-kC`, and `-N` print the exponents for other condition numbers and numbers of sampled gradients, (a) with $\kappa_A = 900$ and $\kappa_C = 1$, and check that the exponents still follow $|1 - qs|$ relative to $\kappa(H) = 900$, to within the sampling floor, although one-sided preconditioning is now exact; (b) with 100, 1000, and 10 000 sampled gradients, and check that the exponent at the bottom of each V falls roughly like the inverse square root of their number, as the sampling error of the factors does (we measured 0.16, 0.044, and 0.015 for $q = 1$, $s = 1$).

7.9 SIMULATION. (Shampoo's transient.) Run plain Shampoo (sums, $\varepsilon = 10^{-12}$, power $\frac{1}{4}$) for 1500 steps on the Kronecker quadratic with $m = 20$, $n = 30$, diagonal A and C whose eigenvalues are geometrically spaced from 1 to $\frac{1}{64}$, a standard Gaussian initial error, and step sizes 1, 3, 10, and 30; clamp the computed eigenvalues of the factors at ε before taking roots, since R_1 is singular up to ε. (a) Observe that for the two larger step sizes the loss first rises (over six seeds, by factors of 2.5 to 4 and of 40 to 54), and that the step size 10 nevertheless reaches $10^{-12}\mathcal{L}(W_0)$ sooner than the step sizes 1 and 3. (b) Explain the transient: a step that is too long makes the gradients grow until the accumulated factors make it stable. (c) Once the gradients have shrunk, the factors stop changing. Check that the late-time rate per step is the rate $\max_k(1 - \eta\mu_k)^2$ of these frozen factors (proof of Proposition 7.11), and compare their κ with the value $\kappa(H)^{1/2} = 64$ of Corollary 7.12 for $q = 2$ (we found κ between 16 and 720, decreasing with the step size). Explain why sums along one run do not have the statistics of isotropic errors.

7.10 RESEARCH. (Are gradient statistics Kronecker?) For layers of a trained network, measure the relative distance $\min_{\Phi,\Psi}\|\Gamma - \Phi \otimes \Psi\|_F / \|\Gamma\|_F$ of the gradient second moment to the nearest Kronecker product (Van Loan and Pitsianis 1993) and the constant $c^\star$. Find conditions on the joint law of inputs and back-propagated gradients, beyond independence, under which the distance is provably small. As far as

we know no general theory exists.

7.11 RESEARCH. (A schedule for the power.) Along a run the statistics pass from error-dominated ($q \approx 2$) to noise-dominated ($q \approx 1$) as the mean gradient shrinks (Proposition 7.13(iii)). In the noisy Kronecker quadratic started from an isotropic error, find the schedule of the power s_t that minimizes the number of steps to a target loss, and compare it with a fixed power with grafting. This is open.

Notes and References

Full-matrix AdaGrad is due to Duchi et al. (2011), who also analyzed the diagonal version (Chapter 6). Gupta et al. (2018) introduced Shampoo for tensors of every order and proved Theorem 7.7 (Lemma 8 in the arXiv version of their paper) and the regret bound; our proof of the bound passes through Lemma 7.6, which yields the middle term of (7.2). Anil et al. (2020) and Shi et al. (2023) made Shampoo practical at scale, with grafting from Agarwal et al. (2020); Kasimbeg et al. (2025) analyze the first AlgoPerf competition (Dahl et al. 2023).

K-FAC (Martens and Grosse 2015) approximates natural gradient descent (Amari 1998), and George et al. (2018) corrected its eigenvalues; Kunstner et al. (2019) and Benzing (2022) question the curvature reading of empirical Fishers and of K-FAC. Morwani et al. (2025) relate Shampoo's factors to the optimal Kronecker approximation of full-matrix AdaGrad's statistics and of the Gauss–Newton matrix, and find the square of Shampoo's approximation, not the approximation itself, close to optimal. SOAP and the identity of Proposition 7.15 with Adafactor (Shazeer and Stern 2018) are due to Vyas et al. (2025), and Eschenhagen et al. (2025) take Shampoo's heuristics apart. That Shampoo without accumulation is the orthogonalized gradient, Muon's update without momentum (Chapter 3), is observed by Bernstein and Newhouse (2024). We know no reference for the exponent law of Corollary 7.12, the exact constants of Proposition 7.9, or Proposition 7.16; all three are elementary consequences of Propositions 7.8 and 7.11 and of Adam's invariance to diagonal rescaling (Kingma and Ba 2015).

TRANSLATING THE LITERATURE. Gupta et al. (2018) vectorize matrices by rows, so their Kronecker products appear in the reverse of our order (Exercise 7.1): the right side of their bound, $(\varepsilon I_m + \sum_t G_t G_t^\top)^{1/2} \otimes (\varepsilon I_n + \sum_t G_t^\top G_t)^{1/2}$, is our $R_T^{1/2} \otimes L_T^{1/2}$; their L_t, R_t, r, and D are ours. Martens and Grosse (2015) write a layer's gradient as $g\bar{a}^\top$ (g is our δ, $\bar{a}$ our a with a coordinate one appended for the bias) and approximate the Fisher block by $\bar{A} \otimes G$, where $\bar{A}$ is our $\mathbb{E}[aa^\top]$ and G our $\mathbb{E}[\delta\delta^\top]$, not a gradient; they stack columns, as we do. Vyas et al. (2025) write L, R, Q_L, Q_R as we do, and their Shampoo with the $1/2$ power is our $s = \frac{1}{2}$.

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Part III

Stability

Practical step sizes sit at the edge of stability. Gradient descent drives the curvature up until the step is barely stable and then holds it there; the first steps of training meet curvature that is too high for the target step size, which is why the step size must start small; and curvature that grows with the gradient is tamed by clipping and by normalized steps.

8 The Edge of Stability

Thresholds, self-stabilization and central flows

IN THE WILD — LEARNING RATES AT THE STABILITY LIMIT

IN THIS CHAPTER. Learning rates are chosen as large as training tolerates, and full-batch training then runs with its curvature held near the optimizer's stability threshold: the sharpness at $2/\eta$ for gradient descent, the preconditioned sharpness at $38/\eta$ for Adam with $\beta_1 = 0.9$. We derive the thresholds from one stability test and study the cubic model of Damian, Nichani and Lee, in which one direction oscillates and the oscillation pushes the curvature down. In its continuous limit every cycle conserves an energy and closes, and over each cycle the curvature averages exactly $2/\eta$ and the squared oscillation exactly twice the rate of progressive sharpening; for the gradient-descent map we prove that the energy increases at every step, so the cycles spiral slowly outward, and derive the rate. Averaging over cycles gives the central flow, gradient flow projected onto the stable region. Progressive sharpening is an input of the model, not explained by it.

8.1 In the wild: the largest learning rate that does not diverge

Gradient descent on a convex quadratic multiplies the component of $\theta_t - \theta^\star$ along an eigenvector of the Hessian with eigenvalue λ_i by $1 - \eta\lambda_i$ at every step, so it converges from every initialization exactly when $\eta\lambda_i < 2$ for every eigenvalue (**??**). The largest eigenvalue, $S(\theta) = \lambda_{\max}(\nabla^2 \mathcal{L}(\theta))$, is the *sharpness*, and $\eta S < 2$ is the classical stability condition.[1] Read as advice, the condition says: estimate the curvature, then choose the step. Networks are not trained that way.

[1] The learning rate and the step size are the same number η; we say learning rate in the practice sections and step size in the theory.

IN THE WILD · folklore. Choose the learning rate by a sweep on a logarithmic grid, neighboring values a factor of two or three apart, and keep the value with the lowest loss after a fixed budget; it is usually among the largest values that train without diverging, and a run that diverges is restarted with a smaller value. The curvature of the loss is measured neither at initialization nor later. Momentum, and Adam's average of the gradient with $\beta_1 = 0.9$ (Kingma and Ba 2015), stay at their defaults during the sweep.

The rule describes how sweeps are commonly run, not a measured

regularity; the *Deep Learning Tuning Playbook* (Godbole et al. 2023) is a systematic guide to running them. What it leaves unsaid is that the curvature is not a fixed property of the loss: it responds to the learning rate.[2]

[2] That the step size limits where gradient descent can be stable, and so selects flat minima, is the dynamical-stability argument of Wu et al. (2018); Jastrzębski et al. (2019) observed that the sharpness reached early in training adapts to the step size.

Observation 8.1 (Progressive sharpening and the edge of stability; Cohen et al. 2021). Train a network by full-batch gradient descent with a fixed step size η. (i) While $S(\theta_t) < 2/\eta$, the sharpness tends to rise. (ii) Once it reaches $2/\eta$, it stops rising and hovers at or just above $2/\eta$, while the training loss keeps decreasing over long time scales, though not monotonically over short ones. (iii) With heavy-ball momentum β the sharpness settles near $(2+2\beta)/\eta$ instead, and with Nesterov momentum near $(2+2\beta)/((1+2\beta)\eta)$: in each case the largest sharpness at which the method is stable on a quadratic.

Cohen and coauthors named the two stages *progressive sharpening* and the *edge of stability*, documented them for fully connected, convolutional and transformer networks with the square and the cross-entropy loss, and derived the momentum thresholds in (iii), the maximum stable sharpness (MSS) in their terms, which the sharpness reached and held in their runs with $\beta = 0.9$. The observation is robust for full batch, with two limits: progressive sharpening is weaker for wide or shallow networks and easy tasks, and with the cross-entropy loss the sharpness typically falls late in training (Cohen et al. 2021).

Observation 8.2 (The adaptive edge of stability; Cohen et al. 2022). Train a network full batch by $\theta_{t+1} = \theta_t - \eta P_t^{-1} m_t$, with $m_t = \beta_1 m_{t-1} + (1-\beta_1)\nabla\mathcal{L}(\theta_t)$ and a diagonal preconditioner P_t: Adam, with $P_t = (1-\beta_1^{t+1})(\mathrm{diag}(\sqrt{\hat{v}_t}) + \epsilon I)$, which absorbs both bias corrections, or RMSProp, with $\beta_1 = 0$ and no bias correction. The *preconditioned sharpness* $S_P = \lambda_{\max}(P_t^{-1}\nabla^2\mathcal{L}(\theta_t))$ rises until it reaches the stability threshold of the same method with P_t frozen, which is $38/\eta$ for Adam with $\beta_1 = 0.9$, and then equilibrates there. The sharpness itself is not held at a threshold: the iterates keep moving into regions of higher curvature, and the preconditioner grows to compensate. Similar effects occur with minibatches, the more so the larger the batch.

The source is an unrefereed preprint (arXiv 2022; a version appeared at a NeurIPS 2023 workshop), but the claim is easy to check: Figure 8.1 reproduces it, with Observation 8.1, in a network that trains in half a minute.[3]

[3] The number is $(2+2\beta_1)/(1-\beta_1)$ at $\beta_1 = 0.9$: the threshold of the moving average through which Adam passes its gradients (Proposition 8.6(iv)).

Observation 8.3 (Minibatches; Jastrzębski et al. 2020, Andreyev and Beneventano 2024). With minibatch stochastic gradient descent the full-batch sharpness levels off below $2/\eta$, at a value set by the step size and the batch size. What settles at $2/\eta$ is the expected curvature of a minibatch loss along its own gradient, which Andreyev and Beneventano call the *batch sharpness*.

The first sentence goes back to Jastrzębski et al. (2019, 2020), and Lee and Jang (2023) proposed an earlier sharpness adapted to the batch; the second rests on an unrefereed preprint (2024, revised in 2025). This chapter is about full batch; Chapter 11 takes up the noise.

The three observations share a shape: a threshold computed from a frozen quadratic model, and training that drives the relevant curvature up to it and holds it there. Why does the iteration not diverge once the threshold is crossed, as the quadratic model says it must? And what does training do while it sits there? Volume 2 of this series, *The Hydrogen Atoms of Deep Learning*, Chapter 15, answers the first question for a single excursion: in a two-parameter product model, a step size above the threshold makes the loss spike and throws the iterates to a flatter minimizer, the *catapult* of Lewkowycz et al. (2020). That model has no progressive sharpening, so after one catapult it stays below the threshold. This chapter adds the sharpening.

8.2 The idealization: one oscillating direction

The quadratic model freezes the Hessian. To see the curvature respond to the step size we need three ingredients: a direction along which the iterates oscillate when the step is too large, a curvature along it that depends on the other parameters, and a training force that pushes the other parameters toward higher curvature. The following family has exactly these.

Assumption 8.4 (One oscillating direction). The parameters split as $\theta = (x, w) \in \mathbb{R} \times \mathbb{R}^{d-1}$; the functions $\mathcal{L}_0$ and s on $\mathbb{R}^{d-1}$ are twice continuously differentiable, with $s(w) > \lambda_{\max}(\nabla^2 \mathcal{L}_0(w))$ for every w; and

$$\mathcal{L}(x, w) = \mathcal{L}_0(w) + \tfrac{1}{2}\, s(w)\, x^2. \tag{8.1}$$

On the floor $x = 0$ of the valley the Hessian is block diagonal, so the sharpness there is $s(w)$, with the x-axis as eigenvector. Gradient descent with step size η is, exactly,

$$x_{t+1} = \big(1 - \eta s(w_t)\big) x_t, \qquad w_{t+1} = w_t - \eta \nabla \mathcal{L}_0(w_t) - \tfrac{\eta}{2}\, x_t^2\, \nabla s(w_t). \tag{8.2}$$

The first equation is the quadratic picture: the oscillating coordinate decays while $\eta s < 2$, and above the threshold it flips sign and grows. The second contains the new force: the oscillation pushes w down the gradient of the curvature, with a strength proportional to x^2. Two numbers measure the competition between the training force and the push,

$$\alpha(w) = -\langle \nabla \mathcal{L}_0(w), \nabla s(w) \rangle, \qquad \gamma(w) = \|\nabla s(w)\|_2^2. \tag{8.3}$$

Along gradient flow $\dot{w} = -\nabla \mathcal{L}_0(w)$ the curvature changes at the rate $\dot{s} = \alpha$, so $\alpha > 0$ is progressive sharpening; along the push, the oscillation lowers s at the rate $\frac{1}{2}\gamma x^2$.

The smallest instance linearizes $\mathcal{L}_0$ and s at a point on the threshold: $\mathcal{L}_0(w) = \mathcal{L}_0(0) + \langle g, w \rangle$ and $s(w) = 2/\eta + \langle c, w \rangle$ with $c \neq 0$ (Assumption 8.4 then holds where $s > 0$). Then $y = \langle c, w \rangle$ is the excess of the curvature over $2/\eta$, and (8.2) gives $x_{t+1} = -(1 + \eta y_t) x_t$ and $y_{t+1} = y_t - \eta \langle c, g \rangle - \frac{\eta}{2} \|c\|_2^2 x_t^2$, while the component of w orthogonal to c performs gradient descent and never feels the oscillation. The pair $(\|c\|_2 x_t, y_t)$ follows the map below with $\alpha = -\langle g, c \rangle$ (Exercise 8.3).

Definition 8.5 (The cubic model). Fix a step size $\eta > 0$ and a rate $\alpha > 0$. The *cubic model* is the loss $\mathcal{L}(x, y) = \frac{1}{2}(2/\eta + y)x^2 - \alpha y$ on $\mathbb{R}^2$, trained by gradient descent with step size η:

$$x_{t+1} = -(1 + \eta y_t)\, x_t, \qquad y_{t+1} = y_t + \eta\Big(\alpha - \tfrac{1}{2}x_t^2\Big). \tag{8.4}$$

The name refers to the cubic coupling $\frac{1}{2}yx^2$. On the line $x = 0$ the sharpness is exactly $2/\eta + y$, and elsewhere it is $2/\eta + y + O(\eta x^2)$. The loss is unbounded below: it models a region in which training raises the curvature, by $\eta\alpha$ per step when $x = 0$ and in general as long as $x^2 < 2\alpha$.

DICTIONARY

IN THE THEORY	IN A TRAINING RUN
the coordinate x, flipping sign at every step	the component of θ_t along the top eigenvector of the Hessian (of $P^{-1}\nabla^2\mathcal{L}$ for Adam)
the other coordinates w, and $\mathcal{L}_0(w)$	all other parameters, and the loss at the center of the oscillation
the curvature $s(w) = 2/\eta + y$	the sharpness $S(\theta_t)$, at the distance y from the threshold
the rate $\alpha = -\langle \nabla\mathcal{L}_0, \nabla s\rangle$	the speed of progressive sharpening along gradient flow
the coupling $\gamma = \|\nabla s\|_2^2$	how fast an oscillation lowers the sharpness, through the third derivative
the time $\tau = \eta t$	steps times learning rate, the clock in which gradient flow has unit speed
the time average of x^2	the variance σ^2 of the oscillation in a central flow
a frozen preconditioner P	Adam's P_t of Observation 8.2, which moves on the time scale $1/(1-\beta_2)$

WHAT WAS THROWN AWAY. The oscillating direction is fixed and alone at the threshold; the loss is quadratic and even in x; the rates α and γ of the cubic model are constant; gradients are exact, without momentum or a preconditioner (Section 8.3 restores both for quadratics). What is kept is the pair of forces whose balance is the phenomenon, and it is what a third-order Taylor expansion of a general loss produces near the threshold (Section 8.4).

WHAT ONE HAD TO NOTICE. In the quadratic theory a growing oscillation is the beginning of divergence. The insight of Damian et al. (2023) is that it is also the restoring force: through the third derivative of the loss, the oscillation produces the gradient $\frac{1}{2}x^2\nabla s$, which lowers the curvature that caused it. This is *self-stabilization*.[4] A technical observation makes the model solvable: in the variables $A = \log x^2$, which forgets the sign flips, and y, a cycle of oscillation is a closed curve.

[4] Damian, Nichani and Lee describe a cycle in four stages: progressive sharpening, blowup, self-stabilization, and return to stability.

8.3 The theory: stability thresholds

All four thresholds come from one test. A recurrence $e_{t+1} = ae_t - be_{t-1}$ with real a, b has the companion matrix $\left(\begin{smallmatrix} a & -b \\ 1 & 0 \end{smallmatrix}\right)$, whose eigenvalues are the roots of $z^2 - az + b$; its solutions tend to zero from every pair of initial values if and only if both roots lie in the open unit disk, which by the Schur–Cohn (Jury) test happens if and only if $|b| < 1$ and $|a| < 1 + b$ (Lemma A.14).

Proposition 8.6 (Stability thresholds). *Let $\mathcal{L}(\theta) = \frac{1}{2}(\theta - \theta^\star)^\top H(\theta - \theta^\star)$ with H symmetric positive semidefinite and $S = \lambda_{\max}(H) > 0$, and let $\eta > 0$ and $0 \le \beta < 1$. Each iteration below converges to a minimizer of $\mathcal{L}$ from every initialization (θ_0, and θ_{-1} or m_{-1}) if and only if the stated inequality holds.*

(i) Gradient descent, $\theta_{t+1} = \theta_t - \eta\nabla\mathcal{L}(\theta_t)$: $\eta S < 2$.

(ii) Heavy ball, $\theta_{t+1} = \theta_t - \eta\nabla\mathcal{L}(\theta_t) + \beta(\theta_t - \theta_{t-1})$: $\eta S < 2 + 2\beta$.

(iii) Moving-average momentum, $m_t = \beta m_{t-1} + (1-\beta)\nabla\mathcal{L}(\theta_t)$, $\theta_{t+1} = \theta_t - \eta m_t$: $\eta S < (2 + 2\beta)/(1 - \beta)$.

(iv) Adam with a frozen preconditioner: m_t as in (iii) with $\beta = \beta_1$ and $\theta_{t+1} = \theta_t - \eta P^{-1} m_t$ with a fixed symmetric $P \succ 0$: $\eta S_P < (2 + 2\beta_1)/(1 - \beta_1)$, where $S_P = \lambda_{\max}(P^{-1}H)$.

For $\beta_1 = 0.9$ the threshold in (iv) is $\eta S_P < 38$.

Proof. The plan is to split each iteration into scalar recursions, one for each eigenvector of H, and to apply the test to each. Let $Hq_i = \lambda_i q_i$ with orthonormal q_i and $e_{t,i} = \langle q_i, \theta_t - \theta^\star \rangle$. Since $\nabla\mathcal{L}(\theta) = H(\theta - \theta^\star)$, each iteration acts on each $e_{t,i}$ separately, and it converges to a minimizer exactly when every $e_{t,i}$ converges, the minimizers being the points $\theta^\star + v$ with $Hv = 0$.

(i) Here $e_{t+1,i} = (1 - \eta\lambda_i)e_{t,i}$. The component is constant when $\lambda_i = 0$, and for $\lambda_i > 0$ it converges from every start (necessarily to 0) if and only if $|1 - \eta\lambda_i| < 1$, that is, $\eta\lambda_i < 2$.

(ii) Here $e_{t+1,i} = (1 + \beta - \eta\lambda_i)e_{t,i} - \beta e_{t-1,i}$. If $\lambda_i = 0$, then the roots are 1 and β, and $e_{t,i} = c_1 + c_2\beta^t$ converges. If $\lambda_i > 0$, then a limit e would satisfy $\eta\lambda_i e = 0$, so convergence from every start means convergence to zero, which by the test holds if and only if $|\beta| < 1$, true by assumption, and $|1 + \beta - \eta\lambda_i| < 1 + \beta$, that is, $0 < \eta\lambda_i < 2 + 2\beta$.

(iii) For $t \ge 1$, $\theta_{t+1} - \theta_t = -\eta\beta m_{t-1} - \eta(1-\beta)\nabla\mathcal{L}(\theta_t) = \beta(\theta_t - \theta_{t-1}) - \eta(1-\beta)\nabla\mathcal{L}(\theta_t)$: the heavy ball with step size $\eta(1-\beta)$, started from θ_0 and $\theta_1 = \theta_0 - \eta\beta m_{-1} - \eta(1-\beta)\nabla\mathcal{L}(\theta_0)$. If $\beta > 0$, then every θ_1 arises from some m_{-1}, so (ii) applies with $\eta(1-\beta)$ in place of η; for $\beta = 0$ the iteration is (i).

(iv) With $\varphi_t = P^{1/2}(\theta_t - \theta^\star)$ and $\mu_t = P^{-1/2} m_t$ the updates become $\mu_t = \beta_1\mu_{t-1} + (1-\beta_1)H_P\varphi_t$ and $\varphi_{t+1} = \varphi_t - \eta\mu_t$, with $H_P = P^{-1/2}HP^{-1/2}$: iteration (iii) on the quadratic with Hessian H_P, which is positive semidefinite and similar to $P^{-1}H = P^{-1/2}H_P P^{1/2}$, so $\lambda_{\max}(H_P) = S_P$; and $\varphi \mapsto \theta^\star + P^{-1/2}\varphi$ maps minimizers to minimizers. Finally $(2 + 1.8)/0.1 = 38$. ∎

The moving-average form (iii), which Adam uses, is the heavy ball with

its step scaled by $1-\beta$, so at a fixed η it tolerates $(1+\beta)/(1-\beta)$ times the curvature of gradient descent, 19 times for $\beta = 0.9$. Adam's bias corrections tend to one and its preconditioner moves, so (iv) concerns a snapshot, and Observation 8.2 says that the snapshot's threshold is where Adam equilibrates. A threshold says when the quadratic model is unstable, not what happens next. Nesterov's threshold is Exercise 8.1. One more fact about snapshots: the preconditioner alone can raise S_P only at a speed that β_2 fixes.

Proposition 8.7 (How fast the preconditioner can sharpen). *Let $v_{t+1} = \beta_2 v_t + (1-\beta_2) g_{t+1}^2$ coordinatewise, with $0 < \beta_2 < 1$ and $v_t > 0$, let $P_t = \operatorname{diag}(\sqrt{v_t}) + \epsilon I$ with $\epsilon \geq 0$, and let H be symmetric with $\lambda_{\max}(P_t^{-1}H) > 0$. Then $P_{t+1} \geq \sqrt{\beta_2}\, P_t$ and $\lambda_{\max}(P_{t+1}^{-1}H) \leq \beta_2^{-1/2}\lambda_{\max}(P_t^{-1}H)$.*

Proof. Coordinatewise $v_{t+1} \geq \beta_2 v_t$ and $\epsilon \geq \sqrt{\beta_2}\,\epsilon$, so $\sqrt{v_{t+1}} + \epsilon \geq \sqrt{\beta_2}(\sqrt{v_t} + \epsilon)$. For $P > 0$, $\lambda_{\max}(P^{-1}H)$ is the largest value of $u^\top Hu / u^\top Pu$ over $u \neq 0$; it is positive exactly when $u^\top Hu > 0$ for some u, for every P. So $\lambda_{\max}(P_{t+1}^{-1}H)$ is attained at a u with $u^\top Hu > 0$, where the quotient for P_{t+1} is at most $\beta_2^{-1/2}$ times the quotient for P_t, which is at most $\lambda_{\max}(P_t^{-1}H)$. ∎

At a fixed Hessian, then, $\log S_P$ grows by at most $\frac{1}{2}\log(1/\beta_2) \approx \frac{1}{2}(1-\beta_2)$ per step: a factor e takes at least about 2000 steps for $\beta_2 = 0.999$. The bound is attained when the new squared gradients are negligible next to v_t. With $\hat{v}_t = v_t/(1-\beta_2^{t+1})$ in place of v_t the factor becomes $\beta_2^{-1/2}\big((1-\beta_2^{t+2})/(1-\beta_2^{t+1})\big)^{1/2}$, appreciably larger only for $t \lesssim 1/(1-\beta_2)$; the factor $1-\beta_1^{t+1}$ of Observation 8.2 grows with t and only slows the climb.

8.4 The theory: self-stabilization

8.4.1 Where the cubic model comes from

The cubic model is what a general loss looks like near the threshold, to leading order; the derivation is due to Damian et al. (2023).

Heuristic derivation of the cubic model. Let $\mathcal{L}$ be three times continuously differentiable, let $S(\bar{\theta}) = 2/\eta$ be a simple eigenvalue with unit eigenvector u at a reference point $\bar{\theta}$, and let $\langle \nabla\mathcal{L}(\bar{\theta}), u\rangle = 0$. Since $dS = u^\top (dH)u$ for a simple eigenvalue, $\nabla S = \nabla^3\mathcal{L}[u,u,\cdot\,]$ at $\bar{\theta}$. Write $\theta = \bar{\theta} + xu + \delta$ with $\delta \perp u$, put $y = \langle \nabla S, \delta\rangle$, and assume $\langle \nabla S, u\rangle = 0$. Expand $\nabla\mathcal{L}(\theta)$ to second order around $\bar{\theta}$. Its component along u is $\frac{2}{\eta}x + xy$ after dropping $\frac{1}{2}x^2\nabla^3\mathcal{L}[u,u,u]$ and $\frac{1}{2}\nabla^3\mathcal{L}[u,\delta,\delta]$; its inner product with ∇S is $-\alpha + \frac{1}{2}\gamma x^2$, with $\alpha = -\langle \nabla\mathcal{L}, \nabla S\rangle$ and $\gamma = \|\nabla S\|_2^2$ at $\bar{\theta}$, after dropping $\langle \nabla S, \nabla^2\mathcal{L}\,\delta\rangle$ and the other terms that involve δ. So one step maps (x,y) to about $\big(-(1+\eta y)x,\ y + \eta(\alpha - \frac{1}{2}\gamma x^2)\big)$, which is (8.4) for $(\sqrt{\gamma}\,x, y)$. ◇

Some dropped terms describe the drift of the reference point, its eigenvector and the rates, which Damian, Nichani and Lee follow along a

constrained trajectory and Section 8.4.6 averages over; others are higher-order terms along the oscillation, which decide what happens over long times (Section 8.6).

8.4.2 Scales, and why two steps

The map (8.4) flips the sign of x at every step, so it is not close to the identity, and no differential equation describes x_t. The squared amplitude does not flip, $x_{t+1}^2 = (1+\eta y_t)^2 x_t^2$; equivalently, the two-step map $x_t \mapsto (1+\eta y_{t+1})(1+\eta y_t)x_t$ has no sign flip. A continuous limit describes the envelope of the oscillation. In the variables $A = \log x^2$ and y the map reads

$$A_{t+1} = A_t + 2\log(1+\eta y_t), \qquad y_{t+1} = y_t + \eta\Big(\alpha - \tfrac{1}{2}e^{A_t}\Big). \tag{8.5}$$

If α, y and $e^A = x^2$ are of order one, then each step changes A and y by $O(\eta)$, and the natural clock is $\tau = \eta t$, the time of gradient flow; $\eta S - 2$ is then of order η, and a cycle lasts a time of order $1/\sqrt{\alpha}$ in τ, or $1/(\eta\sqrt{\alpha})$ steps. The model has one dimensionless parameter: with $a = A - \log(2\alpha)$, $Y = y/\sqrt{\alpha}$ and the clock $\sqrt{\alpha}\,\tau$, the map becomes $a \mapsto a + 2\log(1+\varepsilon Y)$, $Y \mapsto Y + \varepsilon(1-e^a)$, with $\varepsilon = \eta\sqrt{\alpha}$. Its square $\varepsilon^2 = \eta^2\alpha$ is the rise of ηS per step due to sharpening, and the continuous limit $\varepsilon \to 0$ is sharpening that is slow on the scale of a step, the regime of Observation 8.1.

8.4.3 The theorem

The limit of (8.5) is

$$\frac{dA}{d\tau} = 2y, \qquad \frac{dy}{d\tau} = \alpha - \tfrac{1}{2}e^A. \tag{8.6}$$

Let $\phi(A) = \frac{1}{2}e^A - \alpha A$ and $\mathcal{H}(A,y) = y^2 + \phi(A)$. The function ϕ is strictly convex and tends to $+\infty$ as $A \to \pm\infty$, with its minimum $\mathcal{H}_{\min} = \alpha(1-\log 2\alpha)$ at $A^* = \log(2\alpha)$. So $\mathcal{H}$ has the unique minimizer $(A^*, 0)$, which corresponds to the orbit $x = \pm\sqrt{2\alpha}$, $y = 0$ of period two of gradient descent. For $h > \mathcal{H}_{\min}$ let $A_- < A^* < A_+$ be the two solutions of $\phi(A) = h$.

Theorem 8.8 (Self-stabilization in the continuous limit). *Let $\alpha > 0$, let $(A(\tau), y(\tau))$ solve (8.6) from a point other than $(A^*, 0)$, and let $h = \mathcal{H}(A(0), y(0))$.*

(i) The energy $\mathcal{H}$ is constant along the solution.

(ii) The solution is periodic, with period $T(h) = \int_{A_-}^{A_+} dA/\sqrt{h - \phi(A)} < \infty$.

(iii) Over each period $\langle y\rangle = 0$ and $\langle e^A\rangle = 2\alpha$: the curvature along the oscillation averages exactly $2/\eta$, and x^2 averages exactly 2α.

(iv) Gradient descent (8.4) from (x_0, y_0) with $\log x_0^2 = A(0)$ and $y_0 = y(0)$ follows the solution: for every $\tau_{\max} > 0$ there are C and $\eta_0 > 0$, depending on α, $A(0)$, $y(0)$ and $\tau_{\max}$ but not on η, such that for

$0 < \eta \le \eta_0$ *and every integer* $0 \le t \le \tau_{\max}/\eta$,

$$|\log x_t^2 - A(\eta t)| + |y_t - y(\eta t)| \le C\eta.$$

To leading order in ε, an oscillation at the edge is a closed cycle: the curvature rises above $2/\eta$, the oscillation grows and pushes it below, the oscillation dies, and the cycle repeats with the same energy. Part (iii) is exact: the curvature averages exactly the threshold, and the oscillation carries exactly the energy that cancels progressive sharpening, $\frac{1}{2}\gamma\langle x^2\rangle = \alpha$ before the coupling is absorbed.

Proof of (i)–(iii). The plan is to show that $\mathcal{H}$ is conserved, that its level sets are closed curves without equilibria, and that the averages of derivatives over a period vanish. (i) Along a solution, $\frac{d}{d\tau}\mathcal{H} = 2y(\alpha - \frac{1}{2}e^A) + (\frac{1}{2}e^A - \alpha)\,2y = 0$.

(ii) Since $\mathcal{H} \ge y^2 + \mathcal{H}_{\min}$ and ϕ tends to infinity at both ends, the sublevel sets of $\mathcal{H}$ are compact; by (i) the solution stays in one and exists for all τ. The level set $\{\mathcal{H} = h\}$ consists of the arcs $y = \pm\sqrt{h - \phi(A)}$, $A_- \le A \le A_+$, joined at $(A_\pm, 0)$, and the vector field vanishes only at $(A^*, 0)$, which is not on it. On the upper arc $\dot{A} = 2\sqrt{h - \phi(A)} > 0$, so A increases and reaches A_+ after the time $\int dA/(2\sqrt{h - \phi(A)})$, which is finite: ϕ is convex, so it lies below its chord from $(A^*, \mathcal{H}_{\min})$ to (A_+, h), and for $A^* \le A < A_+$

$$h - \phi(A) \ge (h - \mathcal{H}_{\min})\,\frac{A_+ - A}{A_+ - A^*}, \quad \text{so} \quad \int_{A^*}^{A_+} \frac{dA}{\sqrt{h - \phi(A)}} \le \frac{2(A_+ - A^*)}{\sqrt{h - \mathcal{H}_{\min}}},$$

and the same bound holds on $[A_-, A^*]$. At $(A_+, 0)$, $\dot{y} = -\phi'(A_+) < 0$, so the solution passes to the lower arc, along which A decreases to A_-, and at $(A_-, 0)$, $\dot{y} = -\phi'(A_-) > 0$ returns it to the upper arc. After the time $T(h)$ it is back at its starting point, and by uniqueness of solutions it is periodic.

(iii) Over a period A and y return to their values, so integrating (8.6) gives $0 = 2\int y\,d\tau$ and $0 = \int(\alpha - \frac{1}{2}e^A)\,d\tau$. The curvature along the oscillation is $2/\eta + y$, and $e^A = x^2$. ∎

This is the argument that gives the averages of the predator–prey cycles of Lotka and Volterra.[5] Small cycles have the period of the linearization, $T(h) \to 2\pi/\sqrt{2\alpha}$ as $h \downarrow \mathcal{H}_{\min}$ (Exercise 8.2), about $2\pi/(\eta\sqrt{2\alpha})$ steps; larger cycles are slower, because they spend a long time with an oscillation so small that nothing happens but the climb of the curvature. Part (iv) is an error estimate of the kind used for the Euler method, whose step differs from (8.5) by $O(\eta^2)$. Its constant grows exponentially with $\tau_{\max}$, so it covers a fixed number of cycles, not the times τ of order $1/\eta$ of the drift in Proposition 8.11.

[5] The analogy is partial: there both populations are positive and enter through logarithms, while here y changes sign, and (8.6) is the motion of a particle in the potential well ϕ, since $d^2A/d\tau^2 = -2\phi'(A)$.

Lemma 8.9 (The continuous limit). *Let* $z(\tau) = (A(\tau), y(\tau))$ *solve* (8.6) *on* $[0, \tau_{\max}]$, *let* K *be the set of points within* ℓ_1*-distance* 1 *of its orbit, and let* $Y = \max_K |y|$, $E = \max_K e^A$, $L = \max\{4, E/2\}$ *and* $C_1 = \alpha + \frac{1}{2}E + 2Y^2 + \frac{1}{2}EY$. *Let* $z_t = (A_t, y_t)$ *follow* (8.5) *from* $z_0 = z(0)$. *If* $\eta \le 1/(2Y)$ *and* $C_1\eta(e^{L\tau_{\max}} - 1) \le L$, *then for every integer* $0 \le t \le \tau_{\max}/\eta$, $z_t \in K$, $x_t \ne 0$, *and* $\|z_t - z(\eta t)\|_1 \le (C_1/L)(e^{L\eta t} - 1)\eta$.

PROOF. The plan is to bound the error of one step along the exact solution, to show that the map F of (8.5) is $(1+L\eta)$-Lipschitz on K, where $|\eta y| \le \frac{1}{2}$, and to add up the errors with the discrete Grönwall inequality (Theorem A.17).

One step. Along the solution $|y| \le Y$ and $e^A \le E$, so $|\dot y| \le \alpha + \frac{1}{2}E$ and $|\dot A| \le 2Y$. Fix $\tau \le \tau_{\max} - \eta$. The first component of $z(\tau+\eta) - F(z(\tau))$ is $\int_0^\eta 2(y(\tau+r) - y(\tau))\,dr + 2(\eta y - \log(1+\eta y))$ with $y = y(\tau)$; the integral is at most $(\alpha + \frac{1}{2}E)\eta^2$ in absolute value, and the second term lies in $[0, 2\eta^2 Y^2]$ because $0 \le u - \log(1+u) \le u^2$ for $u \ge -\frac{1}{2}$ (the function $u^2 - u + \log(1+u)$ vanishes at 0 and its derivative $u(1+2u)/(1+u)$ has the sign of u). The second component is $-\frac{1}{2}\int_0^\eta (e^{A(\tau+r)} - e^{A(\tau)})\,dr$, at most $\frac{1}{2}EY\eta^2$ because $|e^{A(\tau+r)} - e^{A(\tau)}| \le 2EYr$. So $\|z(\tau+\eta) - F(z(\tau))\|_1 \le C_1\eta^2$.

Lipschitz bound. On a segment in K, $\partial_y\, 2\log(1+\eta y) = 2\eta/(1+\eta y) \le 4\eta$ and $|\partial_A \frac{1}{2}\eta e^A| \le \frac{1}{2}\eta E$, so $\|F(z) - F(z')\|_1 \le (1+L\eta)\|z - z'\|_1$.

Adding up. Let $r_t = \|z_t - z(\eta t)\|_1$, and suppose $r_s \le 1$ for $s \le t$. Then z_t lies in the ℓ_1-ball of radius 1 around $z(\eta t)$, a convex subset of K, and $r_{t+1} \le \|F(z_t) - F(z(\eta t))\|_1 + \|F(z(\eta t)) - z(\eta t + \eta)\|_1 \le (1+L\eta)r_t + C_1\eta^2$. With $r_0 = 0$, induction gives

$$r_{t+1} \le C_1\eta^2 \sum_{s=0}^{t} (1+L\eta)^s \le \frac{C_1\eta}{L}\left(e^{L\eta(t+1)} - 1\right) \le 1$$

as long as $\eta(t+1) \le \tau_{\max}$, by the assumption on η. Finally $z_t \in K$ gives $1 + \eta y_t \ge \frac{1}{2}$, so no step sends x to zero. ∎

PROOF OF THEOREM 8.8(IV). Apply Lemma 8.9 with $C = (C_1/L)(e^{L\tau_{\max}} - 1)$ and $\eta_0 = \min\{1/(2Y), 1/C\}$; by (8.5), $A_t = \log x_t^2$. ∎

8.4.4 What the discrete map does

For the map itself, three identities hold exactly, at every step size. Write $\mathcal{H}_t = \mathcal{H}(A_t, y_t)$.

Lemma 8.10 (Exact identities). *Let (x_t, y_t) follow (8.4) from $x_0 \ne 0$, suppose that $1 + \eta y_t > 0$ for $0 \le t < T$, and write $A_t = \log x_t^2$ and $f_t = \alpha - \frac{1}{2}x_t^2$. Then*

(i) $\frac{1}{T}\sum_{t<T} x_t^2 = 2\alpha - 2(y_T - y_0)/(\eta T)$;

(ii) $\frac{1}{T}\sum_{t<T} \log(1+\eta y_t) = (A_T - A_0)/(2T)$, *and hence* $\frac{1}{T}\sum_{t<T} y_t \ge \left(e^{(A_T - A_0)/(2T)} - 1\right)/\eta$;

(iii) for $t < T$, with equality if and only if $x_t^2 = 2\alpha$ and $y_t = 0$,

$$\mathcal{H}_{t+1} - \mathcal{H}_t = \eta^2 f_t^2 + \tfrac{1}{2}\eta^2 x_t^2 y_t^2 + 2\alpha\left(\eta y_t - \log(1+\eta y_t)\right) \ge 0. \quad (8.7)$$

PROOF. The plan is to sum the two updates for (i) and (ii), and to expand one step of $\mathcal{H}$ for (iii). By (8.4), $x_{t+1}^2 = (1+\eta y_t)^2 x_t^2 > 0$ for $t < T$, so (8.5) holds. Summing the y-update gives $y_T - y_0 = \eta\sum_{t<T}(\alpha - \frac{1}{2}x_t^2)$, which is (i). Summing the A-update gives the identity in (ii), and the concavity of the logarithm, $\log\left(\frac{1}{T}\sum_t (1+\eta y_t)\right) \ge \frac{1}{T}\sum_t \log(1+\eta y_t)$, gives the inequality.

For (iii), drop the index t and write $u = \eta y$, so that $e^{A_{t+1}} = (1+u)^2 e^A$ and $A_{t+1} - A = 2\log(1+u)$. Then

$$\mathcal{H}_{t+1} - \mathcal{H}_t = 2\eta y f + \eta^2 f^2 + \tfrac{1}{2}e^A(2u + u^2) - 2\alpha\log(1+u),$$

and the terms of first order combine, $2\eta y f + e^A u = 2u(\alpha - \frac{1}{2}e^A) + e^A u = 2\alpha u$. What remains is (8.7). Each of its terms is nonnegative, the last because $u - \log(1+u) \geq 0$ for $u > -1$ with equality only at $u = 0$, and the sum vanishes if and only if $f = 0$ and $u = 0$. ∎

Identity (i) is the discrete $\langle x^2 \rangle = 2\alpha$, valid along every orbit, up to a boundary term that is small whenever y stays bounded. Identity (ii) says that the geometric mean of $|x_{t+1}/x_t| = \eta S_t - 1$, with $S_t = 2/\eta + y_t$ the curvature along the oscillation, is $|x_T/x_0|^{1/T}$; if the oscillation neither grows nor decays, then the arithmetic mean of ηS_t is at least 2, the "at or just above $2/\eta$" of Observation 8.1. Identity (iii) has no counterpart in the flow: the energy that the flow conserves increases at every step of the map.

Proposition 8.11 (Gradient descent spirals outward). *Let $\eta > 0$ and $\alpha > 0$, and let F be the map (8.5) on the half-plane $\{1 + \eta y > 0\}$.*

(i) The point $p = (A^*, 0)$, which is the orbit $x = \pm\sqrt{2\alpha}$, $y = 0$ of period two of (8.4), is a fixed point of F, and the Jacobian of F at p has the eigenvalues $1 \pm i\eta\sqrt{2\alpha}$, of modulus $\sqrt{1 + 2\alpha\eta^2} > 1$.

(ii) As long as an orbit of F stays in the half-plane, $\mathcal{H}$ is nondecreasing along it, and it increases strictly at every step taken from a point other than p.

(iii) For every compact set K in the half-plane, every orbit that starts in K at a point other than p leaves K.

Proof. (i) At p, $1 + \eta y = 1$ and $e^A = 2\alpha$, so $F(p) = p$, and the Jacobian $\left(\begin{smallmatrix} 1 & 2\eta/(1+\eta y) \\ -\eta e^A/2 & 1 \end{smallmatrix}\right)$ is $\left(\begin{smallmatrix} 1 & 2\eta \\ -\eta\alpha & 1 \end{smallmatrix}\right)$, with characteristic polynomial $(1-z)^2 + 2\alpha\eta^2$. (ii) is Lemma 8.10(iii). (iii) If an orbit started at $z_0 \in K \setminus \{p\}$ stayed in K, then it would stay in the compact set $K' = K \cap \{\mathcal{H} \geq \mathcal{H}(z_0)\}$, by (ii), which does not contain p because $\mathcal{H}(p) = \mathcal{H}_{\min} < \mathcal{H}(z_0)$. The increment $\mathcal{H}(F(z)) - \mathcal{H}(z)$ is continuous and positive on K', hence at least some $c > 0$ there, and $\mathcal{H}(z_t) \geq \mathcal{H}(z_0) + ct$ would be unbounded on K. ∎

So the period-two orbit that exactly balances progressive sharpening is unstable for every step size, and every other orbit drifts to larger cycles.[6] The drift is slow, and its rate can be computed.

Heuristic derivation of the area law. Let $\mathcal{J}(h)$ be the area enclosed by the level curve $\{\mathcal{H} = h\}$. Over one cycle, gradient descent stays within $O(\eta)$ of the flow (Lemma 8.9) for about $T(h)/\eta$ steps, each raising $\mathcal{H}$ by $\eta^2 Q + O(\eta^3)$ with $Q = f^2 + y^2(\alpha + \frac{1}{2}e^A)$, by (8.7); the level rises by about $\eta \oint Q\, d\tau$. On the cycle $dy = f\, d\tau$ and $dA = 2y\, d\tau$, so $\oint Q\, d\tau = \oint f\, dy + \frac{1}{2}\oint (\alpha + \frac{1}{2}e^A)\, y\, dA$; the cycle runs clockwise, and Green's theorem turns these into the integrals of $\frac{1}{2}e^A$ and of $\frac{1}{2}(\alpha + \frac{1}{2}e^A)$ over the region $\{\mathcal{H} \leq h\}$. On every horizontal

[6] The explicit Euler method does the same to the harmonic oscillator: for $\ddot{q} = -\omega^2 q$ written as a first-order system, one Euler step multiplies the energy by exactly $1 + \eta^2\omega^2$. In (8.5) both variables are updated from the same old values, as in the Euler method.

chord of that region $\mathcal{H} = h$ at both ends, and $\alpha - \frac{1}{2}e^A = -\partial_A \mathcal{H}$, so $\iint (\alpha - \frac{1}{2}e^A)\, dA\, dy = 0$: the average of e^A over the region is 2α, as over the cycle. Hence $\oint Q\, d\tau = 2\alpha \mathcal{J}(h)$. Finally $\mathcal{J}(h) = \int_{A_-}^{A_+} 2\sqrt{h - \phi(A)}\, dA$ has the derivative $T(h)$, so the area grows by $T(h) \cdot 2\alpha\eta \mathcal{J}(h)$ per cycle, that is, by $2\alpha\eta^2 \mathcal{J}$ per step:

$$\mathcal{J}(\mathcal{H}_t) \approx \mathcal{J}(\mathcal{H}_0)\, e^{2\alpha\eta^2 t}. \quad \diamond \tag{8.8}$$

Near the period-two orbit the area law is exact to first order, since the linearization of Proposition 8.11(i) has the determinant $1 + 2\alpha\eta^2$. A small cycle lasts $2\pi/\sqrt{2\alpha}$ in τ and the area grows by the factor e in the time $1/(2\alpha\eta)$, so the drift needs about $1/(2\sqrt{2}\,\pi\varepsilon)$ cycles, some 11 at $\varepsilon = 0.01$: invisible over a few cycles, dominant over many.[7]

[7] The area $\mathcal{J}$ is 2π times the action of the cycle in the sense of classical mechanics. Actions are conserved under slow changes of the parameters (adiabatic invariance), so a slow drift of α along a training run does not by itself change the area of the cycles; the discretization does.

8.4.5 What Damian, Nichani and Lee prove

Their theorem concerns gradient descent on a general loss, three times differentiable. Its hypotheses include progressive sharpening, $\alpha > 0$ along the constrained trajectory (their Assumption 1), an eigengap, the second Hessian eigenvalue below c/η for an absolute constant $c < 2$ (their Assumption 2), and regularity conditions. The constrained trajectory is gradient descent projected onto the stable set $\{S(\theta) \le 2/\eta,\ \langle \nabla \mathcal{L}(\theta), u(\theta) \rangle = 0\}$, with $u(\theta)$ the top eigenvector. Under these conditions they couple gradient descent with it for $O(\varepsilon^{-1})$ steps, with ε a small parameter of their analysis (not ours), and the deviation follows a two-dimensional system with time-dependent coefficients, the counterpart of (8.4).[8] The horizon is finite, as in Theorem 8.8(iv); Proposition 8.11 and (8.8) concern longer times, of order $1/(\alpha\eta^2)$ steps, over which the reduced dynamics itself is unstable; whether real dynamics drifts outward or settles onto a stable period-two orbit is decided by the terms that the reduction drops (Section 8.6).

[8] They note that their conditions are not strictly necessary, and that a local coupling of this kind does not suffice to prove global convergence.

8.4.6 Central flows

Now average over the cycles. In the family (8.1), once $s(w_t)$ has reached $2/\eta$, and as long as $\mathcal{L}_0$ and s change little over a cycle, the pair $(\sqrt{\gamma(w)}\, x,\ s(w) - 2/\eta)$ approximately follows the cubic model with the local rate $\alpha(w)$ over the cycle.

Heuristic derivation of the central flow. If $\alpha(w) > 0$, then $\langle x^2 \rangle = 2\alpha(w)/\gamma(w)$ over a cycle, by Theorem 8.8(iii). Averaging the w-update of (8.2) over the cycle, with $\tau = \eta t$,

$$\frac{dw}{d\tau} = -\nabla \mathcal{L}_0(w) - \tfrac{1}{2}\sigma^2(w)\, \nabla s(w), \qquad \sigma^2(w) = \frac{2 \max\{0, \alpha(w)\}}{\gamma(w)}, \tag{8.9}$$

as long as $s(w) = 2/\eta$, with α and γ as in (8.3). When $\alpha(w) \le 0$, the curvature falls below $2/\eta$, the oscillation decays, and w follows gradient flow. With $\sigma^2 = 2\alpha/\gamma$ the velocity in (8.9) is $-\nabla \mathcal{L}_0$ minus its component along ∇s, tangent to $\{s = 2/\eta\}$: (8.9) is gradient flow projected onto the region $\{s \le 2/\eta\}$. The loss averaged over a cycle is $\mathcal{L}_0(w) + \frac{1}{2}s\langle x^2 \rangle = \mathcal{L}_0(w) + \sigma^2/\eta$. $\diamond$

Cohen et al. (2025) derived this equation for a general loss, with $\mathcal{L}$ and S in place of $\mathcal{L}_0$ and s, and called it a *central flow*, a differential equation for the time-averaged trajectory at the edge. Here σ^2 is the value at which the sharpness reduction caused by the oscillation cancels progressive sharpening, and the flow projects ∇S out of the gradient; they note that this re-derives a main result of Damian et al. (2023) in a simpler, non-rigorous way, extend it to several eigenvalues at the edge, and report that central flows predict long-term trajectories of neural networks with high numerical accuracy. Here and there the status is the same: derived, not proved. In the cubic model the central flow is the point $y = 0$ with $\sigma^2 = 2\alpha$. For RMSProp and Adam the central flow also moves the preconditioner, and Cohen and coauthors find what they call acceleration via regularization: the adaptive optimizers implicitly move toward regions of low curvature, in which they can take larger steps.

8.5 Reading the theory

THRESHOLDS IN NUMBERS. For moving-average momentum, and for Adam with a frozen preconditioner, the largest stable ηS (or ηS_P) is $(2+2\beta)/(1-\beta)$: 2, 6, 38, 78 and 398 for β (or β_1) equal to 0, 0.5, 0.9, 0.95 and 0.99. In the left panel of Figure 8.1, the sharpness of full-batch gradient descent on a small network rises, overshoots, and holds at $2/\eta$, within 1.4% on average over the last 1500 steps for $\eta = 0.4$ and within 0.1% for $\eta = 0.25$. In the right panel, Adam's preconditioned sharpness climbs to $38/\eta$ and stays near it, slightly below: over the last 1500 steps it averages 0.98 and 0.95 of the threshold, with excursions to 1.03 times it. The climb runs at about the limit of Proposition 8.7, because the gradients are already small: for $\eta = 0.003$, S_P grows by the factor $e^{1.80}$ between steps 200 and 1975, where the bound at a fixed Hessian, with the bias correction, allows $e^{1.67}$; the excess comes from the change of the Hessian. Meanwhile the sharpness of Adam's iterates stays between 7.4 and 8.4 from step 300 on, far from a threshold of its own (Observation 8.2).

THE CUBIC MODEL, CYCLE BY CYCLE. On the left of Figure 8.2, gradient descent with $\alpha = 1$ and $\eta = 0.002$ follows the closed level curves of $\mathcal{H}$ for two cycles each: a medium cycle with $\mathcal{H} = 2$, and a large one started below the edge ($y_0 = -0.5$) with a small oscillation $x_0 = 0.01$, as at the onset of the edge of stability. Each large cycle is a catapult: the curvature climbs to $2/\eta + 3.0$, the oscillation grows from 0.01 to about 5, the curvature is pushed down to $2/\eta - 3.0$, and the oscillation dies back. The overshoot equals the undershoot because $\mathcal{H}$ is even in y; with $\alpha = 0$ the same law turns one excursion to $2/\eta + y_0$ into a landing at $2/\eta - y_0$, the reflection law that volume 2 of this series derives for the catapult. A small oscillation at onset means a large cycle, since $\mathcal{H} \approx y_0^2 - \alpha \log x_0^2$. The cycles last $T = 5.08$ and 7.56 in τ (against $2\pi/\sqrt{2} = 4.44$ for the smallest), the large one 3780 steps, and its second loop lies visibly outside the first: the drift. In the middle, the running averages of x^2 and y converge to $2\alpha = 2$ and 0, with oscillations that decay like $1/\tau$; after five periods gradient descent's average of x^2 is 2.04,

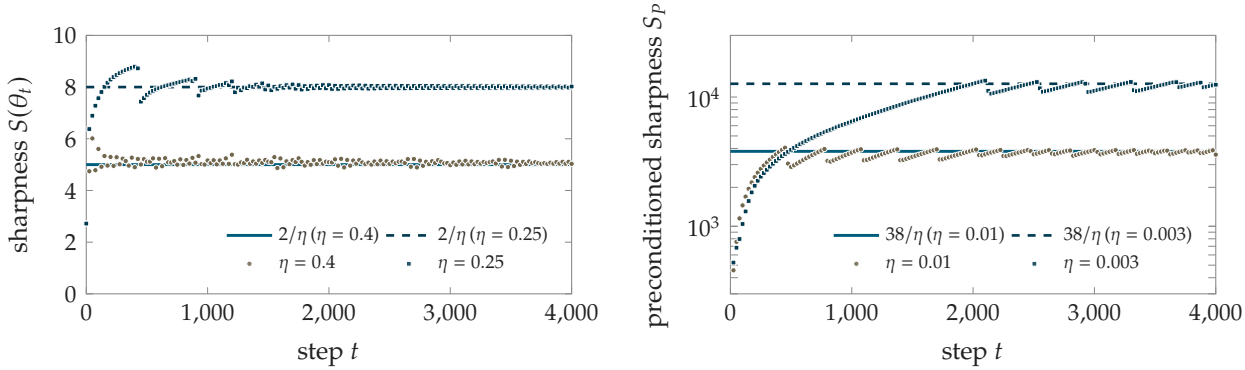


Figure 8.1. The thresholds of Proposition 8.6 (lines) and the measured curvature (points) in a tanh network $8 \to 32 \to 32 \to 1$ (1377 parameters) trained full batch on the square loss, with 256 Gaussian inputs labeled by a random teacher (seeds fixed in `08-eos-network.py`). Left: gradient descent; the sharpness, every 25 steps, by block power iteration with Hessian–vector products. Right: Adam with $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$; the preconditioned sharpness $\lambda_{\max}(P_t^{-1}H_t)$, with P_t as in Observation 8.2, from step 25 on; over the last 1500 steps it averages 0.98 and 0.95 of $38/\eta$.

the difference being exactly the boundary term of Lemma 8.10(i), nonzero because the drifting orbit has fallen slightly out of phase with the flow. On the right, three runs with different (α, η) collapse onto the area law (8.8) while the area grows by the factor e^6; beyond that, ηy is no longer small and the expansion fails.

ADAM'S EDGE WITHOUT SHARPENING. The preconditioned sharpness can reach its threshold with a constant Hessian, because the preconditioner shrinks with the gradients.

EXAMPLE 8.12 (Adam on a parabola). Run Adam with $\beta_1 = 0.9$, $\beta_2 = 0.999$ and $\epsilon = 0$ on $\mathcal{L}(x) = \frac{1}{2}hx^2$. As x converges toward zero, the gradients become negligible next to $\sqrt{v_t}$, and $S_P = h/P_t$ rises at the limit of Proposition 8.7: with $h = 1$, $\eta = 0.01$ and $x_0 = 1$ it grows from $\eta S_P = 0.066$ at step 1000 to the threshold 38 at step 13,254, which is the rate $\frac{1}{2}\log(1/\beta_2)$ per step corrected for the bias correction of $\hat{v}_t$. After a transient that overshoots to 57, it settles into cycles of about 480 steps between about 31 and 40, around an average of 35.4; the root mean square of x is about $0.03\,\eta$. Adam with $\epsilon = 0$ is invariant under rescaling the loss, and rescaling x rescales η, so the range and the average of ηS_P do not depend on h or η, as runs confirm.

The example isolates a mechanism that the network mixes with the cubic one: Adam sharpens through its preconditioner alone, at a speed fixed by β_2, and self-stabilizes through v_t, since a growing oscillation raises the squared gradients, v_t grows, and S_P falls. The average sits about 7% below the threshold, unlike the cubic model's $2/\eta$, because the oscillation grows fast above the threshold but decays slowly below it: the two roots of the momentum recursion have the product β_1, so the amplitude decays no

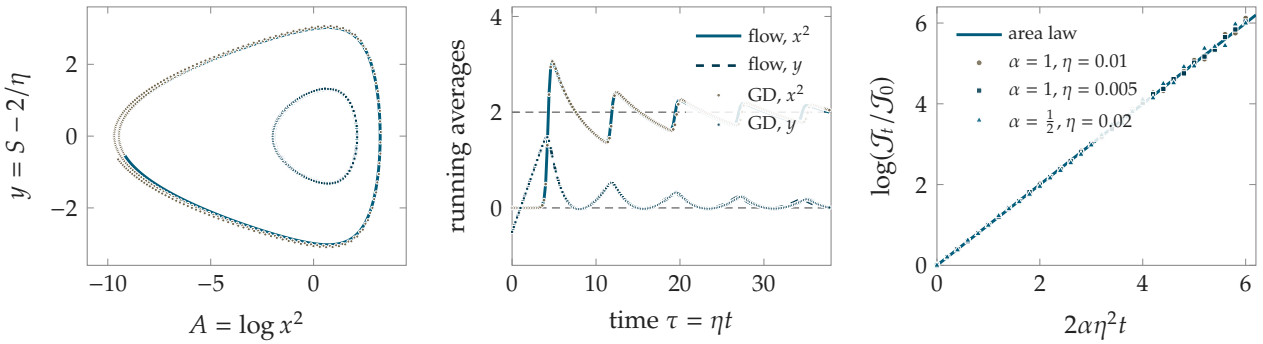


Figure 8.2. The cubic model: theory (curves) against the gradient-descent map (8.4) (points), with no randomness (script `08-eos-cubic.py`). Left: two cycles each of a medium and a large orbit, $\alpha = 1$, $\eta = 0.002$, every 25th iterate, on the level curves of $\mathcal{H}$ through their starting points. Middle: running averages of x^2 and of y along the large orbit over five periods, for the flow (8.6) and for gradient descent. Right: the area enclosed by the level curve through the current iterate, for three runs started on $y = 0$ at $\mathcal{H} = 1$, against the area law (8.8).

faster than $\beta_1^{t/2}$ however far below the threshold ηS_P is, and the cycle spends most of its time below, about 80% in the example (Exercise 8.9).

WHERE THE AGREEMENT MUST FAIL. The theorem concerns $\varepsilon \to 0$ at fixed energy, and large cycles fail first: once ηy is not small the expansion behind (8.8) breaks down. With $\eta = 0.01$, $\alpha = 1$ and $\mathcal{H}_0 = 10$ the drift carries the orbit to cycles in which x falls below 10^{-300}, and in floating point x underflows to zero after about $4 \cdot 10^4$ steps, after which the curvature grows without bound.[9] Networks do not behave this way, and Section 8.6 says why.

[9] At that point $\mathcal{H}$ becomes infinite in floating point, because $\log x^2 = -\infty$; it is this underflow, not an overflow of x, that ends such runs.

8.6 What the theory misses

WHAT WAS IDEALIZED AWAY. The rate of progressive sharpening is an input, constant in the cubic model. One direction is at the edge, with a fixed eigenvector and a gap below it, while in networks the eigenvector turns and several eigenvalues can reach $2/\eta$ together. The loss is quadratic and even along the oscillation, without the terms x^3 and x^4 that decide the long-time behavior. The gradients are exact. The thresholds for momentum and Adam are proved for quadratics with a frozen preconditioner, and self-stabilization for plain gradient descent only.

WHERE IT BREAKS. (1) *Long times.* The cubic model has no stable cycle: its period-two orbit repels at the rate $\alpha\eta^2$ per step. A fourth-order term $\frac{1}{24}\kappa x^4$ changes this at order one in the continuous limit, which becomes $dA/d\tau = 2y + \frac{1}{3}\kappa e^A$, $dy/d\tau = \alpha - \frac{1}{2}e^A$. For $\kappa < 0$, a loss that grows more slowly than quadratically along the oscillation, the cycles spiral in to a stable period-two orbit with $y = -\kappa\alpha/3 > 0$, a curvature at the center of the oscillation slightly above $2/\eta$ (Exercise 8.5). Near that orbit the area changes at the rate $2\alpha(\eta + \kappa/3)$ per unit of τ, so the orbit of the map is

stable for $\kappa < -3\eta$; at $\eta = 0.01$ and $\alpha = 1$ the measured growth rate of the amplitude vanishes at $\kappa = -0.03$ and matches $\alpha(\eta + \kappa/3)$ to within 1% at $\kappa = -0.04$ and -0.06. A cubic term along the oscillation also damps; in one dimension the condition for a stable period-two orbit near the threshold is $3\mathcal{L}'''^2 - \mathcal{L}''\mathcal{L}'''' > 0$ (Exercise 8.5), and Chen and Bruna (2023) analyze such orbits, the flip bifurcation of dynamical systems, through the two-step map. Whether a network's cycles are damped depends on higher derivatives along the top eigenvector, which the cubic model sets to zero; the sustained edge of Observation 8.1 is what damping predicts. (2) *Progressive sharpening.* The model does not say why the sharpness rises, and with the cross-entropy loss it eventually falls (Cohen et al. 2021); for logistic regression on separable data, gradient descent converges at every step size after a phase in which the loss oscillates (Wu et al. 2024, 2023), so the edge is left for good. (3) *Several eigenvalues.* With two directions at the edge, Lemma 8.10(i) becomes a linear system for the two oscillation energies, whose solution can have a negative entry; then one direction cannot stay at the edge (Exercise 8.7), and the eigengap assumption of Damian et al. (2023) fails. (4) *Noise.* With minibatches the full-batch sharpness stays below $2/\eta$ (Observation 8.3); the model has no noise. (5) *Adam's moving preconditioner.* Example 8.12 reaches its edge and stabilizes through v_t alone, 7% below the frozen threshold; the theory covers neither the interaction of this mechanism with the cubic one nor the observation that Adam keeps moving into regions of higher sharpness (Cohen et al. 2022).

Repairs and open problems. Central flows (Cohen et al. 2025) handle several eigenvalues at the edge and adaptive optimizers, at the level of a derivation. Rod flows (Regis and Chewi 2026b) treat two consecutive iterates as the ends of a rod, an extended one-dimensional object with a center and an extent; their authors report that rod flow matches the accuracy of central flows on representative networks and does better on toy examples, and a second paper extends it to heavy ball, Nesterov, RMSProp, Adam and NAdam (Regis and Chewi 2026a); both are unrefereed preprints of 2026. Minimalist models give complete proofs of edge-of-stability behavior in low dimension: a two-parameter example (Zhu et al. 2023), the learning of threshold neurons (Ahn et al. 2023), scalar networks (Kreisler et al. 2023), and a bifurcation analysis (Song and Yun 2023); Arora et al. (2022) prove it near a manifold of minimizers for normalized gradient descent and for gradient descent on $\sqrt{\mathcal{L} - \mathcal{L}^\star}$.[10] Progressive sharpening has partial explanations, through the growth of the output layer (Li et al. 2022) and in second-order regression models (Agarwala et al. 2023). For minibatches the batch sharpness (Andreyev and Beneventano 2024) and the measure of Lee and Jang (2023) replace S. Open: a model in which progressive sharpening is derived and the long phase at the edge is proved; a global-in-time theorem beyond low-dimensional examples; the sign of the damping in networks, which can be measured (Exercise 8.11); and a theory of Adam's edge with the preconditioner moving.

[10] Kalra et al. (2025) analyze the fixed points of a low-dimensional model and its route to chaos.

8.7 Back in the wild

FIELD NOTES	
PRACTICE	Learning rates as large as training tolerates; then the sharpness sits at $2/\eta$ (gradient descent) and the preconditioned sharpness near $38/\eta$ (Adam).
IDEALIZATION	One oscillating direction whose curvature depends on the other parameters; its affine case, the cubic model.
PROVED	Thresholds $2/\eta$, $(2+2\beta)/\eta$, $(2+2\beta)/((1-\beta)\eta)$, $38/\eta$; closed cycles with $\langle y\rangle = 0$, $\langle x^2\rangle = 2\alpha$ in the limit; exact identities and an increasing energy for the map.
PREDICTS	Sharpness averaging $2/\eta$ or slightly more; $\langle x^2\rangle = 2\alpha/\|\nabla S\|^2$; cycles of $2\pi/(\eta\sqrt{2\alpha})$ steps or more; areas growing by e every $1/(2\alpha\eta^2)$ steps unless damped; the central flow (derived).
MISSES	Why the sharpness rises; several eigenvalues; noise; a moving preconditioner; higher-order damping.
TAKEAWAY	The step size sets the curvature (observed; explained in the model). Compare learning rates across β through $(2+2\beta)/(1-\beta)$ (proved for quadratics). Averaging consecutive iterates removes a loss of about σ^2/η (derived in the model; suggested).

The first lesson is a change of viewpoint. The curvature adapts to the learning rate, rising until the step is barely stable and staying there; a sweep that keeps the largest learning rate that does not diverge is choosing the curvature at which training will run, not a point inside a fixed stability region. Training diverges when self-stabilization fails: when the step size grows faster than the curvature can follow, the subject of warmup (Chapter 9), or when one cycle grows too large, one anatomy of a loss spike (Chapter 10). The second lesson is quantitative but modest: at $\beta = 0.9$ moving-average momentum tolerates nineteen times the curvature of gradient descent at the same η, so a learning rate tuned for one β is not right for another (Chapter 5). The oscillation is not a malfunction: it carries exactly the energy that progressive sharpening requires, and its cost in loss, about σ^2/η in the model, is paid by the iterates but not by their average, since $\frac{1}{2}(x_t + x_{t+1}) = -\frac{1}{2}\eta y_t x_t$ (Exercise 8.4). This suggests that averaging iterates (Chapters 5 and 12) helps at the edge; it is a suggestion, not a result about networks.

The theory does not say why the sharpness rises, so it cannot predict when a run reaches the edge. With noise the edge moves (Chapter 11); Adam's edge involves a preconditioner that the frozen threshold ignores; and why Adam beats gradient descent on transformers is a separate question (Chapter 14).

Exercises

8.1 CONSOLIDATION. (Nesterov's threshold.) Nesterov's momentum is $\theta_{t+1} = \vartheta_t - \eta\nabla\mathcal{L}(\vartheta_t)$ with $\vartheta_t = \theta_t + \beta(\theta_t - \theta_{t-1})$. (a) On a quadratic, show that the component along an eigenvector with eigenvalue $\lambda_i > 0$ obeys $e_{t+1} = (1+\beta)(1-\eta\lambda_i)e_t - \beta(1-\eta\lambda_i)e_{t-1}$. (b) Show that it tends to zero from every start if and only if $0 < \eta\lambda_i < (2+2\beta)/(1+2\beta)$, the maximum stable sharpness (times η) that Cohen et al. (2021) give for Nesterov's method. (c) Compare with the heavy ball at $\beta = 0.9$.

8.2 CONSOLIDATION. (Small cycles.) (a) Show that the linearization of (8.6) at $(A^*, 0)$ has the angular frequency $\sqrt{2\alpha}$. (b) Substitute $\phi(A) = \mathcal{H}_{\min} + (h - \mathcal{H}_{\min})\sin^2\vartheta$, with $\vartheta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ of the sign of $A - A^*$, in $T(h)$ and show that $T(h) \to 2\pi/\sqrt{2\alpha}$ as $h \downarrow \mathcal{H}_{\min}$. (c) For $\alpha = 1$, compute $T(2)$ numerically and compare with the value 5.08 of Section 8.5.

8.3 CONSOLIDATION. (The reduction.) In the family (8.1) with $\mathcal{L}_0(w) = \mathcal{L}_0(0) + \langle g, w\rangle$ and $s(w) = 2/\eta + \langle c, w\rangle$, show that $(\|c\|_2 x_t, \langle c, w_t\rangle)$ follows (8.4) with $\alpha = -\langle g, c\rangle$, and that the component of w_t orthogonal to c moves by $-\eta$ times that of g. Deduce that in the continuous limit the central flow (8.9) gives its cycle averages exactly.

8.4 CONSOLIDATION. (Two consecutive iterates.) In the cubic model, show that $\mathcal{L}(\frac{1}{2}(x_t + x_{t+1}), y_t) = \frac{1}{8}\eta^2 y_t^2 x_t^2(2/\eta + y_t) - \alpha y_t$. In the continuous limit, show that over a cycle the loss of the iterates exceeds that of these averages by $2\alpha/\eta + O(1)$ on average.

8.5 VARIATION. (Higher-order terms.) Add $\frac{1}{24}\kappa x^4$ to the cubic model. (a) Show that $x_{t+1} = -(1+\eta y_t)x_t - \frac{1}{6}\eta\kappa x_t^3$, with the same y-update, and derive the limit $dA/d\tau = 2y + \frac{1}{3}\kappa e^A$, $dy/d\tau = \alpha - \frac{1}{2}e^A$. (b) Show that its fixed point $(\log 2\alpha, -\kappa\alpha/3)$ is stable for $\kappa < 0$ and unstable for $\kappa > 0$, a focus when $\kappa^2 < 18/\alpha$, and that $d\mathcal{H}/d\tau = \frac{1}{3}\kappa e^A(\frac{1}{2}e^A - \alpha)$ averages to a negative number over a cycle of the unperturbed flow when $\kappa < 0$ (use $\langle e^A\rangle = 2\alpha$ and Jensen's inequality). (c) Combine with the area law to predict stability of the period-two orbit of the map for $\kappa < -3\eta$; check numerically with $\eta = 0.01$, $\alpha = 1$, $\kappa \in \{-0.02, -0.04\}$. (d) For a one-dimensional loss with $\mathcal{L}'' = 2/\eta$ at a minimum, show that two steps of gradient descent give $x \mapsto x - 2(k_2^2 + k_3)x^3 + O(x^4)$ with $k_2 = -\frac{1}{2}\eta\mathcal{L}'''$ and $k_3 = -\frac{1}{6}\eta\mathcal{L}''''$, so the period-two orbit born at the threshold is stable when $3\mathcal{L}'''^2 - \mathcal{L}''\mathcal{L}'''' > 0$.

8.6 VARIATION. (Momentum does not change the energy.) Run the heavy ball on the cubic model with $2/\eta$ replaced by $(2+2\beta)/\eta$. Show that $\frac{1}{T}\sum_{t<T} x_t^2 = 2\alpha - 2\big((y_T - y_0) - \beta(y_{T-1} - y_{-1})\big)/(\eta T)$, so that $\langle x^2\rangle = 2\alpha$ along every orbit on which y stays bounded, and derive the same for moving-average momentum. What does change with momentum?

8.7 VARIATION. (Two directions at the edge.) Let $\mathcal{L} = \langle g, w\rangle + \frac{1}{2}s_1(w)x_1^2 + \frac{1}{2}s_2(w)x_2^2$ with $s_j(w) = 2/\eta + \langle c_j, w\rangle$, so that $y_j = \langle c_j, w\rangle$ and $\alpha_j = -\langle g, c_j\rangle$. (a) Show that $y_{j,t+1} = y_{j,t} + \eta(\alpha_j - \frac{1}{2}\sum_k G_{jk}x_{k,t}^2)$ with $G_{jk} =$

$\langle c_j, c_k \rangle$, and that if both oscillations and y stay bounded, then $G\langle x^2 \rangle = 2\alpha$. (b) For unit vectors with $\langle c_1, c_2 \rangle = 0.8$ and $\alpha = (1, 0.5)$, $G^{-1}2\alpha = (10/3, -5/3)$. With $\eta = 0.002$, starting below the edge with small oscillations, show numerically that the second oscillation dies out while the first stays at the edge with $\langle x_1^2 \rangle \approx 2$, and that with $\alpha = (1, 0.9)$ both persist, with averages near $G^{-1}2\alpha = (1.56, 0.56)$.

8.8 SIMULATION. (The area law.) Reproduce the right panel of Figure 8.2, computing the area $\mathcal{J}$ of the level curve through each iterate by quadrature. Then start at $\mathcal{H}_0 = 10$ with $\eta = 0.01$ and $\alpha = 1$, observe that x underflows to zero after about $4 \cdot 10^4$ steps, repeat in the variables of (8.5), which cannot underflow, and describe the cycles.

8.9 SIMULATION. (Adam on a parabola.) Reproduce Example 8.12. (a) Measure the rate at which $\log S_P$ grows before the threshold and compare with Proposition 8.7, including the bias correction. (b) Repeat with $\beta_2 = 0.99$ and with $\beta_1 = 0.95$, whose threshold is 78. (c) Explain why the average lies below the threshold, by comparing the growth rate of the oscillation when ηS_P is a few percent above $(2 + 2\beta_1)/(1 - \beta_1)$ with its decay rate when ηS_P is 20% below.

8.10 SIMULATION. (Momentum in the network.) In the setting of Figure 8.1, run Adam with $\eta = 0.01$ and $\beta_1 \in \{0.8, 0.95\}$, and gradient descent with the moving-average momentum of Proposition 8.6(iii) at $\beta = 0.5$ and $\eta = 1$. Check that the preconditioned sharpness settles near $18/\eta$ and somewhat below $78/\eta$, and the sharpness of the momentum run at 6. Then take $\beta = 0.9$ and $\eta = 4.5$, whose threshold $38/\eta \approx 8.4$ is three times the initial sharpness 2.7, and observe divergence within twenty steps. Which hypothesis of Proposition 8.6 fails in the first steps? (Chapter 9 takes up the question.)

8.11 RESEARCH. ★ (Is the edge damped?) For a network at the edge of stability, estimate the third and fourth derivatives of the loss along the top eigenvector by finite differences of Hessian–vector products, compute the damping of Exercise 8.5(d), and compare it with the discretization's anti-damping $2\alpha\eta$, with α estimated from the sharpening rate before the edge. Does the sign of the total predict whether the oscillation settles onto a period-two orbit? As far as we know this is open. *(Hint: what enters is the coefficient of x^2 in the curvature along the oscillation.)*

8.12 RESEARCH. (Sharpening as an output.) Find a model in which the rate α is derived from the data and the architecture and the long phase at the edge, with a decreasing loss, can be proved. Agarwala et al. (2023) derive progressive sharpening in second-order regression models; a model that solves both is open.

Notes and References

Wu et al. (2018) proposed that stability selects the minima that gradient descent can reach, Jastrzębski et al. (2019, 2020) saw the early sharpness adapt to the step size and the batch size, and Lewkowycz et al. (2020) described the catapult, solved in volume 2 of this series. Cohen et al. (2021) named progressive sharpening and the edge of stability and derived the momentum thresholds; Cohen et al. (2022) found the adaptive edge.

Damian et al. (2023) derived the cubic model, identified self-stabilization, and proved the coupling of Section 8.4; an earlier version appeared at a NeurIPS 2022 workshop. Cohen et al. (2025) derived central flows for gradient descent and adaptive optimizers. Theorem 8.8 is the continuous-limit analysis of the cubic model with the proofs written out; we know no reference for the identities of Lemma 8.10, the monotone energy of Proposition 8.11, or the area law (8.8), which are elementary consequences of the model.

TRANSLATING THE LITERATURE. Damian et al. (2023) write x_t, y_t and $\alpha = -\langle \nabla \mathcal{L}, \nabla S \rangle$ as we do, and call the coefficient of the restoring force, built from $\|\nabla S\|^2$, β; it corresponds to our γ, renamed because β is the momentum in this book. Their constrained trajectory $\theta_t^\dagger$ is the discrete counterpart of the central flow. Cohen et al. (2025) write w for the parameters and count time in steps, so their flows carry a factor η that our $\tau = \eta t$ absorbs; their $\sigma^2(t)$ is the cycle average of x^2. Volume 2 of this series writes the reduced dynamics as $dA/dt = 2\varepsilon A$, $d\varepsilon/dt = \gamma - \nu A$ in steps; its $(A, \varepsilon, \gamma, \nu)$ are our $(x^2, \eta y, \eta^2 \alpha, \frac{1}{2}\eta^2 \gamma)$, so its A is our e^A.

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9 Warmup

Why the step size must start small

IN THE WILD — LINEAR WARMUP

IN THIS CHAPTER. Large training runs raise the learning rate from nearly zero to its peak over their first thousands of steps. The smallest idealization that makes this necessary bounds the curvature by an affine function of the loss gap, $S \le K_0 + K_1(\mathcal{L} - \mathcal{L}^\star)$. Under the Polyak–Łojasiewicz inequality we prove that a warmup indexed by the loss reaches accuracy ε in $(K_0 \log(\Delta_0/\varepsilon) + K_1\Delta_0)/\mu$ steps while a constant step pays the initial curvature in every factor of ε; we compute the warmup's profile exactly, and on a model that saturates both hypotheses it is optimal among stable schedules, while a stable linear ramp must be longer by a factor $u_0/(4e)$. We also prove that Adam's first step is a sign step, and derive that a warmup at the edge of stability drives the sharpness down along $2/\eta_t$, as a small network confirms. The curvature bound is not derived for networks, where it fails globally, and the theory misses the catapults through which shorter, unstable warmups succeed and often win.

9.1 In the wild: warming up the learning rate

The learning rate of a modern training run is a curve, and the curve starts low. An early instance is in the paper that introduced residual networks: He et al. (2016) trained a network with 110 layers on CIFAR-10 at the learning rate 0.01 until the training error fell below 80%, about 400 iterations, before switching to 0.1, which they found slightly too large to start converging.[1] Goyal et al. (2017), training a ResNet-50 on ImageNet with minibatches of 8192 images, scaled the learning rate with the minibatch (Chapter 11) and reached it by a linear ramp over five epochs, because the scaling rule fails while the network changes rapidly. The original Transformer (Vaswani et al. 2017) raised its learning rate linearly for 4000 steps and then decayed it as the inverse square root of the step; large language models inherited the ramp.[2]

IN THE WILD · GPT-3, Brown et al. (2020); LLaMA, Touvron et al. (2023); Adam's rule of thumb, Ma and Yarats (2021). Raise the learning rate linearly from (nearly) zero to its peak, over the first 375 million tokens for GPT-3 and the first 2000 steps for LLaMA, then decay it along a cosine to 10% of the peak (Chapter 12); both runs use Adam with $\beta_2 = 0.95$. For Adam a rule of thumb sets the length of a linear warmup

[1] Started at 0.1, they add, the network began to converge after several epochs and reached a similar accuracy.

[2] In this chapter *learning rate*, the word of the practice sections, and *step size*, the word of the theory, mean the same thing.

to $2/(1-\beta_2)$ steps: 2000 for $\beta_2 = 0.999$, 40 for $\beta_2 = 0.95$. Otherwise the length is inherited or tuned by a sweep; we know of no derivation of these numbers.

The observations fall into three groups.

OBSERVATION 9.1 (Warmup and layer normalization; Xiong et al. 2020). In the original Transformer, which normalizes after each residual addition (Post-LN), the gradients of the parameters near the output are large at initialization, so that a large initial learning rate makes training unstable, and warmup avoids this. With the normalization inside the residual branches (Pre-LN) the gradients are well behaved, and Pre-LN Transformers trained without warmup reach comparable results with less training time and less tuning.

The gradient statement is a mean-field calculation at initialization; the rest is experimental.

OBSERVATION 9.2 (Warmup lowers the sharpness; Gilmer et al. 2022, Kalra and Barkeshli 2024). Warmup lets early training avoid or leave regions of high curvature for flatter ones that tolerate larger learning rates, and stabilizes it as much as normalization layers do (Gilmer et al. 2022). When a network starts in a region sharper than the target learning rate tolerates, its sharpness falls during warmup and stays near $2/\eta_t$, and the main benefit of warmup is that larger target learning rates become trainable (Kalra and Barkeshli 2024).

The *sharpness* $S(\theta) = \lambda_{\max}(\nabla^2 \mathcal{L}(\theta))$ is the top eigenvalue of the Hessian, and $2/\eta$ is the stability threshold of gradient descent on a quadratic (Chapter 8). In the small network of Figure 9.2 the sharpness falls from 132 to 20 while the learning rate rises to 0.1; without warmup the same learning rate diverges in six steps.

OBSERVATION 9.3 (Explanations specific to Adam). Liu et al. (2020) attributed Adam's need for warmup to the variance of its adaptive learning rate while the second-moment estimate has averaged few gradients, and proposed RAdam, which rectifies the step; Ma and Yarats (2021) explained it instead by the size of Adam's early updates and found that an untuned linear warmup over $2/(1-\beta_2)$ steps works as well. Kosson et al. (2024) attributed the need in GPT training to large early angular updates of the weights and reduced it by controlling them; Kalra and Barkeshli (2024) initialized Adam's second moment with the first squared gradient (GI-Adam), with benefits similar to those of warmup.

The groups are not rival theories: a transformer trained with Adam has all three mechanisms. What is robust is that some warmup prevents early divergence at the learning rates that practice wants; what is folklore is the length. In Observation 9.2 the curvature falls because the schedule rises; the idealization of this chapter starts from a curvature that falls because the loss does, and asks: why must the step size start small, and how should it grow?

9.2 The idealization: curvature bounded by the loss

Gradient descent on a function whose curvature is at most L everywhere decreases the loss at every step if $\eta < 2/L$ (Lemma A.18). A network has no useful global L (Figure 9.2). The idealization replaces the global bound by one that improves as the loss falls.

Assumption 9.4 (Loss-dependent curvature). The loss $\mathcal{L}\colon \mathbb{R}^d \to \mathbb{R}$ is twice continuously differentiable and bounded below, with $\mathcal{L}^\star = \inf \mathcal{L}$ and gap $\Delta(\theta) = \mathcal{L}(\theta) - \mathcal{L}^\star$. For the initial point θ_0 let $\Theta_0 = \{\theta\colon \mathcal{L}(\theta) \le \mathcal{L}(\theta_0)\}$. There are constants $K_0, K_1 \ge 0$ such that

$$S(\theta) \le K_0 + K_1 \Delta(\theta) \qquad \text{for every } \theta \in \Theta_0. \tag{9.1}$$

The *loss-dependent curvature* condition allows the curvature K_0 at the bottom and the excess $K_1\Delta$ at a gap Δ, and bounds the Hessian from above only. Alimisis et al. (2026) state it with $\|\nabla^2 \mathcal{L}\|_{\mathrm{op}}$ in place of S, which implies (9.1), and Riabinin et al. (2026) state it in a general norm with an extra term $K_\rho \Delta^\rho$ (Remark 9.15).[3] A quadratic satisfies it with $K_1 = 0$, and the function

$$\mathcal{L}_{\mathrm{c}}(z) = \frac{K_0}{K_1}\Big(\cosh\big(\sqrt{K_1}\, z\big) - 1\Big), \qquad z \in \mathbb{R}, \tag{9.2}$$

with equality, $\mathcal{L}_{\mathrm{c}}'' = K_0 + K_1 \mathcal{L}_{\mathrm{c}}$: its curvature grows as fast as its loss allows ($2\cosh z - 2$ is the case $K_0 = 2$, $K_1 = 1$). The second hypothesis says that the gradient is large when the loss is.

Assumption 9.5 (Polyak–Łojasiewicz inequality). There is $\mu > 0$ such that $\|\nabla \mathcal{L}(\theta)\|_2^2 \ge 2\mu \Delta(\theta)$ for every $\theta \in \Theta_0$.

Under the *Polyak–Łojasiewicz inequality* every stationary point in Θ_0 is a global minimizer, and a global curvature bound L gives the rate $1 - \mu/L$ (Lemma A.19).[4] Both hypotheses concern the loss: the first says how large a step the current loss allows, the second how much a step of that size achieves. The extremal member of the class has two kinds of coordinates.

Definition 9.6 (The warmup model). Fix $\mu > 0$, $K_0 \ge 2\mu$ and $K_1 > 0$. The *warmup model* is the loss

$$\mathcal{L}(x, z) = q(x) + \tfrac{1}{2} S(x) \|z\|_2^2, \qquad q(x) = \tfrac{\mu}{2}\|x\|_2^2, \qquad S(x) = K_0 + K_1 q(x),$$

on $\theta = (x, z) \in \mathbb{R}^{d_x} \times \mathbb{R}^{d_z}$, trained by gradient descent from a point (x_0, z_0) with z_0 small.

The slow coordinate x carries the loss, with curvature μ; the stiff coordinate z has the curvature $S(x)$ set by that loss. The model caricatures the start of training, where the loss is high because the network has not fitted the data, and the stiffest curvature is high for the same reason.

Proposition 9.7 (The warmup model saturates both hypotheses). *In the warmup model $\mathcal{L}^\star = 0$, and the following hold.*

[3] Its relative for the gradient, $\|\nabla^2 \mathcal{L}\| \le L_0 + L_1 \|\nabla \mathcal{L}\|$, is the (L_0, L_1)-smoothness of Zhang et al. (2020) and Chapter 10. Alimisis et al. (2026) prefer the loss: their condition is closed under finite sums and affine maps, and a bound in the gradient norm can shrink the step early in training, when warmup should raise it.

[4] The inequality is due to Polyak and to Łojasiewicz, both in 1963; for its uses in machine learning see Karimi et al. (2016).

(i) *Assumptions 9.4 and 9.5 hold on all of* $\mathbb{R}^{d_x+d_z}$ *with the constants* K_0, K_1 *and* μ, *and even* $\|\nabla^2\mathcal{L}(\theta)\|_{\mathrm{op}} \le K_0 + K_1\Delta(\theta)$.

(ii) *On the subspace* $\{z = 0\}$ *both hold with equality:* $S = K_0 + K_1\Delta$ *and* $\|\nabla\mathcal{L}\|_2^2 = 2\mu\Delta$.

(iii) *Gradient descent with steps* η_t *satisfies*

$$x_{t+1} = \Big(1 - \eta_t\mu\big(1 + \tfrac{1}{2}K_1\|z_t\|_2^2\big)\Big)x_t, \qquad z_{t+1} = \big(1 - \eta_t S(x_t)\big)z_t. \tag{9.3}$$

So $\{z = 0\}$ *is invariant, on it* $\Delta_{t+1} = (1 - \mu\eta_t)^2\Delta_t$, *and a perturbation of the stiff coordinate is amplified at step* t *exactly when* $\eta_t S(x_t) > 2$.

PROOF. The plan is to compute the gradient and the Hessian and to bound the Hessian on the one plane where its blocks interact. Write $\varepsilon = \frac{1}{2}K_1\|z\|_2^2$. Then $\nabla_x\mathcal{L} = \mu(1+\varepsilon)x$, $\nabla_z\mathcal{L} = S(x)z$, $\mathcal{L} = q(1+\varepsilon)+\frac{1}{2}K_0\|z\|_2^2$ and $K_0+K_1\mathcal{L} = S(1+\varepsilon)$, which gives (iii); at $z = 0$ the Hessian is block diagonal with blocks μI and $S(x)I$, and $S(x) \ge K_0 \ge \mu$, which gives (ii).

For (i), the Hessian has the diagonal blocks $\mu(1+\varepsilon)I$ and SI and the off-diagonal block $K_1\mu\, xz^\top$. On vectors orthogonal to x in the first block or to z in the second it acts as $\mu(1+\varepsilon)$ or S, both in $[0, S(1+\varepsilon)]$. If $x, z \ne 0$, it remains to bound the 2×2 matrix A that it induces on the span of $(x/\|x\|_2, 0)$ and $(0, z/\|z\|_2)$, with diagonal entries $\mu(1+\varepsilon)$ and S and off-diagonal entry $b = K_1\mu\|x\|_2\|z\|_2$, so that $b^2 = 4\mu K_1 q\varepsilon$. The matrix $S(1+\varepsilon)I - A$ has the nonnegative diagonal entries $(S-\mu)(1+\varepsilon)$ and $S\varepsilon$ and determinant at least $\varepsilon\big((S-\mu)S - 4\mu K_1 q\big)$, and the bracket is nonnegative: $K_0 \ge 2\mu$ gives $S - \mu \ge K_0/2 + K_1 q$, and the inequality of arithmetic and geometric means gives $(K_0/2 + K_1 q)(K_0 + K_1 q) \ge 2\sqrt{2}\,K_0 K_1 q \ge 4\mu K_1 q$. Hence $\lambda_{\max}(A) \le S(1+\varepsilon) = K_0 + K_1\mathcal{L}$. Also $\lambda_{\min}(A) \ge -b$, and $b = 2\sqrt{K_1 q\cdot\mu\varepsilon} \le K_1 q + \mu\varepsilon \le S(1+\varepsilon)$. Finally, since $S^2 \ge K_0^2 \ge \mu K_0$,

$$\|\nabla\mathcal{L}\|_2^2 = 2\mu q(1+\varepsilon)^2 + S^2\|z\|_2^2 \ge 2\mu q(1+\varepsilon) + \mu K_0\|z\|_2^2 = 2\mu\mathcal{L}. \qquad \blacksquare$$

On the line $z = 0$ the model has the most curvature and the least gradient that the hypotheses allow; gradient descent there is a scalar recursion while the stiff coordinate stays quiet, and its instability is the only way a step can fail. The hypothesis $K_0 \ge 2\mu$ is a convenience (Exercise 9.3).

WHAT WAS THROWN AWAY. Gradient noise and the minibatch scaling rule, whose early failure motivated the warmup of Goyal et al. (2017); the response of the sharpness to the step size (Chapter 8); the geometry of networks, where both hypotheses can hold only near the trajectory (Remark 9.14); and every optimizer but gradient descent, until Section 9.4. What remains is the mechanism that the hypothesis isolates: the curvature at the start exceeds what the target step tolerates, and it falls as training proceeds. If warmup were not needed here, it would not be needed for this reason in a network.

WHAT ONE HAD TO NOTICE. The safe step is a property of the current point, and it can be indexed by the loss, which, unlike the gradient norm, does not increase along safe steps; so a step safe now is safe later, a bound in terms

of the loss becomes a bound in terms of time, and an increasing schedule falls out.

DICTIONARY

IN THE THEORY	IN A TRAINING RUN
the gap Δ_t	the training loss minus its floor, which is unknown in practice
$S \le K_0 + K_1\Delta$	the sharpness falls as the loss falls early in training (measured, not derived)
K_0, and the final step $1/K_0$	the curvature late in training, which limits the peak learning rate
the excess $u_0 = K_1\Delta_0/K_0$	how much sharper the initialization is than the trained network
μ, and $\kappa = K_0/\mu$	the slowest direction of descent, and a condition number
slow coordinate x, stiff coordinate z	the bulk of the parameters, and the top eigenvector of the Hessian
$\eta_t S_t > 2$	the edge of stability (Chapter 8): a catapult or a divergence

9.3 The theory: a safe step that grows as the loss falls

The classical descent lemma integrates a curvature bound along the segment of a step. Under Assumption 9.4 the bound is available only where the loss is below its starting value, and the segment is not known in advance to stay there; a continuity argument closes the gap.

Lemma 9.8 (A local descent lemma). *Let $\mathcal{L}$ be twice continuously differentiable, let $\theta, u \in \mathbb{R}^d$ satisfy $\langle \nabla\mathcal{L}(\theta), u\rangle < 0$, and let $M > 0$ be such that $u^\top \nabla^2\mathcal{L}(\theta + su)\, u \le M$ for every $s \ge 0$ with $\mathcal{L}(\theta + su) \le \mathcal{L}(\theta)$. Then for every $s \in \big[0, 2|\langle \nabla\mathcal{L}(\theta), u\rangle|/M\big]$ the segment from θ to $\theta + su$ lies in $\{\mathcal{L} \le \mathcal{L}(\theta)\}$, and*

$$\mathcal{L}(\theta + su) \le \mathcal{L}(\theta) + s\langle \nabla\mathcal{L}(\theta), u\rangle + \tfrac{1}{2}Ms^2.$$

Proof. Let $\varphi(r) = \mathcal{L}(\theta + ru)$, $\gamma = -\varphi'(0) > 0$ and $\bar{s} = 2\gamma/M$, and fix $s \in (0, \bar{s}]$; the case $s = 0$ is trivial. The plan is to show that φ cannot climb back to $\varphi(0)$ before s. The set A of $r \in [0, s]$ with $\varphi \le \varphi(0)$ on $[0, r]$ contains 0, is an interval, and is closed by continuity, so $A = [0, r^*]$. On A the hypothesis gives $\varphi'' \le M$, and Taylor's formula with integral remainder gives, for $r \in A$,

$$\varphi(r) \le \varphi(0) - \gamma r + \tfrac{1}{2}Mr^2 = \varphi(0) - \gamma r\big(1 - r/\bar{s}\big). \tag{9.4}$$

Suppose that $r^* < s$. If $r^* = 0$, then $\varphi'(0) < 0$ gives $\varphi < \varphi(0)$ on some interval $(0, \delta)$; if $r^* > 0$, then $r^* < \bar{s}$ and (9.4) give $\varphi(r^*) < \varphi(0)$, and by continuity $\varphi < \varphi(0)$ on some $[r^*, r^* + \delta)$. Either way A extends beyond r^*, which is absurd. So $r^* = s$, and (9.4) at $r = s$ is the claim. ∎

For gradient descent at $\theta \in \Theta_0$ with gap Δ and gradient $g \neq 0$, take $u = -g$: points of the ray with $\mathcal{L} \leq \mathcal{L}(\theta)$ have gap at most Δ, so Assumption 9.4 gives $M = (K_0 + K_1\Delta)\|g\|_2^2$, and for $0 \leq \eta \leq 2/(K_0 + K_1\Delta)$

$$\mathcal{L}(\theta - \eta g) \leq \mathcal{L}(\theta) - \eta\Big(1 - \frac{\eta(K_0 + K_1\Delta)}{2}\Big)\|g\|_2^2, \tag{9.5}$$

which is at most $\mathcal{L}(\theta) - \frac{\eta}{2}\|g\|_2^2$ if $\eta \leq 1/(K_0 + K_1\Delta)$. This *local descent lemma* replaces the global constant by the curvature that the current loss allows. Since the step does not increase the loss, the step certified at the next point is at least as large: the lemma already contains a warmup.

Theorem 9.9 (Warmup under loss-dependent curvature). *Let Assumptions 9.4 and 9.5 hold with $K_0 > 0$, let $\Delta_0 = \Delta(\theta_0) > 0$, and define $r(D) = \mu D/(K_0+K_1D)$ and the scalar sequence $D_0 = \Delta_0$, $D_{t+1} = D_t - r(D_t)$. Then $\mu \leq K_0$, and the gaps $\Delta_t = \Delta(\theta_t)$ of gradient descent satisfy the following.*

(i) Loss-indexed warmup. *With $\eta_t = 1/(K_0+K_1\Delta_t)$, $\Delta_{t+1} \leq \Delta_t - r(\Delta_t)$ for every t.*

(ii) Scheduled warmup. *With $\eta_t = 1/(K_0 + K_1D_t)$, $\Delta_t \leq D_t$ for every t. This schedule is nondecreasing, from $1/(K_0 + K_1\Delta_0)$ to the limit $1/K_0$.*

(iii) In both cases, for every $t \geq 0$,

$$K_0 \log\frac{\Delta_0}{\Delta_t} + K_1(\Delta_0 - \Delta_t) \geq \mu t, \tag{9.6}$$

so that $\Delta_T \leq \varepsilon$ for every $\varepsilon \in (0, \Delta_0]$ and every $T \geq \big(K_0 \log(\Delta_0/\varepsilon) + K_1(\Delta_0 - \varepsilon)\big)/\mu$.

(iv) Constant step. *With $\eta_t = \eta \leq 1/(K_0 + K_1\Delta_0)$, $\Delta_t \leq (1 - \mu\eta)^t\Delta_0$; at the largest such η this guarantees $\Delta_T \leq \varepsilon$ for $T \geq (K_0 + K_1\Delta_0)\log(\Delta_0/\varepsilon)/\mu$.*

Proof. The plan is to apply (9.5) at every step, after checking that the step is admissible there, and to turn the one-step decrease into (9.6) by comparison with an integral. A point with $\Delta_t = 0$ is a global minimizer, from which gradient descent does not move; at a point of Θ_0 with $\Delta_t > 0$ the gradient is nonzero by Assumption 9.5. We set $\log(\Delta_0/0) = +\infty$.

Part (i). If $\theta_t \in \Theta_0$ and $\Delta_t > 0$, the step $1/(K_0 + K_1\Delta_t)$ is admissible in (9.5), and with Assumption 9.5

$$\Delta_{t+1} \leq \Delta_t - \frac{\|\nabla\mathcal{L}(\theta_t)\|_2^2}{2(K_0 + K_1\Delta_t)} \leq \Delta_t - \frac{\mu\Delta_t}{K_0 + K_1\Delta_t} = \Delta_t - r(\Delta_t);$$

in particular $\theta_{t+1} \in \Theta_0$, and (i) follows by induction.

The inequality $\mu \leq K_0$. The same computation at a point of Θ_0 with gap $\Delta > 0$, with $\mathcal{L}^\star \leq \mathcal{L}(\theta - \eta g)$, gives $\mu \leq K_0 + K_1\Delta$. And Θ_0 contains points of arbitrarily small positive gap: the gaps of (i) satisfy $\Delta_{t+1} \leq (1 - \mu/(K_0 + K_1\Delta_0))\Delta_t$, and if one vanishes, at a first step T, then $\mathcal{L}$ takes

every value in $[\mathcal{L}^\star, \mathcal{L}(\theta_{T-1})]$ on the segment from θ_{T-1} to θ_T, which lies in Θ_0 by Lemma 9.8.

Part (ii). Since $\mu \le K_0$, the map $D \mapsto D\big(1-\mu/(K_0+K_1D)\big)$ sends $[0,\infty)$ into itself, so the D_t are nonnegative and nonincreasing, and they tend to 0 because $D_{t+1} \le (1-\mu/(K_0+K_1\Delta_0))D_t$; hence η_t rises to $1/K_0$. Suppose that $\Delta_t \le D_t$, as at $t=0$, and $\Delta_t > 0$. Then $\theta_t \in \Theta_0$, since $D_t \le \Delta_0$, and $\eta_t \le 1/(K_0+K_1\Delta_t)$ is admissible in (9.5), and since $1-\mu\eta_t \ge 1-\mu/K_0 \ge 0$,

$$\Delta_{t+1} \le \Delta_t - \frac{\eta_t}{2}\|\nabla\mathcal{L}(\theta_t)\|_2^2 \le (1-\mu\eta_t)\Delta_t \le (1-\mu\eta_t)D_t = D_{t+1}.$$

Part (iii). Let $\Phi(D) = K_0\log D + K_1 D$ for $D>0$ and $\Phi(0) = -\infty$; it is increasing, and $\Phi'(D) = (K_0+K_1D)/D$ is decreasing. Let (E_t) be nonnegative with $E_0 = \Delta_0$ and $E_{t+1} \le E_t - r(E_t)$, such as the gaps of (i) or the D_t of (ii). If $E_{t+1} > 0$, then, since Φ' is smallest at the right end of $[E_{t+1}, E_t]$,

$$\Phi(E_t) - \Phi(E_{t+1}) = \int_{E_{t+1}}^{E_t} \Phi'(s)\,\mathrm{d}s \ge (E_t - E_{t+1})\,\Phi'(E_t) \ge r(E_t)\,\frac{K_0+K_1E_t}{E_t} = \mu.$$

Summing, $\Phi(E_t) \le \Phi(\Delta_0) - \mu t$, also when some E_s vanishes. In case (ii), $\Phi(\Delta_t) \le \Phi(D_t)$ because Φ is increasing. So $\Phi(\Delta_t) \le \Phi(\Delta_0) - \mu t$ in both cases, which is (9.6), and if $T \ge (\Phi(\Delta_0) - \Phi(\varepsilon))/\mu$, then $\Phi(\Delta_T) \le \Phi(\varepsilon)$ and $\Delta_T \le \varepsilon$.

Part (iv). If $\Delta_t \le \Delta_0$, then η is admissible in (9.5), and $\Delta_{t+1} \le (1-\mu\eta)\Delta_t$; induct, and use $(1-\mu\eta)^t \le e^{-\mu\eta t}$. ∎

The schedule of (ii) is computed in advance from four numbers; the rule of (i) needs the gap, which in practice is unknown, because $\mathcal{L}^\star$ is. The bound (9.6) is additive: the initial excess curvature costs $K_1\Delta_0/\mu$ steps once and the final curvature K_0/μ steps per factor e of accuracy, whereas the constant step of (iv) pays $(K_0+K_1\Delta_0)/\mu$ per factor e.

Corollary 9.10 (The warmup profile). *In the setting of Theorem 9.9 let $K_1 > 0$ and $u_0 = K_1\Delta_0/K_0$, and for real s let $\omega(s)$ be the positive solution w of $w + \log w = s$. The function*

$$D(t) = \frac{K_0}{K_1}\,\omega\Big(u_0 + \log u_0 - \frac{\mu t}{K_0}\Big), \qquad t \ge 0,$$

solves $\dot D = -r(D)$ with $D(0) = \Delta_0$, and $D_t \le D(t)$ for every t. So the scheduled step $1/(K_0+K_1D_t)$ is at least the warmup profile

$$\eta(t) = \frac{1}{K_0\big(1+\omega(u_0+\log u_0 - \mu t/K_0)\big)},$$

which reaches $1/(2K_0)$, if $u_0 \ge 1$, at the time $T_{1/2} = (K_0/\mu)(u_0 - 1 + \log u_0)$.

Proof. With $u = K_1D/K_0$, $\Phi(D) = K_0(u+\log u) + K_0\log(K_0/K_1)$, so $D(t)$ is the solution of $\Phi(D(t)) = \Phi(\Delta_0) - \mu t$, and differentiating gives $\dot D = -\mu/\Phi'(D) = -r(D)$. The bound $D_t \le D(t)$ is $\Phi(D_t) \le \Phi(\Delta_0) - \mu t$, proved in part (iii) of the theorem. Finally, $\eta(t) = 1/(2K_0)$ means $u_0 + \log u_0 - \mu t/K_0 = 1$. ∎

The function ω is the *Wright omega function*, $\omega(s) = W(e^s)$ with W the principal branch of the Lambert function.[5] While $u = K_1 D/K_0$ is large, $u \approx u_0 - \mu t/K_0$: the gap falls by about μ/K_1 per step, and the step rises along the hyperbola $\eta(t) \approx 1/(K_0 + K_1\Delta_0 - \mu t)$; once u is small, the gap decays at the rate μ/K_0 and the step approaches $1/K_0$ exponentially. The profile is convex until $\eta = 2/(3K_0)$ and concave afterwards, and the discrete schedule follows it up to a relative error μ/K_0 in time (Exercise 9.2). The *warmup* is the first phase; its length $T_{1/2}$ is the initial excess curvature divided by μ, up to a logarithm.

[5] The Lambert function inverts $w \mapsto we^w$, so $w + \log w = s$ means $we^w = e^s$. Both are in SciPy, as `lambertw` and `wrightomega`.

How much does the warmup buy? A constant step large enough for the late phase overshoots the cosh function at the start, and a quadratic punishes every smaller step.

Proposition 9.11 (A constant step pays the initial curvature at every scale). *Let $0 < \mu \le K_0$, $K_1 > 0$, $\Delta_0 > 0$ and $0 < \varepsilon < \Delta_0/2$, and let $\mathcal{L}(x, z) = \frac{\mu}{2}x^2 + \mathcal{L}_c(z)$ on $\mathbb{R}^2$, with $\mathcal{L}_c$ from (9.2), started at (x_0, z_0) with $\frac{\mu}{2}x_0^2 = \mathcal{L}_c(z_0) = \Delta_0/2$ and $z_0 > 0$. Then $\mathcal{L}^\star = 0$, Assumptions 9.4 and 9.5 hold on $\mathbb{R}^2$ with the constants K_0, K_1 and μ, and with $a = \sqrt{K_1}z_0$ and $\eta^* = 2a/(K_0 \sinh a)$, gradient descent with a constant step $\eta > 0$ (i) never reaches $\Delta_t \le \varepsilon$ if $\eta \ge \eta^*$, and (ii) needs at least $(1 - \eta\mu)\log(\Delta_0/(2\varepsilon))/(2\eta\mu)$ steps to reach it if $\eta < \eta^*$. Hence every constant step needs at least*

$$T \ge \left(\frac{(K_0 + K_1\Delta_0/2)\tanh a}{4\mu a} - \frac{1}{2}\right)\log\frac{\Delta_0}{2\varepsilon} \tag{9.7}$$

steps, where $a = \operatorname{arccosh}(1 + K_1\Delta_0/(2K_0)) \approx \log(2 + K_1\Delta_0/K_0)$.

Proof. The plan is to show that steps above η^* make the stiff coordinate grow forever and that steps below it are slow on the quadratic. The Hessian is diagonal with entries $\mu \le K_0$ and $K_0\cosh(\sqrt{K_1}z) = K_0 + K_1\mathcal{L}_c(z)$, both at most $K_0 + K_1\mathcal{L}$. Since $\sinh^2 = (\cosh - 1)(\cosh + 1) \ge 2(\cosh - 1)$, $\mathcal{L}_c'^2 = (K_0^2/K_1)\sinh^2(\sqrt{K_1}z) \ge 2K_0\mathcal{L}_c \ge 2\mu\mathcal{L}_c$, and $(\mu x)^2 = 2\mu \cdot \frac{\mu}{2}x^2$, so $\|\nabla\mathcal{L}\|_2^2 \ge 2\mu\mathcal{L}$.

Gradient descent acts on the coordinates separately, $x_{t+1} = (1 - \eta\mu)x_t$ and $z_{t+1} = \psi(z_t)$ with the odd map $\psi(z) = z - \eta(K_0/\sqrt{K_1})\sinh(\sqrt{K_1}z)$. Let $\eta^*(z) = 2b/(K_0 \sinh b)$ with $b = \sqrt{K_1}z$. For $z > 0$, $|\psi(z)| \ge z$ holds exactly when $\psi(z) \le -z$, that is, when $\eta \ge \eta^*(z)$, and $\eta^*(z)$ decreases in z because $b/\sinh b$ decreases. For (i), if $\eta \ge \eta^* = \eta^*(z_0)$ and $|z_t| \ge z_0$, then $\eta \ge \eta^*(|z_t|)$, so $|z_{t+1}| = |\psi(|z_t|)| \ge |z_t|$; by induction $\Delta_t \ge \mathcal{L}_c(z_0) = \Delta_0/2 > \varepsilon$ for every t. For (ii), $\Delta_t \ge \frac{\mu}{2}x_t^2 = (1 - \eta\mu)^{2t}\Delta_0/2$; if $\eta\mu \ge 1$ the bound claimed is not positive, and otherwise $\Delta_T \le \varepsilon$ forces $2T\log(1/(1-\eta\mu)) \ge \log(\Delta_0/(2\varepsilon))$, and $\log(1/(1-s)) \le s/(1-s)$ gives the claim. Finally, $(1-\eta\mu)/(2\eta\mu)$ decreases in η, so for $\eta < \eta^*$ it exceeds $1/(2\eta^*\mu) - 1/2$, and $K_0 \sinh a = (K_0 + K_1\Delta_0/2)\tanh a$ gives (9.7). ∎

For $K_1\Delta_0 \gg K_0$ the bound (9.7) is about $K_1\Delta_0\log(\Delta_0/(2\varepsilon))/(8\mu a)$: up to the logarithm a and constants, a constant step pays the initial excess curvature in every factor e of accuracy, the warmup once. The logarithm is genuine: the cheapest diverging step exceeds the threshold $2/\mathcal{L}_c''(z_0)$

of the initial curvature by the factor $a \coth a$, since the curvature of $\mathcal{L}_c$ falls along a long step. Alimisis et al. (2026), with H_0, H_1 for K_0, K_1, prove bounds of both kinds: for a warmup step under an aiming condition (a weakening of convexity), an upper bound whose $1/\varepsilon$ term involves H_0 alone; and lower bounds for constant steps that combine, as ours do, a rapidly growing function, which forces the step to be small, with a slowly growing one, on which a small step is slow. Constants matter: for $K_0 = K_1 = 1$, $\mu = 0.02$, $\Delta_0 = 50$ and $\varepsilon = 10^{-6}$, the best constant step on the instance of Proposition 9.11 needs 1396 steps (the bound gives 1392) and the loss-indexed warmup 1051, while Theorem 9.9 guarantees only 3386. The worst cases show the structure; the warmup model shows the size.

Proposition 9.12 (Warmup is optimal among stable schedules). *Consider the warmup model on the line $z = 0$, with $\Delta_0 = q(x_0) > 0$, $S_t = K_0 + K_1\Delta_t$ and $\Phi(D) = K_0 \log D + K_1 D$, and let $0 < \varepsilon < \Delta_0$.*

(i) Every schedule with $0 \le \eta_t S_t \le 2$ for all t, so that no perturbation of the stiff coordinate is ever amplified, needs at least $(1 - 2\mu/K_0)^2\big(\Phi(\Delta_0) - \Phi(\varepsilon)\big)/(4\mu)$ steps to reach $\Delta_T \le \varepsilon$.

(ii) The loss-indexed schedule $\eta_t = c/S_t$ with $0 < c \le 2$ reaches it within $\big(\Phi(\Delta_0) - \Phi(\varepsilon)\big)/\big(c\mu(2 - c\mu/K_0)\big)$ steps.

(iii) Every constant step with $\eta S_0 \le 2$ needs at least $(1 - 2\mu/S_0)\, S_0 \log(\Delta_0/\varepsilon)/(4\mu)$ steps.

Proof. The plan is to bound the decrease of Φ per step, from above in (i) and from below in (ii), using $\Delta_{t+1} = (1 - \mu\eta_t)^2\Delta_t$ from Proposition 9.7(iii). For (i), $\mu\eta_t \le 2\mu/S_t \le 2\mu/K_0 \le 1$, strictly when $\Delta_t > 0$, so the gaps stay positive and $\Delta_t - \Delta_{t+1} = \Delta_t\mu\eta_t(2 - \mu\eta_t) \le 4\mu\Delta_t/S_t$. Since $\Phi'(D) = (K_0 + K_1 D)/D$ is decreasing and $S_{t+1} \le S_t$,

$$\Phi(\Delta_t) - \Phi(\Delta_{t+1}) \le (\Delta_t - \Delta_{t+1})\frac{S_{t+1}}{\Delta_{t+1}} \le \frac{4\mu\Delta_t}{\Delta_{t+1}} = \frac{4\mu}{(1 - \mu\eta_t)^2} \le \frac{4\mu}{(1 - 2\mu/K_0)^2},$$

and summing over $t < T$ with $\Phi(\Delta_T) \le \Phi(\varepsilon)$ gives (i). For (ii), $c\mu/S_t \le 1$ and $\Delta_t - \Delta_{t+1} = \Delta_t(c\mu/S_t)(2 - c\mu/S_t) \ge c\mu(2 - c\mu/K_0)\Delta_t/S_t$, so the integral comparison of part (iii) of Theorem 9.9 gives $\Phi(\Delta_t) \le \Phi(\Delta_0) - c\mu(2 - c\mu/K_0)\, t$. For (iii), $\Delta_T = (1 - \mu\eta)^{2T}\Delta_0 \ge (1 - 2\mu/S_0)^{2T}\Delta_0$, and $\log(1/(1 - s)) \le s/(1 - s)$ with $s = 2\mu/S_0$ finishes as in Proposition 9.11. ∎

So on the warmup model the schedule of Theorem 9.9 ($c = 1$) is within a factor $2 + O(1/\kappa)$ of every stable schedule, riding the stability limit ($c = 2$) is optimal among them up to $1 + O(1/\kappa)$, and the best stable constant step is slower than the latter, up to the same factors, by between $\frac{1}{2}$ and 1 times $\min\{1 + u_0, \log(\Delta_0/\varepsilon)\}$ when $\varepsilon \le \Delta_0/e$. Unstable schedules can do better (Section 9.5).

9.4 The theory: cross-entropy, normalized steps, Adam and the edge

9.4.1 Where loss-dependent curvature comes from

The exponential $e^{-\theta}$ has curvature equal to its value ($K_0 = 0$, $K_1 = 1$) and violates Assumption 9.5; still, the loss-indexed step $1/\Delta_t = e^{\theta_t}$ gives $\theta_{t+1} = \theta_t + 1$, a loss that falls by e per step under a step that grows without bound (Exercise 9.5). For cross-entropy the constant is 2.

Proposition 9.13 (Softmax regression). *For an input $a \in \mathbb{R}^{d_{\text{in}}}$, a class $c \in \{1, \dots, K\}$ and weights $W \in \mathbb{R}^{K \times d_{\text{in}}}$, let $p = \operatorname{softmax}(Wa)$ and $\ell(W) = -\log p_c$. Then*

$$\lambda_{\max}\big(\nabla_W^2 \ell\big) = \|a\|_2^2 \, \lambda_{\max}\big(\operatorname{diag}(p) - pp^\top\big) \le 2\|a\|_2^2 (1 - p_c) \le 2\|a\|_2^2 \, \ell(W),$$

and the constant 2 cannot be lowered. So the average $\mathcal{L}$ of ℓ over data with $\|a_i\|_2 \le R$ satisfies Assumption 9.4 with $K_0 = 2R^2\mathcal{L}^\star$ and $K_1 = 2R^2$, where $\mathcal{L}^\star = 0$ for separable data.

Proof. The plan is to reduce the Hessian to that of the logits and to apply Gershgorin's theorem. With $s = Wa$, $\ell = -s_c + \log \sum_k e^{s_k}$ has Hessian $M = \operatorname{diag}(p) - pp^\top$ in s. Since s is linear in W, the second derivative of ℓ in a direction V is $(Va)^\top M (Va)$, and as V ranges over the unit ball of the Frobenius norm, Va ranges over the ball of radius $\|a\|_2$ (take $V = va^\top / \|a\|_2$); since $M \succeq 0$, the largest eigenvalue is $\|a\|_2^2 \lambda_{\max}(M)$. By Gershgorin's theorem every eigenvalue of M is at most $\max_i \big(M_{ii} + \sum_{j \ne i} |M_{ij}|\big) = \max_i 2p_i(1 - p_i)$, which is at most $2(1 - p_c)$ for $i = c$ and at most $2p_i \le 2(1 - p_c)$ for $i \ne c$; and $1 - p_c \le -\log p_c$. For $K = 2$ the eigenvalue $2p_c(1 - p_c)$ of M has ratio to ℓ tending to 2 as $p_c \to 1$. Averaging over the data gives $\lambda_{\max}(\nabla^2 \mathcal{L}) \le 2R^2 \mathcal{L} = 2R^2 \mathcal{L}^\star + 2R^2 \Delta$. ∎

A bound with $1 - p_c$ in place of $2(1 - p_c)$ fails for two classes as soon as $p_c > \frac{1}{2}$; for binary logistic regression with one weight vector the constant is 1 (Exercise 9.1), the factor 2 coming from the two rows of W moving in opposite directions. So cross-entropy on a linear model, or on the last layer of a network with frozen features, has curvature bounded by the loss. The rest of a network is another matter.

Remark 9.14 (Rescaling symmetries break the global inequality). For the product model $\mathcal{L}(u, v) = \frac{1}{2}(uv - y_\star)^2$ with $y_\star > 0$, every point of the hyperbola $uv = y_\star$ is a global minimizer, with sharpness $u^2 + v^2$, unbounded on it. Every sublevel set with positive gap contains the hyperbola, so no constants satisfy Assumption 9.4. The same holds, for the square loss, in every network whose loss is invariant under multiplying one layer by α and the next by $1/\alpha$, as linear and ReLU networks are. There the assumption can hold only on a bounded region, with constants that depend on it, or along a trajectory.

The product model is solved in volume 2 of this series, *The Hydrogen Atoms of Deep Learning*, Chapter 15, where a large step leaves a sharp minimizer

through a catapult: the dynamics selects the curvature, which no inequality between curvature and loss expresses. Alimisis et al. (2026) report that the condition holds for common architectures, and Riabinin et al. (2026) check it along trajectories; both statements concern the region that training visits.

9.4.2 Normalized steps

For an optimizer whose update has a fixed length in some norm, such as sign descent or Muon (Chapters 2 and 3), the learning rate is a length, and the question changes.

REMARK 9.15 (When normalized steps need warmup). Let normalized steepest descent take steps $\eta_t \operatorname{lmo}(\nabla\mathcal{L}(\theta_t))$ of length η_t in a norm $\|\cdot\|$ (Chapter 1), and suppose that on Θ_0, $u^\top\nabla^2\mathcal{L}(\theta)u \le K(\Delta(\theta))\|u\|^2$ for every u, with K nondecreasing, and $\|\nabla\mathcal{L}(\theta)\|_* \ge c\Delta(\theta)^\nu$ with $c > 0$, $0 < \nu \le 1$. By Lemma 9.8, the loss-indexed length $\eta_t = c\Delta_t^\nu/K(\Delta_t)$ gives $\Delta_{t+1} \le \Delta_t - c^2\Delta_t^{2\nu}/(2K(\Delta_t))$, and it rises as the loss falls exactly where $\Delta^\nu/K(\Delta)$ decreases (Exercise 9.7). For $K = K_0 + K_1\Delta$ and $\nu = 1$ (a convex loss whose sublevel set lies within distance $1/c$ of a minimizer) it never rises; for $K = K_0+K_1\Delta+K_\rho\Delta^\rho$ with $\rho > 1$ and $\nu = 1$ it rises until $\Delta' = \big(K_0/((\rho-1)K_\rho)\big)^{1/\rho}$; for $K = K_0 + K_1\Delta$ and $\nu = \frac{1}{2}$ it rises until $K_1\Delta = K_0$. The middle case is the shape that Riabinin et al. (2026) derive for norm-constrained optimizers such as Muon and Lion, whose learning rate warms up and then decays only when $\rho > 1$, turning at Δ'.

A normalized method thus needs a warmup only when the curvature falls faster than the gradient's dual norm.

9.4.3 Adam's first steps

Adam with bias correction (Kingma and Ba 2015) keeps, from $m_{-1} = v_{-1} = 0$, the coordinatewise averages $m_t = \beta_1 m_{t-1} + (1-\beta_1)g_t$ and $v_t = \beta_2 v_{t-1} + (1-\beta_2)g_t^2$, divides them by $1-\beta_1^{t+1}$ and $1-\beta_2^{t+1}$, and steps $\theta_{t+1} = \theta_t - \eta\,\hat m_t/(\sqrt{\hat v_t}+\epsilon)$. The *bias correction* removes the pull of the zero initial state, and it shapes the first steps.

Proposition 9.16 (Adam's first steps). *Let $\epsilon = 0$, with $0/0 = 0$, and fix a coordinate.*

(i) The first step is a sign step: $\theta_1 - \theta_0 = -\eta\,\operatorname{sign}(g_0)$, whatever the size of g_0.

(ii) For every t, $\hat v_t = \sum_{s=0}^t w_{t,s}g_s^2$ with weights $w_{t,s} = (1-\beta_2)\beta_2^{t-s}/(1-\beta_2^{t+1})$ that sum to 1 and lie in $[\beta_2^t/(t+1), \beta_2^{-t}/(t+1)]$. If $t(1-\beta_2) \le \delta < 1$, then $(1-\delta)\bar v_t \le \hat v_t \le \bar v_t/(1-\delta)$, where $\bar v_t = \frac{1}{t+1}\sum_{s\le t} g_s^2$.

(iii) The effective number of averaged gradients is

$$n_t = \Big(\sum_{s=0}^t w_{t,s}^2\Big)^{-1} = \frac{1+\beta_2}{1-\beta_2}\cdot\frac{1-\beta_2^{t+1}}{1+\beta_2^{t+1}}, \tag{9.8}$$

which increases from 1 to $(1+\beta_2)/(1-\beta_2) = 2/(1-\beta_2) - 1$. If the g_s^2

are independent with common mean σ^2 and variance ς^2, then $\mathbb{E}\hat{v}_t = \sigma^2$ and $\mathrm{Var}(\hat{v}_t) = \varsigma^2/n_t$.

PROOF. At $t = 0$, $\hat{m}_0 = g_0$ and $\hat{v}_0 = g_0^2$, which gives (i). Unrolling, $v_t = (1-\beta_2)\sum_{s\le t}\beta_2^{t-s}g_s^2$ and $\sum_{k\le t}\beta_2^k = (1-\beta_2^{t+1})/(1-\beta_2)$, which gives the weights and their sum; each numerator β_2^{t-s} lies in $[\beta_2^t, 1]$ and the denominator $\sum_{k\le t}\beta_2^k$ in $[(t+1)\beta_2^t, t+1]$, and Bernoulli's inequality $\beta_2^t \ge 1-t(1-\beta_2) \ge 1-\delta$ gives the comparison with $\bar{v}_t$. For (iii), $\sum_s w_{t,s}^2 = (1-\beta_2)^2(1-\beta_2^{2t+2})/\big((1-\beta_2^2)(1-\beta_2^{t+1})^2\big)$, which simplifies to $1/n_t$; the mean and variance follow from linearity and independence. ■

So Adam's first update has size η in every coordinate with a nonzero gradient, however small (Chapter 2), and for $\delta/(1-\beta_2)$ steps its preconditioner is within a factor $(1-\delta)^{-1/2}$ of the root mean square of the gradients seen so far. RAdam (Liu et al. 2020) runs momentum SGD for four steps,[6] then damps Adam's step by a factor of about $\sqrt{k(1-\beta_2)/2}$ after k gradients, for $5 \ll k \ll 1/(1-\beta_2)$ (Exercise 9.4).

[6] The four come from the test $\rho_k > 4$ of the paper, with ρ_k as in Exercise 9.4; some implementations test $\rho_k > 5$.

HEURISTIC DERIVATION: THE $2/(1-\beta_2)$ RULE. Suppose that the gradients of a coordinate in the first steps are independent draws from one distribution, and that a step is safe once the preconditioner is nearly as reliable as it will ever be. By Proposition 9.16(iii) the relative variance of $\hat{v}_t$ is proportional to $1/n_t$, and

$$\frac{n_t}{n_\infty} = \tanh\!\left(\frac{(t+1)\log(1/\beta_2)}{2}\right),$$

which reaches $\tanh 1 \approx 0.76$ when $t+1 = 2/\log(1/\beta_2) \approx 2/(1-\beta_2)$. A warmup that holds the step down until then lasts $2/(1-\beta_2)$ steps, the rule of thumb of Ma and Yarats (2021); RAdam's factor is a square-root warmup on the same time scale. ◇

This time, about the *effective sample size* n_∞ of the second moment, is 2000 steps for $\beta_2 = 0.999$ and 40 for $\beta_2 = 0.95$ (Chapter 6). LLaMA uses $\beta_2 = 0.95$ and warms up for 2000 steps, so long language-model warmups are not explained by Adam's statistics alone.

9.4.4 Warmup at the edge of stability

Assumption 9.4 makes the curvature follow the loss. In networks trained by gradient descent the causality also runs the other way: the sharpness rises to $2/\eta$ and stays there (Chapter 8). What does a rising η_t do to it?

HEURISTIC DERIVATION: THE SHARPNESS FOLLOWS $2/\eta_t$. Suppose that full-batch gradient descent with a slowly increasing schedule has reached the *edge of stability*, $S(\theta_t) = 2/\eta_t$, and that from then on the iterates follow the *central flow* of Chapter 8: in the time $\tau = \sum_s \eta_s$,

$$\frac{d\theta}{d\tau} = -\nabla\mathcal{L} - \tfrac{1}{2}\sigma^2\nabla S,$$

with σ^2 the mean square of the oscillation along the top eigenvector of the Hessian, chosen so that the flow stays on the edge. With $\dot{\eta} = \eta_{t+1} - \eta_t$,

the edge moves at the rate $d(2/\eta)/d\tau = -2\dot{\eta}/\eta^3$, and the flow moves the sharpness at the rate $\langle \nabla S, d\theta/d\tau \rangle = \alpha - \frac{1}{2}\sigma^2 \|\nabla S\|_2^2$, where $\alpha = -\langle \nabla \mathcal{L}, \nabla S \rangle$ is the rate of progressive sharpening. Equating the two,

$$\sigma^2 = \frac{2\,(\alpha + 2\dot{\eta}/\eta^3)}{\|\nabla S\|_2^2}.$$

The edge holds while $\sigma^2 > 0$, that is, while the threshold $2/\eta_t$ falls faster than the sharpness would fall under gradient flow, and then the sharpness follows $2/\eta_t$ down: the schedule, not the loss, sets the curvature. The oscillation costs an excess loss of about σ^2/η, which grows with the speed of the warmup; if σ^2 is not small, the cubic expansion behind the central flow fails, and the iterates catapult or diverge. On the warmup model $\alpha = -2\mu(S - K_0)$ and $\|\nabla S\|_2^2 = 2\mu K_1 (S - K_0)$, so on the edge the condition reads $\dot{\eta} > \mu\eta^2(2 - \eta K_0)$. ◇

If $\alpha > 0$, the edge, once reached, holds for every nondecreasing schedule, and after the warmup the sharpness stays at $2/\eta_{\max}$ (Chapter 8); if $\alpha < 0$, as on the warmup model, it holds only while the warmup outruns the flattening that gradient descent produces on its own. Either way the warmup lowers the curvature that the run will have at its peak learning rate, which is Observation 9.2. The idealizations agree on the direction and differ in the mechanism: in Theorem 9.9 the step waits for the loss to lower the curvature; at the edge the step lowers it, through the oscillation, and the loss pays.

9.5 Reading the theory

THE SHAPE AND THE LENGTH OF THE WARMUP. Corollary 9.10 predicts a hyperbolic rise that turns concave at $2/(3K_0)$ and saturates at $1/K_0$; on the warmup model, where the descent lemma is loose by its factor 2, μ becomes 2μ. Figure 9.1 (right) compares the profile with gradient descent on the full model for $K_0 = K_1 = 1$, $\mu = 0.02$ and $\Delta_0 = 50$: the step reaches $1/(2K_0)$ at step 1324, where the continuum predicts 1323.

WHAT THE WARMUP BUYS. On the same model the loss-indexed warmup reaches $\Delta \le 10^{-8}$ in 1805 steps, within the bound 1827 of Proposition 9.12(ii) (Theorem 9.9 guarantees 3617); riding the limit, $c = 2$, takes 901, against the lower bound 833 for every stable schedule. The constant step $1/S_0$ needs 28469 steps, and the best stable constant step, $2/S_0$, needs 14232, against the lower bound 14226. The ratio of the latter to the loss-indexed warmup, 7.9, is the theory's $(1 + u_0)\log(\Delta_0/\varepsilon)/(2(\log(\Delta_0/\varepsilon) + u_0)) \approx 7.9$, which grows with the excess curvature at initialization and with the accuracy required.

A LINEAR WARMUP IS THE WRONG SHAPE. Practice uses straight lines from zero, and the model says how long a straight line must be to stay stable.

HEURISTIC DERIVATION: THE SHORTEST STABLE LINEAR WARMUP. Take $\eta_t = t/(K_0 T_w)$ for $t \le T_w$ on the line $z = 0$ of the warmup model, and replace the recursion by its continuum $\dot{\Delta} = -2\mu\eta_t\Delta$, so that $\Delta_t = \Delta_0 \exp(-\mu t^2/(K_0 T_w))$.

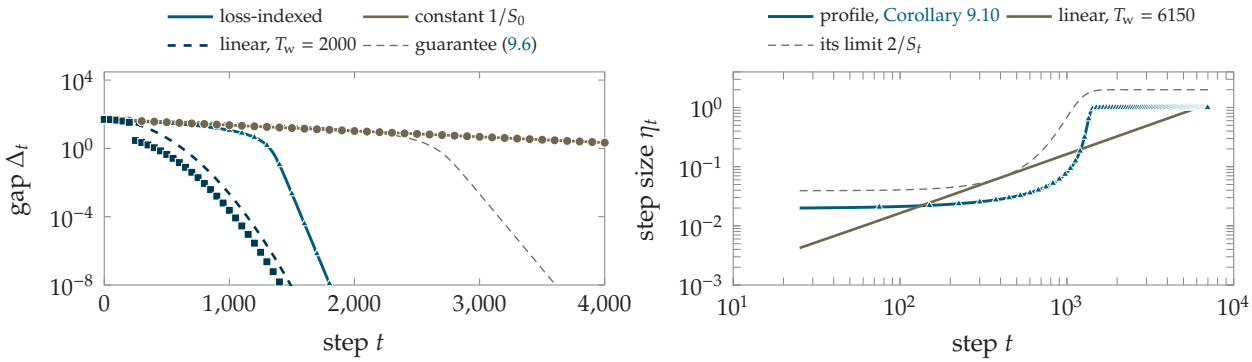


Figure 9.1. The warmup model with $K_0 = K_1 = 1$, $\mu = 0.02$ and $\Delta_0 = 50$ ($\kappa = u_0 = 50$), one slow and one stiff coordinate, the stiff one started at $z_0 = 10^{-6}$; no randomness (script `09-warmup-model.py`). Left: the gap under the loss-indexed warmup $\eta_t = 1/(K_0 + K_1\mathcal{L}(\theta_t))$, under the constant step $1/S_0 = 1/(K_0 + K_1\Delta_0)$, and under a linear warmup to $1/K_0$ over 2000 steps; curves are the recursion $\Delta_{t+1} = (1 - \mu\eta_t)^2\Delta_t$ of Proposition 9.7, points of the same color are gradient descent on the full model. The linear warmup crosses the edge $\eta_t S_t = 2$, its stiff coordinate grows, and a catapult lifts the loss to $6 \cdot 10^3$ (for six steps near $t = 230$, between the points shown) before it converges faster than the recursion. Dashed: the guarantee (9.6), conservative by the factor 2 of the descent lemma. Without warmup ($\eta = 1/K_0$) the loss exceeds 10^{13} after six steps. Right, on logarithmic axes: the step of the loss-indexed warmup (points) against the profile of Corollary 9.10 with 2μ for μ, and the shortest linear warmup to $1/K_0$ that never crosses the edge, which touches its stability limit at $t \approx 400$.

With $s = t/T_w$ and $\chi = \mu T_w/K_0$, the condition $\eta_t S_t \le 2$ reads $s + u_0 s e^{-\chi s^2} \le 2$ for $s \in [0, 1]$. The second term peaks at $s = 1/\sqrt{2\chi}$ with the value $u_0/\sqrt{2e\chi}$, and evaluating the condition there gives $(1 + u_0/\sqrt{e})/\sqrt{2\chi} \le 2$, that is,

$$T_w \ge \frac{K_0}{8\mu}\left(1 + \frac{u_0}{\sqrt{e}}\right)^2.$$

After the warmup S_t decreases, so the condition holds there too once $u_0 e^{-\chi} \le 1$, which is weaker if $u_0 \ge 4$. ◇

For $\kappa = u_0 = 50$ the formula gives 6133 steps and bisection on the exact recursion 6150; for u_0 from 5 to 200 they agree within 6%. A line from the origin must pass below the limit $2/S_t$, which stays near $2/S_0$ until the loss has fallen, so its stable length is quadratic in u_0, while the profile, which starts at the limit, has a length linear in u_0. The ratio is about $u_0/(4e)$, here 4.6, and the stable linear warmup reaches 10^{-8} in 2617 steps instead of 1805.

Shorter warmups and catapults. The theory is silent about schedules that cross the edge, and the model shows that they can win. In a scan of linear warmups to $1/K_0$ with spacing 10, almost all runs shorter than 1500 steps diverge. Between about 1500 and 3000 steps the run crosses the edge, a catapult lifts the loss to between 50 and $1.5 \cdot 10^4$, and the run converges in a number of steps that depends erratically on T_w, from 255 to 1813 with median 1380: all these runs beat the recursion of their schedule, almost

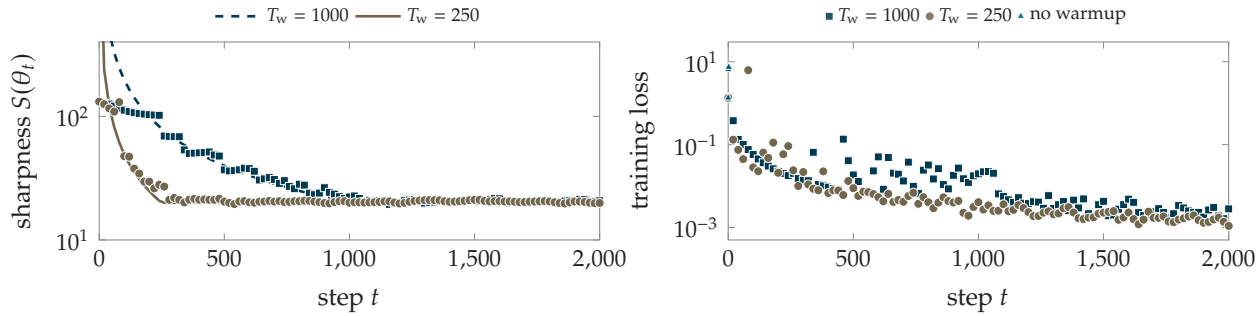


Figure 9.2. Warmup at the edge in a small network: a tanh network with three hidden layers of width 64 and output multiplier 5 (a sharp initialization), trained by full-batch gradient descent on the square loss with 256 Gaussian inputs in $\mathbb{R}^{10}$ labelled by a random tanh teacher, with a linear warmup to $\eta = 0.1$ over 250 or 1000 steps (seeds fixed in `09-warmup-mlp.py`). Left: the sharpness by power iteration with Hessian–vector products every 20 steps (points), and the prediction $2/\eta_t$ of Section 9.4.4 (curves of the same color), which holds once the edge is reached. Right: the training loss; without warmup it exceeds 10^9 after six steps.

all beat the loss-indexed warmup, and one in eight beats the lower bound 833 for stable schedules. From 3000 steps on the run follows the recursion, although it crosses the edge for every $T_w < 6150$: the factor $|1 - \eta_t S_t|$ nearly vanishes when $\eta_t S_t \approx 1$ and crushes the stiff coordinate, which has no time to grow back.

THE EDGE IN A NETWORK. Figure 9.2 trains a small network from a sharp initialization, $S_0 \approx 132$, to the peak learning rate 0.1, with threshold $2/\eta = 20$; before the edge, the sharpness falls slowly on its own. Once $2/\eta_t$ meets it, after about 190 steps of the long warmup and 50 of the short one, the sharpness follows $2/\eta_t$ down, as Section 9.4.4 predicts, within 0.90 to 1.23 times $2/\eta_t$ in the long warmup and, after the catapult, 0.96 to 1.35 in the short one; over the last 500 steps its means are 20.7 and 20.4. The short warmup meets the edge with a catapult: the loss jumps from 0.04 to 6 while the sharpness drops from 130 to 48. Warmups of 50 or 100 steps diverge, 150 suffice. The network holds the edge where the model overshoots it, presumably because each step of a network feeds its top eigendirection, while the model's stiff coordinate is fed only by its initial value.

9.6 What the theory misses

WHAT WAS IDEALIZED AWAY. A global inequality between curvature and loss, which in networks is measured along trajectories and fails on sublevel sets under rescaling symmetries (Remark 9.14); the Polyak–Łojasiewicz inequality, at best local; full batches, although the minibatch scaling rule is another reason to start small (Goyal et al. 2017); a fixed preconditioner, whereas Adam's relevant curvature is a preconditioned sharpness with threshold near $38/\eta$ for $\beta_1 = 0.9$ (Cohen et al. 2022); and normalization layers and weight decay, which make the effective learning rate an angular

speed far from equilibrium early in training (Chapter 13).

WHERE IT BREAKS. First, the causality runs the other way in networks: the sharpness at the end of a warmup is $2/\eta_{\max}$, set by the schedule rather than by the loss (Cohen et al. 2021; Kalra and Barkeshli 2024); in Figure 9.2, at the same loss of about 0.01, the sharpness is 21 after the short warmup and 50 during the long one, so no function of the loss bounds it tightly along both. Second, instability is not failure: constant steps above $2/S_0$ converge through a catapult in wide networks (Lewkowycz et al. 2020) and in the product model of volume 2; for logistic regression, large steps with a non-monotone loss converge faster than every stable step (Wu et al. 2024); and on the warmup model almost every warmup that survives a catapult beats the loss-indexed one. The lower bounds proved here concern schedules that never cross the edge. Third, the mechanism need not be curvature: with Pre-LN, transformers train without warmup (Xiong et al. 2020), and GI-Adam or a control of the angular update recovers much of warmup's benefit (Kalra and Barkeshli 2024; Kosson et al. 2024). Finally, practice fixes the length by rules of thumb (2000 steps in LLaMA, where $2/(1-\beta_2) = 40$ and $K_1\Delta_0/\mu$ is unknown): the theory reproduces the shape of the practice, not its numbers.

REPAIRS AND OPEN PROBLEMS. Alimisis et al. (2026) turn their analysis into a one-parameter warmup that matches a tuned linear warmup on language and vision models; Riabinin et al. (2026) derive a scheduler that sets the warmup length automatically and, in LLaMA pretraining, matches or beats the best tuned warmups; Liu et al. (2025), in an unrefereed preprint, bound the curvature by $K_0 + K_\rho\Delta^\rho$ and show that in some cases warmup makes gradient descent up to $\Theta(T)$ times faster than a non-increasing schedule (Exercise 9.6). GI-Adam, Pre-LN and update controls remove much of the need, each for its own reason. Open: the curvature–loss inequality along the trajectories of a network, even a two-layer one (Exercise 9.10); lower bounds over all schedules, including those that cross the edge, and the fastest schedule on the warmup model; warmup with gradient noise; and a preconditioned version for Adam.

9.7 Back in the wild

FIELD NOTES	
PRACTICE	Linear warmup over a few thousand steps (2000 for LLaMA, five epochs for large-batch ImageNet).
IDEALIZATION	Curvature bounded by the loss and the Polyak–Łojasiewicz inequality; the warmup model; Adam's averages; the central flow.
PROVED	A local descent lemma; a warmup pays the initial curvature once, a constant step at every scale; the exact profile; optimality among stable schedules; softmax curvature at most $2R^2\ell$; Adam's sign-like first step.

PREDICTS	A hyperbolic rise of length about $K_1\Delta_0/\mu$; stable linear ramps $u_0/(4e)$ times longer; the sharpness following $2/\eta_t$ at the edge (derived).
MISSES	The inequality is measured, not derived, and fails globally under rescaling symmetries; catapults; noise, preconditioning, normalization.
TAKEAWAY	Warm up until the curvature allows the peak step, longer for sharper initializations (proved in the idealization); index the warmup by the loss or the sharpness (suggested); long warmups at $\beta_2 = 0.95$ are not explained by Adam's statistics (derived).

The theory makes one firm statement and one suggestion. The statement, proved in the idealization, is that a constant step must be safe at the start and then pays for the initial curvature at every scale of accuracy, while a warmup pays once; the gain is large only when the initialization is much sharper than the trained network, which a power iteration at initialization can compare with $2/\eta_{\max}$. The suggestion is the shape: start at the largest step that the current curvature allows and rise as it falls, a hyperbola rather than a line from zero. The theory gives no length for a language-model warmup and does not forbid short ones: at the edge, and through catapults, networks lower their own curvature faster than the stable theory allows.

The chapter sits between the edge of stability of Chapter 8, which supplies the mechanism of Section 9.4.4, and the clipping of Chapter 10, which tames a curvature that grows with the gradient rather than with the loss. Warmup is the first phase of the schedules of Chapter 12, and the angular update controlled by Kosson et al. (2024) is the effective learning rate of Chapter 13.

Exercises

9.1 CONSOLIDATION. (The softmax constant.) (a) Show that $\lambda_{\max}(\mathrm{diag}(p) - pp^\top) \le 2\max_i p_i(1-p_i) \le 2(1-p_c)$ for every probability vector p and class c. (b) For $K = 2$ show that the top eigenvalue is $2p_c(1-p_c)$, so that the constant 2 of Proposition 9.13 cannot be lowered and $\lambda_{\max} \le 1 - p_c$ fails when $p_c > \frac{1}{2}$. (c) For $\ell(w) = \log(1 + e^{-y\langle w,a\rangle})$ show that $\nabla^2\ell = \sigma(m)\sigma(-m)\,aa^\top \le \ell(w)\,aa^\top$, with $m = y\langle w,a\rangle$ and σ the logistic function.

9.2 CONSOLIDATION. (The profile.) (a) Show that the D_t of Theorem 9.9(ii) satisfy $D_t \ge D(t/(1-\mu/K_0))$ if $\mu < K_0$. *(Hint: bound* $\Phi(D_t) - \Phi(D_{t+1})$ *by* $\Phi'(D_{t+1})(D_t - D_{t+1}) \le \mu D_t/D_{t+1}$.*)* (b) Show that $\eta(t)$ is convex while $\eta(t) < 2/(3K_0)$ and concave afterwards. *(Hint: with* $u = K_1 D/K_0$, $\dot u = -(\mu/K_0)u/(1+u)$, *and* $\ddot\eta$ *has the sign of* $2u - 1$.*)* (c) For $K_0 = K_1 = 1$, $\mu = 0.01$, $\Delta_0 = 100$, find when η reaches $0.9/K_0$.

9.3 CONSOLIDATION. ★ (The hypothesis $K_0 \ge 2\mu$.) Show that the warmup model satisfies $\lambda_{\max}(\nabla^2\mathcal{L}) \le K_0 + K_1\mathcal{L}$ everywhere if and only if $K_0 \ge 25\mu/16$. *(Hint: let* $\varepsilon \to 0$ *in the proof of Proposition 9.7;* $(S-\mu)S \ge 4\mu K_1 q$ *for all* $q \ge 0$ *is quadratic in* $K_1 q$, *with discriminant* $25\mu^2 - 16K_0\mu$.*)*

9.4 CONSOLIDATION. (Adam's effective sample size.) (a) Derive the formula for n_t/n_∞ behind the $2/(1-\beta_2)$ rule. (b) Show that RAdam's $\rho_k = \rho_\infty - 2k\beta_2^k/(1-\beta_2^k)$, $\rho_\infty = 2/(1-\beta_2) - 1$, is the length of the uniform average with the same mean age as Adam's weights $w_{k-1,s}$, that $\rho_\infty = n_\infty$, and that ρ_k/k and n_{k-1}/k tend to 1 as $k(1-\beta_2) \to 0$. (c) With $\epsilon = 0$, show that every step has size η if the gradient is constant, and that the second has size $\eta(1-\beta_1)/(1+\beta_1)$ if $g_1 = -g_0$.

9.5 VARIATION. (No curvature floor: the exponential loss.) Let $\mathcal{L}(w) = \frac{1}{n}\sum_i \exp(-y_i\langle w, a_i\rangle)$ with $\|a_i\|_2 \le R$ and data separable with margin γ: some unit vector $\bar{w}$ has $y_i\langle \bar{w}, a_i\rangle \ge \gamma$ for all i. (a) Show that $\lambda_{\max}(\nabla^2\mathcal{L}(w)) \le R^2\mathcal{L}(w)$ and $\|\nabla\mathcal{L}(w)\|_2 \ge \gamma\mathcal{L}(w)$. (b) Show that gradient descent with $\eta_t = 1/(R^2\mathcal{L}(w_t))$ satisfies $\mathcal{L}(w_{t+1}) \le (1 - \gamma^2/(2R^2))\mathcal{L}(w_t)$, with steps that grow geometrically. (c) Contrast with a constant step, under which the loss decays only polynomially (Chapter 15).

9.6 VARIATION. (Curvature that grows like a power of the gap.) Suppose that $S \le K_0 + K_\rho\Delta^\rho$ on Θ_0, the form of Liu et al. (2025), with $0 < \rho \le 1$, and Assumption 9.5. Show that the step $1/(K_0 + K_\rho\Delta_t^\rho)$ reaches ε within $\big(K_0\log(\Delta_0/\varepsilon) + K_\rho(\Delta_0^\rho - \varepsilon^\rho)/\rho\big)/\mu$ steps. Where is $\rho \le 1$ used?

9.7 VARIATION. (Normalized steps.) (a) Prove the claims of Remark 9.15. *(Hint: the length is admissible because $c\Delta_t^\nu \le \|\nabla\mathcal{L}(\theta_t)\|_*$, and the logarithmic derivative of $\Delta^\nu/K(\Delta)$ has the sign of $\nu K - \Delta K'$.)* (b) On the line $z = 0$ of the warmup model, show that the step length $\eta_t\|\nabla\mathcal{L}(\theta_t)\|_2$ of the loss-indexed warmup rises until $K_1\Delta_t = K_0$ and falls afterwards. (c) Show that sign descent on the stiff coordinate, $z_{t+1} = z_t - h\,\mathrm{sign}(z_t)$, never diverges and eventually adds at most $\frac{1}{2}S(x)h^2$ to the loss.

9.8 SIMULATION. (The warmup model.) With `09-warmup-model.py`: (a) check the heuristic formula for the shortest stable linear warmup for u_0 from 5 to 200; (b) scan the warmup length and locate the window of catapults (T_w between about 1500 and 3000 by default); (c) run the scheduled warmup of Theorem 9.9(ii) and the loss-indexed one with $c = 2$, and explain their step counts by Proposition 9.12.

9.9 SIMULATION. (A network at the edge.) With `09-warmup-mlp.py`: (a) check that $\eta_t S(\theta_t)$ stays near 2 once the edge is reached; (b) find the shortest warmup that trains for the peaks 0.05 and 0.2 (it is between 100 and 150 steps for 0.1); (c) set the output multiplier to 1 ($S_0 \approx 5.7$), the peak to 0.2 and T_w to 250, and observe progressive sharpening: the sharpness rises to $2/\eta_{\max} = 10$ about a thousand steps after the warmup ends.

9.10 RESEARCH. (The inequality for a two-layer network.) Remark 9.14 rules out Assumption 9.4 on sublevel sets of networks with rescaling symmetries. Find conditions (small or balanced initializations, or along trajectories) under which a two-layer network satisfies $S \le K_0 + K_1\Delta$ with explicit constants; we know of none.

Notes and References

Early warmups sped up runs that were slow to start (He et al. 2016); warmup became a linear ramp for large minibatches (Goyal et al. 2017) and Transformers (Vaswani et al. 2017), and took its modern form in GPT-3 and LLaMA (Brown et al. 2020; Touvron et al. 2023). The edge of stability is due to Cohen et al. (2021), the central flow to Cohen et al. (2025), the catapult to Lewkowycz et al. (2020), and Adam's edge to Cohen et al. (2022).

The theory of warmup under loss-dependent curvature is recent: Liu et al. (2025), in a preprint, bound the curvature by a power of the gap, Alimisis et al. (2026) by an affine function of it, and Riabinin et al. (2026) by both, in a general norm. As far as we know, the warmup model, the Wright-omega profile, Proposition 9.12 and the shortest stable linear warmup are new.

TRANSLATING THE LITERATURE. Alimisis et al. (2026) write f for $\mathcal{L}$, w for θ, f^* for $\mathcal{L}^\star$ and H_0, H_1 for K_0, K_1, and bound $\|\nabla^2 f(w)\|$ by $H_0 + H_1(f(w) - f^*)$; their warmup step is $\theta\eta_k$, with θ the constant of their aiming condition, not the parameters. Riabinin et al. (2026) write x for θ and bound the dual norm of gradient differences by $K(x)\|x - y\|$ with $K(x) = K_0 + K_1(f(x) - f^\star) + K_\rho(f(x) - f^\star)^\rho$; with $K_\rho = 0$ and the Euclidean norm their assumption reduces to that of Alimisis et al. Liu et al. (2025) call the bound $K_0 + K_\rho(f(w) - f^*)^\rho$ on the local curvature (ρ, K_0, K_ρ)-smoothness.

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10 Clipping and Spikes

Generalized smoothness and the anatomy of an instability

IN THE WILD — GRADIENT CLIPPING, Z-LOSS, QK-NORM

IN THIS CHAPTER. Large language models are trained with the gradient norm clipped at 1, a penalty on the normalizer of the output softmax and normalized queries and keys, and their losses still spike. Under (L_0, L_1)-smoothness, curvature bounded by L_0 plus L_1 times the gradient norm, we prove that clipped steepest descent reaches a gradient of size ε within $\max\{4L_0\Delta_0/\varepsilon^2, 6L_1\Delta_0/\varepsilon\}$ steps while every fixed step size pays almost linearly for the largest gradient on the way, and that normalized updates are clipped by construction. We then take three instabilities apart exactly: a block of weights that wakes up after a quiet period receives Adam steps that peak above $0.2/\sqrt{1-\beta_2}$ learning rates when $\beta_1 = 0.9$; the cross-entropy has a flat direction along which adaptive updates drift and which the z-loss removes; attention logits grow with the square of the weight scale unless queries and keys are normalized. The theory explains why each defense helps; it does not predict when a spike comes, and real spikes have causes it leaves out.

10.1 In the wild: clipped gradients and spiking losses

As of 2025, most published recipes for large language models rescale the gradient before using it: if the Euclidean norm of the gradient, computed over all parameters at once, exceeds a threshold, then the gradient is multiplied by the threshold over the norm. The rule comes from recurrent networks, whose losses have *cliffs*, thin regions where the gradient and the curvature are both enormous, so that one gradient step throws the parameters far away.[1] Transformers inherited the rule, with a growing list of defenses against the instabilities that clipping does not prevent.

IN THE WILD · GPT-3, Brown et al. (2020); LLaMA, Touvron et al. (2023); PaLM, Chowdhery et al. (2023); ViT-22B, Dehghani et al. (2023); Kimi K2, Kimi Team (2025). Clip the gradient norm at 1.0 before the optimizer step (the global norm in GPT-3 and PaLM; also LLaMA). Use Adam with $\beta_2 = 0.95$ rather than the default 0.999 (GPT-3, LLaMA). Add $10^{-4}(\log Z)^2$ to the cross-entropy, where Z is the normalizer of the softmax over the vocabulary (the *z-loss* of PaLM). Apply a LayerNorm to the queries and the keys before their dot product (*QK-norm*, ViT-22B). With Muon, after each step rescale the query and key weights of every

[1] Pascanu et al. (2013) proposed clipping the norm of the gradient of recurrent networks and explained the need geometrically, building on the analysis of exploding gradients by Bengio et al. (1994).

attention head whose largest logit on the batch exceeds a threshold, 100 for Kimi K2 (*MuonClip*). When the loss spikes anyway, restart from a checkpoint just before the spike and skip the nearby batches (PaLM).

The values pass from recipe to recipe; we know no derivation of the threshold 1.0 or the coefficient 10^{-4}, so both are folklore. Here is what has been observed.

Observation 10.1 (Loss spikes; Chowdhery et al. 2023). While the 540-billion-parameter PaLM model was trained, with the gradient norm clipped at 1.0, the loss spiked roughly 20 times, at irregular intervals and sometimes late in training. Restarting from a checkpoint about 100 steps before a spike and skipping about 200 to 500 batches avoided it. Replaying the batches around a spike from a different, earlier checkpoint did not produce a spike.

So the spike lay in neither the data nor the parameters alone, but in their combination; the report is qualitative and concerns one run. OPT's divergences coincided with a collapse of the dynamic loss scale of mixed precision (Zhang et al. 2022).

Observation 10.2 (Curvature grows with the gradient; Zhang et al. 2020b). Along the training trajectory of an LSTM language model, a local smoothness estimated from the change of the gradient between nearby iterates was positively correlated with the norm of the gradient: where the gradient was large, so was the curvature.

The evidence is a correlation along the trajectories of a few networks, not a law; if it holds, then the step size $1/L$ allowed by a global curvature bound is wasted almost everywhere.

Observation 10.3 (Two instabilities in small models; Wortsman et al. 2024). Small transformers trained at high learning rates reproduce two instabilities first reported for large runs: the growth of the attention logits, which makes the attention weights nearly one-hot (Dehghani et al. 2023), and the divergence of the output logits from log-probabilities (Chowdhery et al. 2023). QK-norm removes the first, and the z-loss the second.

Observation 10.4 (Spikes follow a stale second moment; Wortsman et al. 2023). In training large CLIP models with AdamW, loss spikes followed, within 1 to 8 iterations, steps at which the squared gradients were much larger than AdamW's running estimate of them, mainly in the patch-embedding layer: the second-moment estimate was out of date.

From language models of 7 to 546 billion parameters, Molybog et al. (2023) (unrefereed) argue that Adam can enter a state in which its update is large and nearly uncorrelated with the direction of descent.

Each instability is a step too long for the curvature where it lands, made too long by a quantity that grew without the loss registering it: the gradient near a cliff, the lag of Adam's estimate of the gradient's scale, a common shift of the output logits, the size of queries and keys. Why can a fixed step size not cope with curvature that grows with the gradient, what does clipping buy, and how does a well-tuned run take a step that is far too long?

We answer the first two questions with theorems and the third with three exact mechanisms; we do not predict when a spike happens.

10.2 The idealization: curvature that grows with the gradient

Classical smoothness bounds the curvature by one number everywhere. On a cliff that number is huge, and the step size it allows is wasted on the rest of the landscape. Zhang et al. (2020b) let the bound grow with the gradient.[2]

Assumption 10.5 ((L_0, L_1)-smoothness). Let $\|\cdot\|$ be a norm on $\mathbb{R}^d$ with dual norm $\|\cdot\|_*$, and for a symmetric matrix H let $\|H\| = \max_{\|u\|\le 1}\|Hu\|_*$. The objective $\mathcal{L}\colon \mathbb{R}^d \to \mathbb{R}$ is twice continuously differentiable and bounded below, $\mathcal{L}^\star = \inf \mathcal{L} > -\infty$, and there are constants $L_0 \ge 0$ and $L_1 > 0$ such that

$$\|\nabla^2 \mathcal{L}(\theta)\| \le L_0 + L_1 \|\nabla \mathcal{L}(\theta)\|_* \qquad \text{for every } \theta \in \mathbb{R}^d. \tag{10.1}$$

(L_0, L_1)-*smoothness* allows curvature that grows without bound, provided the gradient grows with it: e^x is $(0, 1)$-smooth, $\cosh x - 1$ is $(1, 1)$-smooth, the logistic loss $\log(1 + e^{-x})$ is $(0, 1)$-smooth and $\frac{1}{4}$-smooth, x^4 is (L_0, L_1)-smooth exactly when $L_0 L_1^2 \ge 16$, and a quadratic with Hessian H is $\|H\|$-smooth and (L_0, L_1)-smooth only if $L_0 \ge \|H\|$ (Exercise 10.1). The extremal example is the *exponential cliff*

$$\mathcal{L}_{\text{cliff}}(x) = \frac{L_0}{L_1^2}\left(e^{L_1|x|} - 1 - L_1|x|\right), \qquad x \in \mathbb{R}, \tag{10.2}$$

with $|\mathcal{L}'_{\text{cliff}}(x)| = \frac{L_0}{L_1}(e^{L_1|x|} - 1)$ and $\mathcal{L}''_{\text{cliff}} = L_0 e^{L_1|x|} = L_0 + L_1|\mathcal{L}'_{\text{cliff}}|$: a valley of curvature L_0 whose walls steepen exponentially, the shape of the exploding gradients of recurrent networks (Bengio et al. 1994; Pascanu et al. 2013).

THE METHOD. Write $g^\sharp = -\|g\|_* \operatorname{lmo}(g)$ for the sharp operator (Chapter 1; Lemma A.5), so that $\langle g, g^\sharp\rangle = \|g\|_*^2$ and $\|g^\sharp\| = \|g\|_*$. *Clipped steepest descent* with step size η (the learning rate; we use the two words as synonyms) and largest step length γ is

$$\theta_{t+1} = \theta_t - \eta_t\, g_t^\sharp, \qquad g_t = \nabla \mathcal{L}(\theta_t), \qquad \eta_t = \min\left\{\eta, \frac{\gamma}{\|g_t\|_*}\right\}, \tag{10.3}$$

with $\eta_t = \eta$ when $g_t = 0$. Its step has length $\min\{\eta\|g_t\|_*, \gamma\}$: below the threshold $\|g_t\|_* = \gamma/\eta$ it is unnormalized steepest descent with step size η, above it normalized steepest descent $\theta_{t+1} = \theta_t + \gamma \operatorname{lmo}(g_t)$. In the Euclidean norm $g^\sharp = g$, and (10.3) is the rule of the practice box, $\theta_{t+1} = \theta_t - \eta \operatorname{clip}_\tau(g_t)$ with $\operatorname{clip}_\tau(g) = \min\{1, \tau/\|g\|_2\} g$ and threshold $\tau = \gamma/\eta$.[3] Clipping is a trust region: the clipped step minimizes $\langle g, u\rangle + \|u\|_2^2/(2\eta)$ over $\|u\|_2 \le \gamma$.

[2] They wrote the condition in the Euclidean norm. A general norm costs nothing in the proofs and is what Proposition 10.10 needs: sign descent and Muon are steepest descent in the ℓ_∞ and spectral norms (Chapters 2 and 3).

[3] Clipping each layer separately is the trust-region step for a product of balls, one per layer, the unit ball of a max norm (Chapter 4); clipping *by value*, coordinate by coordinate, is the trust-region step for a box, and it changes the direction of the step (Exercise 10.3).

DICTIONARY

IN THE THEORY	IN A TRAINING RUN
(L_0, L_1)-smoothness (10.1)	curvature largest where the gradient is: cliffs, the first steps, spikes
L_0; $L_1\|\nabla\mathcal{L}\|_*$; M	the curvature of the flat part, which sets the useful learning rate; the extra curvature near a cliff; the largest gradient norm the run meets
step size η; step length γ; γ/η	learning rate; learning rate times threshold; the threshold `max_norm`
a silent, then loud coordinate; a_k	an embedding row, head or expert whose gradient was near zero and suddenly is not; its Adam update in learning rates
flat direction $\mathbf{1}$; $\alpha(\log Z)^2$	a common shift of all output logits; the z-loss, $\alpha = 10^{-4}$
logit s; gains ω_q, ω_k	an attention logit; the scales in QK-norm

WHAT WAS THROWN AWAY. Noise, except in Theorem 10.11; the network, except for one relation between its curvature and its gradient, with fixed constants that we cannot compute; and finite precision. The failure of fixed step sizes is already present without them, which makes these the right first simplifications.

WHAT ONE HAD TO NOTICE. Classical analyses take a global smoothness constant as given. Once the curvature is tied to the gradient, as Zhang et al. (2020b) saw it is along trajectories, a step whose *length* is bounded is safe everywhere, and bounding the length of the step is what clipping does. For spikes, each defense targets a quantity whose growth the loss cannot see.

10.3 The theory: clipping under generalized smoothness

Everything rests on one lemma: under (10.1) the gradient changes at most exponentially along a segment, at a rate set by L_1.

Lemma 10.6 (Local smoothness). *Let $\mathcal{L}$ satisfy Assumption 10.5, let $\theta, u \in \mathbb{R}^d$, and put $g = \nabla\mathcal{L}(\theta)$ and $r = L_1\|u\|$. Then*

$$\|\nabla\mathcal{L}(\theta+u) - g\|_* \le \big(L_0/L_1 + \|g\|_*\big)\big(e^r - 1\big), \tag{10.4}$$

$$\big|\mathcal{L}(\theta+u) - \mathcal{L}(\theta) - \langle g, u\rangle\big| \le \big(L_0/L_1 + \|g\|_*\big)\big(e^r - 1 - r\big)/L_1. \tag{10.5}$$

Consequently, for $c > 0$, with $\varphi_1(c) = (e^c - 1)/c$ and $\varphi_2(c) = (e^c - 1 - c)/c^2$, if $r \le c$, then

$$\begin{aligned} \|\nabla\mathcal{L}(\theta+u) - g\|_* &\le \varphi_1(c)\,(L_0 + L_1\|g\|_*)\|u\|, \\ \mathcal{L}(\theta+u) &\le \mathcal{L}(\theta) + \langle g, u\rangle + \varphi_2(c)\,(L_0 + L_1\|g\|_*)\|u\|^2. \end{aligned} \tag{10.6}$$

For $c = \frac{1}{3}$, $\varphi_1(c) < 1.19$ and $\varphi_2(c) < \frac{3}{5}$. Both (10.4) and (10.5) hold with equality for the cliff (10.2) at $\theta \ge 0$ and $u \ge 0$.

As $c \to 0$, $\varphi_1 \to 1$ and $\varphi_2 \to \frac{1}{2}$, and (10.6) becomes the descent lemma (Lemma A.18) for the local smoothness constant $L_0 + L_1\|g\|_*$: that constant may be used as long as the step is short compared with $1/L_1$.[4]

[4] These are the functions φ_1, φ_2 of exponential integrators for stiff differential equations, where they arise for the same reason: they integrate the exponential growth of a linear equation over one step.

PROOF. The plan is to bound the growth of the gradient along the segment from θ to $\theta + u$ by a linear differential inequality, solve it with Grönwall's inequality, and integrate. Let $u \neq 0$, the case $u = 0$ being empty. For $s \in [0,1]$ let $\phi(s) = \nabla\mathcal{L}(\theta + su)$, whose derivative $\phi'(s) = \nabla^2\mathcal{L}(\theta + su)u$ is continuous, and $\psi(s) = \int_0^s \|\phi'(\sigma)\|_* \,\mathrm{d}\sigma$, a continuously differentiable function with $\psi(0) = 0$ and $\|\phi(s) - g\|_* \le \psi(s)$. By (10.1) at $\theta + su$, the derivative of ψ is bounded by the current gradient, and the current gradient by $\|g\|_* + \psi(s)$:

$$\psi'(s) \le \big(L_0 + L_1\|\phi(s)\|_*\big)\|u\| \le a + r\,\psi(s), \qquad a = (L_0 + L_1\|g\|_*)\|u\|.$$

Hence $(e^{-rs}\psi(s))' \le a e^{-rs}$, and integrating from 0 to s gives $\psi(s) \le \frac{a}{r}(e^{rs} - 1)$, the differential form of Grönwall's inequality (Theorem A.17). Since $a/r = L_0/L_1 + \|g\|_*$, the case $s = 1$ is (10.4). For (10.5), write the remainder of the first-order expansion as an integral and insert the bound on ψ:

$$\left|\int_0^1 \langle \phi(s) - g, u\rangle \,\mathrm{d}s\right| \le \|u\| \int_0^1 \psi(s)\,\mathrm{d}s \le \frac{a}{r}\|u\|\left(\frac{e^r - 1}{r} - 1\right),$$

which is the right side of (10.5). The series $\varphi_1(c) = \sum_{k\ge 0} c^k/(k+1)!$ and $\varphi_2(c) = \sum_{k\ge 0} c^k/(k+2)!$ have positive coefficients, so both functions increase on $(0,\infty)$; for $r \le c$ this gives $e^r - 1 \le r\varphi_1(c)$ and $e^r - 1 - r \le r^2\varphi_2(c)$, and $r = L_1\|u\|$ turns the two bounds into (10.6). Numerically, $\varphi_1(\frac{1}{3}) = 3(e^{1/3} - 1) = 1.187$, and $\varphi_2(\frac{1}{3}) = 9(e^{1/3} - \frac{4}{3}) < 9(\frac{7}{5} - \frac{4}{3}) = \frac{3}{5}$ because $e < (\frac{7}{5})^3 = 2.744$. For the cliff at $\theta \ge 0$ and $u \ge 0$, both sides of (10.4) equal $\frac{L_0}{L_1}e^{L_1\theta}(e^{L_1 u} - 1)$, and integrating in u gives equality in (10.5). ∎

With the lemma, the main theorem is one application per step.

Theorem 10.7 (Clipped steepest descent). *Let $\mathcal{L}$ satisfy Assumption 10.5, let $\Delta_0 = \mathcal{L}(\theta_0) - \mathcal{L}^\star$, and run (10.3) with $0 < \gamma \le 1/(3L_1)$ and $0 < \eta \le 1/(2L_0)$ (every $\eta > 0$ if $L_0 = 0$). Then:*

(i) $\mathcal{L}(\theta_{t+1}) \le \mathcal{L}(\theta_t) - \frac{1}{2}\min\{\eta\|g_t\|_*^2,\ \gamma\|g_t\|_*\}$ *for every t;*

(ii) for every $\varepsilon > 0$, the number of steps with $\|g_t\|_* > \varepsilon$ is at most $2\Delta_0 \max\{1/(\eta\varepsilon^2), 1/(\gamma\varepsilon)\}$. If $L_0 > 0$, then with $\eta = 1/(2L_0)$ and $\gamma = 1/(3L_1)$ it is at most

$$\max\left\{\frac{4L_0\Delta_0}{\varepsilon^2},\ \frac{6L_1\Delta_0}{\varepsilon}\right\}, \tag{10.7}$$

which is $4L_0\Delta_0/\varepsilon^2$ whenever $\varepsilon \le 2L_0/(3L_1)$.

PROOF. The plan is to apply Lemma 10.6 to one step, whose length is at most γ, and to check that the quadratic remainder costs at most half of the first-order decrease. If $g_t = 0$, then the step is zero and (i) holds. Otherwise put

$u = -\eta_t g_t^\sharp$, so that $\langle g_t, u\rangle = -\eta_t \|g_t\|_*^2$ and $\|u\| = \eta_t \|g_t\|_* = \min\{\eta\|g_t\|_*, \gamma\}$. Since $L_1\|u\| \le \frac{1}{3}$, (10.6) with $c = \frac{1}{3}$ gives

$$\begin{aligned}\mathcal{L}(\theta_{t+1}) &\le \mathcal{L}(\theta_t) - \eta_t\|g_t\|_*^2 + \tfrac{3}{5}\big(L_0 + L_1\|g_t\|_*\big)\eta_t^2\|g_t\|_*^2 \\ &= \mathcal{L}(\theta_t) - \eta_t\|g_t\|_*^2\Big(1 - \tfrac{3}{5}\big(L_0\eta_t + L_1\|u\|\big)\Big).\end{aligned}$$

The two terms in the inner parenthesis are controlled by the two parameters: $L_0\eta_t \le L_0\eta \le \frac{1}{2}$ and $L_1\|u\| \le \frac{1}{3}$. The large parenthesis is therefore at least $1 - \frac{3}{5}\cdot\frac{5}{6} = \frac{1}{2}$, and the decrease is at least $\frac{1}{2}\eta_t\|g_t\|_*^2 = \frac{1}{2}\min\{\eta\|g_t\|_*^2, \gamma\|g_t\|_*\}$, which is (i).

For (ii), sum (i) over $t < T$; the decreases add up to at most $\mathcal{L}(\theta_0) - \mathcal{L}(\theta_T) \le \Delta_0$. Since $x \mapsto \eta x^2$ and $x \mapsto \gamma x$ increase, each step with $\|g_t\|_* > \varepsilon$ contributes more than $\frac{1}{2}\min\{\eta\varepsilon^2, \gamma\varepsilon\}$, so there are at most $2\Delta_0 \max\{1/(\eta\varepsilon^2), 1/(\gamma\varepsilon)\}$ of them, whatever T is. Substituting $\eta = 1/(2L_0)$ and $\gamma = 1/(3L_1)$ gives (10.7), whose first entry is the larger exactly when $\varepsilon \le 2L_0/(3L_1)$. ∎

The bound does not contain the largest gradient met on the way: a clipped step on a cliff is short, but it decreases the loss by $\frac{1}{2}\gamma\|g_t\|_*$, more than an unclipped step near the target does. For a fixed step size the largest gradient is the whole story. Let $M = \sup\{\|\nabla\mathcal{L}(\theta)\|_* : \mathcal{L}(\theta) \le \mathcal{L}(\theta_0)\}$.

Corollary 10.8 (Gradient descent with a step set by M). *Under Assumption 10.5, let $M < \infty$ and $0 < \eta \le \min\{1/(2L_0), 1/(3L_1M)\}$. Then $\theta_{t+1} = \theta_t - \eta g_t^\sharp$ stays in the sublevel set and has $\|g_t\|_* > \varepsilon$ for at most $2\Delta_0/(\eta\varepsilon^2)$ steps, for every $\varepsilon \le M$; with the largest allowed η this is $2\Delta_0 \max\{2L_0, 3L_1M\}/\varepsilon^2$.*

Proof. Put $\gamma = \eta M \le 1/(3L_1)$. As long as θ_t is in the sublevel set, $\|g_t\|_* \le \gamma/\eta$, so the step is the clipped step, and Theorem 10.7(i) gives a decrease of at least $\frac{\eta}{2}\|g_t\|_*^2$, which keeps θ_{t+1} in the sublevel set; by induction this holds for every t, and the count follows as in Theorem 10.7(ii), since $\gamma\varepsilon \ge \eta\varepsilon^2$. ∎

So gradient descent works on the class with a step size of order $1/(L_0 + L_1M)$ and then needs of order $(L_0 + L_1M)\Delta_0/\varepsilon^2$ steps. Up to a logarithm, no fixed step size does better.

Theorem 10.9 (A fixed step pays for the largest gradient). *Let $L_0, L_1, M, \Delta_0 > 0$ and $0 < \varepsilon \le M/2$, with $\Delta_0 \ge M/L_1$, and fix $x_0 \in \mathbb{R}$. Let $\mathcal{F}$ be the set of twice continuously differentiable, (L_0, L_1)-smooth functions $f : \mathbb{R} \to \mathbb{R}$ that are bounded below, with $f(x_0) - \inf f \le \Delta_0$ and $|f'(x)| \le M$ whenever $f(x) \le f(x_0)$. For every step size $\eta > 0$ there is an $f \in \mathcal{F}$ on which gradient descent $x_{t+1} = x_t - \eta f'(x_t)$ has $|f'(x_t)| > \varepsilon$ for every $t < N$, with*

$$N = \frac{L_1M}{\log(1 + L_1M/L_0)} \cdot \frac{\Delta_0 - 4\varepsilon^2/(\sqrt{3}L_0)}{8\varepsilon^2}. \tag{10.8}$$

If $\eta \ge \eta^* = 2\log(1 + L_1M/L_0)/(L_1M)$, then f can be chosen so that

$|f'(x_t)| \geq M$ *for every* t.

When $\varepsilon^2 \leq L_0\Delta_0/4$, the second factor of (10.8) is at least $\Delta_0/(20\varepsilon^2)$; the first factor exceeds L_0, because $\log(1+P) < P$, and is unbounded in M. As functions of the first coordinate, the functions of the proof give the same bound in $\mathbb{R}^d$ with the Euclidean norm.

PROOF. The plan is a dichotomy: a large step diverges on a cliff, a small step crawls on a plateau of slope just above ε. Put $P = L_1M/L_0$ and $y_0 = \log(1+P)$, so that $\eta^* = 2y_0/(L_1M)$.

A large step diverges on the cliff. Let $\eta \geq \eta^*$ and $f(x) = \mathcal{L}_{\text{cliff}}(x-c)$ with $c = x_0 - y_0/L_1$. Then f is (L_0, L_1)-smooth, $|f'(x_0)| = \frac{L_0}{L_1}(e^{y_0}-1) = M$, the sublevel set $\{f \leq f(x_0)\}$ is the interval $|x-c| \leq y_0/L_1$, on which $|f'| \leq M$, and $f(x_0) - \inf f = \frac{L_0}{L_1^2}(P - \log(1+P)) < M/L_1 \leq \Delta_0$; so $f \in \mathcal{F}$. In the variable $z_t = L_1(x_t - c)$, gradient descent reads $z_{t+1} = z_t\big(1 - \eta L_0 \varphi_1(|z_t|)\big)$ with the increasing function φ_1 of Lemma 10.6. At $t=0$, $\eta L_0\varphi_1(y_0) = \eta L_0 P/y_0 = \eta L_1 M/y_0 \geq 2$, hence $|z_1| = |z_0|(\eta L_0\varphi_1(|z_0|) - 1) \geq |z_0|$. Since φ_1 increases, the inequality $\eta L_0\varphi_1(|z_t|) \geq 2$ propagates, and by induction $|z_t| \geq y_0$, that is, $|f'(x_t)| \geq M > \varepsilon$, for every t.

A small step crawls on the plateau. Let $\eta < \eta^*$. If $\Delta_0 \leq 4\varepsilon^2/(\sqrt{3}L_0)$, then $N \leq 0$ and there is nothing to prove, so assume the opposite. With $\sigma = L_0 x/(2\varepsilon)$, let

$$\mathcal{L}_{\text{plat}}(x) = \frac{4\varepsilon^2}{L_0}\big(\sqrt{1+\sigma^2} - 1\big), \qquad \text{so that} \qquad \mathcal{L}'_{\text{plat}}(x) = \frac{2\varepsilon\,\sigma}{\sqrt{1+\sigma^2}}.$$

Then $|\mathcal{L}'_{\text{plat}}| < 2\varepsilon \leq M$ and $0 < \mathcal{L}''_{\text{plat}} = L_0(1+\sigma^2)^{-3/2} \leq L_0$, so $f(x) = \mathcal{L}_{\text{plat}}(x-c)$ is (L_0, L_1)-smooth; choose $c < x_0$ with $f(x_0) = \Delta_0$, which puts f in $\mathcal{F}$. With $a = L_0\Delta_0/(4\varepsilon^2)$, solving $f(x_0) = \Delta_0$ gives $x_0 - c = \frac{2\varepsilon}{L_0}\sqrt{(1+a)^2 - 1} \geq \frac{2\varepsilon}{L_0}a = \frac{\Delta_0}{2\varepsilon}$. On the other hand $|f'(x)| > \varepsilon$ exactly when $|x-c| > r_\varepsilon = 2\varepsilon/(\sqrt{3}L_0)$. Each step has length $\eta|f'(x_t)| < 2\eta\varepsilon$, so $|x_t - c| \geq x_0 - c - 2\eta\varepsilon t$, and $|f'(x_t)| > \varepsilon$ for every $t < (x_0 - c - r_\varepsilon)/(2\eta\varepsilon)$. This number of steps exceeds N:

$$\frac{x_0 - c - r_\varepsilon}{2\eta\varepsilon} \geq \frac{\Delta_0 - 4\varepsilon^2/(\sqrt{3}L_0)}{4\eta\varepsilon^2} > \frac{\Delta_0 - 4\varepsilon^2/(\sqrt{3}L_0)}{4\eta^*\varepsilon^2} = N. \qquad \blacksquare$$

Normalized updates bound their step length by definition, so the same analysis applies to them.

Proposition 10.10 (Normalized steps are clipped by construction). *Let* $\mathcal{L}$ *satisfy Assumption 10.5 and* $0 < \gamma \leq 1/(3L_1)$.

(i) Normalized steepest descent $\theta_{t+1} = \theta_t + \gamma\,\mathrm{lmo}(g_t)$ *satisfies* $\mathcal{L}(\theta_{t+1}) \leq \mathcal{L}(\theta_t) - \frac{4}{5}\gamma\|g_t\|_* + \frac{3}{5}L_0\gamma^2$. *Hence* $\frac{1}{T}\sum_{t<T}\|\nabla\mathcal{L}(\theta_t)\|_* \leq 5\Delta_0/(4\gamma T) + 3L_0\gamma/4$, *and with* $\gamma = \min\{1/(3L_1), 2\varepsilon/(3L_0)\}$ *some* $t < T$ *has* $\|\nabla\mathcal{L}(\theta_t)\|_* \leq \varepsilon$ *once* $T \geq \max\{15L_0\Delta_0/(4\varepsilon^2), 15L_1\Delta_0/(2\varepsilon)\}$.

(ii) This applies to normalized gradient descent (ℓ_2), *to sign descent* (ℓ_∞,

with (10.1) read as $\|\nabla^2\mathcal{L}\|_{\infty\to 1} \le L_0 + L_1\|\nabla\mathcal{L}\|_1$) and to orthogonalized descent on a matrix (spectral norm, nuclear dual).

(iii) *Adam without bias correction, with $\epsilon = 0$ and $\beta_1^2 < \beta_2$, takes steps with $\|\theta_{t+1} - \theta_t\|_\infty \le C\eta$, $C = (1-\beta_1)/\sqrt{(1-\beta_2)(1-\beta_1^2/\beta_2)}$; if $\mathcal{L}$ satisfies Assumption 10.5 in ℓ_∞ and $C\eta \le c/L_1$, then every Adam step satisfies (10.6).*

Proof. The plan is to repeat the proof of Theorem 10.7(i) for a step of fixed length. For (i), $u = \gamma\,\mathrm{lmo}(g_t)$ has $\|u\| \le \gamma$ and $\langle g_t, u\rangle = -\gamma\|g_t\|_*$, also when $g_t = 0$. Lemma 10.6 with $c = \frac{1}{3}$ gives $\mathcal{L}(\theta_{t+1}) \le \mathcal{L}(\theta_t) - \gamma\|g_t\|_* + \frac{3}{5}(L_0 + L_1\|g_t\|_*)\gamma^2$, and $\frac{3}{5}L_1\gamma \le \frac{1}{5}$ gives the one-step bound; summing over $t < T$ gives the average. With the stated γ both terms are at most $\varepsilon/2$ once $T \ge 5\Delta_0/(2\gamma\varepsilon)$, and the smallest gradient norm is at most the average. Part (ii) consists of instances of (i) (Chapters 1 and 3). For (iii), the bound on Adam's update of Chapter 2 (Proposition 2.17) gives $|m_{t,i}|/\sqrt{v_{t,i}} \le C$ in every coordinate, so every step has ℓ_∞-length at most $C\eta$. ∎

Part (iii) does not prove that Adam converges, since its direction need not be a descent direction (Chapter 6); it proves that Adam never steps out of the region where the local curvature bound holds, if $C\eta \lesssim 1/L_1$. Here $C = 7.27$ for $(\beta_1, \beta_2) = (0.9, 0.999)$ and 1.16 for $(0.9, 0.95)$, and Section 10.4 shows that Adam comes close to C when a quiet block of weights wakes up.

The last result concerns noise. Gradient noise in deep networks can have *heavy tails* (Simsekli et al. 2019); Zhang et al. (2020c) found them heavier for BERT than for a convolutional network on ImageNet, and proposed them, with clipping as the remedy, as one reason adaptive methods train attention models faster. With infinite variance, SGD fails at once: on $\mathcal{L}(\theta) = \frac{1}{2}\|\theta\|_2^2$, $\mathbb{E}\mathcal{L}(\theta_1) = \frac{1}{2}\mathbb{E}\|(1-\eta)\theta_0 - \eta\zeta_0\|_2^2 = \infty$ for every $\eta > 0$. Clipping restores a rate.

Theorem 10.11 (Clipped SGD with heavy-tailed noise). *Let $\mathcal{L}$ be L-smooth in the Euclidean norm and bounded below, and let the stochastic gradients be unbiased, $\mathbb{E}[g_t \mid \theta_t] = \nabla\mathcal{L}(\theta_t)$, with a bounded moment of order $p \in (1, 2]$, $\mathbb{E}[\|g_t\|_2^p \mid \theta_t] \le \Gamma^p$. Clipped SGD $\theta_{t+1} = \theta_t - \eta\,\mathrm{clip}_\tau(g_t)$ satisfies*

$$\frac{1}{T}\sum_{t<T}\mathbb{E}\|\nabla\mathcal{L}(\theta_t)\|_2^2 \le \frac{2\Delta_0}{\eta T} + \frac{\Gamma^{2p}}{\tau^{2(p-1)}} + L\eta\Gamma^p\tau^{2-p}. \tag{10.9}$$

If $\Delta_0 > 0$, then with $\tau = \Gamma T^{1/(3p-2)}$ and $\eta = (2\Delta_0/(L\Gamma^2))^{1/2}T^{-p/(3p-2)}$ the right side is $(2\Gamma\sqrt{2L\Delta_0} + \Gamma^2)\,T^{-2(p-1)/(3p-2)}$, so that $\min_{t<T}\mathbb{E}\|\nabla\mathcal{L}(\theta_t)\|_2 \lesssim T^{-(p-1)/(3p-2)}$.

Proof. The plan is to bound the bias and the second moment of a clipped gradient by the pth moment and then to run the descent lemma. Condition on θ_t and write $g = g_t$, $\hat{g} = \mathrm{clip}_\tau(g)$. The two differ only where $\|g\|_2 > \tau$, by $\|g\|_2 - \tau < \|g\|_2 \le \|g\|_2^p\tau^{1-p}$, so the bias is $\|\mathbb{E}\hat{g} - \nabla\mathcal{L}(\theta_t)\|_2 \le \mathbb{E}\|\hat{g} - g\|_2 \le$

$\Gamma^p/\tau^{p-1} =: b$. The second moment: $\|\hat{g}\|_2^2 = \min\{\|g\|_2, \tau\}^2 \le \|g\|_2^p \tau^{2-p}$, as the cases $\|g\|_2 \le \tau$ and $\|g\|_2 > \tau$ show, so $\mathbb{E}\|\hat{g}\|_2^2 \le \Gamma^p \tau^{2-p}$. The descent lemma (Lemma A.18) gives $\mathcal{L}(\theta_{t+1}) \le \mathcal{L}(\theta_t) - \eta\langle \nabla\mathcal{L}(\theta_t), \hat{g}\rangle + \frac{L}{2}\eta^2\|\hat{g}\|_2^2$, and the bias enters through $\langle \nabla\mathcal{L}, \mathbb{E}\hat{g}\rangle \ge \|\nabla\mathcal{L}\|_2^2 - b\|\nabla\mathcal{L}\|_2 \ge \frac{1}{2}\|\nabla\mathcal{L}\|_2^2 - \frac{1}{2}b^2$. Taking expectations,

$$\mathbb{E}\mathcal{L}(\theta_{t+1}) \le \mathbb{E}\mathcal{L}(\theta_t) - \frac{\eta}{2}\mathbb{E}\|\nabla\mathcal{L}(\theta_t)\|_2^2 + \frac{\eta}{2}b^2 + \frac{L}{2}\eta^2\Gamma^p\tau^{2-p}.$$

Summing over $t < T$ and dividing by $\eta T/2$ gives (10.9). With the stated τ and η, the first and third terms both equal $\Gamma\sqrt{2L\Delta_0}\,T^{-2(p-1)/(3p-2)}$ and the second equals $\Gamma^2 T^{-2(p-1)/(3p-2)}$; the last claim follows from $(\mathbb{E}\|\nabla\mathcal{L}\|_2)^2 \le \mathbb{E}\|\nabla\mathcal{L}\|_2^2$. ∎

For $p = 2$ the rate is $T^{-1/4}$, that of SGD with bounded variance. The threshold must grow with the length of the run, since a fixed threshold leaves the bias Γ^p/τ^{p-1} (Exercise 10.7). Zhang et al. (2020c) proved this rate with a matching lower bound, and Koloskova et al. (2023) analyzed fixed thresholds.

10.4 The theory: the anatomy of a spike

A *loss spike* is a step too long for the curvature it lands in. Theorem 10.7 says that a step of bounded length is safe; so how does a clipped, well-tuned run produce steps that are too long, or curvature that is too high? We take apart three mechanisms, one for each further defense in the practice box.

THE OPTIMIZER'S STATE. After a long stretch of tiny gradients, a coordinate whose gradient jumps receives Adam steps that Chapter 6 computes exactly, the quiet-then-loud steps of Proposition 6.13: the first is about $(1-\beta_1)/\sqrt{1-\beta_2}$ learning rates, and the steps then grow. For a spike, what matters is the peak.[5]

[5] Kingma and Ba (2015) observed that Adam's step is roughly bounded by the learning rate, and singled out the most severe case of sparsity, a gradient that was zero at all earlier steps, where the step is $(1-\beta_1)/\sqrt{1-\beta_2}$ learning rates.

Proposition 10.12 (A quiet block wakes up). *Consider one coordinate of Adam without bias correction, with $\beta_1 \in [0,1)$, $\beta_2 \in (0,1)$ and $\epsilon \ge 0$: moments $m_t = \beta_1 m_{t-1} + (1-\beta_1)g_t$ and $v_t = \beta_2 v_{t-1} + (1-\beta_2)g_t^2$ from $m_{-1} = v_{-1} = 0$, and step $-\eta\, m_t/(\sqrt{v_t} + \epsilon)$. Let the coordinate be silent, $g_s = 0$ for $s < t_0$, and then loud, $g_s = \bar{g} \ne 0$ for $s \ge t_0$. At the kth loud step, $t = t_0 + k - 1$, the step is $-\eta\,\mathrm{sign}(\bar{g})\,a_k$ with*

$$a_k = \frac{1-\beta_1^k}{\sqrt{1-\beta_2^k} + \epsilon/|\bar{g}|}. \tag{10.10}$$

With $\epsilon = 0$: $a_1 = (1-\beta_1)/\sqrt{1-\beta_2}$; $a_k \to 1$ as $k \to \infty$; $a_k \le C = (1-\beta_1)/\sqrt{(1-\beta_2)(1-\beta_1^2/\beta_2)}$ for every k if $\beta_1^2 < \beta_2$; and $a_k \ge (1-\beta_1^k)/\sqrt{k(1-\beta_2)}$, so that for $\beta_1 = 0.9$, $\max_k a_k \ge a_{12} \ge 0.207/\sqrt{1-\beta_2}$.

PROOF. The plan is to take the formula from Chapter 6 and then to bound it. Over the loud steps $m_t = (1-\beta_1^k)\bar{g}$ and $v_t = (1-\beta_2^k)\bar{g}^2$, so (10.10) is the

case $\delta = 0$ of Proposition 6.13, whose two bounds then coincide. Its index counts the loud steps from 0: its k is our $k-1$, and its ρ_{k-1} is our a_k. It takes $\bar{g} > 0$; reversing the sign of every gradient reverses m_t and the step and leaves v_t unchanged. Its part (i) gives a_1 and $a_k \to 1$. The bound by C is the bound on $|m_t|/\sqrt{v_t}$ of Chapter 2 (Proposition 2.17), valid for every sequence of gradients, here the one that is zero before t_0 and $\bar{g}$ afterwards. Finally $1-\beta_2^k = (1-\beta_2)\sum_{j<k}\beta_2^j \le k(1-\beta_2)$, and $(1-0.9^{12})/\sqrt{12} = 0.2071$. ∎

In words: after a long silence both moments are empty, and the coordinate goes through a *burst*: m refills in about $1/(1-\beta_1) = 10$ steps and v in about $1/(1-\beta_2)$, and in between the step is too large. If β_2 is close to 1, then the step grows for about $1.26/\log(1/\beta_1) \approx 12$ steps, where the lower bound peaks, and decays to 1 over about $1/(1-\beta_2)$. For $\beta_1 = 0.9$ and $\beta_2 = 0.95$, 0.99, 0.999, the first step is 0.45, 1.00, 3.16, the peak 1.10, 2.13, 6.57 (at $k = 20$, 13, 12), and the worst case C over all gradient sequences 1.16, 2.35, 7.27. A coordinate that was quiet, $|g_s| \le \delta$, rather than silent has its steps between the two bounds of Proposition 6.13, which tend to a_k as $\delta/|\bar{g}| \to 0$. Bias correction counts t from the start of training, and its factors are 1 once $t \gg 1/(1-\beta_2)$: the burst is the transient it removes at the start (Chapter 9), recreated for a coordinate whose memory has emptied (Exercise 10.4). Without momentum the burst is one step of $1/\sqrt{1-\beta_2}$, which is 31.6 learning rates at $\beta_2 = 0.999$; and ϵ caps it only for gradients with $|\bar{g}| \lesssim \epsilon/\sqrt{1-\beta_2^k}$.

Which blocks go silent? Rows of an embedding matrix for tokens absent from recent batches, experts that a router stops choosing, units whose inputs vanish, and, in a milder form, every block whose gradient grows by a large factor within fewer than $1/(1-\beta_2)$ steps (Exercise 10.8). Whether a burst becomes a spike depends on the block's curvature, which the proposition ignores.

The output logits. The second mechanism is a direction in which the loss does not change at all.

Proposition 10.13 (A flat direction and the z-loss). *For logits $z \in \mathbb{R}^K$ let $Z(z) = \sum_k e^{z_k}$ and $\pi = \mathrm{softmax}(z)$, and let $\ell(z) = -\sum_k y_k z_k + \log Z(z)$ be the cross-entropy against a probability vector y (a one-hot label, or the label frequencies of a batch).*

(i) $\ell(z + c\mathbf{1}) = \ell(z)$ for every $c \in \mathbb{R}$; $\langle \nabla\ell(z), \mathbf{1}\rangle = 0$ and $\nabla^2\ell(z)\mathbf{1} = 0$.

(ii) For an output layer $z = Wh + b$ with $W \in \mathbb{R}^{K\times d_{\text{in}}}$, every batch gradient has $\mathbf{1}^\top \nabla_W \ell = 0$ and $\langle \mathbf{1}, \nabla_b \ell\rangle = 0$. Hence an update that is a linear combination of gradients multiplied on the right by matrices (gradient descent, momentum, right preconditioning), and also an orthogonalized update $-\eta\,\mathrm{polar}(G)$, leaves $\mathbf{1}^\top W$ and $\langle \mathbf{1}, b\rangle$ unchanged; a sign step on b changes $\langle \mathbf{1}, b\rangle$ by $-\eta\sum_k \mathrm{sign}(\pi_k - y_k)$.

(iii) With the z-loss, $\ell_\alpha = \ell + \alpha(\log Z)^2$ with $\alpha > 0$, the restriction to the flat line is the parabola $\ell_\alpha(z + c\mathbf{1}) = \ell(z) + \alpha(\log Z(z) + c)^2$, with second

derivative 2α *in* c *and minimum at* $\log Z = 0$. *If* $y_k > 0$ *for every* k, *then the only stationary point of* ℓ_α *is* $z = \log y$ *(coordinatewise), where* $\log Z = 0$ *and* $\pi = y$.

PROOF. The plan is to compute the gradient and the Hessian of ℓ and read everything off them. (i) $Z(z+c\mathbf{1}) = e^c Z(z)$, so $\log Z$ increases by c, as much as $\sum_k y_k z_k$ does. Differentiating, $\nabla\ell = \pi - y$, whose coordinates sum to $1-1=0$, and $\nabla^2\ell = \operatorname{diag}(\pi) - \pi\pi^\top$, which maps $\mathbf{1}$ to $\pi - \pi = 0$. (ii) For one example $\nabla_W \ell = (\pi - y)h^\top$ and $\nabla_b \ell = \pi - y$, so $\mathbf{1}^\top \nabla_W \ell = 0$ by (i), and a batch gradient is an average of such terms. A linear combination $\sum_s c_s G_s R_s$ inherits $\mathbf{1}^\top G_s = 0$. If $\mathbf{1}^\top G = 0$, then $\mathbf{1}$ is orthogonal to the column space of G, which the left singular vectors U span, so $\mathbf{1}^\top \operatorname{polar}(G) = \mathbf{1}^\top U V^\top = 0$. (iii) The parabola follows from $\log Z(z + c\mathbf{1}) = \log Z(z) + c$ and (i). A stationary point has $\pi - y + 2\alpha \log Z\, \pi = 0$, that is, $(1 + 2\alpha \log Z)\pi = y$; summing the coordinates gives $\log Z = 0$, hence $\pi = y$ and $z_k = \log \pi_k + \log Z = \log y_k$. ∎

Gradient descent therefore never moves the output layer along this *flat direction*, nor does Muon (Exercise 10.9), whereas coordinatewise normalization does: the coordinates of $\pi - y$ sum to zero, their signs need not. Decoupled weight decay shrinks the flat component, and the z-loss adds a spring of stiffness 2α that holds $\log Z$ near 0. In exact arithmetic the drift is harmless; in bfloat16, which keeps 8 significant bits, logits near 100 are rounded to multiples of 0.5, which changes the ratios of the probabilities by factors up to $e^{\pm 0.5}$.[6] How strong a spring is needed depends on the speed of the drift, which Section 10.5 measures in a small model.

[6] Zoph et al. (2022) proposed a z-loss on the logits of the router of a mixture of experts, motivated by roundoff in the exponentials of large logits. Whether the output-logit divergence of Observation 10.3 is a precision effect has, as far as we know, not been established.

THE ATTENTION LOGITS. The third mechanism concerns the *attention logits*.

Proposition 10.14 (Attention logits, QK-norm and QK-clip). *For an attention head with query and key weights* $W_Q, W_K \in \mathbb{R}^{d_\mathrm{h} \times d_\mathrm{in}}$ *and inputs* $x, x' \in \mathbb{R}^{d_\mathrm{in}}$, *let* $s(x,x') = \langle W_Q x, W_K x' \rangle / \sqrt{d_\mathrm{h}}$.

(i) The logit is bilinear in (W_Q, W_K): *replacing* (W_Q, W_K) *by* $(\rho W_Q, \rho W_K)$ *multiplies every logit by* ρ^2, *and* $|s(x,x')| \le \|W_Q\|_{\mathrm{op}} \|W_K\|_{\mathrm{op}} \|x\|_2 \|x'\|_2 / \sqrt{d_\mathrm{h}}$.

(ii) With QK-norm, $\tilde{s}(x,x') = \langle \omega_q \odot \mathcal{N}(W_Q x),\ \omega_k \odot \mathcal{N}(W_K x') \rangle / \sqrt{d_\mathrm{h}}$, *where* $\mathcal{N}$ *is RMS normalization or a LayerNorm without bias and without* ϵ, *and* $\omega_q, \omega_k \in \mathbb{R}^{d_\mathrm{h}}$ *are gains,* $|\tilde{s}(x,x')| \le \sqrt{d_\mathrm{h}}\, \|\omega_q\|_\infty \|\omega_k\|_\infty$ *whenever the normalizations are defined, with equality when* $\mathcal{N}(W_Q x) = \mathcal{N}(W_K x')$ *and* $\omega_q = \omega_k$ *is a constant vector.*

(iii) If the logits of a row of attention lie in an interval of length R, *then each of the row's* n_key *attention weights is at most* $1/(1 + (n_\mathrm{key} - 1)e^{-R})$, *and the row's entropy is at least* $\log(1 + (n_\mathrm{key} - 1)e^{-R})$.

(iv) (QK-clip.) If the largest logit of a head on a batch is $s_\mathrm{max} > \bar{s} > 0$, *then replacing* (W_Q, W_K) *by* $(\rho W_Q, \rho W_K)$ *with* $\rho = \sqrt{\bar{s}/s_\mathrm{max}}$ *multiplies every logit of the head by* $\bar{s}/s_\mathrm{max}$, *so the largest becomes* $\bar{s}$.

Proof. The plan is to treat the four parts in turn; only (ii) and (iii) need a computation. (i) Bilinearity is immediate, and Cauchy–Schwarz gives $|\langle W_Q x, W_K x'\rangle| \le \|W_Q\|_{\mathrm{op}}\|x\|_2\|W_K\|_{\mathrm{op}}\|x'\|_2$. (ii) RMS normalization maps v to $\sqrt{d_{\mathrm{h}}}\, v/\|v\|_2$ and the LayerNorm maps it to $\sqrt{d_{\mathrm{h}}}\,(v-\bar v\mathbf{1})/\|v-\bar v\mathbf{1}\|_2$, with $\bar v$ the mean of the coordinates; both images have Euclidean norm $\sqrt{d_{\mathrm{h}}}$. With $a = \mathcal{N}(W_Q x)$ and $b = \mathcal{N}(W_K x')$, Hölder's and the Cauchy–Schwarz inequality give $|\sum_i \omega_{q,i}\omega_{k,i}a_i b_i| \le \|\omega_q\|_\infty\|\omega_k\|_\infty\|a\|_2\|b\|_2 = \|\omega_q\|_\infty\|\omega_k\|_\infty d_{\mathrm{h}}$; divide by $\sqrt{d_{\mathrm{h}}}$. In the stated case every inequality is an equality. (iii) If the logits are $s_1,\dots,s_{n_{\mathrm{key}}}$, then every s_i is at least $s_j - R$, so the weight $e^{s_j}/\sum_i e^{s_i}$ of key j is at most $e^{s_j}/(e^{s_j} + (n_{\mathrm{key}}-1)e^{s_j-R})$, and the entropy $\sum_j p_j \log(1/p_j)$ is at least $\log(1/\max_j p_j)$. (iv) follows from (i). ∎

Part (i) is the source of the growth: the logit is quadratic in the scale of (W_Q, W_K), and under Adam the scale can grow linearly in time, since each coordinate moves by at most $C\eta$ per step. After T steps $\|W_Q - W_Q^{(0)}\|_{\mathrm{op}} \le C\eta T\sqrt{d_{\mathrm{h}} d_{\mathrm{in}}}$, a bound attained by consistent rank-one sign updates, so for inputs with $\|x\|_2 = \sqrt{d_{\mathrm{in}}}$ the bound of (i) grows like $(C\eta T)^2 d_{\mathrm{in}}^2\sqrt{d_{\mathrm{h}}}$; under QK-norm the same argument applied to the gains gives $\sqrt{d_{\mathrm{h}}}(\|\omega^{(0)}\|_\infty + C\eta T)^2$, smaller by the factor d_{in}^2. QK-norm does not forbid sharp attention (with unit gains and $d_{\mathrm{h}} = 128$ the logits may range over $2\sqrt{128} \approx 23$); by (iii), the entropy collapse of Zhai et al. (2023) requires a growth of the logit range, and QK-norm removes the part of it driven by the weight matrices.[7] MuonClip applies a rescaling of the kind in (iv) after each Muon step to every head above the threshold (Kimi Team 2025): the mildest one that brings the batch's largest logit down to $\bar s$.

[7] Dehghani et al. (2023) applied LayerNorms to queries and keys to remove the divergence of ViT-22B; Wortsman et al. (2024) found that the same change removes attention-logit growth in small models.

10.5 Reading the theory

The price of a fixed step. Figure 10.1 puts the two theorems side by side on $f(x,y) = \mathcal{L}_{\mathrm{cliff}}(x) + \mathcal{L}_{\mathrm{plat}}(y)$ with $L_0 = L_1 = 1$, a cliff and a plateau, $(1,1)$-smooth in the Euclidean norm because its Hessian is diagonal. Gradient descent converges exactly below the threshold η^* of the proof of Theorem 10.9, which falls from 1.39 at $M = 1$ to $2.8\cdot 10^{-5}$ at $M = 10^6$. At $0.9\eta^*$ it crawls along the plateau, and its step count, the gradient-flow time of the plateau divided by the step size to within one step, grows like $M/\log M$, from 81 to $4.07\cdot 10^6$. Clipped descent needs 203 steps at $M = 1$ and 241 at $M = 10^6$: about $2\cdot 101$ on the plateau and $3\log\frac{3(1+M)}{5}$ normalized steps down the cliff. For every start, its largest convergent step size lies between $L_1\gamma/(L_0(e^{L_1\gamma/2}-1)) = 1.838$, below which it cannot be caught in a two-cycle across the valley, and $2/L_0$. The agreement is exact because the functions are those of the proofs; it checks the arithmetic, not the idealization.

Where to clip. The threshold of Theorem 10.7, in units of the gradient norm, is $\gamma/\eta = 2L_0/(3L_1)$: clipping should start where $L_1\|\nabla\mathcal{L}\|_*$ begins to dominate L_0. That is a property of the loss; the theory gives no reason for a universal value such as 1.0, and global gradient norms change with the normalization of the loss and the number of parameters. It predicts a threshold that is inactive where the landscape is flat and active on cliffs

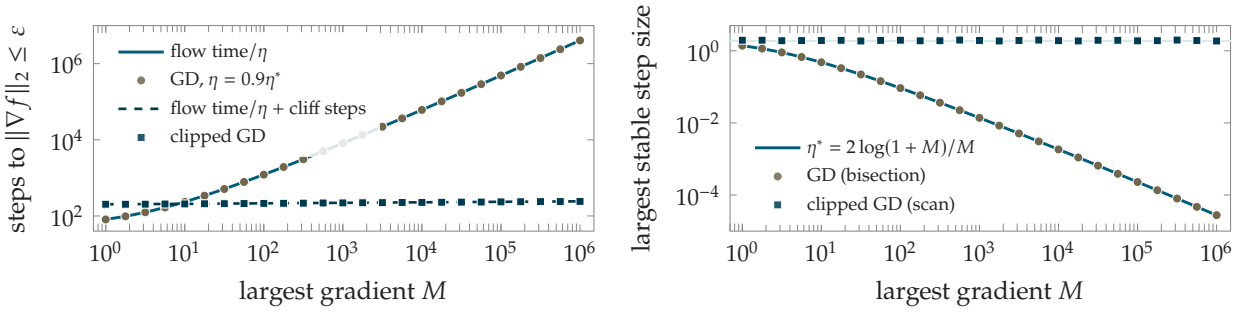


Figure 10.1. Clipped and fixed-step gradient descent on a cliff followed by a plateau, $f(x, y) = \mathcal{L}_{\text{cliff}}(x) + \mathcal{L}_{\text{plat}}(y)$ with $L_0 = L_1 = 1$ and $\varepsilon = 0.05$, started where the cliff's slope is M and the plateau's height is 1. Left: steps until $\|\nabla f\|_2 \le \varepsilon$ for gradient descent at $0.9\eta^*$ (circles) and clipped descent with $\eta = \frac{1}{2}$, $\gamma = \frac{1}{3}$ (squares), against the plateau's gradient-flow time 101.15 divided by the step size, plus $3\log\frac{3(1+M)}{5}$ normalized steps down the cliff for clipped descent (curves). Right: the largest step size with which gradient descent does not diverge (circles, bisection) against η^*, and with which clipped descent converges (squares, scan in steps of 0.0025) against the band $[1.838, 2]$ predicted for every start. Deterministic; `10-spikes-separation.py`.

and during spikes, and a cost, the factor $1/(\gamma\varepsilon)$ of Theorem 10.7(ii), for a threshold set too low. In front of Adam, clipping acts differently: Adam's update is unchanged by a common rescaling of all gradients (up to ϵ) and bounded whatever the gradient (Proposition 10.10(iii)). Clipping therefore does not bound Adam's step; it bounds the weight that one large gradient receives in m and v, which suppresses the loud event of a global jump of the gradient norm. A block that wakes up below the threshold is unaffected: the mechanism of Proposition 10.12 survives global clipping.

THE BURST. In the run of Figure 10.2 the measured steps follow a_k to within 1.5 percent, with peaks of 6.59, 2.13 and 1.09 learning rates for $\beta_2 = 0.999$, 0.99, 0.95 against the predicted 6.57, 2.13, 1.10; minibatch noise, which enters v but on average not m, accounts for the differences. Across β_2 the peak lies above $0.207/\sqrt{1-\beta_2}$, by 18 percent at $\beta_2 = 0.95$ and 0.3 percent at 0.999, and is two to two and a half times a_1; lowering β_2 from 0.999 to 0.95 shrinks it six-fold. This is a mechanism behind the choice $\beta_2 = 0.95$ and behind Observation 10.4; that it is *the* reason for the choice is a suggestion, and a short memory costs a noisier preconditioner (Chapter 6). A larger ϵ damps the burst only where $|\bar{g}|$ is not much larger than $\epsilon/\sqrt{1-\beta_2^k}$ (at $|\bar{g}| = 10\epsilon$ the peak roughly halves near $\beta_2 = 0.999$), and it turns Adam into gradient descent with step size η/ϵ where gradients are far below ϵ.

HOW FAST THE LOGITS DRIFT. Proposition 10.13 does not say whether $\alpha = 10^{-4}$ is enough. Consider a unigram model, a softmax over $K = 1000$ classes with Zipf frequencies whose logits are its only parameters, trained on minibatches of 64 labels (`10-spikes-drift.py`). SGD keeps the mean logit fixed exactly. Under Adam with $\beta_2 = 0.95$ the normalizer drifts

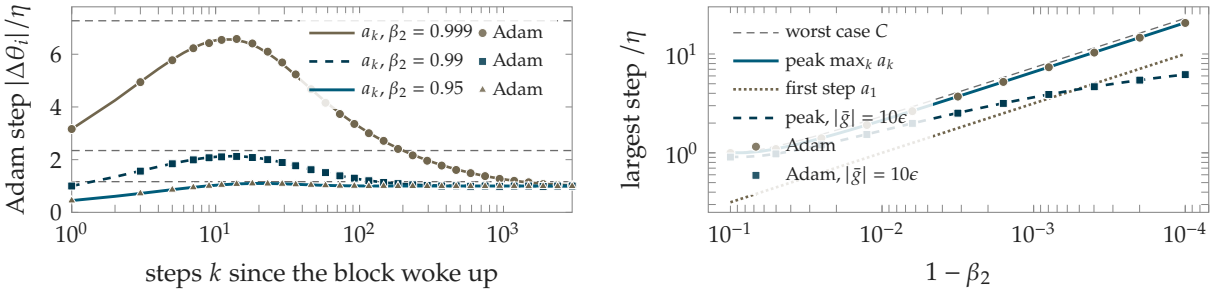


Figure 10.2. Adam's step on a block of 32 weights of a linear regression whose features are absent for the first 10^5 steps, so that the block's moments are zero ($\beta_1 = 0.9$, $\epsilon = 10^{-8}$, bias correction counted from the start, batches of 4096, step size 10^{-6}, labels $\langle \theta_\star, x \rangle$ plus unit Gaussian noise, $\theta_\star \in \{\pm 1\}^{32}$). Left: the median over the block of $|\Delta\theta_i|/\eta$ at the kth step after the features appear (points) against a_k (curves); dashed: the worst case C. Right: the largest such step in the first 80 steps against $1 - \beta_2$ (points), with $\max_k a_k$, a_1 and C (curves); lower curve and squares: the loss scaled by 10^{-7}, so that $|\bar{g}| = 10\epsilon$. One run per point; `10-spikes-burst.py`.

instead: $\log Z$ falls by 0.12 step sizes per step while the cross-entropy stays within 0.14 of its minimum. The direction matches the small transformers of Wortsman et al. (2024), whose output logits become very negative in runs without weight decay. The drift comes from the coordinatewise normalization of the skewed gradient of a rare class, small and positive in most batches and large and negative in the few that contain it; it vanishes in full batch and falls with the batch size (0.21, 0.05, 0.009 for batches of 16, 256, 4096) and with β_2 (0.001 at 0.999). A z-loss of 10^{-4} leaves it unchanged; a z-loss of 0.1 stops it, with an excess cross-entropy of 0.38 instead of 0.14, and decoupled weight decay of 0.1 stops it with an excess of 0.03. In this model 10^{-4} is far too weak, and weight decay does the pinning.

10.6 What the theory misses

WHAT WAS IDEALIZED AWAY. The main theorems have exact gradients; with noise a fixed threshold biases the clipped gradient (Exercise 10.7), and Theorem 10.11 needs a growing threshold and L-smoothness. The constants L_0, L_1 are global and fixed, while a network's curvature changes during training, and twice differentiability excludes ReLU networks. The lower bound concerns a class, not a given network; the burst assumes a constant loud gradient after an exact silence; the propositions on logits are about invariances and bounds, not dynamics; and finite precision, a candidate for the step from a drifted logit to a divergence, appears nowhere.

WHERE IT BREAKS. (1) Assumption 10.5 fails globally for networks with rescaling symmetries. In the product model $\frac{1}{2}(uv - 1)^2$ of volume 2 of this series, *The Hydrogen Atoms of Deep Learning*, Chapter 15, the minimizers $(u, 1/u)$ have zero gradient and sharpness $u^2 + u^{-2}$, so no constants work (Exercise 10.6); the assumption holds only on bounded sets, with an L_0 that grows with the bound. Observation 10.2 is a correlation along trajectories,

not a derivation. (2) Heavy-tailed noise is not the main reason that adaptive methods train transformers faster: Kunstner et al. (2023) found that the gap between SGD and Adam persists in full batch, where there is no noise, and that sign descent with momentum closes most of it (Chapter 14); the case for clipping through Theorem 10.11 is weaker than through Theorem 10.7. (3) Real spikes have interacting causes. In PaLM neither the batches nor the state alone produced a spike (Observation 10.1); OPT's divergences coincided with a collapse of the loss scale (Zhang et al. 2022); Takase et al. (2025) relate spikes to the spectral norms of the sub-layers' Jacobians and remove them through the initialization and the scale of the embeddings, a mechanism of architecture outside all three propositions. (4) In our unigram model the z-loss coefficient of practice does not pin the logits, although it removes the output-logit divergence of Wortsman et al. (2024); its benefit is not explained by Proposition 10.13. (5) In Kimi K2's experiments attention logits exploded more often with Muon than with AdamW (Kimi Team 2025); Proposition 10.14 explains the cure, not the cause. (6) That a threshold of 1.0 is rarely active late in training, except during spikes, is folklore.

REPAIRS AND OPEN PROBLEMS. The theory of generalized smoothness has grown quickly: clipping with momentum and sharper constants (Zhang et al. 2020a); a symmetric form of (10.1) without second derivatives (Chen et al. 2023); curvature bounded by a general function of the gradient norm, under which gradient descent with a small fixed step converges on sublevel sets (Li et al. 2023a); Adam under such assumptions (Li et al. 2023b); and generalized sign descent (Crawshaw et al. 2022). Against the burst the repairs act on the update: clipping it by its root mean square, as Adafactor does (Shazeer and Stern 2018), brought to AdamW as StableAdamW (Wortsman et al. 2023); and spike-aware variants of Adam that clip gradient entries that are large relative to the second-moment estimate and periodically reset the moments (Huang et al. 2025). Against the logits: z-loss, weight decay, QK-norm, QK-clip, and spectral reparametrizations of the weights that prevent the collapse of attention entropy (Zhai et al. 2023). Open: a solvable model in which spikes arise and recur without being put in by hand (Exercise 10.12); (L_0, L_1)-type bounds for a transformer, with constants set by the gains of its normalizations; a principled threshold, for instance $2L_0/(3L_1)$ estimated along the run by regressing local smoothness on the gradient norm, as Zhang et al. (2020b) did; and why Muon grows attention logits.

10.7 Back in the wild

FIELD NOTES	
PRACTICE	Clip the global gradient norm at 1.0; $\beta_2 = 0.95$; z-loss $10^{-4}(\log Z)^2$; QK-norm; MuonClip; after a spike, restart and skip batches.
IDEALIZATION	(L_0, L_1)-smoothness $\|\nabla^2 \mathcal{L}\| \le L_0 + L_1 \|\nabla \mathcal{L}\|_*$; Adam's moments after a quiet period; the flat direction of the softmax; the bilinear attention logit.

PROVED	A sharp local smoothness lemma. Clipped steepest descent: at most $\max\{4L_0\Delta_0/\varepsilon^2, 6L_1\Delta_0/\varepsilon\}$ steps with $\|g\|_* > \varepsilon$; every fixed step: $\gtrsim L_1 M\Delta_0/(\varepsilon^2 \log(1+L_1M/L_0))$ on some function of the class. Normalized steps are clipped by construction. Clipped SGD at $T^{-(p-1)/(3p-2)}$ with pth moments. Exactly: the burst a_k, the pinning by the z-loss, QK-norm's bound by the gains.
PREDICTS	GD's safe step falls like $\log M/M$, clipped descent's does not; clip where $L_1\|\nabla\mathcal{L}\|_* \approx L_0$; Adam's burst peaks above $0.2/\sqrt{1-\beta_2}$ learning rates 12 to 20 steps after a block wakes up, six times lower at $\beta_2 = 0.95$ than at 0.999; global clipping does not stop a block's burst.
MISSES	When spikes happen; interacting causes (data, state, precision); (L_0, L_1)-smoothness fails under rescaling symmetries; why 10^{-4} and 1.0.
TAKEAWAY	Bound the length of every step, by clipping or by normalized updates (proved in the idealization). Prefer a short β_2 or update clipping when blocks can wake up (suggested). Use z-loss, weight decay and QK-norm to remove directions of growth the loss cannot see (proved as mechanisms; their effect on spikes is observed). The values 1.0 and 10^{-4} are folklore.

The theory says one thing with confidence: on landscapes whose curvature grows with the gradient, a step of bounded length is safe and a fixed step size is not. That is why clipping works, why normalized updates (Chapters 1 to 3) do not need it, and why Adam, whose step is bounded, is less exposed to cliffs than gradient descent; in front of Adam, clipping acts on the moments rather than on the step. It does not place the threshold, and it predicts no spike. What it adds to the folklore of spikes is a set of exact mechanisms that make a step too long: a state that lags its gradient, a direction the loss cannot see, a scale that enters the logits quadratically; each defense removes one of them, none removes all.

Part III ends here. The threshold $2/\eta$ of Chapter 8 bounds the curvature a fixed step can stand; warmup (Chapter 9) grows the step as the curvature falls with the loss, and clipping shrinks it as the curvature rises with the gradient, two readings of one idea that both hold for exponential-type losses. Weight decay (Chapter 13) is a defense of this chapter's kind, pinning the flat direction of the logits and bounding the scales in the attention logits; and heavy tails return in Chapter 14, where they turn out not to explain Adam's advantage on transformers.

Exercises

10.1 CONSOLIDATION. (Constants.) (a) Show that x^4 is (L_0, L_1)-smooth if and only if $L_0L_1^2 \geq 16$. (b) Show that e^x is $(0, 1)$-smooth and L-smooth for no L. (c) Show that $\log(1 + e^{-x})$ is $\frac{1}{4}$-smooth and $(0, 1)$-smooth. (d) Show that $\sum_i a_i(\cosh\theta_i - 1)$ with $a_i > 0$ is $(\max_i a_i, 1)$-smooth in the Euclidean norm and $(\sum_i a_i, 1)$-smooth in ℓ_∞. (e) Show that $\frac{1}{2}\theta^\top H\theta$

with $H \neq 0$ is (L_0, L_1)-smooth only if $L_0 \geq \|H\|$. *(Hint: in (a), minimize* $L_0 + 4L_1x^3 - 12x^2$ *over* $x \geq 0$.*)*

10.2 CONSOLIDATION. (The lemma is sharp.) Verify the equality case of Lemma 10.6 and deduce that $\varphi_1(c)$ and $\varphi_2(c)$ in (10.6) are optimal.

10.3 CONSOLIDATION. (Clipping is a trust region.) (a) Show that the clipped step minimizes $\langle g, u\rangle + \|u\|_2^2/(2\eta)$ over the ball $\|u\|_2 \leq \gamma$. (b) Show that clipping by value, $u_i = -\operatorname{sign}(g_i)\min\{\eta|g_i|, \gamma\}$, minimizes the same function over the box $\|u\|_\infty \leq \gamma$. (c) Show that clipping by value can change the direction of the step, and global clipping cannot.

10.4 CONSOLIDATION. ★ (The burst.) (a) Show that a correction of the moments by the age of the coordinate, $m_t/(1-\beta_1^k)$ and $v_t/(1-\beta_2^k)$, would make the step of Proposition 10.12 with $\epsilon = 0$ equal to $\operatorname{sign}(\bar{g})$ at every loud step. (b) For $\beta_1 = 0.9$, show that $(1-\beta_1^k)/\sqrt{k}$ is largest at $k = 12$, and deduce $\max_k a_k \geq 6.55$ at $\beta_2 = 0.999$. (c) For RMSProp, $\beta_1 = 0$, show that the largest a_k is a_1. (d) Show that as $\beta_2 \to 1$, $\sqrt{1-\beta_2}\max_k a_k \to \max_{k\geq 1}(1-\beta_1^k)/\sqrt{k}$ and $\sqrt{1-\beta_2}\,C \to \sqrt{(1-\beta_1)/(1+\beta_1)}$; for $\beta_1 = 0.9$ their ratio is 0.903. *(Hint: in (d), the maximizing* k *stays bounded as* $\beta_2 \to 1$*, so* $1-\beta_2^k \sim k(1-\beta_2)$ *there.)*

10.5 CONSOLIDATION. (The z-loss.) (a) Show that $\nabla(\log Z)^2 = 2(\log Z)\pi$ and $\nabla^2(\log Z)^2 = 2\pi\pi^\top + 2(\log Z)(\operatorname{diag}(\pi) - \pi\pi^\top)$. (b) Show that gradient descent on the bias b of a softmax with the loss $\ell + \alpha(\log Z)^2$ changes $\langle \mathbf{1}, b\rangle$ by exactly $-2\alpha\eta \log Z$ per step, and that without the z-loss decoupled weight decay λ multiplies $\langle \mathbf{1}, b\rangle$ by $1 - \eta\lambda$ per step.

10.6 VARIATION. (No (L_0, L_1) for the product model.) Let $\mathcal{L}(u, v) = \frac{1}{2}(uv - 1)^2$. (a) Show that no constants L_0, L_1 make $\mathcal{L}$ (L_0, L_1)-smooth in the Euclidean norm. (b) Show that $\|\nabla^2\mathcal{L}(u, v)\|_{\mathrm{op}} \leq 2R^2 + 1$ on the disk $u^2 + v^2 \leq R^2$. (c) What does (b) mean for Corollary 10.8 on this model?

10.7 VARIATION. (Clipping with noise is biased.) Let $g = \mu + \zeta$ with $\mu > 0$ and $\zeta = \pm s$ with probability $\frac{1}{2}$ each, $s > \mu$. (a) Compute $\mathbb{E}\,\mathrm{clip}_\tau(g)$ for every $\tau > 0$; show that it is 0 when $\tau \leq s - \mu$ and lies in $[0, \mu]$ always. (b) Deduce that clipped SGD with a fixed threshold can stall where the gradient is not zero, and compare with the bias term of (10.9).

10.8 VARIATION. ★ (A loud jump.) Let $0 < \beta_2 < 1$, and suppose that a coordinate's gradient has been $\delta \neq 0$ so long that $m = \delta$ and $v = \delta^2$ (take $\epsilon = 0$), and then jumps to $F\delta$ with $F > 1$. (a) Show that the kth step after the jump is $\big((1-\beta_1^k) + \beta_1^k/F\big)/\sqrt{(1-\beta_2^k) + \beta_2^k/F^2}$, which tends to a_k as $F \to \infty$. (b) Show that the first step exceeds the steady value 1 for every $F > 1$ if $(1-\beta_1)^2 \geq 1 - \beta_2$, and exactly when $F < (\beta_2 - \beta_1^2)/((1-\beta_2) - (1-\beta_1)^2)$ otherwise; evaluate for $(0.9, 0.95)$. *(Hint: in (b),* $x = 1/F$ *makes the condition a quadratic inequality with the root* $x = 1$*.)*

10.9 VARIATION. (Orthogonalized updates keep the logits centered.) (a) Show that the Newton–Schulz iteration $X \mapsto aX + b(XX^\top)X + c(XX^\top)^2X$ of Chapter 3 preserves $\mathbf{1}^\top X = 0$, so that Muon, applied to an output layer, leaves $\mathbf{1}^\top W$ unchanged. (b) Find a $G \in \mathbb{R}^{3\times 1}$ with $\mathbf{1}^\top G = 0$ and $\mathbf{1}^\top \operatorname{sign}(G) \neq 0$. (c) Show that the mean logit $\frac{1}{K}\mathbf{1}^\top W h$ can still change when $\mathbf{1}^\top W$ is fixed, and say through which parameters.

10.10 SIMULATION. (Heavy tails.) Run SGD and clipped SGD on $\mathcal{L}(\theta) = \frac{1}{2}\|\theta\|_2^2$ in $\mathbb{R}^{10}$ with independent symmetric noise coordinates, $\Pr[|\zeta_i| > x] = (1+x)^{-3/2}$, whose pth moments are finite exactly for $p < \frac{3}{2}$. (a) Observe the rare huge excursions of SGD with a constant step. (b) From $\theta_0 = (1, \dots, 1)$, so $\Delta_0 = 5$, with the schedule of Theorem 10.11 for $p = 1.4$, $L = 1$ and $\Gamma = 22$ (just above $(\mathbb{E}\|g_0\|_2^p)^{1/p} \approx 21.4$), fit the slope of the average of $\|\nabla\mathcal{L}(\theta_t)\|_2^2$ over $t < T$ against $\log T$ for $T = 10^3, \dots, 10^6$, and compare with $-2(p-1)/(3p-2) = -0.36$ (our runs: -0.36, and -0.31 from $\theta_0 = e_1$, where $\eta T < 1$ in the shortest runs). (c) Explain why the bias term of (10.9) is far below its bound here.

10.11 SIMULATION. (Logit drift.) Run `10-spikes-drift.py`. (a) Check that SGD keeps the mean logit fixed to rounding error and that Adam's drift vanishes in full batch. (b) Explain the sign of the drift from the distribution of a rare class's gradient and Adam's coordinatewise normalization. (c) Find, by bisection, the smallest z-loss coefficient for which the drift over the second half of the run is below 10^{-3} step sizes per step at batch size 64, and compare it with 10^{-4}.

10.12 RESEARCH. (A solvable spike.) Find a low-dimensional model in which Adam with fixed hyperparameters produces recurring loss spikes without a switch put in by hand, and compute their period and height as functions of η and β_2. The burst of Proposition 10.12 and Adam's threshold of Chapter 8 are natural ingredients; open is a model whose quiet periods are produced by the dynamics.

Notes and References

Clipping passed from recurrent networks (Pascanu et al. 2013) into transformer recipes with a threshold of 1.0 (Brown et al. 2020; Chowdhery et al. 2023; Touvron et al. 2023), in front of Adam, whose step is bounded anyway (Kingma and Ba 2015). Zhang et al. (2020b) introduced (L_0, L_1)-smoothness and proved the first separation between clipped and fixed-step gradient descent: a bound $O(L_0\Delta_0/\varepsilon^2 + L_1^2\Delta_0/L_0)$ for clipped descent, with larger constants, and a lower bound of the form (10.8) with $\log M + 1$ in place of our logarithm. Zhang et al. (2020a) extended their local smoothness lemma to steps of length up to c/L_1; their descent inequality has the quadratic term $\frac{1}{2}(AL_0 + BL_1\|g\|_*)\|u\|^2$ with $B = \varphi_1(c)$ and $A = 1 + e^c - \varphi_1(c) > B$. Our constants are smaller because integrating (10.4) gives $\varphi_2(c) < \frac{1}{2}\varphi_1(c)$ in the descent inequality, and because counting the steps with $\|g_t\|_* > \varepsilon$ turns their sum into a maximum, never larger up to a constant: $6L_1/\varepsilon \le 4L_0/\varepsilon^2 + \frac{9}{4}L_1^2/L_0$.

The record of spikes lies mostly in technical reports: PaLM's restarts and z-loss (Chowdhery et al. 2023), OPT's restarts (Zhang et al. 2022), ViT-22B's QK-norm (Dehghani et al. 2023) and Kimi K2's QK-clip (Kimi Team 2025). Wortsman et al. (2024) reproduced the logit instabilities in small models and measured stability by the sensitivity of the final loss to the learning rate; Wortsman et al. (2023) tied spikes to an out-of-date second moment, Molybog et al. (2023) proposed a theory of Adam's instability, and Zhai et al. (2023) tied instability to attention entropy. The burst formula (10.10) is the silent case of Proposition 6.13, its first step is in Kingma and Ba (2015), and we know no reference for the bound on its peak or for the logit drift of Section 10.5.

TRANSLATING THE LITERATURE. Zhang et al. (2020b) write f and x for our $\mathcal{L}$ and θ, and clipped gradient descent as $x_{k+1} = x_k - h_c \nabla f(x_k)$ with $h_c = \min\{\eta_c, \gamma\eta_c/\|\nabla f(x_k)\|\}$: their η_c is our η, their γ is a threshold on the gradient norm, our γ/η, and their $\gamma\eta_c$ is our γ; their M is ours. Zhang et al. (2020c) write α for our moment order p. PaLM's z-loss is our $\alpha(\log Z)^2$ with $\alpha = 10^{-4}$. Kimi K2's report writes τ for our $\bar{s}$ and $S^h_{\max}$ for our $s_{\max}$ of head h; its factor $\min\{1, \tau/S^h_{\max}\}$ is our ρ^2. Practitioners' `max_norm` is our $\tau = \gamma/\eta$.

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Part IV

Noise, Schedules and Equilibria

Stochastic gradients set a floor under the loss that rises with the step size. The batch size trades noise for steps, the learning-rate schedule removes the floor at the end of training, and weight decay drives normalized layers to an equilibrium whose angular speed is the effective learning rate.

11 Noise and Batch Size

The noisy quadratic model and the critical batch size

IN THE WILD — BATCH-SIZE SCALING RULES

IN THIS CHAPTER. Up to a critical batch size, doubling the batch and the learning rate together halves the number of steps of a training run; beyond it, larger batches buy almost nothing, and momentum and preconditioners raise the critical batch size. The noisy quadratic model, a quadratic loss whose gradients carry noise of covariance Σ/B, is the smallest idealization with both regimes. We solve it exactly and prove that the steps to a target loss, at the best constant step size, are asymptotic to $E_{\min}/B$ at small batch sizes, with a step size proportional to B, and equal a constant $T_{\min}$ at large ones, with a step size near the stability limit $2/S$; the critical batch size $E_{\min}/T_{\min}$ is of order $\operatorname{tr}\Sigma/(S\varepsilon)$ for a target ε. Preconditioning raises this scale, and when the noise lives in the bulk of the spectrum no diagonal preconditioner lowers the small-batch cost by much. With its fixed landscape and noise, the model cannot say how the critical batch size moves with the data and the model.

11.1 In the wild: batch sizes and scaling rules

A data-parallel run averages the gradients of B examples per step; the average is unbiased, with covariance Σ/B for the per-example covariance Σ. While the batch fits the hardware a step takes about the same time whatever B is, so the number of steps T measures wall-clock time and the number of examples $E = BT$ the cost in data and computation.[1] A larger batch pays off only through fewer steps. How should the learning rate change with the batch size, and how large can the batch be before it stops buying steps? We say *learning rate* in the practice sections and *step size* in the theory; both name η.

IN THE WILD · ResNet-50 on ImageNet, Goyal et al. (2017); Adam, Malladi et al. (2022); GPT-3, Brown et al. (2020). For SGD, the *linear scaling rule*: when the batch size is multiplied by k, multiply the learning rate by k. Goyal et al. trained ResNet-50 with the learning rate $0.1 \cdot B/256$ for batch sizes up to 8192, after a linear warmup over five epochs. For Adam, the *square-root rule*: multiply the learning rate by $\sqrt{k}$, and $1-\beta_1$ and $1-\beta_2$ by k. Both are used up to a *critical batch size*, beyond which larger batches save few steps, and batch sizes are chosen near it. Language models train on batches of millions of tokens

[1] For a language model an example is a sequence or a token, and batch sizes are quoted in tokens. Nothing here depends on the unit, as long as B and E are counted in the same one.

(3.2 million for the largest GPT-3 model, in 2020), often reached by increasing the batch size early in training.

The linear rule is discussed by Krizhevsky (2014); Goyal et al. (2017) made it the recipe that trained ResNet-50 on ImageNet in an hour. Hoffer et al. (2017) argued for a learning rate proportional to $\sqrt{B}$ for SGD, which keeps the covariance of the update fixed, Malladi et al. (2022) derived the square-root rule for adaptive methods, and Smith et al. (2018) replaced a decay of the learning rate by an increase of the batch size.

OBSERVATION 11.1 (Three regimes of data parallelism; Shallue et al. 2019). With the learning rate and the other hyperparameters retuned for every batch size, the number of steps needed to reach a fixed out-of-sample error falls like $1/B$ at small batch sizes (*perfect scaling*), then more slowly, and finally not at all. The shape is the same across the image and text workloads studied; where perfect scaling ends differs widely between workloads and depends on the model, the data set, and the optimizer.

OBSERVATION 11.2 (The gradient noise scale; McCandlish et al. 2018). Across tasks from image classification to reinforcement learning, the critical batch size is roughly predicted by the gradient noise scale $\operatorname{tr}\Sigma/\|\nabla\mathcal{L}\|_2^2$ measured during training, which grows as the loss falls and is larger for more complex tasks. The steps T and examples E needed to reach a target fit the relation $(T/T_{\min}-1)(E/E_{\min}-1)=1$.

OBSERVATION 11.3 (Optimizers; Zhang et al. 2019). On image classification and language modeling workloads, the preconditioned optimizers Adam and K-FAC (Chapter 7) reach much larger critical batch sizes than momentum SGD.

OBSERVATION 11.4 (Language models; Kaplan et al. 2020, Zhang et al. 2025, Marek et al. 2025). In language-model pretraining the critical batch size grows as the loss falls, depending on the loss rather than directly on the model size (Kaplan et al.), and in controlled sweeps it grows with the amount of data more than with the model size (Zhang et al.). Batches down to one sequence train stably, with Adam's β_2 set to keep its averaging horizon fixed in tokens, and at such batch sizes plain SGD comes close to Adam; the gap grows with the batch size (Marek et al.).

These observations rest on expensive sweeps over a few workloads each;[2] the noise scale predicts the critical batch size only roughly, and McCandlish et al.'s paper is an unrefereed preprint. Why does doubling the batch halve the steps up to some batch size and not beyond, what sets that batch size, and how do momentum and preconditioners move it?

[2] At batch size one, the half-life rule of Marek et al. (2025) gives $\beta_2 = 0.9999$ in their runs.

11.2 The idealization: the noisy quadratic model

Assumption 11.5 (The noisy quadratic model). (i) The loss is $\mathcal{L}(\theta) = \frac{1}{2}\theta^\top H\theta$ on $\mathbb{R}^d$, with H symmetric positive definite, eigenvalues $h_1 \geq \cdots \geq h_d > 0$, and sharpness $S = h_1 = \lambda_{\max}(H)$. We work in an orthonormal eigenbasis, so $H = \operatorname{diag}(h_1, \ldots, h_d)$, $\theta^\star = 0$, and $\mathcal{L}^\star = 0$.

(ii) At step t the optimizer receives $g_t = H\theta_t + \zeta_t$, where, conditionally on the σ-field $\mathcal{F}_t$ generated by $\zeta_0, \ldots, \zeta_{t-1}$, $\mathbb{E}[\zeta_t \mid \mathcal{F}_t] = 0$ and $\mathbb{E}[\zeta_t\zeta_t^\top \mid \mathcal{F}_t] = \Sigma/B$, for a fixed positive semidefinite $\Sigma \neq 0$ and a batch size $B > 0$. We write $c_i = \Sigma_{ii}$ in the eigenbasis of H.

(iii) The initialization θ_0 is deterministic; $\Delta_{0,i} = \frac{1}{2}h_i\theta_{0,i}^2$ is the initial loss of mode i, and $\Delta_0 = \mathcal{L}(\theta_0) = \sum_i \Delta_{0,i}$.

The average of B independent per-example gradients satisfies (ii); in the theory B is a positive real number that enters only through Σ/B, and the noise need be neither Gaussian nor diagonal in the eigenbasis of H. The standard choice is noise proportional to curvature, $\Sigma = \sigma^2 H$, as in Zhang et al. (2019): at the minimizer of a well-specified likelihood the covariance of the per-example gradients is the Fisher information, which equals the Hessian of the population loss there.[3] The running example, the *harmonic model*, has $d = 1000$, $h_i = 1/i$, $\Sigma = H$, $\theta_0 = (1, \ldots, 1)$ and the target $\varepsilon = 0.05\,\Delta_0$, with $\Delta_0 \approx 3.74$ and effective rank $\operatorname{tr} H/S \approx 7.49$.

[3] The Hessian spectra of trained networks have a few outlier eigenvalues above a bulk (Ghorbani et al. 2019). Proposition 11.14 uses this shape.

DICTIONARY

IN THE THEORY	IN A TRAINING RUN
the quadratic $\frac{1}{2}\theta^\top H\theta$, eigenvalues h_i	the loss near the region the run is in; its Hessian spectrum (a bulk and outliers)
the sharpness $S = h_1$, the limit $\eta < 2/S$	the largest stable learning rate; the edge of stability (Chapter 8)
the covariance Σ, often $\sigma^2 H$	the per-example gradient covariance, close to the Fisher information near a minimum
$T(B)$ and $E(B) = B\,T(B)$	steps and examples to a target loss, at a tuned constant learning rate
the floor $\mathcal{R}_\infty \approx \eta \operatorname{tr}\Sigma/(4B)$	the plateau at a constant learning rate, removed by a decay (Chapter 12)
the noise-limited regime, $\eta \approx \gamma^\star B$	perfect scaling and the linear scaling rule
the curvature-limited regime, η near $2/S$	maximal data parallelism
$B_{\text{crit}} = E_{\min}/T_{\min}$	the critical batch size
a preconditioner P, $S_P = \lambda_{\max}(P^{-1}H)$	Adam's $\sqrt{\hat{v}}$, Shampoo, or K-FAC, frozen as a function of H

WHAT WAS THROWN AWAY. The landscape is fixed, so there is no feature learning and no progressive sharpening; the noise does not depend on the parameters, although near interpolation it shrinks with the loss (Exer-

cise 11.5); every step sees fresh examples, so there are no epochs and no generalization gap; the step size is constant; and the object is the expected loss. What remains is the smallest model with the two effects of the step size that the question is about: curvature, which limits the step size through stability, and noise, whose effect grows with the step size and falls with B.

WHAT ONE HAD TO NOTICE. The batch size enters only through the noise level, while the step size enters twice: in the speed ηh_i of every mode and in a noise floor proportional to η/B. Stability, $\eta < 2/S$, does not depend on B; the floor, about $\eta \operatorname{tr}\Sigma/(4B)$, must lie below the target, which caps η at a multiple of B. When the floor binds, the best step size is proportional to B and the steps to $1/B$; when stability binds, the steps do not depend on B. The critical batch size is where the two meet.

11.3 The theory: exact dynamics

Theorem 11.6 (Exact dynamics of SGD in the noisy quadratic model). *Let Assumption 11.5 hold, let $\theta_{t+1} = \theta_t - \eta g_t$ with a constant step size $\eta > 0$, and write $m_{t,i} = \mathbb{E}\theta_{t,i}^2$ and $\mathcal{R}_t(\eta, B) = \mathbb{E}\mathcal{L}(\theta_t)$.*

(i) For every $t \geq 0$ and i, $m_{t+1,i} = (1-\eta h_i)^2 m_{t,i} + \eta^2 c_i/B$.

(ii) If $0 < \eta < 2/S$, then for every $t \geq 0$

$$\mathcal{R}_t(\eta, B) = \sum_{i=1}^{d} \Big[F_i + (\Delta_{0,i} - F_i)(1-\eta h_i)^{2t} \Big], \qquad F_i = \frac{\eta c_i}{2B(2-\eta h_i)}, \tag{11.1}$$

which converges geometrically to the noise floor

$$\mathcal{R}_\infty(\eta, B) = \frac{\eta}{2B} \sum_{i=1}^{d} \frac{c_i}{2-\eta h_i} \in \Big[\frac{\eta \operatorname{tr}\Sigma}{4B}, \frac{\eta \operatorname{tr}\Sigma}{2B(2-\eta S)} \Big]. \tag{11.2}$$

(iii) If $\eta S \geq 2$ and $c_1 > 0$, or if $\eta S > 2$ and $\theta_{0,1} \neq 0$, then $\mathcal{R}_t(\eta, B) \to \infty$.

(iv) The expected loss depends on the noise only through the conditional means and variances of its coordinates in the eigenbasis of H.

PROOF. The plan is to show that each coordinate is a scalar linear recursion driven by noise uncorrelated with its past, and to solve it. In the eigenbasis, $\theta_{t+1,i} = (1-\eta h_i)\theta_{t,i} - \eta \zeta_{t,i}$. The coordinate $\theta_{t,i}$ is a function of $\zeta_0, \dots, \zeta_{t-1}$, hence $\mathcal{F}_t$-measurable, and Assumption 11.5(ii) gives $\mathbb{E}[\zeta_{t,i} \mid \mathcal{F}_t] = 0$ and $\mathbb{E}[\zeta_{t,i}^2 \mid \mathcal{F}_t] = c_i/B$. Squaring the update and conditioning on $\mathcal{F}_t$, the cross term vanishes:

$$\mathbb{E}\big[\theta_{t+1,i}^2 \,\big|\, \mathcal{F}_t\big] = (1-\eta h_i)^2 \theta_{t,i}^2 + \eta^2 c_i/B.$$

Taking expectations gives (i), and by induction every $m_{t,i}$ is finite. For (ii), put $\rho_i = (1-\eta h_i)^2$. Since $0 < \eta h_i \leq \eta S < 2$, we have $0 \leq \rho_i < 1$ and $1 - \rho_i = \eta h_i(2 - \eta h_i) > 0$, so (i) has the fixed point $m_i^\infty = \eta^2 c_i/(B(1-\rho_i)) = \eta c_i/(Bh_i(2-\eta h_i))$, and $m_{t,i} - m_i^\infty = \rho_i^t (m_{0,i} - m_i^\infty)$. Multiplying by $h_i/2$, with

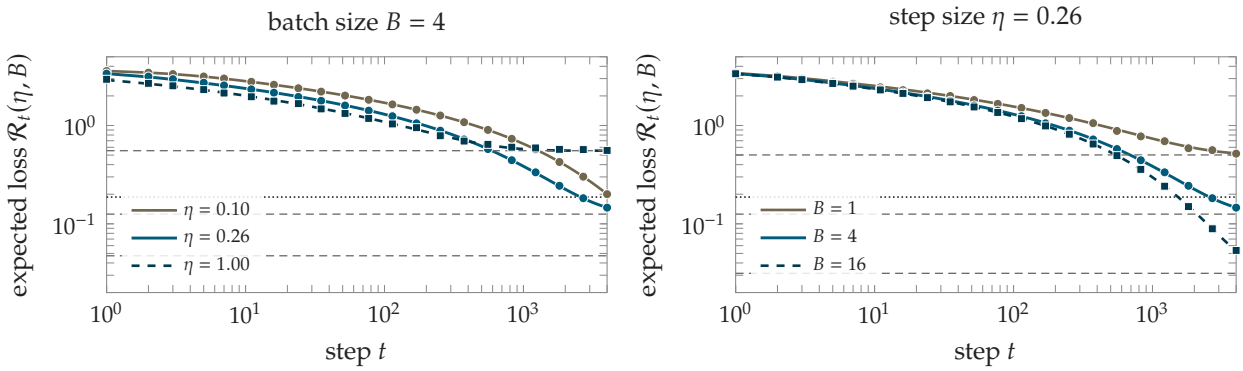


Figure 11.1. The expected loss of SGD in the harmonic model ($d = 1000$, $h_i = 1/i$, $\Sigma = H$, $\theta_0 = (1, \dots, 1)$). Curves: the exact formula (11.1); dashed: the floors (11.2); dotted: the target $\varepsilon = 0.05\,\Delta_0$. Points: averages of 64 independent runs with Gaussian noise of covariance Σ/B (seed fixed in `11-noise-curves.py`), within 4.4% of the curves, as the Monte Carlo error allows. Left: at a fixed batch size a larger step is faster and has a higher floor; $\eta = 0.26$ is near the step size that reaches the target first. Right: at a fixed step size the batch size changes only the floor.

$\frac{1}{2}h_i m_i^\infty = F_i$ and $\frac{1}{2}h_i m_{0,i} = \Delta_{0,i}$, gives the ith term of (11.1); the convergence is geometric with ratio $\max_i \rho_i < 1$, and the bounds in (11.2) follow from $2-\eta S \le 2-\eta h_i \le 2$. For (iii), $\rho_1 \ge 1$ when $\eta S \ge 2$, so (i) gives $m_{t,1} \ge t\eta^2 c_1/B$ and also $m_{t,1} \ge \rho_1^t \theta_{0,1}^2$; in both cases $\mathcal{R}_t \ge \frac{1}{2}S m_{t,1} \to \infty$. Part (iv) holds because the argument used nothing else about ζ_t. ∎

Each mode relaxes geometrically from its initial loss to its share F_i of the floor, at the rate $(1-\eta h_i)^2 \approx e^{-2\eta h_i}$ per step; when $\Delta_{0,i} > F_i$ for every i, as at all step sizes that matter here, the expected loss decreases strictly. The step size sets both the speed and the floor, which for $\eta S \ll 1$ is about $\eta \operatorname{tr}\Sigma/(4B)$; the batch size sets only the floor (Figure 11.1).

A preconditioner that is a function of H changes only the coordinates.

Lemma 11.7 (Preconditioning is a change of variables). *Let $P = \operatorname{diag}(1/\omega_1, \dots, 1/\omega_d)$ with every $\omega_i > 0$, and let $\theta_{t+1} = \theta_t - \eta P^{-1} g_t$. Then $\theta'_t = P^{1/2}\theta_t$ is SGD with step size η on the noisy quadratic model with Hessian $H' = P^{-1/2}HP^{-1/2}$ and noise covariance $\Sigma' = P^{-1/2}\Sigma P^{-1/2}$, with $\mathcal{L}(\theta_t) = \frac{1}{2}\theta_t'^{\top} H' \theta'_t$; in the eigenbasis, $h'_i = \omega_i h_i$, $c'_i = \omega_i c_i$, and $\Delta'_{0,i} = \Delta_{0,i}$. In particular Theorem 11.6 holds with these substitutions: preconditioned SGD is stable for $\eta < 2/S_P$, with the preconditioned sharpness $S_P = \lambda_{\max}(P^{-1}H) = \max_i \omega_i h_i$, and its floor lies between $\eta \operatorname{tr}(P^{-1}\Sigma)/(4B)$ and $\eta \operatorname{tr}(P^{-1}\Sigma)/(2B(2-\eta S_P))$.*

Proof. Multiplying the update by $P^{1/2}$ and writing $\theta_t = P^{-1/2}\theta'_t$ gives $\theta'_{t+1} = \theta'_t - \eta(H'\theta'_t + \zeta'_t)$ with $\zeta'_t = P^{-1/2}\zeta_t$, whose conditional mean is 0 and conditional covariance Σ'/B; and $\frac{1}{2}\theta_t^{\top} H \theta_t = \frac{1}{2}\theta_t'^{\top} H' \theta'_t$. Since $P^{-1/2} = \operatorname{diag}(\sqrt{\omega_i})$, we get $h'_i = \omega_i h_i$, $c'_i = \omega_i c_i$, $\theta'_{0,i} = \theta_{0,i}/\sqrt{\omega_i}$, so $\Delta'_{0,i} = \Delta_{0,i}$, and $\operatorname{tr}\Sigma' = \sum_i \omega_i c_i = \operatorname{tr}(P^{-1}\Sigma)$. ∎

The powers $P = H^p$, $0 \le p \le 1$, interpolate between SGD and Newton's method; $p = \frac{1}{2}$ is what Adam's denominator becomes when the noise dominates and $\Sigma \propto H$ (Section 11.4). A preconditioner that does not commute with H also yields a noisy quadratic model (Exercise 11.3).

Momentum couples each coordinate to its previous value; Chapter 5 computes the stationary variance of the heavy ball on one quadratic direction, and summed over the modes it gives the floor.

Proposition 11.8 (Momentum in the noisy quadratic model). *Let Assumption 11.5 hold, and let $\theta_{t+1} = \theta_t - \eta g_t + \beta(\theta_t - \theta_{t-1})$ with $\eta > 0$, $0 \le \beta < 1$, and $\theta_{-1} = \theta_0$.*

(i) The expected loss converges for every θ_0 and every Σ if and only if $\eta S < 2 + 2\beta$, and then, with the effective step size $\eta_e = \eta/(1-\beta)$, its limit is

$$\mathcal{R}_\infty^\beta(\eta, B) = \sum_{i=1}^{d} \frac{\eta_e c_i}{2B\left(2 - \frac{1-\beta}{1+\beta}\eta_e h_i\right)}.$$

Each term is at most the corresponding term of SGD's floor at step size η_e, when that step is stable, and agrees with it to first order in η_e.

(ii) (Proof sketched.) *For fixed β, let $T_\beta(B)$ be the steps to the target ε at the best $\eta \in (0, (2+2\beta)/S)$. Then $B\,T_\beta(B) \to E_{\min}$ as $B \to 0$, the same limit as for SGD in Theorem 11.11.*

Proof. The plan for (i) is to treat each mode as a second-order linear recursion. Mode i evolves by $x_{t+1} = a_i x_t - \beta x_{t-1} - \eta\zeta_{t,i}$ with $a_i = 1 + \beta - \eta h_i$. By the Jury test (Lemma A.14) both roots of $z^2 - a_i z + \beta$ lie in the open unit disk if and only if $|a_i| < 1 + \beta$, that is, $0 < \eta h_i < 2 + 2\beta$. If this holds for every i, then the means tend to zero and the second moments converge. If it fails for $i = 1$, then a root of modulus at least 1 exists, the impulse response of mode 1 does not tend to zero, and the variance of mode 1, which is $\eta^2 c_1/B$ times the sum of its squared impulse response, diverges when $c_1 > 0$. For a stable mode, the stationary moments $\gamma_0 = \mathbb{E}x_t^2$ and $\gamma_1 = \mathbb{E}x_t x_{t-1}$ satisfy $\gamma_1 = a_i\gamma_0 - \beta\gamma_1$ and $\gamma_0 = (a_i^2 + \beta^2)\gamma_0 - 2a_i\beta\gamma_1 + \eta^2 c_i/B$ (Lemma A.15), because the noise is uncorrelated with x_t and x_{t-1}; eliminating $\gamma_1 = a_i\gamma_0/(1+\beta)$ and using $(1+\beta)^2 - a_i^2 = \eta h_i(2 + 2\beta - \eta h_i)$,

$$\tfrac{1}{2} h_i \gamma_0 = \frac{\eta c_i (1+\beta)}{2B(1-\beta)\big(2(1+\beta) - \eta h_i\big)},$$

which is the stated term after $\eta = (1-\beta)\eta_e$. Since $(1-\beta)/(1+\beta) \le 1$, its denominator is at least $2B(2 - \eta_e h_i)$, and both equal $\eta_e c_i/(4B)$ to first order. ∎

Sketch of the proof of (ii). The variance of mode i is $\eta^2 c_i/B$ after one step and never decreases, so a step size that reaches the target has $\eta^2\,\mathrm{tr}(H\Sigma)/(2B) < \varepsilon$, and $\eta \to 0$. For small η the roots are $\mu_+ = 1 - \eta h_i/(1-\beta) + O(\eta^2)$ and $\mu_- = \beta + O(\eta)$; with $\theta_{-1} = \theta_0$ the mean of mode i is $\theta_{0,i}\mu_+^t(1 + O(\eta))$, and the squared impulse response sums to

the variance $\eta c_i(1-\mu_+^{2t})(1+O(\eta))/(2Bh_i(1-\beta)) + O(\eta^2 c_i/B)$, uniformly in t. With $\eta = (1-\beta)\gamma B$ and $t = E/B$ the expected loss is thus SGD's limit $D(\gamma E) + \frac{\gamma}{4}Q(\gamma E)$ (Definition 11.10) up to relative errors $O(\eta)$ in γE and in both terms and an absolute error $O(\gamma\eta)$, and the argument of Theorem 11.11(iii) applies. Omitted is the bookkeeping of these errors, which replaces facts (a)–(c); we know of no published proof, and Zhang et al. (2019) state the conclusion as a prediction of the model. ■

The benefit of momentum is at large batch sizes, where effective step sizes up to $(2+2\beta)/((1-\beta)S)$, which is $38/S$ for $\beta = 0.9$, are stable and the iteration accelerates (Chapter 5). The one-step model from which McCandlish et al. (2018) derived their noise scale is exact in the noisy quadratic model.

Proposition 11.9 (The one-step model is exact). *Let Assumption 11.5 hold, fix θ with $G = \nabla\mathcal{L}(\theta) = H\theta \neq 0$, and let $g = G + \zeta$ with $\mathbb{E}\zeta = 0$ and $\mathbb{E}\zeta\zeta^\top = \Sigma/B$. Then*

$$\mathbb{E}\mathcal{L}(\theta - \eta g) = \mathcal{L}(\theta) - \eta\|G\|_2^2 + \frac{\eta^2}{2}\left(G^\top H G + \frac{\operatorname{tr}(H\Sigma)}{B}\right), \tag{11.3}$$

which is smallest at $\eta = \eta_{\max}/(1 + B_{\text{noise}}/B)$, where it equals $\mathcal{L}(\theta) - \Delta_{\max}/(1 + B_{\text{noise}}/B)$, with $\eta_{\max} = \|G\|_2^2/G^\top H G$, $\Delta_{\max} = \|G\|_2^4/(2G^\top H G)$ and $B_{\text{noise}} = \operatorname{tr}(H\Sigma)/G^\top H G$.

(i) The gradient noise scale *$B_{\text{simple}} = \operatorname{tr}\Sigma/\|G\|_2^2$ is at least $\operatorname{tr}\Sigma/(2S\mathcal{L}(\theta))$, with equality if and only if $H\theta = S\theta$; if $H = hI$, then $B_{\text{noise}} = B_{\text{simple}} = \operatorname{tr}\Sigma/(2h\mathcal{L}(\theta))$.*

(ii) $1/S \le \eta_{\max} \le 1/h_d$, so the one-step optimum can exceed the stability limit $2/S$.

(iii) At the stationary second moments of SGD with $\eta < 2/S$, the measured scales $\operatorname{tr}\Sigma/\mathbb{E}\|G\|_2^2$ and $\operatorname{tr}(H\Sigma)/\mathbb{E}G^\top H G$ equal $2B\operatorname{tr}\Sigma/(\eta\operatorname{tr}(H\Sigma))$ and $2B\operatorname{tr}(H\Sigma)/(\eta\operatorname{tr}(H^2\Sigma))$ up to factors in $[1-\eta S/2, 1]$.

Proof. The plan is to expand the quadratic loss exactly and to minimize over η. Since $\mathcal{L}(\theta - \eta g) = \mathcal{L}(\theta) - \eta G^\top g + \frac{1}{2}\eta^2 g^\top H g$, with $\mathbb{E}G^\top g = \|G\|_2^2$ and $\mathbb{E}g^\top H g = G^\top H G + \operatorname{tr}(H\Sigma)/B$, we get (11.3), a quadratic in η minimized at $\|G\|_2^2/(G^\top H G + \operatorname{tr}(H\Sigma)/B)$ with the stated minimum. (i) From $H^2 \preceq SH$, $\|G\|_2^2 = \theta^\top H^2\theta \le 2S\mathcal{L}(\theta)$, with equality if and only if $H(S-H)\theta = 0$, that is, $H\theta = S\theta$; for $H = hI$ substitute $G = h\theta$. (ii) $1/\eta_{\max}$ is the Rayleigh quotient of H at G. (iii) At stationarity $\mathbb{E}\theta_i^2 = \eta c_i/(Bh_i(2-\eta h_i))$, so $\mathbb{E}\|G\|_2^2 = \frac{\eta}{B}\sum_i h_i c_i/(2-\eta h_i)$ and $\mathbb{E}G^\top H G = \frac{\eta}{B}\sum_i h_i^2 c_i/(2-\eta h_i)$; use $2 - \eta S \le 2 - \eta h_i \le 2$. ■

By (iii), a noise scale measured at the floor of a run reports the run's own ratio B/η, up to factors set by the spectrum (Section 11.5).

11.4 The theory: the critical batch size

Definition 11.10 (Steps and examples to a target). Let Assumption 11.5 hold and fix a target $0 < \varepsilon < \Delta_0$. For $B > 0$, the *steps to target* $T(B)$ is the smallest $t \in \mathbb{N}$ for which some $\eta \in (0, 2/S)$ gives $\mathcal{R}_t(\eta, B) < \varepsilon$, and $E(B) = B\,T(B)$. Without noise, $D_t(\eta) = \sum_i \Delta_{0,i}(1 - \eta h_i)^{2t}$ is the loss of gradient descent, and $T_{\min}$ is the smallest t for which some $\eta \in (0, 2/S)$ gives $D_t(\eta) < \varepsilon$. In continuous time let

$$D(\tau) = \sum_{i=1}^{d} \Delta_{0,i} e^{-2h_i\tau}, \qquad Q(\tau) = \sum_{i=1}^{d} c_i\big(1 - e^{-2h_i\tau}\big),$$

let τ_ε solve $D(\tau_\varepsilon) = \varepsilon$, and let

$$E_{\min} = \min_{\tau > \tau_\varepsilon} \frac{\tau\, Q(\tau)}{4\big(\varepsilon - D(\tau)\big)}, \tag{11.4}$$

attained at some $\tau^\star$, with $\gamma^\star = 4(\varepsilon - D(\tau^\star))/Q(\tau^\star)$, so that $E_{\min} = \tau^\star/\gamma^\star$. The *critical batch size* is $B_{\text{crit}} = E_{\min}/T_{\min}$.

The function D is the loss of gradient flow at time τ, and Q the noise accumulated by then. Since D decreases continuously and strictly from Δ_0 to 0, τ_ε is well defined, and the objective in (11.4) tends to infinity as $\tau \downarrow \tau_\varepsilon$ and as $\tau \to \infty$, so the minimum exists. It is a budget: the flow time τ leaves $\varepsilon - D(\tau)$ for the noise, the largest step per example whose noise $\frac{\gamma}{4}Q(\tau)$ fits is $4(\varepsilon - D(\tau))/Q(\tau)$, and τ divided by it is the number of examples. In the harmonic model $\tau_\varepsilon = 350$, $\tau^\star = 677$, $\gamma^\star = 0.0678$ and $E_{\min} = 9987$; two thirds of the budget go to noise.

Theorem 11.11 (Two regimes and the critical batch size). *Let Assumption 11.5 hold, fix a target $0 < \varepsilon < \Delta_0$, and use the notation of Definition 11.10.*

(i) (Monotonicity.) *$T(B)$ is finite and nonincreasing in B, and $T(B) \geq T_{\min}$ for every B.*

(ii) (Curvature-limited regime.) *There is $B_1 < \infty$ such that $T(B) = T_{\min}$ for every $B \geq B_1$.*

(iii) (Noise-limited regime.) *$B\,T(B) \to E_{\min}$ as $B \to 0$. Quantitatively, if $\gamma^\star BS \leq \frac{1}{2}$, then $B\,T(B) \leq E_{\min}(1 + \gamma^\star BS)^2(1 + \gamma^\star B/\tau^\star) + B$; and if $8BS^2\varepsilon \leq \operatorname{tr}(H\Sigma)$, then $B\,T(B) \geq E_{\min}\big(1 + S\sqrt{2B\varepsilon/\operatorname{tr}(H\Sigma)}\big)^{-2}$.*

(iv) (Linear scaling.) *Let $\eta_B \in (0, 2/S)$ satisfy $\mathcal{R}_{T(B)}(\eta_B, B) < \varepsilon$. Every limit point of η_B/B as $B \to 0$ equals $4(\varepsilon - D(\tau))/Q(\tau)$ for a minimizer τ of (11.4); if the minimizer is unique, then $\eta_B/B \to \gamma^\star$.*

Proof. The plan is to split the expected loss into a deterministic part and a noise part proportional to $1/B$: at large batch sizes the noise part vanishes, and at small ones the step size is forced to zero, so that the discrete recursion approaches its continuous-time version, which is (11.4). For $0 < \eta < 2/S$,

Theorem 11.6(ii) gives

$$\mathcal{R}_t(\eta, B) = D_t(\eta) + \frac{M_t(\eta)}{B}, \qquad M_t(\eta) = \sum_{i=1}^{d} BF_i\big(1 - (1-\eta h_i)^{2t}\big) \geq 0,$$

and $BF_i = \eta c_i/(2(2-\eta h_i))$ does not depend on B. We use three facts: (a) $(1-x)^2 \leq e^{-2x}$ for $0 \leq x \leq 1$; (b) $(1-x)^2 \geq e^{-2x(1+x)}$ for $0 \leq x \leq \frac{1}{2}$, since $\log(1-x) \geq -x - x^2$ there; (c) Q is nondecreasing and concave with $Q(0) = 0$, so $Q(a\tau) \leq aQ(\tau)$ and $Q(\tau/a) \geq Q(\tau)/a$ for $a \geq 1$.

(i) As $\eta \to 0$ the floor, at most $\eta \operatorname{tr}\Sigma/(2B(2-\eta S))$, tends to 0; since $\mathcal{R}_t$ converges to the floor, $T(B)$ is finite. Since $M_t \geq 0$, $\mathcal{R}_t(\eta, B)$ is nonincreasing in B, so a pair (η, t) that reaches the target at batch size B reaches it at every larger one, and T is nonincreasing. Since $\mathcal{R}_t \geq D_t$, every such pair has $D_t(\eta) < \varepsilon$, so $T(B) \geq T_{\min}$; and $T_{\min}$ is finite: $D_t(1/S) \leq D(t/S)$ by (a).

(ii) Choose $\eta_\infty \in (0, 2/S)$ with $\delta = \varepsilon - D_{T_{\min}}(\eta_\infty) > 0$, and let $B_1 = 1 + M_{T_{\min}}(\eta_\infty)/\delta$. For $B \geq B_1$, $\mathcal{R}_{T_{\min}}(\eta_\infty, B) < \varepsilon$, so $T(B) \leq T_{\min}$, and (i) gives equality.

(iii) *Upper bound.* Let $\gamma^\star BS \leq \frac{1}{2}$ and $0 < \gamma \leq \gamma^\star$, put $\eta = \gamma B$, so $\eta S \leq \frac{1}{2}$, and let $t = \lceil \tau^\star/\eta \rceil$, so $\tau^\star \leq \eta t < \tau^\star + \eta$. By (a), $D_t(\eta) \leq D(\eta t) \leq D(\tau^\star)$. In the noise part, $BF_i \leq \frac{1}{4}\eta c_i(1+\eta h_i)$ because $(1+x)(2-x) \geq 2$ on $[0,1]$, and by (b) and the concavity of $u \mapsto 1 - e^{-u}$,

$$1 - (1-\eta h_i)^{2t} \leq 1 - e^{-2\eta h_i t(1+\eta h_i)} \leq (1+\eta h_i)\big(1 - e^{-2h_i\eta t}\big).$$

Summing over i and using (c) in the form $Q(\eta t) \leq Q(\tau^\star + \eta) \leq (1 + \eta/\tau^\star)Q(\tau^\star)$,

$$\frac{M_t(\eta)}{B} \leq (1+\eta S)^2\,\frac{\gamma}{4}\,Q(\eta t) \leq (1+\gamma^\star BS)^2\Big(1 + \frac{\gamma^\star B}{\tau^\star}\Big)\frac{\gamma}{4}\,Q(\tau^\star).$$

Since $\varepsilon - D(\tau^\star) = \frac{1}{4}\gamma^\star Q(\tau^\star)$, the target is reached at step t whenever $\gamma(1+\gamma^\star BS)^2(1+\gamma^\star B/\tau^\star) < \gamma^\star$. For every such γ, $T(B) \leq \tau^\star/(\gamma B) + 1$; letting γ increase to $\gamma^\star(1+\gamma^\star BS)^{-2}(1+\gamma^\star B/\tau^\star)^{-1}$ gives the upper bound, since $\tau^\star/\gamma^\star = E_{\min}$.

Lower bound. Let $t = T(B)$ and $\eta \in (0, 2/S)$ with $\mathcal{R}_t(\eta, B) < \varepsilon$, and put $\gamma = \eta/B$ and $\tau = \eta t$. Since $\mathcal{R}_0 = \Delta_0 > \varepsilon$, $t \geq 1$, and Theorem 11.6(i) gives $m_{t,i} \geq \eta^2 c_i/B$; so $\varepsilon > \mathcal{R}_t \geq \eta^2 \operatorname{tr}(H\Sigma)/(2B)$, where $\operatorname{tr}(H\Sigma) = \sum_i h_i c_i > 0$ because $\Sigma \neq 0$. If $8BS^2\varepsilon \leq \operatorname{tr}(H\Sigma)$, then $\eta S \leq S\sqrt{2B\varepsilon/\operatorname{tr}(H\Sigma)} \leq \frac{1}{2}$, and (b), (a) and $BF_i \geq \frac{1}{4}\eta c_i$ give, with $\tau' = \tau(1+\eta S)$,

$$\varepsilon > \mathcal{R}_t(\eta, B) \geq D(\tau') + \frac{\gamma}{4}\,Q(\tau).$$

Hence $\tau' > \tau_\varepsilon$ and $\gamma < 4(\varepsilon - D(\tau'))/Q(\tau)$. Using $Q(\tau) = Q(\tau'/(1+\eta S)) \geq Q(\tau')/(1+\eta S)$,

$$B\,T(B) = \frac{\tau}{\gamma} > \frac{\tau\,Q(\tau)}{4\big(\varepsilon - D(\tau')\big)} \geq \frac{\tau'\,Q(\tau')}{4(1+\eta S)^2\big(\varepsilon - D(\tau')\big)} \geq \frac{E_{\min}}{(1+\eta S)^2}, \quad (11.5)$$

which is the lower bound. Together the two bounds give $B\,T(B) \to E_{\min}$.

(iv) Let $B_k \to 0$ and write $\eta_k = \eta_{B_k}$, $t_k = T(B_k)$, $\gamma_k = \eta_k/B_k$, $\tau_k = \eta_k t_k$, and $\tau_k' = \tau_k(1+\eta_k S)$. For large k the lower-bound argument applies: $\eta_k \to 0$, $\tau_k' > \tau_\varepsilon$, $\gamma_k < 4(\varepsilon - D(\tau_k'))/Q(\tau_k)$, and $\tau_k/\gamma_k = B_k t_k \to E_{\min}$. Since $\tau_k \ge \frac{2}{3}\tau_\varepsilon$ and $B_k t_k > \tau_k Q(\tau_k)/(4\varepsilon) \ge \tau_k Q(\frac{2}{3}\tau_\varepsilon)/(4\varepsilon)$, the sequences τ_k, γ_k, and $1/\gamma_k$ are bounded. Along a subsequence $\gamma_k \to \bar\gamma > 0$ and $\tau_k \to \bar\tau = \bar\gamma E_{\min}$, and in the limit $\bar\gamma \le 4(\varepsilon - D(\bar\tau))/Q(\bar\tau)$, which forces $D(\bar\tau) < \varepsilon$; therefore

$$E_{\min} = \frac{\bar\tau}{\bar\gamma} \ge \frac{\bar\tau\, Q(\bar\tau)}{4\big(\varepsilon - D(\bar\tau)\big)} \ge E_{\min}.$$

Equality holds throughout: $\bar\tau$ is a minimizer and $\bar\gamma = 4(\varepsilon - D(\bar\tau))/Q(\bar\tau)$. If the minimizer is unique, then every convergent subsequence has the limit $\gamma^\star$, so the bounded sequence γ_k converges to $\gamma^\star$. ∎

Parts (iii) and (iv) are the *noise-limited regime*: the examples needed become independent of the batch size and the best step size approaches $\gamma^\star B$, the *linear scaling rule*; the bounds show that the regime lasts while $\gamma^\star BS \ll 1$. Part (ii) is the *curvature-limited regime*, where stability alone limits the step size. The critical batch size is the ratio of the two asymptotes.

Corollary 11.12 (The scale of the critical batch size). *In the setting of Theorem 11.11, let τ_δ solve $D(\tau_\delta) = \delta$ for $0 < \delta < \Delta_0$; let D^{sl} be the sum defining D restricted to the modes with $h_i \le S/4$, and $\tau_\varepsilon^{\mathrm{sl}} = \inf\{\tau \ge 0 : D^{\mathrm{sl}}(\tau) < \varepsilon\}$. Then*

$$\frac{\tau_\varepsilon Q(\tau_\varepsilon)}{4\varepsilon} \le E_{\min} \le \frac{\tau_{\varepsilon/2} Q(\tau_{\varepsilon/2})}{2\varepsilon} \qquad \text{and} \qquad \frac{S\tau_\varepsilon^{\mathrm{sl}}}{4} \le T_{\min} \le S\tau_\varepsilon + 1.$$

Consequently, with $B_\varepsilon = \operatorname{tr}\Sigma/(S\varepsilon)$, the ratio $B_{\mathrm{crit}}/B_\varepsilon$ lies between $\frac{Q(\tau_\varepsilon)}{4\operatorname{tr}\Sigma} \cdot \frac{S\tau_\varepsilon}{S\tau_\varepsilon+1}$ and, when $\tau_\varepsilon^{\mathrm{sl}} > 0$, $2\tau_{\varepsilon/2}/\tau_\varepsilon^{\mathrm{sl}}$.

Proof. For $\tau > \tau_\varepsilon$ we have $Q(\tau) \ge Q(\tau_\varepsilon)$ and $0 < \varepsilon - D(\tau) < \varepsilon$, so the objective of (11.4) is at least $\tau_\varepsilon Q(\tau_\varepsilon)/(4\varepsilon)$; at $\tau = \tau_{\varepsilon/2}$ its denominator is 2ε. The step size $1/S$ gives $D_t(1/S) \le D(t/S)$ by fact (a) of the previous proof, which is below ε at the smallest integer $t > S\tau_\varepsilon$. Conversely, let $\eta \in (0, 2/S)$ and $D_t(\eta) < \varepsilon$. For the modes with $h_i \le S/4$, $x = \eta h_i < \frac{1}{2}$ and $\log(1-x) \ge -2x$ give $(1-\eta h_i)^{2t} \ge e^{-2h_i(4t/S)}$, so $\varepsilon > D^{\mathrm{sl}}(4t/S)$ and $4t/S \ge \tau_\varepsilon^{\mathrm{sl}}$. The bounds on B_{crit} follow by division, with $Q \le \operatorname{tr}\Sigma$ in the upper one. ∎

Up to these shape factors, $B_{\mathrm{crit}} \asymp \operatorname{tr}\Sigma/(S\varepsilon)$: the critical batch size grows like $1/\varepsilon$ as the target falls, grows with the noise, and falls with the sharpness that caps the curvature-limited step; for $\Sigma = \sigma^2 H$ it is σ^2/ε times the effective rank $\operatorname{tr} H/S$. In the harmonic model $B_\varepsilon = 40$, the corollary places B_{crit} between 9.5 and 125, and the exact value is $9987/177 = 56$.

Theorem 11.13 (Preconditioning raises the critical batch size). *Let Assumption 11.5 hold, let $P = \operatorname{diag}(1/\omega_1, \dots, 1/\omega_d)$ with every $\omega_i > 0$, and define $T^P(B)$, $E^P_{\min}$, $T^P_{\min}$, and $B^P_{\mathrm{crit}} = E^P_{\min}/T^P_{\min}$ for $\theta_{t+1} = \theta_t - \eta P^{-1} g_t$*

as in Definition 11.10, with $\eta \in (0, 2/S_P)$.

(i) Theorem 11.11 and Corollary 11.12 hold after the substitutions $h_i \to \omega_i h_i$, $c_i \to \omega_i c_i$, $S \to S_P = \max_i \omega_i h_i$, and $\operatorname{tr}\Sigma \to \operatorname{tr}(P^{-1}\Sigma)$; so D becomes $D_P(\tau) = \sum_i \Delta_{0,i} e^{-2\omega_i h_i \tau}$, Q becomes $Q_P(\tau) = \sum_i \omega_i c_i (1 - e^{-2\omega_i h_i \tau})$, the linear rule holds at small batch sizes, and B^P_{crit} is within the shape factors of Corollary 11.12 of $\operatorname{tr}(P^{-1}\Sigma)/(S_P \varepsilon)$.

(ii) The advantage of preconditioning, $T(B)/T^P(B)$, tends to $E_{\min}/E^P_{\min}$ as $B \to 0$ and equals $T_{\min}/T^P_{\min}$ for every large B.

(iii) For $P = H^p$ with $0 \le p \le 1$ and $\Sigma = \sigma^2 H$, the scale $\operatorname{tr}(P^{-1}\Sigma)/(S_P\varepsilon) = (\sigma^2/\varepsilon)\sum_i (h_i/S)^{1-p}$ is nondecreasing in p; it is σ^2/ε times the effective rank $\operatorname{tr} H/S$ at $p = 0$ and $\sigma^2 d/\varepsilon$ at $p = 1$.

Proof. By Lemma 11.7, preconditioned SGD in the coordinates $P^{1/2}\theta$ is SGD on a noisy quadratic model with $h'_i = \omega_i h_i$, $c'_i = \omega_i c_i$, the same initial losses, and the same loss values, whose stable step sizes are those in $(0, 2/S_P)$; apply Theorem 11.11 and Corollary 11.12 to it. For (ii), $T(B)/T^P(B) = B\,T(B)/(B\,T^P(B))$ tends to $E_{\min}/E^P_{\min}$ by Theorem 11.11(iii), applied to both models, and both step counts equal their limits beyond the two thresholds of Theorem 11.11(ii). For (iii), $\omega_i = h_i^{-p}$ gives $\omega_i h_i = h_i^{1-p}$, with maximum S^{1-p}, and $\omega_i c_i = \sigma^2 h_i^{1-p}$; each $(h_i/S)^{1-p}$ is nondecreasing in p because $h_i/S \le 1$. ∎

Preconditioning by H^{-p} divides the stiffest curvature by more than the others, which narrows the spread of curvatures that one stable step size must serve and raises the scale (iii) of the critical batch size. Its effect on $E^P_{\min}$ depends on the problem: in the harmonic model $p = \frac{1}{2}$ gives $E^P_{\min} = 3390$ against 9987, an advantage of 2.95 that persists as $B \to 0$, and 10 steps instead of 177 at large batch sizes. At small batch sizes a preconditioner can only reallocate time among the modes: mode i is SGD with step size $\eta\omega_i$ (Lemma 11.7), and with $\Sigma = \sigma^2 H$ every mode accumulates the same noise per unit of its own relaxation time. SGD gives the stiff modes more relaxation time than they need and pays for it in noise; when they carry little of the noise, as in a spiked spectrum, there is little to gain.

Proposition 11.14 (A spiked spectrum: little gain at small batch sizes). *Let H have k eigenvalues κh, the outliers, and m eigenvalues h, the bulk, with $\kappa \ge 4$. Let $\Sigma = \sigma^2 H$ and $\theta_0 = \bar\theta \mathbf{1}$, so that the bulk starts with the loss $L_{\text{b}} = \frac{1}{2} m h \bar\theta^2$ and the outliers with ρL_{b}, where $\rho = k\kappa/m$; and let $\varepsilon = \phi L_{\text{b}}$ with $0 < \phi < 1$. With $\psi(x) = -x\log(1-x)$ and $\chi(x) = -\log(1-x)$, let $V(\phi)$ be the infimum of $\psi(x)/(x - 1 + \phi)$ over $1 - \phi < x < 1$, let $W_\rho(\phi)$ be the infimum of $(\psi(x) + \rho\,\chi(x))/(\phi - (1+\rho)(1-x))$ over $1 - \phi/(1+\rho) < x < 1$, and let $R(\rho, \phi) = W_\rho(\phi)/V(\phi)$, which is at least 1 and tends to 1 as $\rho \to 0$.*

(i) (Small batch sizes.) *Every diagonal preconditioner has $E^P_{\min} \ge \frac{\sigma^2}{4h\bar\theta^2} V(\phi)$, and every power $P = H^p$, $0 \le p \le 1$, SGD included, has $E^{H^p}_{\min} \le \frac{\sigma^2}{4h\bar\theta^2} W_\rho(\phi)$. Hence $E_{\min}/E^P_{\min} \le R(\rho, \phi)$ for every diagonal*

P, and $E_{\min}/E_{\min}^{H^p} \ge 1/R(\rho,\phi)$.

(ii) (Large batch sizes.) *For $0 \le p \le 1$, $T_{\min}/T_{\min}^{H^p} \ge \kappa \log(1/\phi)/\big(8 + 4\kappa^{1-p}\log((1+\rho)/\phi)\big)$, which for large κ is about $\frac{1}{4}\kappa^p \log(1/\phi)/\log((1+\rho)/\phi)$.*

PROOF. The plan is to write the objective of (11.4), for the model of Theorem 11.13(i), in terms of the fraction of its initial loss that each mode has lost. For a diagonal P and a time τ put $u_i = 2\omega_i h_i \tau$ and $x_i = 1 - e^{-u_i}$. Because $c_i = \sigma^2 h_i$, $\tau\omega_i c_i(1-e^{-u_i}) = \frac{1}{2}\sigma^2 u_i(1-e^{-u_i}) = \frac{1}{2}\sigma^2\psi(x_i)$ and $\Delta_{0,i}e^{-u_i} = \frac{1}{2}h_i\bar{\theta}^2(1-x_i)$, so

$$\tau Q_P(\tau) = \frac{\sigma^2}{2}\sum_i \psi(x_i), \qquad D_P(\tau) = \frac{\bar{\theta}^2}{2}\sum_i h_i(1-x_i).$$

(i) *Lower bound.* Dropping the outliers from both sums decreases the objective. The function ψ is convex on $[0,1)$, since $\psi''(x) = (1-x)^{-1} + (1-x)^{-2} > 0$; so by Jensen's inequality the bulk contributes at least $m\,\psi(\bar{x})$ to $\sum_i \psi(x_i)$, with $\bar{x}$ the mean of its x_i, while it contributes exactly $mh(1-\bar{x})$ to $\sum_i h_i(1-x_i)$. If $D_P(\tau) < \varepsilon$, then $1-\bar{x} < \phi$ and

$$\frac{\tau Q_P(\tau)}{4(\varepsilon - D_P(\tau))} \ge \frac{\frac{1}{2}\sigma^2 m\,\psi(\bar{x})}{4L_{\mathrm{b}}(\bar{x}-1+\phi)} = \frac{\sigma^2}{4h\bar{\theta}^2}\cdot\frac{\psi(\bar{x})}{\bar{x}-1+\phi} \ge \frac{\sigma^2}{4h\bar{\theta}^2}V(\phi).$$

Upper bound. For $P = H^p$ the bulk modes share $u = 2h^{1-p}\tau = \chi(x)$, and the outliers have $u_{\mathrm{o}} = \kappa^{1-p}u \ge u$. Then $\psi(x_{\mathrm{o}}) \le u_{\mathrm{o}} \le \kappa\,\chi(x)$ and $1 - x_{\mathrm{o}} \le 1 - x$, so, with $k\kappa = \rho m$, $\tau Q_P(\tau) \le \frac{1}{2}\sigma^2 m(\psi(x) + \rho\,\chi(x))$ and $D_P(\tau) \le (1+\rho)L_{\mathrm{b}}(1-x)$. As τ runs over $(0,\infty)$, x runs over $(0,1)$, which gives $E_{\min}^{H^p} \le \frac{\sigma^2}{4h\bar{\theta}^2}W_\rho(\phi)$, and SGD is the case $p = 0$. As for R: the numerator of W_ρ is at least $\psi(x)$ and its denominator at most $x - 1 + \phi$, so $W_\rho \ge V$. The infimum $V(\phi)$ is attained at some x_ϕ, because the quotient tends to infinity at both ends of the interval; for small ρ, $(1+\rho)(1-x_\phi) < \phi$, and $W_\rho(\phi) \le (\psi(x_\phi) + \rho\,\chi(x_\phi))/(\phi - (1+\rho)(1-x_\phi)) \to V(\phi)$.

(ii) Apply Corollary 11.12 to SGD and, through Theorem 11.13(i), to H^p. For SGD, $S = \kappa h$ and the modes with $h_i \le S/4$ are the bulk, since $\kappa \ge 4$; so $D^{\mathrm{sl}}(\tau) = L_{\mathrm{b}}e^{-2h\tau}$, $\tau_\varepsilon^{\mathrm{sl}} = \log(1/\phi)/(2h)$, and $T_{\min} \ge \frac{1}{8}\kappa\log(1/\phi)$. For H^p, $S_P = (\kappa h)^{1-p}$ and $D_P(\tau) \le (1+\rho)L_{\mathrm{b}}e^{-2h^{1-p}\tau}$, so $T_{\min}^{H^p} \le \frac{1}{2}\kappa^{1-p}\log((1+\rho)/\phi) + 1$. ∎

For the spiked models of Figure 11.3 ($\kappa = 100$, $k = 10$, target 5% of the initial loss), R is 2.39 for $\rho = 1$ and 1.12 for $\rho = 0.1$, against exact small-batch advantages of $p = \frac{1}{2}$ of 1.83 and 1.09; the large-batch advantages are 9.1 and 9.8, close to $\kappa^{1/2} = 10$, so bound (ii) is pessimistic by a factor of four to five. When the noise lives in the bulk, the bulk is already well conditioned; at large batch sizes the outliers set the stable step size.

HEURISTIC DERIVATION OF THE SCALING RULES. *SGD.* In the noise-limited regime the expected loss after E examples at step size $\eta = \gamma B$ approaches $D(\gamma E) + \frac{\gamma}{4}Q(\gamma E)$, the expected loss at time $\tau = \gamma E$ of the Ornstein–Uhlenbeck process $d\theta = -H\theta\,d\tau + \sqrt{\gamma}\,\Sigma^{1/2}dW_\tau$, which is the *stochastic differential equation*

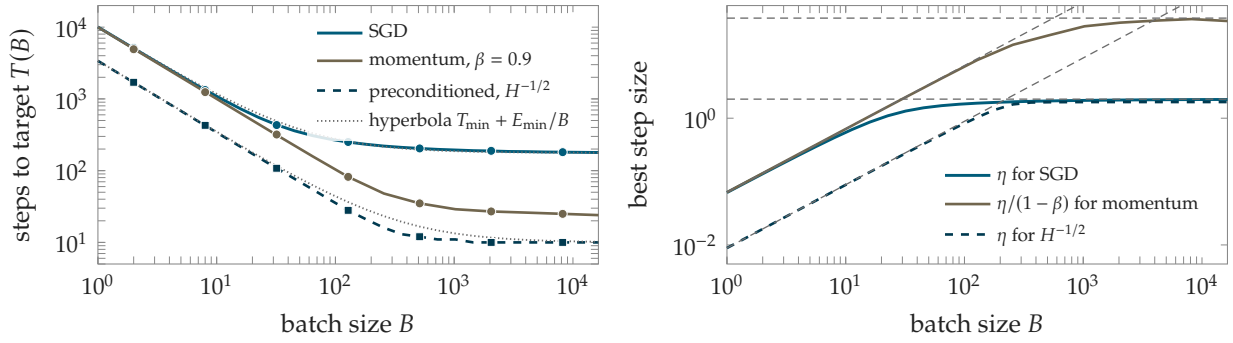


Figure 11.2. Steps to the target $\varepsilon = 0.05\,\Delta_0$ in the harmonic model, at the best constant step size. Left: curves are exact (Theorem 11.6 minimized over stable step sizes; the heavy ball by its exact second-moment recursion); points are simulations, the first step at which the average loss of 64 runs (256 for $B \geq 128$) at the theory's step size, smoothed over 2% of the predicted time, is below the target (seeds in `11-noise-steps.py`); all agree within 1.6%. Dotted: the hyperbolas $T_{\min} + E_{\min}/B$ built from the exact limits. Right: the best step sizes (for momentum, the one minimizing the loss at step $T(B)$), with the linear rules $\gamma^\star B$ of Theorem 11.11(iv), the limit $2/S = 2/S_P = 2$, and momentum's limit 38 for $\eta/(1-\beta)$ (dashed).

of Li et al. (2017) for the model. Runs with equal η/B have the same loss per example, which is the linear rule, and Theorem 11.11(iv) shows that the tuned step size follows it. It fails when ηS is no longer small: $(1-\eta h_i)^2$ departs from $e^{-2\eta h_i}$, and stability caps η below $2/S$.

Adam. Take H diagonal in Adam's coordinates, $\beta_1 = 0$, $\epsilon = 0$, and β_2 close to 1, over a stretch where the gradient statistics change slowly, so that $\hat{v}_{t,i} \approx (\partial_i \mathcal{L})^2 + c_i/B$. While noise dominates coordinate i, $B(\partial_i \mathcal{L})^2 \ll c_i$, the step $\eta g_{t,i}/\sqrt{\hat{v}_{t,i}}$ is about $\eta\sqrt{B}\, g_{t,i}/\sqrt{c_i}$: preconditioned SGD with $P^{-1} = \operatorname{diag}(c_i^{-1/2})$ and step size $\eta\sqrt{B}$, which for $\Sigma = \sigma^2 H$ is the power $p = \frac{1}{2}$ up to the factor σ. The linear rule for it asks for $\eta\sqrt{B} \propto B$, that is, $\eta \propto \sqrt{B}$, the *square-root scaling rule*. For $\hat{v}$ to average over a fixed number of examples, $1-\beta_2 \propto B$, and likewise $1-\beta_1$; since $\sqrt{\hat{v}}$ scales like $B^{-1/2}$, so must ϵ. The rule fails once $B(\partial_i \mathcal{L})^2$ is comparable to c_i, where the step approaches the sign step of Chapter 2, whose size does not depend on B, and when the effective step approaches $2/S_P$. ◇

Malladi et al. (2022) derived the square-root rule from a stochastic differential equation for adaptive methods, and Li et al. (2021) studied when such equations describe SGD.

11.5 Reading the theory

STEPS AGAINST BATCH SIZE. Figure 11.2 is the theory's version of Observation 11.1. For SGD the steps fall like $1/B$ up to $B \approx 10$ and level off near $T_{\min} = 177$; the best step size follows the rule $\gamma^\star B$ while $\gamma^\star BS$ is small, falls below it as the rule approaches $1/S$ at $B = 1/(\gamma^\star S) \approx 15$, and creeps up to 1.975 at $B = 16384$, within 1.3% of $2/S$, so that in the curvature-limited regime the tuned learning rate sits at the edge of stability (Chapter 8). The

critical batch size is 56. Momentum with $\beta = 0.9$ follows SGD at small batch sizes, as Proposition 11.8(ii) requires (9989 steps at $B = 1$, against 10063), and levels off at 23 steps, a critical batch size of 434. Preconditioning by $H^{-1/2}$ lowers the small-batch line by the factor 2.95 and the large-batch level to 10 steps, a critical batch size of 339. Measured by where $T(B) = 2T_{\min}$, the knee moves from 46 to 186, a factor of four rather than six.[4]

[4] On the hyperbola $T = T_{\min} + E_{\min}/B$ the batch size B_{crit} is exactly where $T = 2T_{\min}$ and $E = 2E_{\min}$, so the two definitions of the knee would agree if the hyperbola were exact.

THE HYPERBOLA. If $T(B) = T_{\min} + E_{\min}/B$ held exactly, then McCandlish et al.'s relation would follow. The exact $T(B)$ lies between 0.89 and 1.04 times this hyperbola for SGD and between 0.67 and 1.00 times it for $H^{-1/2}$, but the product $(T/T_{\min} - 1)(E/E_{\min} - 1)$ is a more sensitive test. With the real-valued crossing times, which $T(B)$ and $T_{\min}$ round up (rounding moves E by up to B, the order the product measures), it rises from 0.42 as $B \to 0$ to 3.5 at $B = 16384$ for SGD, and stays between 0.11 and 0.25 for $H^{-1/2}$. As $B \to 0$ it tends to $E_1/T_{\min}$, where $E(B) = E_{\min} + E_1 B + o(B)$ ($E_1 \approx 75$ for SGD), and the relation assumes $E_1 = T_{\min}$, which nothing in the model enforces. A least-squares fit of the hyperbola gives $B_{\text{crit}} \approx 52$ (Exercise 11.11).

NOISE SCALES. On the noise-free path of gradient flow at the target, McCandlish et al.'s scales are $B_{\text{noise}} \approx 9.5 \times 10^5$ and $B_{\text{simple}} \approx 1.1 \times 10^4$, orders of magnitude above $B_{\text{crit}} = 56$: the one-step model lets the step grow to $\eta_{\max}$, close to $1/h$ for the slow modes that still carry loss and far beyond $2/S$ (Proposition 11.9(ii)). Measured as in practice, on the noisy iterates of the tuned run at $B = 1$, they are $B_{\text{simple}} \approx 132$ and $B_{\text{noise}} \approx 40$: near the floor the full-batch gradient is dominated by the fluctuations of the stiff modes, and by Proposition 11.9(iii) the measured scales approach $2\operatorname{tr}\Sigma/(\gamma^\star \operatorname{tr}(H\Sigma)) = 134$ and $2\operatorname{tr}(H\Sigma)/(\gamma^\star \operatorname{tr}(H^2\Sigma)) = 40$: they measure the inverse of the tuned step size per example. For $H = hI$ the agreement is close: with $\Delta_0/\varepsilon = 20$, B_{crit} is 1.42 times B_{simple} at loss ε (Exercise 11.2).

THE ADVANTAGE OF PRECONDITIONING. Figure 11.3 shows the batch-size dependence that Chapter 14 uses to compare Adam with SGD. The advantage grows with the batch size from its small-batch limit to about its large-batch limit (two curves overshoot slightly near the knee). The small-batch limit is a property of the problem: 2.95 for the harmonic spectrum, 1.83 for the spiked spectrum with $\rho = 1$, and 1.09 for $\rho = 0.1$, where the bulk carries 91% of the noise. It depends on the spectrum, the noise, and the initialization through (11.4), and tends to 1 with the outliers' share of the noise (Proposition 11.14).

WHERE THE AGREEMENT MUST FAIL. The residual differences in the figures are Monte Carlo error and the bias of locating a first passage on a noisy curve. A single run fluctuates more: at the target of the harmonic model with $B = 4$ its loss has a standard deviation of 17% of the target, and since the stiff modes decorrelate within a few steps, one run at the best step size first touches the target well before the expected loss reaches it (median about 2000 steps in 200 runs, against $T(4) = 2577$). The theory describes the expected loss, or a loss averaged over runs or steps.

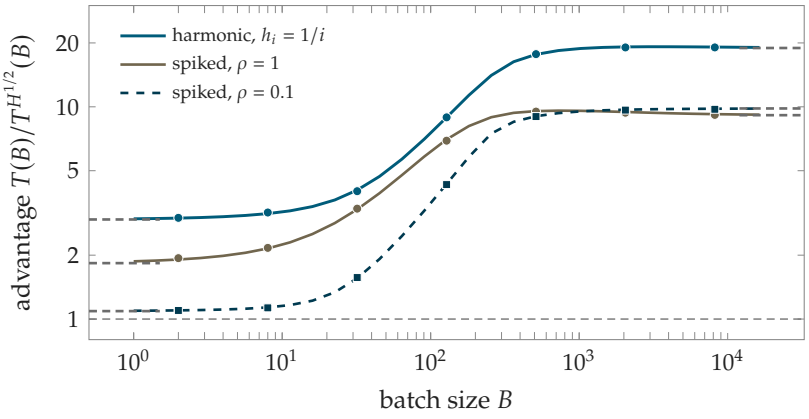


Figure 11.3. The advantage of preconditioning by $H^{-1/2}$, the ratio of the steps to target of SGD and of preconditioned SGD, with $\Sigma = H$, $\theta_0 = (1, \dots, 1)$ and the target 5% of the initial loss: the harmonic model, and spiked models with 10 outliers of curvature 1 above $m = 1000$ ($\rho = 1$) or $m = 10^4$ ($\rho = 0.1$) bulk modes of curvature 0.01. Curves: exact real-valued first-passage times; short dashes at the ends: the limits of Theorem 11.13(ii). Points: simulations as in Figure 11.2, with the crossing located by linear interpolation and the bulk simulated by 500 representative modes (`11-noise-precond.py`).

11.6 What the theory misses

WHAT WAS IDEALIZED AWAY. The landscape and the noise are fixed, while in a network the sharpness rises toward the stability threshold (Chapter 8) and the noise covariance changes with the parameters; near interpolation it shrinks with the loss, which turns the critical batch size into the target-independent effective rank $\operatorname{tr} H/S$ (Exercise 11.5). The model is online, without epochs or a generalization gap. The step size is constant; with averaged iterates every diagonal preconditioner has the same asymptotic cost $\operatorname{tr}(H^{-1}\Sigma)/(2E)$ (Polyak and Juditsky 1992) (Exercise 11.7). Adam's preconditioner is estimated from noisy gradients and lags them (Chapter 6).

WHERE IT BREAKS. (1) The critical batch size of a real run moves: the noise scale grows during training (McCandlish et al. 2018), the critical batch size of language models is a power of the loss (Kaplan et al. 2020), and in pretraining it grows mainly with the amount of training data, not with the model size (Zhang et al. 2025). The model's $1/\varepsilon$ gives the direction of the first two but not their rate, and it cannot express the third. (2) Shallue et al. (2019) found that the optimal learning rate followed neither the linear nor the square-root rule consistently across workloads. (3) Large batches can generalize worse at a fixed number of epochs (Hoffer et al. 2017; Keskar et al. 2017), which an online model cannot show. (4) At very small batch sizes plain SGD comes close to Adam on language models (Marek et al. 2025); with a constant step size the model reproduces this when the noise lives in the bulk of a spiked spectrum (Proposition 11.14) but not in the harmonic model, where the gap stays near 3, and averaging the iterates closes the gap for small targets in every case (Exercise 11.7).

Which case describes a transformer, whose Hessian is heterogeneous across blocks (Zhang et al. 2024), is not settled (Chapter 14). (5) The diffusion approximation behind the scaling rules has conditions of validity that Li et al. (2021) test experimentally, and heavy-tailed noise (Chapter 10) is outside the model.

REPAIRS AND OPEN PROBLEMS. Batch-size schedules exploit the growth of the critical batch size: Smith et al. (2018) increase the batch size instead of decaying the learning rate, GPT-3 ramped its batch size up (Brown et al. 2020),[5] and Merrill et al. (2025) measure the critical batch size of language models over a run and propose a batch-size warmup.[6] Noise-scale estimators make the critical batch size measurable (McCandlish et al. 2018), with the caveat of Proposition 11.9(iii). Parameter-dependent noise gives the interpolation regime its own critical batch size (Jain et al. 2018; Ma et al. 2018). Zhang et al. (2025) justify the growth of the critical batch size with the data by infinite-width and least-squares analyses. Open: a model of a network that derives the measured dependence on the data-set and model sizes; a theory for normalized optimizers such as sign descent and Muon, whose step does not shrink with the noise (Chapter 5); and measurements placing transformer spectra and noise between the harmonic and spiked cases.

[5] Linearly, from 32 thousand tokens to the full batch over the first 4 to 12 billion tokens, depending on the model size.

[6] For OLMo models of 1B and 7B parameters they find the critical batch size near zero at initialization, rising quickly and then leveling off; with the warmup, OLMo 1B reached a slightly better loss with 43% fewer gradient steps.

11.7 Back in the wild

FIELD NOTES	
PRACTICE	Scale the learning rate linearly (SGD) or as the square root (Adam) with the batch size, up to a critical batch size; train language models with batches of millions of tokens, ramped up early in training.
IDEALIZATION	The noisy quadratic model: a fixed quadratic, noise of covariance Σ/B, a constant step size, the expected loss.
PROVED	Exact loss curves and floors; $T(B) = T_{\min}$ for large B, and $B\,T(B) \to E_{\min}$ with $\eta \approx \gamma^\star B$ as $B \to 0$; $B_{\text{crit}} \asymp \operatorname{tr}\Sigma/(S\varepsilon)$ up to shape factors; preconditioning raises this scale; when a spiked spectrum's outliers carry little noise, diagonal preconditioners save few examples at small batch sizes.
PREDICTS	Perfect scaling below B_{crit} and saturation above it, with the step size near $2/S$; a critical batch size growing like $1/\varepsilon$ as the target falls; momentum and preconditioners help mainly above the knee; noise scales measured on a run reflect its own ratio B/η.
MISSES	Curvature and noise that change during training, noise that depends on the parameters, finite data and epochs, schedules, heavy tails, generalization.
TAKEAWAY	Scale $\eta \propto B$ below the knee and expect diminishing returns above it (proved in the idealization); expect larger critical batch sizes late in training and for preconditioned optimizers (suggested); expect little gain from preconditioning

at very small batch sizes when the noise lives in the bulk (suggested).

For a practitioner the theory gives the scaling rules a range of validity. The linear rule is a theorem of the model while the step it prescribes is small compared with $1/S$; when the tuned learning rate approaches the stability limit, the rule and the returns of larger batches stop together. The critical batch size is of order $\operatorname{tr}\Sigma/(S\varepsilon)$: it grows as the loss falls, which supports ramping the batch size up, grows with the noise, and falls with the sharpness. The theory supplies neither the constant in front nor the evolution of Σ and S in a network, so the batch size is still chosen by measurement; where exactly to sit near the critical batch size depends on the relative cost of steps and examples, and is folklore.

The stability constraint is the threshold $2/S$ of Chapter 8; the floor is where Chapter 12 starts, and the equilibrium of Chapter 13 is another floor. The batch-size dependence of the advantage of preconditioning (Theorem 11.13 and Proposition 11.14) is one of the explanations that Chapter 14 offers for why Adam beats SGD on transformers, and the square-root heuristic bears on the choice of β_2 (Chapter 6).

Exercises

11.1 CONSOLIDATION. (Momentum's floor.) With $\eta_e = \eta/(1-\beta)$ fixed and $\eta_e S < 2$, show that the floor of Proposition 11.8 decreases in β and differs from SGD's floor at η_e by a relative amount at most $\eta_e S/(2-\eta_e S)$.

11.2 CONSOLIDATION. (One mode.) Let $d = 1$, with curvature h, noise c, and $r = \Delta_0/\varepsilon > 1$. (a) Show that $T_{\min} = 1$, at $\eta = 1/h$, and that $T(B) = 1$ if and only if $B > (1-1/r)c/(2h\varepsilon)$. (b) Show that $E_{\min} = \frac{c}{8h\varepsilon} V_1(r)$ with $V_1(r) = \min_{x>\log r} x(1-e^{-x})/(1-re^{-x})$, and that $V_1(r) = \log r + \log\log r + O(1)$. (c) For $r = 20$, where $V_1 = 5.69$, show that a batch size just above $(1-1/r)c/(2h\varepsilon)$ reaches the target with about two thirds of $E_{\min}$ examples, so $E \geq E_{\min}$ can fail. (d) For $H = hI$, show that $B_{\text{crit}} = E_{\min}$ is $V_1(r)/4$ times B_{simple} at loss ε. *(Hint: in (b), put $x = \log r + s$.)*

11.3 CONSOLIDATION. (A preconditioner that does not commute with H.) For an arbitrary $P > 0$ and the update $\theta_{t+1} = \theta_t - \eta P^{-1} g_t$, show that $P^{1/2}\theta_t$ performs SGD on a noisy quadratic model with Hessian $P^{-1/2}HP^{-1/2}$, whose top eigenvalue is S_P, and noise covariance $P^{-1/2}\Sigma P^{-1/2}$, so that Theorem 11.6 applies in its eigenbasis.

11.4 CONSOLIDATION. (The diffusion limit.) Let θ_τ solve $d\theta_\tau = -H\theta_\tau\, d\tau + \sqrt{\gamma}\,\Sigma^{1/2} dW_\tau$, started at θ_0, with W a standard Brownian motion in $\mathbb{R}^d$. Show that $\mathbb{E}\mathcal{L}(\theta_\tau) = D(\tau) + \frac{\gamma}{4}Q(\tau)$, and that the infimum over $\gamma > 0$ of the examples τ/γ after which this is below ε is $E_{\min}$. Why does this limit say nothing about step sizes with ηS near 2?

11.5 VARIATION. (Interpolation.) Least squares with inputs $x \sim \mathcal{N}(0, H)$

and noiseless labels has per-example gradient covariance $\Sigma(\theta) = 2\mathcal{L}(\theta)H + H\theta\theta^\top H$. (a) Show that $m_t = (\mathbb{E}\theta_{t,i}^2)_i$ satisfies $m_{t+1} = Am_t$ with $A = \operatorname{diag}(a_i) + (\eta^2/B)hh^\top$, $a_i = (1-\eta h_i)^2 + \eta^2 h_i^2/B$ and $h = (h_1, \dots, h_d)$. (b) Show that $\mathbb{E}\mathcal{L}(\theta_t) \to 0$ from every θ_0 if and only if $\max_i a_i < 1$ and $\frac{\eta}{B}\sum_i h_i/(2-\eta h_i(1+1/B)) < 1$. (c) Deduce that the largest stable step size lies between $\min\{B/\operatorname{tr} H, B/((B+1)S)\}$ and $\min\{2B/\operatorname{tr} H, 2B/((B+1)S)\}$, so that the critical batch size is about the effective rank $\operatorname{tr} H/S$, whatever the target. *(Hint: A is entrywise positive, and its largest eigenvalue $\mu > \max_i a_i$ solves $1 = (\eta^2/B)\sum_i h_i^2/(\mu - a_i)$.)*

11.6 VARIATION. (Isotropic noise.) Redo the proof of Proposition 11.14(i) for $\Sigma = \sigma^2 I$: show that $\tau\omega_i c_i(1-e^{-u_i}) = \frac{\sigma^2}{2h_i}\psi(x_i)$, that the outliers contribute at most $\frac{\sigma^2}{2h}k\kappa^{-p}\chi(x)$ to τQ_P for $P = H^p$, and that as $k\kappa/m \to 0$ the ratio $E_{\min}/E_{\min}^P$ is at most $1 + o(1)$ for every diagonal P and tends to 1 for every power $P = H^p$. Where does H^{-p} now raise the floor?

11.7 VARIATION. ⋆ (Averaging.) Let $\bar\theta_n$ be the mean of n consecutive stationary iterates of SGD with $\eta < 2/S$. (a) Show that $\mathbb{E}\mathcal{L}(\bar\theta_n) = \frac{1}{2nB}\operatorname{tr}(H^{-1}\Sigma)(1+o(1))$ as $n \to \infty$, whatever η is, and that the same holds after every diagonal preconditioning. (b) Conclude that with averaging, for small targets, every diagonal preconditioner needs about $\operatorname{tr}(H^{-1}\Sigma)/(2\varepsilon)$ examples, as in Polyak and Juditsky (1992). What is left for a preconditioner to improve? *(Hint: each mode is an autoregression of order one; sum its autocovariances.)*

11.8 VARIATION. (The square-root rule for sign descent.) Let $\theta_{t+1,i} = \theta_{t,i} - \eta\operatorname{sign}(g_{t,i})$ with Gaussian noise, $\mu = \partial_i\mathcal{L}(\theta_t)$ and $s^2 = c_i/B$. (a) Show that $\mathbb{E}[\operatorname{sign} g_{t,i} \mid \mathcal{F}_t] = \operatorname{erf}(\mu/(s\sqrt{2})) = \sqrt{2/\pi}\,\mu/s + O((\mu/s)^3)$. (b) Deduce that to leading order the sign step has the conditional mean of SGD preconditioned by $\omega_i = \sqrt{2B/(\pi c_i)}$, and $\pi/2$ times its conditional variance. (c) For $\Sigma = \sigma^2 H$, identify the power p, derive $\eta \propto \sqrt{B}$ in the noise-dominated regime, and explain why the rule fails once $B\mu^2 \gtrsim c_i$.

11.9 SIMULATION. (The momentum coefficient.) In `11-noise-steps.py`, replace $\beta = 0.9$ by 0.5, 0.8, 0.85, and 0.95. The small-batch part should not move (Proposition 11.8(ii)), and the large-batch level should become 59, 22, 21, and 41 steps, against 23. Compare the best β with Polyak's choice, about 0.86 for the condition number $h_1/h_{700} = 700$ of the modes that must converge (Chapter 5), and explain why too large a β hurts.

11.10 SIMULATION. (The outliers' share.) With `11-noise-precond.py`, compute the small-batch advantage of $H^{-1/2}$ for spiked spectra with $\kappa = 100$, $k = 10$, and $\rho \in \{1, 0.1, 0.01\}$, and compare with the bound $R(\rho, \phi)$ of Proposition 11.14 at $\phi = 0.05(1+\rho)$: you should find 1.83, 1.09, and 1.009 against 2.39, 1.12, and 1.012. With $\kappa = 1000$ and $\rho = 0.1$, check that the small-batch advantage stays near 1 (1.10) while the large-batch one grows to 31.

11.11 SIMULATION. (Fitting the hyperbola.) Fit $T(B) = a + b/B$ to the exact

SGD curve of Figure 11.2 by least squares on $\log T$. You should find $a \approx 180$ and $b \approx 9300$, so $b/a \approx 52$, within 8.3% of the curve everywhere, against $T_{\min} = 177$ and $E_{\min} = 9987$. For $H^{-1/2}$ the best fit is off by 31% near the knee; explain why from Figure 11.2.

11.12 RESEARCH. (The critical batch size and the data.) Zhang et al. (2025) report that the critical batch size grows with the amount of data rather than the model size. Formulate a solvable model with a finite data set, such as linear regression on a finite sample trained for several epochs, derive how B_{crit} depends on the data and model sizes, and compare with their least-squares analysis. Open: a model of a network whose derived dependence matches the measurements.

Notes and References

Zhang et al. (2019) used the noisy quadratic model, with the per-mode recursion of Theorem 11.6, its floor, and the change of variables of Lemma 11.7, to predict how the critical batch size depends on the optimizer, including preconditioning by powers of H and momentum; their solutions show perfect scaling, its end, and how preconditioning and momentum move the end. A floor proportional to the step size is standard (Bottou et al. 2018), and averaging removes it (Polyak and Juditsky 1992). Smith et al. (2018) argued that the ratio of learning rate to batch size controls the noise of SGD.

McCandlish et al. (2018) named and measured the critical batch size and derived the one-step model of Proposition 11.9; Shallue et al. (2019) mapped the three regimes across workloads, and Kaplan et al. (2020) folded the critical batch size into the scaling laws of language models. For least squares in the interpolation regime, Ma et al. (2018) and Jain et al. (2018) found a critical batch size of the order of an effective rank (Exercise 11.5). The two limits of Theorem 11.11 with the formula (11.4), the bounds of Corollary 11.12, the spiked-spectrum proposition, and the stationary noise scales of Proposition 11.9(iii) are, as far as we know, not stated in this form elsewhere; the regimes themselves, and the numerical evidence that preconditioning moves the knee, are in Zhang et al. (2019).

TRANSLATING THE LITERATURE. McCandlish et al. (2018) write ϵ for the learning rate (our η), S and E for the steps and examples to a target (our T and E; our S is the sharpness), G for the true gradient (our $\nabla\mathcal{L}$), and $S_{\min}$, $E_{\min}$ for the limits (our $T_{\min}$, $E_{\min}$); their $B_{\text{crit}} = E_{\min}/S_{\min}$, B_{noise}, and B_{simple} are ours. Zhang et al. (2019) call the noise covariance C (our Σ), with diagonal entries c_i in the eigenbasis of H as here, and precondition by $P = H^p$, as in Theorem 11.13.

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12 Schedules

Cosine, warmup–stable–decay and the river valley

IN THE WILD — COSINE AND WSD SCHEDULES

IN THIS CHAPTER. Large models are trained with a learning rate that rises briefly and then decays, along a cosine, along a straight line, or in a short cooldown after a long plateau, and the loss falls fastest exactly when the learning rate falls. For convex Lipschitz losses we prove a bound on the last iterate of stochastic gradient descent under an arbitrary schedule, show that linear decay to zero attains the rate of iterate averaging exactly, and evaluate the bound for the schedules in use. In a river valley, a slowly descending direction flanked by stiff and noisy ones, we compute the expected loss exactly for every schedule: it is a term that depends on the area under the schedule plus a term that remembers only the recent schedule, which is the form of the empirical laws that predict loss curves across schedules, and it predicts how the best cooldown scales with the length of the run. Neither idealization has a landscape that changes during training, and the river valley's predictions depend on constants that must be measured.

12.1 In the wild: the shape of the learning rate

Ask what learning rate a large language model was trained with and the answer is a curve. Most published pretraining recipes since GPT-3 warm the learning rate up over a small part of the run and then decay it to a small value at the planned end, and the shape of the decay is one of the few hyperparameters still argued about in public. As of 2025 three shapes dominate the open recipes.

IN THE WILD · GPT-3, Brown et al. (2020); LLaMA, Touvron et al. (2023); Chinchilla, Hoffmann et al. (2022); MiniCPM, Hu et al. (2024); Hägele et al. (2024); Bergsma et al. (2025). Warm the learning rate up linearly from zero to its peak $\eta_{\max}$, over the first 375 million tokens (GPT-3) or 2000 steps (LLaMA). Then do one of three things. *Cosine*: decay along half a period of a cosine to a tenth of the peak, $\eta_t = \eta_{\max}\big(0.1 + 0.45(1 + \cos(\pi t/T))\big)$; LLaMA and Chinchilla reach the tenth at the end of the run, GPT-3 after 260 of its 300 billion tokens. *Linear decay to zero*, $\eta_t = \eta_{\max}(1 - t/T)$ (Bergsma et al. 2025; Defazio et al. 2023). *Warmup–stable–decay* (WSD): hold the peak, then decay to zero or nearly zero over the last 10–20% of the steps (MiniCPM; Hägele et al.

(2024) recommend about 20%, with the shape $1 - \sqrt{\cdot}$). The original transformer used the inverse square root, $\eta_t \propto \min(t^{-1/2}, t\,T_{\mathrm{w}}^{-3/2})$ with $T_{\mathrm{w}} = 4000$ (Vaswani et al. 2017).

The cosine came from vision, where Loshchilov and Hutter (2017) introduced cosine annealing with periodic restarts; language models kept a single half period.[1] The shapes look very different, yet their final losses in tuned runs are close, and which is best is still debated.

[1] Earlier vision recipes divided the learning rate by 10 at two or three fixed epochs, a *step decay*. The classical rule of stochastic approximation is $\eta_t \propto 1/t$, which satisfies the conditions $\sum_t \eta_t = \infty$ and $\sum_t \eta_t^2 < \infty$ of Robbins and Monro (1951). Every schedule in the box violates the second: at a fixed peak, $\sum_{t \le T} \eta_t^2$ grows without bound with the horizon T, linearly for the decaying shapes and like $\log T$ for the inverse square root.

Observation 12.1 (The cooldown drop; Hu et al. 2024, Hägele et al. 2024). Train with WSD. During the stable phase the loss decreases slowly; when the decay begins it falls much faster, and after a decay over 10–20% of the steps it matches, or slightly beats, a cosine schedule of the same length. Longer decays help a little more, with diminishing returns.

Observation 12.2 (The horizon; Hoffmann et al. 2022). A cosine schedule planned for more steps than the run takes, by more than about 25%, gives a worse model than one planned for exactly the steps taken. WSD runs of different lengths coincide until their cooldowns, so one stable phase can be branched into cooldowns of several lengths, which makes scaling experiments cheaper (Hägele et al. 2024).

The next two observations are of a different kind: fits and analogies that describe whole loss curves.

Observation 12.3 (Loss curves across schedules; Tissue et al. 2025, Luo et al. 2025). Fitted to the loss curves of one or two schedules, a law of the form

$$\mathcal{L}(t) \approx L_\infty + C\,S_1(t)^{-\alpha} - \mathrm{LD}(t), \qquad S_1(t) = \sum_{s \le t} \eta_s,$$

predicts the loss curves of the same model under other schedules. The *forward area* $S_1(t)$ is the area under the schedule so far; the *loss drop* $\mathrm{LD}(t)$ adds, for each decrease of the learning rate, a term proportional to the decrease that grows and then saturates with the number of steps (Tissue et al. 2025) or with the area (Luo et al. 2025) since the decrease.

Observation 12.4 (Convex theory agrees; Schaipp et al. 2025). Evaluate, for each schedule and each step t, the bound on $\mathbb{E}\mathcal{L}(\theta_t) - \mathcal{L}^\star$ that convex optimization gives for stochastic gradient descent on a convex Lipschitz loss. The curves look like language-model loss curves: WSD plateaus and then drops suddenly, and cosine and WSD end close together. Schaipp et al. (2025) report that the bound reproduces the matching performance of cosine and WSD and the necessity and length of the cooldown, and that learning rates chosen with it improved Llama-type models with 124M and 210M parameters.

Language-model training is far from convex, so Observation 12.4 suggests that the schedule acts through something simpler than the landscape; in the experiments of Defazio et al. (2023), too, warmup followed by linear decay matched or beat cosine almost everywhere, as their worst-case theory

predicts. Why does the loss fall when the learning rate falls, why do such different shapes end so close together, and how long must the decay be?

12.2 The idealization: two pictures of a schedule

Throughout, θ_1 is the initial point and stochastic gradient descent with step sizes $\eta_1, \eta_2, \dots$ is

$$\theta_{t+1} = \theta_t - \eta_t\, g_t, \qquad t = 1, 2, \dots, \tag{12.1}$$

with g_t a stochastic gradient at θ_t; "learning rate" (in practice) and "step size" (in the theory) are synonyms. Let $\mathcal{F}_t$ be the σ-algebra generated by $g_1, \dots, g_{t-1}$, so that θ_t is $\mathcal{F}_t$-measurable. A *schedule* is the sequence (η_t), often written $\eta_t = \eta\, \varphi(t/T)$ with a peak η and a *shape* $\varphi\colon [0,1] \to [0,1]$. The *area* before step t is $A_t = \sum_{s=1}^{t-1} \eta_s$, the time of the gradient flow that the run follows when its steps are small.

Convex Lipschitz losses. The first picture keeps nothing of the landscape except convexity.

Assumption 12.5 (Convex Lipschitz stochastic optimization). The loss $\mathcal{L}\colon \mathbb{R}^d \to \mathbb{R}$ is convex and has a minimizer $\theta^\star$, and $D = \|\theta_1 - \theta^\star\|_2$. The stochastic gradients satisfy $\mathbb{E}[g_t \mid \mathcal{F}_t] \in \partial\mathcal{L}(\theta_t)$ and $\mathbb{E}\big[\|g_t\|_2^2 \mid \mathcal{F}_t\big] \le G^2$.

The constant G bounds subgradients and noise together, so the subgradient method on a G-Lipschitz function and stochastic gradient descent on a smooth function with $\|\nabla\mathcal{L}\|_2^2 + \mathbb{E}\|\zeta_t\|_2^2 \le G^2$ are both covered; the norm is Euclidean because stochastic gradient descent is steepest descent in ℓ_2 (Chapter 1). A statement about the whole class cannot see the scales of a particular landscape, and this will show in the numbers.

The river valley. The second picture keeps the simplest landscape with two time scales. Wen et al. (2025) proposed that the pretraining loss of a language model is shaped like a *river valley*: a deep valley whose floor descends slowly along a river, between steep walls. With a large learning rate the iterate oscillates between the walls while it moves quickly along the river; when the learning rate decays the oscillation dies out, and the progress made along the river shows in the loss.[2] We make the walls quadratic, the oscillation noise-driven, and the river straight or weakly curved.

[2] Wen et al. (2025) derive the picture in a toy model trained on bigram data and use it to design a WSD variant that reuses decayed checkpoints. The quadratic walls and the noise below are our simplification.

Assumption 12.6 (The river valley). The parameter is $\theta = (x, y)$ with river coordinates $x \in \mathbb{R}^p$ and hill coordinates $y \in \mathbb{R}^m$, and

$$\mathcal{L}(x, y) = r(x) + \frac{1}{2}\sum_{i=1}^{m} h_i\, y_i^2, \qquad h_i > 0.$$

The stochastic gradient is $g_t = \nabla\mathcal{L}(\theta_t) + \zeta_t$ with $\mathbb{E}[\zeta_t \mid \mathcal{F}_t] = 0$ and $\mathbb{E}[\zeta_{t,i}^2 \mid \mathcal{F}_t] = \sigma_i^2$ for each hill coordinate y_i. The river r is linear, with a constant gradient ∇r, or a quadratic whose curvatures are

much smaller than the h_i, with noise of fixed variance in each of its eigendirections.

With a quadratic river the model is the noisy quadratic model of Chapter 11, its spectrum split into hills, whose curvature times the area of the run is large, and river directions, for which it is small. In a hill the iterate soon forgets its past and sits in a noise-driven equilibrium; along the river it never forgets, and it is still descending when the run ends.

DICTIONARY

IN THE THEORY	IN A TRAINING RUN
step size η_t, peak η, shape φ	the schedule $\eta_{\max}\varphi(t/T)$
area $A_t = \sum_{s<t}\eta_s$	the forward area S_1 of the empirical laws; how far the run has travelled along the river
river $r(x)$	the part of the loss that keeps falling at a constant learning rate, roughly a power of the area
hill y_i, curvature h_i	a sharp direction of the Hessian, along which the iterate oscillates at a high learning rate
noise variance σ_i^2	gradient noise along that direction, Σ_{ii}/B at batch size B
noise floor $\eta\sigma_i^2/(2(2-\eta h_i))$	the excess loss of the stable phase, removed by the cooldown
relaxation time $1/(2\eta h_i)$ steps	how long the loss takes to respond to a change of the learning rate
annealing term $\sum_k(\eta_{k-1}-\eta_k)\mathcal{G}(A_T-A_k)$	the loss drop $\mathrm{LD}(t)$ of the empirical laws
D and G^2	the distance to a good solution; the mean squared gradient

WHAT WAS THROWN AWAY. The convex picture throws away curvature, and with it every distinction between fast and slow directions, as well as nonconvexity and every optimizer but plain stochastic gradient descent; it comes first because in it a statement about every schedule and every loss of a class can be proved. The river valley throws away the coupling of river and hills (in the picture of Wen et al. (2025) the river bends and the walls turn with it), changes of curvature, noise that depends on the position, and momentum, Adam and weight decay. It keeps what the convex picture lacks, two time scales, and it is solvable exactly.

WHAT ONE HAD TO NOTICE. The loss at the end of a run is a functional of the whole schedule, and in both pictures it depends on the schedule through two features with very different memories: the area, which remembers everything, and an average of the step sizes over the last few relaxation times, which remembers almost nothing. Large steps buy area; small steps at the end erase the short memory. A schedule is a compromise between

two terms, and the shapes in use are different compromises between the same two terms.

12.3 The theory: the last iterate under an arbitrary schedule

The classical analysis bounds the average iterate: by the online-gradient-descent bound of Theorem A.21 and Jensen's inequality, the η-weighted average of the iterates is within $(D^2 + G^2 \sum_t \eta_t^2)/(2\sum_t \eta_t)$ of the optimum, which is $DG/\sqrt{T}$ for the constant step $D/(G\sqrt{T})$ (Nemirovski et al. 2009). Practitioners use the *last iterate*.[3] The tool for θ_T is the same expansion with the minimizer replaced by an earlier iterate.[4]

Lemma 12.7 (Anchored regret). *Let Assumption 12.5 hold and let $\eta_1, \dots, \eta_T > 0$. For each $k \in \{1, \dots, T\}$ and for $u = \theta^\star$ or $u = \theta_k$,*

$$\sum_{t=k}^{T} \eta_t \, \mathbb{E}\big[\mathcal{L}(\theta_t) - \mathcal{L}(u)\big] \le \frac{1}{2}\,\mathbb{E}\|\theta_k - u\|_2^2 + \frac{G^2}{2}\sum_{t=k}^{T} \eta_t^2.$$

Proof. The plan is to expand the distance of θ_{t+1} to u and telescope from $t = k$ on. Let $s_t = \mathbb{E}[g_t \mid \mathcal{F}_t] \in \partial\mathcal{L}(\theta_t)$; by Jensen's inequality $\|s_t\|_2 \le G$. All expectations are finite: $(\mathbb{E}\|\theta_t\|_2^2)^{1/2} \le \|\theta_1\|_2 + G\sum_{s<t}\eta_s$ by Minkowski's inequality, and $0 \le \mathcal{L}(\theta_t) - \mathcal{L}^\star \le \mathcal{L}(\theta_1) - \mathcal{L}^\star + G\|\theta_t - \theta_1\|_2$ by the subgradient inequality at θ_t. By (12.1),

$$\|\theta_{t+1} - u\|_2^2 = \|\theta_t - u\|_2^2 - 2\eta_t\langle g_t, \theta_t - u\rangle + \eta_t^2\|g_t\|_2^2.$$

For $t \ge k$ both θ_t and u are $\mathcal{F}_t$-measurable, so conditioning on $\mathcal{F}_t$ replaces g_t by s_t in the middle term, and $\langle s_t, \theta_t - u\rangle \ge \mathcal{L}(\theta_t) - \mathcal{L}(u)$. Hence $2\eta_t\mathbb{E}[\mathcal{L}(\theta_t) - \mathcal{L}(u)] \le \mathbb{E}\|\theta_t - u\|_2^2 - \mathbb{E}\|\theta_{t+1} - u\|_2^2 + \eta_t^2 G^2$. Summing over $t = k, \dots, T$ and dropping $-\mathbb{E}\|\theta_{T+1} - u\|_2^2 \le 0$ gives the claim. ∎

Theorem 12.8 (Last iterate under an arbitrary schedule). *Let Assumption 12.5 hold, let $\eta_1, \dots, \eta_T > 0$, and let $\theta_1, \dots, \theta_T$ be generated by (12.1). Write $R_k = \sum_{t=k}^{T}\eta_t$ and $V_k = \sum_{t=k}^{T}\eta_t^2$. Then*

$$\mathbb{E}\mathcal{L}(\theta_T) - \mathcal{L}^\star \;\le\; \frac{D^2 + G^2 V_1}{2R_1} \;+\; \frac{G^2}{2}\sum_{k=1}^{T-1}\frac{\eta_k}{R_{k+1}}\cdot\frac{V_k}{R_k}. \tag{12.2}$$

Proof. The plan is to compare, for each k, the η-weighted mean of the expected losses from step k on with the loss at step k, using θ_k as the comparator in Lemma 12.7, and to telescope from the mean over the whole run down to the last iterate. Write $\ell_t = \mathbb{E}\mathcal{L}(\theta_t)$ and $\bar{\ell}_k = R_k^{-1}\sum_{t=k}^{T}\eta_t\ell_t$, so that $\bar{\ell}_T = \ell_T$.

The whole run. Lemma 12.7 with $k = 1$ and $u = \theta^\star$ gives $R_1(\bar{\ell}_1 - \mathcal{L}^\star) \le \frac{1}{2}(D^2 + G^2V_1)$.

[3] Averaging in stochastic approximation goes back to Polyak and Juditsky (1992); in deep learning it survives as the moving average of the weights and in schedule-free methods (Chapter 5).

[4] The proofs use one more stochastic gradient g_T at θ_T and the point θ_{T+1}. Neither affects $\theta_1, \dots, \theta_T$, so the step size η_T is a free parameter of the bounds.

One step of the telescope. Let $1 \le k \le T-1$. Lemma 12.7 with $u = \theta_k$ gives $R_k(\bar\ell_k - \ell_k) \le \frac{1}{2}G^2V_k$. By definition $R_k\bar\ell_k = \eta_k\ell_k + R_{k+1}\bar\ell_{k+1}$; subtracting $R_{k+1}\bar\ell_k = (R_k - \eta_k)\bar\ell_k$ from both sides,

$$R_{k+1}\big(\bar\ell_{k+1} - \bar\ell_k\big) = \eta_k\big(\bar\ell_k - \ell_k\big) \le \frac{G^2}{2}\,\frac{\eta_k V_k}{R_k}.$$

Summation. Divide by $R_{k+1} > 0$ and add the identity $\ell_T - \mathcal{L}^\star = (\bar\ell_1 - \mathcal{L}^\star) + \sum_{k=1}^{T-1}(\bar\ell_{k+1} - \bar\ell_k)$. ∎

The first term of (12.2) is the averaging bound. The second, the price of the last iterate, averages the step sizes near the end: V_k/R_k is the mean step size from step k on, and its weight η_k/R_{k+1}, the current step over the area still to come, is negligible early and of order one at the end. The two terms are the convex counterparts of the river and the hills of Section 12.4: one decreases with the area, the other remembers the recent schedule. Constant steps make the weights $1/(T-k)$, whose harmonic sum costs a logarithm; a decay to zero makes the step sizes small where the weights are large.[5]

For linear decay the theorem gives about $2DG/\sqrt{T}$ (Corollary 12.10), twice the averaging bound. A different argument closes the gap.

Proposition 12.9 (Linear decay matches averaging). *Let Assumption 12.5 hold, let $\gamma > 0$, and run (12.1) with $\eta_t = \gamma\,(T-t)/T$ for $t = 1, \dots, T-1$. Then*

$$\mathbb{E}\mathcal{L}(\theta_T) - \mathcal{L}^\star \le \frac{D^2}{2\gamma T} + \frac{\gamma G^2}{2},$$

which is $DG/\sqrt{T}$ for $\gamma = D/(G\sqrt{T})$, the bound for the average iterate of stochastic gradient descent with the constant step γ.

Proof. The plan is to write the last iterate as the average of a sequence that takes constant steps along the same stochastic gradients, and to show by convexity that the regret of that sequence controls the last iterate. Let $z_1 = \theta_1$ and $z_{t+1} = z_t - \gamma g_t$ for $t = 1, \dots, T$, with g_t the stochastic gradient at θ_t.

The last iterate is an average. We claim that $T\theta_t = \sum_{s<t} z_s + (T-t+1)z_t$ for $1 \le t \le T$. It holds for $t = 1$, and if it holds for t, the claimed values at $t+1$ and t differ by $T^{-1}\big(z_t + (T-t)z_{t+1} - (T-t+1)z_t\big) = \frac{T-t}{T}(z_{t+1} - z_t) = -\eta_t g_t$, which is the step (12.1). In particular $\theta_T = T^{-1}\sum_{s=1}^{T} z_s$.

A convexity inequality. Let $p_t = t^{-1}\sum_{s\le t} z_s$, so that $p_1 = \theta_1$ and $p_T = \theta_T$. For $2 \le t \le T$ the claim reads $T\theta_t = (t-1)p_{t-1} + (T-t+1)z_t$; eliminating z_t with $tp_t = (t-1)p_{t-1} + z_t$ gives

$$z_t - \theta_t = \frac{t-1}{T-t+1}\,(\theta_t - p_{t-1}), \qquad p_t = \frac{(t-1)(T-t)\,p_{t-1} + T\theta_t}{t\,(T-t+1)}.$$

The second identity makes p_t a convex combination of p_{t-1} and θ_t, so with $F_1 = \mathcal{L}(\theta_1)$ and $F_t = \big((t-1)(T-t)F_{t-1} + T\mathcal{L}(\theta_t)\big)/(t(T-t+1))$, convexity and induction give $\mathcal{L}(p_t) \le F_t$; note that $F_T = \mathcal{L}(\theta_T)$. Let $s_t \in \partial\mathcal{L}(\theta_t)$ and

[5] The comparison with an earlier iterate is the argument of Shamir and Zhang (2013); Orabona (2020) gave a version without a bounded domain, which we follow. Defazio et al. (2023) prove a last-iterate bound for general schedules by a reduction from regret, and Schaipp et al. (2025) evaluate such a bound along training runs.

$u \in \mathbb{R}^d$. The first identity and the subgradient inequality at θ_t, applied to u and to p_{t-1}, give

$$\langle s_t, z_t - u\rangle \geq \mathcal{L}(\theta_t) - \mathcal{L}(u) + \frac{t-1}{T-t+1}\big(\mathcal{L}(\theta_t) - F_{t-1}\big) = tF_t - (t-1)F_{t-1} - \mathcal{L}(u),$$

the equality being the definition of F_t rearranged (for $t = 1$ read $(t-1)F_{t-1} = 0$). The right side telescopes, so $\sum_{t=1}^{T}\langle s_t, z_t - u\rangle \geq T\big(\mathcal{L}(\theta_T) - \mathcal{L}(u)\big)$.

Conclusion. Take $u = \theta^\star$ and $s_t = \mathbb{E}[g_t \mid \mathcal{F}_t]$. Since z_t is $\mathcal{F}_t$-measurable, $\mathbb{E}\langle g_t, z_t - \theta^\star\rangle = \mathbb{E}\langle s_t, z_t - \theta^\star\rangle$. Since $z_{t+1} = z_t - \gamma g_t$, the expansion in the proof of Lemma 12.7, with z_t in place of θ_t (it uses only the form of the step, not where g_t was computed), gives $\sum_{t=1}^{T}\mathbb{E}\langle g_t, z_t - \theta^\star\rangle \leq D^2/(2\gamma) + \gamma T G^2/2$. Divide by T. ∎

The gradient g_t enters θ_T with the weight $\gamma(T-t)/T$, exactly its total weight in the average of $z_1, \dots, z_T$: the last iterate of linear decay *is* an average, computed without storing one. Defazio et al. (2023) derive the schedule from a conversion of this kind with general weights (Exercise 12.6).

Corollary 12.10 (Constant steps and linear decay). *Under the hypotheses of Theorem 12.8, let* $H_n = \sum_{j=1}^{n} 1/j$.

(i) For $\eta_t = \eta$ *the right side of* (12.2) *is* $D^2/(2\eta T) + \frac{1}{2}G^2\eta\,(1 + H_{T-1})$. *Its minimum over* η *is* $DG\sqrt{(1+H_{T-1})/T}$; *the step* $\eta = D/(G\sqrt{T})$ *gives* $(DG/\sqrt{T})(1 + \frac{1}{2}H_{T-1})$.

(ii) For $\eta_t = \eta\,(T+1-t)/(T+1)$ *it is* $D^2/(\eta T) + G^2\eta\,(T - \frac{1}{2} + H_{T-1})/(T+1)$, *with minimum* $(2DG/\sqrt{T})(1 + O(T^{-1}\log T))$.

Proof. (i) Here $V_k/R_k = \eta$ and $\eta_k/R_{k+1} = 1/(T-k)$, so the sum is ηH_{T-1}, and the first term is $D^2/(2\eta T) + G^2\eta/2$; the minimum of $a/\eta + b\eta$ is $2\sqrt{ab}$. (ii) Put $m = T + 1 - k$ and $\delta = \eta/(T+1)$. Then $\eta_k = \delta m$, $R_k = \delta m(m+1)/2$, $V_k = \delta^2 m(m+1)(2m+1)/6$ and $R_{k+1} = \delta(m-1)m/2$, so the kth term of the sum is $\frac{2\delta}{3}\cdot\frac{2m+1}{m-1}$, and $\sum_{m=2}^{T}\frac{2m+1}{m-1} = 2(T-1) + 3H_{T-1}$. With $R_1 = \eta T/2$ and $V_1 = \eta^2 T(2T+1)/(6(T+1))$, collecting terms gives the formula. ∎

With the best constant step the bound exceeds the averaging bound by the factor $\sqrt{1 + H_{T-1}} \approx \sqrt{\ln T}$, and with the step $D/(G\sqrt{T})$ by about $\frac{1}{2}\ln T$. Its constant is not sharp: Zamani and Glineur (2025) computed the exact worst case of the last iterate of the subgradient method with constant steps, and for the best constant step it is at most $DG\sqrt{1 + \log(N)/4}/\sqrt{N+1}$ after N steps.[6] For other shapes the sums in (12.2) are Riemann sums. For a shape φ of bounded variation on $[0,1]$ and positive on $[0,1)$, put $\Phi(s) = \int_s^1 \varphi$, $\Psi(s) = \int_s^1 \varphi^2$ and

$$J[\varphi] = \frac{\Psi(0)}{\Phi(0)} + \int_0^1 \frac{\varphi(s)\,\Psi(s)}{\Phi(s)^2}\,ds. \tag{12.3}$$

[6] Zamani and Glineur (2025) also show that linear decay attains the lower bound $DG/\sqrt{N+1}$ that holds for all first-order methods, and that no single sequence of step sizes attains it at every N: the optimal schedule must know its horizon, which is Observation 12.2 in the worst case. For $\eta_t \propto 1/\sqrt{t}$, which does not, the last iterate loses a logarithmic factor that cannot be removed (Harvey et al. 2019).

shape	$\Phi(0)$	$J[\varphi]$	best bound $\times\sqrt{T}/(DG)$
constant	1	$1 + H_{T-1}$ (exact)	$\sqrt{1 + H_{T-1}}$, which is 3.28 at $T = 10^4$
linear decay to zero	1/2	2	2
cosine to zero	1/2	2.122	2.060
cosine to $\rho = 0.1$	0.55	$1.775 + 0.1 \ln T$	2.21 at $T = 10^4$; 2.31 at $T = 10^5$
WSD, linear cooldown of fraction c	$1 - c/2$	$2 + \ln \frac{2-c}{c}$	2.28, 2.16, 2.03 for $c = 0.1, 0.2, 0.5$
WSD, $1 - \sqrt{\cdot}$ cooldown, $c = 0.2$	0.867	3.92	2.13

Table 12.1. The last-iterate bound (12.2) at its best peak (Proposition 12.11); the constant and floor rows are exact discrete values, whose J grows with T. A floor ρ adds $\rho \ln T$ to J (Exercise 12.3).

Proposition 12.11 (The bound for a shape). *Let φ be a shape as above such that $\varphi(s)/(1-s)^p$ has a positive limit as $s \to 1$, for some $p > 0$. Let $\eta_t = \eta\,\varphi(t/(T+1))$ for $t \le T$, with $\eta = \gamma D/(G\sqrt{T})$ for a fixed $\gamma > 0$. Then $J[\varphi] < \infty$; as $T \to \infty$ the right side of (12.2) is $(DG/\sqrt{T})\big(1/(2\gamma\Phi(0)) + \gamma J[\varphi]/2 + o(1)\big)$, and its minimum over γ is $DG\sqrt{J[\varphi]/(\Phi(0)T)}\,(1 + o(1))$.*

Proof sketch. Near $s = 1$ the hypothesis gives $\Phi \asymp (1-s)^{p+1}$, $\Psi \asymp (1-s)^{2p+1}$ and $\varphi\Psi/\Phi^2 \asymp (1-s)^{p-1}$, which is integrable, so $J[\varphi] < \infty$. With $s = k/(T+1)$, the sums R_k and V_k are Riemann sums of $\eta T\Phi(s)$ and $\eta^2 T\Psi(s)$, so for $1 - s \gg 1/T$ the kth term of the sum in (12.2) is $(\eta/T)\,\varphi\Psi/\Phi^2$ at s up to a factor $1 + o(1)$: the sum is a Riemann sum of the integral in (12.3). The power law at $s = 1$ makes the terms with $1 - s \le \varepsilon$ contribute $O(\eta\varepsilon^p)$ in total, uniformly in T; we omit this estimate, which for the shapes of Table 12.1 can be done by exact summation as in Corollary 12.10(ii). The hypothesis cannot be dropped: if φ falls faster than every power of $1 - s$, as $e^{-0.02/(1-s)}$ does, the last term alone is close to η, and the bound converges to the formula with $J[\varphi] + 1$ in place of $J[\varphi]$. At $T = 10^5$ the discrete bound agrees with the limit within 0.02% for the shapes of Table 12.1. ∎

Corollary 12.12 (The cooldown in the bound). *For WSD with peak η and a linear cooldown to zero over the last fraction $c \in (0, 1]$ of the steps, the right side of (12.2) is, as $T \to \infty$ with $\eta\sqrt{T}$ fixed, $U(c)\,(1 + o(1))$ with*

$$U(c) = \frac{D^2}{(2-c)\,\eta T} + \frac{G^2\eta}{2}\Big(2 + \ln\frac{2-c}{c}\Big). \tag{12.4}$$

(i) With the peak tuned for each c, the bound is

$$\min_{\eta>0} U(c) = DG\sqrt{\frac{2\left(2 + \ln\frac{2-c}{c}\right)}{(2-c)\,T}},$$

which decreases in c: the bound prefers to decay over the whole run.

(ii) With the peak fixed, let $\mu = G\eta\sqrt{T}/D$ be the peak in units of the best peak for linear decay. Then U is minimized at $c^\star = \min\{1, 2\mu^2/(1+\mu^2)\}$: a peak below the best one asks for a shorter cooldown.

PROOF. The plan is to compute J for this shape and apply Proposition 12.11. Here $\Phi(0) = 1 - c/2$ and $\Psi(0) = 1 - 2c/3$. On the cooldown $\Phi(s) = (1-s)^2/(2c)$ and $\Psi(s) = (1-s)^3/(3c^2)$, so the integrand of (12.3) is $4/(3c)$ and contributes $4/3$. On the stable phase, with $w = 1 - s - c/2$, the integrand is $(w - c/6)/w^2$, with integral $\ln\frac{2-c}{c} - \frac{1}{3} + \frac{c}{3(2-c)}$ over $w \in [c/2, 1 - c/2]$. Adding $\Psi(0)/\Phi(0) = \frac{2(3-2c)}{3(2-c)}$ gives $J = 2 + \ln\frac{2-c}{c}$. (i) The tuned value is $DG\sqrt{2F(c)/T}$ with $F(c) = (2 + \ln r)/(2-c)$ and $r = (2-c)/c \geq 1$, and $F'(c) = (1 - r + \ln r)/(2-c)^2 \leq 0$ because $\ln r \leq r - 1$. (ii) The derivative $U'(c) = D^2/((2-c)^2\eta T) - G^2\eta/(c(2-c))$, multiplied by $c(2-c)^2\eta T/D^2 > 0$, is $c(1+\mu^2) - 2\mu^2$, which changes sign from negative to positive at $2\mu^2/(1+\mu^2)$; if that exceeds 1, then U decreases on $(0, 1]$. ■

So the bound reproduces Observation 12.1 in a weak form: a cooldown removes the logarithm of the constant schedule, and its benefit saturates. It does not single out 20%; with tuned peaks it prefers a decay over the whole run, as in the experiments of Defazio et al. (2023) and Bergsma et al. (2025), and it prefers the $1 - \sqrt{\cdot}$ cooldown of Hägele et al. (2024) to a linear one.

12.4 The theory: the river valley

In the river valley the expected loss can be computed exactly for every schedule. For a hill i write $q_{i,t} = (1 - \eta_t h_i)^2$ and $P_i(k, T) = \prod_{t=k}^{T-1} q_{i,t}$ (an empty product is 1), the contraction of its second moment between steps k and T. When $0 \leq \eta h_i < 2$, let

$$v_i(\eta) = \frac{\eta\,\sigma_i^2}{h_i\,(2 - \eta h_i)},$$

the stationary second moment of the hill at the constant step η (Lemma A.15).

Theorem 12.13 (Stochastic gradient descent in the river valley). *Let Assumption 12.6 hold and let (η_t) be a schedule. For each hill i and each $T \geq 1$:*

(i) $\mathbb{E}\,y_{T,i}^2 = P_i(1,T)\,y_{1,i}^2 + \sigma_i^2 \sum_{k=1}^{T-1} \eta_k^2\, P_i(k+1, T)$.

(ii) If $0 \leq \eta_t h_i < 2$ for $t < T$, then

$$\mathbb{E}\,y_{T,i}^2 = v_i(\eta_1) + P_i(1,T)\big(y_{1,i}^2 - v_i(\eta_1)\big) - \sum_{k=2}^{T-1} \big(v_i(\eta_{k-1}) - v_i(\eta_k)\big)\big(1 - P_i(k,T)\big). \quad (12.5)$$

(iii) If $0 \leq \eta_t h_i \leq \frac{1}{2}$ for $k \leq t < T$, then $e^{-2h_i(A_T - A_k) - 2h_i^2 \sum_{t=k}^{T-1}\eta_t^2} \leq P_i(k,T) \leq e^{-2h_i(A_T - A_k)}$.

(iv) If the river is linear, then $\mathbb{E}\,r(x_T) = r(x_1) - \|\nabla r\|_2^2\, A_T$. If it is quadratic, each of its eigendirections satisfies (i)–(iii) with its own curvature and noise.

Consequently $\mathbb{E}\,\mathcal{L}(\theta_T) = \mathbb{E}\,r(x_T) + \frac{1}{2}\sum_i h_i\,\mathbb{E}\,y_{T,i}^2$ is given in closed form by the schedule.

PROOF. The plan is to derive the second-moment recursion of each coordinate and to rewrite it around the moving equilibrium $v_i(\eta_t)$. (i) The hill coordinate moves by $y_{t+1,i} = (1-\eta_t h_i)y_{t,i} - \eta_t \zeta_{t,i}$. Square it and condition on $\mathcal{F}_t$; the cross term vanishes because $y_{t,i}$ is $\mathcal{F}_t$-measurable and $\mathbb{E}[\zeta_{t,i} \mid \mathcal{F}_t] = 0$, so $\mathbb{E}[y_{t+1,i}^2 \mid \mathcal{F}_t] = q_{i,t} y_{t,i}^2 + \eta_t^2 \sigma_i^2$. Take expectations and unroll from $t = 1$. (ii) Write $u_t = \mathbb{E} y_{t,i}^2$ and $v_t = v_i(\eta_t)$; the step size η_T does not affect $y_{T,i}$, so we may take $\eta_T = \eta_{T-1}$ and v_T is defined. A direct computation gives $q_{i,t} v_t + \eta_t^2 \sigma_i^2 = v_t$, so subtracting it from the recursion of (i) gives $u_{t+1} - v_t = q_{i,t}(u_t - v_t)$, that is, $u_{t+1} - v_{t+1} = q_{i,t}(u_t - v_t) + (v_t - v_{t+1})$. Unrolling, $u_T - v_T = P_i(1,T)(u_1 - v_1) + \sum_{k=2}^{T} (v_{k-1} - v_k) P_i(k,T)$. The term $k = T$ has $P_i(T,T) = 1$ and cancels the unused v_T, and $v_{T-1} = v_1 - \sum_{k=2}^{T-1}(v_{k-1} - v_k)$ gives (12.5). (iii) For $0 \le z \le \frac{1}{2}$, $e^{-z-z^2} \le 1 - z \le e^{-z}$: the right inequality is convexity of the exponential, and $\ln(1-z) + z + z^2$ vanishes at 0 and has derivative $z(1-2z)/(1-z) \ge 0$. Apply it with $z = \eta_t h_i$, square, and multiply. (iv) A linear river has $x_{t+1} = x_t - \eta_t(\nabla r + \zeta_{t,x})$, so $\mathbb{E} x_T = x_1 - A_T \nabla r$, and r is affine. The eigendirections of a quadratic river are decoupled coordinates of the same form as the hills. ∎

Part (ii), the *annealing decomposition*, is the structure behind the chapter: the second moment of a hill at step T is its equilibrium at the initial step size, plus the fading memory of the start, minus one term for each decrease of the step size, the decrease of the equilibrium times the fraction $1 - P_i(k,T)$ of it realized since. By (iii) that fraction depends on the area since the decrease, $1 - e^{-2h_i(A_T - A_k)}$ up to relative corrections of order ηh_i; an increase of the step size, as in warmup, enters with the opposite sign.

Corollary 12.14 (Noise floor and relaxation). *Under Assumption 12.6, let $0 < \eta h_i < 2$ for every hill.*

(i) At the constant step η the hill part of the expected loss converges, at the rate $(1-\eta h_i)^2$ per step in hill i, to the noise floor $\mathcal{L}_{\text{hill}}(\eta) = \sum_i \eta\sigma_i^2/(2(2-\eta h_i))$, *which lies between $\frac{\eta}{4}\sum_i \sigma_i^2$ and $\eta \sum_i \sigma_i^2/(4 - 2\eta \max_i h_i)$.*

(ii) If $\mathbb{E} y_{k,i}^2 = v_i(\eta)$ and the step size is $\eta' \in [0, \eta)$ from step k on, then $\mathbb{E} y_{T,i}^2 = v_i(\eta') + \big(v_i(\eta) - v_i(\eta')\big)(1-\eta' h_i)^{2(T-k)}$ for $T \ge k$.

PROOF. (i) In (12.5) all differences vanish and $P_i(1,T) = (1-\eta h_i)^{2(T-1)}$; multiply by $h_i/2$ and sum, and use $2 - \eta \max_i h_i \le 2 - \eta h_i \le 2$. (ii) The recursion $u_{t+1} - v_i(\eta') = (1-\eta' h_i)^2 (u_t - v_i(\eta'))$ of the proof of Theorem 12.13(ii) holds for $t \ge k$. ∎

So at a constant learning rate each hill adds about $\eta\sigma_i^2/4$ to the loss, a floor proportional to the learning rate and to the noise, hence to $1/B$. A drop of the learning rate removes the excess on the *relaxation time* $1/(\eta' h_i(2 - \eta' h_i)) \approx 1/(2\eta' h_i)$ of the *new* step size, so a decay to zero never quite finishes.

THE COOLDOWN LAW. Take one hill in equilibrium at the peak η, and a linear cooldown to zero over the last K steps. Measure time by the area: to

first order in ηh the recursion of Theorem 12.13(i) is $dv/dA = -2hv + \eta(A)\sigma^2$, because a step changes v by $-(2\eta_t h - \eta_t^2 h^2)v + \eta_t^2\sigma^2$ and the area by η_t. The cooldown has area $\eta K/2$, and with ϱ the area that remains before the end the step size is $\eta(\varrho) = \sqrt{2\eta\varrho/K}$. Solving the linear equation from $v = \eta\sigma^2/(2h)$,

$$v_{\text{end}} = \frac{\eta\sigma^2}{2h} e^{-\eta h K} + \sigma^2 \int_0^{\eta K/2} e^{-2h\varrho} \sqrt{2\eta\varrho/K}\, d\varrho,$$

an incomplete gamma function. With $\kappa = \eta h K$ the two terms combine into

$$\frac{h}{2}\, v_{\text{end}} = \frac{\eta\sigma^2}{4}\, \chi(\kappa), \qquad \chi(\kappa) = \frac{\sqrt{\pi}}{2}\, \frac{\operatorname{erf}\sqrt{\kappa}}{\sqrt{\kappa}}. \tag{12.6}$$

A short cooldown has $\chi(\kappa) = 1 - \kappa/3 + O(\kappa^2)$ and removes almost nothing; a long one leaves $\chi(\kappa) \approx \frac{\sqrt{\pi}}{2}\kappa^{-1/2}$. The residual noise after a linear cooldown of K steps decays like $K^{-1/2}$, not exponentially: the noise injected within the last area $1/(2h)$, at step sizes of order $(\eta/(hK))^{1/2}$, has no area left in which to be forgotten. ◇

The exact recursion confirms (12.6) within relative errors of order ηh: started at the equilibrium $v(\eta)$, with the steps $\eta(1 - j/K)$ for $j = 1, \dots, K$, for $\eta h = 0.01$ and $K = 100, 1000, 10^4$ it leaves 0.746, 0.279 and 0.0882 of the floor, against $\chi = 0.747$, 0.280 and 0.0886. The cooldown that removes the most noise in K steps, ignoring the river, is a $1/t$ decay, with a residual of order $1/K$ (Exercise 12.8).[7]

[7] The error function enters as $\int_0^x \sqrt{u}\, e^{-u} du = \frac{\sqrt{\pi}}{2} \operatorname{erf}\sqrt{x} - \sqrt{x}\, e^{-x}$. A cooldown whose step size vanishes like the pth power of the remaining steps leaves a tail $\kappa^{-p/(p+1)}$: $\kappa^{-1/2}$ for the linear and $1 - \sqrt{\cdot}$ shapes, $\kappa^{-2/3}$ for a cosine.

THE BEST COOLDOWN. Let the river be linear, so that by Theorem 12.13(iv) it loses $\omega = \|\nabla r\|_2^2$ per unit of area, and replace the last K steps of a constant schedule by a linear cooldown. The cooldown gives up the area $\eta K/2$ and removes noise by (12.6), so the final loss is, up to a constant, $\omega\eta K/2 + \sum_i \frac{1}{4}\eta\sigma_i^2\chi(\eta h_i K)$. For one hill with $\eta h K \gg 1$ the derivative vanishes at

$$K^\star = \left(\frac{\sqrt{\pi}\,\sigma^2}{8\,\omega}\right)^{2/3} (\eta h)^{-1/3}, \tag{12.7}$$

where the residual noise is twice the river progress given up. The best cooldown is a fixed number of steps, and its *fraction* of the run falls like $1/T$. A river that flattens changes this. If the river loss is a power of the area, $\bar r(A) = L_\infty + CA^{-\alpha}$, then its slope at the end of a run of area ηT is $\alpha C(\eta T)^{-1-\alpha}$, and the same balance, with $1 - c/2 \approx 1$, gives the best cooldown fraction

$$c^\star \approx \left(\frac{\sqrt{\pi}}{8\alpha C} \sum_i \frac{\sigma_i^2}{\sqrt{h_i}}\right)^{2/3} \eta^{(1+2\alpha)/3}\, T^{(2\alpha-1)/3}. \tag{12.8}$$

A fixed fraction is optimal at every horizon exactly when $\alpha = \frac{1}{2}$; a river that falls faster than the inverse square root of the area asks for a longer cooldown in a longer run, a slower one for a shorter cooldown. ◇

For one hill with $\eta h = 0.01$, $\sigma^2 = 1$ and $\omega = 10^{-4}$, the exact recursion puts the best linear cooldown at 790 steps, (12.7) at 789; the asymptotic form fails, as it must, when $\eta h K^\star$ is of order one. The decomposition also explains the laws of Observation 12.3.

FROM THE RIVER VALLEY TO THE EMPIRICAL LAWS. Multiply (12.5) by $h_i/2$, sum over the hills, and keep the first order in ηh_i, where $\frac{h_i}{2} v_i(\eta) \approx \frac{1}{4}\eta\sigma_i^2$ and $1 - P_i(k,T) \approx 1 - e^{-2h_i(A_T - A_k)}$. Let $\mathcal{G}(u) = \sum_i \frac{1}{4}\sigma_i^2\big(1 - e^{-2h_i u}\big)$, a saturation function built from the Laplace transform of the noise-weighted hill spectrum. Once the initial transient has faded,

$$\mathbb{E}\mathcal{L}(\theta_T) \approx \mathbb{E}r(x_T) + \frac{\eta_1}{4}\sum_i \sigma_i^2 - \sum_{k=2}^{T-1}(\eta_{k-1} - \eta_k)\,\mathcal{G}(A_T - A_k). \tag{12.9}$$

This is the form of Observation 12.3: an area term, close to $L_\infty + CA_T^{-\alpha}$ for a quadratic river with a power-law spectrum, and a loss drop that adds, for each decrease of the learning rate, the decrease times a saturating function of the area since. If the noise is spread over many hills and its distribution over their curvatures, with weights σ_i^2, is a gamma distribution with shape ν and scale $\bar{h}$, then $\mathcal{G}(u) = \frac{1}{4}\big(\sum_i \sigma_i^2\big)\big(1 - (1 + 2\bar{h}u)^{-\nu}\big)$, the power-law saturation of Luo et al. (2025). A single hill at a nearly constant step $\bar{\eta}$ after the decreases gives the saturation $1 - e^{-2\bar{\eta}h(T-k)}$, counted in steps, which is the annealing area of Tissue et al. (2025). Finally, the continuum form of Theorem 12.13(i) writes the noise term as $\sum_k \eta_k^2 \mathcal{K}(A_T - A_{k+1})$ with the kernel $\mathcal{K} = \mathcal{G}'$, a convolution of the squared schedule with a kernel in area-time; Li et al. (2025) derive a functional scaling law of this kind for stochastic gradient descent on kernel regression with power-law spectra. ◇

The model predicts what the laws leave open: the loss relaxes after a decrease in *area*, not in steps, because a quadratic hill contracts by $(1 - \eta h)^2 \approx e^{-2h\eta}$ per step, that is, by e^{-2h} per unit of area. Tissue et al. (2025) count the relaxation in steps, Luo et al. (2025) in area with a fitted correction that depends on the learning rate, which measures a departure from a fixed quadratic landscape.

REMARK 12.15 (Branching, rewarming and averaging). After a long stable phase the hills have forgotten everything but the peak, so by (12.5) the loss after a cooldown depends on the stable phase only through its area (Observation 12.2); rewarming a cooled checkpoint refills the hills within a relaxation time and keeps the river progress. Averaging can replace the decay, as in schedule-free methods (Defazio et al. 2024) (Chapter 5): the mean of the last K iterates at a constant step gives up the same area as a linear cooldown of K steps and leaves a residual of order $1/(\eta h K)$, not $(\eta h K)^{-1/2}$ (Exercise 12.9).

12.5 Reading the theory

LOSS CURVES. Figure 12.1 runs the four schedules in a river valley built to look like a language model: a quadratic river in 200 directions whose loss falls like the inverse square root of the area, and 40 noisy hills with curvatures between 0.2 and 2. The points, means of 100 runs, agree with the exact curves within their sampling error (final values 0.203, 0.156, 0.158, 0.131 against 0.207, 0.157, 0.158, 0.131). The constant schedule reaches the floor $\mathcal{L}_{\text{hill}}(0.04) = 0.102$ within a few hundred steps and follows the river

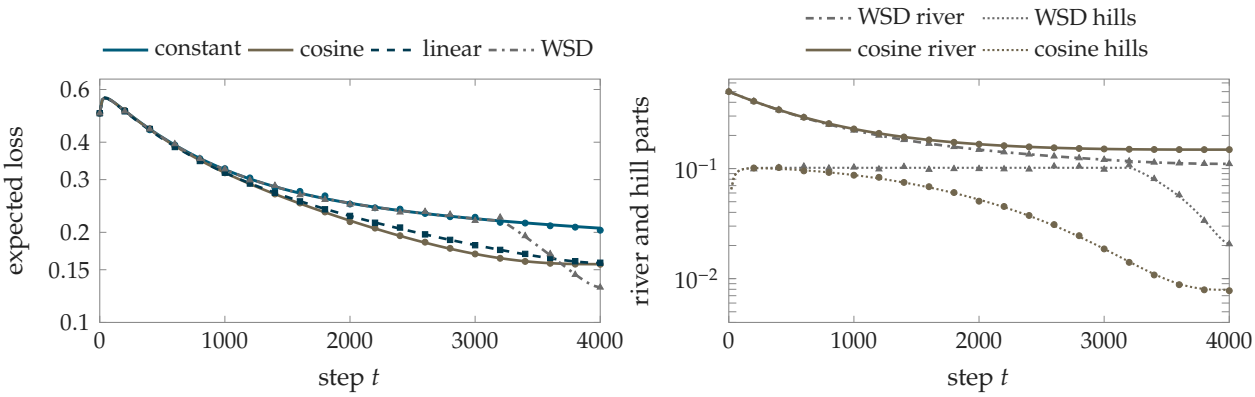


Figure 12.1. Four schedules in a river valley: curves are the exact expected loss (Theorem 12.13), points the mean of 100 runs of stochastic gradient descent with Gaussian noise (seed 12), every 200 steps. River: 200 directions with curvatures $0.02\,j^{-2}$, initial losses proportional to j^{-2} (total 0.5), no noise. Hills: 40, curvatures log-spaced in $[0.2, 2]$, started at 0, noise variance $1/4$ each. Peak 0.04, $T = 4000$, no warmup; WSD cools down linearly over the last 800 steps. Left: total loss. Right: river part (upper curves) and hill part (lower curves). The points agree with the curves within sampling error (at most 2.7% apart, less than three standard errors); the river carries no noise, so its points coincide with its curves. Script: `12-schedules-river.py`.

down at that height. In the cooldown of WSD the hill part falls from 0.102 to 0.021 while the river part continues from 0.117 to 0.110: four fifths of the floor removed for a tenth of the area. Cosine and linear decay remove almost all of the floor but cover half the area, and their river part ends at 0.149. At a common peak the model ranks WSD first, cosine and linear decay together, the constant schedule last.

That ranking is not a prediction about tuned runs. With each peak tuned separately among the peaks at which every step is stable, $\eta \max_i h_i < 2$, the model raises the peaks of the decaying schedules far above the common one (cosine to that limit, linear decay to 0.64, WSD to 0.21) and ranks cosine first (0.052, against 0.074 for linear decay, 0.096 for WSD and 0.195 for constant steps): a larger peak buys area, and a long decay removes the larger floor. Cosine even gains from peaks up to nearly twice that limit (0.050 at the peak 1.9): the stiff hills grow during its first, unstable steps, and the contraction during the later steps undoes the growth, which only exactly quadratic hills allow. Nothing in the model penalizes a large peak short of divergence (Chapters 8 and 10). What is robust are the decomposition and its time scales: the cooldown drop is the noise floor at the peak, about $\eta\,\mathrm{tr}(\Sigma_{\mathrm{hill}})/(4B)$, so it should double when the peak doubles or the batch size halves; it takes a few relaxation times of the slowest noisy direction; and it leaves a residual decaying only like $(\eta h K)^{-1/2}$. The curves are exact for this model, so their agreement with the points is guaranteed up to sampling error; it must fail once the curvature changes along the run, the noise depends on the position, or the optimizer is not plain stochastic gradient

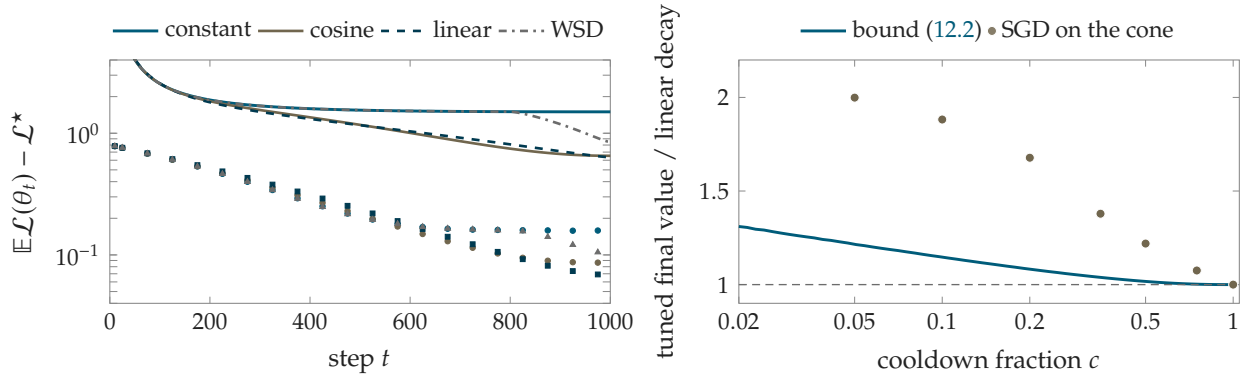


Figure 12.2. The last-iterate bound against stochastic gradient descent on the cone $\sqrt{2/\pi}\,\|\theta - \theta^\star\|_2$ in $d = 100$ dimensions, $D = 1$, $G^2 = d$. Left: the bound (12.2) for θ_t, computed from $\eta_1, \dots, \eta_t$ (curves), and the mean of 200 runs (points, seed 21), $T = 1000$, common peak $D/(G\sqrt{T})$; WSD cools down over the last 200 steps. At $t = T$ the bound is 7.6 to 9.5 times the simulated value. Right: WSD with cooldown fraction c, peak tuned for each c, final value relative to linear decay; the bound is minimized exactly, the simulation takes the best of 17 peaks between a quarter of and four times $D/(G\sqrt{T})$, 40 runs each (seed 22). Script: `12-schedules-bound.py`.

descent.

THE CONVEX BOUND ALONG A RUN. Figure 12.2 (left) evaluates (12.2) at every step on the *cone* $\mathcal{L}(\theta) = \mathbb{E}_a|\langle a, \theta - \theta^\star\rangle| = \sqrt{2/\pi}\,\|\theta - \theta^\star\|_2$ with $a \sim \mathcal{N}(0, I_{100})$, which is convex with a nonsmooth minimum and has $G^2 = d$ exactly. The shapes agree with the simulation and with Observation 12.4: constant steps plateau, WSD plateaus and then drops, cosine and linear decay descend smoothly, and both rank the final values linear decay, cosine, WSD, constant. The values do not agree: the bound is 7.6 to 9.5 times the simulated suboptimality at the end. On the cone the last iterate of constant steps settles exactly at the averaging floor $\eta G^2/2$, with no logarithm, and it forgets a change of the step size within about 130 steps whatever the step size, so its final loss is set by the last steps. Tuned comparisons (right panel) suffer: the bound charges a 20% cooldown 8% over linear decay, the cone 68%; cosine 3%, the cone 31%. The bound covers every convex Lipschitz function, and the functions on which the last iterate comes within a constant factor of it are specially constructed (Harvey et al. 2019; Zamani and Glineur 2025); it agrees with training curves in shapes and rankings, not in values.

HOW LONG TO COOL DOWN. Figure 12.3 tests the cooldown heuristics in the valley of Figure 12.1 and in one whose river falls like A^{-1}. For $\alpha = \frac{1}{2}$ the best cooldown fraction is 0.24, 0.28 and 0.27 at $T = 1000$, 4000 and 16000, fixed as (12.8) predicts; for $\alpha = 1$ it is 0.21, 0.35 and 0.48, a factor 2.3 over a sixteenfold increase of T, against $16^{1/3} = 2.5$. The continuum minimizer coincides with the exact one within the grid spacing, and the minimum is flat, as in Observation 12.1: at $\alpha = \frac{1}{2}$ and $T = 4000$ every fraction between 0.18 and 0.41 is within 1% of the best. At fractions this large the peak

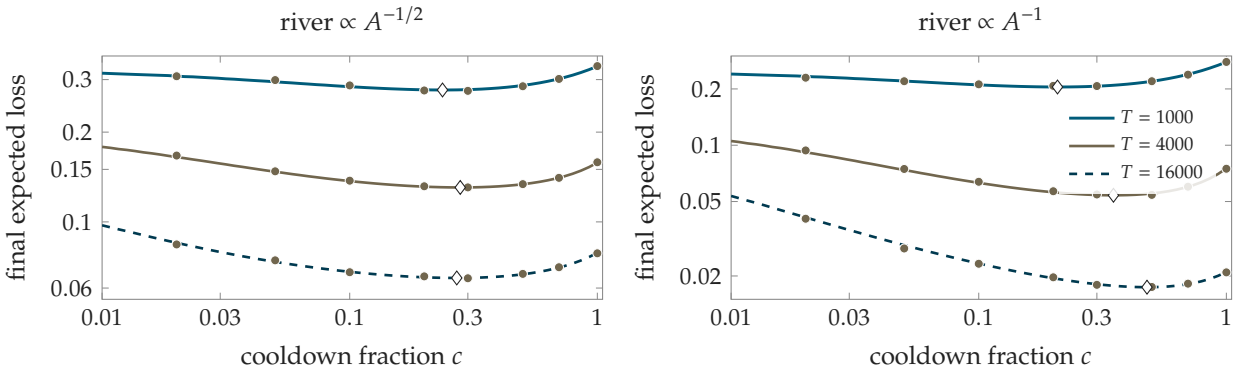


Figure 12.3. The final loss of WSD against its cooldown fraction c, in the valley of Figure 12.1 (left, river loss $\propto A^{-1/2}$) and in the same valley with initial river losses $\propto j^{-3}$ (right, $\propto A^{-1}$), at the peak 0.04, for $T = 1000$, 4000, 16000 (top to bottom). Curves: exact expected loss (Theorem 12.13); points: mean of 40 runs of stochastic gradient descent (seed 31), within sampling error of the curves (at most 4% apart, about two standard errors); diamonds: minimizer of the continuum approximation, with the river loss evaluated at the area of the schedule and the hills by (12.6), plotted at the exact loss. Script: `12-schedules-cooldown.py`.

dependence of (12.8) is approximate: doubling the peak multiplies $c^\star$ by 1.42 to 1.55 for $\alpha = \frac{1}{2}$ and by 1.56 to 1.72 for $\alpha = 1$, against $2^{2/3} \approx 1.59$ and 2. The value $c^\star \approx 0.27$ depends on the noise, the curvatures and the river; the scaling with T and its dependence on α are the prediction.

12.6 What the theory misses

WHAT WAS IDEALIZED AWAY. The convex bound holds for every convex Lipschitz loss and so cannot see the scales of a particular one (Figure 12.2). The river valley keeps two time scales but fixes everything else: quadratic hills of constant curvature, decoupled from the river; additive noise with a fixed covariance, whereas in a network the gradient covariance follows the Hessian and changes along the run; and plain stochastic gradient descent, without momentum, Adam's normalization, clipping or weight decay. The river's exponent α and the spread of the noise over the hills are inputs, not outputs, and they set the constants of every prediction. Neither model needs warmup, which Chapter 9 explains by the curvature that the early steps meet.

WHERE IT BREAKS. (1) In the river valley only stability limits the peak of a long decay, so with tuned peaks the model ranks cosine well ahead of WSD, whereas WSD matches cosine in language models (Observation 12.1); what limits the peak in practice, instabilities and loss spikes, is outside the model (Chapters 8 and 10). (2) The landscape moves with the schedule. In full-batch training the sharpness rises while it is below $2/\eta$ and then stays near $2/\eta$ (Cohen et al. 2021), so a decaying learning rate lets the sharpest directions sharpen again, and part of the oscillation that a cooldown removes is the oscillation at the edge of stability (Chapter 8), whose size is set by the

dynamics rather than by noise. The two mechanisms differ in a testable way: the noise floor of Corollary 12.14 scales like $1/B$, the oscillation at the edge does not. (3) The empirical laws relax in steps, or in area with a fitted correction, not exactly in area as a fixed quadratic hill does (Section 12.4). (4) With AdamW the norms of normalized weights equilibrate on the time scale $1/(\eta\lambda)$, which a cooldown lengthens (Chapter 13), so part of what a cooldown does acts through weight norms that the river valley does not have. (5) The convex bound prefers a decay over the whole run, as do the experiments of Defazio et al. (2023) and Bergsma et al. (2025), but it underestimates the penalty of short cooldowns on the cone by an order of magnitude, so its agreement with practice is one of direction. And the river valley is itself a conjecture about language models, supported by a toy derivation and by the laws it implies, not derived from a network (Wen et al. 2025).

REPAIRS AND OPEN PROBLEMS. The empirical laws are repairs of the kind the model suggests: Tissue et al. (2025) and Luo et al. (2025) keep the area term and the annealing structure of (12.9) and fit the saturation, and Li et al. (2025) derive a functional law from kernel regression with power-law spectra, a noisy quadratic model whose river and hills are the two ends of one spectrum. Defazio et al. (2023) refine the schedule with observed gradient norms and obtain a warmup and a fast final decay; schedule-free methods (Defazio et al. 2024) average instead of decaying, and Hägele et al. (2024) compare weight averaging with cooldowns. For least squares the last iterate under step decay (Ge et al. 2019) and under decaying steps in the overparameterized regime (Wu et al. 2022) is understood. Open: a river valley derived from a network (Exercise 12.12); a solved model whose hills sit at the edge of stability; a mechanism for the learning-rate dependence of the relaxation (Exercise 12.13); and a derivation of the cooldown fraction near 20% that practitioners use.

12.7 Back in the wild

FIELD NOTES	
PRACTICE	Warmup, then cosine to a tenth of the peak, linear decay to zero, or a constant peak with a cooldown over the last 10–20% (WSD).
IDEALIZATION	Convex Lipschitz SGD; the river valley, a slow river between stiff noisy hills.
PROVED	A last-iterate bound for every schedule; linear decay attains the averaging rate $DG/\sqrt{T}$ exactly; exact second moments and the annealing decomposition in the river valley.
PREDICTS	Loss = area term + recent-schedule term; a noise floor $\approx \eta \operatorname{tr}\Sigma_{\text{hill}}/(4B)$ that a cooldown removes, leaving $\propto (\eta h K)^{-1/2}$; best cooldown fraction $\propto T^{(2\alpha-1)/3}$; in the bound, a floor ρ costs $\rho \ln T$.
MISSES	A landscape that moves with the schedule; position-dependent noise; Adam, momentum, weight decay; tuned

peaks; why 20%.

TAKEAWAY

Decay to (nearly) zero (proved: linear decay is worst-case optimal, and in the river valley a final step $\rho\eta$ keeps about a fraction ρ of the floor); expect the cooldown drop to scale with the peak and $1/B$, and cool down for longer than the slowest relaxation time (derived); the exact 20% is folklore backed by experiments.

The two idealizations agree on three points. The learning rate should end at or near zero: the bound charges a final floor ρ a term $\rho \ln T$, and the river valley leaves a fraction ρ of the noise floor in place. The loss at the end is an area term minus a recent-schedule term, which is why very different shapes end close together and why the empirical laws work. And the benefit of a longer cooldown saturates. They disagree on its length: the bound, with a tuned peak, wants a decay over the whole run; the river valley wants a cooldown set by the relaxation times and the river's slope, a fixed fraction of the run only if the river falls like the inverse square root of the area. A short cooldown branched from a stable checkpoint measures the noise floor, the size of the prize, before the shape is tuned. That 20% is enough is supported by experiments (Hägele et al. 2024), not by these theories.

The chapter closes a thread from **??** : sign descent never converges at a fixed step size, and only a schedule removes its two-cycle (Exercise 12.7). Momentum and schedule-free methods remove such floors by averaging instead (Chapter 5); the river valley is the noisy quadratic model of Chapter 11 at the time scale of a run; and with decoupled weight decay the schedule is the step size of a Frank–Wolfe method, a cooldown its decaying step (Chapter 13).

Exercises

12.1 CONSOLIDATION. (Two exact bounds.) Prove Corollary 12.10, and show that at $T = 10^4$ the constant step $D/(G\sqrt{T})$ gives a bound 1.79 times that of the best constant step.

12.2 CONSOLIDATION. (Shapes of the cooldown.) For $\varphi = 1$ on $[0, 1-c]$ and $\varphi(s) = ((1-s)/c)^p$ after, show that $\Phi(0) = 1 - cp/(p+1)$ and $J[\varphi] = 1 + 1/p + \ln((p+1-cp)/c)$. For $c = 0.2$, find the p that minimizes $\sqrt{J/\Phi(0)}$ (about 1.35) and compare with the $1-\sqrt{\cdot}$ cooldown of Table 12.1. *(Hint: with $u = (1-s)/c$, the integrand of (12.3) on the cooldown is $(p+1)^2 u^{p-1}/(c(2p+1))$.)*

12.3 CONSOLIDATION. (A floor costs a logarithm.) Let $\eta_t = \eta\varphi(t/(T+1))$ with φ positive and Lipschitz on $[0,1]$ and $\varphi(1) = \rho$. Show that the last-iterate term of (12.2) is $\frac{1}{2}G^2\eta(\rho \ln T + O(1))$, and check the entry $1.775 + 0.1 \ln T$ of Table 12.1. *(Hint: near the end, $R_k \approx \rho\eta(T-k+1)$ and $V_k \approx \rho^2\eta^2(T-k+1)$.)*

12.4 CONSOLIDATION. (Step decay.) A hill starts at $v_i(\eta)$ and the step size

drops to $\eta/10$. (a) Show that the excess over the new floor is below 1% of the old floor after about $22.5/(\eta h)$ steps. (b) Show that an increase of the step size enters (12.5) with a positive sign; describe the hill during a linear warmup.

12.5 CONSOLIDATION. (The cooldown law.) (a) Derive (12.6) from $dv/dA = -2hv+\eta(A)\sigma^2$. (b) Show that $\chi(\kappa) = 1-\kappa/3+O(\kappa^2)$ and $\chi(\kappa) \sim \frac{\sqrt{\pi}}{2}\kappa^{-1/2}$. (c) Derive (12.7), and show that at $K^\star$ the residual noise is twice the area cost $\omega\eta K^\star/2$.

12.6 VARIATION. (Weighted conversions; Defazio et al. (2023).) Let $w_t > 0$, $W = \sum_{t\le T} w_t$, $W_t = \sum_{s=t}^{T} w_s$, $z_{t+1} = z_t - \gamma w_t g_t$ and $\eta_t = \gamma w_t W_{t+1}/W$. (a) Show that $W\theta_T = \sum_t w_t z_t$, and generalize the proof of Proposition 12.9 to $\mathbb{E}\mathcal{L}(\theta_T) - \mathcal{L}^\star \le W^{-1}\big(D^2/(2\gamma) + \frac{\gamma}{2}G^2\sum_t w_t^2\big)$. (b) Show by the Cauchy–Schwarz inequality that uniform weights, that is linear decay, are best. (c) If $\mathbb{E}\|g_t\|_2^2 \le G_t^2$, show that the best weights are $w_t \propto G_t^{-2}$; what does $\eta_t \propto G_t^{-2}\sum_{s>t} G_s^{-2}$ look like when gradient norms fall?

12.7 VARIATION. (Normalized steps need a schedule.) Sign descent on $\frac{1}{2}hx^2$ is $x_{t+1} = x_t - \eta_t \operatorname{sign}(x_t)$ (**??**). (a) With a constant step, show that the iterates end on a two-cycle with $|x| \le \eta$, or at the fixed point 0 when x_1/η is an integer. (b) Let $\eta_t \ge \eta_{t-1}/2$ for all t; show that $|x_t| \le \eta_{t-1}$ implies $|x_{t+1}| \le \eta_t$, so the loss then stays below $h\eta_t^2/2$. (c) Compare this floor, quadratic in the step and present without noise, with the noise floor of Corollary 12.14.

12.8 VARIATION. (The noise-optimal cooldown.) In one hill, choose $\eta_t \ge 0$ to minimize v_K in $v_{t+1} = (1-\eta_t h)^2 v_t + \eta_t^2\sigma^2$. (a) Show that the step minimizing v_{t+1} given v_t is $hv_t/(\sigma^2 + h^2 v_t)$ and gives $1/v_{t+1} = 1/v_t + h^2/\sigma^2$. (b) Show that this greedy choice is optimal. *(Hint: $v \mapsto \min_\eta\{(1-\eta h)^2 v + \eta^2\sigma^2\}$ is increasing.)* (c) Deduce $v_K = \sigma^2/(h^2(K+n_0))$ with $n_0 = \sigma^2/(h^2 v_0)$, reached by $\eta_t = 1/(h(t+n_0+1))$, and compare with (12.6).

12.9 VARIATION. (Averaging instead of cooling.) In one hill at equilibrium at the step η, let $\bar{y}$ average the last K iterates; put $b = 1-\eta h$ and $v = v_i(\eta)$. (a) Show that $\mathbb{E}\bar{y}^2 = \frac{v}{K}\big(\frac{1+b}{1-b} - \frac{2b(1-b^K)}{K(1-b)^2}\big)$, so that the hill loss at $\bar{y}$ is about $2/(\eta hK)$ times the floor. (b) Show that with a linear river the average gives up the same area as a linear cooldown of K steps, up to one step. (c) Conclude that tail averaging beats a linear cooldown when ηhK is large; compare with Chapter 5 and with Hägele et al. (2024), who compare averaging with cooldowns.

12.10 SIMULATION. (Shapes and peaks.) In `12-schedules-river.py`: (a) replace the linear cooldown by a $1-\sqrt{\cdot}$ cooldown of the same length and check that the final loss falls from 0.131 to 0.129; (b) raise the peak to 0.2, check that cosine (0.080) now beats WSD (0.096), and explain why with the river and hill parts; (c) add a linear warmup over 100 steps and check that, at the end of the run, only the river parts change.

12.11 SIMULATION. (The cone.) In `12-schedules-bound.py`: (a) check that constant steps settle at $\eta G^2/2$, and that after the step is halved the excess decays by a factor e in about 130 steps, at $\eta = 0.0032$ and at $\eta = 0.01$; (b) set $d = 400$ and explain, with the first term of (12.2), why cosine and linear decay now end above constant steps.

12.12 RESEARCH. (Derive the river.) Find a network model on structured data in which a river valley, with its exponent α and hill spectrum, can be derived from the data. Wen et al. (2025) give a bigram example; a model with learned features is open.

12.13 RESEARCH. (Steps or area?) The river valley predicts that the loss relaxes after a decrease of the learning rate in area, the cone that it relaxes in steps. Which mechanism (a nonsmooth minimum, curvature adapting at the edge of stability, Adam's normalization) produces the learning-rate dependence that the empirical laws fit, and with what exponent? This is open, as far as we know.

Notes and References

The step-size conditions of Robbins and Monro (1951) and the averaging of Polyak and Juditsky (1992) are the classical theory of schedules, and Nemirovski et al. (2009) showed that constant steps with averaging are optimal for convex Lipschitz problems. For the last iterate, Shamir and Zhang (2013) proved the $\log T/\sqrt{T}$ rate of the steps $1/\sqrt{t}$, Harvey et al. (2019) showed it tight, Jain et al. (2019) designed a horizon-dependent schedule with the optimal rate, Orabona (2020) removed the bounded domain, and Liu and Zhou (2024) extended the analysis further. Zamani and Glineur (2025) found linear decay optimal for the subgradient method, and Defazio et al. (2023) for stochastic gradient descent. Our proof of Theorem 12.8 follows the comparison argument; our proof of Proposition 12.9 is a conversion of the kind used by Defazio et al. (2023), written out for uniform weights.

WSD was introduced for MiniCPM (Hu et al. 2024) and studied by Hägele et al. (2024), and Schaipp et al. (2025) found its agreement with the convex bound; the river valley is due to Wen et al. (2025), the laws of loss curves across schedules to Tissue et al. (2025) and Luo et al. (2025). We have not found the cooldown law (12.6), the optimal cooldowns (12.7) and (12.8), or the gamma-type saturation in the literature we could check; all are derived here, not proved.

TRANSLATING THE LITERATURE. Defazio et al. (2023) and Schaipp et al. (2025) write x_t for our θ_t and f for $\mathcal{L}$, and start at x_1 as we do. Schaipp et al. (2025) write the schedule as a base learning rate γ times a shape η_t, our peak η times $\varphi(t/T)$, call the bound Ω_t, allow a bound $G_t^2 \geq \mathbb{E}\|g_t\|^2$ that varies with t, and write $D = \|x_1 - x_\star\|$; their Theorem 3.1, which they present as a version of Theorem 10 of Defazio et al. (2023), is built from the same sums as (12.2), a term in D^2 and the squared steps and a term with the steps still to come, with each η_t^2 weighted by its own G_t^2. Defazio et al. (2023) work with weights w_t; their schedule $\eta_t \propto w_t \sum_{s>t} w_s$ is that of

Exercise 12.6. In Tissue et al. (2025) and Luo et al. (2025) the forward area $S_1(t) = \sum_{s \le t} \eta_s$ is our A_{t+1}, the area $\sum_{s=k}^{t} \eta_s$ since a decrease at step k is our $A_{t+1} - A_k$, the loss drop $\mathrm{LD}(t)$ is our annealing sum in (12.9), their L_0 and A are our L_∞ and C, and their α is our river exponent. The decay factor λ (typically 0.999) in the annealing area of Tissue et al. (2025) is our $e^{-2\bar{\eta}h}$, and the exponent of the power-law saturation of Luo et al. (2025) plays the role of our gamma shape ν.

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13 Weight Decay

Frank–Wolfe, rotational equilibrium and the effective learning rate

IN THE WILD — ADAMW'S DECOUPLED WEIGHT DECAY

IN THIS CHAPTER. Pretraining recipes multiply the weights by $1-\eta\lambda$ before every AdamW step, with $\lambda = 0.1$, although a weight matrix followed by a normalization layer computes the same function at every scale. When the step has a fixed length in a norm, decoupled decay is exactly the Frank–Wolfe method on the ball of radius $1/\lambda$: we prove that the ball attracts every initialization at rate $(1-\eta\lambda)^t$, that the learning-rate schedule acts as a Frank–Wolfe step-size schedule, and where in the ball consistent and noisy coordinates settle. When the loss is scale invariant, we prove that the norm converges from every initialization to an equilibrium at which the weights rotate by $\arccos(1-\eta\lambda) \approx \sqrt{2\eta\lambda}$ per step whatever the size of the gradients, so that only the product $\eta\lambda$ matters, and that the rotation lags behind a changing step size. Momentum, preconditioning, gradients that respond to the effective learning rate, and the effects of decay on generalization are what these idealizations miss.

13.1 In the wild: a regularizer that does not regularize

For stochastic gradient descent, weight decay and an L_2 penalty are the same thing: adding $\frac{\lambda}{2}\|\theta\|_2^2$ to the loss adds $\lambda\theta$ to the gradient, and $\theta \leftarrow \theta - \eta(g + \lambda\theta)$ is $\theta \leftarrow (1-\eta\lambda)\theta - \eta g$. For Adam they differ, because the gradient of the penalty is divided by $\sqrt{\hat{v}}$ with the rest. Loshchilov and Hutter (2019) applied the shrinkage outside the adaptive step and found that this *decoupled weight decay* makes the best decay less dependent on the learning rate and improves Adam's generalization; the result is AdamW. Every optimizer of this chapter has the form

$$\theta_{t+1} = (1-\eta_t\lambda)\,\theta_t - \eta_t\, u_t, \tag{13.1}$$

where u_t is the optimizer's step direction: the gradient for SGD, $\hat{m}_t/(\sqrt{\hat{v}_t}+\epsilon)$ for AdamW, the orthogonalized momentum for Muon. We say learning rate for η_t here and step size in the theory; the two are the same.

IN THE WILD · GPT-3, Brown et al. (2020); LLaMA, Touvron et al. (2023); Moonlight, Liu et al. (2025). Train with AdamW and decoupled weight decay $\lambda = 0.1$. The peak learning rates of the GPT-3 models range from 6×10^{-5} (the largest) to 6×10^{-4} (the smallest), so each step shrinks the weights by a factor between $1-6\times10^{-6}$ and $1-6\times10^{-5}$;

the factor follows the schedule, while λ stays fixed. Decay the weight matrices; by a convention of open training code (folklore), biases and normalization gains are not decayed. When Muon replaces AdamW on the hidden matrices, keep a decoupled decay of the same form. PyTorch's AdamW multiplies the weights by $1 - \eta_t\lambda$, with the default $\lambda = 10^{-2}$.

This does not fit the picture of a penalty.[1] AdamW does not minimize the loss plus $\frac{\lambda}{2}\|\theta\|_2^2$, which for billions of weights would dwarf the loss; it keeps them in the ball $\|\theta\|_\infty \le 1/\lambda = 10$, up to a factor near one for Adam's memory (Section 13.3). And most weight matrices of a transformer feed a normalization layer, directly or through the residual stream, which erases the scale of its input.

[1] The same argument name means different things: in PyTorch's Adam, `weight_decay` adds $\lambda\theta$ to the gradient, an L_2 penalty; in AdamW it is decoupled.

OBSERVATION 13.1 (van Laarhoven 2017; Zhang et al. 2019). If the output of a weight matrix W passes through batch, layer, or RMS normalization, then the network computes the same function with W and cW for every $c > 0$, so an L_2 penalty on W does not constrain the function; it changes the scale of W, and with it the size of each step relative to W, an effective learning rate. Zhang et al. (2019) identified this as one of three mechanisms by which weight decay improves the test error of vision networks.

OBSERVATION 13.2 (Wan et al. 2021; Kosson et al. 2024). With normalization and weight decay, the norm of each scale-invariant weight and its angle of rotation per step settle to a steady state. Wan et al. (2021) derived the steady-state angle $\sqrt{2\eta\lambda}$ for SGD and confirmed their predictions on ImageNet and MS COCO; Kosson et al. (2024) called the state *rotational equilibrium*, studied it for single neurons under AdamW, Lion, and SGD with momentum, and argued that its balance across layers and neurons is part of why AdamW beats Adam with an L_2 penalty.

OBSERVATION 13.3 (Loshchilov and Hutter 2019; Wang and Aitchison 2025). The more batch passes, the smaller the optimal λ. Loshchilov and Hutter (2019) proposed $\lambda = \lambda_{\text{norm}}\sqrt{b/(BT)}$ for batch size b, B training points, and T epochs, "merely one possibility informed by few experiments". Wang and Aitchison (2025) read AdamW's weights as an exponential moving average of its updates over $1/(\eta\lambda)$ iterations, and report that the optimal timescale in epochs is roughly the same across model and dataset sizes.

Liu et al. (2025) name the addition of weight decay as one of the two changes that made Muon work at scale. A counter-observation: Lobacheva et al. (2021) found that with batch normalization and weight decay, training can be periodic rather than stationary; the norm shrinks, the effective learning rate grows, training destabilizes without diverging, and a new period begins.[2] What, then, does λ set in a training run? We give two answers, one per idealization: the radius of a ball the weights cannot leave, and the speed at which scale-invariant weights rotate.

[2] D'Angelo et al. (2024) argue that for language models trained for about one epoch, weight decay acts on optimization rather than generalization: it lowers the training loss and stabilizes training in low precision. The mechanisms of this chapter are optimization mechanisms.

13.2 The idealization: normalized steps and scale invariance

The first idealization keeps the geometry of the step and nothing else.

Assumption 13.4 (Steepest descent with decoupled decay). Fix a norm $\|\cdot\|$ on $\mathbb{R}^d$ with dual norm $\|\cdot\|_*$, a decay $\lambda > 0$, and step sizes $\eta_t > 0$ with $\eta_t\lambda \le 1$. The iteration is

$$\theta_{t+1} = (1 - \eta_t\lambda)\,\theta_t + \eta_t\,\mathrm{lmo}(g_t),$$

that is, (13.1) with $u_t = -\mathrm{lmo}(g_t)$, where g_t is whatever vector the optimizer feeds to its linear minimization oracle: a gradient, a stochastic gradient, or a moving average of gradients.

AdamW without memory ($\beta_1 = \beta_2 = 0$, $\epsilon = 0$) is the ℓ_∞ case, $\mathrm{lmo}(g) = -\operatorname{sign}(g)$ (Chapter 2); Lion with decay is the ℓ_∞ case applied to an interpolation of momentum and gradient; Muon with decay is the spectral case, $\mathrm{lmo}(G) = -\operatorname{polar}(G)$, applied to the momentum (Chapter 3). Norms, oracles, and dual norms are those of Chapter 1. The second idealization keeps only the symmetry that normalization layers create.

Definition 13.5 (Scale invariance). A function $\mathcal{L}\colon \mathbb{R}^d \setminus \{0\} \to \mathbb{R}$ is *scale invariant* if $\mathcal{L}(c\theta) = \mathcal{L}(\theta)$ for every $\theta \neq 0$ and $c > 0$.

A layer normalization without gain has $\mathrm{LN}(cWx) = \mathrm{LN}(Wx)$ for $c > 0$ (up to the constant inside its square root), so a loss that depends on W only through $\mathrm{LN}(Wx)$ is scale invariant in W; batch, RMS, and weight normalization and the normalization of queries and keys act alike. Here $\|\theta\|_2$ is the Euclidean norm of all entries of the weight in question and $\hat{\theta} = \theta/\|\theta\|_2$ its direction.

Assumption 13.6 (Rotation on a sphere). The loss $\mathcal{L}$ and each per-example loss $\ell(\cdot;\xi)$ are scale invariant and differentiable; the step direction u_t in (13.1) is orthogonal to θ_t; and, in the main theorem, the stochastic gradient has a constant length on the unit sphere, $\|g_t\|_2\|\theta_t\|_2 = \Gamma$.

WHAT WAS THROWN AWAY. For the first idealization, everything except that each step has a fixed length in a norm; the rates also need a convex loss. For the second, everything except that the loss ignores the layer's scale; the main theorem also freezes the gradient size on the sphere and assumes steps orthogonal to the weights, exact for SGD and false for Adam's preconditioned and momentum steps. Each simplification isolates one mechanism: the ball holds for every sequence of bounded steps, whatever produced them, and the equilibrium needs only orthogonality and a balance of two forces.

WHAT ONE HAD TO NOTICE. In (13.1) the factor $1-\eta\lambda$ is a weight of a convex combination, $(1-\eta\lambda)\theta + \eta\lambda\cdot(-u/\lambda)$, and for steps of fixed length this is a

Frank–Wolfe step. For a scale-invariant layer the gradient step is tangent to the sphere and can only enlarge the norm, while the decay shrinks it; their balance fixes the norm, and the norm fixes how far each step turns the weights.

DICTIONARY

IN THE THEORY	IN A TRAINING RUN
the ball $\mathcal{B}_\lambda = \{\theta : \|\theta\| \le 1/\lambda\}$	the set AdamW's weights are drawn into: $\|\theta_i\| \le 10$ at $\lambda = 0.1$ (11.65 with memory)
Frank–Wolfe step size $\gamma_t = \eta_t\lambda$	the learning-rate schedule times the decay
the vertex $\mathrm{lmo}(g_t)/\lambda$	where a step points: $-\operatorname{sign}(g_t)/\lambda$ for AdamW without memory
scale invariance, $\mathcal{L}(c\theta) = \mathcal{L}(\theta)$	a weight matrix followed by a normalization layer
$\Gamma = \|g\|_2\|\theta\|_2$	the gradient norm of the layer rescaled to unit norm
the angle ω_t between θ_t and θ_{t+1}	the rotation of the weight matrix in one step
$\arccos(1-\eta\lambda) \approx \sqrt{2\eta\lambda}$	equilibrium rotation: 0.0077 rad per step at $\eta = 3\times10^{-4}$, $\lambda = 0.1$
the timescale $1/(\eta\lambda)$	the number of recent updates the weights average over

FIELD GUIDE · ADAMW *Loshchilov and Hutter (2019)*

UPDATE	$m_t = \beta_1 m_{t-1} + (1-\beta_1)g_t$, $v_t = \beta_2 v_{t-1} + (1-\beta_2)g_t^2$, $\theta_{t+1} = (1-\eta_t\lambda)\theta_t - \eta_t\,\hat{m}_t/(\sqrt{\hat{v}_t}+\epsilon)$
NORM	ℓ_∞ when $\beta_1 = \beta_2 = 0$ (Chapter 2); with the decay, Frank–Wolfe on $\|\theta\|_\infty \le 1/\lambda$ (Theorem 13.7)
PRECONDITIONER	$\mathrm{diag}(\sqrt{\hat{v}_t}) + \epsilon$ (Chapter 6)
AVERAGES	EMAs of g and g^2; through the decay, θ_t is an EMA of $-u_s/\lambda$ over $1/(\eta\lambda)$ steps (Proposition 13.16)
STATE	two numbers per parameter
IN THE WILD	$\lambda = 0.1$, peak η from 6×10^{-5} to 6×10^{-4}; matrices decayed, gains and biases usually not
CAUTION	codebases differ in what multiplies λ (see the notes); Adam with an L_2 penalty is a different method (Exercise 13.4)

FIELD GUIDE · MUONW *Liu et al. (2025)*

UPDATE	$M_t = \beta M_{t-1} + G_t$, $O_t \approx \mathrm{polar}(M_t)$ by Newton–Schulz, $W_{t+1} = (1-\eta_t\lambda)W_t - \eta_t\, s\, O_t$
NORM	spectral (Chapter 3); with the decay, Frank–Wolfe on $\|W\|_{\mathrm{op}} \le s/\lambda$
AVERAGES	momentum on G; the decay averages updates over $1/(\eta\lambda)$ steps

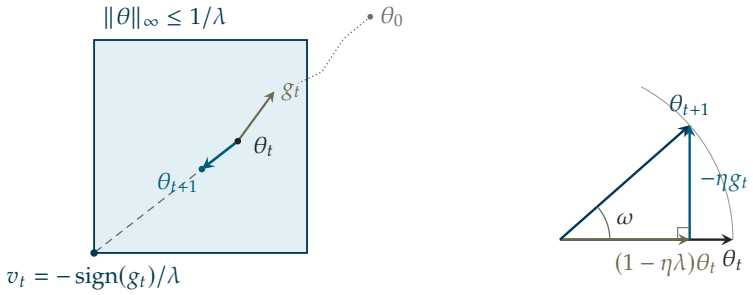


Figure 13.1. Two pictures of decoupled weight decay, drawn with $\eta\lambda = \frac{1}{4}$. Left: a step of AdamW without memory moves θ_t a fraction $\eta\lambda$ of the way to the vertex $-\operatorname{sign}(g_t)/\lambda$ of the ball $\|\theta\|_\infty \le 1/\lambda$; an initialization outside the ball is drawn in (Theorem 13.7). Right: for a scale-invariant layer the gradient is orthogonal to the weights; in equilibrium $\|\theta_{t+1}\|_2 = \|\theta_t\|_2$, and the right triangle gives $\cos\omega = 1-\eta\lambda$ (Theorem 13.12).

IN THE WILD	hidden matrices only, AdamW elsewhere; scale s set to match AdamW's update size; decoupled decay as in AdamW (Liu et al. 2025)

FIELD GUIDE · HYPERBALL — *Wen et al. (2026)*

UPDATE	$W_{t+1} = R\,(\hat{W}_t - \alpha_t \hat{U}_t)/\|\hat{W}_t - \alpha_t \hat{U}_t\|_F$, with $\hat{W} = W/R$, $\|W_0\|_F = R$, and $\hat{U}_t$ the update of a base optimizer (Adam, Muon) scaled to unit norm
NORM	the base optimizer's, on the sphere $\|W\|_F = R$; no decay
IN THE WILD	unrefereed 2026 preprint; reported speedups over weight decay up to 1.2B parameters (Section 13.6)

13.3 The theory: decoupled weight decay is Frank–Wolfe

To minimize a smooth convex f over a compact convex set $\mathcal{K}$, the *Frank–Wolfe method* (Frank and Wolfe 1956), or conditional gradient method, minimizes the linearization of f over $\mathcal{K}$ and moves a fraction γ_t of the way to the minimizer:

$$v_t \in \arg\min_{v\in\mathcal{K}} \langle \nabla f(x_t), v\rangle, \qquad x_{t+1} = (1-\gamma_t)\,x_t + \gamma_t\, v_t.$$

It needs only a linear minimization oracle for $\mathcal{K}$, and its iterates stay in $\mathcal{K}$.[3] For the ball of a norm the oracle is lmo rescaled (Figure 13.1, left). Write $\mathcal{B}_\lambda = \{\theta : \|\theta\| \le 1/\lambda\}$, $\gamma_t = \eta_t\lambda$ and $\Pi_t = \prod_{s<t}(1-\gamma_s)$, with $\Pi_0 = 1$.

Theorem 13.7 (Decoupled weight decay is Frank–Wolfe). *Let the iteration satisfy Assumption 13.4.*

(i) With $v_t = \operatorname{lmo}(g_t)/\lambda$ we have $v_t \in \arg\min_{v\in\mathcal{B}_\lambda}\langle g_t, v\rangle$ and $\theta_{t+1} =$

[3] For the ℓ_∞ ball a projection is a clipping and as cheap as the oracle. For the spectral ball a projection clips the singular values and needs a singular value decomposition, while the oracle needs only the polar factor, which Muon approximates by Newton–Schulz iterations (Chapter 3).

$(1-\gamma_t)\theta_t + \gamma_t v_t$: *every step is a Frank–Wolfe step on* $\mathcal{B}_\lambda$ *with step size* γ_t *for the linear function* $\langle g_t, \cdot \rangle$.

(ii) *For every* t, $\theta_t - \Pi_t \theta_0 \in (1-\Pi_t)\,\mathcal{B}_\lambda$. *Consequently the distance from* θ_t *to* $\mathcal{B}_\lambda$ *satisfies*

$$\big(\|\theta_t\| - 1/\lambda\big)_+ \le \Pi_t \,\big(\|\theta_0\| - 1/\lambda\big)_+,$$

with equality for every t *when* $\operatorname{lmo}(g_s) = \theta_0/\|\theta_0\|$ *for all* s. *So* $\mathcal{B}_\lambda$ *is invariant and, at a constant step size, attracts every initialization at the rate* $(1-\eta\lambda)^t$.

(iii) *Let* $\mathcal{L}$ *be convex and L-smooth with respect to* $\|\cdot\|$, *let* $g_t = \nabla\mathcal{L}(\theta_t)$, *and let* $h_t = \mathcal{L}(\theta_t) - \mathcal{L}_\lambda^\star$ *with* $\mathcal{L}_\lambda^\star = \min_{\mathcal{B}_\lambda} \mathcal{L}$. *Then for every* t

$$h_{t+1} \le (1-\gamma_t)\, h_t + \tfrac{L}{2}\,\gamma_t^2\,\|v_t - \theta_t\|^2. \tag{13.2}$$

For $\gamma_t = 2/(t+2)$, *that is* $\eta_t = 2/(\lambda(t+2))$, *and* $D = 1/\lambda + \max\{1/\lambda, \|\theta_0\|\}$, *we have* $h_t \le 2LD^2/(t+2)$ *for* $t \ge 1$, *which is* $8L/(\lambda^2(t+2))$ *if* $\theta_0 \in \mathcal{B}_\lambda$. *For a constant* $\gamma_t = \gamma$, $\limsup_t h_t \le 2L\gamma/\lambda^2 = 2L\eta/\lambda$, *and* $h_t \le (1-\gamma)^t h_0 + 2L\gamma/\lambda^2$ *for every* t *if* $\theta_0 \in \mathcal{B}_\lambda$.

(iv) *If instead* $g_t = \nabla\mathcal{L}(\theta_t) + \zeta_t$ *with an error* ζ_t, *then* (13.2) *holds with the additional term* $2\gamma_t\|\zeta_t\|_*/\lambda$ *on its right side.*

Proof. The plan is to read (i) off the definition of lmo, to unroll the recursion for (ii), and to repeat the classical Frank–Wolfe argument for (iii) and (iv), keeping track of iterates that start outside the ball.

(i) The map $u \mapsto u/\lambda$ is a bijection from the unit ball onto $\mathcal{B}_\lambda$ and multiplies $\langle g_t, u\rangle$ by $1/\lambda > 0$, so it maps minimizers of $\langle g_t, \cdot\rangle$ over the unit ball onto those over $\mathcal{B}_\lambda$; and $\eta_t \operatorname{lmo}(g_t) = \eta_t\lambda \cdot \operatorname{lmo}(g_t)/\lambda = \gamma_t v_t$.

(ii) Write $\Pi_{s,t} = \prod_{s\le k<t}(1-\gamma_k)$, so that $\Pi_{0,t} = \Pi_t$ and $\Pi_{t,t} = 1$. By induction on t, (i) gives

$$\theta_t = \Pi_t\theta_0 + \sum_{s<t} w_s\, v_s, \qquad w_s = \gamma_s \Pi_{s+1,t}.$$

The weights are nonnegative because $0 < \gamma_k \le 1$, and they telescope, $w_s = \Pi_{s+1,t} - \Pi_{s,t}$, to $1 - \Pi_t$. If $\Pi_t < 1$, then $\sum_s w_s v_s/(1-\Pi_t)$ is a convex combination of points of $\mathcal{B}_\lambda$, so $\sum_s w_s v_s \in (1-\Pi_t)\mathcal{B}_\lambda$; if $\Pi_t = 1$, then the sum is zero. The triangle inequality gives $\|\theta_t\| - 1/\lambda \le \Pi_t(\|\theta_0\| - 1/\lambda)$; take positive parts, as $\Pi_t \ge 0$. If $\operatorname{lmo}(g_s) = \theta_0/\|\theta_0\|$ for all s, then every v_s is a positive multiple of θ_0 and the triangle inequality is an equality. Finally, in every norm the distance from θ to $\mathcal{B}_\lambda$ is $(\|\theta\| - 1/\lambda)_+$: if $\|\theta\| > 1/\lambda$, then the point $\theta/(\lambda\|\theta\|)$ of the ball is at distance $\|\theta\| - 1/\lambda$, and every $x \in \mathcal{B}_\lambda$ has $\|\theta - x\| \ge \|\theta\| - \|x\| \ge \|\theta\| - 1/\lambda$.

(iii) The minimum $\mathcal{L}_\lambda^\star$ is attained at some $\theta^\star \in \mathcal{B}_\lambda$, since $\mathcal{B}_\lambda$ is compact. The descent lemma (Lemma A.18), applied to the step $\gamma_t(v_t - \theta_t)$, gives

$$\mathcal{L}(\theta_{t+1}) \le \mathcal{L}(\theta_t) + \gamma_t\langle \nabla\mathcal{L}(\theta_t), v_t - \theta_t\rangle + \tfrac{L}{2}\gamma_t^2\|v_t - \theta_t\|^2.$$

Because v_t minimizes $\langle \nabla\mathcal{L}(\theta_t), \cdot\rangle$ over $\mathcal{B}_\lambda \ni \theta^\star$, and $\mathcal{L}$ is convex,

$$\langle \nabla\mathcal{L}(\theta_t), v_t - \theta_t\rangle \le \langle \nabla\mathcal{L}(\theta_t), \theta^\star - \theta_t\rangle \le \mathcal{L}(\theta^\star) - \mathcal{L}(\theta_t) = -h_t.$$

Substituting gives (13.2), which does not use the sign of h_t (negative values occur while θ_t is outside the ball). By (ii), $\|v_t - \theta_t\| \le 1/\lambda + \|\theta_t\| \le 2/\lambda + \Pi_t e_0$ with $e_0 = (\|\theta_0\| - 1/\lambda)_+$.

For $\gamma_t = 2/(t+2)$ we have $\gamma_0 = 1$, so the first step lands on the vertex v_0 and $\Pi_t = 0$ for $t \ge 1$; hence $\|v_t - \theta_t\| \le 2/\lambda \le D$ for $t \ge 1$, and $\|v_0 - \theta_0\| \le 1/\lambda + \|\theta_0\| \le D$. We show $h_t \le 2LD^2/(t+2)$ by induction. At $t = 1$, (13.2) with $\gamma_0 = 1$ gives $h_1 \le \frac{L}{2}D^2 \le \frac{2}{3}LD^2$. If the bound holds at $t \ge 1$, then

$$h_{t+1} \le \frac{t}{t+2}\cdot\frac{2LD^2}{t+2} + \frac{L}{2}\cdot\frac{4}{(t+2)^2}D^2 = \frac{2LD^2(t+1)}{(t+2)^2} \le \frac{2LD^2}{t+3},$$

because $(t+1)(t+3) \le (t+2)^2$. If $\theta_0 \in \mathcal{B}_\lambda$, then $D = 2/\lambda$.

For a constant γ, unrolling (13.2) gives

$$h_t \le (1-\gamma)^t h_0 + \frac{L}{2}\gamma^2 \sum_{s<t}(1-\gamma)^{t-1-s}\left(\frac{2}{\lambda} + (1-\gamma)^s e_0\right)^2.$$

Expanding the square, the sum is at most $4/(\lambda^2\gamma) + (4e_0/\lambda)\, t(1-\gamma)^{t-1} + e_0^2(1-\gamma)^{t-1}/\gamma$, whose last two terms tend to zero, and vanish if $\theta_0 \in \mathcal{B}_\lambda$.

(iv) Now v_t minimizes $\langle g_t, \cdot\rangle$, so $\langle g_t, v_t\rangle \le \langle g_t, \theta^\star\rangle$, and with $\nabla\mathcal{L}(\theta_t) = g_t - \zeta_t$,

$$\langle \nabla\mathcal{L}(\theta_t), v_t - \theta_t\rangle \le \langle \nabla\mathcal{L}(\theta_t), \theta^\star - \theta_t\rangle + \langle \zeta_t, \theta^\star - v_t\rangle.$$

The first term is at most $-h_t$ as in (iii); the second is at most $\|\zeta_t\|_*\|\theta^\star - v_t\| \le 2\|\zeta_t\|_*/\lambda$ by Hölder's inequality (Lemma A.3). ■

So the schedule is a Frank–Wolfe step-size schedule, $\gamma_t = \eta_t\lambda$, on a ball of radius $1/\lambda$.[4] The classical $\gamma_t = 2/(t+2)$ is a learning rate that starts at $1/\lambda$ and decays like $1/t$; every schedule with $\gamma_t \to 0$ and $\sum_t \gamma_t = \infty$ gives $\limsup_t h_t \le 0$ (Exercise 13.3), while a constant one leaves a neighborhood of size at most $2L\eta/\lambda$, which a decaying schedule removes (Chapter 12). These are worst cases: on the quadratic of Figure 13.3 the gap falls like $1/t^2$, and the constant-step excess is of order γ^2 (Exercise 13.1). Part (iv) adds a floor that no schedule removes, $2\sigma_*/\lambda$ when $\mathbb{E}\|\zeta_t\|_* \le \sigma_*$, for the error of a stochastic gradient or of a momentum average (Chapter 5), which is one reason to feed a momentum to the oracle, as Lion and Muon do.

For AdamW with memory and full-batch gradients, Xie and Li (2024) proved that if the iterates converge under a nonincreasing schedule with divergent sum, then the limit is a Karush–Kuhn–Tucker point of $\min \mathcal{L}$ subject to $\|\theta\|_\infty \le 1/\lambda$; Chen et al. (2024) showed with a Lyapunov function that Lion with decay solves the same constrained problem, and Pethick et al. (2025) built optimizers, Muon-like ones among them, directly as conditional-gradient methods. The proof of (ii) uses only that $\|\mathrm{lmo}(g_t)\| \le 1$: steps with $\|u_t\| \le C$ are drawn into the ball of radius C/λ at the same rate. Adam's step

[4] The radius depends on what multiplies λ. In PyTorch the decay per step is $\eta_t\lambda$ and the radius $1/\lambda$; in the AdamW of Loshchilov and Hutter (2019) the decay is the schedule multiplier times λ while the step also carries the base learning rate α, and the radius is α/λ.

is bounded coordinatewise by $(1-\beta_1)/\sqrt{(1-\beta_2)(1-\beta_1^2/\beta_2)}$ when $\beta_1^2 < \beta_2$ (Chapter 2), which is 1.165 for $(\beta_1, \beta_2) = (0.9, 0.95)$, so AdamW's weights are drawn into the ℓ_∞ ball of radius $1.165/\lambda$.[5] Where in the ball do the weights sit? One coordinate of sign descent answers exactly.

[5] The bound is proved in Chapter 2 without bias correction. With bias correction, the Cauchy–Schwarz bound after t steps increases with t to the same limit, as we checked numerically for $(0.9, 0.95)$ and $(0.9, 0.999)$.

Proposition 13.8 (Consistent and noisy coordinates). *Let $0 < \gamma = \eta\lambda < 1$, $x_0 \in \mathbb{R}$, and $x_{t+1} = (1-\gamma)x_t - \eta s_t$, where $|s_t| \le 1$ and $\mathbb{E}[s_t \mid \mathcal{F}_t] = \mu$ for every t, with $\mathcal{F}_t$ generated by $s_0, \dots, s_{t-1}$.*

(i) $\mathbb{E}x_t = -\mu/\lambda + (1-\gamma)^t(x_0 + \mu/\lambda)$; if $s_t = \mu \in \{-1, 1\}$ is deterministic, then x_t follows this curve to the vertex $-\mu/\lambda$.

(ii) If $\mu = 0$ and $\mathbb{E}[s_t^2 \mid \mathcal{F}_t] = 1$, as for random signs, then $\mathbb{E}x_t^2 = (1-\gamma)^{2t}x_0^2 + \eta^2\big(1-(1-\gamma)^{2t}\big)/\big(1-(1-\gamma)^2\big)$, which tends to $\eta/(\lambda(2-\eta\lambda))$: the coordinate settles at root mean square $\lambda^{-1}\sqrt{\gamma/(2-\gamma)} \approx \lambda^{-1}\sqrt{\eta\lambda/2}$.

Proof. Taking expectations, $\mathbb{E}x_{t+1} = (1-\gamma)\mathbb{E}x_t - \eta\mu$, whose solution is (i). For (ii), $\mathbb{E}[x_t s_t] = \mathbb{E}[x_t\,\mathbb{E}[s_t \mid \mathcal{F}_t]] = 0$ because x_t is determined by $\mathcal{F}_t$, so $\mathbb{E}x_{t+1}^2 = (1-\gamma)^2\mathbb{E}x_t^2 + \eta^2$. ∎

Under sign descent, a coordinate whose gradient has mean $\bar{g}_i$ and Gaussian noise of standard deviation σ_i has $|\mu| = \operatorname{erf}(|\bar{g}_i|/(\sigma_i\sqrt{2}))$: it settles at the vertex when the gradient is consistent and deep inside when it is noise, at 0.2% to 0.5% of the radius for the GPT-3 values $\eta\lambda \in [6\times10^{-6}, 6\times10^{-5}]$. The ball constrains only coordinates whose updates keep one sign for about $1/(\eta\lambda)$ steps; for the others the decay balances noise against shrinkage, and for scale-invariant weights this balance is the whole story.

13.4 The theory: scale invariance and the rotational equilibrium

Gradients in this section are Euclidean. Two facts make scale-invariant weights special.

Lemma 13.9 (Scale invariance). *Let $\mathcal{L}$ be scale invariant and twice differentiable on $\mathbb{R}^d \setminus \{0\}$. For every $\theta \neq 0$ and $c > 0$: (i) $\langle \nabla\mathcal{L}(\theta), \theta\rangle = 0$; (ii) $\nabla\mathcal{L}(c\theta) = c^{-1}\nabla\mathcal{L}(\theta)$, so $\nabla\mathcal{L}(\theta) = \nabla\mathcal{L}(\hat{\theta})/\|\theta\|_2$; (iii) $\nabla^2\mathcal{L}(c\theta) = c^{-2}\nabla^2\mathcal{L}(\theta)$, so the sharpness is $S(\theta) = S(\hat{\theta})/\|\theta\|_2^2$.*

Proof. Differentiate $\mathcal{L}(c\theta) = \mathcal{L}(\theta)$ in c at $c = 1$ for (i), and in θ, once and twice, for $c\nabla\mathcal{L}(c\theta) = \nabla\mathcal{L}(\theta)$ and $c^2\nabla^2\mathcal{L}(c\theta) = \nabla^2\mathcal{L}(\theta)$; then take $c = \|\theta\|_2$ and replace θ by $\hat{\theta}$. ∎

So a gradient step is tangent to the sphere, and gradient and curvature fall as the norm grows; by (iii), the stability threshold $\eta S < 2$ of Chapter 8 reads $(\eta/\|\theta\|_2^2)\,S(\hat{\theta}) < 2$. If each per-example loss is scale invariant, then (i) holds for every stochastic gradient, and a step of (13.1) is a right triangle (Figure 13.1, right).

Lemma 13.10 (One step). *Let $\theta \neq 0$, $0 < \eta\lambda < 1$, $\theta' = (1-\eta\lambda)\theta - \eta u$ with $\langle u, \theta\rangle = 0$, and let ω be the angle between θ and θ'. Then*

(i) $\|\theta'\|_2^2 = (1-\eta\lambda)^2\|\theta\|_2^2 + \eta^2\|u\|_2^2$;

(ii) $\cos\omega = (1-\eta\lambda)\|\theta\|_2/\|\theta'\|_2$ *and* $\tan\omega = \eta\|u\|_2/((1-\eta\lambda)\|\theta\|_2)$;

(iii) if $u \neq 0$, then $\hat\theta' = (\hat\theta - \alpha\hat u)/\|\hat\theta - \alpha\hat u\|_2$ with $\hat u = u/\|u\|_2$ and $\alpha = \tan\omega$;

(iv) $\|\theta' - \theta\|_2^2/\|\theta\|_2^2 = \eta^2\lambda^2 + \eta^2\|u\|_2^2/\|\theta\|_2^2$.

Proof. Pythagoras' theorem for the orthogonal vectors θ and u, with $\langle\theta, \theta'\rangle = (1-\eta\lambda)\|\theta\|_2^2$ for (ii); for (iii), divide θ' by $(1-\eta\lambda)\|\theta\|_2 > 0$; for (iv), $\theta' - \theta = -\eta\lambda\theta - \eta u$. ∎

Write $a = (1-\eta\lambda)^2$; then $\cos\omega = \sqrt{a}$ exactly when the norm does not change. The weakest form of the equilibrium needs nothing more; we call ω_t the *angular update*.

Proposition 13.11 (The equilibrium angle is a time average). *Let $\theta_{t+1} = (1-\eta\lambda)\theta_t - \eta u_t$ with $\langle u_t, \theta_t\rangle = 0$, $\theta_t \neq 0$, constant η and λ with $0 < \eta\lambda < 1$, and $q_t = \|\theta_t\|_2^2$. For all $T_1 < T_2$,*

$$\sum_{t=T_1}^{T_2-1} q_t \tan^2\omega_t = \frac{1-a}{a}\sum_{t=T_1}^{T_2-1} q_t + \frac{q_{T_2} - q_{T_1}}{a}.$$

If $q_{T_2} = q_{T_1}$, then the q-weighted mean of $\tan^2\omega_t$ over the window is exactly $(1-a)/a = \tan^2\big(\arccos(1-\eta\lambda)\big)$, whatever the sizes of the steps.

Proof. By Lemma 13.10(i) and (ii), $a q_t \tan^2\omega_t = \eta^2\|u_t\|_2^2 = q_{t+1} - a q_t = (q_{t+1} - q_t) + (1-a)q_t$. Sum over the window and divide by a. ∎

Gradients may be large in some steps and small in others, but once decay and growth cancel on average, the mean rotation is fixed by $\eta\lambda$ alone; this is why measured angles are robust. With a constant gradient size on the sphere, the norm is driven to that balance from every start.

Theorem 13.12 (Rotational equilibrium of SGD). *Let $\theta_0 \neq 0$, $0 < \eta\lambda < 1$, and $\theta_{t+1} = (1-\eta\lambda)\theta_t - \eta g_t$ with $\langle g_t, \theta_t\rangle = 0$ and $\|g_t\|_2\|\theta_t\|_2 = \Gamma > 0$ for every t. Let $q_t = \|\theta_t\|_2^2$ and*

$$q_\infty = \frac{\eta\Gamma}{\sqrt{1-a}} = \Gamma\sqrt{\frac{\eta}{\lambda(2-\eta\lambda)}} \approx \Gamma\sqrt{\frac{\eta}{2\lambda}}. \tag{13.3}$$

(i) $q_{t+1} = a q_t + \eta^2\Gamma^2/q_t$, and $q_t \to q_\infty$ for every $\theta_0 \neq 0$; the distance $|\log(q_t/q_\infty)|$ never increases, and decreases strictly unless $q_t = q_\infty$.

(ii) Suppose $\eta\lambda \leq 1 - 1/\sqrt{2}$, so that $a \geq \frac{1}{2}$. If $q_t \geq q_\infty$, then $0 \leq q_{t+1} - q_\infty \leq a\,(q_t - q_\infty)$. If $q_t < q_\infty$, then $q_{t+1} > q_t$, and either $q_{t+1} \geq q_\infty$ or $0 < q_\infty - q_{t+1} \leq (2a-1)(q_\infty - q_t)$. Near q_∞ the error is multiplied by $2a - 1 = 1 - 4\eta\lambda + 2\eta^2\lambda^2$ per step.

(iii) As $t \to \infty$: the angle ω_t between θ_t and θ_{t+1} tends to $\omega_\infty = \arccos(1-\eta\lambda) = \sqrt{2\eta\lambda}\,(1 + O(\eta\lambda))$; the relative step $\|\theta_{t+1} - \theta_t\|_2/\|\theta_t\|_2$ tends to $\sqrt{2\eta\lambda}$ exactly; and the direction moves on the unit sphere by $\hat\theta_{t+1} = (\hat\theta_t - \eta_{\mathrm{eff},t}\hat g_t)/\|\hat\theta_t - \eta_{\mathrm{eff},t}\hat g_t\|_2$ with $\hat g_t = \|\theta_t\|_2\, g_t$, which is the gradient at $\hat\theta_t$ when $g_t = \nabla\ell(\theta_t;\xi_t)$ for a scale-invariant ℓ; the effective step size *$\eta_{\mathrm{eff},t} = \eta/((1-\eta\lambda)q_t)$ tends to $\tan(\omega_\infty)/\Gamma$.*

(iv) If instead $\Gamma_t = \|g_t\|_2\|\theta_t\|_2 \in [\Gamma_1, \Gamma_2]$ for every t, with $a \ge \frac12$ and $\Gamma_1/\Gamma_2 > (1-a)/a$, then $q_\infty(\Gamma_1) \le \liminf_t q_t \le \limsup_t q_t \le q_\infty(\Gamma_2)$, where $q_\infty(\Gamma)$ is given by (13.3).

Proof. The plan is to reduce the dynamics to one scalar map, to show that the map moves every point toward its fixed point on a logarithmic scale, and to read the angles off Lemma 13.10.

(i) Lemma 13.10(i) with $\|g_t\|_2^2 = \Gamma^2/q_t$ gives the recursion, and $q_{t+1} \ge aq_t > 0$. Put $x_t = q_t/q_\infty$. Since $\eta^2\Gamma^2 = (1-a)q_\infty^2$, dividing by q_∞ gives $x_{t+1} = \varphi(x_t)$ with

$$\varphi(x) = a\,x + (1-a)\,\frac{1}{x},$$

a convex combination, with both weights in $(0,1)$, of x and $1/x$. If $x \ne 1$, then these two numbers lie on opposite sides of 1 and $\varphi(x)$ lies strictly between them, so $|\log\varphi(x)| < |\log x|$. The numbers $\ell_t = |\log x_t|$ therefore decrease to a limit $\ell \ge 0$. Suppose $\ell > 0$. The continuous function $y \mapsto |y| - |\log\varphi(e^y)|$ is positive on the compact set $\{y : \ell \le |y| \le \ell_0\}$, which contains every $\log x_t$, so it has a positive minimum δ there, and $\ell_t \le \ell_0 - t\delta$, which contradicts $\ell_t \ge \ell$ for $t > \ell_0/\delta$. Hence $\ell = 0$.

(ii) Factor the distances:

$$\varphi(x) - 1 = (x-1)\Big(a - \frac{1-a}{x}\Big), \qquad \varphi(x) - x = (1-a)\,\frac{1-x^2}{x}. \tag{13.4}$$

If $x \ge 1$, then the second factor of the first identity lies in $[2a-1, a) \subseteq [0, a)$ because $a \ge \frac12$, so $0 \le \varphi(x) - 1 \le a(x-1)$. If $x < 1$, then the second identity gives $\varphi(x) > x$, and the second factor of the first is less than $2a-1$: either it is at most zero and $\varphi(x) \ge 1$, or it is positive and $0 < 1 - \varphi(x) < (2a-1)(1-x)$. Multiply by q_∞. The local factor is $\varphi'(1) = 2a - 1$.

(iii) Since $q_{t+1}/q_t \to 1$, Lemma 13.10(ii) gives $\cos\omega_t = (1-\eta\lambda)\sqrt{q_t/q_{t+1}} \to 1 - \eta\lambda$, and Lemma 13.10(iv) gives $\|\theta_{t+1} - \theta_t\|_2^2/q_t = \eta^2\lambda^2 + \eta^2\Gamma^2/q_t^2 \to \eta^2\lambda^2 + (1-a) = 2\eta\lambda$; the expansion $\arccos(1-z) = \sqrt{2z}\,(1 + z/12 + O(z^2))$ gives the size of ω_∞. For the direction, Lemma 13.10(iii) gives the update with $\alpha_t\hat u_t = \eta g_t/((1-\eta\lambda)\|\theta_t\|_2) = \eta_{\mathrm{eff},t}\hat g_t$, and Lemma 13.9(ii) for $\ell(\cdot;\xi_t)$ gives $\hat g_t = \nabla\ell(\hat\theta_t;\xi_t)$. The limit of $\eta_{\mathrm{eff},t}$ is $\eta/((1-\eta\lambda)q_\infty) = \sqrt{1-a}/(\sqrt{a}\,\Gamma)$, and $\sqrt{1-a}/\sqrt{a} = \tan\omega_\infty$ because $\cos\omega_\infty = \sqrt{a}$.

(iv) Let $b_t = \eta^2\Gamma_t^2$, $\underline b = \eta^2\Gamma_1^2$, $\bar b = \eta^2\Gamma_2^2$, $\underline q = q_\infty(\Gamma_1) = \sqrt{\underline b/(1-a)}$ and $\bar q = q_\infty(\Gamma_2) = \sqrt{\bar b/(1-a)}$, so that $q_{t+1} = aq_t + b_t/q_t$ and $\underline b \le b_t \le \bar b$. *Lower bound.* If $q_t < \underline q$, then $q_{t+1} - q_t \ge \underline b/q_t - (1-a)q_t = (1-a)(\underline q^2 - q_t^2)/q_t \ge (1-a)(\underline q - q_t)$, so $\underline q - q_{t+1} \le a(\underline q - q_t)$; if $q_t \ge \underline q$, then $q_{t+1} \ge a\underline q_t + \underline b/q_t \ge \underline q$ by (ii) with $\Gamma = \Gamma_1$. Hence $(\underline q - q_t)_+ \le a^t(\underline q - q_0)_+$. *Upper bound.* If $q_t \ge \bar q$,

then $q_{t+1} \le aq_t + \bar{b}/\bar{q} = aq_t + (1-a)\bar{q}$, so $q_{t+1} - \bar{q} \le a(q_t - \bar{q})$. If $q_t < \bar{q}$, then $q_{t+1} \le aq_t + \bar{b}/q_t$, which by (13.4) with $x = q_t/\bar{q}$ is at most $\bar{q}$ once $q_t \ge (1-a)\bar{q}/a$. Since $q/\bar{q} = \Gamma_1/\Gamma_2 > (1-a)/a$, the lower bound gives $q_t \ge (1-a)\bar{q}/a$ for all large t, and from then on $(q_{t+1} - \bar{q})_+ \le a(q_t - \bar{q})_+$. ■

The time scales of the *equilibrium norm* differ from a natural guess. Far above equilibrium the squared norm decays like $a^t \approx e^{-2\eta\lambda t}$, as if the gradient were absent; near equilibrium it relaxes at the rate $4\eta\lambda$, twice as fast, because a smaller norm means a larger gradient (Lemma 13.9(ii)); from far below it rises like $q_\infty\sqrt{2(1-a)t}$ and reaches $0.8\,q_\infty$ after about $1/(4\eta\lambda)$ steps. Optimizers whose step has a fixed size, such as AdamW, Lion, and Muon, lack this restoring force, and their recursion is linear.[6]

[6] At $a = \frac{1}{2}$, that is $\eta\lambda = 1 - 1/\sqrt{2}$, the map $\varphi(x) = \frac{1}{2}(x + 1/x)$ is Newton's iteration for the square root of 1, and the local factor $2a - 1$ vanishes: convergence is then quadratic.

Proposition 13.13 (Steps of fixed size: AdamW, Lion, Muon). *Let $\theta_{t+1} = (1 - \eta_t\lambda)\theta_t - \eta_t u_t$ with $0 < \eta_t\lambda < 1$ and $q_t = \|\theta_t\|_2^2$.*

(i) If $\langle u_t, \theta_t\rangle = 0$ and $\|u_t\|_2^2 = \rho^2 d$ with $\rho > 0$ for every t, then $q_{t+1} = (1-\eta_t\lambda)^2 q_t + \eta_t^2\rho^2 d$ exactly. At a constant step size, $q_t - q_\infty = a^t(q_0 - q_\infty)$ with $q_\infty = \eta^2\rho^2 d/(1-a) = \rho^2\eta d/(\lambda(2-\eta\lambda))$, and $\cos\omega_t \to 1 - \eta\lambda$.

(ii) If instead $\mathbb{E}[\langle u_t, \theta_t\rangle \mid \mathcal{F}_t] = 0$ and $\mathbb{E}[\|u_t\|_2^2 \mid \mathcal{F}_t] = \rho^2 d$, with $\mathcal{F}_t$ the past up to θ_t, then the same recursion holds for $\mathbb{E}q_t$.

Proof. Part (i) is Lemma 13.10(i) and (ii) and the solution of a linear recursion. For (ii), expand $q_{t+1} = (1-\eta_t\lambda)^2 q_t - 2\eta_t(1-\eta_t\lambda)\langle u_t, \theta_t\rangle + \eta_t^2\|u_t\|_2^2$ and take conditional expectations. ■

In equilibrium the entries have root mean square about $\rho\sqrt{\eta/(2\lambda)}$, the noisy-coordinate value of Proposition 13.8, and the angle is SGD's. Sign steps have $\rho = 1$, and an orthogonalized $d_{\text{out}} \times d_{\text{in}}$ step scaled by s has $\rho^2 d = s^2 \min\{d_{\text{out}}, d_{\text{in}}\}$. Orthogonality is exact for normalized gradient steps $\rho\sqrt{d}\, g_t/\|g_t\|_2$, holds on average for sign steps when the gradient on the sphere is symmetric noise, and fails for preconditioned and momentum steps (Section 13.5). For momentum we have a derivation, not a proof.

Heuristic derivation. *Momentum and the angle per step.* Model AdamW's step, when the gradient is noise, by $u_t = m_t = \beta_1 m_{t-1} + (1-\beta_1)\xi_t$ with independent ξ_t of mean zero and $\mathbb{E}\|\xi_t\|_2^2 = d$, the role of $g/\sqrt{\hat{v}}$. Then $\mathbb{E}\|u_t\|_2^2 \to d(1-\beta_1)/(1+\beta_1)$, while a sum of $k \gg 1/(1-\beta_1)$ consecutive steps has second moment about kd, because the weights $(1-\beta_1)\beta_1^j$ sum to one: momentum smooths the walk without slowing its diffusion. If $1/(\eta\lambda) \gg 1/(1-\beta_1)$, then the weights average about $1/(\eta\lambda)$ steps, so their squared norm keeps the value $\eta d/(2\lambda)$ of Proposition 13.13 (Exercise 13.5), and the angle per step is

$$\omega \approx \sqrt{\frac{\eta^2\mathbb{E}\|u_t\|_2^2}{\eta d/(2\lambda)}} = \sqrt{2\eta\lambda\,\frac{1-\beta_1}{1+\beta_1}}, \tag{13.5}$$

0.23 times $\sqrt{2\eta\lambda}$ for $\beta_1 = 0.9$, while over k steps the direction still turns by about $\sqrt{2\eta\lambda k}$. In the balance behind Proposition 13.11, the rest of the decay is offset by the radial component of the step $-\eta m_t$: averaged past gradients were tangent to past directions and point partly inward at the present one, so the step points partly outward. ◇

When the step size changes, the equilibrium moves and the norm follows with a lag.

Proposition 13.14 (The rotation lags the schedule). *In Proposition 13.13(i), let $q_\infty(\eta) = \rho^2\eta d/(\lambda(2-\eta\lambda))$ and $\omega_\infty(\eta) = \arccos(1-\eta\lambda)$. If η_t is nonincreasing and $q_0 \geq q_\infty(\eta_0)$, then $q_t \geq q_\infty(\eta_t)$ and $\omega_t \leq \omega_\infty(\eta_t)$ for every t. If η_t is nondecreasing and $q_0 \leq q_\infty(\eta_0)$, then both inequalities reverse.*

Proof. The function q_∞ increases on $(0, 1/\lambda)$, and $q_\infty(\eta_t)$ is the fixed point of the increasing affine map $q \mapsto (1-\eta_t\lambda)^2 q + \eta_t^2\rho^2 d$. So $q_t \geq q_\infty(\eta_t)$ implies $q_{t+1} \geq q_\infty(\eta_t) \geq q_\infty(\eta_{t+1})$; induct. By Lemma 13.10(ii), $\tan\omega_t = \eta_t\rho\sqrt{d}/((1-\eta_t\lambda)\sqrt{q_t})$, at most the same expression with $q_\infty(\eta_t)$ in place of q_t, which is $\tan\omega_\infty(\eta_t)$. The nondecreasing case reverses every inequality. ■

A decaying schedule therefore cools a scale-invariant layer faster than the quasi-static $\sqrt{2\eta_t\lambda}$: the relaxation time $1/(2\eta_t\lambda)$ grows as η_t falls, and near the end of a decay to zero the rotation falls like η_t rather than $\sqrt{\eta_t}$ (Exercise 13.7); a warmup from a small norm rotates faster than its equilibrium. Since only the direction matters, this can be said with a learning rate on the sphere, which is what Hyperball sets.

Proposition 13.15 (Weight decay is a Hyperball schedule). *Let $\mathcal{L}$ and each $\ell(\cdot;\xi)$ be scale invariant, and let the base step be $u_t = U(\hat\theta_t, \xi_t)$, a function of the direction and of the randomness ξ_t only, with $\langle u_t, \theta_t\rangle = 0$ and $\|u_t\|_2^2 = \rho^2 d$, $\rho > 0$. Let q_t solve the recursion of Proposition 13.13(i) from $q_0 = \|\theta_0\|_2^2$, and run Hyperball on the unit sphere, $\hat w_{t+1} = (\hat w_t - \alpha_t\hat U_t)/\|\hat w_t - \alpha_t\hat U_t\|_2$ with $\hat U_t = U(\hat w_t, \xi_t)/\|U(\hat w_t, \xi_t)\|_2$, from $\hat w_0 = \hat\theta_0$, with the same ξ_t and*

$$\alpha_t = \frac{\eta_t\rho\sqrt{d}}{(1-\eta_t\lambda)\sqrt{q_t}}.$$

Then $\hat w_t = \hat\theta_t$ for every t, and the two runs have the same losses. At a constant step size, $\alpha_t \to \tan\omega_\infty \approx \sqrt{2\eta\lambda}$.

Proof. By Proposition 13.13(i), $\|\theta_t\|_2^2 = q_t$. If $\hat w_t = \hat\theta_t$, then $\hat U_t = \hat u_t$, and Lemma 13.10(iii) gives $\hat\theta_{t+1} = \hat w_{t+1}$. Induct; scale invariance gives equal losses; the limit follows from $q_t \to q_\infty$. ■

The hypotheses hold exactly for normalized gradient steps,[7] whose direction depends on θ only through $\hat\theta$ by Lemma 13.9(ii), and approximately for sign, orthogonalized, and momentum steps, up to a radial component

[7] So a Hyperball learning rate α is an angle per step: $\alpha = 0.01$ matches the equilibrium of weight decay with $\eta\lambda \approx \alpha^2/2 = 5\times10^{-5}$.

that averages out and a norm that changes slowly. The last result looks at the weights themselves.

Proposition 13.16 (AdamW is an exponential moving average of its updates). *For* (13.1) *with* $0 < \eta_t\lambda \le 1$ *and* $\Pi_{s,t} = \prod_{s\le k<t}(1-\eta_k\lambda)$,

$$\theta_t = \Pi_{0,t}\,\theta_0 - \sum_{s<t}\eta_s\,\Pi_{s+1,t}\,u_s, \qquad \Pi_{s,t} \le \exp\Big(-\lambda\sum_{s\le k<t}\eta_k\Big).$$

At a constant step size, $\theta_t = (1-\eta\lambda)^t\theta_0 + \sum_{s<t}\eta\lambda(1-\eta\lambda)^{t-1-s}(-u_s/\lambda)$, *an exponential moving average of the points* $-u_s/\lambda$ *with timescale* $\tau = 1/(\eta\lambda)$ *steps, in which updates older than* k *steps carry total weight at most* $(1-\eta\lambda)^k$.

PROOF. Induction on t; $1-x \le e^{-x}$; and $\sum_{s<t-k}\eta\lambda(1-\eta\lambda)^{t-1-s} \le (1-\eta\lambda)^k$. ∎

This is Theorem 13.7(ii) read as an average, and τ is the *timescale of weight decay*. A run covers $\lambda\sum_t\eta_t$ timescales,[8] $\eta\lambda T$ at a constant step size, and if the right timescale τ is a fixed fraction of the run, as the evidence of Wang and Aitchison (2025) suggests (Observation 13.3), then $\eta\lambda T$ should be held fixed: at fixed learning rate and batch size, λ should fall like the inverse of the number of training tokens, not its inverse square root as in the rule of Loshchilov and Hutter (2019). When μP scales AdamW's hidden-layer learning rate like $1/d_{\text{in}}$ (Chapter 4), holding $\eta\lambda$ fixed requires $\lambda \propto d_{\text{in}}$; with a fixed λ, wider models rotate more slowly and reach equilibrium later. Both rules follow from the identity and the hypothesis; neither is proved optimal.

[8] This average is over updates, with weights fixed by the decay; the model averages of Chapter 5 average the iterates, with weights chosen freely.

13.5 Reading the theory

ONLY THE PRODUCT. In equilibrium the rotation of a scale-invariant weight depends on η and λ only through $\eta\lambda$ (Theorem 13.12 and Proposition 13.13); the quotient η/λ sets the norm, which the function ignores. So $(\eta,\lambda) \mapsto (c\eta,\lambda/c)$ leaves the equilibrium dynamics of every scale-invariant layer unchanged (Figure 13.2, middle), while embeddings, output layers, gains, and biases see η and λ separately.

SIZES AND TIMES. At $\eta = 3\times10^{-4}$ and $\lambda = 0.1$ the equilibrium angle is 7.75×10^{-3} radians per step, 1.8×10^{-3} with $\beta_1 = 0.9$ by (13.5), while over 1000 steps the direction turns by about 0.24 radians either way; the norm relaxes by a factor e every $1/(2\eta\lambda) \approx 1.7\times10^4$ steps. The number of timescales $\lambda\sum_t\eta_t$ varies widely: with the hyperparameters of Brown et al. (2020) (300B tokens, cosine decay to 10%), the smallest GPT-3 model (6×10^{-4}, batches of 0.5M tokens) covers about 18 and the largest (6×10^{-5}, 3.2M tokens) about 0.3; if its layers behaved like the idealization, the largest never reached equilibrium.

THE FIGURE. In Figure 13.2 (left) the norm relaxes as the recursions predict, and single runs stay within 4% (SGD) and 5.5% (sign descent) of them; from $q_\infty/16$, SGD reaches $0.8\,q_\infty$ by $\eta\lambda t \approx 0.25$, sign descent only by 0.8. In the

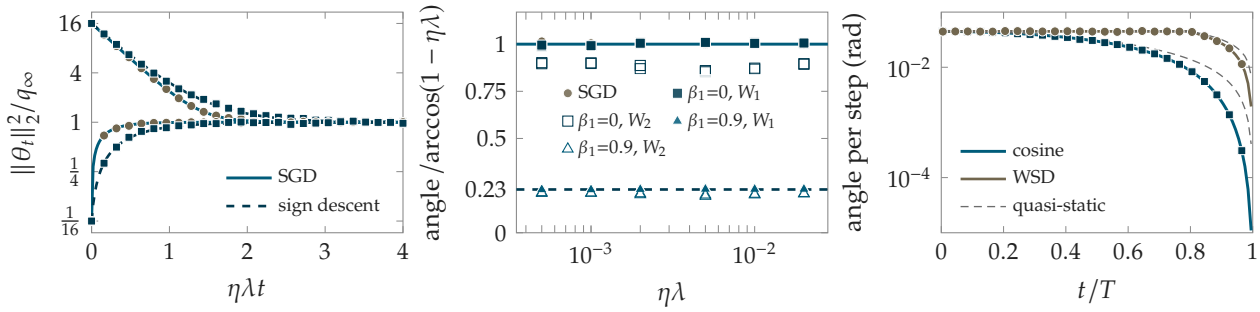


Figure 13.2. The rotational equilibrium: exact predictions (curves) and runs (marks). Left: squared norm of the linear model $x \mapsto \langle x, \hat{\theta} \rangle$ in $\mathbb{R}^{1024}$ under SGD (circles) and sign descent, AdamW without memory (squares), $\eta\lambda = 10^{-3}$, batch 8, labels independent of the inputs ($\Gamma^2 = 1023 \cdot 2/8$), one run per start (seeds 11, 12); curves from Theorem 13.12(i) and Proposition 13.13. Middle: root-mean-square angle per step over the last 4000 of 10^4 steps, over $\arccos(1 - \eta\lambda)$, for the two scale-invariant matrices of a network with two hidden layers of width 32 and layer normalization, teacher labels, batch 16, seed 0: SGD (circles), AdamW with $\beta_2 = 0.95$ and $\beta_1 = 0$ (squares) or 0.9 (triangles), filled W_1, open W_2; two (η, λ) pairs per product; lines at 1 and at $\sqrt{(1-\beta_1)/(1+\beta_1)}$. Right: the left model ($d = 2048$, sign steps) under cosine (squares) and WSD (circles) schedules, $\eta_{\max}\lambda = 10^{-3}$, $T = 2 \times 10^4$: recursion of Proposition 13.13, one run each (seeds 21, 22; every fourth window of 200 steps), quasi-static $\arccos(1 - \eta_t\lambda)$ (dashed). Scripts `13-decay-relaxation.py`, `13-decay-sweep.py`, `13-decay-schedule.py`.

middle, SGD's angle lies within 1.5% of $\arccos(1-\eta\lambda)$ for every product and within 0.1% for $\eta\lambda \geq 2 \times 10^{-3}$; the excess at the smallest products is the drift term of Proposition 13.11, as the norm still grows slowly there. AdamW without momentum matches on W_1 and falls 10 to 15% short on W_2, where its preconditioned step has an outward component that we measured to carry about a fifth of the decay budget. With $\beta_1 = 0.9$ the ratio is 0.23 on W_1, as (13.5) predicts, and about 0.2 on W_2. On the right the recursion follows the run within 2.2%, and the angle falls to 0.36 of its quasi-static value at 90% of the cosine schedule and 0.06 at 99%, as Proposition 13.14 requires.

WHERE THE WEIGHTS SIT. In Figure 13.3 (left), coordinates with a consistent gradient are pulled to the face of the ball exactly along $1 + (1-\eta\lambda)^t(\lambda\theta_0 - 1)$, so the rate of Theorem 13.7(ii) is attained; coordinates with signal-to-noise ratio one settle at $0.683/\lambda$ under sign descent and $0.705/\lambda$ under AdamW, whose averaged step is slightly larger; noise coordinates settle at $0.0224/\lambda$ under both. On the right the guarantees hold with room to spare: under $2/(t+2)$ the gap falls like a few times $1/t^2$, and at a constant step it fluctuates around $0.4\gamma^2$, far below $2L\gamma/\lambda^2$.

WHERE THE AGREEMENT MUST FAIL. Where steps are not orthogonal (momentum and preconditioning, the middle panel), where the gradient on the sphere changes faster than the norm can follow, and where training is shorter than $1/(\eta\lambda)$. The Frank–Wolfe rates are worst cases over convex losses; what transfers to a network is the ball.

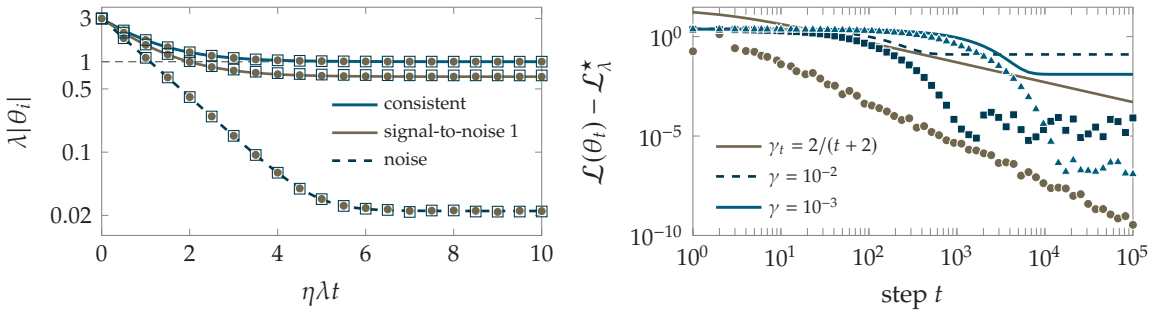


Figure 13.3. AdamW and the Frank–Wolfe ball $\|\theta\|_\infty \le 1/\lambda$. Left: sign descent (circles) and AdamW with $(\beta_1, \beta_2) = (0.9, 0.95)$ (open squares), $\eta = 0.01$, $\lambda = 0.1$, linear loss with Gaussian gradient noise, all coordinates started at $3/\lambda$ (seed 31): 20 with a consistent gradient (largest value; curve: equality in Theorem 13.7(ii)), 200 with signal-to-noise ratio 1 (mean; Proposition 13.8(i)), 2000 with pure noise (root mean square; Proposition 13.8(ii)). Right: the constrained gap of sign descent with decay on a convex quadratic in $\mathbb{R}^{10}$, $\lambda = 1$, five active constraints at the optimum (seed 7; $L = 6.35$ by enumerating sign vectors); marks: runs with the step rule of the same color; curves: guarantees of Theorem 13.7(iii). Script `13-decay-frankwolfe.py`.

13.6 What the theory misses

WHAT WAS IDEALIZED AWAY. *Exact scale invariance:* embeddings, output layers, gains, and biases are not scale invariant, and the matrices that write into the residual stream are so only jointly, not one at a time; for these, η and λ act separately and the decay does regularize. *A fixed gradient size on the sphere:* in a network Γ_t falls with the loss and depends on the effective step itself, since a larger step keeps the direction farther from the minimizers on the sphere; in a teacher–student version of Figure 13.2 (left), with a unit teacher and label noise of standard deviation 1, the student's wandering raised Γ^2 by about 26% above its value at the teacher, and the squared norm settled about 12% above the prediction made with that value, at the one made with the measured Γ. Theorem 13.12(iv) brackets such effects but does not model the feedback. *Orthogonal steps:* preconditioners and momentum add to Proposition 13.11 the radial term $-2\eta(1-\eta\lambda)\sum_t \langle u_t, \theta_t\rangle$, which in our network carried a fifth of the decay budget without momentum and most of it with momentum. *Convexity:* the rates of Theorem 13.7 need a convex loss with a useful smoothness constant; the ball statements do not.

WHERE IT BREAKS. (1) The equilibrium need not be stable. Lobacheva et al. (2021) found periodic cycles with batch normalization and weight decay instead of a steady state. By Lemma 13.9(iii) a shrinking norm raises the curvature the step meets, so the instability of Chapter 8 can return in this form, which our theorem, with Γ fixed, cannot see. (2) In vision networks weight decay does more than set a rotation speed (Zhang et al. 2019), and nothing here predicts test error. (3) Pretraining does not end at the constrained solutions of the Frank–Wolfe view: at $\lambda = 0.1$ the ball is far larger than the weights, and the limit theorem of Xie and Li (2024)

assumes that the iterates converge; the constraint matters for consistent coordinates and for the implicit bias of long training (Chapter 15). (4) At scale the effective-learning-rate view is supported but not settled, by four unrefereed 2026 preprints. Loss curves of language models with different learning rates and norms collapse, below the variation between seeds, when indexed by the effective learning rate, and weight decay and Hyperball act mainly through the effective-learning-rate schedules they induce (Liu et al. 2026). Fixing the norms with Hyperball speeds up Muon by 20 to 30% in token-equivalent terms over weight-decay baselines up to 1.2B parameters and improves the transfer of the learning rate across widths and depths (Wen et al. 2026), which suggests that the equilibrium is not the best way to set the effective learning rate; the difference has been attributed to the time course of the effective step size rather than to a better direction (Xiao, Sun, et al. 2026), and for scale-invariant networks the two methods are dynamically equivalent (Yuan et al. 2026), as in Proposition 13.15. (5) For the largest runs the transient, not the equilibrium, may be the regime of interest (Section 13.5).

Repairs and open problems. Fixing the norm removes the transient and makes the angle a hyperparameter: Hyperball, and the rotational variants of AdamW, SGD, and Lion of Kosson et al. (2024), which control the rotation explicitly and reduce the need for warmup (Chapter 9). Proposition 13.15 says what such a method must do to imitate weight decay, and Proposition 13.14 what it gains: an angle that follows the schedule without lag. Scaling rules for λ (Loshchilov and Hutter 2019; Wang and Aitchison 2025) rest on hypotheses about the right number of timescales that no theory derives. Open: a model in which Γ depends on the effective step and the stability of the equilibrium, or the period of the cycles, can be computed (Exercise 13.10); a theory of decay for parameters that are not scale invariant; and decay and schedule together in the river valley of Chapter 12 (Exercise 13.11).

13.7 Back in the wild

FIELD NOTES	
PRACTICE	AdamW with decoupled decay $\lambda = 0.1$ at $\eta \sim 10^{-4}$; decay for Muon; rescaling λ with training length and width.
IDEALIZATION	(a) Steps of fixed length in a norm with decoupled decay. (b) A scale-invariant loss, steps orthogonal to the weights, constant gradient size on the sphere.
PROVED	Frank–Wolfe on the ball of radius $1/\lambda$, attracting at rate $(1-\eta\lambda)^t$; global convergence of the norm; angle $\arccos(1-\eta\lambda)$, exact on average at a stationary norm; lag under schedules; Hyperball equivalence; AdamW is an EMA of its updates.
PREDICTS	Only $\eta\lambda$ matters in equilibrium: 7.7×10^{-3} rad per step at $\eta\lambda = 3 \times 10^{-5}$, after about $1/(2\eta\lambda)$ steps; momentum shrinks the angle per step by $\sqrt{(1-\beta_1)/(1+\beta_1)}$ (derived).

MISSES	Layers that are not scale invariant; gradients that respond to the effective step; radial components of Adam and momentum steps; generalization; periodic instabilities.
TAKEAWAY	Treat $\eta\lambda$ as the hyperparameter of scale-invariant layers (proved in the idealization); keep $\eta\lambda T$ fixed as runs lengthen (suggested); expect the effective learning rate to lag the schedule (proved in the model).

The theory tells a practitioner what λ does, not what it should be. For layers that feed a normalization the decay is a speed: once the norm has settled, the weights turn by about $\sqrt{2\eta\lambda}$ radians per step whatever the gradients, and a change of learning rate that keeps $\eta\lambda$ fixed changes nothing after the transient. The ball of radius $1/\lambda$ is real but, at $\lambda = 0.1$, far from the weights of a language model; it shapes where long training ends (Chapter 15) more than how pretraining proceeds. Which product is best, how many timescales a run should cover, and whether an equilibrium beats a fixed norm are empirical questions, and the 2026 evidence on them is unrefereed.

The Frank–Wolfe identity is the normalized steepest descent of Chapter 1, in the norms of Chapters 2 and 3, made convergent by its schedule, the thread of Chapters 5 and 12; the equilibrium angle is the per-layer learning rate of μP (Chapter 4) and the angular update of warmup (Chapter 9) seen from the sphere; and by Lemma 13.9(iii) the stability threshold of Chapter 8 for a scale-invariant layer is a threshold on $\eta/\|\theta\|_2^2$, which the decay controls.

Exercises

13.1 CONSOLIDATION. (Frank–Wolfe on an interval.) Let $f(\theta) = \frac{h}{2}\theta^2$ and $\theta_{t+1} = (1-\gamma_t)\theta_t - \gamma_t\lambda^{-1}\operatorname{sign}(\theta_t)$. (a) For a constant $\gamma \in (0,1)$, show that the two-cycle $\pm\gamma/(\lambda(2-\gamma))$ attracts every start whose orbit avoids 0 (a fixed point, as $\operatorname{sign} 0 = 0$), so that f settles at $\frac{h}{2}\gamma^2/(\lambda^2(2-\gamma)^2)$, about $\gamma/16$ times the guarantee $2h\gamma/\lambda^2$. (b) For $\gamma_t = 2/(t+2)$ and $\theta_0 \neq 0$, show that $|\theta_t| = 1/(\lambda(2\lfloor t/2\rfloor + 1))$ for $t \geq 1$.

13.2 CONSOLIDATION. (The Frank–Wolfe gap.) For convex differentiable f and $\theta \in \mathcal{B}_\lambda$, show that $\mathcal{G}(\theta) = \langle \nabla f(\theta), \theta\rangle + \|\nabla f(\theta)\|_*/\lambda$ is at least $f(\theta) - \min_{\mathcal{B}_\lambda} f$ and vanishes exactly at the minimizers over $\mathcal{B}_\lambda$. For the ℓ_∞ ball, deduce that at a minimizer every coordinate has zero partial derivative or sits at $-\operatorname{sign}(\partial_i f)/\lambda$.

13.3 CONSOLIDATION. (Schedules for Frank–Wolfe.) Let $h_{t+1} \leq (1-\gamma_t)h_t + C\gamma_t^2$ with $\gamma_t \in [0,1]$, $\gamma_t \to 0$ and $\sum_t \gamma_t = \infty$. Show that $\limsup_t h_t \leq 0$. Compute the bound on h_T for the linear decay $\gamma_t = \gamma_0(1 - t/T)$ and compare it with the constant step γ_0.

13.4 VARIATION. (Adam with an L_2 penalty.) In Adam with $\beta_1 = 0$, replace decoupled decay by a penalty, $u_t = (g_t + \lambda\theta_t)/\sqrt{\hat{v}_t}$ and $\theta_{t+1} = \theta_t - \eta u_t$, on a scale-invariant layer whose gradient on the unit sphere is noise of variance σ^2 per coordinate. Assuming $\hat{v}_t \approx \sigma^2/\|\theta_t\|_2^2$, show that the decay per step is $\eta\lambda\|\theta_t\|_2/\sigma$ and that the norm settles at $\|\theta\|_2^3 \approx$

$\eta\sigma d/(2\lambda)$, so that the equilibrium angle depends on the size of the gradient, unlike AdamW's (Kosson et al. 2024). (A simulation with $\beta_2 = 0.999$ agrees to 0.3%.)

13.5 VARIATION. (Momentum, exactly.) Let $\theta_{t+1} = c\theta_t - \eta m_t$ and $m_t = \beta m_{t-1} + (1-\beta)\xi_t$ with $c = 1 - \eta\lambda$ and independent ξ_t of mean zero and covariance I_d. (a) Using Lemma A.15 for $(\theta_{t,i}, m_{t-1,i})$, show that the stationary $\mathbb{E}\|\theta\|_2^2$ is $\eta^2 d\,(1-\beta)(1+c\beta)/((1+\beta)(1-c\beta)(1-c^2)) = \frac{\eta^2 d}{1-c^2}\big(1 + O(\frac{\eta\lambda}{1-\beta})\big)$. (b) Show that $\mathbb{E}\|m\|_2^2 = d(1-\beta)/(1+\beta)$; deduce (13.5). (c) For PyTorch's SGD with momentum, $m_t = \beta m_{t-1} + g_t + \lambda\theta_t$ and $\theta_{t+1} = \theta_t - \eta m_t$, show in the same model that the angle per step is about $\sqrt{2\eta\lambda/(1+\beta)}$.

13.6 VARIATION. (MuonW.) A scale-invariant $d_{\text{out}} \times d_{\text{in}}$ matrix is trained by $W_{t+1} = (1-\eta\lambda)W_t - \eta s\,\operatorname{polar}(G_t)$ with full-rank gradients G_t. (a) Show with a 2×2 example that $\langle \operatorname{polar}(G_t), W_t\rangle$ need not vanish although $\langle G_t, W_t\rangle = 0$. (b) Assuming it averages to zero, compute the equilibrium $\|W\|_F$; for $s = 0.2\sqrt{\max\{d_{\text{out}}, d_{\text{in}}\}}$, $\eta = 10^{-3}$, $\lambda = 0.1$ and $d_{\text{in}} = d_{\text{out}} = 1024$, compare the spectral norm of a Gaussian matrix of that Frobenius norm with the radius s/λ.

13.7 VARIATION. (The lag under linear decay.) In Proposition 13.13(i), let $\eta_t = \eta_0(1 - t/T)$ with $\eta_0\lambda T \gg 1$, started at $q_\infty(\eta_0)$. Treating t as continuous, show that the norm tracks $q_\infty(\eta_t)$ while $\eta_t\lambda(T-t) \gg 1$ and then freezes, with a remaining fraction of the run of about $(\eta_0\lambda T)^{-1/2}$, after which the angle falls like η_t.

13.8 SIMULATION. (Universality.) Run `13-decay-sweep.py` with $\beta_1 = 0.5$, with $\beta_2 = 0.999$, and with width 64. Check that SGD's angle stays at $\arccos(1-\eta\lambda)$ and that AdamW's with $\beta_1 = 0.5$ is near $1/\sqrt{3}$ of it on the first layer, as (13.5) predicts; measure the radial component of AdamW's step on each layer.

13.9 SIMULATION. (Where in the ball.) In `13-decay-frankwolfe.py`, vary the signal-to-noise ratio of the middle group over $\{0.25, 0.5, 1, 2, 4\}$; check that sign descent settles at $\operatorname{erf}(\text{snr}/\sqrt{2})/\lambda$, and measure where AdamW settles.

13.10 RESEARCH. (Feedback and stability.) Let the gradient size on the sphere depend on the effective step, $\Gamma^2 = \Gamma_0^2 + \kappa\,\eta_{\text{eff}}$, as for a student that wanders around a teacher. Find the equilibrium of Theorem 13.12 and its stability. Is there a natural model, for instance one in which $\eta_{\text{eff}} S(\hat\theta)$ is capped at 2 as in Chapter 8, where the equilibrium loses stability and the cycles of Lobacheva et al. (2021) appear? As far as we know this is open.

13.11 RESEARCH. (Weight decay in the river valley.) Combine the river-valley model of Chapter 12 with a scale-invariant layer whose direction carries the river and the hills. For a given schedule, which λ minimizes the

final loss, and how does the answer depend on the length of training? Compare with the timescale rule of Proposition 13.16.

Notes and References

Decoupled weight decay is due to Loshchilov and Hutter (2019), and the effective-learning-rate view of decay to van Laarhoven (2017). Wan et al. (2021) derived the equilibrium angle for SGD with momentum and proved convergence to it at a linear rate under their assumptions; Kosson et al. (2024) studied it per neuron for several optimizers; Lobacheva et al. (2021) found the periodic alternative. Theorem 13.12 is the deterministic core of these analyses; we know no earlier statement of Proposition 13.11 in this form.

The Frank–Wolfe method was introduced by Frank and Wolfe (1956) for quadratic programs; Hazan (2016) treats it among projection-free methods. The identification of decoupled decay with Frank–Wolfe on a norm ball runs through Xie and Li (2024) for AdamW, Chen et al. (2024) for Lion, and Pethick et al. (2025) for general oracles. The moving-average reading of AdamW and its scaling rules are those of Wang and Aitchison (2025). The 2026 literature on fixed norms and effective learning rates is unrefereed (Liu et al. 2026; Wen et al. 2026; Xiao, Sun, et al. 2026; Yuan et al. 2026). The network of Figure 13.2 is small; its agreement with the theory checks the identities, not the behavior of large models.

TRANSLATING THE LITERATURE. Loshchilov and Hutter (2019) write AdamW as $\theta_t = \theta_{t-1} - \eta_t\big(\alpha\hat{m}_t/(\sqrt{\hat{v}_t}+\epsilon) + \lambda\theta_{t-1}\big)$, with α the learning rate and η_t a schedule multiplier: their $\alpha\eta_t$ is our η_t and their λ is our $\alpha\lambda$. PyTorch's AdamW decays by `lr·weight_decay` per step and is our (η, λ) (Schaipp 2024). Wang and Aitchison (2025) write $\tau_{\text{iter}} = 1/(\eta\lambda)$ for our τ and $\tau_{\text{epoch}} = \tau_{\text{iter}}/M$ for M steps per epoch. Kosson et al. (2024) follow one neuron's weight vector, a row of our W; their angular update, like that of Wan et al. (2021), is our ω_t. Hyperball's step of norm $\eta_t R$ on the sphere of radius R makes its η_t our α_t of Proposition 13.15. Liu et al. (2026) define the effective learning rate as the learning rate divided by the parameter norm, proportional to our $\tan\omega_t$ for steps of fixed size; for SGD our effective step size divides by the squared norm. In the Frank–Wolfe literature γ_t is the step size and $\mathcal{K}$ the constraint set, here $\eta_t\lambda$ and $\mathcal{B}_\lambda$.

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Part V

Choosing an Optimizer

The two lines of the book meet in two questions: why Adam trains transformers so much faster than gradient descent does, and where each optimizer ends up when it runs for a long time. Both answers depend on the geometry of the problem as much as on the optimizer.

Why Adam Beats SGD on Transformers

14

Heterogeneous curvature, heavy-tailed classes and the batch-size regime

IN THE WILD — ADAM FOR TRANSFORMERS, SGD FOR CONVNETS

IN THIS CHAPTER. Transformers are trained with Adam because stochastic gradient descent trains them far more slowly, while on convolutional networks the two are close; the gap survives full batches and widens with the batch size. We study three solvable idealizations, a table whose rows are used with Zipfian frequencies, a quadratic whose blocks sit at different curvature scales, and the noisy quadratic model with a diagonal preconditioner, connected by one symmetry: gradient descent turns the weight of a row or a block into a clock, and sign descent and Adam do not see it. We prove that under Zipf's law gradient descent removes only the fraction $\ln 10/Z_K \approx \ln 10/\ln K$ of the loss per decade of steps, that blockwise step sizes replace the global condition number by the worst block's, and that a preconditioner's advantage grows with the batch size from a noise-weighted constant to its condition-number factor. The models leave out parameter sharing and the interplay of the mechanisms; a heuristic derivation, checked numerically, shows the preconditioner's small-batch advantage vanishing for averaged iterates, as plain SGD's deficit at batch size one reportedly does in practice.

14.1 In the wild: Adam for language, SGD for vision

The transformer was trained with Adam from the start (Kingma and Ba 2015; Vaswani et al. 2017), and as of 2025 most large language models are trained with Adam, AdamW or a close relative. Convolutional networks for image classification are often still trained with stochastic gradient descent with momentum, which there reaches test accuracy as good as that of adaptive methods or better (Wilson et al. 2017). In this chapter *learning rate*, the practitioners' word, and *step size*, the theorist's, mean the same thing.

IN THE WILD · GPT-3, Brown et al. (2020); LLaMA, Touvron et al. (2023); ResNets on ImageNet, Goyal et al. (2017). Train a transformer language model with Adam or AdamW, $\beta_1 = 0.9$, $\beta_2 = 0.95$, weight decay 0.1, a warmup and a cosine decay (GPT-3 also sets $\epsilon = 10^{-8}$); SGD is not used. Train a ResNet on ImageNet with SGD with momentum 0.9, learning rate $0.1 \cdot B/256$ at batch size B, weight decay 10^{-4} and a step

> decay; Adam is not needed.

The rule is folklore, supported by training runs rather than by a theorem. What has been measured:

OBSERVATION 14.1 (The gap; Kunstner et al. 2023; Zhang et al. 2020; Zhang et al. 2024). On transformers, and most of all on language models, tuned SGD with momentum reduces the training loss much more slowly than Adam; on convolutional networks for image classification the two are close.

An early explanation was the noise: Zhang et al. (2020) found the stochastic gradients of attention models heavy-tailed, argued that adaptive methods and clipping cope with such noise where SGD does not, and proposed an adaptive clipping method that they tested on BERT.

OBSERVATION 14.2 (Not the noise; Kunstner et al. 2023). As the batch size grows, up to the full training set, the gap between SGD and Adam on transformers does not close: SGD gains less from larger batches than Adam. With full batches Adam behaves much like sign descent with momentum, which closes most of the gap.

Sign descent is steepest descent in the ℓ_∞ norm (Chapter 2), so the question moves from the noise to the geometry. Two answers have been proposed and tested.

OBSERVATION 14.3 (Rare classes are slow; Kunstner et al. 2024). Group the tokens into *frequency buckets* of similar frequency and follow the training loss of each bucket. Under gradient descent the loss of the frequent tokens falls first, and that of the infrequent ones, which together account for most of the examples, much more slowly; under Adam and sign descent the buckets progress at much more similar speeds. The same happens in vision and linear models with heavy-tailed classes.

OBSERVATION 14.4 (Block heterogeneity; Zhang et al. 2024). The spectra of the Hessian restricted to the parameter blocks (embeddings, attention, feed-forward layers, normalization gains, output layer) differ widely from block to block in transformers and are similar in convolutional networks. SGD performs on par with Adam on problems without this *block heterogeneity*, including quadratics built for the test, and worse when it is present.

Kunstner et al. (2024) trace the slowness of rare classes to gradients and Hessians that are imbalanced across classes and correlated with each other, and Zhang et al. (2024) trace the block effect to the single step size that SGD uses for all blocks.[1] Zhao et al. (2025) report that adaptivity in the last layer and the normalization parameters alone keeps most of Adam's advantage, and Xie et al. (2025) that rotating the parameters of GPT-2 slows Adam but not SGD: the advantage lives in the coordinate system. The last observation brings the batch size back.

OBSERVATION 14.5 (Small batches; Marek et al. 2025). Language models train stably with batch sizes down to one when Adam's β_2 is rescaled to keep its half-life fixed in tokens, and small batches are more robust to hyperparameters and as efficient per FLOP as large ones or more. At small

[1] Frederik Kunstner's PhD thesis, *Why Do Machine Learning Optimizers That Work, Work?* (University of British Columbia, 2024), asks why such heuristics beat classical methods and formalizes what makes per-coordinate step sizes adaptive.

batch sizes even vanilla SGD, without momentum, trains them stably and comes close to Adam; the gap grows with the batch size.

Marek et al. (2025) is refereed; a preprint reaches a similar conclusion for SGD with momentum (Srećković et al. 2025). This chapter models three of the explanations exactly and asks: *which structure of the loss makes a per-coordinate normalization worth an order of magnitude in steps, why is the benefit specific to transformers, and why does it depend on the batch size?*

14.2 The idealization: frequencies, blocks and batches

Definition 14.6 (Table models). Let $q_1 \geq \cdots \geq q_K > 0$, $\sum_j q_j = 1$, be the frequencies of K items (tokens or classes). A *table model* has one row $w_j \in \mathbb{R}^r$ per item and the loss

$$\mathcal{L}(W) = \sum_{j=1}^{K} q_j \, \ell_j(w_j), \qquad W = (w_1, \ldots, w_K), \tag{14.1}$$

with differentiable row losses ℓ_j. The *square-loss table* has $\ell_j(w) = \frac{1}{2}\|w - y_j\|_2^2$, the *softmax table* has $\ell_j(w) = -\sum_k v_{jk} \log \operatorname{softmax}(w)_k$ for probability vectors v_j. The frequencies follow *Zipf's law* if $q_j = 1/(jZ_K)$, where $Z_K = 1 + \frac{1}{2} + \cdots + \frac{1}{K}$ is the K-th *harmonic number*.

The softmax table is a bigram language model: row j holds the logits of the token after token j, v_j is the conditional distribution of that token, and q_j is the frequency of token j as an input. An embedding table has the same structure, since its row j gets a gradient only from examples that contain token j. The square-loss table with $y_j = v_j$ is the bigram model trained with the square loss on one-hot targets e_Y: given $X = j$, the expectation of $\frac{1}{2}\|w_X - e_Y\|_2^2$ is $\frac{1}{2}\|w_j - v_j\|_2^2 + \frac{1}{2}(1 - \|v_j\|_2^2)$. It is the model of Kunstner and Bach (2025). In the output layer of a classifier the frequencies are those of the labels, and near the optimum they enter as curvatures (Proposition 14.13).

The second model is a quadratic $\mathcal{L}(\theta) = \frac{1}{2}(\theta - \theta^\star)^\top H(\theta - \theta^\star)$ with block-diagonal $H = \operatorname{diag}(H_1, \ldots, H_N) \succ 0$, whose blocks stand for the parameter groups of Observation 14.4; a table is the case of blocks that are rows. The third is the *noisy quadratic model* of Chapter 11.

Assumption 14.7 (Noisy quadratic model; Zhang et al. 2019). The loss is $\mathcal{L}(\theta) = \frac{1}{2}\sum_{i=1}^{d} h_i \theta_i^2$ with $h_1 \geq \cdots \geq h_d > 0$. At step t the optimizer receives $g_t = H\theta_t + \zeta_t$, where the ζ_t are independent of each other and of the past, with mean zero and covariance $\operatorname{diag}(c_1, \ldots, c_d)/B$ at batch size B, a positive real number that enters only through the noise. Preconditioned SGD with multipliers $a_1, \ldots, a_d > 0$ updates $\theta_{t+1,i} = \theta_{t,i} - \eta a_i g_{t,i}$; SGD is $a_i = 1$. We write $\lambda_i = a_i h_i$ for the curvatures that the preconditioned method sees.

Near a minimum of a well-specified model the gradient covariance is close to the Hessian, which suggests $c_i \propto h_i$; our example takes $h_i = 1/i$

and $c_i = h_i$. Adam is modelled by $a_i = h_i^{-1/2}$: its second moment estimates $\mathbb{E} g_{t,i}^2 = h_i^2 \theta_{t,i}^2 + c_i/B$, in which the noise dominates at small batch sizes, so that its preconditioner is $\mathrm{diag}(c)^{1/2}/\sqrt{B}$, a constant times $H^{1/2}$ when $c \propto h$ (Exercise 14.7). At large batch sizes the gradient dominates, and Adam is closer to sign descent.[2] One symmetry of optimizers connects the three models.

[2] Zhang et al. (2019) use the same device, preconditioning by a power of the curvature, to stand for adaptive and second-order optimizers. Adam's preconditioner is not the Hessian; see Chapter 6.

Definition 14.8 (Scale invariance). An optimizer computes its update u_t from the gradients $g_0, \ldots, g_t$ and its hyperparameters, and sets $\theta_{t+1} = \theta_t + u_t$. Partition the coordinates into blocks. The optimizer *acts blockwise* if the update of each block is computed from the gradients of that block by the rule that the optimizer follows when run on that block alone, and it is *scale invariant* if multiplying the gradients of every block, at all steps, by a positive constant of that block leaves all updates unchanged. With blocks of one coordinate we say *coordinatewise*, for both properties.

Sign descent $u_t = -\eta \,\mathrm{sign}(g_t)$ is coordinatewise scale invariant, and so are signum and Lion, which take signs of moving averages (Chapters 2 and 5), and Adam, RMSProp and AdaGrad with $\epsilon = 0$: multiplying every $g_{t,i}$ by $s_i > 0$ multiplies $m_{t,i}$ by s_i and $v_{t,i}$ by s_i^2, and leaves $m_{t,i}/\sqrt{v_{t,i}}$ unchanged (we read $0/0$ as 0). Adalayer (Zhao et al. 2025) and Adam-mini (Zhang et al. 2025), which keep one second moment or one learning rate per block, are blockwise scale invariant. Gradient descent is not; normalized gradient descent is invariant only under a common scale for all blocks; and Adam with $\epsilon > 0$ only for gradients much larger than ϵ, below which it is SGD with momentum and step size η/ϵ (Chapter 6).[3]

[3] With $\epsilon = 10^{-8}$ the condition is mild: in a Zipfian table with $K = 10^5$ the rarest row carries the weight $q_K \approx 10^{-6}$, so its gradients exceed ϵ until its error falls below about 10^{-2}.

DICTIONARY

IN THE THEORY	IN A TRAINING RUN
the frequency q_j of row j	how often token j occurs as an input (embedding, bigram rows) or as a label (output rows)
gradient descent's clock $\eta q_j t$	the learning rate times the number of times token j was seen, divided by the batch size
Zipf's law; $Z_K \approx \ln K + 0.58$	token frequencies of text; $Z_K \approx 11.4$ for 50,000 tokens
a scale-invariant optimizer	Adam with small ϵ, sign descent, Lion; Adam-mini blockwise
blocks H_b at different scales	parameter groups of Observation 14.4
small and large batch limits	noise-limited and curvature-limited training
the multipliers $a_i = h_i^{-1/2}$	Adam at small batch sizes, whose second moment is mostly noise

WHAT WAS THROWN AWAY. Shared parameters (in a transformer the embedding rows feed shared layers), nonlinearity, changing curvature, and noise that is heavy-tailed or depends on the parameters. These go first

because what remains can be solved exactly, so that the signature of each mechanism, a logarithmic loss curve, a rate set by blockwise spectra, a gap depending on the batch size, can be held against the observations one at a time.

WHAT ONE HAD TO NOTICE. That the frequency of a token is a curvature: the Hessian of a table is block diagonal with blocks $q_j \nabla^2 \ell_j$, so Zipf's law becomes the spectrum $1/j$, the hardest power law for gradient descent (Kunstner and Bach 2025); and that per-coordinate normalization is exactly the operation that removes such a weight.

14.3 The theory: frequencies are clocks

Proposition 14.9 (Weights are clocks). *Let $\mathcal{L}(\theta) = \sum_{b=1}^{N} s_b \, \ell_b(\theta_b)$ with weights $s_b > 0$ and functions ℓ_b with locally Lipschitz gradients, $\theta = (\theta_1, \dots, \theta_N)$ split into blocks.*

(i) Gradient descent with step size η on $\mathcal{L}$ moves each block θ_b as gradient descent with step size ηs_b on ℓ_b alone. Gradient flow on $\mathcal{L}$ is $\theta_b(t) = \varphi_b(s_b t)$ as long as these exist, where φ_b is gradient flow on ℓ_b from $\theta_b(0)$.

(ii) An optimizer that acts blockwise and is scale invariant moves each block exactly as when run on ℓ_b alone, from the same point with the same hyperparameters. Its iterates do not depend on $s_1, \dots, s_N$.

PROOF. The plan is to compare each block with its solo run, step by step. The gradient of $\mathcal{L}$ with respect to θ_b is $s_b \nabla \ell_b(\theta_b)$, which gives (i) for gradient descent; for the flow, $\psi(t) = \varphi_b(s_b t)$ satisfies $\psi'(t) = -s_b \nabla \ell_b(\psi(t))$, the equation of block b under gradient flow on $\mathcal{L}$, with the same initial point, and solutions are unique because the gradients are locally Lipschitz. For (ii), let $\tilde{\theta}_{b,t}$ be the iterates of the solo run on ℓ_b, and suppose that $\theta_{b,\tau} = \tilde{\theta}_{b,\tau}$ for $\tau \le t$, as holds for $t = 0$. The gradients of block b seen so far are then s_b times those of the solo run; the update of block b is computed from them by the rule of the solo run, and by scale invariance it equals the update of the solo run. Hence $\theta_{b,t+1} = \tilde{\theta}_{b,t+1}$. ∎

For gradient descent the step size must be stable for every block, so on a table whose row losses all have the Hessian ΛI it needs $\eta q_1 \Lambda < 2$: the most frequent row caps the step size, and the rarest then needs about $1/(\eta q_K \Lambda) > q_1/(2q_K)$ steps. The ratio q_1/q_K is a condition number made of frequencies alone, and under Zipf's law it is K. What this costs is the content of the next theorem. Write $\gamma = 0.5772\dots$ for Euler's constant and $E_1(z) = \int_z^\infty u^{-1} e^{-u} \, \mathrm{d}u$.

Lemma 14.10 (Exponentially weighted harmonic sums). *If a function $f \ge 0$ on $[1, K]$ is nondecreasing up to some point and nonincreasing after it, then $|\sum_{j=1}^{K} f(j) - \int_1^K f| \le \max f$. For $b > 0$ and every integer $K \ge 1$,*

$J_K(b) = \sum_{j=1}^{K} j^{-1} e^{-b/j}$ satisfies

$$-\frac{1}{eb} - \frac{e^{-b}}{b} \le J_K(b) - \ln\frac{K}{b} + \gamma \le \frac{1}{eb} + \frac{b}{K}.$$

Proof. The plan is to compare sums with integrals on the two monotone pieces, and then to evaluate the integral. Let f increase on $[1, m]$ and decrease on $[m, K]$, and let $p = \lfloor m \rfloor$. On $[j, j+1]$ the integral of f lies between $f(j)$ and $f(j+1)$, the smaller being $f(j)$ for $j < p$ and $f(j+1)$ for $j > p$; on $[p, p+1]$, when $p < K$, it lies between $\min(f(p), f(p+1))$ and $\max f$. Adding the lower bounds gives $\int_1^K f \ge \sum_j f(j) - \max(f(p), f(p+1))$, and adding the upper bounds gives $\int_1^K f \le \sum_j f(j) + \max f - f(1) - f(K)$.

For the second statement, $f(x) = x^{-1}e^{-b/x}$ increases up to $x = b$ and decreases afterwards, with maximum $1/(eb)$, and $u = b/x$ gives $\int_1^K f = E_1(b/K) - E_1(b)$. The function $E_1(z) + \ln z$ has derivative $(1 - e^{-z})/z$ and the classical limit $-\gamma$ at $z = 0$, so $E_1(z) = -\gamma - \ln z + \mathrm{Ein}(z)$ with $\mathrm{Ein}(z) = \int_0^z (1 - e^{-u})u^{-1}\,\mathrm{d}u \in [0, z]$; and $0 \le E_1(b) \le b^{-1}\int_b^\infty e^{-u}\,\mathrm{d}u = e^{-b}/b$. Hence $\int_1^K f - \ln(K/b) + \gamma = \mathrm{Ein}(b/K) - E_1(b) \in [-e^{-b}/b,\, b/K]$, and the first statement adds $\pm 1/(eb)$. ∎

Theorem 14.11 (Zipf frequencies). *Consider the square-loss table with Zipfian frequencies $q_j = 1/(jZ_K)$, $K \ge 2$, started from rows with $\|w_j(0) - y_j\|_2 = 1$, and run gradient descent with step size $\eta \in (0, 2/q_1)$; write $\mathcal{L}_t$ for the loss after t steps, so $\mathcal{L}_0 = \frac{1}{2}$.*

(i) For every η, $\mathcal{L}_t = \frac{1}{2}\sum_{j=1}^{K} q_j(1 - \eta q_j)^{2t}$.

(ii) For $\eta = 1/q_1$ and every $t \ge 1$,

$$\frac{\mathcal{L}_t}{\mathcal{L}_0} = 1 - \frac{\mathcal{A}(t) - R_K(t)}{Z_K}, \qquad \mathcal{A}(t) = \sum_{j \ge 1} \frac{1}{j}\left[1 - \left(1 - \frac{1}{j}\right)^{2t}\right],$$

with $0 \le R_K(t) \le 2t/K$ and $|\mathcal{A}(t) - \ln(2t) - 2\gamma| \le \frac{3}{2t}$. Also, $\mathcal{L}_t/\mathcal{L}_0 = \left(\ln\frac{K}{2t} - \gamma + \delta\right)/Z_K$ with $|\delta| \le \frac{3}{2t} + \frac{2t}{K}$.

(iii) For every $\eta \in (0, 2/q_1)$ and every $t \ge 1$, $\mathcal{L}_t/\mathcal{L}_0 \ge \left(\ln\frac{K}{4t} - \gamma - \frac{3}{2t}\right)/Z_K$.

(iv) For every $\varepsilon \in (0, 1)$ and every stable step size, the number T_ε of steps needed to reach $\mathcal{L}_t \le \varepsilon\mathcal{L}_0$ satisfies $T_\varepsilon \ge \frac{K}{4}e^{-\gamma - \varepsilon Z_K - 3/(2T_\varepsilon)}$; as $K \to \infty$ with ε fixed, $T_\varepsilon \ge (1 - o(1))\frac{e^{-\gamma}}{4}Ke^{-\varepsilon Z_K}$, which is of order $K^{1-\varepsilon}$.

The amount $\mathcal{A}(t)$ does not depend on K and grows by $\ln 10$ per decade: *between steps t and $10t$, for $1 \ll t \ll K$, gradient descent removes the fraction $\ln 10/Z_K$ of the initial loss*, the same in every decade, until t approaches K.

Proof. The plan is to reduce the loss to a harmonic sum with geometric weights, compare those weights with exponential ones, and apply Lemma 14.10. By Proposition 14.9(i), row j runs gradient descent with step

size ηq_j on $\frac{1}{2}\|w - y_j\|_2^2$, which multiplies its error by $1 - \eta q_j$ at each step; this is (i).

For (ii), $\eta = 1/q_1 = Z_K$ gives $\eta q_j = 1/j$, so $\mathcal{L}_t/\mathcal{L}_0 = F_K(t)/Z_K$ with $F_K(t) = \sum_{j\le K} j^{-1}(1-1/j)^{2t}$, and since $Z_K = \sum_{j\le K} 1/j$,

$$F_K(t) = Z_K - \mathcal{A}(t) + R_K(t), \qquad R_K(t) = \sum_{j>K} \frac{1}{j}\Big[1 - \Big(1 - \frac{1}{j}\Big)^{2t}\Big].$$

Bernoulli's inequality gives $0 \le 1-(1-1/j)^{2t} \le 2t/j$, so the series converge and $0 \le R_K(t) \le \sum_{j>K} 2t/j^2 \le 2t/K$. To size $F_K(t)$, compare it with $J_K(2t)$. For $j \ge 2$ put $x = 1/j \le \frac{1}{2}$; the inequalities $-x - x^2 \le \ln(1-x) \le -x$ on $[0, \frac{1}{2}]$ give

$$e^{-2t/j}\Big(1 - \frac{2t}{j^2}\Big) \le e^{-2t/j - 2t/j^2} \le \Big(1 - \frac{1}{j}\Big)^{2t} \le e^{-2t/j},$$

and the term $j = 1$ of F_K vanishes. Hence $0 \le J_K(2t) - F_K(t) \le e^{-2t} + \sum_{j\ge 2} g(j)$ with $g(x) = 2t x^{-3} e^{-2t/x}$, which increases up to $x = 2t/3$ and decreases afterwards, has maximum $\frac{27}{4t^2}e^{-3} < 0.34/t^2$, and has $\int_0^\infty g = 1/(2t)$; by the lemma, $\sum_{j\ge 2} g(j) \le \frac{1}{2t} + \frac{0.34}{t^2}$. With the lemma at $b = 2t$,

$$-\frac{1}{2et} - \frac{e^{-2t}}{2t} - e^{-2t} - \frac{1}{2t} - \frac{0.34}{t^2} \le F_K(t) - \ln\frac{K}{2t} + \gamma \le \frac{1}{2et} + \frac{2t}{K}.$$

For $t \ge 1$ the left side is at least $-1.25/t$, since $e^{-2t} \le e^{-2}$ and $te^{-2t} \le e^{-2}$ there. This is the second form of (ii). For the first, let $K \to \infty$ in $\mathcal{A}(t) = Z_K - F_K(t) + R_K(t)$: since $Z_K - \ln K \to \gamma$ and $R_K(t) \to 0$, the bounds on $F_K(t) - \ln K$ pass to the limit and give $|\mathcal{A}(t) - \ln(2t) - 2\gamma| \le 1.25/t$.

For (iii), let $j \ge 2$; then $0 < \eta q_j < 2/j \le 1$, so $(1-\eta q_j)^{2t} \ge (1-2/j)^{2t} \ge 0$. Drop the terms $j \le 3$, and for $j \ge 4$ use $\ln(1-x) \ge -x - x^2$ with $x = 2/j \le \frac{1}{2}$, to get

$$\frac{\mathcal{L}_t}{\mathcal{L}_0} \ge \frac{1}{Z_K}\Big[J_K(4t) - \sum_{j\le 3}\frac{e^{-4t/j}}{j} - \sum_{j\ge 4}\frac{8t}{j^3}e^{-4t/j}\Big].$$

By the lemma applied to $8tx^{-3}e^{-4t/x}$ the last sum is at most $\frac{1}{2t} + \frac{0.17}{t^2}$; the middle sum is at most $\frac{11}{6}e^{-4t/3} \le 0.49/t$ for $t \ge 1$; and the lemma at $b = 4t$ gives $J_K(4t) \ge \ln\frac{K}{4t} - \gamma - \frac{1}{4et} - \frac{e^{-4t}}{4t}$. The errors add up to less than $1.3/t$.

For (iv), if $\mathcal{L}_t \le \varepsilon\mathcal{L}_0$, then (iii) gives $\ln\frac{K}{4t} - \gamma - \frac{3}{2t} \le \varepsilon Z_K$, that is, $t \ge \frac{K}{4}e^{-\gamma - \varepsilon Z_K - 3/(2t)}$; at $t = T_\varepsilon$ this is the first claim. Since $T_\varepsilon \ge 1$, it gives $T_\varepsilon \ge \frac{K}{4}e^{-\gamma-\varepsilon Z_K - 3/2} \to \infty$, so $e^{-3/(2T_\varepsilon)} \to 1$. Finally $Z_K = \ln K + \gamma + O(1/K)$, so $Ke^{-\varepsilon Z_K} = e^{-\varepsilon\gamma}K^{1-\varepsilon}(1+o(1))$. ∎

The constant $\frac{3}{2}$ is not sharp: numerically $\mathcal{A}(t) - \ln(2t) - 2\gamma$ is close to $1/(4t)$. By (iii) the best stable step gains at most about a factor two in time. The asymptotic form of (iv) converges slowly, with relative errors of order $K^{-\varepsilon}$: for $K = 10^4$ and $\varepsilon = 0.1$, formula (i) gives 1361 steps at $\eta = 1/q_1$ and 683 at the best constant step size, against the lower bound 526 of (iv). A scale-invariant optimizer sees none of this.

Corollary 14.12 (Sign descent and Adam on a table). *Start the square-loss table, with arbitrary frequencies, from the same error $e = w_j(0) - y_j \in \mathbb{R}^r$ in every row. For every coordinatewise scale-invariant optimizer, $\mathcal{L}_t/\mathcal{L}_0$ does not depend on the frequencies or on K: it is the normalized loss of the optimizer on the single row $\frac{1}{2}\|w - y\|_2^2$ from the error e. Under sign descent with step size η, the i-th coordinate of the error has size $|e_i| - t\eta$ for $t \le \lfloor |e_i|/\eta \rfloor$, and afterwards it alternates between two values whose sum is η, or stays at 0; so $\mathcal{L}_t/\mathcal{L}_0 \le r\eta^2/\|e\|_2^2$ for $t > \max_i |e_i|/\eta$.*

Proof. The plan is to apply Proposition 14.9(ii) with the rows as blocks, which is allowed because a coordinatewise scale-invariant optimizer acts blockwise for every partition and is invariant under rescalings that are constant on rows. Every row then follows the solo run from e, so $\mathcal{L}_t = \sum_j q_j \ell(t)$ for the solo loss $\ell(t)$, and $\mathcal{L}_t/\mathcal{L}_0 = \ell(t)/\ell(0)$. Under sign descent the solo gradient is the error, updated by $e_i \leftarrow e_i - \eta \operatorname{sign}(e_i)$: its size falls by η while it is at least η; then each step flips its sign and maps the size ρ to $\eta - \rho$, unless it is exactly 0, where it stays. ∎

So on the Zipf table with $K = 10^4$, gradient descent needs 683 steps at its best constant step size to reach a tenth of the initial loss, and sign descent with step size 0.005 needs 55 for every K (Figure 14.1). The corollary removes the frequencies, not the spread of the targets within a row: when the targets are themselves Zipfian, as in a bigram model, a constant step must suit the smallest targets and the run must be long enough for the largest. On Zipf-distributed data Kunstner and Bach (2025) find that sign descent needs a number of steps growing only like the square root of the dimension; Exercise 14.2 derives this for one row, for the loss at consecutive steps, since sign descent oscillates.

For the softmax table the frequency enters in the same way, exactly; for the output layer, where the frequencies are those of the labels, it enters as the curvature at the optimum.

Proposition 14.13 (Frequencies in the softmax). *(i) For the softmax table, with $s_j = \operatorname{softmax}(w_j)$,*

$$\nabla_{w_j}\mathcal{L} = q_j\,(s_j - \nu_j), \qquad \nabla^2_{w_j}\mathcal{L} = q_j\big(\operatorname{diag}(s_j) - s_j s_j^\top\big),$$

the Hessian is block diagonal over the rows, and $0 \le \operatorname{diag}(s) - ss^\top \le \frac{1}{2}I$ for every probability vector s. Hence Proposition 14.9 applies with the weights q_j.

(ii) For logits $z \in \mathbb{R}^K$ of classes with frequencies $q_1 > \cdots > q_K > 0$, the loss $\ell(z) = -\sum_k q_k \log \operatorname{softmax}(z)_k$ is minimized exactly at $z = \log q + c\mathbf{1}$, $c \in \mathbb{R}$, where its Hessian is $\operatorname{diag}(q) - qq^\top$. Its eigenvalues are 0, with eigenvector $\mathbf{1}$, and $\mu_1 \ge \cdots \ge \mu_{K-1}$ with $q_{k+1} \le \mu_k \le q_k$. For Zipfian q the condition number μ_1/μ_{K-1} on the subspace orthogonal to $\mathbf{1}$ lies between $(K-1)/2$ and K.

Proof. The plan is to compute the derivatives and to read the spectrum from the Courant–Fischer characterization. (i) Since $\sum_k v_{jk} = 1$, $\ell_j(w) = -\sum_k v_{jk} w_k + \log \sum_k e^{w_k}$, whose gradient is $s_j - v_j$, and $\partial s_k / \partial w_l = s_k(\delta_{kl} - s_l)$ gives the Hessian. The form $u^\top(\mathrm{diag}(s) - ss^\top)u = \sum_k s_k u_k^2 - (\sum_k s_k u_k)^2$ is the variance of u_k for k drawn from s, nonnegative and, by Popoviciu's inequality, at most $\frac{1}{4}(\max_k u_k - \min_k u_k)^2 \le \frac{1}{2}\|u\|_2^2$.

(ii) The gradient $\mathrm{softmax}(z) - q$ vanishes exactly when $z - \log q \in \mathbb{R}\mathbf{1}$, and ℓ is convex by (i), so these points are the minimizers. If $(\mathrm{diag}(q) - qq^\top)u = 0$, then $q_k u_k = q_k \langle q, u \rangle$ for every k, so $u \propto \mathbf{1}$. By Courant–Fischer, μ_k is the largest value, over subspaces U of dimension k, of the least value of the form on unit vectors of U. Since $-qq^\top \preceq 0$, $\mu_k \le q_k$; and the vectors of $\mathrm{span}(e_1, \dots, e_{k+1})$ orthogonal to q form a subspace of dimension at least k on which the form is $\sum_{i \le k+1} q_i u_i^2 \ge q_{k+1}\|u\|_2^2$, so $\mu_k \ge q_{k+1}$. For Zipfian q this puts μ_1/μ_{K-1} between $q_2/q_{K-1} = (K-1)/2$ and $q_1/q_K = K$. ∎

Near its optimum the output layer is a quadratic whose curvatures interlace the frequencies, and at uniform initialization the gradient $1/K - q_k$ of a frequent class carries its frequency: the imbalanced, correlated gradients and Hessians that Kunstner et al. (2024) find in linear models. In a table, by Proposition 14.9, gradient descent fits a bucket of frequency q at the time $1/(\eta q)$, a decade of time per decade of frequency, while a scale-invariant optimizer fits all buckets together (Figure 14.1, right).[4]

[4] Yadav et al. (2025), a workshop paper, prove in a minimal next-token-prediction model with heavy-tailed classes that sign descent converges faster than normalized gradient descent.

14.4 The theory: blocks and batches

Rows are blocks that do not interact, each with its own weight; the block quadratic keeps the first property and replaces the weights by curvature scales. For a block write $\mu_b = \lambda_{\min}(H_b)$, $L_b = \lambda_{\max}(H_b)$ and $\kappa_b = L_b/\mu_b$; globally $L = \max_b L_b$, $\mu = \min_b \mu_b$ and $\kappa = L/\mu$; and call $\omega = \max_b L_b / \min_b L_b$ the *scale spread*.

Theorem 14.14 (Block heterogeneity). *Let $\mathcal{L}(\theta) = \frac{1}{2}(\theta - \theta^\star)^\top H(\theta - \theta^\star)$ with $H = \mathrm{diag}(H_1, \dots, H_N) \succ 0$.*

(i) Let $\rho(\eta) = \max(|1 - \eta L|, |1 - \eta\mu|)$. Gradient descent with step size η has $\|\theta_t - \theta^\star\|_2 \le \rho(\eta)^t \|\theta_0 - \theta^\star\|_2$, with equality for some $\theta_0 \ne \theta^\star$, and $\min_\eta \rho(\eta) = (\kappa - 1)/(\kappa + 1)$, attained at $\eta = 2/(L + \mu)$.

(ii) With a step size η_b for each block, the best worst-case rate is the largest of the rates $(\kappa_b - 1)/(\kappa_b + 1)$.

(iii) $\max(\omega, \max_b \kappa_b) \le \kappa \le \omega \max_b \kappa_b$.

(iv) If $H_b = s_b \bar{H}_b$ with $s_b > 0$, every blockwise scale-invariant optimizer produces the same iterates for all choices of the scales s_b.

Proof. The plan is to diagonalize and treat each block as in (i). (i) In an orthonormal eigenbasis of H, the component along an eigenvector with eigenvalue λ_k is multiplied by $1 - \eta\lambda_k$ at each step, so the norm ratio is at most $\max_k |1 - \eta\lambda_k|^t$, with equality for an eigenvector attaining the maximum. The function $\lambda \mapsto |1 - \eta\lambda|$ is convex, so the maximum over the

spectrum is at μ or L, and the branches $1-\eta\mu$ and $\eta L-1$ cross at $\eta = 2/(L+\mu)$, where both equal $(\kappa-1)/(\kappa+1)$. (ii) The blocks evolve independently, so the worst case is the worst block, and each block's rate is minimized separately by (i). (iii) Since $\mu_b \le L_b$, $\kappa \ge \max_b L_b/\min_b L_b = \omega$, and $\kappa \ge L_b/\mu_b$ for every b; if b^* attains $\min_b \mu_b$, then $\kappa = (\max_b L_b/L_{b^*})(L_{b^*}/\mu_{b^*}) \le \omega\kappa_{b^*}$. (iv) is Proposition 14.9(ii) with $\ell_b(\theta_b) = \frac{1}{2}(\theta_b - \theta_b^\star)^\top \bar{H}_b(\theta_b - \theta_b^\star)$. ∎

Gradient descent needs about $\frac{\kappa}{2}\ln\frac{1}{\varepsilon}$ steps to reduce the error by a factor ε, blockwise step sizes about $\frac{1}{2}\max_b \kappa_b \ln\frac{1}{\varepsilon}$, and by (iii) the gain $\kappa/\max_b \kappa_b$ is at most the scale spread ω, with equality when every $\kappa_b = 1$. Part (iii) also says what block heterogeneity is. Arrange one spectrum in two ways: each block a narrow band at its own scale, or every block the whole range. Gradient descent cannot tell the two apart; blockwise step sizes gain ω in the first arrangement and nothing in the second, where $\omega = 1$. In the language of Observation 14.4, transformers are closer to the first arrangement and convolutional networks to the second. By (iv), a blockwise scale-invariant optimizer, Adam among them, does not see the scales: its iterates are those of the problem rescaled to $\omega = 1$.

Both invariances are tied to the coordinate system. Gradient descent is *rotation equivariant*: for an orthogonal Q, gradient descent on $\theta \mapsto \mathcal{L}(Q\theta)$ from $Q^\top\theta_0$ produces $Q^\top$ times the iterates on $\mathcal{L}$. Scale invariance does not survive a rotation that mixes rows or blocks, and for sign descent the ℓ_∞ smoothness constant of a quadratic can grow from $\operatorname{tr} H$ to about $\frac{2}{\pi}d\lambda_{\max}(H)$ after a random rotation (Chapter 2), which is the effect that Xie et al. (2025) measure in GPT-2 (Exercise 14.9).[5]

[5] SOAP runs Adam in the eigenbasis of Shampoo's Kronecker factors (Chapter 7): it chooses the coordinate system in which a diagonal, scale-invariant method acts, instead of inheriting the parameters' own.

With noise, the advantage of a preconditioner has different limits at large and small batch sizes, the two regimes of Chapter 11. The small-batch limit is a number of samples, defined through the bias and the noise per unit of η/B of the flow at time $\tau = \eta t$:

$$\mathcal{B}_a(\tau) = \frac{1}{2}\sum_{i=1}^{d} h_i\theta_{0,i}^2 e^{-2\lambda_i\tau}, \qquad \mathcal{V}_a(\tau) = \frac{1}{4}\sum_{i=1}^{d} a_i c_i\big(1-e^{-2\lambda_i\tau}\big). \tag{14.2}$$

We write $\Phi_a(\tau,\alpha) = \mathcal{B}_a(\tau) + \alpha\mathcal{V}_a(\tau)$, $\Delta_{0,i} = \frac{1}{2}h_i\theta_{0,i}^2$, so that $\sum_i \Delta_{0,i} = \mathcal{L}(\theta_0)$, and $V_a = \sum_i a_i c_i$.

Lemma 14.15 (Small steps). *In the noisy quadratic model let $\lambda_{\max} = \max_i \lambda_i$, $\eta \in (0, 2/\lambda_{\max})$ and $\alpha = \eta/B$. Then, with $u_i = (1-\eta\lambda_i)^{2t}$,*

$$\mathbb{E}\mathcal{L}(\theta_t) = \sum_{i=1}^{d}\Big[\Delta_{0,i}u_i + \frac{\alpha a_i c_i}{4}\,\frac{2(1-u_i)}{2-\eta\lambda_i}\Big],$$

and if $\eta \le 1/(2\lambda_{\max})$, then $|\mathbb{E}\mathcal{L}(\theta_t) - \Phi_a(\eta t,\alpha)| \le \eta\lambda_{\max}\big(\mathcal{L}(\theta_0) + \alpha V_a\big)$ for every t.

Proof. The plan is to solve the recursion of the second moments and to compare it term by term with the flow. Squaring $\theta_{t+1,i} = (1-\eta\lambda_i)\theta_{t,i} - \eta a_i\zeta_{t,i}$ and taking expectations gives $\mathbb{E}\theta_{t+1,i}^2 = (1-\eta\lambda_i)^2\mathbb{E}\theta_{t,i}^2 + \eta^2 a_i^2 c_i/B$

(Chapter 11), so $\mathbb{E}\theta_{t,i}^2 = u_i\theta_{0,i}^2 + (\eta^2 a_i^2 c_i/B)(1-u_i)/(1-(1-\eta\lambda_i)^2)$; multiplying by $\frac{1}{2}h_i$ and using $1-(1-x)^2 = x(2-x)$ with $x = \eta\lambda_i$ gives the formula. Now let $x \le \frac{1}{2}$ and $\bar{u} = e^{-2xt}$. The inequalities $-x - x^2 \le \ln(1-x) \le -x$ give $\bar{u}e^{-2tx^2} \le u_i \le \bar{u}$, hence $0 \le \bar{u} - u_i \le 2tx^2\bar{u} = 2x\,(xt)e^{-2xt} \le x/e$. The term of Φ_a for coordinate i is $\Delta_{0,i}\bar{u} + \frac{1}{4}\alpha a_i c_i(1-\bar{u})$, and $\frac{2(1-u_i)}{2-x} - (1-\bar{u}) = \frac{x}{2-x}(1-u_i) + (\bar{u}-u_i) \in [0, 2x]$. So the two terms differ by at most $x\Delta_{0,i}/e + x\alpha a_i c_i/2$, and summing over i with $x \le \eta\lambda_{\max}$ gives the bound. ∎

Theorem 14.16 (The gap and the batch size). *In the noisy quadratic model let $\theta_{0,i} \neq 0$ and $c_i > 0$ for all i, let $\lambda_{\min}$ and $\lambda_{\max}$ be the extreme effective curvatures, and let $0 < \varepsilon < \mathcal{L}(\theta_0)$. Let $T_a(B)$ be the least t such that some $\eta \in (0, 2/\lambda_{\max})$ gives $\mathbb{E}\mathcal{L}(\theta_t) \le \varepsilon$, and $T_a(\infty)$ the same count for exact gradients ($c = 0$).*

(i) (Large batches.) $T_a(B)$ is nonincreasing in B and at least $T_a(\infty)$. Unless ε is one of the countably many numbers $\inf_\eta \mathcal{L}(\theta_t)$, $t \in \mathbb{N}$, for exact gradients, $T_a(B) = T_a(\infty)$ for all large B.

(ii) (Small batches.) As $B \to 0$, $B\,T_a(B)$ converges to the number of samples

$$n_a(\varepsilon) = \min\left\{\frac{\tau\,\mathcal{V}_a(\tau)}{\varepsilon - \mathcal{B}_a(\tau)} : \tau > 0,\ \mathcal{B}_a(\tau) < \varepsilon\right\}.$$

(iii) (High accuracy.) If $\lambda_{\min} < \lambda_{\max}$, then as $\varepsilon \to 0$,

$$T_a(\infty) \sim \frac{\ln(1/\varepsilon)}{4\,\operatorname{artanh}(\lambda_{\min}/\lambda_{\max})}, \qquad n_a(\varepsilon) \sim \frac{V_a}{8\lambda_{\min}} \cdot \frac{\ln(1/\varepsilon)}{\varepsilon}.$$

Proof. The plan is to read the large-batch limit off the exact formula of Lemma 14.15, to match the discrete dynamics to its flow from above and from below for the small-batch limit, and to estimate directly for (iii).

(i) The noise term of the formula is nonnegative and proportional to $1/B$, which gives the first two claims. Let $t^* = T_a(\infty)$; unless ε is exceptional, some η^* gives a deterministic loss below ε at step t^*, the noise term at (η^*, t^*) tends to 0 as $B \to \infty$, and $T_a(B) \le t^*$ for large B.

(ii) The minimum exists: the admissible τ form an interval $(\tau_\varepsilon, \infty)$, since $\mathcal{B}_a$ decreases continuously from $\mathcal{L}(\theta_0)$ to 0; the quotient is continuous there and tends to ∞ at both ends, because its denominator vanishes at τ_ε and $\mathcal{V}_a(\tau) \to V_a/4 > 0$.

Upper bound. Let τ be admissible and $0 < \alpha < (\varepsilon - \mathcal{B}_a(\tau))/\mathcal{V}_a(\tau)$, so that $\Phi_a(\tau, \alpha) < \varepsilon$. Put $\eta = \alpha B$ and $t = \lceil \tau/\eta \rceil$. As $B \to 0$, $\eta t \to \tau$, $\Phi_a(\eta t, \alpha) \to \Phi_a(\tau, \alpha)$, and the error bound of the lemma tends to 0; so for small B, $\mathbb{E}\mathcal{L}(\theta_t) \le \varepsilon$ and $B\,T_a(B) \le Bt \le \tau/\alpha + B$. Letting $B \to 0$, then $\alpha \uparrow (\varepsilon - \mathcal{B}_a(\tau))/\mathcal{V}_a(\tau)$, and minimizing over τ gives $\limsup_{B\to 0} B\,T_a(B) \le n_a(\varepsilon)$.

Lower bound. Let η_B attain the count $t_B = T_a(B)$, and $\alpha_B = \eta_B/B$. Since $2/(2-\eta\lambda_i) \ge 1$, the term of coordinate i in the formula of the lemma is a convex combination of $\Delta_{0,i}$ and a number at least $\frac{1}{4}\alpha a_i c_i$, hence at least $\min(\Delta_{0,i}, \frac{1}{4}\alpha a_i c_i)$; these bounds sum to more than ε for every α above some $\bar{\alpha}$, since their sum tends to $\mathcal{L}(\theta_0) > \varepsilon$. So $\alpha_B \le \bar{\alpha}$, $\eta_B \le \bar{\alpha}B$, and for small B

the lemma applies with $\tau_B = \eta_B t_B$:

$$\varepsilon \geq \mathbb{E}\mathcal{L}(\theta_{t_B}) \geq \Phi_a(\tau_B, \alpha_B) - \delta_B, \qquad \delta_B = \bar{\alpha} B \lambda_{\max}\big(\mathcal{L}(\theta_0) + \bar{\alpha} V_a\big) \to 0.$$

Hence $B\, t_B = \tau_B/\alpha_B \geq \tau_B \mathcal{V}_a(\tau_B)/(\varepsilon + \delta_B - \mathcal{B}_a(\tau_B)) \geq n_a(\varepsilon + \delta_B)$. It remains to show $\liminf_{\delta \downarrow 0} n_a(\varepsilon + \delta) \geq n_a(\varepsilon)$. Let $\delta_k \downarrow 0$ and let τ_k attain $n_a(\varepsilon + \delta_k) \leq n_a(\varepsilon)$. Along a subsequence with $\tau_k \to \infty$ or $\tau_k \to \tau_\varepsilon$ the quotients would tend to ∞, as above; so the τ_k have a limit point in $(\tau_\varepsilon, \infty)$, where the quotient is continuous in (τ, ε), and the limit of $n_a(\varepsilon + \delta_k)$ along it is at least $n_a(\varepsilon)$.

(iii) Let $\kappa_a = \lambda_{\max}/\lambda_{\min}$ and let $\rho = (\kappa_a - 1)/(\kappa_a + 1)$ be the rate of Theorem 14.14(i), so that $\ln(1/\rho) = 2\,\mathrm{artanh}(1/\kappa_a)$. As in that theorem, every η has $|1 - \eta\lambda_i| \geq \rho$ for some i with $\lambda_i \in \{\lambda_{\min}, \lambda_{\max}\}$, so the deterministic loss at step t is at least $m\rho^{2t}$, with m the least $\Delta_{0,i}$ among those i, while $\eta = 2/(\lambda_{\max} + \lambda_{\min})$ gives a loss at most $\mathcal{L}(\theta_0)\rho^{2t}$. Hence $T_a(\infty)$ lies between $\frac{\ln(m/\varepsilon)}{2\ln(1/\rho)}$ and $\frac{\ln(\mathcal{L}(\theta_0)/\varepsilon)}{2\ln(1/\rho)} + 1$. For n_a, let m_1 be the sum of the $\Delta_{0,i}$ with $\lambda_i = \lambda_{\min}$. Every admissible τ has $m_1 e^{-2\lambda_{\min}\tau} < \varepsilon$, that is, $\tau > \tau_1 = \ln(m_1/\varepsilon)/(2\lambda_{\min})$, and $\tau\mathcal{V}_a(\tau)$ increases in τ, so $n_a(\varepsilon) \geq \tau_1 \mathcal{V}_a(\tau_1)/\varepsilon$ with $\mathcal{V}_a(\tau_1) \to V_a/4$. Conversely, at $\tau = (\ln(\mathcal{L}(\theta_0)/\varepsilon) + s)/(2\lambda_{\min})$ with $s = \ln\ln(1/\varepsilon)$ we have $\mathcal{B}_a(\tau) \leq \varepsilon e^{-s}$ and $\mathcal{V}_a \leq V_a/4$, so $n_a(\varepsilon) \leq \frac{V_a}{8\lambda_{\min}\varepsilon} \frac{\ln(\mathcal{L}(\theta_0)/\varepsilon)+s}{1-e^{-s}}$. Both bounds are $\frac{V_a}{8\lambda_{\min}} \frac{\ln(1/\varepsilon)}{\varepsilon}(1 + o(1))$. ∎

By (14.2), coordinate i of the preconditioned method at time τ and noise level α is coordinate i of SGD at time $a_i\tau$ and noise level $a_i\alpha$, with the same number of samples τ/α. *In the small-batch limit a diagonal preconditioner does not create samples; it reallocates time among the coordinates at a fixed number of samples.* This helps when SGD's single time is wrong for some coordinates, and less than at large batch, where the preconditioner shortens the time itself.

Corollary 14.17 (The two ends of the gap). *Compare SGD with the preconditioner $a_i = h_i^{-p}$, $0 \leq p < 1$, under the hypotheses of Theorem 14.16, and write $\kappa = h_1/h_d > 1$. As $\varepsilon \to 0$,*

$$\frac{T_1(\infty)}{T_a(\infty)} \to \frac{\mathrm{artanh}(\kappa^{p-1})}{\mathrm{artanh}(\kappa^{-1})} \geq \kappa^p, \qquad \frac{n_1(\varepsilon)}{n_a(\varepsilon)} \to \frac{\sum_i c_i}{\sum_i (h_d/h_i)^p c_i} \in [1, \kappa^p].$$

For $h_i = 1/i$, $c_i = h_i$ and $0 < p < 1$ the small-batch limit is $pZ_d(1 + o(1))$ as $d \to \infty$, the large-batch limit at least d^p.

Proof. For SGD, $\lambda_{\min}/\lambda_{\max} = 1/\kappa$, $\lambda_{\min} = h_d$ and $V_1 = \sum_i c_i$; for the preconditioner, $\lambda_{\min}/\lambda_{\max} = \kappa^{p-1}$, $\lambda_{\min} = h_d^{1-p}$ and $V_a = \sum_i h_i^{-p} c_i$. Both limits follow from Theorem 14.16(iii); the first is at least κ^p because $\mathrm{artanh}(x)/x$ increases on $(0, 1)$, and $(h_d/h_i)^p \in [\kappa^{-p}, 1]$ gives the bounds on the second. For $h_i = 1/i$ and $c_i = h_i$ the numerator is Z_d and the denominator $d^{-p}\sum_{i \leq d} i^{p-1}$ tends to $1/p$, by comparison with $\int_0^d x^{p-1}\,\mathrm{d}x = d^p/p$. ∎

So the gap opens with the batch size, in the high-accuracy limit from $pZ_d \approx p \ln d$ to d^p, and the preconditioner moves the *critical batch size*

$n_a(\varepsilon)/T_a(\infty)$, where the two asymptotes cross, to the right by the ratio of the two ends (Chapter 11).[6] The small-batch end is close to 1 only if the noise sits mostly in the flattest directions: with a constant step size the model does not give parity at batch size one. Averaging the iterates changes this.

[6] McCandlish et al. (2018) fit steps-to-target curves of the form $S_{\min}(1 + B_{\text{crit}}/B)$; in our notation $S_{\min} = T_a(\infty)$ and $S_{\min}B_{\text{crit}} = n_a(\varepsilon)$.

HEURISTIC DERIVATION: AVERAGING. In the small-batch limit, coordinate i of preconditioned SGD in the time $\tau = \eta t$ follows, to leading order, the Ornstein–Uhlenbeck process $\mathrm{d}\theta_i = -\lambda_i\theta_i\,\mathrm{d}\tau + a_i\sqrt{\alpha c_i}\,\mathrm{d}W_\tau$, with $\alpha = \eta/B$. Its stationary autocovariance is $(\alpha a_i^2 c_i/(2\lambda_i))e^{-\lambda_i|s|}$, with integral $\alpha a_i^2 c_i/\lambda_i^2$ over $s \in \mathbb{R}$. The average of the iterates over a window of length $\tau_w \gg 1/\lambda_i$, which uses $n_w = \tau_w/\alpha$ samples, therefore has variance about $\alpha a_i^2 c_i/(\lambda_i^2\tau_w) = c_i/(h_i^2 n_w)$, and contributes $c_i/(2h_i n_w)$ to the loss. Neither a_i nor the step size appears: this is the variance of the best estimate of $\theta_i^\star = 0$ from n_w samples with noise variance c_i and curvature h_i. Once every direction has relaxed, averaged SGD and averaged preconditioned SGD reach the same loss with the same samples; the preconditioner helps only through the burn-in and the relaxation of the flattest directions, whose cost grows like $\ln(1/\varepsilon)$, against $1/\varepsilon$ for the variance. So the small-batch advantage of averaged iterates tends to one. ◇

14.5 Reading the theory

LOGARITHMIC LOSS CURVES. Theorem 14.11 predicts that on Zipfian data gradient descent removes the same fraction $\ln 10/Z_K$ of the initial loss in every decade of steps: against $\log t$ its loss is a straight line, flatter for larger vocabularies, with slopes 0.44, 0.31 and 0.24 per decade for $K = 10^2$, 10^3 and 10^4, and about 0.20 for 50,000 tokens (Figure 14.1, left). For $K = 10^4$ the asymptote is within 0.003 of the exact loss at $t = 10$ and $t = 100$, and falls below it as t approaches K, where $R_K(t)$ matters and the curve turns into the exponential tail of the rarest rows. Sign descent and Adam fall on one curve for every K, as Corollary 14.12 requires: runs with $K = 10^4$ and $K = 10^2$ agree with the single-row predictions to rounding error. Sign descent shrinks each error by η per step and stops at its floor; Adam with $\beta_2 = 0.999$, once its second moment stops growing, acts like heavy ball with a slowly varying diagonal preconditioner and converges linearly.

THE STAIRCASE. The right panel of Figure 14.1 is the frequency-bucket picture of Observation 14.3 in a model where it is exact: gradient descent fits the bucket of frequency near q around $t = 1/(\eta q)$, a decade after the previous one, and Adam fits all buckets together. Each bucket $[10^k, 10^{k+1})$ with $k \geq 1$ carries about the fraction $\ln 10/Z_K$ of the data, so the average loss is dominated by buckets that gradient descent has not reached yet. The disadvantage concentrates in the rows that serve rare tokens; the role of the normalization gains in Zhao et al. (2025) is a block effect.

THE GAP AND THE BATCH SIZE. In Figure 14.2, SGD needs 10063, 723, 220 and 184 steps at $B = 1, 16, 256$ and 4096, and the preconditioner $a_i = h_i^{-1/2}$ needs 3392, 214, 16 and 10. At small batches both follow their asymptotes $n_a(\varepsilon)/B$, with $n_1(\varepsilon) = 9987$ and $n_a(\varepsilon) = 3390$ samples; at large batches they

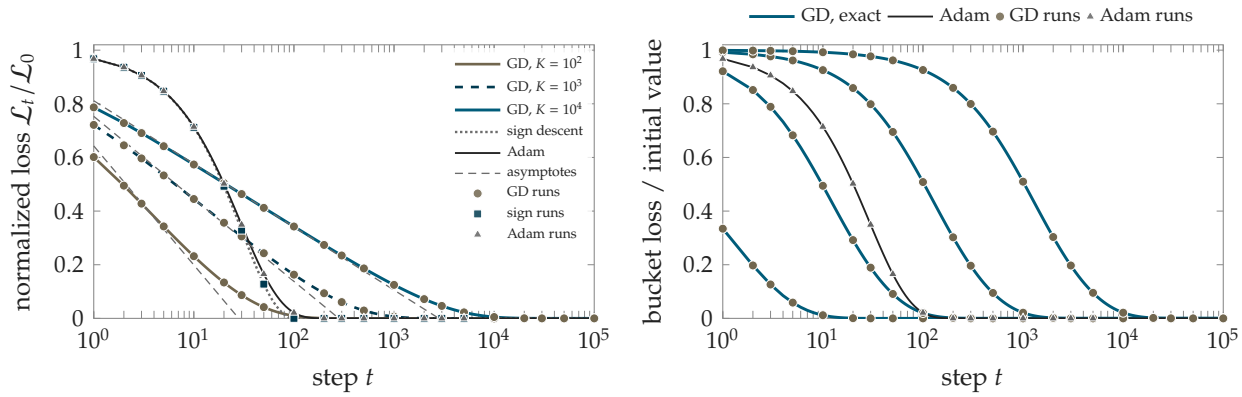


Figure 14.1. The square-loss table with Zipfian frequencies, every row starting at distance one from its target (rows of 16 Gaussian quantiles rescaled to unit norm). Left: curves are the exact loss of Theorem 14.11(i) for gradient descent with $\eta = 1/q_1$ ($K = 10^2, 10^3, 10^4$ from left to right), its asymptote (dashed), and the frequency-free predictions of Corollary 14.12: the exact sign-descent loss of one row and Adam on one row ($\eta = 0.005$; Adam $\beta = (0.9, 0.999)$, $\epsilon = 0$); points are runs on the full table ($K = 10^4$ for sign descent and Adam; gradient descent runs use one coordinate per row, which gives the same loss). Right: for $K = 10^4$, the buckets $j \in [1, 10), \ldots, [10^3, 10^4]$ (left to right), each relative to its initial loss; Adam's four coincide. Deterministic; script `14-transformers-zipf.py`.

level off near the deterministic counts 177 and 10. The advantage grows from 2.95 to 17.7, the critical batch size from 56 to 339; in the high-accuracy limit of Corollary 14.17 the ends would be 3.83 and 31.6. Monte Carlo runs agree with the exact counts to within 2% or one step. RMSProp, Adam with $\beta_1 = 0$ and $\beta_2 = 0.999$, needs 3402, 865, 212 and 54 steps at $B = 1, 4, 16, 64$, the counts of the $H^{1/2}$ preconditioner; at larger batches the gradient dominates its second moment, and the model stops applying.

WHAT CLOSES THE GAP AT SMALL BATCH. With a constant step size the small-batch advantage does not tend to one: it is 2.95 here and tends to 3.83, close to $pZ_d = 3.74$, as the target falls. It is near one in two cases (Figure 14.2, right). With isotropic noise it is 1.11, tending to $1 + p = 1.5$ (Exercise 14.6). With averaged iterates it is 0.99 at $B = 1$, where both methods need about 2400 steps against 10063 and 3392 for their last iterates, and it stays at one for lower targets (0.998 at 1%, 1.000 at 0.1%, where both need 133,300 steps, close to the $d/(2\varepsilon) \approx 133{,}600$ samples of the efficient estimator in the heuristic derivation). A linear decay of the step size to zero, which acts much like averaging (Chapter 12), gives 1.32 at $B = 1$ and 4. In every case the advantage grows with the batch size to the curvature factor, 18 to 22. So the model reproduces the trend of Observation 14.5, with parity at batch size one under averaging and most of the way to it under a decaying step, which recipes do use. This partly reconciles the batch-size observation with the curvature explanations (Section 14.7).

WHERE THE AGREEMENT MUST FAIL. The small-batch limit needs $\eta\lambda_{\max}$ small, which here holds already at $B = 1$ but fails for much stiffer problems,

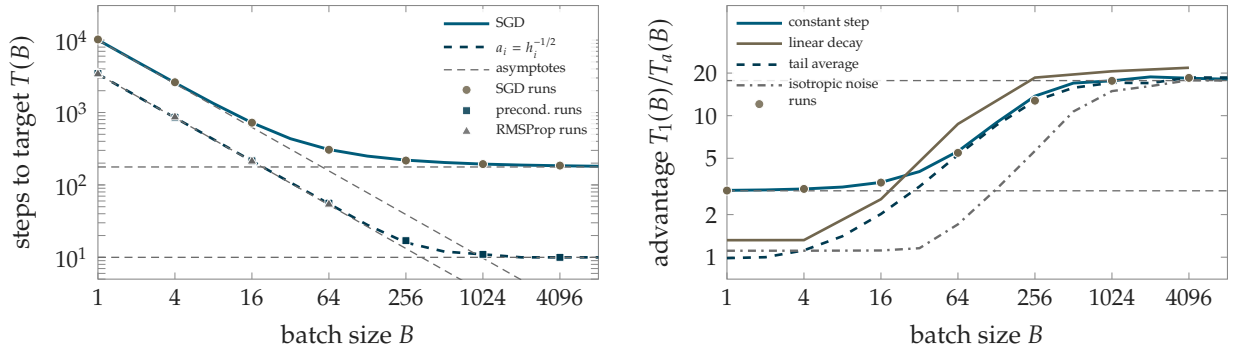


Figure 14.2. Steps to target in the noisy quadratic model ($d = 1000$, $h_i = 1/i$, $c_i = h_i$, $\theta_0 = \mathbf{1}$, target 5% of the initial loss), SGD against $a_i = h_i^{-1/2}$, each at its best constant step size. Left: exact counts (curves; 400 step sizes tried), the asymptotes of Theorem 14.16 (dashed), and Monte Carlo runs (points; 32 runs, first step at which the run average, smoothed over the last 2% of the steps, is below the target; unsmoothed, the stiff coordinates' fluctuations make SGD's first passage 5 to 12% early); triangles: RMSProp (16 runs, four learning rates) where its second moment is noise-dominated. Right: the exact advantage for a constant step (dashed: its two limits), a linear decay to zero, the average of the last iterates, and isotropic noise $c_i = \frac{1}{d}\sum_j h_j$. Script `14-transformers-batch.py`, seeds fixed.

and Adam is $H^{1/2}$ preconditioning only while noise dominates its second moment; with $\beta_1 > 0$ it is, at small batch, to first order the same method with the same step size, since Adam's momentum is an average (Chapter 5).

14.6 What the theory misses

WHAT WAS IDEALIZED AWAY. *Parameter sharing:* rare tokens' embedding rows feed layers that every token shares, so the table model is exact for a bigram model, and for the embedding and output rows of a network it captures only the frequency factor of their gradients. *Separate mechanisms:* heavy-tailed classes produce block heterogeneity, since embedding and output rows are blocks whose scales are frequencies. *Fixed curvature and noise:* in training both change, and the noise of language models is heavy-tailed (Zhang et al. 2020). *Adam's state:* with $\epsilon > 0$ the rarest rows fall back to momentum SGD with step size η/ϵ, and the lag of the preconditioner is ignored. *Schedules* enter only through the heuristic and the numerics.

WHERE IT BREAKS. (1) No single mechanism accounts for the gap: Kunstner et al. (2024) reproduce it with class imbalance in linear and vision models, Zhang et al. (2024) with block heterogeneity in quadratics, which accompanies it in networks; neither excludes the other, and these models cannot weigh them. (2) Measured gaps depend on tuning, as Wen et al. (2026) show for newer optimizers against AdamW, so the predicted sizes are hard to test. (3) The batch-size picture rests on one refereed study (Marek et al. 2025) and one preprint (Srećković et al. 2025); with constant steps the noisy quadratic model predicts a residual small-batch advantage of about $p \ln d$, and it gives parity only with averaging. (4) Outside the noise-dominated

regime Adam is not $H^{1/2}$ preconditioning; at full batch it is closer to sign descent, whose speed on a table is set by the spread of the targets within rows (Kunstner and Bach 2025). (5) The block theorem says that blockwise steps can gain the scale spread, not that Adam's per-coordinate steps realize the gain; they can also lose, for instance after a rotation (Chapter 2), as Xie et al. (2025) find in GPT-2. (6) For convolutional networks the explanation is negative: SGD does not lose if their blockwise spectra are homogeneous, which is observed (Zhang et al. 2024), not derived.

REPAIRS AND OPEN PROBLEMS. Blockwise optimizers keep the invariance that matters at a fraction of Adam's memory: Adalayer, with one second moment per layer (Zhao et al. 2025), and Adam-mini, with one learning rate per block (Zhang et al. 2025). Shampoo, SOAP and Muon do not depend on the bases of the rows and columns of each weight matrix (Chapters 3 and 7). The scaling laws of Kunstner and Bach (2025) extend Theorem 14.11 to every power-law exponent and to sign descent. Open are a model in which the three mechanisms act together, with shared parameters and sampled tokens, and a theory of Adam at small batch that includes the estimation noise of its second moment and the half-life rule of Marek et al. (2025) (Exercises 14.11 and 14.12).

14.7 Back in the wild

FIELD NOTES	
PRACTICE	Adam(W) for transformers, SGD with momentum for convnets; the gap survives full batch, grows with the batch size, and nearly vanishes at batch size one.
IDEALIZATION	A table with Zipfian row frequencies; a block-diagonal quadratic with blocks at different scales; the noisy quadratic model with $H^{1/2}$ for Adam.
PROVED	Weights are clocks for gradient descent and invisible to scale-invariant methods; on Zipf tables gradient descent removes $\ln 10/Z_K$ of the loss per decade and needs order $K^{1-\varepsilon}$ steps; blockwise steps replace κ by $\max_b \kappa_b$; at high accuracy the gap tends to at least κ^p at large batch and to a noise-weighted constant at small batch.
PREDICTS	Logarithmic loss curves and a bucket staircase for GD on heavy-tailed data; Adam's progress independent of the vocabulary size; a gap opening with the batch size.
MISSES	Parameter sharing, interacting mechanisms, Adam's ϵ and memory, schedules, the size of the gap at scale.
TAKEAWAY	Expect Adam's advantage to be largest with heavy-tailed vocabularies and large batches (proved in the idealizations, suggested for networks), and smallest at batch size one with averaging (derived and computed in the noisy quadratic model); blockwise learning rates should keep most of it (suggested).

In these idealizations the gap has nothing to do with attention: it is what a

scale invariance buys on a loss whose rows and blocks live at different scales, and Zipfian vocabularies and heterogeneous parameter blocks make such losses. An optimizer that keeps one statistic per block should keep most of the advantage, as Adalayer and Adam-mini report; that it keeps all of it is not proved. The batch size changes the answer: at large batch training is limited by curvature, and the preconditioner pays its full condition-number factor; at small batch it is limited by samples, and with averaging SGD is as fast as its preconditioned version in the noisy quadratic model, in agreement with Marek et al. (2025). The rule "SGD for convnets" agrees with the block theorem if convnets are block-homogeneous, which is observed, not derived. The scale invariance is the coordinatewise geometry of Chapters 2 and 6, its loss under rotation motivates the optimizers of Chapters 3 and 7, and averaging ties the batch-size theorem to schedules (Chapter 12); where each optimizer ends up is the subject of Chapter 15.

Exercises

14.1 CONSOLIDATION. (Other Zipf exponents.) Let $q_j \propto j^{-\alpha}$, rows of unit initial error, and $\eta = 1/q_1$. (a) Show that $\mathcal{L}_t/\mathcal{L}_0 = \sum_j j^{-\alpha}(1 - j^{-\alpha})^{2t}/\sum_j j^{-\alpha}$. (b) For $\alpha > 1$ and $K = \infty$, show that $\mathcal{L}_t/\mathcal{L}_0 \sim \Gamma(1 - \frac{1}{\alpha})(2t)^{-(\alpha-1)/\alpha}/(\alpha\zeta(\alpha))$ as $t \to \infty$. (c) For $\alpha < 1$, show that $\mathcal{L}_t/\mathcal{L}_0$ stays above $\frac{1}{2}$ until t is of order K^α. Conclude that only $\alpha = 1$ gives a fixed decrease per decade over a range of times that grows with K. *(Hint: in (b) compare the sum with* $\int_1^\infty x^{-\alpha} e^{-2tx^{-\alpha}}\,\mathrm{d}x$ *and substitute* $u = 2tx^{-\alpha}$*.)*

14.2 CONSOLIDATION. ★ (Zipfian targets.) A single row has targets $y_i = 1/(iZ_r)$, $i \le r$, starts at $w = 0$, and runs sign descent with a constant step size η. (a) Show that every coordinate with $y_i < \eta/2$ has error y_i at even steps and $\eta - y_i$ at odd steps, so that the loss at every odd step is at least $\frac{1}{8}\eta^2 \cdot \#\{i : y_i < \eta/2\}$. (b) Deduce that if the loss is at most ε times its initial value at two consecutive steps, and r is large compared with $1/\varepsilon$, then η is at most of order $\sqrt{\varepsilon/r}/Z_r$ and the number of steps at least of order $\sqrt{r/\varepsilon}$: the square-root law of Kunstner and Bach (2025). (c) Show that the loss at even steps alone falls below ε times its initial value within a number of steps of order $1/\varepsilon$, whatever r, and say what this means for measuring an oscillating method. *(Hint: the initial loss is* $\frac{1}{2}\|y\|_2^2 \approx \pi^2/(12Z_r^2)$*; at even steps the at most* $1/(\eta Z_r)$ *coordinates with* $y_i \ge \eta$ *have errors at most* η*.)*

14.3 CONSOLIDATION. (The softmax Hessian.) (a) Show that $\lambda_{\max}(\mathrm{diag}(s) - ss^\top) = \frac{1}{2}$ for a probability vector s if and only if s has two coordinates equal to $\frac{1}{2}$. (b) Deduce from the descent lemma (Lemma A.18) that gradient descent with every step size below $4/q_1$ decreases the loss of the softmax table at every step. (c) For $K = 1000$ Zipfian frequencies, check the interlacing of Proposition 14.13(ii) numerically; the condition number is 885.

14.4 CONSOLIDATION. (The bounds of Theorem 14.14(iii).) Give two-block quadratics with (a) $\kappa = \max_b \kappa_b$, (b) $\kappa = \omega$, (c) $\kappa = \omega \max_b \kappa_b$, and compute in each case the gain of blockwise step sizes.

14.5 VARIATION. (Momentum on the Zipf table.) Run heavy ball with momentum β and $\eta = 1/q_1$ on the Zipf table. (a) For rows with $\eta q_j \ll (1-\beta)^2$, show that the error contracts by about $1 - \eta q_j/(1-\beta)$ per step. (b) Deduce that for $t \gg 1/(1-\beta)$ heavy ball at step t is about gradient descent at step $t/(1-\beta)$: momentum shifts the curve by $\log_{10}(1/(1-\beta))$ decades without changing its slope. Check that for $K = 10^4$ and $\beta = 0.9$ heavy ball at steps 100 and 1000 matches gradient descent at steps 1000 and 10^4 to within 3%.

14.6 VARIATION. (Noise not proportional to curvature.) In Corollary 14.17 let $h_i = 1/i$ and $c_i = i^{-\vartheta}$. Show that the small-batch limit tends, as $d \to \infty$, to $1 + p/(1-\vartheta)$ for $0 \le \vartheta < 1$, and grows like $p \ln d$ for $\vartheta = 1$. Compare the value 1.5 for isotropic noise with the 1.11 of Figure 14.2, whose target is not small.

14.7 VARIATION. (Adam in the noisy quadratic model.) (a) Show that $\mathbb{E}[g_{t,i}^2 \mid \theta_t] = h_i^2 \theta_{t,i}^2 + c_i/B$. (b) Deduce that RMSProp with β_2 close to one behaves near θ_t like preconditioned SGD with $a_i = \sqrt{B}/(c_i + B h_i^2 \theta_{t,i}^2)^{1/2}$. (c) For $c_i = c_0 h_i$ and small B, show that RMSProp with learning rate η is the preconditioner $h_i^{-1/2}$ with step size $\eta\sqrt{B/c_0}$, and that keeping its small-batch dynamics fixed as B changes requires $\eta \propto \sqrt{B}$, the square-root rule of Malladi et al. (2022). (d) What does RMSProp become at large B?

14.8 SIMULATION. (The softmax table.) Train a softmax bigram table with $K = 1000$ Zipfian input frequencies and Zipfian conditional distributions, from $W = 0$, by gradient descent with $\eta = 1/q_1$ and by Adam with $\epsilon = 0$. Check that under gradient descent row j after t steps has the loss of the single-row problem after t steps of size q_j/q_1 (Proposition 14.13(i)), and that under Adam all input-frequency buckets coincide.

14.9 SIMULATION. (Rotating the table.) In the setting of Figure 14.1 with $K = 1000$, replace W by QW for a random orthogonal Q. Check that gradient descent is unchanged, that sign descent stalls near 0.15 of the initial loss, and that Adam's advantage in steps to 10^{-3} of it falls from about 11 to about 2.4; explain with Proposition 14.9 and the rotation equivariance of gradient descent.

14.10 SIMULATION. (Averaging and decay.) With the script of Figure 14.2, reproduce the advantages 2.97 (constant step), 1.32 (linear decay) and 0.99 (tail average) at $B = 1$. Lower the target to 1% and 0.1% of the initial loss, and check that the averaged advantage stays near one while the constant-step advantage rises toward 3.83 (3.24 and 3.43).

14.11 RESEARCH. (One model for three mechanisms.) Find a model in which heavy-tailed classes, block heterogeneity and the batch size act together, such as a softmax bigram model with a shared embedding of dimension $k \ll K$ trained on sampled tokens, and derive the advantage of a blockwise scale-invariant optimizer over SGD as a function of K, k and B. As far as we know none is solved.

14.12 RESEARCH. (Adam at batch size one.) Extend Theorem 14.16 to RMSProp with $1-\beta_2 \propto B$, the half-life rule of Marek et al. (2025), including the estimation noise of the second moment, and decide whether Adam keeps an advantage over SGD with a decaying schedule as $B \to 0$.

Notes and References

The Zipf table is the linear bigram model of Kunstner and Bach (2025), whose scaling laws for gradient descent and sign descent cover every power-law exponent. Theorem 14.11 isolates the input frequencies by giving every row the same initial error, and the constants then come out exactly: the slope $\ln 10/Z_K$ per decade and the offset $\ln 2 + \gamma$. Proposition 14.9 and Corollary 14.12 are elementary, and the interlacing in Proposition 14.13 is the classical one for a rank-one perturbation of a diagonal matrix.

The noisy quadratic model is due to Zhang et al. (2019), who found that preconditioning raises the critical batch size (McCandlish et al. 2018) and, unlike momentum, helps at small batch sizes too, and who also studied averaged iterates and learning-rate decay in it. Theorem 14.16(i)–(ii) are the two regimes of Chapter 11, and the averaging heuristic is the efficiency of averaged iterates (Chapter 11); we have not seen part (iii) and Corollary 14.17 stated elsewhere.

TRANSLATING THE LITERATURE. Kunstner and Bach (2025) write π_k for the token frequencies, our q_k, and α for the exponent of the power law, as in Exercise 14.1. In the noisy quadratic model of Zhang et al. (2019), as in Chapter 11, h_i and c_i are the curvatures and noise variances along the eigendirections, and their preconditioner H^p, which multiplies the gradient by H^{-p}, is our $a_i = h_i^{-p}$. McCandlish et al. (2018) write $S_{\min}$ and $E_{\min}$ for the least numbers of steps and of examples, our $T_a(\infty)$ and $n_a(\varepsilon)$, and $B_{\text{crit}} = E_{\min}/S_{\min}$ for the critical batch size. What Zhang et al. (2024) call Adam's coordinate-wise learning rates is, for deterministic gradients, our scale invariance: each coordinate's learning rate is set by that coordinate's own gradients.

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15 Where Steepest Descent Ends

Implicit bias and the margin of a norm

IN THE WILD — ADAPTIVE METHODS AND GENERALIZATION

IN THIS CHAPTER. When a model can fit its training data, training past zero training error keeps growing the weights, and the optimizer decides the direction in which they grow; adaptive methods have been reported to choose directions that generalize worse than those of SGD. For a linear classifier on separable data with the exponential loss we prove that normalized steepest descent in a norm maximizes the margin measured in that norm, with an explicit bound for every step-size sequence: gradient descent finds the ℓ_2 maximum-margin classifier, sign descent and Adam without memory the ℓ_∞ one, boosting with small steps the ℓ_1 one, orthogonalized updates the spectral one, and decoupled weight decay stops the march within $\lambda \log n$ of the maximum. On a dataset in the spirit of Wilson et al. the ℓ_∞ bias misclassifies every negative test example and the ℓ_2 bias none; on another the roles reverse. The idealization misses depth, noise, Adam's ϵ, finite training, and the evidence that well-tuned adaptive methods need not generalize worse.

15.1 In the wild: the solutions optimizers choose

Two optimizers that reach the same training loss need not reach the same network. Most of this book compares optimizers by speed; this chapter asks where they end, and whether the end point explains a difference in test error. The question became pressing when Wilson et al. (2017) compared AdaGrad, RMSProp and Adam with stochastic gradient descent and concluded that practitioners should reconsider the use of adaptive methods for training neural networks.[1]

> IN THE WILD · ResNets, He et al. (2016); Wilson et al. (2017); Keskar and Socher (2017). Train convolutional networks for image classification with SGD with momentum 0.9 and weight decay; for ResNets on ImageNet, a learning rate of 0.1 for batches of 256, divided by 10 whenever the error plateaus, and weight decay 10^{-4}. When test accuracy is what counts, prefer SGD to Adam, or start with Adam and switch to SGD later in training. Train transformers with Adam or AdamW.

The first sentence is a documented recipe; the second is partly folklore, resting on comparisons whose outcome depends on tuning; the third is the

[1] As everywhere in the book, "learning rate" (in the practice sections) and "step size" (in the theory) are synonyms.

subject of Chapter 14.

OBSERVATION 15.1 (Wilson et al. 2017). On image classification (CIFAR-10), character-level language modeling and parsing, the solutions of AdaGrad, RMSProp and Adam generalized worse than those of well-tuned SGD, sometimes even with a lower training loss. On a constructed linear problem trained with the square loss, the adaptive methods provably converge to a classifier that labels every new example positive, gradient descent to one that labels every new example correctly.

The constructed problem is the clean part of this evidence; Example 15.14 rebuilds it for margins. A second kind of evidence is that training past zero training error still changes the solution: for a convolutional network on CIFAR-10 trained with SGD and momentum, Soudry et al. (2018) saw the training loss decay roughly like $1/t$ and the last layer's norm grow like $\log t$, while the validation loss rose and the validation error kept improving.

OBSERVATION 15.2 (Choi et al. 2019; Loshchilov and Hutter 2019). With every hyperparameter tuned, ϵ and momentum included, adaptive methods matched or beat momentum SGD on every workload of Choi et al. (2019) (an unrefereed preprint), which they explain by inclusion: Adam with a large ϵ behaves like momentum SGD. With decoupled weight decay, Adam's test error on image classification nears SGD's (Loshchilov and Hutter 2019).

The third kind of evidence is geometric: AdamW without memory is the Frank–Wolfe method on an ℓ_∞ ball of radius $1/\lambda$ (Chapter 13; Xie and Li (2024)), so optimizers move in different geometries. The status, then: that adaptive methods generalize worse is observed in some settings and not in others; what is solid is that the solution reached at a given training loss depends on the optimizer. The question for the theory is: *when the training data can be fit exactly, where does steepest descent in a given norm end up, and can the answer explain why two optimizers with the same training loss generalize differently?*

15.2 The idealization: separable data and the margin of a norm

Assumption 15.3 (Separable linear classification). The examples $x_1, \dots, x_n$ lie in a finite-dimensional real inner-product space V: $\mathbb{R}^d$ with $\langle \theta, x \rangle = \theta^\top x$, or $\mathbb{R}^{K\times d}$ with $\langle W, X \rangle = \operatorname{tr}(W^\top X)$. The labels are $y_i \in \{-1, +1\}$. A norm $\|\cdot\|$ on V has dual norm $\|\cdot\|_*$, and $R = \max_i \|x_i\|_*$. The data are *linearly separable* through the origin: some θ has $y_i \langle \theta, x_i \rangle > 0$ for every i. The classifier is $x \mapsto \operatorname{sign}\langle \theta, x \rangle$, trained on the exponential loss

$$\mathcal{L}(\theta) = \frac{1}{n} \sum_{i=1}^{n} \exp\bigl(-y_i \langle \theta, x_i \rangle\bigr). \tag{15.1}$$

Only the signed examples $z_i = y_i x_i$ matter. We write $q_i(\theta) = \langle \theta, z_i \rangle$ for the margin of example i, positive when the example is classified correctly,

and $q_{\min}(\theta) = \min_i q_i(\theta)$.

Definition 15.4 (The margin of a norm). For $\theta \neq 0$ the *normalized margin* is $\bar{\gamma}(\theta) = q_{\min}(\theta)/\|\theta\|$. The *maximum margin* of the data in the norm $\|\cdot\|$, and the *max-margin set* of the maximum-margin classifiers, are

$$\gamma = \max_{\|u\| \le 1} \min_{i \le n} \langle u, z_i \rangle, \qquad \mathcal{U}^\star = \{u : \|u\| \le 1,\ q_{\min}(u) = \gamma\}.$$

When several norms are in play we write $\gamma_2, \gamma_\infty, \gamma_1, \gamma_{\mathrm{op}}$.

The maximum exists by compactness, $\gamma > 0$ exactly when the data are separable, every $u \in \mathcal{U}^\star$ then has $\|u\| = 1$, and $\bar{\gamma}(\theta) \le \gamma$ always. Since the dual-norm distance from x to the hyperplane $\{\langle \theta, \cdot \rangle = 0\}$ is $|\langle \theta, x \rangle|/\|\theta\|$ (Hölder's inequality and its equality case), the ℓ_2 maximum-margin classifier is the hard-margin support vector machine through the origin, and the ℓ_∞ and ℓ_1 ones maximize the ℓ_1 and ℓ_∞ distances to the nearest example.

The optimizers are the normalized steepest descent methods of Chapter 1,

$$\theta_{t+1} = \theta_t + \eta_t \,\mathrm{lmo}\big(\nabla \mathcal{L}(\theta_t)\big), \qquad \mathrm{lmo}(g) \in \arg\min_{\|u\| \le 1} \langle g, u \rangle, \tag{15.2}$$

with step sizes $\eta_t > 0$ that may depend on the iterates so far, and every choice of lmo where the minimizer is not unique; each step has length at most η_t, and $\langle g, \mathrm{lmo}(g) \rangle = -\|g\|_*$. In the ℓ_2 norm this is normalized gradient descent; in the ℓ_∞ norm, sign descent, which is Adam with $\beta_1 = \beta_2 = 0$ and $\epsilon = 0$ (Chapter 2);[2] in the ℓ_1 norm, greedy coordinate descent; in the spectral norm on matrices, the orthogonalized update of Muon without momentum (Chapter 3).

[2] With bias correction and $\epsilon = 0$, the first step of Adam is a sign step for all β_1, β_2 (Chapter 2).

DICTIONARY

IN THE THEORY	IN A TRAINING RUN
margin $q_i(\theta) = \langle \theta, z_i \rangle$	the correct logit minus the largest other logit
separable data, $\inf \mathcal{L} = 0$	training past zero training error
the norm of the step	ℓ_2 for SGD, ℓ_∞ for sign descent and Adam, spectral for Muon, ℓ_1 for boosting
$\bar{\gamma}(\theta)$ and its maximum γ	the worst logit gap per unit of weight norm
softmax weights $p_i(\theta)$	gradient weights, dominated by hard examples
distance $\tau_T = \sum_{t<T} \eta_t$	the area under the learning-rate schedule
the ball $\|\theta\| \le 1/\lambda$	decoupled weight decay λ (Chapter 13)

WHAT WAS THROWN AWAY, AND WHY. A linear model in fixed features, like a last layer on frozen features, is the smallest model in which the loss has no minimizer and the optimizer must choose a direction (homogeneous networks are in Section 15.4); separability is the interpolation regime of overparameterized classifiers. The exponential loss has the tail of the logistic and cross-entropy losses, gradient descent has the same limit on the

logistic loss (Ji and Telgarsky 2021; Soudry et al. 2018), and the exponential makes the gradient an exact softmax-weighted sum. Gradients are full-batch, without momentum or second moments (Section 15.4 collects what is known about those). The theorem describes the end of training, with explicit rates, and the rates turn out to be the crux.

One optimizer of this chapter is new to the book. With features given by weak classifiers, $x_{ij} = h_j(x_i) \in \{-1, 1\}$, boosting is ℓ_1 steepest descent on the exponential loss: AdaBoost (Freund and Schapire 1997) weights the examples by $p_i \propto e^{-y_i\langle\theta, x_i\rangle}$, picks the weak classifier with the largest weighted correlation $|\sum_i p_i y_i h_j(x_i)|$ with the labels, and moves its coefficient by an exact line search; gradient boosting shrinks each step by a small factor (Friedman 2001).[3]

[3] That AdaBoost is coordinate descent on the exponential loss was noticed by Breiman and by Friedman, Hastie and Tibshirani; Friedman (2001) generalized it to gradient boosting.

WHAT ONE HAD TO NOTICE. On separable data the loss has no minimizer, a method that drives it to zero sends $\|\theta_t\|$ to infinity, and only the direction can converge. The gradient is a softmax over the examples, concentrated on the smallest margins, so late in training the optimizer pushes in the direction that raises the smallest margins fastest *per unit of its own norm*. Soudry et al. (2018) made this precise for gradient descent, and Gunasekar et al. (2018) saw that the norm of the step decides which margin is maximized.

15.3 The theory: the margin of a norm

Throughout this section Assumption 15.3 holds, so $\gamma > 0$, and $\mathcal{P}_n = \{p \in \mathbb{R}^n : p_i \geq 0,\ \sum_i p_i = 1\}$ is the probability simplex.

15.3.1 Duality, and a gradient that never vanishes

Lemma 15.5 (Duality of margins). *Let $C = \operatorname{conv}\{z_1, \dots, z_n\}$. Then*

$$\gamma = \min_{p \in \mathcal{P}_n} \left\| \sum_{i=1}^{n} p_i z_i \right\|_* = \min_{c \in C} \|c\|_*,$$

the dual-norm distance from the origin to the convex hull of the signed examples; the data are separable if and only if $0 \notin C$. If $u^\star \in \mathcal{U}^\star$ and $p^\star$ attains the minimum, with $c^\star = \sum_i p_i^\star z_i$, then (i) $\langle u^\star, c^\star\rangle = \|c^\star\|_* = \gamma$, so $u^\star$ maximizes $\langle u, c^\star\rangle$ over the unit ball, that is, $u^\star$ is a value of $\operatorname{lmo}(-c^\star)$; and (ii) $p_i^\star = 0$ for every i with $q_i(u^\star) > \gamma$.

PROOF. The plan is to prove the two inequalities between γ and $\delta = \min_{c \in C}\|c\|_*$ separately, by Hölder's inequality and by the separating hyperplane theorem; δ is attained because C is compact. For $\|u\| \leq 1$ and $p \in \mathcal{P}_n$, a minimum is at most an average, and Hölder's inequality (Lemma A.3) bounds the average:

$$q_{\min}(u) \leq \sum_i p_i q_i(u) = \left\langle u, \sum_i p_i z_i \right\rangle \leq \left\| \sum_i p_i z_i \right\|_*,$$

so $\gamma \leq \delta$. Conversely, $\gamma \geq q_{\min}(0) = 0$, so let $\delta > 0$. The open set $\{v : \|v\|_* < \delta\}$ and C are disjoint convex sets, so some $\theta \neq 0$ and $b \in \mathbb{R}$ have

$\langle\theta, v\rangle \le b \le \langle\theta, c\rangle$ whenever $\|v\|_* < \delta$ and $c \in C$. The supremum of the left side is $\delta\|\theta\|$ by the bidual formula $\|\theta\| = \max_{\|v\|_*\le 1}\langle\theta, v\rangle$ (Lemma A.3), so $u = \theta/\|\theta\|$ has $q_i(u) \ge \delta$ for every i, and $\gamma \ge \delta$. Separability means $\gamma > 0$, that is, $\delta > 0$, that is, $0 \notin C$. For (i) and (ii), the same chain at $u^\star$ and $p^\star$ reads $\gamma = q_{\min}(u^\star) \le \sum_i p_i^\star q_i(u^\star) = \langle u^\star, c^\star\rangle \le \|c^\star\|_* = \gamma$. The second equality is (i). The first says that $\sum_i p_i^\star(q_i(u^\star) - \gamma) = 0$, a sum of nonnegative terms, which is (ii). ∎

The examples with $p_i^\star > 0$ are *support vectors*. By (i), if the optimizer's weights on the examples settled at $p^\star$, its step could point along a maximum-margin classifier; the rest of the section makes this quantitative.

Lemma 15.6 (The gradient is at least the margin times the loss). *For every $\theta \in V$, with the softmax weights $p_i(\theta) = e^{-q_i(\theta)}/\sum_j e^{-q_j(\theta)}$,*

$$\nabla\mathcal{L}(\theta) = -\mathcal{L}(\theta)\sum_{i=1}^{n} p_i(\theta)\, z_i, \qquad \gamma\,\mathcal{L}(\theta) \le \|\nabla\mathcal{L}(\theta)\|_* \le R\,\mathcal{L}(\theta).$$

In particular $\nabla\mathcal{L}$ never vanishes, $\mathcal{L}$ has no minimizer, and $\inf\mathcal{L} = 0$.

Proof. Differentiating (15.1) gives the formula, since $\frac{1}{n}\sum_j e^{-q_j} = \mathcal{L}$. As $p(\theta) \in \mathcal{P}_n$, Lemma 15.5 gives $\|\sum_i p_i z_i\|_* \ge \gamma$, and the triangle inequality gives $\|\sum_i p_i z_i\|_* \le R$. A convex differentiable function without stationary points has no minimizer, and $\mathcal{L}(su^\star) \le e^{-s\gamma} \to 0$ for $u^\star \in \mathcal{U}^\star$. ∎

We call $r(\theta) = \|\nabla\mathcal{L}(\theta)\|_*/\mathcal{L}(\theta) \in [\gamma, R]$ the *edge* at θ, as in boosting.[4] The lemma is the key inequality of the chapter: the gradient never becomes small faster than the loss.

[4] In boosting the edge of a weak classifier with weighted error ε is $1 - 2\varepsilon$, its weighted correlation with the labels.

Lemma 15.7 (One normalized step). *For every $\theta \in V$ and $\eta > 0$, with $u = \mathrm{lmo}(\nabla\mathcal{L}(\theta))$ and $r = r(\theta)$,*

$$\mathcal{L}(\theta + \eta u) \le \Big(\cosh(\eta R) - \frac{r}{R}\sinh(\eta R)\Big)\mathcal{L}(\theta) \le \exp\Big(-\eta r + \frac{\eta^2 R^2}{2}\Big)\mathcal{L}(\theta).$$

Proof. The plan is to write the ratio of the losses as an average of exponentials of bounded numbers with mean ηr, and to bound it by a chord and then by a Gaussian bound. Put $s_i = \langle u, z_i\rangle$; Hölder's inequality gives $|s_i| \le R$, and Lemma 15.6 with $\langle g, \mathrm{lmo}(g)\rangle = -\|g\|_*$ gives $\sum_i p_i(\theta)s_i = -\langle u, \nabla\mathcal{L}(\theta)\rangle/\mathcal{L}(\theta) = r$. Since $q_i(\theta + \eta u) = q_i(\theta) + \eta s_i$, the ratio $\mathcal{L}(\theta + \eta u)/\mathcal{L}(\theta)$ is $\sum_i p_i(\theta)e^{-\eta s_i}$. On $[-R, R]$ the convex function $s \mapsto e^{-\eta s}$ lies below its chord, $e^{-\eta s} \le \cosh(\eta R) - (s/R)\sinh(\eta R)$, and averaging gives the first inequality. For the second, fix $\rho \in [-1, 1]$ and put $D = \cosh x - \rho\sinh x \ge e^{-x} > 0$ and $N = \sinh x - \rho\cosh x$, so that $D' = N$ and $N' = D$. The function $f(x) = \log D(x) + \rho x$ has $f(0) = 0$, $f'(0) = N(0)/D(0) + \rho = 0$, and $f'' = (D^2 - N^2)/D^2 \le 1$, so $f(x) \le x^2/2$ for $x \ge 0$ by Taylor's theorem. With $x = \eta R$ and $\rho = r/R$ this is the second inequality. ∎

The second bound is Hoeffding's lemma, the step behind the Azuma–Hoeffding inequality of Theorem A.24;[5] the first is attained when every $s_i = \pm R$, as in AdaBoost with weak classifiers valued in $\{-1, 1\}$.

[5] For a random variable X with values in $[a, b]$, Hoeffding's lemma says $\log \mathbb{E}e^{\kappa(X-\mathbb{E}X)} \le \kappa^2(b-a)^2/8$; here X takes the value $-\eta s_i$ with probability p_i.

15.3.2 The theorem

The proof follows the *smoothed margin* $\psi(\theta) = -\log(n\mathcal{L}(\theta)) = -\log \sum_i e^{-q_i(\theta)}$, a soft minimum of the margins: since $e^{-q_{\min}} \le \sum_i e^{-q_i} \le n e^{-q_{\min}}$,

$$q_{\min}(\theta) - \log n \le \psi(\theta) \le q_{\min}(\theta). \tag{15.3}$$

We write $\tau_T = \sum_{t<T} \eta_t$ for the distance travelled in T steps.

Theorem 15.8 (The margin of a norm; after Gunasekar et al. 2018). *Assume Assumption 15.3, let $\theta_0 = 0$, and let θ_t follow (15.2) with step sizes $\eta_t > 0$, which may depend on $\theta_0, \dots, \theta_t$, and every choice of* lmo. *Then for every $T \ge 1$:*

(i) $\mathcal{L}(\theta_T) \le \exp\left(-\gamma\tau_T + \frac{R^2}{2}\sum_{t<T}\eta_t^2\right)$ *and* $\|\theta_T\| \le \tau_T$;

(ii) if $\gamma\tau_T > \log n + \frac{R^2}{2}\sum_{t<T}\eta_t^2$, *then θ_T classifies every training example correctly and*

$$\gamma - \frac{\log n + \frac{R^2}{2}\sum_{t<T}\eta_t^2}{\tau_T} \le \bar{\gamma}(\theta_T) \le \gamma. \tag{15.4}$$

Proof. The plan is to show that the smoothed margin grows by at least $\gamma\eta_t - R^2\eta_t^2/2$ per step while the norm grows by at most η_t. By Lemma 15.7 at θ_t and $r(\theta_t) \ge \gamma$ (Lemma 15.6),

$$\psi(\theta_{t+1}) - \psi(\theta_t) = -\log\frac{\mathcal{L}(\theta_{t+1})}{\mathcal{L}(\theta_t)} \ge \eta_t r(\theta_t) - \frac{R^2\eta_t^2}{2} \ge \gamma\eta_t - \frac{R^2\eta_t^2}{2}.$$

Summing over $t < T$, with $\psi(0) = -\log n$,

$$\psi(\theta_T) \ge \gamma\tau_T - \frac{R^2}{2}\sum_{t<T}\eta_t^2 - \log n, \tag{15.5}$$

which is the bound on the loss in (i), as $n\mathcal{L} = e^{-\psi}$; the bound on the norm is the triangle inequality. For (ii), the right side A_T of (15.5) is positive, and (15.3) gives $q_{\min}(\theta_T) \ge A_T > 0$: every example is classified correctly and $\theta_T \ne 0$. Dividing by $\|\theta_T\| \le \tau_T$ gives $\bar{\gamma}(\theta_T) \ge A_T/\tau_T$. The upper bound holds because $\theta_T/\|\theta_T\|$ is in the unit ball. ∎

The bound has two error terms. The term $\log n/\tau_T$ is the price of starting from uniform weights on the examples, and it is all that remains when the steps shrink at a fixed distance τ_T, in the limit of a flow. The term $R^2\sum_t \eta_t^2/(2\tau_T)$ is the price of finite steps.

Corollary 15.9 (Step sizes). *In the setting of Theorem 15.8:*

(i) If $\eta_t \to 0$ and $\sum_t \eta_t = \infty$, then $\bar{\gamma}(\theta_t) \to \gamma$, every limit point of

$\theta_t/\|\theta_t\|$ *lies in* $\mathcal{U}^\star$, *and if* $\mathcal{U}^\star = \{u^\star\}$, *then* $\theta_t/\|\theta_t\| \to u^\star$.

(ii) *For* $\eta_t = c/(R\sqrt{t+1})$: $\gamma - \bar\gamma(\theta_T) \le R\big(\log n + \frac{c^2}{2}(1 + \log T)\big)/\big(2c(\sqrt{T+1}-1)\big)$.

(iii) *For a constant step* η: $\gamma - \bar\gamma(\theta_T) \le \log n/(\eta T) + R^2\eta/2$, *so* $\liminf_t \bar\gamma(\theta_t) \ge \gamma - R^2\eta/2$ *if* $R^2\eta < 2\gamma$; *for a known horizon,* $\eta = \sqrt{2\log n/T}/R$ *gives* $\gamma - \bar\gamma(\theta_T) \le R\sqrt{2\log n/T}$.

In (ii) and (iii) the bounds hold whenever their right sides are below γ.

PROOF. The plan is to show that the error term of (15.4) vanishes and to use compactness for the directions. (i) Given $\varepsilon > 0$, choose t_0 with $\eta_t \le \varepsilon$ for $t \ge t_0$; then $\sum_{t<T}\eta_t^2 \le \sum_{t<t_0}\eta_t^2 + \varepsilon\tau_T$, and since $\tau_T \to \infty$ the error term has limit superior at most $R^2\varepsilon/2$. The directions $u_t = \theta_t/\|\theta_t\|$ lie on the compact unit sphere, and if $u_{t_k} \to \bar u$, then $q_{\min}(\bar u) = \lim_k \bar\gamma(\theta_{t_k}) = \gamma$ by continuity, so $\bar u \in \mathcal{U}^\star$; if $\mathcal{U}^\star$ is a single point, every subsequence has a further subsequence converging to it. (ii) Insert $\sum_{k\le T} k^{-1/2} \ge 2(\sqrt{T+1}-1)$ and $\sum_{k\le T} k^{-1} \le 1 + \log T$ in (15.4). (iii) Here $\tau_T = \eta T$ and $\sum_t \eta_t^2 = \eta^2 T$, and the stated η equalizes the two terms. ∎

A constant step is guaranteed only to come within $R^2\eta/2$; whether it reaches the maximum is left open by the theorem, and Section 15.5 shows that in our experiments it does.

REMARK 15.10 (Hedge against a best response). The weights evolve by multiplicative weights, $p_{t+1,i} \propto p_{t,i}e^{-\eta_t\langle u_t, z_i\rangle}$, against the best response $u_t = \mathrm{lmo}(\nabla\mathcal{L}(\theta_t))$ in the game $\max_u \min_p \langle u, \sum_i p_i z_i\rangle$ of value γ: (15.5) is Hedge's regret bound, in the game of Freund and Schapire (1999).

Gunasekar et al. (2018) proved, for every norm and initialization, that the normalized margin of the unnormalized method $\theta_{t+1} = \theta_t - \eta_t\nabla\mathcal{L}(\theta_t)^\sharp$ (Chapter 1) converges to γ for steps $\eta_t \le \min\{\eta_+, 1/(R^2\mathcal{L}(\theta_t))\}$, η_+ a constant.[6] A constant step from $\theta_0 = 0$ follows from Theorem 15.8, with a rate.

[6] So reads the arXiv version; the proceedings print max for min. The condition has the same form for a summed or an averaged loss, and a constant $\eta \le 1/R^2$ from $\theta_0 = 0$ meets it, since the loss stays below 1 (see the proof below).

Corollary 15.11 (Unnormalized steepest descent). *Assume Assumption 15.3, let* $0 < \eta \le 1/R^2$, $\theta_0 = 0$ *and* $\theta_{t+1} = \theta_t - \eta\nabla\mathcal{L}(\theta_t)^\sharp$. *Then* $\mathcal{L}(\theta_T) \le 1/(1 + \eta\gamma^2 T/2)$, *and whenever the right side is below* γ,

$$\gamma - \bar\gamma(\theta_T) \le R\,\frac{\log n + \frac{1}{2} + R^2/\gamma^2}{\log(1 + \eta\gamma^2 T/2)};$$

in particular $\bar\gamma(\theta_T) \to \gamma$, *and the limit points of* $\theta_T/\|\theta_T\|$ *lie in* $\mathcal{U}^\star$.

PROOF. The plan is to read the method as (15.2) with the adaptive step $\eta_t = \eta\|\nabla\mathcal{L}(\theta_t)\|_*$, which is what $g^\sharp = -\|g\|_*\,\mathrm{lmo}(g)$ gives, and to control τ_T and $\sum_t\eta_t^2$ through the loss $\mathcal{L}_t = \mathcal{L}(\theta_t)$. With $r_t = r(\theta_t)$ we have $\eta_t = \eta r_t\mathcal{L}_t$, and Lemma 15.7 gives $\mathcal{L}_{t+1} \le \mathcal{L}_t\exp\big(-\eta r_t^2\mathcal{L}_t(1 - \eta R^2\mathcal{L}_t/2)\big)$. If $\mathcal{L}_t \le 1$, as at $t = 0$, the inner factor is at least $\frac{1}{2}$, so $\mathcal{L}_{t+1} \le \mathcal{L}_t e^{-\eta\gamma^2\mathcal{L}_t/2} \le \mathcal{L}_t$; by induction this holds for all t, and $e^{-x} \le 1/(1+x)$ gives $1/\mathcal{L}_{t+1} \ge 1/\mathcal{L}_t + \eta\gamma^2/2$, hence

the bound on $\mathcal{L}_T$. Since $q_i(\theta) \le R\|\theta\|$, we have $\mathcal{L}(\theta) \ge e^{-R\|\theta\|}$, and so $\tau_T \ge \|\theta_T\| \ge \log(1+\eta\gamma^2 T/2)/R$. Since $\eta_t \le \eta R \mathcal{L}_t$ and $\eta R^2 \le 1$,

$$\frac{R^2}{2}\sum_{t\ge 0}\eta_t^2 \le \frac{\eta^2 R^4}{2}\sum_{t\ge 0}\frac{1}{(1+\eta\gamma^2 t/2)^2} \le \frac{\eta^2 R^4}{2}\Big(1+\frac{2}{\eta\gamma^2}\Big) \le \frac{1}{2}+\frac{R^2}{\gamma^2}.$$

Inserting both bounds in (15.4) gives the rate; the limit points are handled as in Corollary 15.9. ■

The rate is logarithmic, as for the direction of gradient descent (Soudry et al. 2018): an unnormalized step shrinks with the loss, so the distance travelled grows only like $\log t$, while a normalized method with a constant step travels a distance proportional to t. This is the margin side of the book's theme of normalized updates (Chapter 1).

15.3.3 Four geometries

Corollary 15.12 (What each optimizer maximizes). *Let $\theta_0 = 0$, $\eta_t \to 0$ and $\sum_t \eta_t = \infty$. (i)* Normalized gradient descent ($\mathrm{lmo}(g) = -g/\|g\|_2$, $R = \max_i \|x_i\|_2$) *converges in direction to the hard-margin support vector machine, the unique element of $\mathcal{U}^\star$. (ii)* Sign descent ($\mathrm{lmo}(g) = -\operatorname{sign}(g)$, *every value in $[-1,1]$ allowed where $g_j = 0$; $R = \max_i \|x_i\|_1$) has ℓ_∞ margin tending to γ_∞, the value of the linear program $\max\{s : \min_i \langle u, z_i\rangle \ge s,\ \|u\|_\infty \le 1\}$. (iii)* Greedy coordinate descent ($\mathrm{lmo}(g) = -\operatorname{sign}(g_j)e_j$, $j \in \arg\max_j |g_j|$; $R = \max_i \|x_i\|_\infty$), *which is boosting with small steps, has ℓ_1 margin tending to γ_1, the value of a linear program. (iv)* Orthogonalized updates *on $\mathbb{R}^{K\times d}$* ($\mathrm{lmo}(G) = -\operatorname{polar}(G)$, $R = \max_i \|X_i\|_{\mathrm{nuc}}$) *have spectral margin tending to $\gamma_{\mathrm{op}} = \max\{\min_i \langle W, Z_i\rangle : \|W\|_{\mathrm{op}} \le 1\}$, the value of a semidefinite program. In (ii)–(iv) limit points of the directions are maximum-margin classifiers, and the directions converge when $\mathcal{U}^\star$ is a single point.*

Proof. The oracles and dual norms are computed in Chapters 1 to 3 (the spectral one through Theorem A.7), so all but the uniqueness in (i) is Corollary 15.9(i). If $u \ne u'$ were both in $\mathcal{U}^\star$ for ℓ_2, then $v = (u+u')/2$ would have $q_{\min}(v) \ge \gamma$ and $\|v\|_2 < 1$, so $v/\|v\|_2$ would have a margin above γ. ■

A multiclass classifier $W \in \mathbb{R}^{K\times d}$ with rows w_c fits case (iv): its margin on example i is $\min_{c\ne y_i}(w_{y_i} - w_c)^\top x_i = \min_{c\ne y_i}\langle W, Z_{ic}\rangle$ with $Z_{ic} = (e_{y_i} - e_c)x_i^\top$, of nuclear norm $\sqrt{2}\|x_i\|_2$, and the multiclass exponential loss $\frac{1}{n(K-1)}\sum_i \sum_{c\ne y_i} e^{-\langle W, Z_{ic}\rangle}$, with the tail of the cross-entropy, is (15.1) for the signed examples Z_{ic}. In practice Muon trains hidden layers and Adam the output layer (Chapter 3): a model, not a description.

Remark 15.13 (Sparse and dense maximum margins). For $V = \mathbb{R}^d$ the ℓ_∞ problem is the linear program $\max s$ subject to $\langle u, z_i\rangle \ge s$ and $-1 \le u_j \le 1$, whose feasible set contains no line; so it has an optimal vertex, where $d+1$ independent constraints are active, at most n of them margin constraints. Hence some ℓ_∞-maximum-margin classifier has at least $d+1-n$ coordinates equal to ± 1. Writing $u = u^+ - u^-$ with $u^\pm \ge 0$ and $\sum_j (u_j^+ + u_j^-) \le 1$, the

same count shows that some ℓ_1-maximum-margin classifier has at most n nonzero coordinates. If $\mathcal{U}^\star$ is a single point, it is that vertex.

When $d \gg n$ and the maxima are unique, sign descent thus ends with all but at most $n-1$ weights at ± 1:[7] the ℓ_∞ geometry charges nothing for full weight, so every feature that helps the worst-case margin gets it. Coordinate descent ends with at most n features, gradient descent in the span of the support vectors ($u^\star = c^\star/\|c^\star\|_2$ by Lemma 15.5): three priors on the interpolating solution.

[7] That a linear program over a polyhedron without lines attains its optimum at a vertex is the fundamental theorem of linear programming.

EXAMPLE 15.14 (A dataset on which the ℓ_∞ bias fails). In the spirit of the construction of Wilson et al. (2017), take $n_+ \geq 1$ positive and $n_- \geq 1$ negative examples. Let $e_{(i)}$ be the indicator vector of the features owned by example i, one for a positive and five for a negative example, disjoint across examples, and let

$$x_i = \left(y_i,\ 1,\ 1,\ e_{(i)}\right) \in \mathbb{R}^{3+n_++5n_-}.$$

The first feature is the label, the next two are constant, and the owned features let each example be memorized. A new example with label y has the shared features $(y, 1, 1)$ and features of its own that no training example has; every classifier trained on the sample puts weight zero on those (steepest descent from zero, with $\operatorname{sign} 0 = 0$, never moves a coordinate in which every gradient vanishes), so it scores the new example by $y\theta_1 + \theta_2 + \theta_3$. New labels are $+1$ with probability $p \in (0,1)$.

Proposition 15.15 (The two biases on the example). *On the data of Example 15.14, for every $n_+, n_- \geq 1$:*

(i) The ℓ_∞ max-margin set is the single point u_∞ with $u_1 = u_2 = u_3 = 1$, $u = 1$ on the features of positive examples and $u = -1$ on those of negative ones, and $\gamma_\infty = 4$. It scores every new example by $y + 2 > 0$: its test error is $1 - p$.

(ii) Let $P = 3 + 1/n_+$, $Q = 3 + 5/n_-$, $A_+ = (Q+1)/(PQ-1)$, $A_- = (P+1)/(PQ-1)$. The ℓ_2 max-margin classifier is the direction of $\theta_2 = \sum_i \alpha_i z_i$ with $\alpha_i = A_+/n_+$ for positive and A_-/n_- for negative examples; every example is a support vector, and θ_2 is the minimum-norm solution of $\langle \theta, x_i \rangle = y_i$ $(i \leq n)$. It scores new positives by $3A_+ - A_- > 0$ and new negatives by $A_+ - 3A_- < 0$: its test error is 0.

(iii) The ℓ_1 max-margin set is $\{e_1\}$, $\gamma_1 = 1$, and its test error is 0.

Hence from $\theta_0 = 0$ with $\eta_t \to 0$ and $\sum_t \eta_t = \infty$, the test error of sign descent tends to $1-p$ and those of normalized gradient descent and coordinate descent tend to 0. If $n_+ > n_-$, sign descent with all step sizes stays on the ray $\{s u_\infty : s > 0\}$ from the first step, with test error $1-p$ at every step.

PROOF. The plan is to bound the ℓ_∞ and ℓ_1 margins directly and to verify the KKT conditions for ℓ_2. (i) If $\|u\|_\infty \leq 1$, a positive example has $q_i(u) = u_1 + u_2 + u_3 + u_{(i)} \leq 4$, with equality only if all four equal 1; then a negative example has $q_j(u) = u_1 - u_2 - u_3 - \sum_{k \in (j)} u_k = -1 - \sum_{k \in (j)} u_k$, which is at least 4 only if its five features carry -1. Since u_∞ gives every example the margin 4, $\mathcal{U}^\star = \{u_\infty\}$.

(ii) The direction of the ℓ_2 classifier is that of the minimizer of $\frac{1}{2}\|\theta\|_2^2$ subject to $q_i(\theta) \geq 1$, and the KKT conditions $\theta = \sum_i \alpha_i z_i$, $\alpha \geq 0$, $\alpha_i(q_i(\theta) - 1) = 0$, $q_i(\theta) \geq 1$ are sufficient for this convex problem. With the stated α, θ has the entries $a = A_+ + A_-$ in the first feature, $c = A_+ - A_-$ in the second and third, A_+/n_+ in the features of positive examples and $-A_-/n_-$ in those of negative ones, so the margins of positive and of negative examples are

$$a + 2c + A_+/n_+ = PA_+ - A_- \qquad \text{and} \qquad a - 2c + 5A_-/n_- = QA_- - A_+.$$

The stated $A_\pm$ make both equal to 1 and are positive, as $P, Q > 3$. So the KKT conditions hold with all constraints active, θ_2 is the unique minimizer by strict convexity, and it also minimizes the norm subject to $q_i(\theta) = 1$, that is, to $\langle \theta, x_i \rangle = y_i$. New examples get $a+2c = 3A_+ - A_- = (3Q+2-P)/(PQ-1) > 0$, as $3Q > 9 > P$, and $-a + 2c = A_+ - 3A_- = (Q - 3P - 2)/(PQ - 1) < 0$, as $Q \leq 8 < 3P + 2$.

(iii) The average $\bar{z} = \frac{1}{n}\sum_i z_i$ has first entry 1 and all others of absolute value below 1, so $\gamma_1 \leq \|\bar{z}\|_\infty = 1$ by Lemma 15.5, and e_1 attains 1. For $u \in \mathcal{U}^\star$, a positive i and a negative j give $2 \leq q_i(u) + q_j(u) = 2u_1 + u_{(i)} - \sum_{k\in(j)} u_k \leq 2|u_1| + |u_{(i)}| + \sum_{k\in(j)}|u_k| \leq 2\|u\|_1 \leq 2$, with equality only if $u = e_1$.

The limits follow from Corollary 15.12, because the score of a new example is continuous in the direction and does not vanish at the three limits. Finally, if $n_+ > n_-$ and $\theta = su_\infty$, all margins equal $4s$, the weights p_i are uniform, and $\nabla\mathcal{L}(\theta) = -\mathcal{L}(\theta)\bar{z}$ with $\operatorname{sign}(\bar{z}) = u_\infty$, since the second and third entries of $\bar{z}$ are $(n_+ - n_-)/n > 0$; so sign descent moves along u_∞. ∎

The ℓ_∞ geometry gives the constant features full weight because they help the positive examples, which own one feature each, keep up with the negative ones, which own five; on new examples they outvote the label. With a dense ground truth the roles reverse (Exercise 15.6). Wilson et al.'s adaptive methods, trained on the square loss, converge to a multiple of $\operatorname{sign}(X^\top y)$, which for $n_+ > n_-$ is u_∞, and gradient descent to the minimum-norm interpolant θ_2: here the square and exponential losses agree.

15.4 The theory: weight decay, adaptive methods and networks

15.4.1 Weight decay stops the march

Normalized steepest descent with decoupled weight decay, $\theta_{t+1} = (1 - \eta_t\lambda)\theta_t + \eta_t \operatorname{lmo}(\nabla\mathcal{L}(\theta_t))$, is the Frank–Wolfe method on the ball $\{\|\theta\| \leq 1/\lambda\}$; with $\eta_t\lambda = 2/(t+2)$ it minimizes a convex objective that is smooth there (Chapter 13), such as $\mathcal{L}$. So the limit points of AdamW without memory minimize $\mathcal{L}$ over the ℓ_∞ ball of radius $1/\lambda$.

Proposition 15.16 (The end point at radius B). *Assume Assumption 15.3, let $B > 0$, and let θ_B minimize $\mathcal{L}$ over $\{\|\theta\| \leq B\}$. Then $\|\theta_B\| = B$ and $\mathcal{L}(\theta_B) \leq e^{-\gamma B}$, and if $B > \log(n)/\gamma$, then $\gamma - \log(n)/B \leq \bar{\gamma}(\theta_B) \leq \gamma$.*

PROOF. If $\|\theta_B\| < B$, then θ_B would be a local, hence by convexity a global, minimizer of $\mathcal{L}$, which does not exist (Lemma 15.6). For $u^\star \in \mathcal{U}^\star$, $\mathcal{L}(\theta_B) \le \mathcal{L}(Bu^\star) \le e^{-\gamma B}$. Then $e^{-q_{\min}(\theta_B)} \le n\mathcal{L}(\theta_B) \le ne^{-\gamma B}$, so $q_{\min}(\theta_B) \ge \gamma B - \log n$; divide by $\|\theta_B\| = B$. ■

So decoupled weight decay replaces the endless march along a direction by a point at radius $1/\lambda$, within $\lambda \log n$ of the maximum margin. That constrained and regularized minimizers approach the maximum-margin direction as the constraint is relaxed is the *regularization path* picture of Rosset et al. (2004); the proposition gives it a rate and covers the ℓ_2 penalty (Exercise 15.7).

15.4.2 Adam, AdamW and Muon

For the optimizers with memory we state known results without proof. *Momentum*: heavy-ball momentum, with full or stochastic gradients, leaves the limit at the ℓ_2 maximum margin (Wang et al. 2022); averaging changes the speed, not the norm. *Adam*: on separable data with the logistic loss and decaying steps, and with the ϵ of the denominator neglected, Adam converges in direction to the ℓ_∞ maximum margin, in polynomial time for a general class of steps (Zhang et al. 2024). With a fixed $\epsilon > 0$ the limit changes:[8] the gradients tend to zero, $\sqrt{\hat{v}_t}$ falls below ϵ, Adam becomes gradient descent with momentum, and RMSProp and Adam tend to the ℓ_2 maximum margin (to KKT points for homogeneous networks), while AdaGrad does not (Wang et al. 2021, 2022). *AdamW*: if it converges with non-increasing steps whose sum diverges, its limit is a KKT point of minimizing the loss subject to $\|\theta\|_\infty \le 1/\lambda$ (Xie and Li 2024), for linear models the end point of Proposition 15.16. *Spectral descent and Muon*: on multiclass separable data with the cross-entropy loss they converge to the spectral-norm maximum-margin classifier (Fan et al. 2025).

[8] A rough estimate from Lemma 15.6: the gradient entries are of the order of the loss, so with $\epsilon = 10^{-8}$ the switch comes once the training loss is near 10^{-8}.

15.4.3 Homogeneous networks

A ReLU network without biases, of depth a, satisfies $f(x; c\theta) = c^a f(x; \theta)$ for $c > 0$. With $q_i(\theta) = y_i f(x_i; \theta)$, define $\mathcal{L}$ and ψ as before. The witness $u^\star$ of the linear case is no longer available, but Euler's identity for homogeneous functions takes its place.

Proposition 15.17 (The smoothed margin never decreases). *Let $f(x; \theta)$ be continuously differentiable in $\theta \in V$ and homogeneous of degree $a > 0$. Let $\theta(t)$, $t \ge t_0$, be absolutely continuous with $\|\dot\theta(t)\| \le 1$ and $\langle \nabla\mathcal{L}(\theta(t)), \dot\theta(t)\rangle = -\|\nabla\mathcal{L}(\theta(t))\|_*$ for almost every t (a normalized steepest flow), and let $n\mathcal{L}(\theta(t_0)) < 1$. Then $\tilde\gamma(t) = \psi(\theta(t))/\|\theta(t)\|^a$ is nondecreasing, and $q_{\min}(\theta(t))/\|\theta(t)\|^a \ge \tilde\gamma(t_0) > 0$ for $t \ge t_0$.*

PROOF. The plan is to show that $\log\psi$ grows at least as fast as $a \log\|\theta\|$. Differentiating $f(x; c\theta) = c^a f(x; \theta)$ at $c = 1$ gives $\langle \nabla q_i(\theta), \theta\rangle = a\, q_i(\theta)$, so, with (15.3),

$$\langle -\nabla\mathcal{L}(\theta), \theta\rangle = \frac{a}{n}\sum_i e^{-q_i(\theta)} q_i(\theta) \ge a\mathcal{L}(\theta) q_{\min}(\theta) \ge a\mathcal{L}(\theta)\psi(\theta),$$

and $\|\nabla\mathcal{L}(\theta)\|_* \geq \langle -\nabla\mathcal{L}(\theta), \theta\rangle / \|\theta\|$. Along the flow, for almost every t,

$$\frac{d}{dt}\psi(\theta(t)) = \frac{\|\nabla\mathcal{L}(\theta)\|_*}{\mathcal{L}(\theta)} \geq \frac{a\,\psi(\theta)}{\|\theta\|}.$$

So ψ stays at least $\psi(\theta(t_0)) > 0$, and $\theta(t)$ stays away from 0, because $|q_i(\theta)| \leq M\|\theta\|^a$ with M the largest $|f(x_i;\cdot)|$ on the unit sphere. As $t \mapsto \|\theta(t)\|$ is absolutely continuous with derivative at most $\|\dot\theta\| \leq 1$, for almost every t

$$\frac{d}{dt}\log\psi(\theta(t)) \geq \frac{a}{\|\theta(t)\|} \geq \frac{a}{\|\theta(t)\|}\frac{d}{dt}\|\theta(t)\| = \frac{d}{dt}\,a\log\|\theta(t)\|,$$

so $\log\tilde\gamma$ is absolutely continuous with a nonnegative derivative almost everywhere, hence nondecreasing. The last claim follows from $q_{\min} \geq \psi$. ■

For linear models ($a = 1$) this is the flow version of Theorem 15.8. Lyu and Li (2020) proved, for homogeneous networks (ReLU networks included, through Clarke subdifferentials) trained by gradient flow or gradient descent with small steps on the exponential or logistic loss from a small enough loss ($n\mathcal{L} < 1$ for the exponential loss), that the smoothed margin increases, the loss tends to zero, and every limit point of $\theta/\|\theta\|_2$ is the direction of a KKT point of $\min \frac{1}{2}\|\theta\|_2^2$ subject to $q_i(\theta) \geq 1$; Tsilivis et al. (2025) proved the analogue for steepest flows in other norms.

Proof sketch of the KKT limit. Proposition 15.17 gives the monotonicity for differentiable networks. A KKT point $\bar\theta$ with $q_{\min}(\bar\theta) = 1$ is a nonnegative combination of the $\nabla q_i(\bar\theta)$ of its minimum-margin examples; $-\nabla\mathcal{L} = \frac{1}{n}\sum_i e^{-q_i}\nabla q_i$ is such a combination, with weights concentrating on the smallest margins, so it suffices that θ and $-\nabla\mathcal{L}$ align along a sequence of times. The inequality $\|\nabla\mathcal{L}\|_* \geq \langle -\nabla\mathcal{L}, \theta\rangle/\|\theta\|$ used above is an equality only under alignment, and since $\tilde\gamma$ is bounded the integrated deficit is finite, which forces alignment along a sequence. Omitted: the quantitative version, the core of Lyu and Li (2020), and its extension to dual norms (Tsilivis et al. 2025). ■

A KKT point is a stationary point of the margin problem, not necessarily a maximizer: for ReLU networks the limit need not be even a local maximizer of the margin (Vardi et al. 2022).

15.5 Reading the theory

Three norms, three classifiers. Figure 15.1 shows Lemma 15.5 on six examples in the plane: the ℓ_1 classifier uses one feature, the ℓ_∞ classifier both at full weight, and the ℓ_2 classifier lies between, fixed by two support vectors; each is the normal of the line that separates the hull of the signed examples from the largest dual ball around the origin.

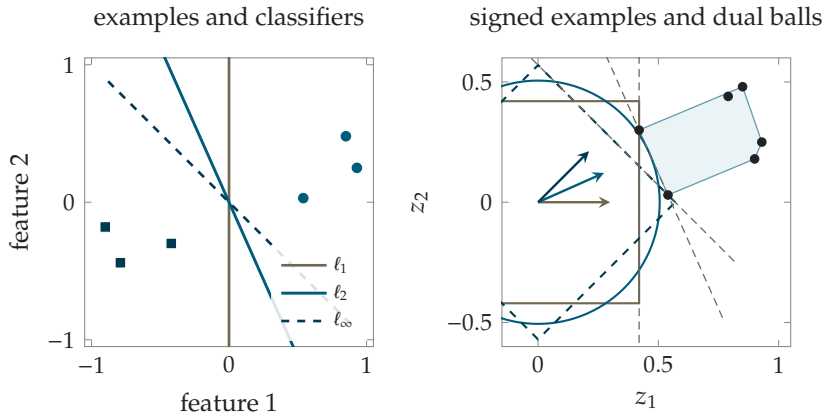


Figure 15.1. Three geometries on one dataset. Left: six examples (circles positive, squares negative) and the maximum-margin classifiers through the origin for ℓ_1, ℓ_2 and ℓ_∞ (dashed). Right: the signed examples $z_i = y_i x_i$, their convex hull, and the largest dual balls outside it: the ℓ_∞ square for the ℓ_1 margin ($\gamma_1 = 0.42$), the circle for ℓ_2 ($\gamma_2 = 0.506$), the ℓ_1 diamond (dashed) for ℓ_∞ ($\gamma_\infty = 0.57$); thin lines $\langle u^\star, c\rangle = \gamma$ separate them, and the arrows are the directions $u^\star$. Primal and dual programs agree to 10^{-6} (script `15-bias-geometry.py`).

end point of	normalized margin measured in		
	ℓ_2	ℓ_∞	ℓ_1
normalized gradient descent	**0.3108**	0.3666	0.1673
sign descent	0.3020	**0.3750**	0.1598
coordinate descent	0.2921	0.3350	**0.1709**
maximum margin γ	0.3115	0.3766	0.1717

Table 15.1. End points of the vector runs of Figure 15.2 (decaying steps, $T = 10^5$) measured in every norm. Each optimizer approaches the maximum margin of its own norm (bold), not the others.

EACH OPTIMIZER MAXIMIZES ITS OWN MARGIN. In Figure 15.2 the four optimizers of Corollary 15.12 run from zero, and in each panel the points rise to the maximum margin of the optimizer's own norm (linear or quadratic programs; for the spectral norm, cutting planes on the dual of Lemma 15.5). The decaying-step bound lies below its points and reaches the line slowly; the constant-step bound stalls at $\gamma - R^2\eta/2$, while the constant-step points do not. Table 15.1 measures the end points in every norm: each comes within $2 \cdot 10^{-3}$ of the maximum margin of its own norm and stays below the others; coordinate descent, for instance, ends 11% below γ_∞ in the ℓ_∞ margin. On the matrix data the spectral margins of the end points of orthogonalized updates, normalized gradient descent and sign descent are 0.2276, 0.2231 and 0.2099, against $\gamma_{\mathrm{op}} = 0.2286$; their Frobenius margins are 0.2067, 0.2085 and 0.1855, against 0.2094.

THE DISTANCE TRAVELLED IS THE CLOCK. The runs obey a simpler law than the theorem. At $t = 10^5$ the gap times the distance travelled, $(\gamma - \bar{\gamma}(\theta_t))\,\tau_t$,

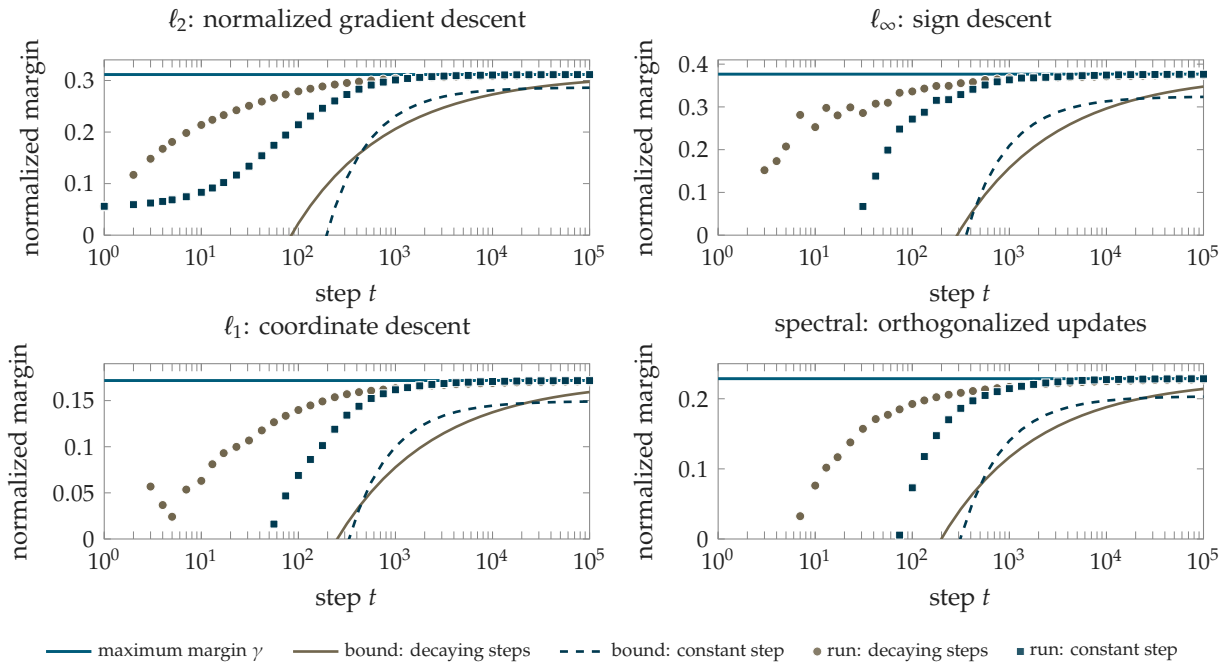


Figure 15.2. Normalized margins of normalized steepest descent in four norms from $\theta_0 = 0$, with steps $\eta_t = 1/(R\sqrt{t+1})$ (circles) and $\eta = 0.05/R$ (squares), $R = \max_i \|z_i\|_*$; lines: the exact maximum margins; curves: the bound (15.4). Vector panels: 16 separable unit vectors in $\mathbb{R}^5$; spectral panel: $W \in \mathbb{R}^{3\times4}$ on 12 examples in three classes (24 signed matrices of nuclear norm one). One run each, fixed seeds (script `15-bias-margins.py`).

is 0.46 for ℓ_2, 0.49 to 0.51 for ℓ_∞, 0.57 for ℓ_1 and 0.68 for the spectral norm, the same for both schedules within 4%, although their distances differ eightfold. So the margin is a function of the distance travelled, as for a flow, and approaches the maximum like C/τ_t with C well below the flow bound $\log n$ (2.77; 3.18 for the spectral data): these runs do not pay the finite-step term of (15.4). The reason, as far as we can see, is that near the maximum-margin direction a step changes the small margins by nearly the same amount, so the variance that Hoeffding's lemma charges for is small, and the rest cancels over consecutive steps. Whether this holds for every dataset and norm is open as far as we know (Exercise 15.10); for ℓ_2, whose dual problem is smooth, Ji and Telgarsky (2021) prove an $O(1/t)$ margin rate with aggressive steps.

The law gives time scales: within ε of the maximum after about $C/(\varepsilon\eta)$ steps for a constant step η, $(C/(2c\varepsilon))^2$ for $\eta_t = c/\sqrt{t+1}$, and exponentially many in $1/\varepsilon$ for unnormalized gradient descent, which travels a distance of order $\log t$ (Corollary 15.11). Sign descent, Adam and Muon reach the regime of their bias in polynomially many steps; gradient descent creeps.

WHERE ADAMW ENDS. In Figure 15.3 the exact end point at radius $B = 1/\lambda$, computed by constrained minimization, and AdamW without memory with the Frank–Wolfe schedule agree to $2 \cdot 10^{-4}$. At $\lambda = 0.1$, the usual value

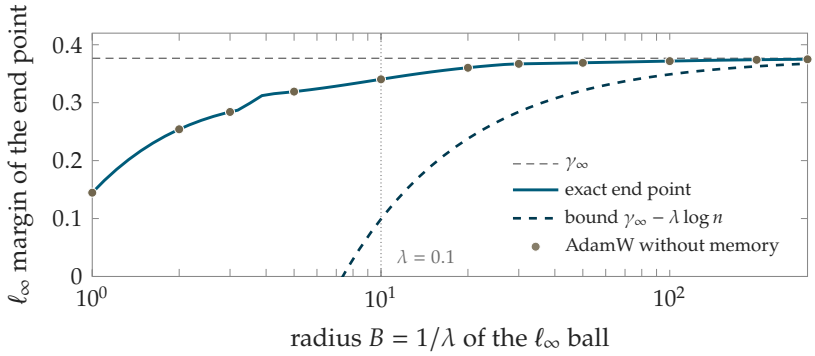


Figure 15.3. Where AdamW without memory ends, on the vector data of Figure 15.2: the ℓ_∞ margin of the minimizer of the loss over $\{\|\theta\|_\infty \le B\}$ (curve), the bound of Proposition 15.16 (dashed), and AdamW with $\beta_1 = \beta_2 = \epsilon = 0$, $\eta_t\lambda = 2/(t+2)$, after $2 \cdot 10^4$ steps (points). Kinks: changes of the saturated coordinates or of the example with the smallest margin. Script `15-bias-decay.py`.

in language-model training (Chapter 13), the end point's ℓ_∞ margin is 0.340, 10% below $\gamma_\infty = 0.377$; the guarantee $\gamma_\infty - \lambda \log n$ is 0.099. For $5 \le B \le 300$ the exact gap lies between $0.28/B$ and $0.49/B$, so $\log n/B$ has the right order in B and a constant six to ten times too large.

WHERE AGREEMENT MUST FAIL. Where the direction is not determined: if $\mathcal{U}^\star$ has more than one point, as can happen for ℓ_1 and ℓ_∞, the theorem fixes the margin but not the limit inside $\mathcal{U}^\star$, which then depends on the step sizes and even on floating-point arithmetic (Exercise 15.4).[9] And it says nothing about the steps before $\gamma\tau_t$ exceeds $\log n$.

[9] In double precision e^{-x} underflows to zero above $x \approx 745$, in single precision above $x \approx 104$; an implementation must compute the weights relative to the smallest margin.

15.6 What the theory misses

WHAT WAS IDEALIZED AWAY. *Fixed features.* In a network the features move, and the bias depends on the parametrization as well as on the optimizer: for diagonal linear networks gradient descent behaves like an ℓ_1 method from a small initialization and like an ℓ_2 method from a large one (Woodworth et al. 2020). *Separability and long training.* The theory needs a run that fits its data and lasts until $\gamma\tau_t \gg \log n$; language-model pretraining makes about one pass over its data and never interpolates, so the theory does not apply there, and vision training often stops soon after zero training error. *Full batch and memory.* Minibatch noise changes the trajectory and the time scales; momentum leaves the limit unchanged, Adam's ϵ does not. *The example* shows that an implicit bias *can* decide the test error, not that it does on real data.

WHERE IT BREAKS. (1) *The generalization gap is not robust.* With every hyperparameter tuned, adaptive methods matched or beat momentum SGD (Choi et al. 2019); decoupled weight decay closed most of the gap on image classification (Loshchilov and Hutter 2019); on transformers adaptive methods are preferred, for reasons of speed (Chapter 14). The max-margin mechanism predicts a gap only if the data are interpolated, the run lasts

into the regime of the bias, and the data favour one geometry, and the comparisons, with or without gaps, do not check these conditions. (2) *Adam's ϵ decides the limit.* With a fixed $\epsilon > 0$ the long-run bias of Adam is the ℓ_2 margin (Wang et al. 2021): the ℓ_∞ bias lasts only while the gradients, which decay like the loss, are large compared with ϵ (Exercise 15.8). (3) *KKT is not maximality.* For ReLU networks the limit direction can fail to be even a local maximizer of the margin (Vardi et al. 2022). (4) *Margins rank deep networks poorly.* In a large comparison of complexity measures, norm- and margin-based ones predicted generalization less well than sharpness-based ones (Jiang et al. 2020). (5) *The step-size conditions are too cautious, and the loss matters.* Gradient descent on the logistic loss with an arbitrary constant step, even one that makes the loss oscillate, still converges in direction to the ℓ_2 maximum margin, while on the exponential loss large steps can diverge (Wu et al. 2023); and constant normalized steps converge in our runs, where Corollary 15.9 guarantees only $\gamma - R^2\eta/2$.

REPAIRS AND OPEN PROBLEMS. Faster margin rates with aggressive and normalized steps (Ji and Telgarsky 2021; Nacson et al. 2019); steepest flows on homogeneous networks (Tsilivis et al. 2025); Adam, AdamW and Muon (Section 15.4). Open: (a) whether constant normalized steps reach the maximum margin in the ℓ_1, ℓ_∞ and spectral norms, at the rate $1/\tau_t$ observed in Figure 15.2; (b) the implicit bias of Muon, with momentum and a norm per layer, in a deep network; (c) a theory that predicts from the data which geometry generalizes better, and so the sign of the difference between Adam and SGD.

15.7 Back in the wild

FIELD NOTES	
PRACTICE	Prefer SGD to adaptive methods when test accuracy counts; partly folklore, with mixed evidence.
IDEALIZATION	Linear classifier, separable data, exponential loss; normalized steepest descent in a norm; weight decay as a norm ball.
PROVED	Margin $\geq \gamma - (\log n + \frac{R^2}{2}\sum \eta_t^2)/\sum \eta_t$ for every step-size sequence; AdamW ends within $\lambda \log n$; a dataset where the ℓ_∞ bias errs on all negatives, the ℓ_2 bias on none.
PREDICTS	SGD $\to \ell_2$ SVM; sign descent, Adam with negligible $\epsilon \to \ell_\infty$; boosting $\to \ell_1$; orthogonalized updates $\to$ spectral; gaps $1/\log t$ (plain GD), and C/τ_t observed (normalized).
MISSES	Depth and moving features, noise, Adam's ϵ, finite training, no interpolation in language-model pretraining.
TAKEAWAY	The optimizer's norm is a prior on the interpolating solution (proved in the idealization); which prior generalizes depends on the data (examples both ways); "SGD generalizes better" is folklore.

What should a practitioner take from this? When a model interpolates its training data and trains far past zero training error, the choice of optimizer

is also a choice of solution: the classifier that maximizes the worst-case margin in the optimizer's own norm (proved in the idealization). Nothing makes the ℓ_2 prior the better one: Example 15.14 penalizes the ℓ_∞ prior, a dense ground truth the ℓ_2 prior (Exercise 15.6), and the rule that SGD generalizes better is folklore. The knobs that decide the end point are the norm of the update, Adam's ϵ, the weight decay, and the distance travelled.

The chapter closes the unification line on the side of solutions: the norms of Chapters 1 to 3 reappear as the norms whose margins are maximized, the averages of Chapter 5 leave the limit unchanged, a preconditioner (Chapter 6) changes it through its norm (Exercise 15.5), and the ball of decoupled weight decay (Chapter 13) turns the limit into a point. The other half of Part V, Chapter 14, is about speed.

Exercises

15.1 CONSOLIDATION. (Three margins of two examples.) For the signed examples $z_1 = (5, 1)$ and $z_2 = (4, 3)$, show that $\gamma_1 = 4$ with $\mathcal{U}^\star = \{e_1\}$, $\gamma_2 = 11/\sqrt{5}$ with $\mathcal{U}^\star = \{(2,1)/\sqrt{5}\}$, and $\gamma_\infty = 6$ with $\mathcal{U}^\star = \{(1,1)\}$; check Lemma 15.5 with the points of $[z_1, z_2]$ nearest the origin in the dual norms, and name the support vectors.

15.2 CONSOLIDATION. (Every initialization.) Prove that from an arbitrary θ_0, whenever the right side is below γ, $\gamma - \bar{\gamma}(\theta_T) \le \big((R+\gamma)\|\theta_0\| + \log n + \frac{R^2}{2}\sum_{t<T}\eta_t^2\big)/(\|\theta_0\| + \tau_T)$. *(Hint: $\psi(\theta_0) \ge -R\|\theta_0\| - \log n$ and $\|\theta_T\| \le \|\theta_0\| + \tau_T$.)*

15.3 CONSOLIDATION. ⋆ (AdaBoost's margin.) Let every entry of every z_i be ± 1, as for weak classifiers valued in $\{-1, 1\}$, so that $R = 1$ in the ℓ_1 geometry, and let every edge r_t be below 1. (a) Show that $\eta = \operatorname{artanh}(r)$ minimizes the first bound of Lemma 15.7 and that with it $\mathcal{L}(\theta_{t+1}) = \sqrt{1 - r_t^2}\,\mathcal{L}(\theta_t)$ exactly: this is AdaBoost. (b) Deduce $\liminf_t \bar{\gamma}(\theta_t) \ge -\frac{1}{2}\log(1-\gamma^2)/\operatorname{artanh}(\gamma)$, about $\gamma/2$ for small γ. *(Hint: $r \mapsto -\frac{1}{2}\log(1-r^2)/\operatorname{artanh}(r)$ increases on $(0,1)$, and $r_t \ge \gamma$.)*

15.4 CONSOLIDATION. (A tie.) Let $z_1 = (2, 0)$ and $z_2 = (1, 2)$. (a) Show that $\gamma_\infty = 2$ and that the ℓ_∞ max-margin set is $\{(1, s) : \frac{1}{2} \le s \le 1\}$. (b) Show that sign descent from 0 moves exactly along $(1, 1)$. (c) A run that computes the weights as $p_i \propto e^{-(q_i - q_{\min})}$ in double precision, where e^{-x} underflows to 0 above $x \approx 745$, leaves the ray once τ_t exceeds about 745; explain why, and why its margin stays 2.

15.5 VARIATION. (Preconditioning changes the norm.) For a fixed $P \succ 0$, show that normalized steepest descent in $\|\theta\|_P = \sqrt{\theta^\top P \theta}$ takes the step $-\eta_t P^{-1} g/\|g\|_{P^{-1}}$, and that its limit direction is the ℓ_2 maximum-margin direction of the examples $P^{-1/2}x_i$, mapped back by $P^{-1/2}$. Do the same for $\|D\theta\|_\infty$ with a diagonal $D \succ 0$: a frozen preconditioner changes the implicit bias.

15.6 VARIATION. (A dense ground truth.) Let $x \sim \mathcal{N}(0, I_2)$ have the label $\operatorname{sign}(x_1 + x_2)$, and train on the one example $x = (1, \frac{1}{2})$, label $+1$. Show that $u_\infty = (1, 1)$ and $u_2 = (2, 1)/\sqrt{5}$, that a classifier's test error is its angle to $(1, 1)$ over π, and so that the ℓ_∞ classifier has test error 0 and the ℓ_2 one $\arccos(3/\sqrt{10})/\pi \approx 0.102$.

15.7 VARIATION. (Coupled weight decay.) Let θ_λ minimize $\mathcal{L}(\theta) + \frac{\lambda}{2}\|\theta\|_2^2$. Show that θ_λ minimizes $\mathcal{L}$ over the ball of radius $\|\theta_\lambda\|_2$, so that Proposition 15.16 applies, and, comparing with $Bu^\star$ for $B = \log(1/\lambda)/\gamma_2$, that $\mathcal{L}(\theta_\lambda) \to 0$, $\|\theta_\lambda\|_2 \to \infty$ and $\bar{\gamma}(\theta_\lambda) \to \gamma_2$ as $\lambda \to 0$.

15.8 SIMULATION. (Adam's ϵ on the example.) On the data of Example 15.14 with $n_+ = 6$, $n_- = 4$, run Adam ($\beta_1 = 0.9$, $\beta_2 = 0.999$, bias correction, steps $0.05/\sqrt{t+1}$) from zero for 10^5 steps with $\epsilon \in \{0, 10^{-8}, 10^{-3}\}$, recording the score $(\theta_2 + \theta_3 - \theta_1)/\|\theta\|_\infty$ of a new negative example. (a) Show that for $\epsilon = 0$ the iterates stay on the ray of u_∞, so the score stays 1 (along the ray all gradients are positive multiples of one vector). (b) Observe that the score leaves 1 near $t = 2 \cdot 10^4$ for $\epsilon = 10^{-8}$, and at once for $\epsilon = 10^{-3}$, where it is negative by $t = 10^5$; explain this by comparing ϵ with each coordinate of $\sqrt{\hat{v}_t}$, and why the score must eventually become negative.

15.9 SIMULATION. (The distance is the clock.) Rerun `15-bias-margins.py` with constant steps $0.2/R$ and $0.01/R$ and decaying steps $0.3/(R\sqrt{t+1})$. Check that $(\gamma - \bar{\gamma}(\theta_t))\,\tau_t$ tends to the same constant for every schedule, compare it with $\log n$, and the gap with the asymptote $R^2\eta/2$ of the constant-step bound.

15.10 RESEARCH. (Constant steps.) Prove that normalized steepest descent with a constant step in the ℓ_∞ or ℓ_1 norm satisfies $\bar{\gamma}(\theta_t) \to \gamma$, ideally at the rate $O(1/\tau_t)$ of Figure 15.2, or find a dataset on which it does not. For ℓ_2 the smooth dual gives rates (Ji and Telgarsky 2021); for ℓ_1 and ℓ_∞ the step jumps between vertices of the ball. AdaBoost, whose steps do not shrink, can stop short (Rudin et al. 2004).

15.11 RESEARCH. (Muon in depth.) For a two-layer linear network $x \mapsto W_2W_1x$ trained with orthogonalized updates on each layer (a shared step length in a max norm over the layers, Chapter 4), Proposition 15.17 applies with $a = 2$. Characterize the KKT points of the max-margin problem through W_2W_1, compare with gradient descent, and find the effect of momentum; as far as we know this is open.

Notes and References

Margins in norms other than ℓ_2 entered machine learning with boosting: Freund and Schapire (1997) introduced AdaBoost, Schapire et al. (1998) explained its resistance to overfitting by the growth of the ℓ_1 margins, and Freund and Schapire (1999) analyzed multiplicative weights in repeated games, the view of Remark 15.10. Friedman (2001) read boosting as steepest

descent in function space, Rosset et al. (2004) showed that boosting with small steps approximately follows a regularization path ending at the ℓ_1 maximum margin, and Rudin et al. (2004) that AdaBoost can cycle and stop short of it: boosting was known to maximize a margin long before gradient descent was.

Soudry et al. (2018) proved that gradient descent on separable data converges in direction to the ℓ_2 support vector machine at the rate $1/\log t$, and Gunasekar et al. (2018) that steepest descent in a norm converges to the maximum margin in that norm; Theorem 15.8 is a version for normalized steps, through Hoeffding's lemma and the regret argument of Remark 15.10, with an explicit bound for every step-size sequence. Rates were improved with aggressive steps (Nacson et al. 2019) and by a primal–dual analysis in which gradient descent induces mirror descent on the example weights (Ji and Telgarsky 2021). The companion volume's chapter on margins, in *The Hydrogen Atoms of Deep Learning*, treats the Euclidean case for gradient flow. The empirical debate began with Wilson et al. (2017), was continued by Keskar and Socher (2017) and Loshchilov and Hutter (2019), and was reframed by Choi et al. (2019).

TRANSLATING THE LITERATURE. Gunasekar et al. (2018) write w for our θ and N for our n; their steepest descent step $\Delta w_t \in \arg\min_v \langle \nabla \mathcal{L}(w_t), v\rangle + \frac{1}{2}\|v\|^2$ is our $-\nabla\mathcal{L}^\sharp$, so their method is the unnormalized one of Corollary 15.11, and their step size η_t corresponds to our normalized step $\eta_t \|\nabla\mathcal{L}(w_t)\|_*$. Soudry et al. (2018) write $\hat{w}$ for our $u^\star/\gamma_2$ in ℓ_2. Lyu and Li (2020) write L for the order of homogeneity (our a), $\bar{\gamma}$ and $\tilde{\gamma}$ for the normalized and smoothed margins, and sum the losses, so their condition $\mathcal{L} < 1$ is our $n\mathcal{L} < 1$. In boosting, the example weights D_t are our $p(\theta_t)$, the edge $1 - 2\varepsilon_t$ of the chosen weak classifier with weighted error ε_t is our $r(\theta_t)$, and AdaBoost's step $\alpha_t = \frac{1}{2}\log((1-\varepsilon_t)/\varepsilon_t)$ is our $\operatorname{artanh}(r(\theta_t))$.

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Epilogue

What theory owes practice

The prologue listed the decisions that a practitioner makes before a training run and promised a theory of each. This epilogue keeps the account. It records, practice by practice, what the chapters proved and in which idealization, what they predicted, and what they left unexplained; it says what the idealizations miss when they are taken together; and it lists the questions that the book raises and cannot answer.

A ledger

Table 15.1 has one row for each practice. The verdict is one of three. A practice is *explained* when a theorem in a faithful idealization accounts for it and the prediction survives a comparison with simulation or with the evidence from training runs. It is *partly explained* when a mechanism has been identified and proved in an idealization, but either the idealization is far from training or the prediction is only qualitative. It is *unexplained* when the book has nothing to say beyond a description, or when the practice is folklore. The last column says what the verdict rests on, which is where a reader who doubts it should look.

Three things stand out. First, the update rules themselves are well explained: the first line of the book, that optimizers are steepest descent in norms with preconditioning and moving averages, accounts for the form of every rule in the field guide, and it predicts how learning rates must scale with width. Second, almost every practice of the second line is partly explained, and in the same way: a mechanism is proved in an idealization, and the idealization contains one or two parameters, a curvature bound, a noise level, a rate of sharpening, that are measured rather than derived. Third, the numbers that practitioners actually set, $\beta_1 = 0.9$, $\beta_2 = 0.95$, weight decay 0.1, a warmup of two thousand steps, a cooldown of a fifth of training, are almost all unexplained. The theory says which combinations matter and how they should scale; it does not say why these values.

What the idealizations miss together

Each chapter listed what its own idealization misses. Read together, the lists repeat five omissions.

The landscape is fixed. Almost every idealization in the book is an optimizer running on a fixed objective: a quadratic, a convex function, a cubic model of the edge of stability, a valley with a river. The objective of a network changes as it trains. Sharpness rises, features form, norms grow and settle, and the gradient noise changes with them. Where the book needed such a change, it put it in by hand, as the rate of progressive sharpening in Chapter 8 or the loss-dependent curvature in Chapter 9.

PRACTICE	CH.	VERDICT	WHAT THE VERDICT RESTS ON
Normalized updates	1	explained	Every update in use is a steepest-descent step in some norm; rates depend on the norm only through a smoothness constant and a dual norm
Adam's sign-like steps	2	explained	Memoryless Adam is sign descent; with memory its step is at most $C(\beta_1, \beta_2)$ times the learning rate, 7.27 at $(0.9, 0.999)$ and 1.16 at $(0.9, 0.95)$
When ℓ_∞ steps help	2	partly	Dense gradients against curvature concentrated on the coordinate axes; the comparison is between guarantees, and it depends on the coordinate system
Muon's orthogonalization	3	explained	The steepest step in the spectral norm is the polar factor; Muon's quintic keeps singular values in a band near one
When Muon helps	3	partly	Nuclear rank of the gradient against stable rank of the inputs, for guarantees; shrinking gains at scale are not explained
μP learning-rate rules	4	partly	Exponents of width follow from rank-one gradients and the RMS→RMS norm, and match measured slopes; constants and long-run transfer do not
Momentum	5	partly	Stability regions and noise averaging are exact on quadratics; normalized steps need momentum when the noise is skewed; why $\beta = 0.9$ is not explained
Adam's β_2	6	partly	$1/(1-\beta_2)$ is the memory of the preconditioner and sets the size of a step after a quiet stretch; the values used in practice are not derived
Shampoo, K-FAC, SOAP	7	partly	The Kronecker bound and Shampoo's regret are proved; conditioning improves by an exponent that depends on the noise model; empirical wins are not derived
Training at $2/\eta$	8	partly	Thresholds for gradient descent, momentum and frozen Adam are exact; self-stabilization is proved in a cubic model, in continuous time; the rise of sharpness is an input
Linear warmup	9	partly	If curvature is bounded by the loss, then a rising step size is optimal among stable schedules; the bound is measured, not derived, and long warmups are not explained
Gradient clipping	10	explained	Under (L_0, L_1)-smoothness clipping removes the dependence on the largest gradient, and no fixed step can; the threshold 1.0 is folklore
z-loss and QK-norm	10	partly	Each removes a mechanism of instability (a flat direction of the softmax; quadratic growth of attention logits); real spikes have interacting causes
Batch-size rules	11	partly	In the noisy quadratic model the critical batch size is exact and preconditioning moves it; the scaling rules are heuristic
Cosine and WSD	12	partly	Convex last-iterate bounds for every schedule, and an exact river-valley model with a noise floor and a cooldown law; the model does not robustly reproduce WSD's parity with cosine
Decoupled weight decay	13	explained	Steepest descent with decoupled decay is Frank–Wolfe on a norm ball; normalized layers reach a rotational equilibrium with angle $\arccos(1-\eta\lambda)$ per step
The value $\lambda = 0.1$	13	unexplained	Only the product $\eta\lambda$ and the timescale $1/(\eta\lambda)$ are fixed by the theory
Adam over SGD for transformers	14	partly	Three solvable mechanisms: heavy-tailed classes slow gradient descent to a logarithmic decay; block curvature; and a gap that grows with the batch size; their interaction is open
Adaptive methods and generalization	15	partly	Steepest descent reaches the maximum margin of its own norm on separable data; deep networks are not covered

Table 15.1. The practices of the book, with the verdict of the chapter that treats them and what the verdict rests on.

One mechanism at a time. Warmup is explained by curvature in Chapter 9, by the statistics of Adam's first steps in Chapters 2 and 6, and by the size of early angular updates in Chapter 13; the advantage of Adam on transformers by three mechanisms in Chapter 14. In a training run they all operate at once, and nothing in the book says how they combine.

Guarantees, not performance. Several comparisons in the book are comparisons of worst-case guarantees: which norm gives the better descent bound (Chapters 1 to 3), which schedule gives the better last-iterate bound (Chapter 12). The chapters checked how far the guarantees are from what the optimizers actually do, and the answer was often a factor of two or more. A guarantee can rank two methods correctly for the wrong reason.

Simple noise. The noise in most chapters has bounded variance, often Gaussian, often proportional to the curvature. Heavy tails appear only in Chapter 10, and the structure of data, such as the Zipf law of token frequencies, only in Chapter 14. Both turned out to matter where they appeared.

No scale. The most consistent finding of careful benchmarks is that the speedups of newer optimizers over a well-tuned AdamW shrink as models grow (Wen et al. 2026b). No idealization in this book has a size parameter that reproduces this; the noisy quadratic model of Chapters 11 and 14 comes closest, through the dependence of the critical batch size on the spectrum.

What is open

The chapters end with research exercises. Here are the questions that recur across them, in roughly the order of the book.

(i) *The right norm.* The norm of a network is assembled from norms on its layers, but the norms for embeddings, for attention, for normalization gains and for the output layer are chosen by analogy (Chapter 4). Is there a principle, perhaps from the statistics of activations and gradients, that selects them?

(ii) *The averaging constants.* Why do $\beta_1 = 0.9$ and $\beta_2 \in [0.95, 0.999]$ work so widely (Chapters 5 and 6)? The theory explains which timescales these constants set, not why those timescales are right.

(iii) *Progressive sharpening.* Gradient descent raises the sharpness until the step is barely stable (Chapter 8). Why does the sharpness rise in the first place? A model in which progressive sharpening is an output rather than an input would change the second half of the book.

(iv) *Long warmups.* Every explanation of warmup in Chapter 9 predicts a warmup of tens or hundreds of steps for the settings it treats, while recipes use thousands. What sets the length?

(v) *Cooldowns at scale.* The river-valley model predicts the noise floor and the shape of the loss during a cooldown (Chapter 12); it does not reproduce the robust parity of warmup–stable–decay with cosine schedules, or the dependence of the best cooldown on scale.

(vi) *Shrinking gains.* Why do the gains of Muon, SOAP and other matrix methods over AdamW shrink with model size, and do methods that fix the weight norms directly, such as Hyperball (Wen et al. 2026a), keep them (Chapters 3, 7 and 13)?

(vii) *Hyperparameter scaling laws.* How should the learning rate, the batch size and the weight decay scale with model size and data size together? The book gives the exponent of width (Chapter 4), the critical batch size of a quadratic (Chapter 11) and the timescale of weight decay (Chapter 13); a joint theory is open.

(viii) *Spikes.* A solvable model of a loss spike in a transformer, with the optimizer state, the data and the architecture all playing their parts, does not exist (Chapter 10).

(ix) *Beyond pretraining.* The optimizers of this book are tuned for pretraining. Their behavior in mixtures of experts, in reinforcement learning from feedback, in low precision and in fine-tuning raises questions that the book does not touch.

How to turn a practice into a theorem

The chapters followed one method, and it may be useful to state it. Begin with the practice as practitioners state it, with its numbers and its sources, and with the evidence for and against it; the first section of every chapter is an attempt at this, and the attempt often showed that the practice was folklore. Find the smallest idealization in which the practice has a reason: one coordinate for sign descent, one matrix for Muon, a cubic for the edge of stability, a valley with a river for schedules, a sphere for weight decay. Prove what the idealization implies, completely, so that its consequences are not in doubt. Extract a number from the proof: a threshold, an exponent, a rate, a timescale, an equilibrium. Compare the number with a simulation, and then with training runs, and report the disagreement. Finally, list what the idealization threw away, and look in that list for the next idealization.

The method has a limit that the ledger makes plain: it explains mechanisms one at a time, in models that are much smaller than the networks they describe. Whether the mechanisms add up to an account of large-scale training is the question this book leaves for the next one.

Notes and References

The benchmark of Wen et al. (2026b) compares ten optimizers on language models of up to 1.2 billion parameters with careful tuning of each, and finds that the speedups over AdamW shrink with scale; its follow-up (Wen et al. 2026a), a preprint, proposes fixing the norms of weights and updates directly. The open problems above are discussed, from the practitioner's side, in the *Deep Learning Tuning Playbook* (Godbole et al. 2023). The individual chapters give the sources for each entry of the ledger.

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A Toolkit

A

Duality, matrices, recurrences and online learning

This appendix collects the mathematics that the chapters share, in the form in which they use it, with a proof wherever one fits in a page; it is a reference, not a chapter to be read in order. Norms and duality (Appendix A.1) are the language of Chapters 1 to 4 and return in Chapters 5, 13 and 15. Matrix analysis (Appendix A.2) supplies the polar factor of Chapter 3 and the Löwner order, Kronecker products, and geometric mean behind the preconditioners of Chapters 6 and 7. Linear recurrences (Appendix A.3) give the stability thresholds of Chapters 5 and 8, the noise floors of Chapters 5, 11 and 12, and, through Grönwall's inequality, the local smoothness bounds of Chapter 10. The descent lemma and the Polyak–Łojasiewicz inequality (Appendix A.4) start the convergence proofs of the prologue and of Chapters 1, 2 and 9; online convex optimization (Appendix A.5) is the setting of Chapters 6, 7 and 12; and martingale and tail bounds (Appendix A.6) control the noise in Chapter 5 and the sign errors in Chapter 2.

Conventions. Vectors live in $\mathbb{R}^d$. A linear functional, such as a gradient, is identified with a vector z through the pairing $\langle z, x\rangle = \sum_i z_i x_i$,[1] and matrices carry the pairing $\langle A, B\rangle = \operatorname{tr}(A^\top B)$. An unadorned $\|\cdot\|$ is an arbitrary norm. For symmetric matrices, $A \preceq B$ means that $B - A$ is positive semidefinite, written $B - A \succeq 0$, and $A \prec B$ that it is positive definite.

[1] The identification uses the pairing, not the norm. Changing the norm changes which step is steepest; it does not change the gradient.

A.1 Norms and duality

A norm measures steps. A gradient turns a step into a rate of change of the loss, and the dual norm measures it by the largest rate it offers per unit step.

Definition A.1 (Dual norm). The *dual norm* of a norm $\|\cdot\|$ on $\mathbb{R}^d$ is $\|z\|_* = \max_{\|x\|\le 1}\langle z, x\rangle$. The maximum exists because the unit ball is closed and bounded, hence compact.

One fact from convex geometry is used twice, here and as the engine of online gradient descent (Theorem A.21). For $P \succ 0$ write $\|x\|_P = \sqrt{x^\top P x}$.

Lemma A.2 (Projection onto a convex set). *Let $P \succ 0$ and let $\mathcal{K} \subseteq \mathbb{R}^d$ be nonempty, closed, and convex. Every $y \in \mathbb{R}^d$ has a unique nearest point $\Pi(y) = \arg\min_{x\in\mathcal{K}}\|x - y\|_P$, and for every $u \in \mathcal{K}$,*

$$\langle P(y-\Pi(y)), u-\Pi(y)\rangle \le 0, \qquad \|\Pi(y)-u\|_P^2 \le \|y-u\|_P^2 - \|y-\Pi(y)\|_P^2.$$

Proof. A minimizer p exists because $x \mapsto \|x - y\|_P$ is continuous and, for a fixed $x_0 \in \mathcal{K}$, the points of $\mathcal{K}$ with $\|x - y\|_P \le \|x_0 - y\|_P$ form a

compact set. For $u \in \mathcal{K}$ and $s \in (0,1]$ the point $p + s(u-p)$ lies in $\mathcal{K}$, so $0 \le \|y-p-s(u-p)\|_P^2 - \|y-p\|_P^2 = -2s\langle P(y-p), u-p\rangle + s^2\|u-p\|_P^2$; dividing by s and letting $s \downarrow 0$ gives the first inequality. Expanding $\|y-u\|_P^2 = \|(y-p)+(p-u)\|_P^2$ and using the first inequality gives the second, and the second with $u = p'$ shows that every minimizer p' equals p. ∎

Lemma A.3 (Dual norms). *Let $\|\cdot\|$ be a norm on $\mathbb{R}^d$ and $\|z\|_* = \max_{\|x\|\le 1}\langle z, x\rangle$ its dual (Definition A.1).*

(i) The dual norm is a norm, and Hölder's inequality *$|\langle z, x\rangle| \le \|z\|_*\|x\|$ holds for all z and x.*

(ii) (Bidual.) For every x, $\|x\| = \max_{\|z\|_*\le 1}\langle z, x\rangle$.

(iii) Let $1 \le p \le \infty$ and $1/p + 1/q = 1$, with $q = \infty$ for $p = 1$ and $q = 1$ for $p = \infty$. The dual of $\|\cdot\|_p$ is $\|\cdot\|_q$. For $z \ne 0$ the maximum of $\langle z, x\rangle$ over $\|x\|_p \le 1$ is attained at $x = \operatorname{sign}(z)$ if $q = 1$, at $x = \operatorname{sign}(z_i)e_i$ with $|z_i| = \|z\|_\infty$ if $q = \infty$, and at $x_i = \operatorname{sign}(z_i)|z_i|^{q-1}/\|z\|_q^{q-1}$ if $1 < q < \infty$.

Proof. Parts (i) and (iii) are computations; for (ii) the plan is to separate a point of the unit sphere from the unit ball by a hyperplane, obtained as a limit of projections. (i) By the equivalence of norms on $\mathbb{R}^d$, $c\|x\|_2 \le \|x\| \le C\|x\|_2$ for some $0 < c \le C$, so $\|z\|_2/C \le \|z\|_* \le \|z\|_2/c$. The dual norm is positively homogeneous by definition, even because the unit ball is symmetric, and subadditive because a maximum of a sum is at most the sum of the maxima. Applying the definition to $\pm x/\|x\|$ gives Hölder's inequality.

(ii) Hölder's inequality gives $\max_{\|z\|_*\le 1}\langle z, x\rangle \le \|x\|$; for the reverse we may assume $\|x\| = 1$. Let B be the unit ball, let p_k be the Euclidean projection of $y_k = (1+1/k)x \notin B$ onto B (Lemma A.2 with $P = I$), and let $z_k = (y_k - p_k)/\|y_k - p_k\|_2$. By the lemma $\langle z_k, u - p_k\rangle \le 0$ for $u \in B$, while $\langle z_k, y_k - p_k\rangle > 0$, so $\langle z_k, u\rangle < \langle z_k, y_k\rangle$ for every $u \in B$. A subsequence of the unit vectors z_k converges to a unit vector z with $\langle z, u\rangle \le \langle z, x\rangle$ for all $u \in B$, that is, $0 < \|z\|_* \le \langle z, x\rangle$; so $z/\|z\|_*$ has dual norm 1 and pairs with x to at least $1 = \|x\|$.

(iii) For $p \in \{1, \infty\}$, $|\langle z, x\rangle| \le \sum_i |z_i||x_i|$, which is at most $\|z\|_\infty\|x\|_1$ and at most $\|z\|_1\|x\|_\infty$, and the stated vectors attain these bounds. For $1 < p < \infty$, the concavity of the logarithm gives Young's inequality $ab \le a^p/p + b^q/q$ for $a, b \ge 0$; summed over the coordinates of x and z with $\|x\|_p = \|z\|_q = 1$ it gives $|\langle z, x\rangle| \le 1$, hence $|\langle z, x\rangle| \le \|z\|_q\|x\|_p$ by homogeneity. Since $(q-1)p = q$, the stated x has $\|x\|_p^p = \sum_i |z_i|^q/\|z\|_q^q = 1$ and $\langle z, x\rangle = \|z\|_q$. ∎

For a convex function f on $\mathbb{R}^d$, a vector z is a subgradient of f at x if $f(y) \ge f(x) + \langle z, y - x\rangle$ for all y; the set $\partial f(x)$ of subgradients is the *subdifferential*. A convex function is differentiable at x if and only if $\partial f(x)$ has a single element, which is then $\nabla f(x)$ (Rockafellar 1970, Section 25).

Lemma A.4 (The subdifferential of a norm and the linear minimization oracle). *For every $x \in \mathbb{R}^d$, $\partial\|\cdot\|(x) = \{z : \|z\|_* \le 1,\ \langle z, x\rangle = \|x\|\}$. In particular $\partial\|\cdot\|(0)$ is the dual unit ball, and every subgradient at $x \neq 0$ has dual norm 1. Applied to the dual norm, this says that for every $g \in \mathbb{R}^d$*

$$\arg\min_{\|u\|\le 1}\langle g, u\rangle = -\partial\|\cdot\|_*(g) = \{u : \|u\| \le 1,\ \langle g, u\rangle = -\|g\|_*\}. \quad \text{(A.1)}$$

Hence every linear minimization oracle $\mathrm{lmo}(g)$ *satisfies $\langle g, \mathrm{lmo}(g)\rangle = -\|g\|_*$, and $\|\mathrm{lmo}(g)\| = 1$ when $g \neq 0$, while every point of the unit ball is a minimizer when $g = 0$. The oracle is unique exactly when $\|\cdot\|_*$ is differentiable at g, and then $\mathrm{lmo}(g) = -\nabla\|\cdot\|_*(g)$.*

Proof. If $\|z\|_* \le 1$ and $\langle z, x\rangle = \|x\|$, then $\|x\| + \langle z, y - x\rangle = \langle z, y\rangle \le \|y\|$ for every y, so z is a subgradient. Conversely, for a subgradient z the subgradient inequality at $y = 0$ and at $y = 2x$ gives $\langle z, x\rangle = \|x\|$, after which it reads $\langle z, y\rangle \le \|y\|$ for all y, that is, $\|z\|_* \le 1$; and at $x \neq 0$, $\|x\| = \langle z, x\rangle \le \|z\|_*\|x\|$ forces $\|z\|_* = 1$. By Lemma A.3(ii) the norm $\|\cdot\|$ is the dual of $\|\cdot\|_*$, so the first part applied to $\|\cdot\|_*$ at g identifies $\partial\|\cdot\|_*(g)$ with the maximizers of $\langle g, u\rangle$ over the unit ball; since the ball is symmetric, the minimizers are their negatives. The last sentence is the criterion for differentiability quoted above. ∎

The *convex conjugate* of a function f on $\mathbb{R}^d$ is $f^*(z) = \sup_x\{\langle z, x\rangle - f(x)\}$: at a gradient, the best first-order gain of a step minus its cost.

Lemma A.5 (The conjugate of half a squared norm, and the sharp operator). *The conjugate of $f(x) = \frac{1}{2}\|x\|^2$ is $f^*(z) = \frac{1}{2}\|z\|_*^2$, and the supremum defining $f^*(z)$ is attained exactly at the points $x = -\|z\|_* u$ with $u \in \arg\min_{\|u\|\le 1}\langle z, u\rangle$. Consequently the* sharp operator

$$g^\sharp \in \arg\max_{u\in\mathbb{R}^d}\left\{\langle g, u\rangle - \tfrac{1}{2}\|u\|^2\right\}$$

is $g^\sharp = -\|g\|_*\,\mathrm{lmo}(g)$, one value for each choice of the oracle; every value satisfies $\|g^\sharp\| = \|g\|_*$ and $\langle g, g^\sharp\rangle = \|g\|_*^2$; and for every $L > 0$,

$$\min_{u\in\mathbb{R}^d}\left\{\langle g, u\rangle + \frac{L}{2}\|u\|^2\right\} = -\frac{\|g\|_*^2}{2L}, \qquad \text{attained at } u = -\frac{g^\sharp}{L}. \quad \text{(A.2)}$$

Proof. For every x, Hölder's inequality and $ab \le \frac{1}{2}a^2 + \frac{1}{2}b^2$ give $\langle z, x\rangle - \frac{1}{2}\|x\|^2 \le \|z\|_*\|x\| - \frac{1}{2}\|x\|^2 \le \frac{1}{2}\|z\|_*^2$, with equality in both steps if and only if $\|x\| = \|z\|_*$ and $\langle z, x\rangle = \|z\|_*\|x\|$. If $z = 0$, then $x = 0$. If $z \neq 0$, write $x = -\|z\|_* u$; the conditions become $\|u\| = 1$ and $\langle z, u\rangle = -\|z\|_*$, which by Lemma A.4 describe the minimizers of $\langle z, u\rangle$ over the unit ball, the set (A.1). Such minimizers exist, so the supremum is attained and equals $\frac{1}{2}\|z\|_*^2$; the claims about $g^\sharp$ are the case $z = g$. For (A.2), the substitution $u = -v/L$ gives $\langle g, u\rangle + \frac{L}{2}\|u\|^2 = -\frac{1}{L}\left(\langle g, v\rangle - \frac{1}{2}\|v\|^2\right)$, whose minimum is $-\frac{1}{2L}\|g\|_*^2$, attained at $v = g^\sharp$. ∎

norm of a step	dual norm	an oracle $\mathrm{lmo}(g)$	sharp operator $g^\sharp$
$\|u\|_2$	$\|g\|_2$	$-g/\|g\|_2$	g
$\|u\|_\infty$	$\|g\|_1$	$-\operatorname{sign}(g)$	$\|g\|_1 \operatorname{sign}(g)$
$\|u\|_1$	$\|g\|_\infty$	$-\operatorname{sign}(g_i)\, e_i$	$g_i\, e_i$
$\|u\|_P$	$\|g\|_{P^{-1}}$	$-P^{-1}g/\|g\|_{P^{-1}}$	$P^{-1}g$
$\|U\|_{\mathrm{op}}$	$\|G\|_{\mathrm{nuc}}$	$-\operatorname{polar}(G)$	$\|G\|_{\mathrm{nuc}} \operatorname{polar}(G)$
$\|U\|_F$	$\|G\|_F$	$-G/\|G\|_F$	G
$\|U\|_{\mathrm{nuc}}$	$\|G\|_{\mathrm{op}}$	$-u_1 v_1^\top$	$\sigma_1(G)\, u_1 v_1^\top$

Table A.1. Duality maps, for $g \neq 0$ and $G \neq 0$. In the ℓ_1 row, $|g_i| = \|g\|_\infty$; in the nuclear row, (u_1, v_1) is a top singular pair of G. The oracle is unique except in four cases, where the table lists one choice: ℓ_∞ when some $g_i = 0$, ℓ_1 when the largest $|g_i|$ occurs at two or more coordinates, the spectral norm when G does not have full rank, and the nuclear norm when $\sigma_1(G) = \sigma_2(G)$. Proofs: Lemma A.3(iii) and Chapter 1 for vectors, Example A.6 for $\|\cdot\|_P$, and Theorem A.7 with Chapter 3 for matrices.

When the oracle is not unique, $g^\sharp$ stands for each maximizer; all of them have the same norm and the same pairing with g, which is all that the convergence proofs use. In the Euclidean norm $g^\sharp = g$, and preconditioning is the same construction for a quadratic norm; Table A.1 lists the duality maps of the norms used in this book.

Example A.6 (Quadratic norms and preconditioning). Let $P \succ 0$. The dual of $\|x\|_P$ is $\|z\|_{P^{-1}}$, $g^\sharp = P^{-1}g$, and for $g \neq 0$ the oracle is unique, $\mathrm{lmo}(g) = -P^{-1}g/\|g\|_{P^{-1}}$. Indeed $\langle z, x\rangle = \langle P^{-1/2}z, P^{1/2}x\rangle \leq \|z\|_{P^{-1}}\|x\|_P$ by the Cauchy–Schwarz inequality, with equality exactly when x is a nonnegative multiple of $P^{-1}z$. So preconditioned gradient descent $\theta \leftarrow \theta - \eta P^{-1}g$ is unnormalized steepest descent in $\|\cdot\|_P$: choosing a preconditioner is choosing a quadratic norm (Chapter 6).

A.2 Matrix analysis

Singular values. Let $G \in \mathbb{R}^{m\times n}$ and $k = \min\{m, n\}$. A singular value decomposition $G = \sum_{i=1}^k \sigma_i u_i v_i^\top$ has $\sigma_1 \geq \cdots \geq \sigma_k \geq 0$ and orthonormal families (u_i) in $\mathbb{R}^m$ and (v_i) in $\mathbb{R}^n$; it exists by the spectral theorem for $G^\top G$ (Horn and Johnson 2013), and $\sigma(G) = (\sigma_1, \ldots, \sigma_k)$ is the vector of singular values. A function acts on a symmetric $A = \sum_i \lambda_i a_i a_i^\top$, a spectral decomposition, by $f(A) = \sum_i f(\lambda_i) a_i a_i^\top$, a polynomial in A that does not depend on the decomposition; this defines A^s for $A \succeq 0$ and $s \geq 0$, with $A^0 = I$, and for $A \succ 0$ and all real s, and $A^{1/2}$ is the only positive semidefinite square root of A (Horn and Johnson 2013). Keeping the $r = \operatorname{rank} G$ terms with $\sigma_i > 0$ gives the compact decomposition $G = U \operatorname{diag}(\sigma) V^\top$ with $U^\top U = V^\top V = I_r$, and $\operatorname{polar}(G) = UV^\top$, which does not depend on the decomposition: $\operatorname{polar}(G) = Gf(G^\top G)$ for $f(x) = x^{-1/2}$ $(x > 0)$ and $f(0) = 0$. The *Schatten norm* $\|G\|_{S_p} = \|\sigma(G)\|_p$ is the spectral norm for $p = \infty$, the Frobenius norm for $p = 2$, and the nuclear norm for $p = 1$. A norm is

unitarily invariant if $\|Q_1 G Q_2\| = \|G\|$ for all orthogonal Q_1, Q_2; Schatten norms are, and that they are norms is part of the next theorem.

Theorem A.7 (von Neumann's trace inequality and Schatten duality). *For all $A, B \in \mathbb{R}^{m\times n}$,*

$$|\operatorname{tr}(A^\top B)| \le \sum_{i=1}^{k} \sigma_i(A)\,\sigma_i(B), \tag{A.3}$$

with equality when $A = \sum_i a_i u_i v_i^\top$ and $B = \sum_i b_i u_i v_i^\top$ are singular value decompositions with the same vectors. Consequently, for $1 \le p \le \infty$ and $1/p + 1/q = 1$, $\max_{\|B\|_{S_p}\le 1} \operatorname{tr}(A^\top B) = \|A\|_{S_q}$: the Schatten functions are norms, and the dual of $\|\cdot\|_{S_p}$ is $\|\cdot\|_{S_q}$. In particular the spectral and nuclear norms are dual to each other, the Frobenius norm is its own dual, and $\operatorname{tr}(A^\top \operatorname{polar}(A)) = \|A\|_{\mathrm{nuc}}$ with $\|\operatorname{polar}(A)\|_{\mathrm{op}} \le 1$.

Proof. The plan is to reduce (A.3) to an inequality between the two vectors of singular values, weighted by a matrix whose rows and columns sum to at most one, and to prove that inequality by summation by parts. Let $a = \sigma(A)$, $b = \sigma(B)$, and take full decompositions $A = U_1 D_a V_1^\top$, $B = U_2 D_b V_2^\top$ with orthogonal U_1, U_2, V_1, V_2, where $D_a \in \mathbb{R}^{m\times n}$ has a_i in position (i,i) and zeros elsewhere, and likewise D_b. With the orthogonal matrices $X = U_1^\top U_2$ and $Y = V_2^\top V_1$, the cyclic property of the trace gives

$$\operatorname{tr}(A^\top B) = \operatorname{tr}(D_a^\top X D_b Y) = \sum_{i,j\le k} a_i X_{ij} b_j Y_{ji},$$

so $|\operatorname{tr}(A^\top B)| \le \sum_{i,j\le k} a_i w_{ij} b_j$ with $w_{ij} = \frac{1}{2}(X_{ij}^2 + Y_{ji}^2) \ge |X_{ij}Y_{ji}|$. Rows and columns of orthogonal matrices are unit vectors, so every row and every column of $(w_{ij})_{i,j\le k}$ sums to at most 1. Put $a_{k+1} = b_{k+1} = 0$, $\alpha_l = a_l - a_{l+1} \ge 0$ and $\gamma_l = b_l - b_{l+1} \ge 0$, so that $a_i = \sum_{l\ge i} \alpha_l$ and $b_j = \sum_{l\ge j} \gamma_l$. Exchanging the order of summation,

$$\sum_{i,j\le k} a_i w_{ij} b_j = \sum_{l,l'\le k} \alpha_l \gamma_{l'} \sum_{i\le l,\, j\le l'} w_{ij} \le \sum_{l,l'\le k} \alpha_l \gamma_{l'} \min\{l, l'\} = \sum_{i\le k} a_i b_i,$$

because the inner sum is at most l by the row sums and at most l' by the column sums, and because the last identity is the first one with w replaced by the identity. With common singular vectors, $\operatorname{tr}(A^\top B) = \sum_{i,j} a_i b_j (u_i^\top u_j)(v_j^\top v_i) = \sum_i a_i b_i$.

For the duality, (A.3) and Hölder's inequality for vectors bound $\operatorname{tr}(A^\top B)$ by $\|\sigma(A)\|_q \|\sigma(B)\|_p$. Let $A = \sum_i a_i u_i v_i^\top \ne 0$, and let c maximize $\langle \sigma(A), c\rangle$ over the unit ball of ℓ_p, as in Lemma A.3(iii), with $c = e_1$ if $p = 1$. Its entries are nonnegative and nonincreasing, so $B = \sum_i c_i u_i v_i^\top$ has $\sigma(B) = c$ and $\operatorname{tr}(A^\top B) = \|\sigma(A)\|_q$. Hence $\|A\|_{S_q}$ is the maximum of the linear functions $A \mapsto \operatorname{tr}(A^\top B)$ over $\{\|\sigma(B)\|_p \le 1\}$, so it is subadditive, hence a norm, and it is the dual of $\|\cdot\|_{S_p}$. For $p = \infty$, $c = \operatorname{sign}(\sigma(A))$ and $B = \operatorname{polar}(A)$. ∎

Applied to the spectral norm, the theorem says that $-\operatorname{polar}(G)$ is a steepest unit step for a matrix gradient G, which is the update of Muon (Chapter 3).[2]

[2] von Neumann (1937) proved the inequality for square matrices, in the paper that identified the unitarily invariant norms with the symmetric gauge functions of the singular values.

When G has full rank, the polar factor is also the semi-orthogonal matrix closest to G.

Theorem A.8 (The polar factor). *Let $G \in \mathbb{R}^{m\times n}$ have rank r and compact decomposition $G = U\operatorname{diag}(\sigma)V^\top$.*

(i) $G = \operatorname{polar}(G)(G^\top G)^{1/2} = (GG^\top)^{1/2}\operatorname{polar}(G)$. *The matrix* $\operatorname{polar}(G)$ *has r singular values equal to 1 and the others 0; it is semi-orthogonal* ($\operatorname{polar}(G)^\top\operatorname{polar}(G) = I_n$ *if* $m \geq n$, $\operatorname{polar}(G)\operatorname{polar}(G)^\top = I_m$ *if* $m \leq n$) *if and only if* $r = \min\{m, n\}$, *and then* $\operatorname{polar}(G) = G(G^\top G)^{-1/2}$ *for* $m \geq n$ *and* $\operatorname{polar}(G) = (GG^\top)^{-1/2}G$ *for* $m \leq n$.

(ii) Let $m \geq n$. Every $Q \in \mathbb{R}^{m\times n}$ with $Q^\top Q = I_n$ satisfies

$$\|G - Q\|_F^2 \geq \sum_{i=1}^{r}(\sigma_i - 1)^2 + (n - r),$$

with equality if and only if $QV = U$. If $r = n$, then $\operatorname{polar}(G)$ *is the unique nearest semi-orthogonal matrix in the Frobenius norm; if $r < n$, then the nearest ones are the semi-orthogonal Q with $QV = U$. For $m < n$ the same holds with $QQ^\top = I_m$, $U^\top Q = V^\top$, and $m - r$ for $n - r$.*

(iii) The semi-orthogonal matrices nearest to G in the Frobenius norm are also nearest in every unitarily invariant norm.

Proof of (i) and (ii). The plan is to read (i) off the compact decomposition and to reduce (ii) to a trace that the Cauchy–Schwarz inequality bounds term by term. (i) From $G^\top G = V\operatorname{diag}(\sigma^2)V^\top$ we get $(G^\top G)^{1/2} = V\operatorname{diag}(\sigma)V^\top$ and $\operatorname{polar}(G)(G^\top G)^{1/2} = UV^\top V\operatorname{diag}(\sigma)V^\top = G$; the other identity is the same computation with $GG^\top$. The expression $UV^\top$ is a compact decomposition with r singular values 1, and $\operatorname{polar}(G)^\top\operatorname{polar}(G) = VV^\top$, the projection onto the column span of V, equals I_n exactly when $r = n$; then $G(G^\top G)^{-1/2} = U\operatorname{diag}(\sigma)V^\top V\operatorname{diag}(\sigma)^{-1}V^\top = UV^\top$. The case $m \leq n$ is the transpose.

(ii) Since $\operatorname{tr}(Q^\top Q) = n$, $\|G - Q\|_F^2 = \|G\|_F^2 - 2\operatorname{tr}(G^\top Q) + n$ with $\operatorname{tr}(G^\top Q) = \sum_{i\leq r}\sigma_i u_i^\top Q v_i$. Because $\|Qv\|_2^2 = v^\top Q^\top Q v = \|v\|_2^2$, the Cauchy–Schwarz inequality gives $u_i^\top Q v_i \leq 1$, with equality if and only if $Qv_i = u_i$. Hence $\|G - Q\|_F^2 \geq \sum_i \sigma_i^2 - 2\sum_i \sigma_i + n$, which is the bound, with equality if and only if $QV = U$. Such Q exist: complete $(v_i)_{i\leq r}$ to an orthonormal basis $(v_i)_{i\leq n}$ of $\mathbb{R}^n$ and $(u_i)_{i\leq r}$ to an orthonormal family $(u_i)_{i\leq n}$ in $\mathbb{R}^m$, possible as $m \geq n$, and put $Q = \sum_{i\leq n} u_i v_i^\top$. If $r = n$, then V is orthogonal and $QV = U$ gives $Q = UV^\top$. For $m < n$ apply this to $G^\top$. ∎

Part (iii) is due to Fan and Hoffman (1955) for square matrices; Higham (2008, Chapter 8) treats the rectangular case. It is not used in this book.

The Löwner order. The *Löwner order* is preserved by congruence: $A \preceq B$ implies $CAC^\top \preceq CBC^\top$ for every C of compatible size, with the converse when C is invertible. A symmetric N satisfies $N \preceq I$ if and only if $\lambda_{\max}(N) \leq 1$, and for every matrix X, $\|X\|_{\mathrm{op}} \leq 1$ if and only if $X^\top X \preceq I$, if

and only if $XX^\top \preceq I$. A function f on $[0,\infty)$ is *operator monotone* if $0 \preceq A \preceq B$ implies $f(A) \preceq f(B)$ for symmetric matrices of every size.

Theorem A.9 (Löwner–Heinz inequality). *Let $0 \preceq A \preceq B$ and $0 \le r \le 1$. Then $A^r \preceq B^r$, and if $A \succ 0$, then also $B^{-r} \preceq A^{-r}$. For every $r > 1$ the first statement fails for some pair; for $r = 2$, $A = \left(\begin{smallmatrix}1&1\\1&1\end{smallmatrix}\right) \preceq B = \left(\begin{smallmatrix}2&1\\1&1\end{smallmatrix}\right)$, but $B^2 - A^2 = \left(\begin{smallmatrix}3&1\\1&0\end{smallmatrix}\right)$ has determinant -1.*

Proof. The plan is to turn each inequality $A^s \preceq B^s$ into an operator-norm bound, to derive the case $r = \frac{1}{2}$ from the case $r = 1$, and to obtain every $r \in [0,1]$ by repeating that step. Assume first that $B \succ 0$. For $s \ge 0$, congruence with $B^{-s/2}$ shows that $A^s \preceq B^s$ is equivalent to $B^{-s/2}A^sB^{-s/2} \preceq I$, that is, to $\|A^{s/2}B^{-s/2}\|_{\mathrm{op}} \le 1$. Let $\mathcal{S}$ be the set of $s \in [0,1]$ for which $A^s \preceq B^s$ for all such pairs, of every size; it contains 0 and 1. If $r, s \in \mathcal{S}$, then $X = A^{r/2}B^{-r/2}$ and $Y = A^{s/2}B^{-s/2}$ have operator norm at most 1, and so does $X^\top Y = B^{-r/2}A^{(r+s)/2}B^{-s/2}$. This matrix is similar to $N = B^{(r-s)/4}(X^\top Y)B^{(s-r)/4} = B^{-(r+s)/4}A^{(r+s)/2}B^{-(r+s)/4}$, so the eigenvalues of the positive semidefinite N are at most 1, and $N \preceq I$, which by congruence is $A^{(r+s)/2} \preceq B^{(r+s)/2}$. With $r = 0$ and $s = 1$ this is the case $\frac{1}{2}$: $\|A^{1/2}B^{-1/2}\|_{\mathrm{op}} \le 1$ gives $B^{-1/4}A^{1/2}B^{-1/4} \preceq I$. By induction $\mathcal{S}$ contains the dyadic rationals of $[0,1]$. For $r \in (0,1]$ and positive dyadic $r_j \to r$, $x^{r_j} \to x^r$ for all $x \ge 0$, so $A^{r_j} \to A^r$ and $B^{r_j} \to B^r$, and $r \in \mathcal{S}$ as the positive semidefinite cone is closed. If B is singular, apply the result to $A \preceq B + \varepsilon I$ and let $\varepsilon \downarrow 0$.

For the inverses, let $0 \prec A \preceq B$, so that $X = A^{1/2}B^{-1/2}$ has $\|X\|_{\mathrm{op}} \le 1$. Then $XX^\top = A^{1/2}B^{-1}A^{1/2} \preceq I$, and congruence with $A^{-1/2}$ gives $B^{-1} \preceq A^{-1}$; applied to $A^r \preceq B^r$, this gives $B^{-r} \preceq A^{-r}$. In the example, $B - A = \mathrm{diag}(1,0)$. The case $r = 2$ rules out every $r > 1$: if $A^r \preceq B^r$ held for all pairs with $B \succ 0$, then so would $A^{r^j} \preceq B^{r^j}$ for all j (apply it repeatedly), and halving, which never used $s \le 1$, and continuity would give every exponent in $[0, r^j]$, which contains 2 for large j. ∎

So $t \mapsto t^r$ is operator monotone for $0 \le r \le 1$ and $t \mapsto t^{-r}$ reverses the order; this keeps the square roots and fourth roots of AdaGrad and Shampoo monotone as gradients accumulate (Chapters 6 and 7).[3]

[3] The inequality is due to Löwner (1934), who characterized all operator monotone functions, and to Heinz (1951). The proof by repeated halving is one of several; Bhatia (1997, Chapter V) gives the general theory.

Kronecker products. For $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{p\times q}$, the *Kronecker product* $A \otimes B \in \mathbb{R}^{mp\times nq}$ is the block matrix with blocks $A_{ij}B$, and $\mathrm{vec}\colon \mathbb{R}^{m\times n} \to \mathbb{R}^{mn}$ stacks the columns, $\mathrm{vec}(X) = (X_{11}, \dots, X_{m1}, X_{12}, \dots, X_{mn})^\top$.

Lemma A.10 (Kronecker products and vectorization). *For matrices of compatible sizes:*

(i) $(A\otimes B)(C\otimes D) = (AC)\otimes(BD)$, $(A\otimes B)^\top = A^\top \otimes B^\top$, *and* $\mathrm{tr}(A\otimes B) = \mathrm{tr}(A)\,\mathrm{tr}(B)$; *in particular $A \otimes I$ and $I \otimes B$ commute, with product $A \otimes B$.*

(ii) With column stacking,

$$\mathrm{vec}(AXB) = (B^\top \otimes A)\,\mathrm{vec}(X), \qquad \mathrm{vec}(uw^\top) = w \otimes u,$$

and $\langle \mathrm{vec}(X), \mathrm{vec}(Y)\rangle = \mathrm{tr}(X^\top Y)$. *For symmetric $A \in \mathbb{R}^{m\times m}$ and*

$B \in \mathbb{R}^{n\times n}$, *the matrix* $B\otimes A$ *acts on* $\operatorname{vec}(X)$, $X \in \mathbb{R}^{m\times n}$, *as* $X \mapsto AXB$, *and* $\operatorname{vec}(X)^\top (B\otimes A)\operatorname{vec}(X) = \operatorname{tr}(X^\top AXB)$.

(iii) *If* $A = \sum_i \lambda_i(A) a_i a_i^\top$ *and* $B = \sum_j \lambda_j(B) b_j b_j^\top$ *are spectral decompositions, then* $A\otimes B = \sum_{i,j} \lambda_i(A)\lambda_j(B)(a_i\otimes b_j)(a_i\otimes b_j)^\top$ *is one; so the eigenvalues of* $A\otimes B$ *are the products* $\lambda_i(A)\lambda_j(B)$, *and* $A\otimes B \succeq 0$ *when* $A, B \succeq 0$.

(iv) *For* $A, B \succeq 0$ *and* $r > 0$, $(A\otimes B)^r = A^r \otimes B^r$; *for* $A, B \succ 0$ *this holds for every real* r.

(v) *If* $0 \preceq A \preceq A'$ *and* $0 \preceq B \preceq B'$, *then* $A\otimes B \preceq A'\otimes B'$.

Proof. The idea is to check each identity on blocks or on rank-one matrices and to extend it by linearity. (i) Block (i,j) of $(A\otimes B)(C\otimes D)$ is $\sum_l A_{il}C_{lj}BD$; the transpose is taken blockwise; the diagonal blocks of $A\otimes B$ are $A_{ii}B$. (ii) Column l of $uw^\top$ is $w_l u$, so $\operatorname{vec}(uw^\top) = w\otimes u$. With $X = \sum_j x_j e_j^\top$ we have $\operatorname{vec}(X) = \sum_j e_j \otimes x_j$ and $AXB = \sum_j (Ax_j)(B^\top e_j)^\top$, so by the mixed product $\operatorname{vec}(AXB) = \sum_j (B^\top e_j)\otimes(Ax_j) = (B^\top \otimes A)\operatorname{vec}(X)$. The last claims from $(B\otimes A)\operatorname{vec}(X) = \operatorname{vec}(AXB^\top)$ and $B = B^\top$. (iii) By the mixed product the vectors $a_i \otimes b_j$ are orthonormal and $(a_i a_i^\top)\otimes(b_j b_j^\top) = (a_i\otimes b_j)(a_i\otimes b_j)^\top$; expand $A\otimes B$ bilinearly. (iv) Apply $x \mapsto x^r$ to (iii), using $(xy)^r = x^r y^r$ for $x, y \geq 0$. (v) $A'\otimes B' - A\otimes B = (A'-A)\otimes B' + A\otimes(B'-B)$, a sum of products of positive semidefinite matrices. ∎

Under column stacking, the matrix that multiplies X from the right comes first: Shampoo's step $L^{-1/4}GR^{-1/4}$, with $L \in \mathbb{R}^{m\times m}$ accumulating $GG^\top$ and $R \in \mathbb{R}^{n\times n}$ accumulating $G^\top G$, is $(R^{-1/4}\otimes L^{-1/4})\operatorname{vec}(G)$.[4]

[4] Some papers stack rows. With $\operatorname{rvec}(X) = \operatorname{vec}(X^\top)$ the identity reads $\operatorname{rvec}(AXB) = (A\otimes B^\top)\operatorname{rvec}(X)$, and the factors trade places; check the convention before comparing formulas from different sources.

The matrix geometric mean. The geometric mean of two positive numbers is the solution $x > 0$ of $x\,a^{-1}x = b$. For matrices that do not commute, AB is not even symmetric, and the right generalization keeps the equation. It rests on Schur complements.

Lemma A.11 (Schur complements). *If* $A \succ 0$ *and* C *is symmetric, then* $\left(\begin{smallmatrix} A & X \\ X^\top & C \end{smallmatrix}\right) \succeq 0$ *if and only if* $C \succeq X^\top A^{-1} X$.

Proof. For the invertible matrix $E = \left(\begin{smallmatrix} I & -A^{-1}X \\ 0 & I \end{smallmatrix}\right)$, a direct multiplication gives $E^\top \left(\begin{smallmatrix} A & X \\ X^\top & C \end{smallmatrix}\right) E = \operatorname{diag}\big(A,\ C - X^\top A^{-1} X\big)$, and congruence by an invertible matrix preserves positive semidefiniteness in both directions. ∎

Theorem A.12 (The matrix geometric mean). *For positive definite* $A, B \in \mathbb{R}^{n\times n}$ *let* $A\#B = A^{1/2}\big(A^{-1/2}BA^{-1/2}\big)^{1/2}A^{1/2}$.

(i) $A\#B$ *is the unique positive definite solution* X *of* $XA^{-1}X = B$. *Hence* $A\#B = B\#A$ *and* $(sA)\#(sB) = s(A\#B)$ *for* $s > 0$.

(ii) *(Variational characterization.)* $A\#B$ *is the largest symmetric* X, *in the*

Löwner order, such that

$$\begin{pmatrix} A & X \\ X & B \end{pmatrix} \succeq 0 :$$

it satisfies the condition, and every symmetric X that satisfies it has $X \preceq A\#B$. Consequently $A\#B$ is monotone in each argument, and $A\#B \preceq \frac{1}{2}(A+B)$.

(iii) (Joint concavity.) For positive definite A_1, A_2, B_1, B_2 and $s \in [0,1]$, $A_1\#B_1 + A_2\#B_2 \preceq (A_1+A_2)\#(B_1+B_2)$ and

$$s(A_1\#B_1) + (1-s)(A_2\#B_2) \preceq \big(sA_1 + (1-s)A_2\big)\#\big(sB_1 + (1-s)B_2\big).$$

(iv) (Commuting case.) If $AB = BA$, then $A\#B = (AB)^{1/2} = A^{1/2}B^{1/2}$.

Proof. The plan is to derive everything from the equation in (i) and from the block test of Lemma A.11. (i) Put $C = A^{-1/2}BA^{-1/2} \succ 0$. The matrix $X = A^{1/2}C^{1/2}A^{1/2}$ is positive definite with $XA^{-1}X = A^{1/2}CA^{1/2} = B$. Conversely, if $X \succ 0$ solves the equation, then $Y = A^{-1/2}XA^{-1/2} \succ 0$ satisfies $Y^2 = C$, so $Y = C^{1/2}$ and $X = A\#B$. If $Z = B\#A$, then $ZB^{-1}Z = A$; inverting gives $Z^{-1}BZ^{-1} = A^{-1}$, that is, $ZA^{-1}Z = B$, so $Z = A\#B$. And sX solves the equation for (sA, sB).

(ii) By Lemma A.11 the condition is $XA^{-1}X \preceq B$, which after congruence with $A^{-1/2}$ reads $Y^2 \preceq C$ for the symmetric $Y = A^{-1/2}XA^{-1/2}$. Then $(Y^2)^{1/2} \preceq C^{1/2}$ by Theorem A.9 with $r = \frac{1}{2}$, and $Y \preceq (Y^2)^{1/2}$ because each eigenvalue y of Y satisfies $y \le |y|$; so $Y \preceq C^{1/2}$, and congruence with $A^{1/2}$ gives $X \preceq A\#B$. The choice $X = A\#B$, that is $Y = C^{1/2}$, satisfies the condition. If $A \preceq A'$ and $B \preceq B'$, then adding $\operatorname{diag}(A'-A, B'-B) \succeq 0$ shows that $A\#B$ satisfies the condition for (A', B'), so $A\#B \preceq A'\#B'$. For every admissible X and every x, the vector $(x, -x)$ gives $x^\top Ax - 2x^\top Xx + x^\top Bx \ge 0$, that is, $X \preceq \frac{1}{2}(A+B)$.

(iii) The block matrices of (ii) for $(A_1, B_1, A_1\#B_1)$ and $(A_2, B_2, A_2\#B_2)$ are positive semidefinite, hence so is their sum, the block matrix for $(A_1+A_2, B_1+B_2, A_1\#B_1 + A_2\#B_2)$; now apply (ii). For $0 < s < 1$ the second inequality follows from the first, applied to $sA_1, sB_1, (1-s)A_2, (1-s)B_2$, and from (i); for $s \in \{0,1\}$ it is an equality.

(iv) The matrices $A^{1/4}$ and $A^{1/2}$ are polynomials in A, and $B^{1/2}$ is a polynomial in B, so they commute. Hence $X = A^{1/2}B^{1/2} = A^{1/4}B^{1/2}A^{1/4}$ is positive definite and $XA^{-1}X = A^{1/2}B^{1/2}A^{-1/2}B^{1/2} = B$, so $X = A\#B$ by (i); and $X^2 = AB$. ∎

Part (ii) turns an inequality about square roots, which are hard to handle, into one about a block matrix, which is linear in its entries. The next corollary is the form in which Chapter 7 uses it.

Corollary A.13 (Kronecker square roots are jointly concave). *For posi-*

tive semidefinite $A_1, \dots, A_T \in \mathbb{R}^{m\times m}$ and $B_1, \dots, B_T \in \mathbb{R}^{n\times n}$,

$$\sum_{t=1}^{T} A_t^{1/2} \otimes B_t^{1/2} \preceq \Big(\sum_{t=1}^{T} A_t\Big)^{1/2} \otimes \Big(\sum_{t=1}^{T} B_t\Big)^{1/2}.$$

Proof. Suppose first that all A_t and B_t are positive definite. The matrices $A_t \otimes I$ and $I \otimes B_t$ commute, with product $A_t \otimes B_t$, so by Theorem A.12(iv) and Lemma A.10(iv) their geometric mean is

$$(A_t \otimes I)\#(I \otimes B_t) = (A_t \otimes B_t)^{1/2} = A_t^{1/2} \otimes B_t^{1/2}.$$

Summing over t and using the first inequality of Theorem A.12(iii) $T-1$ times gives $\sum_t A_t^{1/2} \otimes B_t^{1/2} \preceq \big((\sum_t A_t) \otimes I\big)\#\big(I \otimes \sum_t B_t\big)$, which is the right side by the same computation. In general, replace A_t and B_t by $A_t + \varepsilon I$ and $B_t + \varepsilon I$ and let $\varepsilon \downarrow 0$. ∎

For $T = 1$ the corollary is an equality; its content is that $(A, B) \mapsto A^{1/2} \otimes B^{1/2}$ is superadditive, a case of a theorem of Ando (1979).[5] It underlies the comparison, in the analysis of Shampoo, between the Kronecker-factored preconditioner and the full-matrix AdaGrad preconditioner (Gupta et al. 2018), which Chapter 7 proves.

[5] Ando (1979) proved the variational characterization of $A\#B$, its joint concavity, and the joint concavity of $(A, B) \mapsto A^{1-s} \otimes B^{s}$ for $0 \le s \le 1$. Bhatia (2007, Chapter 4) treats matrix means.

A.3 Linear recurrences, Lyapunov equations, and Grönwall's inequality

On a quadratic, gradient descent, the heavy-ball method, and Adam with a frozen preconditioner are linear recurrences, one for each eigendirection of the (preconditioned) Hessian, of order one or two and possibly driven by noise. Stability thresholds are statements about the roots of their characteristic polynomials, and noise floors are their stationary variances. The *spectral radius* $\rho(A)$ of a square matrix is the largest absolute value of its complex eigenvalues.

Lemma A.14 (Stability of linear recurrences). *Let $A \in \mathbb{R}^{d\times d}$, and let a_0, a_1, a, and β be real numbers.*

(i) For every $\varepsilon > 0$ there is C_ε with $\|A^t\|_{\mathrm{op}} \le C_\varepsilon(\rho(A) + \varepsilon)^t$ for all $t \ge 0$. Hence $x_{t+1} = Ax_t$ converges to 0 from every x_0 if and only if $\rho(A) < 1$, and then geometrically fast; if $\rho(A) \ge 1$, then some real x_0 gives iterates that do not converge to 0, and if $\rho(A) > 1$, then some real x_0 gives unbounded iterates.

*(ii) (*Schur–Cohn–Jury test.*) Both roots of $z^2 + a_1 z + a_0$ lie in the open unit disk if and only if $|a_0| < 1$ and $|a_1| < 1 + a_0$.*

(iii) (Momentum form.) Both roots of $z^2 - (1 + \beta - a)z + \beta$ lie in the open unit disk if and only if $|\beta| < 1$ and $0 < a < 2 + 2\beta$. Equivalently, $y_{t+1} = (1 + \beta - a)y_t - \beta y_{t-1}$ converges to 0 from every initial pair (y_0, y_{-1}) if and only if these conditions hold.

Proof. (i) The plan is to make A almost diagonal by a change of basis. By Schur's triangularization $A = QRQ^*$ with Q unitary and R upper triangular with the eigenvalues on its diagonal (Horn and Johnson 2013). For $D = \operatorname{diag}(1, \delta, \dots, \delta^{d-1})$ with $\delta \in (0, 1]$, the matrix $D^{-1}RD$ has entries $\delta^{j-i}R_{ij}$: the diagonal of R, and above it entries of absolute value at most $\delta|R_{ij}|$. So $\|D^{-1}RD\|_{\mathrm{op}} \le \rho(A) + \delta\|R\|_F \le \rho(A) + \varepsilon$ for $\delta = \min\{1, \varepsilon/\|R\|_F\}$ (every δ works if $R = 0$). With $S = QD$, $A^t = S(D^{-1}RD)^t S^{-1}$ and $\|A^t\|_{\mathrm{op}} \le \|S\|_{\mathrm{op}}\|S^{-1}\|_{\mathrm{op}}(\rho(A) + \varepsilon)^t \le \delta^{1-d}(\rho(A) + \varepsilon)^t$, the norms being over complex vectors. If $\rho(A) < 1$, take $\rho(A) + \varepsilon < 1$. Conversely, let λ_1 be an eigenvalue with $|\lambda_1| = \rho(A) \ge 1$ and eigenvector $v = a + ib \ne 0$, a, b real. Then $\|A^t a\|_2 + \|A^t b\|_2 \ge |\lambda_1|^t \|v\|_2$, so one of the real sequences $A^t a$, $A^t b$ does not tend to 0, and if $|\lambda_1| > 1$, then one of them is unbounded.

(ii) Let $p(z) = z^2 + a_1 z + a_0 = (z - z_1)(z - z_2)$. If $|z_1|, |z_2| < 1$, then $|a_0| = |z_1 z_2| < 1$, and $p(1) = (1 - z_1)(1 - z_2)$ and $p(-1) = (1 + z_1)(1 + z_2)$ are positive, as products of positive real factors or as $|1 \mp z_1|^2$ for a conjugate pair; so $1 \pm a_1 + a_0 > 0$, which is $|a_1| < 1 + a_0$. Conversely, let $|a_0| < 1$ and $|a_1| < 1 + a_0$, so that $p(\pm 1) > 0$. A conjugate pair has $|z_1|^2 = z_1 z_2 = a_0 < 1$. Real roots $z_1 \le z_2$ lie on either side of the vertex $-a_1/2 \in (-1, 1)$ (as $|a_1| < 2$); since p increases on $[-a_1/2, \infty)$ and $p(1) > 0 = p(z_2)$, $z_2 < 1$, and symmetrically $z_1 > -1$.

(iii) With $a_0 = \beta$ and $a_1 = -(1 + \beta - a)$ the conditions of (ii) are $|\beta| < 1$ and $|1 + \beta - a| < 1 + \beta$, that is, $0 < a < 2 + 2\beta$. The recurrence is $x_{t+1} = Ax_t$ for $x_t = (y_t, y_{t-1})$ with the companion matrix $A = \left(\begin{smallmatrix} 1+\beta-a & -\beta \\ 1 & 0 \end{smallmatrix}\right)$, whose characteristic polynomial is $z^2 - (1 + \beta - a)z + \beta$, so (i) applies. ∎

Part (iii) is the form used in Chapters 5 and 8. On a mode of curvature h, the heavy-ball method $\theta_{t+1} = \theta_t - \eta h\theta_t + \beta(\theta_t - \theta_{t-1})$ is (iii) with $a = \eta h$, so it is stable if and only if $|\beta| < 1$ and $0 < \eta h < 2 + 2\beta$. The moving-average form, $m_t = \beta m_{t-1} + (1 - \beta)h\theta_t$ and $\theta_{t+1} = \theta_t - \eta m_t$, satisfies

$$\theta_{t+1} - \theta_t = \beta(\theta_t - \theta_{t-1}) - \eta(1 - \beta)h\theta_t,$$

which is (iii) with $a = \eta(1 - \beta)h$; for $0 \le \beta < 1$ it is stable if and only if $0 < \eta h < (2 + 2\beta)/(1 - \beta)$, a threshold equal to 38 at $\beta = 0.9$.[6]

Lemma A.15 (Linear recursions driven by noise). *Let $x_{t+1} = A_t x_t + \xi_t$, with $\mathbb{E}\|x_0\|_2^2 < \infty$, deterministic A_t, and noise with $\mathbb{E}[\xi_t \mid \mathcal{F}_t] = 0$ and $\mathbb{E}[\xi_t \xi_t^\top] = Q_t$, where $\mathcal{F}_t$ is generated by $x_0, \xi_0, \dots, \xi_{t-1}$. Let $C_t = \mathbb{E}[x_t x_t^\top]$.*

(i) $\mathbb{E}x_{t+1} = A_t \mathbb{E}x_t$ and $C_{t+1} = A_t C_t A_t^\top + Q_t$, and the covariance $C_t - (\mathbb{E}x_t)(\mathbb{E}x_t)^\top$ obeys the same recursion as C_t.

(ii) If $A_t = A$ with $\rho(A) < 1$ and $Q_t = Q$, then the discrete Lyapunov equation $\Sigma = A\Sigma A^\top + Q$ has the unique solution $\Sigma_\infty = \sum_{j \ge 0} A^j Q (A^\top)^j \ge 0$, and $C_t - \Sigma_\infty = A^t(C_0 - \Sigma_\infty)(A^\top)^t \to 0$ geometrically fast.

(iii) (Order one.) If $x_{t+1} = a_t x_t + \xi_t$ is scalar and $q_t = \mathbb{E}\xi_t^2$, then

$$\mathbb{E}x_T^2 = \Big(\prod_{t<T} a_t^2\Big)\mathbb{E}x_0^2 + \sum_{s<T} q_s \prod_{s<u<T} a_u^2;$$

for constant $|a| < 1$ and q this is $a^{2T}\mathbb{E}x_0^2 + q(1 - a^{2T})/(1 - a^2) \to$

[6] Test (ii) is the quadratic case of the criterion of Schur and Cohn for the roots of a polynomial to lie in the unit disk, which Jury put in the tabular form used in control engineering; his book (Jury 1964) presents it.

$q/(1-a^2)$.

(iv) *(Order two.) Let* $y_{t+1} = c_1 y_t + c_2 y_{t-1} + e_t$ *with square-integrable initial values,* $\mathbb{E}[e_t \mid \mathcal{F}_t] = 0$ *and* $\mathbb{E}e_t^2 = s^2$ *for a filtration with* y_t, y_{t-1} *measurable for* $\mathcal{F}_t$, *and stable coefficients,* $|c_2| < 1$ *and* $|c_1| < 1 - c_2$. *Then* $\mathbb{E}y_t^2 \to \gamma_0$ *and* $\mathbb{E}[y_t y_{t-1}] \to \gamma_1$, *with*

$$\gamma_0 = \frac{(1-c_2)\,s^2}{(1+c_2)\big((1-c_2)^2 - c_1^2\big)}, \qquad \gamma_1 = \frac{c_1\gamma_0}{1-c_2}. \tag{A.4}$$

Proof. The plan is to expand the second moment, in which the noise enters only through its covariance, and to solve the resulting linear recursion. (i) In the expansion of $x_{t+1}x_{t+1}^\top$ the cross terms vanish in expectation, because x_t is $\mathcal{F}_t$-measurable and so $\mathbb{E}[x_t\xi_t^\top] = \mathbb{E}\big[x_t\mathbb{E}[\xi_t \mid \mathcal{F}_t]^\top\big] = 0$; the mean and the covariance follow in the same way. (ii) By Lemma A.14(i), with $\rho(A) + \varepsilon < 1$, the terms of the series are bounded in norm by $C_\varepsilon^2(\rho(A)+\varepsilon)^{2j}\|Q\|_{\mathrm{op}}$, so it converges; it is positive semidefinite and solves the equation. Subtracting the equation from (i) gives $C_{t+1} - \Sigma_\infty = A(C_t - \Sigma_\infty)A^\top$, hence the formula; and two solutions differ by some $Z = AZA^\top = A^tZ(A^\top)^t \to 0$. (iii) Unroll (i). (iv) The vector (y_t, y_{t-1}) obeys (i) with the companion matrix $A = \left(\begin{smallmatrix} c_1 & c_2 \\ 1 & 0 \end{smallmatrix}\right)$ and $Q = \mathrm{diag}(s^2, 0)$; the polynomial $z^2 - c_1 z - c_2$ has its roots in the open unit disk by Lemma A.14(ii) with $a_1 = -c_1$, $a_0 = -c_2$, so (ii) applies. The second row of A is $(1, 0)$, so the $(2,2)$ entry of the Lyapunov equation makes both diagonal entries of Σ_∞ equal, $\Sigma_\infty = \left(\begin{smallmatrix} \gamma_0 & \gamma_1 \\ \gamma_1 & \gamma_0 \end{smallmatrix}\right)$, and its $(1,2)$ and $(1,1)$ entries read $\gamma_1 = c_1\gamma_0 + c_2\gamma_1$ and $\gamma_0 = (c_1^2 + c_2^2)\gamma_0 + 2c_1c_2\gamma_1 + s^2$. Substituting the first into the second and using $(1-c_2)(1-c_1^2-c_2^2) - 2c_1^2c_2 = (1+c_2)\big((1-c_2)^2 - c_1^2\big)$ gives (A.4); the denominators are positive under the stability conditions. ■

Example A.16 (Heavy ball in noise). Let $\theta_{t+1} = \theta_t - \eta(h\theta_t + \zeta_t) + \beta(\theta_t - \theta_{t-1})$ with $h > 0$, $\mathbb{E}[\zeta_t \mid \mathcal{F}_t] = 0$, and $\mathbb{E}\zeta_t^2 = \sigma^2$, for a filtration as in Lemma A.15(iv). Part (iv) applies with $c_1 = 1 + \beta - \eta h$, $c_2 = -\beta$, and $s^2 = \eta^2\sigma^2$, and $(1-c_2)^2 - c_1^2 = \eta h(2 + 2\beta - \eta h)$; so for $|\beta| < 1$ and $0 < \eta h < 2 + 2\beta$,

$$\lim_{t\to\infty} \mathbb{E}\theta_t^2 = \frac{\eta\sigma^2(1+\beta)}{h(1-\beta)(2+2\beta-\eta h)}.$$

At $\beta = 0$ this is the stationary variance $\eta\sigma^2/(h(2-\eta h))$ of stochastic gradient descent, also given by (iii); to first order in η, heavy ball has the variance of stochastic gradient descent with step $\eta/(1-\beta)$ (Chapters 5 and 11).

Differential and difference inequalities. The last tool, named after Gronwall (1919), compares a function that grows at most linearly in itself with the exponential solving the corresponding equation.

Theorem A.17 (Grönwall's inequality). *Let* $T > 0$.

(i) *(Differential form.) If* u *is absolutely continuous on* $[0, T]$ *and* $u' \le$

$bu + a$ almost everywhere, for integrable a and b, then with $B(t) = \int_0^t b$,

$$u(t) \le u(0)e^{B(t)} + \int_0^t a(s)e^{B(t)-B(s)}\,\mathrm{d}s;$$

for constants $b > 0$ and a, $u(t) \le \big(u(0) + a/b\big)e^{bt} - a/b$.

(ii) (Integral form.) If u, a, and b are continuous on $[0,T]$, $b \ge 0$, and $u(t) \le a(t) + \int_0^t b(s)u(s)\,\mathrm{d}s$ for all t, then $u(t) \le a(t) + \int_0^t a(s)b(s)e^{\int_s^t b}\,\mathrm{d}s$; if moreover a is nondecreasing, then $u(t) \le a(t)\exp\big(\int_0^t b\big)$.

(iii) (Discrete form.) If $b_t \ge 0$ and $u_{t+1} \le (1 + b_t)u_t + a_t$ for all $t \ge 0$, then $u_t \le u_0 \prod_{s<t}(1+b_s) + \sum_{s<t} a_s \prod_{s<r<t}(1+b_r)$, which is at most $\big(u_0 + \sum_{s<t} a_s\big)\exp\big(\sum_{s<t} b_s\big)$ when u_0 and all a_s are nonnegative.

Proof. (i) The function $v = ue^{-B}$ is absolutely continuous with $v' = (u' - bu)e^{-B} \le ae^{-B}$ almost everywhere; integrate and multiply by $e^{B(t)}$. For constants, $\int_0^t ae^{b(t-s)}\,\mathrm{d}s = (a/b)(e^{bt} - 1)$. (ii) For $R(t) = \int_0^t bu$, $R' = bu \le ba + bR$ because $b \ge 0$, and (i) with $R(0) = 0$ bounds R; then use $u \le a + R$. If a is nondecreasing, then the integral is at most $a(t)\int_0^t b(s)e^{\int_s^t b}\,\mathrm{d}s = a(t)\big(e^{\int_0^t b} - 1\big)$. (iii) Induction on t, and $1 + b \le e^b$. ∎

Chapter 10 applies the differential form to a function of s built from $\nabla\mathcal{L}(\theta + su)$, which is Lipschitz, hence absolutely continuous and differentiable almost everywhere, under a Hessian bound growing with the gradient.

A.4 Smoothness and the Polyak–Łojasiewicz inequality

A differentiable $\mathcal{L}\colon \mathbb{R}^d \to \mathbb{R}$ is L-smooth with respect to $\|\cdot\|$, for a constant $L > 0$, if $\|\nabla\mathcal{L}(\theta) - \nabla\mathcal{L}(\theta')\|_* \le L\|\theta - \theta'\|$ for all θ, θ'; the gradient is a functional, so it is measured in the dual norm.

Lemma A.18 (Descent lemma in a general norm). *Let $\mathcal{L}$ be L-smooth with respect to $\|\cdot\|$.*

(i) For all θ and u, $|\mathcal{L}(\theta + u) - \mathcal{L}(\theta) - \langle\nabla\mathcal{L}(\theta), u\rangle| \le \frac{L}{2}\|u\|^2$; in particular

$$\mathcal{L}(\theta + u) \le \mathcal{L}(\theta) + \langle\nabla\mathcal{L}(\theta), u\rangle + \frac{L}{2}\|u\|^2. \tag{A.5}$$

(ii) If $\mathcal{L}$ is bounded below by $\mathcal{L}^\star$, then $\|\nabla\mathcal{L}(\theta)\|_*^2 \le 2L\big(\mathcal{L}(\theta) - \mathcal{L}^\star\big)$ for every θ.

(iii) (Converse under convexity.) If $\mathcal{L}$ is convex and differentiable and (A.5) holds for all θ, u, then $\mathcal{L}$ is L-smooth with respect to $\|\cdot\|$, and for all

θ, θ',

$$\mathcal{L}(\theta') \geq \mathcal{L}(\theta) + \langle \nabla\mathcal{L}(\theta), \theta' - \theta\rangle + \frac{1}{2L}\|\nabla\mathcal{L}(\theta') - \nabla\mathcal{L}(\theta)\|_*^2,$$

$$\langle \nabla\mathcal{L}(\theta') - \nabla\mathcal{L}(\theta), \theta' - \theta\rangle \geq \frac{1}{L}\|\nabla\mathcal{L}(\theta') - \nabla\mathcal{L}(\theta)\|_*^2.$$

In the Euclidean norm the second inequality is called co-coercivity.

(iv) *(Hessian test.) A twice continuously differentiable $\mathcal{L}$ is L-smooth with respect to $\|\cdot\|$ if and only if $u^\top \nabla^2\mathcal{L}(\theta)v \leq L$ for all θ and all $\|u\|, \|v\| \leq 1$; where $\nabla^2\mathcal{L}(\theta) \geq 0$ it suffices to take $u = v$. For a quadratic with Hessian $H \geq 0$ the best constant is $\max_{\|u\|\leq 1} u^\top H u$: $\lambda_{\max}(H)$ in the Euclidean norm and $\max_{s\in\{\pm 1\}^d} s^\top H s$ in $\|\cdot\|_\infty$.*

Proof. The plan is to integrate along segments for (i) and (iv) and to minimize the right side of (A.5) for (ii) and (iii). (i) Since the gradient is continuous,

$$\mathcal{L}(\theta + u) - \mathcal{L}(\theta) - \langle \nabla\mathcal{L}(\theta), u\rangle = \int_0^1 \langle \nabla\mathcal{L}(\theta + su) - \nabla\mathcal{L}(\theta), u\rangle \, \mathrm{d}s,$$

and by Hölder's inequality and smoothness the integrand is at most $Ls\|u\|^2$ in absolute value. (ii) By (A.2), the right side of (A.5) at $u = -g^\sharp/L$, $g = \nabla\mathcal{L}(\theta)$, is $\mathcal{L}(\theta) - \|g\|_*^2/(2L)$, and it is at least $\mathcal{L}^\star$.

(iii) We apply the argument of (ii) to $\mathcal{L}$ minus its linearization. Fix θ and let $\phi(y) = \mathcal{L}(y) - \langle \nabla\mathcal{L}(\theta), y\rangle$, a convex function with $\nabla\phi(\theta) = 0$, hence minimized at θ, which satisfies (A.5) with the same L. For every y, by (A.2),

$$\phi(\theta) \leq \min_u \phi(y + u) \leq \phi(y) - \frac{1}{2L}\|\nabla\phi(y)\|_*^2,$$

which at $y = \theta'$ is the first inequality. Adding it to the same inequality with θ and θ' exchanged gives the second, and Hölder's inequality bounds its left side by $\|\nabla\mathcal{L}(\theta') - \nabla\mathcal{L}(\theta)\|_*\|\theta' - \theta\|$, which gives L-smoothness.

(iv) For a symmetric M, $\|Mv\|_* = \max_{\|u\|\leq 1} u^\top M v$. If the condition holds, then $\nabla\mathcal{L}(\theta') - \nabla\mathcal{L}(\theta) = \int_0^1 \nabla^2\mathcal{L}(\theta + s(\theta' - \theta))(\theta' - \theta)\,\mathrm{d}s$ has dual norm at most $L\|\theta' - \theta\|$; conversely, $\nabla^2\mathcal{L}(\theta)v$ is the limit of $(\nabla\mathcal{L}(\theta + sv) - \nabla\mathcal{L}(\theta))/s$. For $M \geq 0$, the Cauchy–Schwarz inequality gives $u^\top M v \leq \max\{u^\top M u, v^\top M v\}$, and the convex function $u \mapsto u^\top H u$ attains its maximum over the cube at a vertex. ∎

Without convexity the converse fails: $-\frac{1}{2}\|\theta\|_2^2$ satisfies (A.5) for every $L > 0$, however small. Two consequences of (A.5), with $g = \nabla\mathcal{L}(\theta)$, start the convergence theory of Chapter 1: the unnormalized step $-g^\sharp/L$, which minimizes its right side, and the normalized step $\eta\,\mathrm{lmo}(g)$ of length at most η (Lemma A.4) guarantee

$$\mathcal{L}\big(\theta - g^\sharp/L\big) \leq \mathcal{L}(\theta) - \frac{\|g\|_*^2}{2L},$$

$$\mathcal{L}\big(\theta + \eta\,\mathrm{lmo}(g)\big) \leq \mathcal{L}(\theta) - \eta\|g\|_* + \frac{L\eta^2}{2}.$$

The only traces of the norm in these bounds are the constant L and the dual norm of the gradient.

Lemma A.19 (The Polyak–Łojasiewicz inequality). *Let $\mathcal{L}$ be differentiable with $\mathcal{L}^\star = \inf \mathcal{L} > -\infty$, let $\Delta(\theta) = \mathcal{L}(\theta) - \mathcal{L}^\star$ and $\mu > 0$. The function $\mathcal{L}$ satisfies the PL inequality with constant μ with respect to $\|\cdot\|$ if $\frac{1}{2}\|\nabla\mathcal{L}(\theta)\|_*^2 \geq \mu\,\Delta(\theta)$ for all θ.*

(i) If a sequence satisfies $\mathcal{L}(\theta_{t+1}) \leq \mathcal{L}(\theta_t) - c_t\|\nabla\mathcal{L}(\theta_t)\|_*^2$ with $c_t \geq 0$, then $\Delta_{t+1} \leq (1 - 2\mu c_t)\Delta_t$ and $\Delta_T \leq \exp\bigl(-2\mu \sum_{t<T} c_t\bigr)\Delta_0$, where $\Delta_t = \Delta(\theta_t)$.

(ii) If $\mathcal{L}$ is moreover L-smooth with respect to $\|\cdot\|$ and not constant, then $\mu \leq L$, and $\theta_{t+1} = \theta_t - \eta\nabla\mathcal{L}(\theta_t)^\sharp$ with $0 < \eta < 2/L$ satisfies $\Delta_t \leq \bigl(1 - 2\mu\eta(1 - L\eta/2)\bigr)^t\Delta_0$, which is $(1 - \mu/L)^t\Delta_0$ at $\eta = 1/L$. In the Euclidean norm this is gradient descent.

(iii) A function that is μ-strongly convex with respect to $\|\cdot\|$, that is, $\mathcal{L}(\theta + u) \geq \mathcal{L}(\theta) + \langle\nabla\mathcal{L}(\theta), u\rangle + \frac{\mu}{2}\|u\|^2$ for all θ, u, satisfies the PL inequality with constant μ.

Proof. (i) The PL inequality gives $\|\nabla\mathcal{L}(\theta_t)\|_*^2 \geq 2\mu\Delta_t$, hence the first claim. Since $\Delta_{t+1} \geq 0$, either $\Delta_t = 0$ or $1 - 2\mu c_t \geq 0$, so $\Delta_{t+1} \leq \max\{0, 1 - 2\mu c_t\}\Delta_t \leq e^{-2\mu c_t}\Delta_t$; multiply. (ii) At a point with $\Delta > 0$, the PL inequality and Lemma A.18(ii) give $2\mu\Delta \leq \|\nabla\mathcal{L}\|_*^2 \leq 2L\Delta$. By (A.5) with $u = -\eta g^\sharp$ and Lemma A.5, $\mathcal{L}(\theta_{t+1}) \leq \mathcal{L}(\theta_t) - \eta(1 - L\eta/2)\|g\|_*^2$, so (i) applies with $c_t = \eta(1 - L\eta/2) \leq 1/(2L)$, and $1 - 2\mu c_t \geq 1 - \mu/L \geq 0$. (iii) By strong convexity and (A.2) with $L = \mu$, $\mathcal{L}^\star \geq \inf_u\{\mathcal{L}(\theta) + \langle g, u\rangle + \frac{\mu}{2}\|u\|^2\} = \mathcal{L}(\theta) - \|g\|_*^2/(2\mu)$. ∎

The PL inequality asks for neither convexity nor a unique minimizer, only that the gradient be large wherever the loss is far from its infimum; least squares satisfies it even when the minimizer is not unique (Exercise A.4). Part (i) applies to every step-size rule that guarantees a decrease proportional to the squared gradient, which is how Chapter 9 combines it with a curvature bound that grows with the loss.[7]

[7] Polyak (1963) introduced the inequality and proved linear convergence of gradient descent under it; Łojasiewicz (1963) proved an inequality of this type for real-analytic functions. Karimi et al. (2016) compared it with other conditions for linear convergence and popularized the name.

A.5 Online convex optimization

In online convex optimization a learner picks x_t in a closed convex set $\mathcal{K} \subseteq \mathbb{R}^d$ at each round $t = 1, \dots, T$, then observes a convex function f_t and a subgradient $g_t \in \partial f_t(x_t)$. Its *regret* against $u \in \mathcal{K}$ is $\sum_t (f_t(x_t) - f_t(u))$, which by convexity is at most the linearized regret $\sum_t \langle g_t, x_t - u\rangle$. The bounds below hold for every sequence of functions, in particular for the stochastic gradients of a training run. Adaptive methods normalize each gradient by the accumulated size of the gradients so far, and the following lemma controls the resulting sums.

Lemma A.20 (Sums of normalized increments). *Let $a_1, \dots, a_T \geq 0$, $S_0 = 0$, and $S_t = \sum_{s \leq t} a_s$.*

(i) $\sum_{t=1}^{T} a_t/\sqrt{S_t} \leq 2\sqrt{S_T}$, where terms with $S_t = 0$ are read as 0.

(ii) *For $\varepsilon > 0$, $\sum_{t=1}^{T} a_t/(\varepsilon + S_t) \le \log(1 + S_T/\varepsilon)$.*

(iii) *(Matrix form.) If $G_0 \succeq 0$ and $G_1, \dots, G_T \succ 0$, then $\sum_{t=1}^{T} \operatorname{tr}\big((G_t - G_{t-1})G_t^{-1/2}\big) \le 2\operatorname{tr}(G_T^{1/2}) - 2\operatorname{tr}(G_0^{1/2})$. In particular, if $G_t = G_{t-1} + g_t g_t^{\top}$ and $G_0 \succ 0$, then $\sum_t g_t^{\top} G_t^{-1/2} g_t \le 2\operatorname{tr}(G_T^{1/2})$.*

Proof. (i) If $a_t > 0$, then $a_t/\sqrt{S_t} = (\sqrt{S_t} - \sqrt{S_{t-1}})(\sqrt{S_t} + \sqrt{S_{t-1}})/\sqrt{S_t} \le 2(\sqrt{S_t} - \sqrt{S_{t-1}})$, and if $a_t = 0$ both sides vanish; the sum telescopes. (ii) Since $1/(\varepsilon + x) \ge 1/(\varepsilon + S_t)$ on $[S_{t-1}, S_t]$, $a_t/(\varepsilon + S_t) \le \log \frac{\varepsilon + S_t}{\varepsilon + S_{t-1}}$, and the sum telescopes. (iii) For $A \succ 0$ and $B \succeq 0$, the Cauchy–Schwarz inequality for $\langle X, Y\rangle = \operatorname{tr}(X^{\top}Y)$ with $X = A^{-1/4}B^{1/2}$ and $Y = A^{1/4}$, followed by $\sqrt{xy} \le \frac{1}{2}(x + y)$, gives

$$\operatorname{tr}(B^{1/2}) = \operatorname{tr}(X^{\top}Y) \le \operatorname{tr}(BA^{-1/2})^{1/2} \operatorname{tr}(A^{1/2})^{1/2} \le \tfrac{1}{2}\operatorname{tr}(BA^{-1/2}) + \tfrac{1}{2}\operatorname{tr}(A^{1/2}),$$

that is, $\operatorname{tr}\big((A - B)A^{-1/2}\big) \le 2\operatorname{tr}(A^{1/2}) - 2\operatorname{tr}(B^{1/2})$. Take $A = G_t$ and $B = G_{t-1}$ and sum. ∎

The constant 2 is best possible, as $a_t = 1$ gives $\sum_{t \le T} t^{-1/2} \ge 2\sqrt{T + 1} - 2$, and the current term must be included: with S_{t-1} for S_t, $a_1 = 1$ and $a_2 = M$ give at least M, which no multiple of $\sqrt{S_T} = \sqrt{1 + M}$ bounds.[8]

[8] The scalar lemma appears in the analysis of self-confident learning rates by Auer et al. (2002). Adaptive per-coordinate steps of this kind were introduced independently by McMahan and Streeter (2010) and by Duchi et al. (2011), who named the method AdaGrad and also analyzed its full-matrix version.

Theorem A.21 (Online gradient descent). *Let $\mathcal{K} \subseteq \mathbb{R}^d$ be nonempty, closed, and convex, $f_1, \dots, f_T$ convex with subgradients $g_t \in \partial f_t(x_t)$, $P_1, \dots, P_T \succ 0$, $x_1 \in \mathcal{K}$, and $x_{t+1} = \arg\min_{x \in \mathcal{K}} \|x - (x_t - P_t^{-1} g_t)\|_{P_t}$.*

(i) *(General preconditioners.) For every $u \in \mathcal{K}$, with $P_0 = 0$,*

$$\begin{aligned}\sum_{t=1}^{T} \big(f_t(x_t) - f_t(u)\big) &\le \sum_{t=1}^{T} \langle g_t, x_t - u\rangle \\ &\le \frac{1}{2}\sum_{t=1}^{T} (x_t - u)^{\top}(P_t - P_{t-1})(x_t - u) \\ &\quad + \frac{1}{2}\sum_{t=1}^{T} g_t^{\top} P_t^{-1} g_t.\end{aligned}$$

(ii) *(Step sizes.) If $P_t = I/\eta_t$ with $\eta_1 \ge \cdots \ge \eta_T > 0$ and $\mathcal{K}$ has Euclidean diameter D, then for every $u \in \mathcal{K}$*

$$\sum_{t=1}^{T} \big(f_t(x_t) - f_t(u)\big) \le \frac{D^2}{2\eta_T} + \frac{1}{2}\sum_{t=1}^{T} \eta_t \|g_t\|_2^2.$$

If $\|g_t\|_2 \le G$, the constant step $\eta = D/(G\sqrt{T})$ gives $DG\sqrt{T}$ and $\eta_t = D/(G\sqrt{t})$ gives $\frac{3}{2}DG\sqrt{T}$. With $S_t = \sum_{s \le t} \|g_s\|_2^2$, the adaptive step $\eta_t = D/\sqrt{2S_t}$ gives $\sqrt{2}\,D\sqrt{S_T}$, and the constant step $D/\sqrt{S_T}$, which needs S_T in advance, gives $D\sqrt{S_T}$.

(iii) *(Weighted form.) If $P_t = I/\eta_t$ with arbitrary $\eta_t > 0$, then for every*

$u \in \mathcal{K}$, with $\mathcal{K}$ possibly unbounded,

$$\sum_{t=1}^{T} \eta_t \big(f_t(x_t) - f_t(u)\big) \le \frac{1}{2}\|x_1 - u\|_2^2 - \frac{1}{2}\|x_{T+1} - u\|_2^2 + \frac{1}{2}\sum_{t=1}^{T} \eta_t^2 \|g_t\|_2^2,$$

and the corresponding bound on $\sum_t \eta_t \langle g_t, x_t - u\rangle$ is an equality when $\mathcal{K} = \mathbb{R}^d$.

Proof. The plan is to bound the progress toward u in one round and to sum. By Lemma A.2 with $P = P_t$ and $y = x_t - P_t^{-1} g_t$, $\|x_{t+1} - u\|_{P_t}^2 \le \|y - u\|_{P_t}^2 = \|x_t - u\|_{P_t}^2 - 2\langle g_t, x_t - u\rangle + g_t^\top P_t^{-1} g_t$, that is,

$$\langle g_t, x_t - u\rangle \le \tfrac{1}{2}\big(\|x_t - u\|_{P_t}^2 - \|x_{t+1} - u\|_{P_t}^2\big) + \tfrac{1}{2} g_t^\top P_t^{-1} g_t. \qquad \text{(A.6)}$$

(i) Summing (A.6) and regrouping the squared distances by the point x_t instead of the matrix, $\sum_t (\|x_t - u\|_{P_t}^2 - \|x_{t+1} - u\|_{P_t}^2) = \sum_t (x_t - u)^\top (P_t - P_{t-1})(x_t - u) - \|x_{T+1} - u\|_{P_T}^2$; drop the last term, and use $f_t(x_t) - f_t(u) \le \langle g_t, x_t - u\rangle$.

(ii) Here $(x_t - u)^\top (P_t - P_{t-1})(x_t - u) = (1/\eta_t - 1/\eta_{t-1})\|x_t - u\|_2^2 \le (1/\eta_t - 1/\eta_{t-1})D^2$, with $1/\eta_0 = 0$, and the sum telescopes to D^2/η_T. For the constant step both terms are $\frac{1}{2}DG\sqrt{T}$. For $\eta_t = D/(G\sqrt{t})$ the terms are $\frac{1}{2}DG\sqrt{T}$ and $\frac{1}{2}DG\sum_t t^{-1/2} \le DG\sqrt{T}$, by Lemma A.20(i) with $a_t = 1$. For the adaptive step both terms are at most $D\sqrt{S_T/2}$, by Lemma A.20(i) with $a_t = \|g_t\|_2^2$; rounds with $S_t = 0$ have $g_t = 0$, leave the iterate in place and contribute nothing, so we read $1/\eta_t = 0$ for them. The last choice makes both terms $\frac{1}{2}D\sqrt{S_T}$.

(iii) With $P_t = I/\eta_t$, multiply (A.6) by η_t and sum; the distances telescope whatever the steps. When $\mathcal{K} = \mathbb{R}^d$ there is no projection, and the first inequality of the proof is an equality. ∎

The preconditioners pay for their growth through the first sum in (i) and are rewarded through the second. The pieces of this appendix combine as follows for full-matrix AdaGrad, and in the same way for Shampoo in Chapter 7.

Remark A.22 (Full-matrix AdaGrad). Let $\varepsilon > 0$, $G_t = \varepsilon I + \sum_{s \le t} g_s g_s^\top$, and $P_t = G_t^{1/2}/\eta$ on a set $\mathcal{K}$ of Euclidean diameter D. By Theorem A.9, $P_t - P_{t-1} \succeq 0$, so $(x_t - u)^\top (P_t - P_{t-1})(x_t - u) \le D^2 \operatorname{tr}(P_t - P_{t-1})$, which sums to $D^2 \operatorname{tr}(P_T)$; and $\sum_t g_t^\top P_t^{-1} g_t \le 2\eta \operatorname{tr}(G_T^{1/2})$ by Lemma A.20(iii). So Theorem A.21(i) bounds the regret by $\big(\frac{D^2}{2\eta} + \eta\big)\operatorname{tr}(G_T^{1/2})$, which is $\sqrt{2}\, D \operatorname{tr}(G_T^{1/2})$ at $\eta = D/\sqrt{2}$.

Averaging turns regret bounds into bounds for stochastic optimization (Cesa-Bianchi et al. 2004); part (iii) allows an arbitrary schedule.

Corollary A.23 (Stochastic gradient descent on a convex function). *Let $\mathcal{L}$ be convex on a closed convex $\mathcal{K}$, $x_1 \in \mathcal{K}$, and $x_{t+1} = \arg\min_{x \in \mathcal{K}} \|x - (x_t - \eta_t g_t)\|_2$ with deterministic $\eta_t > 0$, where $\mathbb{E}[g_t \mid g_1, \dots, g_{t-1}] \in \partial \mathcal{L}(x_t)$ and*

$\mathbb{E}\|g_t\|_2^2 \le G^2$. *Then for every* $u \in \mathcal{K}$ *and* $\bar{x}_T = \sum_t \eta_t x_t / \sum_t \eta_t$,

$$\mathbb{E}\mathcal{L}(\bar{x}_T) - \mathcal{L}(u) \le \frac{\|x_1 - u\|_2^2 + G^2 \sum_{t\le T} \eta_t^2}{2 \sum_{t\le T} \eta_t}.$$

Proof. Apply Theorem A.21(iii) to the linear functions $f_t(x) = \langle g_t, x\rangle$. Since x_t is a function of $g_1, \dots, g_{t-1}$, conditioning on them and the subgradient inequality give $\mathbb{E}\langle g_t, x_t - u\rangle \ge \mathbb{E}[\mathcal{L}(x_t) - \mathcal{L}(u)]$. Take expectations and use Jensen's inequality $\mathcal{L}(\bar{x}_T) \le \sum_t \eta_t \mathcal{L}(x_t)/\sum_t \eta_t$. ∎

A.6 Martingales and tail bounds

Gradient noise is a sequence of martingale differences: given the past, a stochastic gradient is unbiased. The theorem below controls sums of such differences; the lemma after it bounds the probability that a noisy estimate has the wrong sign, which is what sign descent pays for noise.

Theorem A.24 (Doob's and the Azuma–Hoeffding inequalities). *Let* $(M_t)_{t=0}^T$ *be a martingale with respect to a filtration* $(\mathcal{F}_t)_{t=0}^T$.

(i) (Doob's maximal inequality.) *If* $\mathbb{E}M_T^2 < \infty$, *then for every* $a > 0$,

$$\Pr\Big(\max_{t\le T}|M_t| \ge a\Big) \le \frac{\mathbb{E}M_T^2}{a^2}, \qquad \mathbb{E}\Big[\max_{t\le T} M_t^2\Big] \le 4\,\mathbb{E}M_T^2.$$

The same holds in $\mathbb{R}^d$, *with* $|\cdot|$ *replaced by* $\|\cdot\|_2$, *and then* $\mathbb{E}\|M_T - M_0\|_2^2 = \sum_{t=1}^T \mathbb{E}\|M_t - M_{t-1}\|_2^2$.

(ii) (Azuma–Hoeffding inequality.) *If* $|M_t - M_{t-1}| \le c_t$ *almost surely for constants* $c_t > 0$, *and* $V = \sum_{t=1}^T c_t^2$, *then for every* $a > 0$,

$$\Pr\Big(\max_{t\le T}(M_t - M_0) \ge a\Big) \le e^{-a^2/(2V)}$$

and $\Pr(|M_T - M_0| \ge a) \le 2e^{-a^2/(2V)}$.

Proof. The plan is to prove a maximal inequality for nonnegative submartingales, processes with $\mathbb{E}[X_{t+1} \mid \mathcal{F}_t] \ge X_t$, by stopping at the first crossing, and to apply it to $\|M_t\|_2^2$ and to $e^{\kappa(M_t - M_0)}$. Let $X_t \ge 0$ be an integrable submartingale, $X^* = \max_{t\le T} X_t$ and $b > 0$. The events $E_t = \{X_t \ge b,\ X_s < b \text{ for } s < t\}$ are disjoint, lie in $\mathcal{F}_t$ and have union $\{X^* \ge b\}$, and $b\Pr(E_t) \le \mathbb{E}[X_t; E_t] \le \mathbb{E}[X_T; E_t]$. Summing over t,

$$b\,\Pr(X^* \ge b) \le \mathbb{E}\big[X_T;\, X^* \ge b\big]. \tag{A.7}$$

(i) By the conditional Jensen inequality, $\|M_t\|_2^2 \le \mathbb{E}[\|M_T\|_2^2 \mid \mathcal{F}_t]$ is integrable, and $\|M_t\|_2$ and $\|M_t\|_2^2$ are nonnegative submartingales. Then (A.7) for $\|M_t\|_2^2$ with $b = a^2$ gives the first bound. For $X_t = \|M_t\|_2$, (A.7) and the Cauchy–Schwarz inequality give $\mathbb{E}[(X^*)^2] = \int_0^\infty 2b\,\Pr(X^* \ge b)\,db \le 2\mathbb{E}[X_T X^*] \le 2(\mathbb{E}X_T^2)^{1/2}(\mathbb{E}(X^*)^2)^{1/2}$, and $\mathbb{E}(X^*)^2 \le \sum_t \mathbb{E}X_t^2 < \infty$, which gives

the second. The increments are orthogonal, $\mathbb{E}\langle M_t - M_{t-1}, M_s - M_{s-1}\rangle = 0$ for $s < t$, by conditioning on $\mathcal{F}_{t-1}$.

(ii) For $\kappa > 0$ and $|x| \le c$, convexity gives $e^{\kappa x} \le \cosh(\kappa c) + (x/c)\sinh(\kappa c)$, the chord through $x = \pm c$. As $\mathbb{E}[M_t - M_{t-1} \mid \mathcal{F}_{t-1}] = 0$, this yields $\mathbb{E}[e^{\kappa(M_t - M_{t-1})} \mid \mathcal{F}_{t-1}] \le \cosh(\kappa c_t) \le e^{\kappa^2 c_t^2/2}$, comparing the series of $\cosh y$ and $e^{y^2/2}$ term by term, and conditioning successively gives $\mathbb{E}e^{\kappa(M_T - M_0)} \le e^{\kappa^2 V/2}$. The nonnegative submartingale $e^{\kappa(M_t - M_0)}$ and (A.7) with $b = e^{\kappa a}$ give $\Pr(\max_t(M_t - M_0) \ge a) \le e^{-\kappa a + \kappa^2 V/2}$, and $\kappa = a/V$ gives the first bound; the second applies it to M and $-M$. ∎

For independent summands in intervals centered at their means, (ii) is Hoeffding's inequality (Hoeffding 1963), which Azuma (1967) extended to martingale differences. Doob's inequalities are from Doob (1953); Williams (1991) gives textbook proofs, and Boucheron et al. (2013) develop the exponential method much further.

Lemma A.25 (Sign errors: Markov, Chebyshev, and Gauss). *Let Y have mean $g \neq 0$ and variance $\sigma^2 < \infty$, for instance a stochastic estimate of one coordinate g of a gradient.*

(i) $\Pr(\operatorname{sign} Y \neq \operatorname{sign} g) \le \Pr(|Y - g| \ge |g|) \le \min\{\sigma/|g|, \sigma^2/g^2\}$, *so that* $|g| \Pr(\operatorname{sign} Y \neq \operatorname{sign} g) \le \sigma$. *If the law of Y is symmetric about g, then the first probability is half the second.*

(ii) (Gauss's inequality.) *Suppose that the law of Y is unimodal about v: apart from a possible atom at v, it has a density that is nondecreasing on $(-\infty, v)$ and nonincreasing on (v, ∞). Let $\tau^2 = \mathbb{E}(Y - v)^2$. Then for every $k > 0$,*

$$\Pr\big(|Y - v| \ge k\big) \le \begin{cases} 4\tau^2/(9k^2) & \text{if } k \ge 2\tau/\sqrt{3}, \\ 1 - k/(\sqrt{3}\,\tau) & \text{if } k \le 2\tau/\sqrt{3}. \end{cases}$$

(iii) If Y has a density that is symmetric about g and nonincreasing in $|y - g|$, then $\Pr(\operatorname{sign} Y \neq \operatorname{sign} g)$ *is at most $2\sigma^2/(9g^2)$ if $|g| \ge 2\sigma/\sqrt{3}$ and at most $\frac{1}{2} - |g|/(2\sqrt{3}\,\sigma)$ otherwise. Consequently* $|g| \Pr(\operatorname{sign} Y \neq \operatorname{sign} g) \le \frac{\sqrt{3}}{8}\sigma$, *with equality for the uniform law at $|g| = \frac{\sqrt{3}}{2}\sigma$.*

Proof. Part (i) is Markov's and Chebyshev's inequalities, and (iii) combines (i) and (ii); for (ii) the plan is to slice a unimodal density horizontally into uniform densities on intervals that end at the mode, and to check the inequality for each. (i) Let $g > 0$, replacing Y by $-Y$ if necessary. A sign error means $Y - g \le -g$, which implies $|Y - g| \ge g$; apply Markov's inequality with $\mathbb{E}|Y - g| \le \sigma$, and Chebyshev's. Under symmetry the disjoint events $\{Y - g \le -g\}$ and $\{Y - g \ge g\}$ have equal probabilities and make up $\{|Y - g| \ge g\}$.

(ii) Let $v = 0$, and let p be the density away from 0. For $s > 0$ the set $\{y > 0 : p(y) > s\}$ is an interval with left end 0, as p is nonincreasing; let $z(s)$ be its length, finite because $s\,z(s) \le 1$. Writing $p(y) = \int_0^\infty \mathbf{1}\{p(y) > s\}\,ds$,

Tonelli's theorem gives $\int_0^\infty \phi\, p = \int_0^\infty \int_0^{z(s)} \phi(y)\,\mathrm{d}y\,\mathrm{d}s$ for measurable $\phi \ge 0$, so that

$$\Pr(Y \ge k) = \int_0^\infty (z(s)-k)_+\,\mathrm{d}s, \qquad \Pr(Y>0) = \int_0^\infty z(s)\,\mathrm{d}s,$$

and $\mathbb{E}[Y^2; Y>0] = \int_0^\infty \frac{1}{3} z(s)^3\,\mathrm{d}s$. If $\alpha \ge 0$ and $\gamma > 0$ satisfy $(1-k/z)_+ \le \alpha + \gamma z^2$ for all $z > 0$, then $(z-k)_+ \le \alpha z + \gamma z^3$, and integrating gives $\Pr(Y \ge k) \le \alpha \Pr(Y>0) + 3\gamma \mathbb{E}[Y^2; Y>0]$; with the same bound for $-Y$, and since the atom at 0 contributes to neither side, $\Pr(|Y| \ge k) \le \alpha + 3\gamma\tau^2$. Take $z_0 = \max\{3k/2, \sqrt{3}\,\tau\}$, $\gamma = k/(2z_0^3)$ and $\alpha = 1 - 3k/(2z_0) \ge 0$. Then $q(z) = \alpha + \gamma z^2$ and $f(z) = 1 - k/z$ have the same value and slope at z_0; as f is concave and q convex, $f \le q$ on $(0,\infty)$, and $q \ge 0$, so $(1-k/z)_+ \le q(z)$. The bound $\alpha + 3\gamma\tau^2 = 1 - 3k/(2z_0) + 3k\tau^2/(2z_0^3)$ equals $4\tau^2/(9k^2)$ for $z_0 = 3k/2$, that is, $k \ge 2\tau/\sqrt{3}$, and $1 - k/(\sqrt{3}\,\tau)$ for $z_0 = \sqrt{3}\,\tau$.

(iii) Such a law is unimodal about g with $\tau = \sigma$, and by (i) and (ii) with $k = |g|$ the probability of a sign error is at most half the bound of (ii). With $x = |g|/\sigma$, the function $x(\frac{1}{2} - x/(2\sqrt{3}))$ is at most $\frac{\sqrt{3}}{8}$, its value at $x = \frac{\sqrt{3}}{2}$, and $x \cdot 2/(9x^2) \le \frac{\sqrt{3}}{9}$ for $x \ge 2/\sqrt{3}$. The uniform law on $[g - \sqrt{3}\sigma, g + \sqrt{3}\sigma]$ errs with probability $\frac{1}{2} - g/(2\sqrt{3}\sigma)$ for $0 < g \le \sqrt{3}\sigma$, which is $\frac{1}{4}$ at $g = \frac{\sqrt{3}}{2}\sigma$. ∎

For Gaussian noise, $\max_g |g| \Pr(\operatorname{sign} Y \ne \operatorname{sign} g)$ is about 0.17σ, against the unimodal bound $\frac{\sqrt{3}}{8}\sigma \approx 0.22\sigma$ and Markov's σ.[9] Since $\mathbb{E}[g \operatorname{sign} Y] \ge |g|\big(1 - 2\Pr(\operatorname{sign} Y \ne \operatorname{sign} g)\big)$, part (i) gives $\mathbb{E}\langle \nabla\mathcal{L}, \operatorname{sign}\tilde{g}\rangle \ge \|\nabla\mathcal{L}\|_1 - 2\sum_i \sigma_i$ for a stochastic gradient $\tilde{g}$ with coordinatewise noise levels σ_i, the bound used in Chapter 2; under unimodal symmetric noise, (iii) replaces $2\sum_i \sigma_i$ by $\frac{\sqrt{3}}{4}\sum_i \sigma_i$.

[9] Gauss (1823) proved the inequality in his work on least squares; Pukelsheim (1994) reviews it and gives an elementary proof of the Vysochanskiĭ–Petunin inequality, its analogue for deviations from the mean, and Bernstein et al. (2018) use it for signSGD under unimodal symmetric noise.

Exercises

A.1 CONSOLIDATION. Let $\theta = (\theta_1, \dots, \theta_N)$ be split into blocks with norms $\|\cdot\|_{(l)}$, and $\|\theta\| = \max_l \|\theta_l\|_{(l)}$. Show that $\|z\|_* = \sum_l \|z_l\|_{(l),*}$ and that an oracle can be computed block by block, $\operatorname{lmo}(g)_l = \operatorname{lmo}_{(l)}(g_l)$. What are the dual norm and an oracle of $\sum_l \|\theta_l\|_{(l)}$?

A.2 CONSOLIDATION. ⋆ Show that $g \mapsto g^\sharp$ is single-valued and linear if and only if $\|\cdot\| = \|\cdot\|_P$ for some $P \succ 0$. *(Hint: If $g^\sharp = Mg$, then $\|g\|_*^2 = \langle g, Mg\rangle$ by Lemma A.5; use Example A.6 and Lemma A.3(ii).)*

A.3 CONSOLIDATION. For $A, B \succ 0$, show that $(A\#B)^{-1} = A^{-1}\#B^{-1}$, and deduce $2(A^{-1} + B^{-1})^{-1} \preceq A\#B \preceq \frac{1}{2}(A+B)$.

A.4 CONSOLIDATION. Show that $\mathcal{L}(\theta) = \frac{1}{2}\|X\theta - y\|_2^2$ satisfies the PL inequality in the Euclidean norm with μ the smallest nonzero eigenvalue of $X^\top X$, even when $X^\top X$ is singular. *(Hint: $\Delta(\theta) = \frac{1}{2}\|\Pi(X\theta - y)\|_2^2$ for the orthogonal projection Π onto the range of X.)*

A.5 SIMULATION. For Gaussian, Laplace, uniform, and Student-t_3 noise of variance 1, compute $\max_{g>0} g \Pr(\operatorname{sign} Y \ne \operatorname{sign} g)$ by quadrature

and compare it with 1, with $\frac{1}{2}$, and with $\frac{\sqrt{3}}{8}$. You should find about 0.170, 0.130, 0.217, and 0.113; explain why the heavy-tailed law does best at fixed variance.

Notes and References

Rockafellar (1970) is the reference for dual norms, subdifferentials, and conjugates. Nesterov (2018, Section 2.1) proves the characterizations of smooth convex functions in the Euclidean norm, and Beck (2017) works with a general norm and its dual, as this book does; the notation $g^\sharp$ follows Kelner et al. (2014). For matrix analysis, Bhatia (1997) covers unitarily invariant norms, von Neumann's inequality, and operator monotone functions, Bhatia (2007) matrix means, and Horn and Johnson (2013) everything else; Horn and Johnson (1991, Chapter 4) treats Kronecker products and vectorization.

Online gradient descent is due to Zinkevich (2003), whose regret bound with $\eta_t \propto t^{-1/2}$ is a case of Theorem A.21(ii). Hazan (2016) and Orabona (2019) are modern introductions to online learning; Orabona's notes treat adaptive and parameter-free methods in the language of this appendix. For martingales and concentration see Williams (1991) and Boucheron et al. (2013).

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How This Book Was Made

Colophon

Roles

Optimizers in the Wild was written by Claude, an AI model made by Anthropic, in October 2026. Xiaoyu Li conceived the series *The Missing Textbooks* and proposed this volume: its subject, its gap (optimization textbooks stop at convex analysis and stationary points, while practice is decided by Adam, warmup, weight decay, learning-rate schedules, Muon and the edge of stability), its two lines (the unification of optimizers as steepest descent under norms with preconditioning and moving averages, and the dynamics of training), and its position relative to existing work. Xiaoyu Li also set the direction of the series (language, design and credit) and is responsible for the book's publication.

Process

A written plan fixed the thesis, the fifteen chapters, the global notation, the labels of the shared appendix, and a specification for each chapter, including the editor's plan for each main theorem; several of these plans were checked numerically before writing began, and two were corrected as a result. A style guide fixed the anatomy of a chapter, the voice, a proof style that follows Halmos, the distinction between proofs, sketches, heuristic derivations, observations and folklore, and the rule that every chapter compares an exact prediction with a simulation. Separate instances of the same model then drafted the chapters and the appendix in parallel, each re-deriving its results, checking them numerically, verifying its references by web search, and writing the scripts behind its figures. A coordinating instance wrote the prologue, preface, notation, epilogue and this colophon, designed the volume's additions to the series typography, and integrated the whole.

Each chapter then received an independent referee pass from a separate instance, which re-derived the proofs, re-ran the figure scripts, checked constants and exercises numerically, and checked historical claims and references against sources wherever these could be reached. The referees corrected the errors they found directly.

Limits

No human has yet checked every proof. Several results are presented as heuristic derivations, marked with ◇, because no proof is known to the author or because a proof would not fit; they are checked against simulations but not proved. In the environment in which the book was written, references could be checked only through web search, not by reading the papers themselves, so bibliographic details were included only

where a search confirmed them. Results from 2025 and 2026 are cited from preprints where no refereed version could be found, and are marked as such where a claim rests on them. Practices change quickly, and the recipes quoted in the book are dated. Corrections are welcome and will be recorded in the repository of *The Missing Textbooks*.

Typeset with pdfLaTeX in `missingbook`, the series class built on KOMA-Script, with the volume style `wild`; Palatino (newpx) for text and mathematics, biblatex and biber for references, Ti*k*Z and pgfplots for figures, whose data were computed with Python, NumPy, SciPy and PyTorch. The emblem on the title page shows one gradient and its three steepest-descent directions, the points of the ℓ_1, ℓ_2 and ℓ_∞ unit balls that make the largest inner product with it; the wave at the foot of each opening page rises from chapter to chapter.